

Microbiology AIIMS 2017

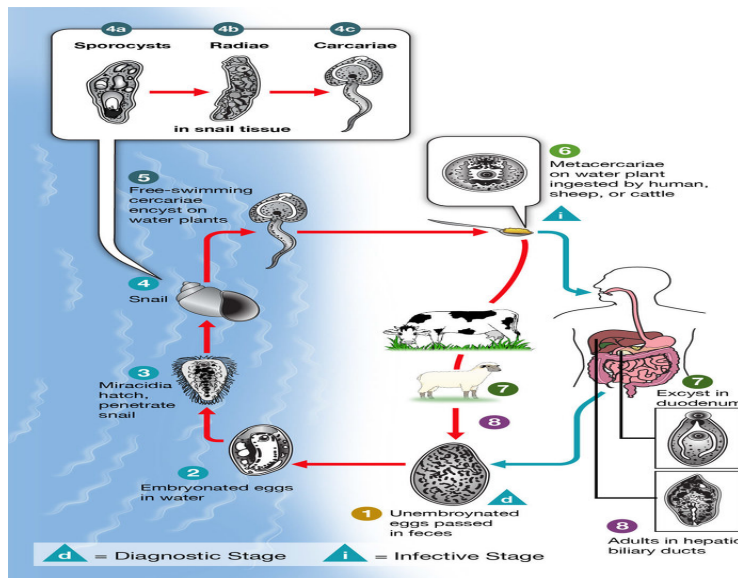
Question 1:

A 28 year old male admitted with history of altered sensorium and hydrophobia. A suspicion of rabies was made and you want to diagnose the condition. You have taken corneal scrapings from the patient and which of the following tests will you do to confirm your diagnosis

- a) Indirect immunofluorescence
- b) Seller stain for negri bodies
- c) Viral culture
- d) Real time PCR

Question 2:

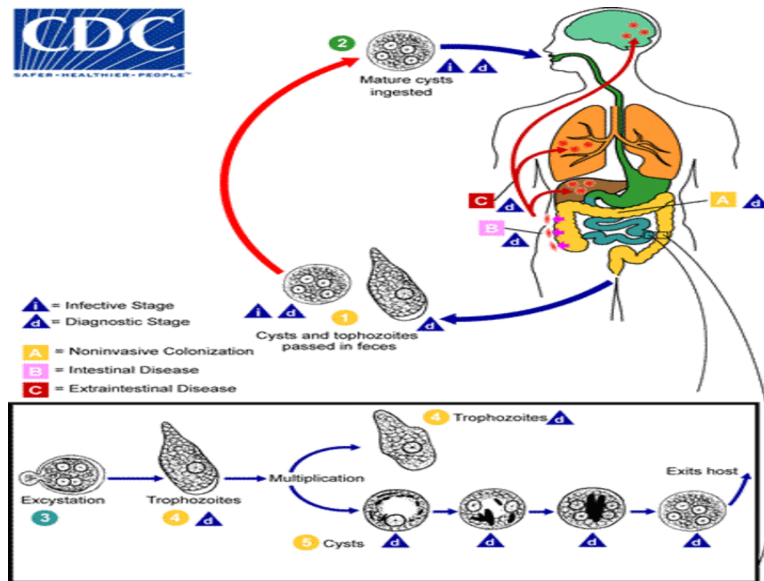
Identify the parasite from the given life cycle.



- a) Fasciola hepatica
- b) Fasciolopsis buski
- c) Pargaoimus westermani
- d) Clonorchis sinensis

Question 3:

The life cycle given below is of which of the following:



- a) *Entamoeba vermicularis*
- b) *Ancylostoma duodenal*
- c) *Balantidium coli*
- d) *Entamoeba histolytica*

Question 4:

A 25 year old male with history of fever and enlarged lymph nodes and sore throat was suspected to have infectious mono-nucleosis and was ordered a Paul Bunnell test. Which of the following is the principle of the test?

- a) Complement mediated cell damage
- b) Agglutination reaction
- c) Heterophile antibody test
- d) Neutralisation reaction

Question 5:

A blood spill in the operation theatre must be cleaned with _____.

- a) Chlorine compound
- b) Phenolic compound
- c) Quarternary ammonium compounds

- d) Alcoholic compounds

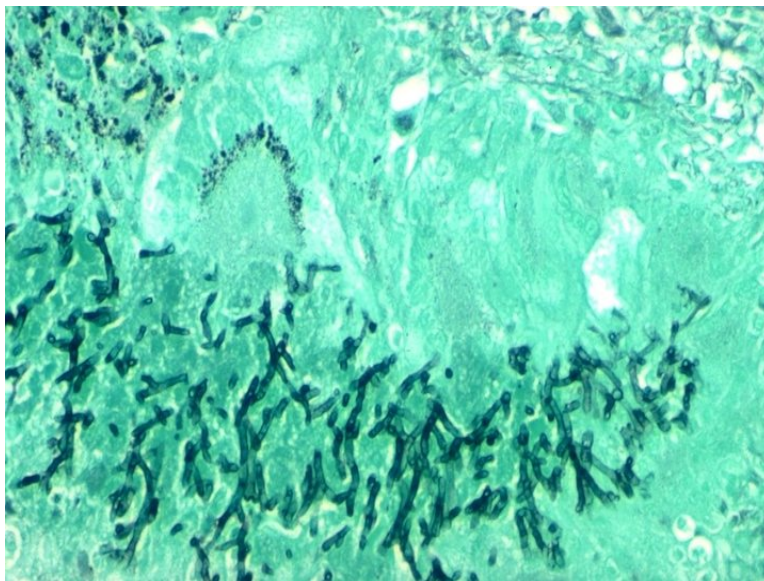
Question 6:

Urine LAM is used for the diagnosis of _____

- a) *M. tuberculosis*
- b) *M. leprae*
- c) *Mycoplasma*
- d) *Listeria monocytogenes*

Question 7:

Identify the organism in the given slide stained with Gomori methenamine silver stain.



- a) Obtuse angled mucor
- b) Acute angled penicillium
- c) Acute angled branching aspergillus
- d) Obtuse angled rhizopus

Question 8:

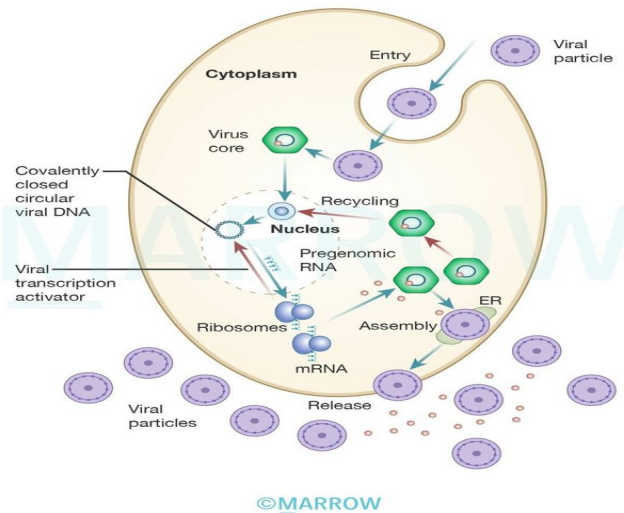
A confirmed case of which of the following viral haemorrhagic fevers was found in India?

- a) Ebola hemorrhagic fever

- b) Yellow fever
- c) Crimean congo hemorrhagic fever
- d) Marburg hemorrhagic fever

Question 9:

The given life cycle is suggestive of which of the following viruses?



- a) HIV
- b) Hepatitis B
- c) Influenza
- d) Herpes simplex

Question 10:

1,3 beta D- glucan is helpful for identification of?

- a) Invasive candidiasis
- b) Rhizopus
- c) Cryptococcus
- d) Mucormycosis

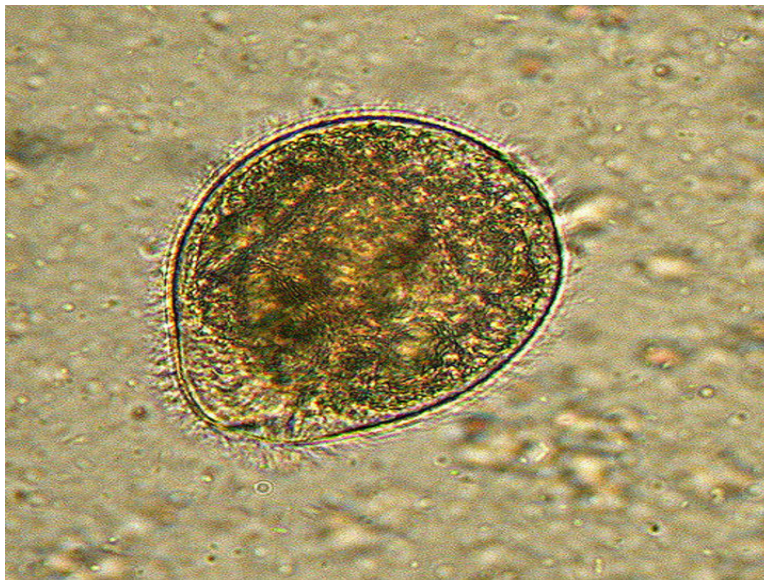
Question 11:

Relapse is seen in ?

- a) *Plasmodium vivax* and *falciparum*
- b) *Plasmodium ovale* and *falciparum*
- c) *Plasmodium vivax* and *ovale*
- d) *Plasmodium vivax* and *malariae*

Question 12:

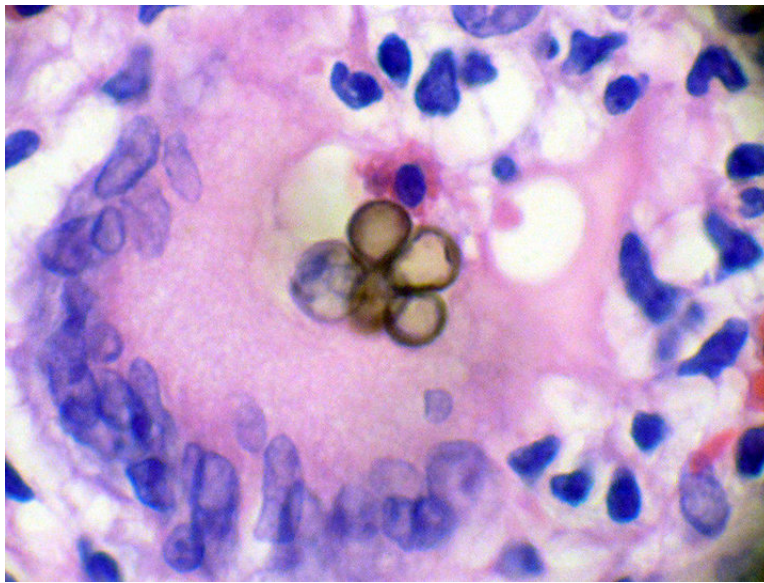
Identify the organism.



- a) *Balantidium coli*
- b) *Entamoeba dispar*
- c) *Cryptosporidium*
- d) *Giardia*

Question 13:

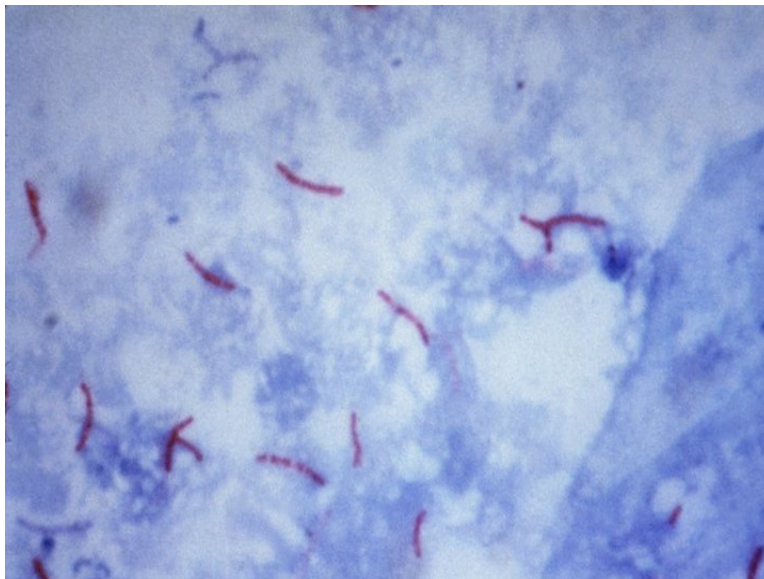
A female from Himachal Pradesh presented with H/O thorn prick, a year back has a verrucous lesion in the skin with the following microscopic findings. Identify the agent?



- a) Blastomycosis
- b) Pheohyphomycosis
- c) Chromoblastomycosis
- d) Sporotrichosis

Question 14:

The following slide is given as a spotter in your microbiology practical examination. Identify the following organism.

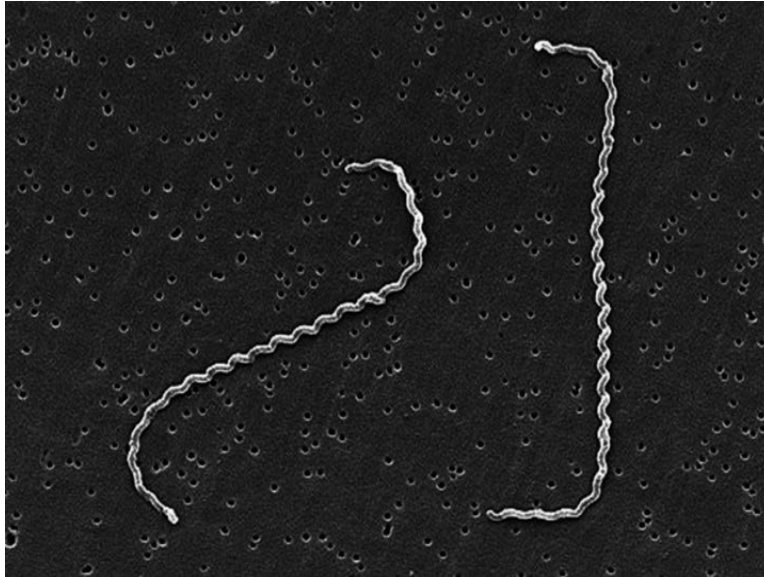


- a) Mycobacteria

- b) Nocardia
- c) Actinomyces
- d) Corynebacterium

Question 15:

Identify the organism from the given image?



- a) Vibrio
- b) Leptospira
- c) Helicobacter
- d) Salmonella

Question 16:

Which of the following organisms causes the following disease?



- a) Coxsackievirus
- b) Human herpes virus 7
- c) Pox virus
- d) Molluscum contagiosum

Question 17:

Which of the following is not a dimorphic fungus?

- a) *Blastomyces dermatidis*
- b) *Histoplasma capsulatum*
- c) *Pneumocystis jirovecii*
- d) *Penicillium marneffi*

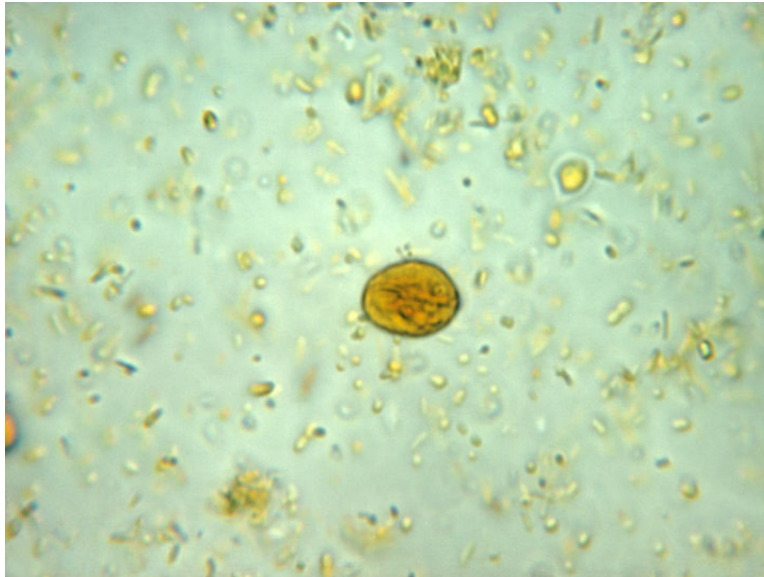
Question 18:

There are 3 to 5% healthy hepatitis B carriers in India who are asymptomatic. They are at the risk of developing hepatocellular carcinoma in the future because _____

- a) They are unable to mount inflammation against the virus
- b) Virus can integrate with host DNA and form complementary DNA
- c) There is a risk of elevation of transaminases
- d) Liver parenchymal cells are in a state of high proliferation

Question 19:

Identify the following?



- a) Entamoeba dispar
- b) Giardia lamblia
- c) Balantidium coli
- d) Entamoeba histolytica

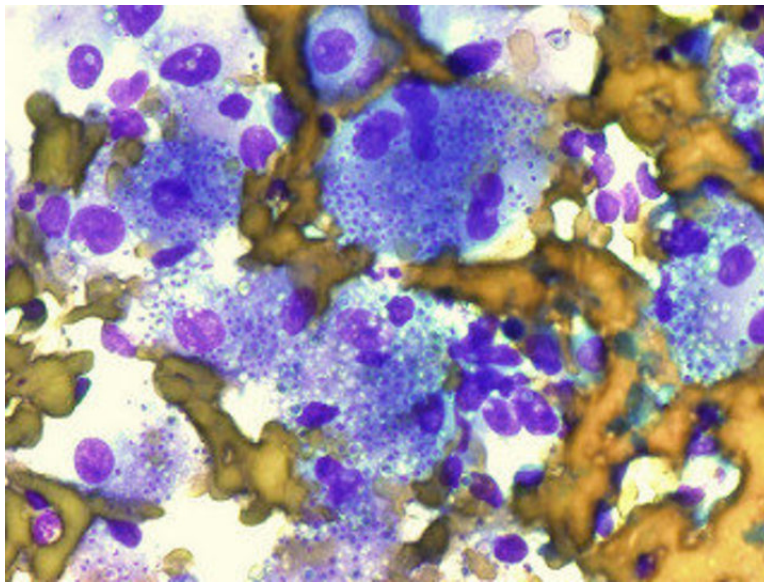
Question 20:

Best method of diagnosis for Clostridium difficile?

- a) Pure strain isolation from culture
- b) Immunofluorescence
- c) Toxin gene detection by PCR
- d) Toxin Detection by ELISA

Question 21:

A patient with HIV develops diarrhoea and faecal examination shows Isospora belli. He was treated with cotrimoxazole. Diarrhoea subsided but fever persisted. Bone marrow examination of this patient is shown in the given image. Which of the following statement is false?



- a) It cannot be grown in sabouraud dextrose agar
- b) Spores are infective form
- c) It is intracellular budding yeast
- d) It can cause systemic disease

Question 22:

Bio-film confer resistance to antibiotics by all except _____

- a) Mechanical barrier
- b) Increased excretion of antibiotics
- c) Slow metabolism
- d) Adherence

Question 23:

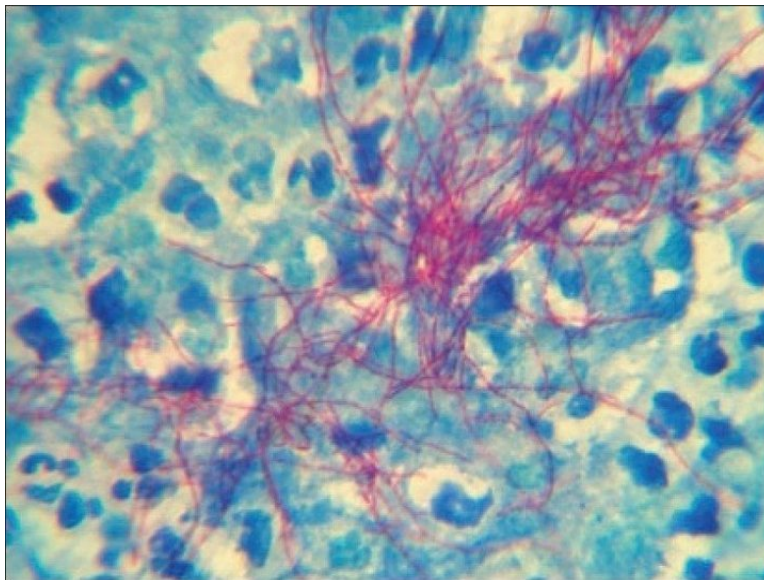
Identify the egg in the picture given below:



- a) *A. lumbricoides*
- b) *Strongyloides stercoralis*
- c) *E. vermicularis*
- d) *Ancylostoma duodenale*

Question 24:

Identify the organism from the given staining technique.



- a) *Mycobacterium leprae*
- b) *Mycobacterium tuberculosis*

- c) *Actinomyces israeli*
- d) *Nocardia asteroides*

Question 25:

Which of the following staphylococcal infections is not toxin mediated?

- a) Food poisoning
- b) Toxic shock syndrome
- c) Septicemia
- d) Staphylococcal scalded skin syndrome

Question 26:

All are features of Zika Fever except

- a) Petechial rash
- b) Guillan barre syndrome
- c) Fever with arthritis
- d) Conjunctivitis

Question 27:

(1–3)- β -D glucan assay can not be used in

- a) Invasive cryptococcosis
- b) Invasive aspergillosis
- c) *Pneumocystis jiroveci* pneumonia
- d) Invasive candidiasis and candidemia

Question 28:

A patient was sputum negative for tuberculosis, but chest X-ray was suggestive of tuberculosis. Which is the next step according to RNTCP ?

- a) Line probe assay
- b) CB - NAAT

- c) Liquid culture
d) None of the above

Answer Key

Question No.	Correct Option
1	c
2	a
3	d
4	c
5	a
6	a
7	c
8	c
9	b
10	a
11	c
12	a
13	c
14	a
15	b
16	a
17	c
18	b
19	b
20	c
21	a
22	d
23	d
24	d
25	c
26	a
27	a
28	b

Detailed Explanations

Solution to Question 1:

Viral culture from corneal scrapings is used to confirm diagnosis.

Multiple tests in succession/simultaneously are the best. Usually Direct Fluorescent Antibody test followed by virus isolation/culture is useful.

Virus isolation through culture: Best specimens are brain and saliva. CSF, corneal smears and centrifuged urine can also be used.

Option A: Serological tests like indirect immunofluorescence are insensitive.

Option B: Negri bodies can only be demonstrated in brain (pyramidal cells of Ammon's horn, and the Purkinje cells of the cerebellum are the most common site).

Option D: Real time-PCR Amplification Detection of rabies virus RNA is highly sensitive and specific. This technique can detect virus in fresh saliva samples, skin, CSF, and brain tissues. However it is not confirmatory test.

Solution to Question 2:

The given life cycle belongs to *Fasciola hepatica*.

Fasciola hepatica (sheep liver fluke) causes disease primarily in sheep and other domestic animals. Humans are infected by eating watercress (or other aquatic plants) contaminated by larvae (metacercariae) that excyst in the duodenum, penetrate the gut wall, and reach the liver, where they mature into adults.

The adult flukes in the bile ducts produce eggs, which are excreted in the feces. The eggs hatch in freshwater and miracidia enter the snails. Miracidia develop into cercariae, which then encyst on aquatic vegetation. Sheep and humans eat the plants, thus completing the life cycle. Most infections are asymptomatic. Halzoun is a painful pharyngitis caused by the presence of adult flukes on the posterior pharyngeal wall after eating raw sheep liver. Obstructive jaundice can occur months or years later.

Diagnosis is made by identification of the eggs in the feces.

Solution to Question 3:

The given life cycle is of *Entamoeba histolytica*.

Its life cycle includes 2 stages - the motile trophozoite and the non-motile cyst, both of which are seen in the image shown above. The transmission is primarily by the fecal-oral route. There is no animal reservoir. The most frequent site of systemic disease is the liver.

The ingested cysts differentiate into trophozoites in the ileum which invade the colonic epithelium. The trophozoites penetrate the submucosal layer and form abscesses which break down to form flask-shaped ulcers.

The trophozoites multiply by binary fission and produce cysts, and both stages are passed in the feces. Trophozoites are rapidly destroyed outside the body. Cysts survive in the external environment and are responsible for transmission.

Note: The genus *Entamoeba* has four morphologically identical species - *E. histolytica*, *E. dispar*, *E. bangladeshi*, and *E. moshkovskii*. *E. moshkovskii* which was initially thought to be non-pathogenic has recently been shown to be a causative agent of non-invasive diarrhea.

Solution to Question 4:

Paul Bunnell test is a heterophile antibody reaction test and is used in glandular fever like Infectious Mononucleosis.

Heterophile antibodies have the ability to agglutinate RBCs of different animal species.

Paul Bunnell test uses sheep erythrocytes for agglutination and Monospot test uses horse erythrocytes for agglutination.

In Infectious Mononucleosis, IgM heterophile antibodies are detectable for the first 3 months of infection and they can agglutinate sheep erythrocytes.

Type of test	Name of the test	Test used for
Tube agglutination	Widal	Typhoid
	Standard agglutination test	Brucellosis
	Microscopic agglutination test	Leptospirosis
	Paul-Bunnell test	Infectious mononucleosis
	Weil-felix test	Rickettsial infection
	Streptococcal MG agglutination test	Primary atypical pneumonia
Antiglobulin test	Direct coomb's test	Detecting fetal RH Antibody
	Indirect Coomb's test	Detecting maternal Rh antibody
Passive agglutination test	Hemagglutination	Rose Waaler test
	Latex agglutination test	Detection of ASO, CRP, RA factor, HCG
	Coagglutination test	Diagnosis of legionellosis, Gonorrhea

Solution to Question 5:

Sodium hypochlorite, a compound of chlorine, is the best disinfectant for surface blood spills. It is used with a concentration of 0.5-5%.

If the spill contains large amounts of blood or body fluids, clean the visible matter with disposable absorbent material and discard the contaminated materials in appropriate, labelled containment.

A 1:10–1:100 dilution of 5.25%–6.15% sodium hypochlorite (i.e., household bleach) or an EPA-registered tuberculocidal disinfectant has been recommended for decontaminating blood spills.

Solution to Question 6:

Urine lipoarabinomannan (LAM) assay is used for the diagnosis of tuberculosis.

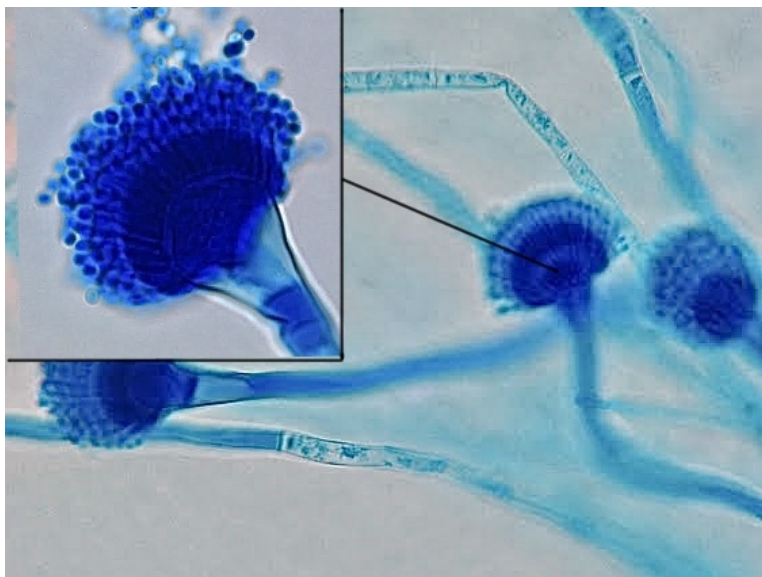
Lipoarabinomannan (LAM) antigen is a lipopolysaccharide present in mycobacterial cell wall. It is released from metabolically active cells or degenerating cells of patients having tuberculosis.

Urine LAM testing is especially useful in diagnosing TB in patients with HIV and low CD 4 counts.

Solution to Question 7:

The given slide stained with GMS shows acute-angled (45°) uniform, branching septate hyphae which is suggestive of *Aspergillus* species.

Aspergillus species grow rapidly, producing aerial hyphae that bear conidia in chains. Long aseptate conidiophores (specialised conidia-bearing hyphae) are seen with terminal vesicles. Phialides (vase-shaped cells producing conidia) with conidia are arranged in basipetal chains on the terminal vesicle. The following image shows the lactophenol cotton blue solution (LPCB) mount of *Aspergillus fumigatus* showing the long conidiophores with terminal vesicles.



Option A: *Mucor* produces nonseptate, broad hyphae with aerial sporangiophores which end in a sporangium.

Option B: *Penicillium* produces septate hyphae with branched, septate and brush-like conidiophores.

Option D: Rhizopus appears similar to Mucor but it has rhizoids with sporangiophores arising in groups directly above the rhizoids.

Solution to Question 8:

Crimean Congo hemorrhagic fever has been found in India.

Four cases were first described in India in 2011 in the state of Gujarat. It is a severe form of viral hemorrhagic fever caused by negative sense ssRNA Nairovirus species. It is zoonotic in origin with Ixodes ticks as vectors with livestock as primary reservoirs. The outbreak was caused due to person-to-person transmission in a tertiary care hospital where the index patient (first identified case) was admitted.

Solution to Question 9:

The given life cycle is of the hepatitis B virus.

Steps:

- The infectious virion attaches to cells and uncoating occurs.
- Cell enzymes in the hepatocytes convert dsDNA to covalently closed circular double-stranded DNA (cccDNA)
- HBV mRNA and pre-genome RNA is synthesized using cccDNA as a template.
- pre-genome RNA is encapsidated by packaging signal into new core particles.
- These new core particles are templates for HBV reverse transcriptase.
- Negative-strand DNA is synthesized by viral polymerase by reverse transcription.
- The cores bud from the pre-Golgi membranes, acquiring HBsAg-containing envelopes.
- Once viral particles are assembled, they either exit the cell or cause reinfection of the same cell.

Solution to Question 10:

1,3 beta D- glucan is helpful for the identification of Invasive candidiasis (invasive fungal infections).

1-3 beta D-glucan assay test detects the presence of (1-3)- β -d-glucan (BDG) which is an abundant cell wall polysaccharide found in most fungi except

- Cryptococci
- Zygomycetes
- Blastomyces dermatitidis

Solution to Question 11:

Relapse is seen in cases of *Plasmodium vivax* and *ovale*.

Malaria relapse :

In case of *Plasmodium vivax* and *ovale* two kinds of sporozoites are seen, some of which multiply inside hepatic cells to form schizonts and others persist and remain dormant as hypnozoites.

True relapse is caused due to hypnozoites (resting forms). Some get activated to become schizonts and release merozoites, which go on infecting RBCs producing clinical relapse.

In case of *Plasmodium falciparum* and *malariae*, the initial tissue phase disappears completely and no hypnozoites are found, hence relapse is not seen.

Solution to Question 12:

The given image is the trophozoite form of the *Balantidium coli*.

Balantidium coli occurs in two stages

- Trophozoites are oval in shape and contain two nuclei. There is the presence of cilia over the entire surface and are larger around the mouth. Its anterior end is narrow and the posterior end is broad. At the anterior end, there is a groove leading to the mouth (cytostome). Posteriorly, there is a small anal pore (cytopyge).
- Cysts are smaller, round with a micronucleus, a macronucleus and many contractile vacuoles.

Living trophozoites/cysts are yellowish or greenish in color on wet mount.

Solution to Question 13:

The given clinical history and histopathological image are suggestive of Chromoblastomycosis

Chromoblastomycosis (chromomycosis) is a subcutaneous mycotic infection that is usually caused by traumatic inoculation.

Over months to years, the primary lesion becomes verrucous and wart-like with extension along the draining lymphatics.

Histologically, the lesions are granulomatous and the dark sclerotic bodies may be seen within leukocytes or giant cells (medlar/copper penny bodies)

The following are the various dematiaceous species of fungi

- *Phialophora verrucosa*
- *Fonsecaea pedrosoi*
- *Fonsecaea compacta*
- *Rhinocladiella aquaspersa*
- *Cladophialophora carrionii*

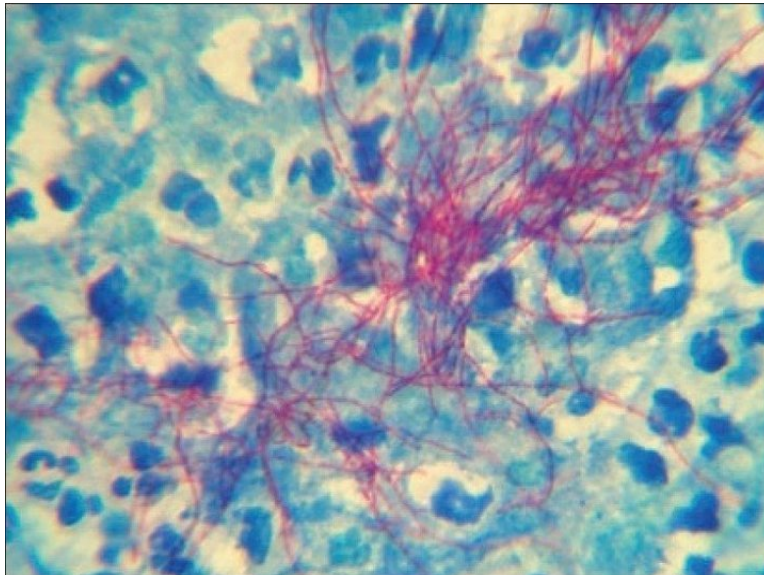
Treatment: Surgical excision with wide margins is the treatment of choice for small lesions. Chemotherapy with flucytosine or itraconazole may be efficacious for larger lesions. The application of local heat is also beneficial.

Solution to Question 14:

The image shows pink non-branching acid-fast bacilli on a blue background. These findings are consistent with the staining characteristics of mycobacteria. The stain used is the Ziehl-Neelsen stain.

Other options:

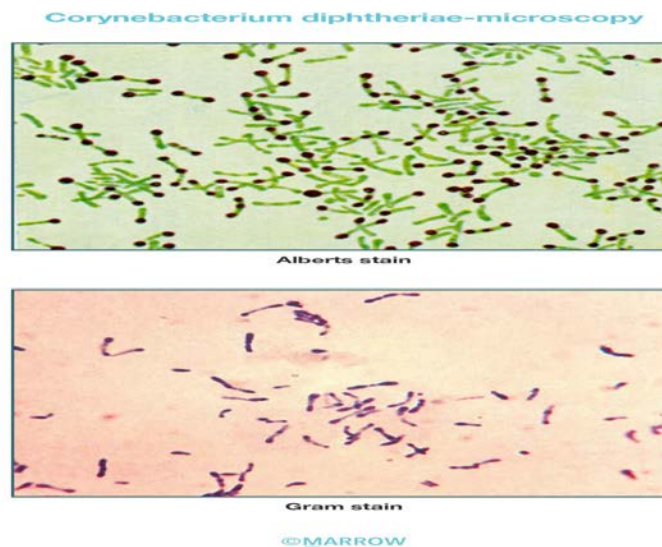
Option B: The image below shows *Nocardia* stained using an acid-fast staining technique. *Nocardia* is partially acid-fast and appears as branching filaments.



Option C: The image given below is of a gram-stained specimen of *Actinomyces israelii*. *A. israelii* is non acid-fast



Option D: The images below show the staining characteristics of *Corynebacterium diphtheriae*. It is stained with Albert's stain which shows the needle appearance.



Solution to Question 15:

The organisms in the image are leptospira.

In spirochetes, the presence of two axial filaments (periplasmic flagella) with polar insertions are located in periplasmic space called as endoflagella.

Endoflagella are seen in :

- Treponema
- Borrelia

- Leptospira

Solution to Question 16:

The image shows vesicles on the palms, soles, and oral mucosa, consistent with hand-foot-and-mouth disease. It is caused by coxsackievirus.

Hand-foot-and-mouth disease (HFMD) is an exanthematous fever affecting mainly young children, characterized by oral and pharyngeal ulcerations and a vesicular rash of the palms and soles that may spread to the arms and legs. Vesicles heal without crusting, which clinically differentiates them from the vesicles of herpesviruses and poxviruses. It is usually a benign illness resolving in 1-2 weeks.

Coxsackie A16, A9, and B 1-3 are common causative agents.

Other diseases caused by Coxsackie A virus:

- Aseptic meningitis is associated with a maculopapular rash.
- Herpangina
- Minor respiratory infections

Note: Conditions associated with the rash on palms and soles: Secondary syphilis, Hand foot, and mouth disease, Rickettsial fever caused by *Rickettsia rickettsiae*

Solution to Question 17:

Pneumocystis jirovecii is not a dimorphic fungi, it is a yeast like organism.

Dimorphic fungi :

- *Blastomyces dermatitidis*
- *Histoplasma capsulatum*
- *Paracoccidioides brasiliensis*
- *Penicillium marneffei*
- *Sporothrix schenckii*
- *Coccidioides immitis*
- *Coccidioides posadasii*

Solution to Question 18:

In most tumors, HBV DNA is integrated into the cell genome, and in most cases, the virus DNA has undergone rearrangements, including deletions. After integration, it promotes the overexpression of cell genes that can also contribute to oncogenesis.

The only viral gene product that is consistently present in the tumor cells is the X protein.

Solution to Question 19:

The above image shows giardia lamblia.

Cysts are resistant forms and are responsible for transmission of giardiasis. Both cysts and trophozoites can be found in the feces (diagnostic stages). The cysts are hardy and can survive several months in cold water.

Infection occurs by the ingestion of cysts in contaminated water, food, or by the fecal-oral route (hands or fomites).

In the small intestine, excystation releases trophozoites (each cyst produces two trophozoites).

Trophozoites multiply by longitudinal binary fission, remaining in the lumen of the proximal small bowel where they can be free or attached to the mucosa by a ventral sucking disk.

Encystation occurs as the parasites transit toward the colon. The cyst is the stage found most commonly in nondiarrheal feces. Because the cysts are infectious when passed in the stool or shortly afterward, person-to-person transmission is possible.

Giardia lamblia exists in :

- Infective form – Quadrinucleate cyst
- Tennis racket shaped trophozoite is actively motile with central axostyle with pair of nuclei
- Oval shaped cyst denoted with cyst wall

Solution to Question 20:

Nucleic acid amplification tests, including PCR assays, for C. difficile toxin A or B gene in stool have been approved for diagnostic purposes of Clostridium difficile infection due to high sensitivity, specificity and rapid turnaround time.

The diagnosis of CDI is based on a combination of clinical criteria:

- (1) diarrhea (≥ 3 unformed stools per 24 h for ≥ 2 days) with no other recognized cause plus
- (2) toxin A or B detected in the stool, toxin-producing C. difficile detected in the stool by PCR or culture, or pseudomembranes seen in the colon.

Other diagnostic methods include :

- Stool Culture in selective media
- Demonstrating Toxin by Hep-2 cell cultures
- Toxin gene identification by Enzyme immunoassay
- Colonoscopy or sigmoidoscopy

Despite the array of tests available for C. difficile and its toxins, no single traditional test has high sensitivity, high specificity, and rapid turnaround.

Most laboratory tests for toxins, including enzyme immunoassays, lack sensitivity.

Nucleic acid amplification tests, including PCR assays, for *C. difficile* toxin A or B gene in stool have now been approved for diagnostic purposes and appear to be both rapid and sensitive while retaining high specificity.

Solution to Question 21:

The given clinical findings and bone marrow examination is suggestive of *Histoplasma capsulatum*.

It can be grown in sabouraud dextrose agar and blood agar.

In SDA – white cottony mycelia growth can be seen.

Mode of infection – Inhalation of spores

Yeast cells occur within budding yeasts as intracellular fungi

Disseminated infection can occur in immunocompromised patients with reticuloendothelial system being affected the most.

Solution to Question 22:

Adherence is a mechanism of bacteria to attach to host cell for penetration. It is not a mechanism of resistance.

Mechanisms of biofilm-mediated resistance to antibiotics:

- Mechanical barrier: The lipopolysaccharide biofilms prevent entry of water-soluble drugs.
- Efflux pumps: These pumps on the biofilms actively excrete drug out.
- Slow metabolism: Decreased oxygen and nutrients passage through the biofilms slows down the metabolism of bacteria.
- Anaerobic pathway: Because of the above reason bacteria follow anaerobic metabolism.
- Persisters: Some bacteria undergo phenotypic modification into dormant, spore-like structure and survive antibiotic insult. These are called as persisters which are activated later.
- Enzymatic inactivation of drugs by biofilms
- Genetic adaptation to antibiotic insults
- Quorum sensing: It is cell to cell signaling among the bacteria to coordinate among themselves for contribution in biofilm synthesis.

Solution to Question 23:

The egg in the given picture is that of a hookworm. Eggs of both *Ancylostoma Duodenale* (old world hookworm) and *Necator Americanus* (new world hookworm) are identical.

The eggs of hookworm are :

- Oval or elliptical
- Colorless
- Not bile stained
- With a thin transparent hyaline shell membrane
- Thin shell
- Segmented ovum - 4 to 8 blastomeres
- There is clear space between segmented ovum and egg shell.
- Eggs float in saturated salt solution.

Humans are infected when filariform larvae in moist soil penetrate the skin.

Solution to Question 24:

The organism shown above is the *Nocardia* species. The above image shows acid-fast staining, where the organisms are shown appear as acid-fast (stained with fuschia) and branching rods or filaments. They are aerobes found particularly in the soil and are found to produce lung infections in the immunocompromised.

Options A, B: *Mycobacteria* species are acid-fast but do not have branching filaments.

Option C: *Actinomyces* species is gram-positive but not acid-fast.

Solution to Question 25:

Staphylococcal septicemia is not toxin mediated.

Septicemia occurs due to the invasion of bacterial organisms into the bloodstream. This causes an inflammatory response to bacterial products, leading to various manifestations seen in sepsis - fever, hypotension, alkalosis.

Toxin mediated staphylococcal infections include food poisoning caused by enterotoxin, toxic shock syndrome caused by toxic shock syndrome toxin and staphylococcal scalded skin syndrome caused by exfoliative/epidermolytic toxin.

Solution to Question 26:

Petechial rash is not a feature of Zika Virus. Though petechiae may rarely appear in Zika virus disease it is however not included in the CDC case definition of Zika virus disease. Petechial rash points towards a diagnosis of Dengue fever while a maculopapular rash suggests a Zika virus

disease.

Zika virus infection is characterised by influenza-like clinical signs including fever, headache and malaise. A maculopapular rash, conjunctivitis, myalgia and arthralgia usually accompany or follow those manifestations.

The complications of Zika virus are:

- Guillain Barre Syndrome
- Fetal microcephaly in pregnant women.

Solution to Question 27:

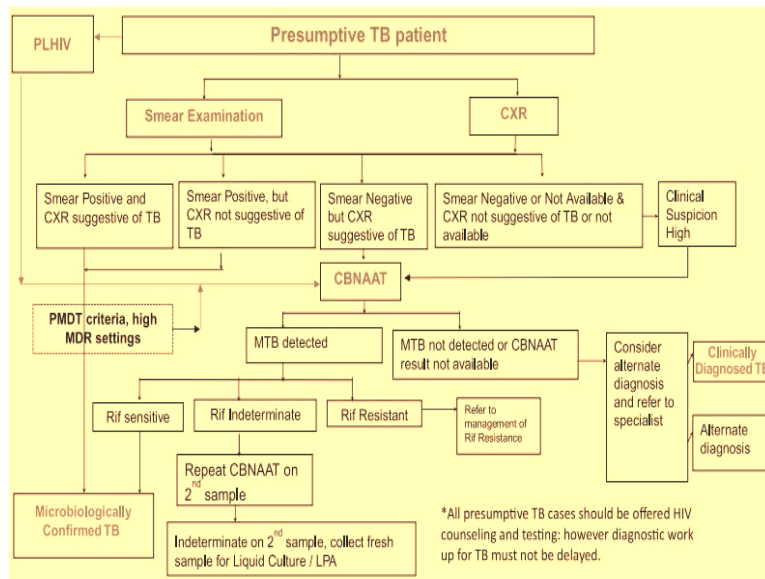
(1-3) - β - D glucan assay is used for diagnosis of candida infections, invasive aspergillosis and Pneumocystis jirovecii pneumonia.

It is not useful in case of invasive cryptococcosis.

Solution to Question 28:

- CB-NAAT and Line probe assay are rapid molecular diagnostic techniques used for diagnosis of tuberculosis and also to assess resistance to isoniazid and rifampicin.
- CB NAAT (Cartridge Based Nucleic Acid Amplification Test) provides rapid and accurate diagnosis of TB by detecting Mycobacterium tuberculosis and mutations responsible for rifampicin resistance.
- It can be used for sputum samples as well as samples from extrapulmonary sites.
- It is mainly used for diagnosis of DR-TB in presumptive DR-TB patients and TB in key populations such as pediatric population, PLHIV and extrapulmonary TB.
- It is done when smear is negative for TB but chest X-ray is suggestive of tuberculosis and also in cases smear is negative and chest X-ray is not suggestive of TB but there is high index of clinical suspicion.

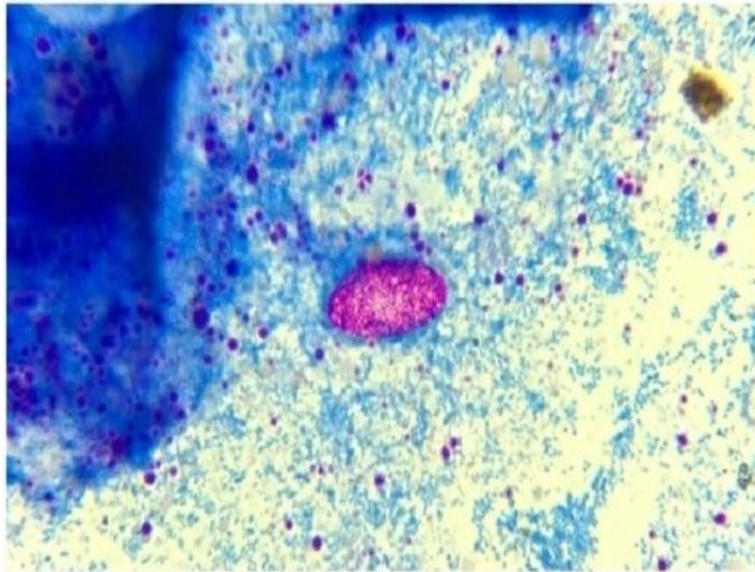
Diagnostic algorithm for pulmonary TB:



Microbiology AIIMS 2018

Question 1:

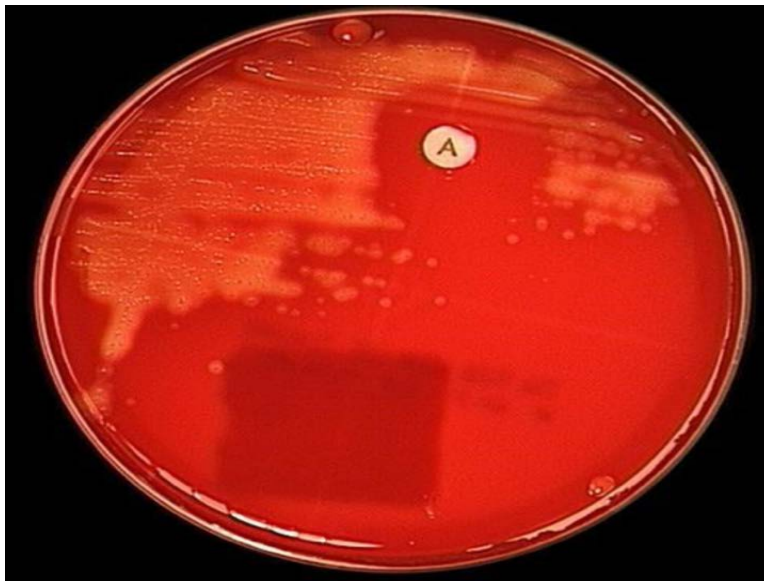
A kinyoun positive oocyst (25 μm) was isolated from an HIV patient with diarrhea. Identify the organism.



- a) Cystoisospora
- b) Cyclospora
- c) Cryptosporidium
- d) Microspora

Question 2:

The test shown below, which uses a bacitracin disk, is useful in identifying _____.



- a) *Staphylococcus aureus*
- b) *Streptococcus pyogenes*
- c) *Clostridium botulinum*
- d) *Bacillus*

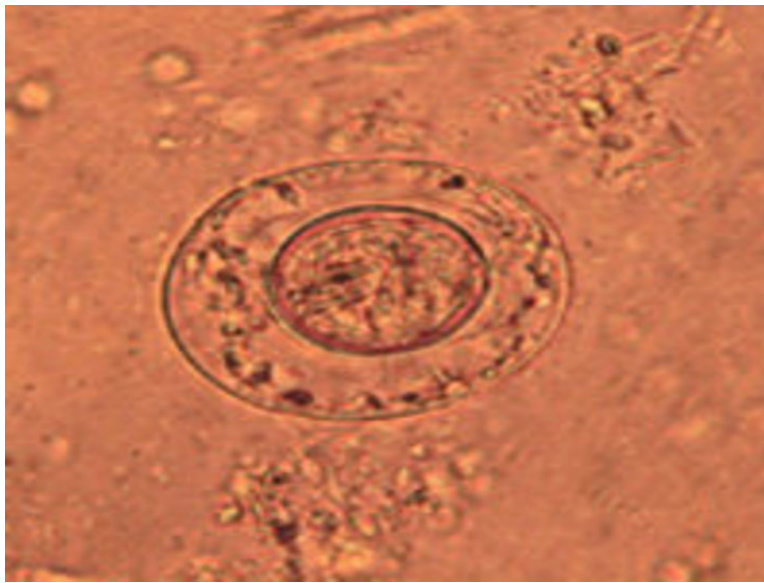
Question 3:

What is the incubation period of lymphogranuloma venereum?

- a) 1-5 days
- b) 7-10 days
- c) 10-30 days
- d) 30-60 days

Question 4:

Which is the agent used to treat the infection caused by the cestode egg in the image?



- a) Praziquantel
- b) Ivermectin
- c) Albendazole
- d) Diethylcarbamazine

Question 5:

Human herpesvirus 8 is associated with all except _____.

- a) Kaposi sarcoma
- b) Castleman's disease
- c) Adult T-cell leukemia
- d) Primary effusion lymphoma

Question 6:

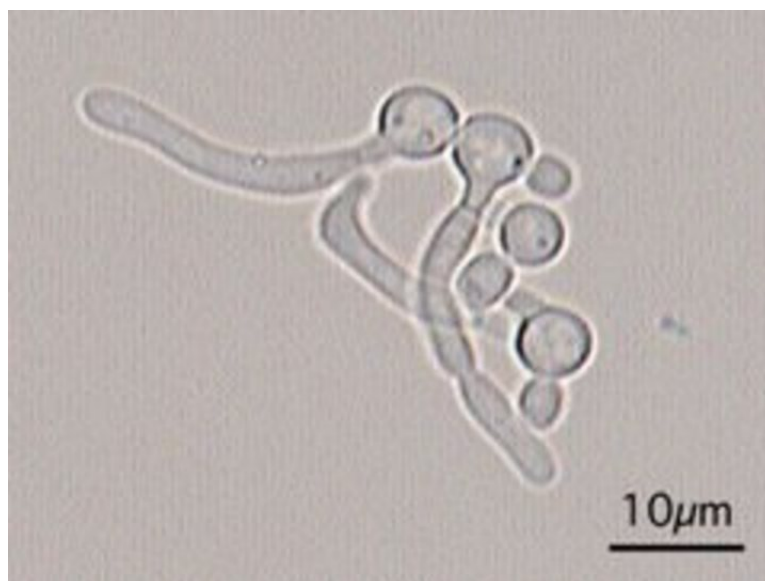
Identify the condenser in the following image.



- a) Above the stage
- b) Below the stage
- c) Near the eyepiece
- d) Above the objective lenses

Question 7:

The following test is positive for identification of _____



- a) *Candida albicans*
- b) *Candida glabrata*

- c) *Candida parapsilosis*
- d) *Candida tropicalis*

Question 8:

Choose the correct statement regarding CMV.

- a) Ganciclovir resistance is due to mutation in UL97 phosphotransferase gene
- b) CMV can be cultured in vitro in any mammalian cell
- c) Commonest manifestation of CMV associated disease is watery diarrhea
- d) UL54 mutated CMV strains do not show cross resistance to foscarnet

Question 9:

Solvent/Detergent (SD) plasma destroys lipid-enveloped viruses. So, on SD plasma transfusion, which of the following infection is likely to occur?

- a) Human Immunodeficiency Virus (HIV)
- b) Hepatitis A Virus (HAV)
- c) Hepatitis B Virus (HBV)
- d) Hepatitis C Virus (HCV)

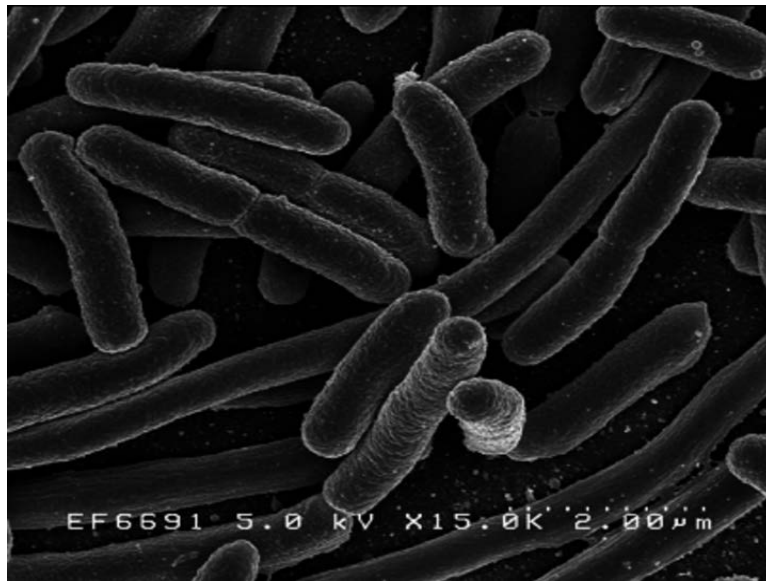
Question 10:

Plasma sterilization makes use of _____.

- a) Hydrogen peroxide
- b) Ethylene oxide
- c) Gamma rays
- d) UV rays

Question 11:

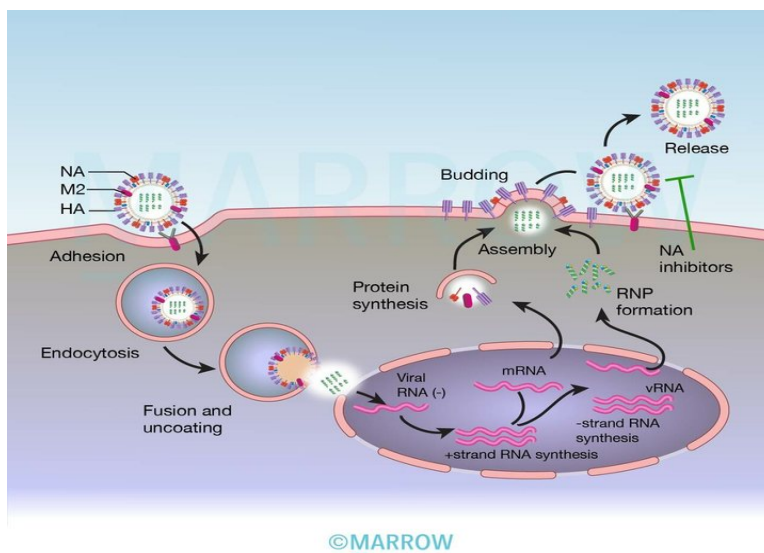
Identify the organism below that causes effacement and attachment of intestinal cells.



- a) ETEC
- b) EIEC
- c) EPEC
- d) EAEC

Question 12:

Which of the following organisms is represented by the given form of replication?

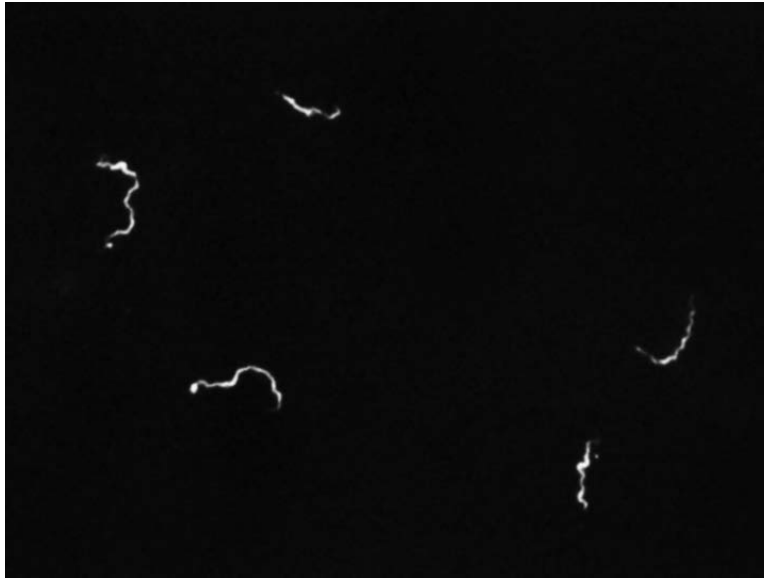


- a) Hepatitis virus
- b) Influenza virus

- c) HIV
- d) Herpes virus

Question 13:

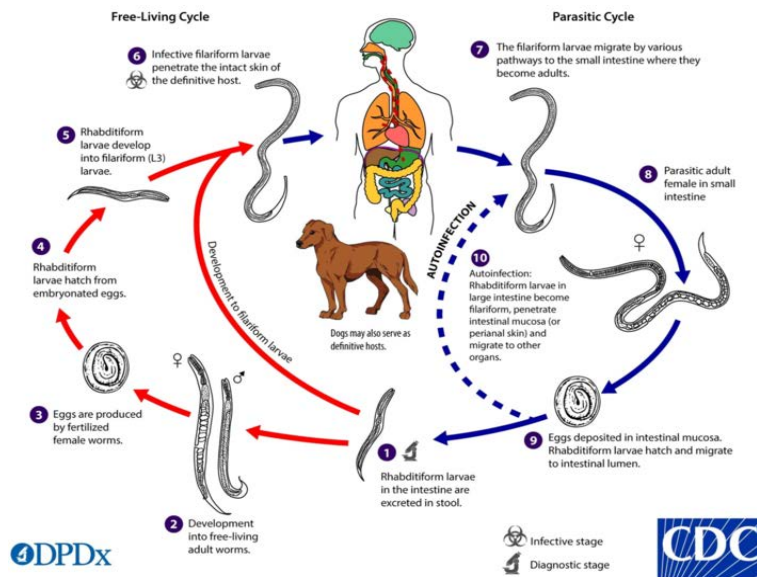
Which method of diagnosis is shown in the image?



- a) Darkfield microscopy
- b) Phase contrast microscopy
- c) Fluorescent microscopy
- d) Electron microscopy

Question 14:

Identify the helminth from the life cycle given below.



- a) *Ancylostoma duodenale*
- b) *Strongyloides stercoralis*
- c) *Ascaris lumbricoides*
- d) *Enterobius vermicularis*

Question 15:

Northern blotting is used for detection of which of the following?

- a) DNA
- b) RNA
- c) Proteins
- d) None

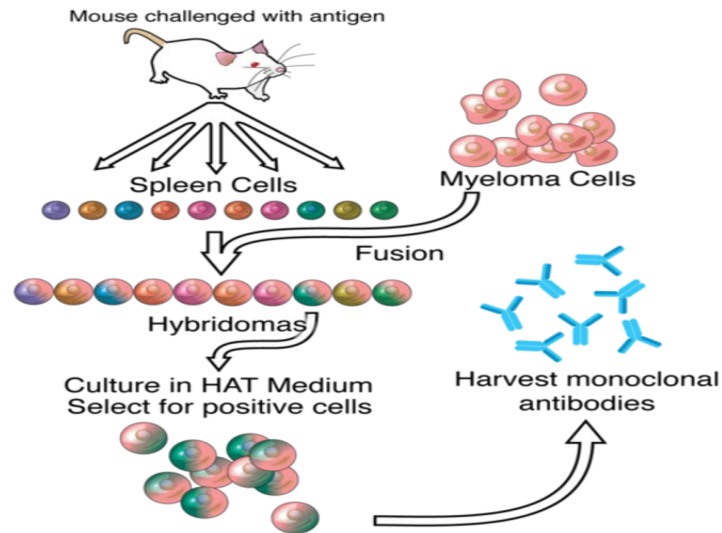
Question 16:

A cook prepares sandwiches for 10 people going for picnic. Eight out of them develop severe gastroenteritis within 4-6 hrs of consumption of the sandwiches. It is likely that on investigations the cook is found to be the carrier of:

- a) *Salmonella typhi*
- b) *Vibrio cholera*
- c) *Entamoeba histolytica*
- d) *Staphylococcus aureus*

Question 17:

The following image depicts:



- a) Production of monoclonal antibodies
- b) Cell lines for viral culture
- c) Vaccine synthesis
- d) Process of genetic engineering

Question 18:

A patient has complains of fever with chills. A throat examination reveals swollen glands and a thick, whitish membrane over his tonsils. Albert staining is to be done. What are the constituents of Albert's stain?

- a) Carbol fuschin and methylene blue
- b) Carbol fuschin and malachite green
- c) Toluidine and malachite green
- d) Toluidine and methylene blue

Question 19:

Which of the following infection presents with mucus in stools and abdominal pain?

- a) Giardia
- b) Entamoeba
- c) Campylobacter
- d) Staphylococcus aureus

Question 20:

A patient's plasma has high levels of interferon beta. He most likely has an infection due to

- a) Virus
- b) Bacteria
- c) Fungi
- d) Mycoplasma

Question 21:

Which of the following is the most potent stimulator of naive T-cells?

- a) Mature dendritic cells
- b) Follicular dendritic cells
- c) B lymphocytes
- d) NK cells

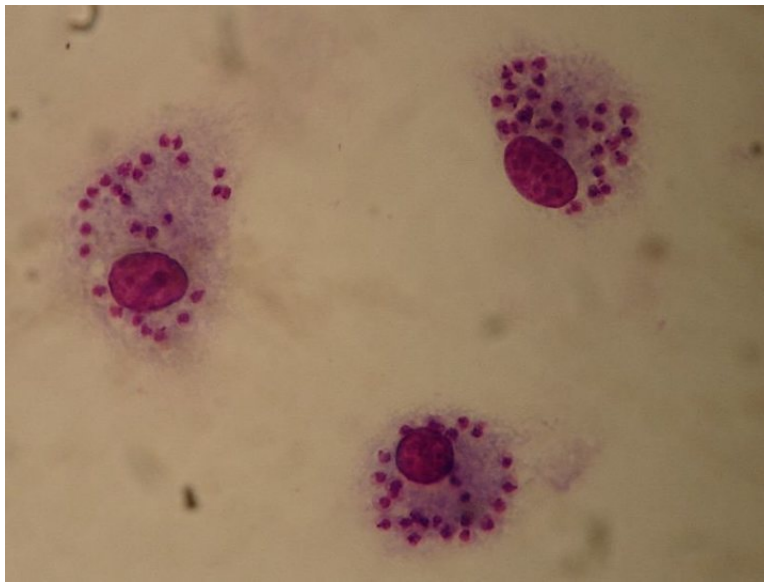
Question 22:

Which of the following is an immunologically privileged site?

- a) Seminiferous tubule
- b) Area postrema
- c) Optic nerve
- d) Loop of Henle

Question 23:

Bone marrow aspirate of a patient shows the following features. Which of the following is true?



- a) Buffy coat is used to diagnose the parasite
- b) It is an obligate extracellular parasite
- c) It is transmitted from sheep
- d) Stibogluconate is the first line of treatment in India.

Question 24:

Which of the following drugs is not used for typhoid fever in adults?

- a) Azithromycin
- b) Ciprofloxacin
- c) Ceftriaxone
- d) Amikacin

Answer Key

Question No.	Correct Option
1	a
2	b
3	c
4	a
5	c

6	b
7	a
8	a
9	b
10	a
11	c
12	b
13	a
14	b
15	b
16	d
17	a
18	c
19	b
20	a
21	a
22	a
23	a
24	d

Detailed Explanations

Solution to Question 1:

Diarrhea in an HIV patient is suggestive of a coccidian parasitic infestation. Kinyoun stain positive oocysts in the image are identified as *Cystoisospora*.

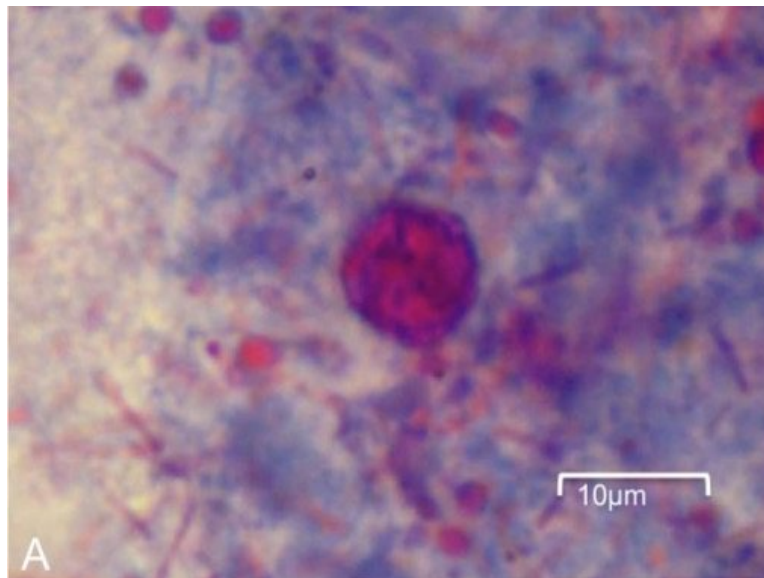
Points in favor:

- HIV patient with diarrhea
- Oocysts are long and oval-shaped, which stain pink to red with Kinyoun staining.

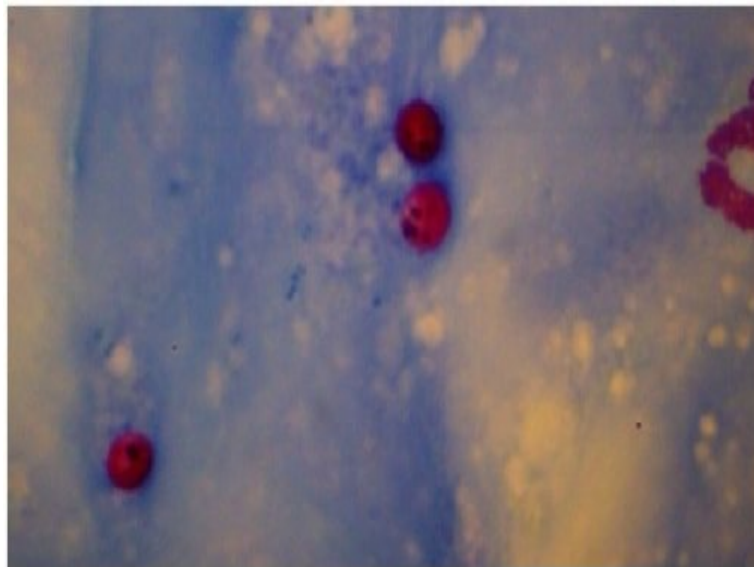
Comparison of coccidian parasites causing diarrhea in an immunocompromised host:

Property	<i>Cryptosporidium</i>	<i>Cyclospora</i>	<i>Cystoisospora</i>
Sporulated oocyst	4-6 μm , round, contains four sporozoites	8-12 μm , round, contains 2 sporocyst each having two sporozoites	23-36 μm , oval, contain two sporocysts, each having four sporozoites.
Acid-fast detection	Uniformly acid-fast	Variable acid-fast	Uniformly acid-fast
Sporulation of the oocyst	Occurs inside the host cell	Occurs in soil	Occurs in soil

Property	Cryptosporidium	Cyclospora	Cystoisospora
Diagnostic form	Sporulated cyst	Unsporulated cyst	Unsporulated oocyst



Modified acid-fast staining (like Kinyoun stain) showing oocysts of *Cyclospora cayetanensis*.



Modified acid-fast staining (like Kinyoun stain) showing oocysts of *Cryptosporidium* species.

In immunocompromised patients, the infection causes severe diarrhea similar to cholera and require hospitalization.

The diagnosis is made with stool microscopy. The cyst wall and sporoblasts are dyed red with the Ziehl–Neelsen (warm) or Kinyoun (cold) staining methods. Both methods have good specificity and therefore are useful as confirmatory tests.

The treatment is fluid replacement and cotrimoxazole.

Solution to Question 2:

The given image shows blood agar culture with bacitracin disk with a zone of inhibition around the disk, which means that the organism is sensitive to bacitracin.

Among the options given, *Streptococcus pyogenes* is susceptible to bacitracin.

Streptococci belonging to group A are identified (presumptively) by inhibition of growth by bacitracin.

Solution to Question 3:

The incubation period of Lymphogranuloma venereum ranges from 10-30 days.

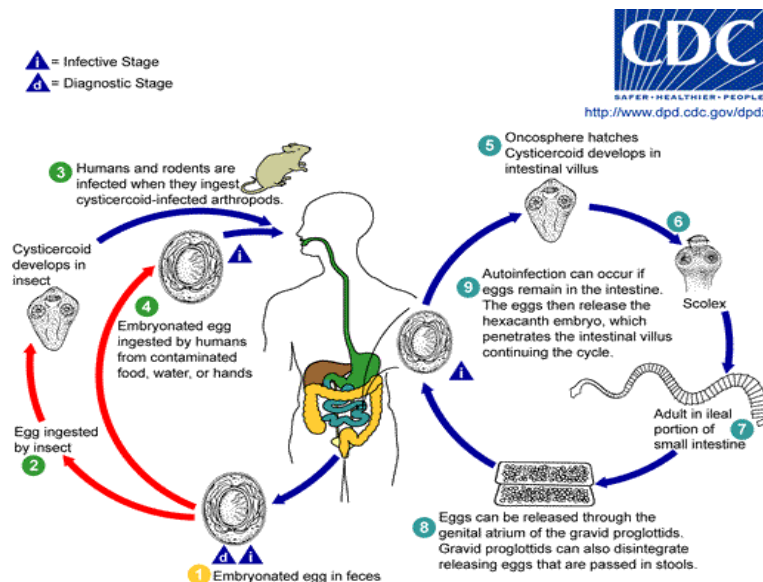
Chlamydia trachomatis serovars L1, L2, and L3 cause lymphogranuloma venereum (LGV). The most common presentation is the inguinal syndrome, which is characterized by painful inguinal lymphadenopathy.

Solution to Question 4:

The egg in the image is of *Hymenolepis nana* (dwarf tapeworm) and it is treated by praziquantel (5–10 mg/kg once).

The egg is oval, colorless, non-bile stained, and has a single shell with two distinct membranes. Inner membrane has two small knobs or poles from which 4 to 8 polar filaments arise and spread out between the two membranes. It is embryonated and the oncosphere has six hooks (three pairs of hooklets).

Life cycle of *Hymenolepis nana*:



Hymenolepiasis is the most common of all the cestode infections and is usually asymptomatic.

Solution to Question 5:

Adult T-cell leukemia is caused by HTLV-1 (and not HHV8).

Human Herpesvirus 8 (HHV8), also known as Kaposi sarcoma-associated herpesvirus (KSHV), is lymphotropic virus and is closely related to Epstein–Barr virus (EBV).

Diseases caused by HHV:

- Kaposi sarcomas
- Vascular tumors of mixed cellular composition
- Body cavity-based lymphomas occurring in AIDS patients (Primary effusion lymphoma)
- Multicentric Castleman disease

Solution to Question 6:

This is a modern compound microscope, and the condenser is located below the stage.



The purpose of the condenser lens is to focus the light from the illuminator onto the specimen. Microscopes with a stage condenser lens render a sharper image than those with no condenser lens.

The simple microscope was invented first by Antonie van Leeuwenhoek, the father of microscopy and microbiology. It uses 1 lens for magnification. The compound microscope, introduced by Hans and Zacharias Janssen uses 2 lenses, an objective lens and an eyepiece, to provide better magnification power than the simple microscope.

Solution to Question 7:

The image shows the germ tube test used for identifying *Candida albicans*.

Candida species, when cultured, grow as oval, budding yeast cells with pseudohyphae. Unlike other species of *Candida*, *Candida albicans* can also produce true hyphae or germ tubes. This can be demonstrated after incubation in serum maintained at 37°C for about 90 minutes. This is known as the Reynolds-Braude phenomenon. This test is positive for *Candida dubliniensis* as well. Another distinguishing feature of *C. albicans* is that it produces large, spherical chlamydospores on nutritionally deficient media.

Solution to Question 8:

Resistance to ganciclovir most commonly occurs due to mutations in the UL97 phosphotransferase gene.

Cytomegalovirus (CMV) is a beta-herpes virus. It is the most common congenital infection. In adults, its disease is mostly subclinical (option C). An apparent CMV infection resembles infectious mononucleosis (fever, sore throat, enlarged lymph nodes and splenomegaly). It is the most common infection in the donor organ seen 1-4 months after solid organ transplantation. In immunosuppressed hosts, it causes pneumonia and colitis.

Diagnosis is by PCR assays. Human CMV replicates in vitro only in human fibroblasts (option B) and can thus be cultured in them, although the virus is often isolated from epithelial cells of the host. The cultured fibroblasts show owl-eye inclusion bodies after 2-3 weeks.

UL54 (DNA polymerase) gene mutations can confer CMV resistance to ganciclovir, cidofovir and foscarnet (option D).

Solution to Question 9:

Solvent detergent plasma is a form of blood plasma made from plasma collected from many people which is then processed with solvents to try to get rid of viruses. HAV is a non-enveloped virus, and hence it's not affected by solvent/detergent (SD) plasma.

Only the viruses with an envelope are susceptible to SD plasma destruction.

Hepatitis B, C, and HIV are enveloped.

Solution to Question 10:

A variety of hydrogen peroxide gas-based processes that use plasma as part of the sterilization process are known as "plasma sterilization."

The mechanisms of antimicrobial action in these processes are primarily due to hydrogen peroxide gas (or condensed gas) and not plasma, although the plasma generation phase of these processes is known to assist in the breakdown of residual peroxide in the chamber load.

Solution to Question 11:

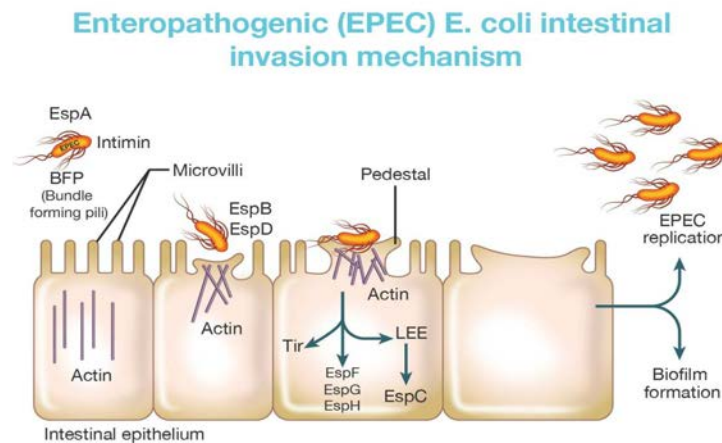
Attachment and effacement lesions on intestinal cells are caused by enteropathogenic *Escherichia coli* (EPEC).

E. coli is the second most common cause of diarrhea in the pediatric age group. Rotavirus is the most common cause. EPEC causes watery diarrhea which is self-limiting.

EPEC adheres to the mucosal cells of the small bowel and causes loss of microvilli (effacement). This leads to the formation of filamentous actin pedestals or cuplike structures and occasionally may cause entry of the EPEC into the mucosal cells.

Pathogenicity requires the presence of two important factors:

- Bundle-forming pilus encoded by plasmid containing EPEC adherence factor (EAF).
- Chromosomal locus of enterocyte effacement (LEE) pathogenicity island.



The image below shows an electron microscopy image of *E. coli*.



Solution to Question 12:

The given image shows generation of mRNA from negative sense RNA within the nucleus. This occurs during the replication of influenza virus.

Orthomyxoviruses (influenza virus) contain segmented (8 segments) negative sense ssRNA that is replicated in the nucleus. Stages of viral replication include the following:

- Attachment, penetration, uncoating: The virus attaches to cell-surface sialic acid via hemagglutinin. It is then internalized by receptor-mediated endocytosis. There is fusion of the viral envelope and cell membrane, resulting in uncoating or release of nucleocapsid into the host cell cytoplasm.
- Transcription and translation: mRNAs are produced from nucleocapsids via a virus-encoded polymerase. This process occurs in the nucleus. The mRNAs thus produced are translated into viral proteins in the cytoplasm.
- Viral RNA replication: The viral genome is replicated by the same virus-encoded polymerase involved in transcription. The negative sense RNA is converted to its positive sense form. The positive sense copies serve as templates for replication of the viral genome.
- Maturation: The virus matures by budding from the surface of the cell.

Solution to Question 13:

The given image is of darkfield microscopy as it has a dark background with a refractile organism. The dark ground microscope has an optical system that enhances the contrast of unstained bodies.

Dark ground microscopy is the most specific and sensitive technique to diagnose syphilis when an active chancre or condyloma lata is present to visualize *Treponema pallidum*.

Solution to Question 14:

The image shows the life cycle of *Strongyloides stercoralis*.

Strongyloides stercoralis is the smallest nematode known to cause human infection. It has both a free-living and a parasitic life cycle. Its natural host is man. The infective form to man is the filariform larva. Infection occurs through penetration of the skin by larvae in the soil. Then, the larvae migrate to the small intestine. They molt twice and become adult female worms in the small intestine. The eggs laid in the mucosa of the intestine hatch immediately releasing the rhabditiform larva (L1 stage). Thus, it is ovoviviparous. These rhabditiform larvae in the bowel can develop directly into filariform larvae that penetrate the colonic wall or perianal skin and enter the circulation to repeat the migration. This establishes ongoing internal reinfection.

The rhabditiform larva can also be passed in the stool. It may directly metamorphose to the infective filariform larva in the soil, or develop into adult worms. The free-living adult worms then pass eggs, which hatch and mature to form infective filariform larvae.

In *Ancylostoma duodenale* infestation, eggs containing segmented ova (not rhabditiform larvae) are passed in the stool. The rhabditiform larva hatches 2 days after the eggs are passed in the soil.

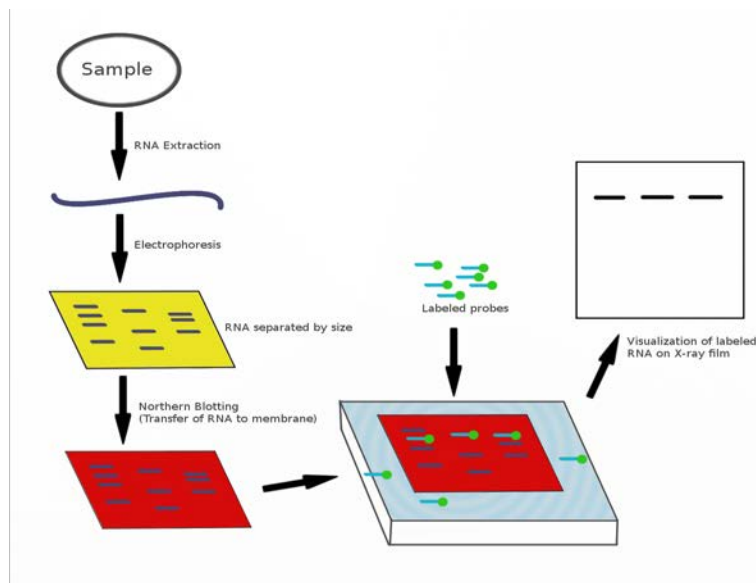
Note: Based on the method of reproduction, nematodes are classified as follows:

- Oviparous - They lay eggs:
- Unsegmented eggs: *Ascaris lumbricoides*, *Trichuris trichuria*
- Segmented eggs: *Ancylostoma duodenale*, *Necator americanus*
- Eggs containing larva: *Enterobius vermicularis*
- Viviparous - They produce larvae. e.g., *Trichinella spiralis*, *Wuchereria bancrofti*, *Brugia malayi*
- Ovoviviparous – They lay eggs containing fully formed larvae that hatch immediately. e.g., *Strongyloides stercoralis*

Solution to Question 15:

Northern blotting is used for the detection of RNA.

Southern blotting is used for the detection of DNA and Western blotting is used for the detection of protein.



Solution to Question 16:

The cook is most likely to be carrier of *Staph. aureus*.

The onset of food poisoning with in 4-6 hrs of food is mainly due to:

- *Staphylococcus aureus*
- *Bacillus cereus* (has no carriers)

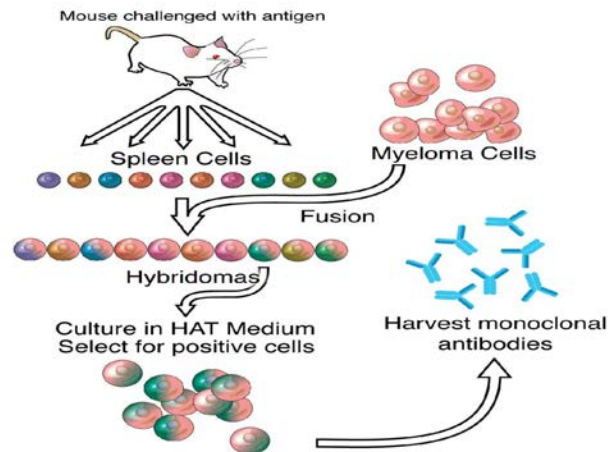
Solution to Question 17:

The image depicts the production of monoclonal antibodies by hybridoma technology.

The following are the steps involved in hybridoma technology:

- Mice are immunized with an antigen. After several weeks of immunization, blood samples are obtained from mice for measurement of serum antibodies.
- When the antibody titer is high enough, mice are commonly boosted by injecting antigen, without adjuvant, intraperitoneally or intravenously. Then the mice are euthanized and their spleens removed for in vitro hybridoma cell production.
- Fusing antibody-producing spleen cells, which have a limited life span, with myeloma cells (derived from an immortal tumor of lymphocytes) results in a hybridoma that is capable of unlimited growth.
- The cells are cultured in a medium containing hypoxanthine, aminopterin, and thymidine (HAT medium) which allows only the fused cells to survive in culture.
- Monoclonal antibodies are then harvested.

Hybridoma monoclonal antibody production



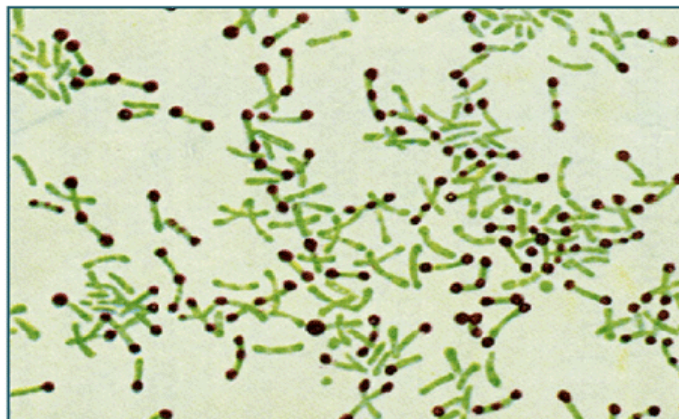
Solution to Question 18:

The given scenario is suggestive of diphtheria caused by *Corynebacterium diphtheria*. This organism is stained using Albert's stain which contains toluidine blue 'O' and malachite green. Both these dyes have an affinity for acidic cytoplasm as they are basic dyes.

Albert's, Neisser's, and Ponder's stain are used to demonstrate the metachromatic granules (Volutin / Babes Ernst granules) of *Corynebacterium diphtheria*.

Corynebacterium diphtheria appears as green-coloured bacilli with purple-blue-coloured metachromatic granules in cuneiform arrangement.

Corynebacterium diphtheriae-microscopy



Alberts stain

Solution to Question 19:

Stools with mucus and abdominal pain are seen in infection due to *Entamoeba*.

Clinical features of intestinal amebiasis:

- Acute intestinal amebiasis presents as dysentery accompanied by lower abdominal discomfort and flatulence. Stools usually have blood and mucus.
- Chronic amebiasis presents with low-grade symptoms such as occasional diarrhea, weight loss, and fatigue.
- A granulomatous lesion called an ameboma may form in the cecal or rectosigmoid areas of the colon which can resemble an adenocarcinoma of the colon.

Other options:

Option a - *Giardia* is often asymptomatic, but in some cases can lead to fat malabsorption, dull epigastric pain, and flatulence.

Option c - *Campylobacter* presents with fever, pain and bloody or watery diarrhea. Stool contains leucocytes and blood.

Option d - Staphylococcal food poisoning is toxin-mediated and presents with nausea, vomiting, and diarrhea.

Solution to Question 20:

Interferon-beta is secreted during infection due to viruses.

Interferons are classified as cytokines. Interferons are non-toxic, non-antigenic and species-specific molecules. They are inactivated by proteases. They are stable at pH over 2-10, except $\text{INF-}\gamma$ (labile at pH 2) and are non-dialysable and non-sedimentable.

There are three types of interferons - α , β and γ . $\text{INF-}\alpha$ or leukocyte interferon is produced by leucocytes. It has antiviral activity, is immunoregulatory and antiproliferative. $\text{INF-}\beta$ or fibroblast interferon is produced by fibroblasts and epithelial cells. It has antiviral activity and is thus increased in viral infections. It enhances MHC class I and NK cell activity. $\text{INF-}\gamma$ or immune interferon is produced by T-lymphocytes, NK cells. It activates macrophages, increases neutrophil function, enhances both MHC class I and class II expression and has anti-proliferative (anti-tumour) functions.

Solution to Question 21:

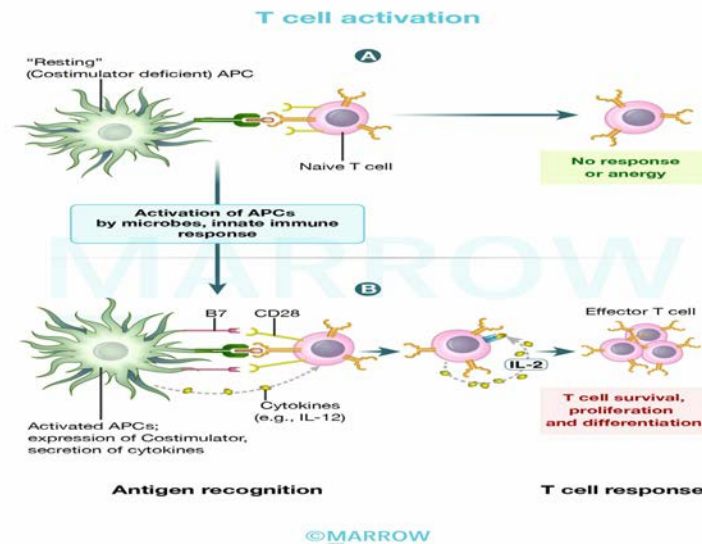
Mature dendritic cells are the most potent stimulator of naive T cells.

Naive CD4^+ T cells are activated when they come in contact with two ligands - MHC class II and a costimulatory membrane protein (eg: CD28). Both of these are present in antigen-presenting cells, the most potent of which is the mature dendritic cell. Other professional antigen-presenting cells include macrophages and B lymphocytes (option C).

Other options:

Option B: Follicular dendritic cells are functionally different from dendritic cells. They are not antigen-presenting-cells and do not activate naive T cells.

Option D: NK cells do not stimulate naive T cells.



Solution to Question 22:

The seminiferous tubules are an immunologically privileged site.

Immunologically privileged sites can tolerate the introduction of antigens without eliciting an immune response. This allows a higher likelihood of graft survival, even without perfect HLA matching. As a result, an allograft can be placed in these sites without endangering a rejection reaction.

The unique immune tolerance in these regions is primarily due to the lack of effective lymphatic drainage, which prevents antigen presentation to the immune system, and the presence of blood-tissue barriers that restrict immune cell infiltration. Low vascularity contributes to immune privilege by minimising antigen exposure and immune cell recruitment.

Examples of immune-privileged sites are the anterior chamber of the eye, cornea, testes, and brain.

Note: Although part of the brain, the blood-brain barrier is absent at the area postrema, therefore it is not an immune-privileged site.

Solution to Question 23:

The given image shows the amastigote form of leishmaniasis located intracellularly in macrophages. Buffy coat is used to diagnose the parasite.

Visceral leishmaniasis (kala-azar) is a slowly progressing indigenous disease caused by a protozoan parasite of genus *Leishmania*. Leishmaniasis is a parasitic disorder transmitted by the

bite of an infected female *Phlebotomus* sandfly in developing countries. The disease is transmitted by sand flies, which inoculate the flagellated promastigotes into the skin of the host.

In human beings, the disease presents in four different forms with a broad range of clinical manifestation:

- Visceral leishmaniasis, or kala-azar,
- Cutaneous leishmaniasis,
- Mucocutaneous leishmaniasis, and
- Diffuse cutaneous leishmaniasis.

The life cycle of *Leishmania* involves two forms,

- The promastigote which develops and lives extracellularly in the sandfly vector and
- The amastigote which multiplies intracellularly in the reticuloendothelial cells of the host.

It lowers immunity, causes persistent fever, anemia, liver and spleen enlargement, and if left untreated, death may occur due to opportunistic infection.

Diagnosis:

- Rapid diagnostic test: rK39 Rapid diagnostic test and Kala azar dipstick test strip.
- Parasitological diagnosis: Demonstration of LD bodies in buffy coat smear - confirmatory.
- Aldehyde test of Napier: Widely used for surveillance.
- Serological tests for diagnosis: Rapid dipstick test, Direct agglutination test, ELISA, Indirect Fluorescent Antibody Test
- Leishmanin (Montenegro) test: Induration of 5mm or more is considered positive, usually positive 4-6 weeks after onset in cutaneous leishmaniasis, valuable for distinguishing immune from nonimmune subjects and endemicity.

Drugs used in the treatment of visceral leishmaniasis are sodium stibogluconate, miltefosine, and amphotericin. Liposomal amphotericin single dose 10mg/Kg IV is the first line of treatment in India.

Solution to Question 24:

Amikacin is not used in the treatment of typhoid fever.

Amikacin is an aminoglycoside. Despite showing efficient in vitro killing of *Salmonella*, first- and second-generation cephalosporins and aminoglycosides have proven to be ineffective in actual clinical cases.

Antibiotic therapy for enteric fever in adults			
Indication	Agent	Dosage (Route)	Duration, Days
Empirical Treatment			

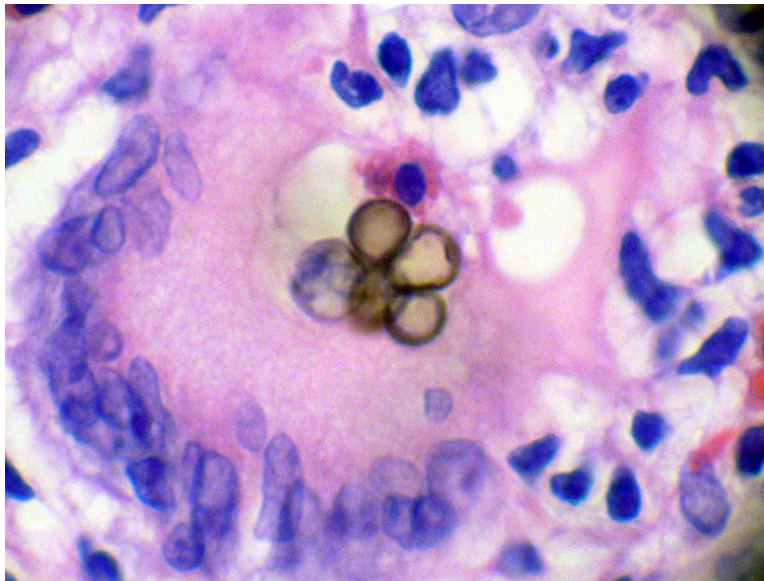
Antibiotic therapy for enteric fever in adults

	Ceftriaxone	2 g/d (IV)	10-14
Azithromycin	1 g/d (PO)	5	
Fully Susceptible			
Optimal treatment	Ciprofloxacin	500 mg bid (PO) or 400 mg q12 h (IV)	5-7
Azithromycin	1 g/d (PO)	5	
Alternative treatment	Amoxicillin	1 g tid (PO) or 2 g q6h (IV)	14
Chloramphenicol	25 mg/kg tid (PO or IV)	14-21	
Trimethoprim-sulfamethoxazole	160/800 mg bid (PO)	7-14	
Multidrug-resistant, fluoroquinolone-susceptible			
Optimal treatment	Ceftriaxone	2 g/d (IV)	10-14
Azithromycin	1 g/d (PO)	5	
Alternative treatment	Ciprofloxacin	500 mg bid (PO) or 400 mg q12h (IV)	5-14
Quinolone-resistant			
Optimal treatment	Ceftriaxone	2 g/d (IV)	10-14
Azithromycin	1 g/d (PO)	5	
Alternative treatment	High dose ciprofloxacin	750 mg bid (PO) or 400 mg q12h (IV)	10-14
Ceftriaxone-resistant			
Optimal treatment	Meropenem	1 g q8h (IV)	10-14
Azithromycin	1 g/d (PO)	5	

Microbiology AIIMS 2019 (May & Nov)

Question 1:

A farmer from the sub-Himalayan belt had a thorn prick injury a year back and developed a warty, verrucous growth at the site of prick. A skin biopsy sample was taken and sent for histopathological examination. The characteristic microscopic body as shown in the image was seen. Which of the following could be the most likely diagnosis?



- a) Blastomycosis
- b) Pheohyphomycosis
- c) Chromoblastomycosis
- d) Sporotrichosis

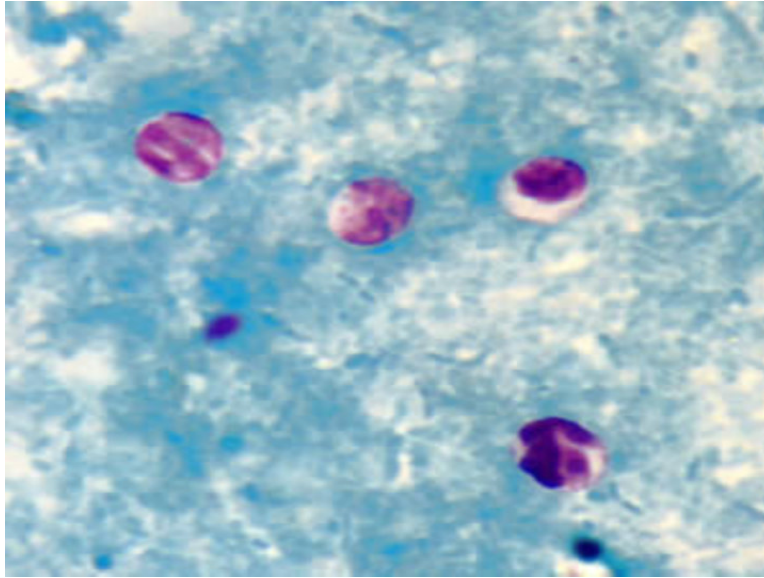
Question 2:

Choose the wrong statement about Plasmodium species:

- a) Drug resistant falciparum is seen in India
- b) Presence of duffy blood group antigen is protective against vivax
- c) Severe malaria with shock is called algid malaria
- d) Relapse is seen in P. ovale

Question 3:

A HIV patient with fever, chronic diarrhea presented to OPD. His stool was examined which showed the following oocysts. Identify the components used for staining the stool.



- a) Crystal violet, iodine, acid alcohol, carbol fuchsin
- b) Methanol, Kinyoun's carbol fuchsin, acid alcohol, methylene blue
- c) Methanol, dilute carbol fuchsin, acid alcohol, crystal violet
- d) Methylene blue, iodine, acid alcohol, carbol fuchsin

Question 4:

Which immunoglobulin is elevated in a case of chronic allergy?

- a) IgA
- b) IgM
- c) IgE
- d) IgG

Question 5:

A patient is diagnosed with Hepatitis B infection. Which of the following markers will always be present in his serum, even during recurrences, or if he develops chronic hepatitis?

- a) HBc Ag

- b) Anti HBs Ab
- c) Anti HBc Ab
- d) HBe Ag

Question 6:

Which among the following occupation is a risk factor for this presenting illness?



- a) Kennel Worker
- b) Lifeguard in a swimming pool
- c) Veterenarian
- d) Poultry worker

Question 7:

Transmission assessment survey is carried out for:

- a) *P. vivax*
- b) *P. falciparum*
- c) Filariasis
- d) Leishmania

Question 8:

Which among the following is a subcutaneous fungal agent?

- a) *Cryptococcus neoformans*
- b) *Histoplasma capsulatum*
- c) *Sporothrix schenckii*
- d) *Talaromyces marneffe*

Question 9:

A hospital has reported an outbreak of MRSA infection. On investigation, it was found out that a staff nurse had nasal carriage of MRSA. Which of the following drugs helps in the removal of colonization of bacteria?

- a) Oral vancomycin
- b) Inj cephalosporin
- c) Topical mupirocin
- d) Inhaled colistin

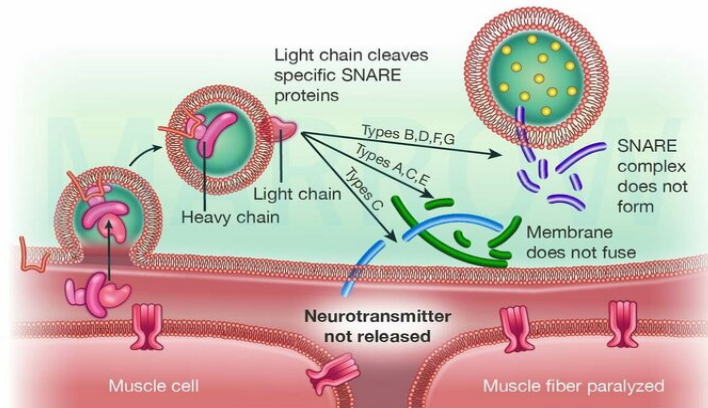
Question 10:

Mw vaccine is extracted from which bacteria, mentioned below:

- a) *M. indicus pranii*
- b) *M. welchii*
- c) *M. leprae*
- d) *M. bovis*

Question 11:

Which toxin acts via the mechanism depicted in the image below?



©MARROW

- a) Botulinum
- b) Tetanus
- c) Diphtheria
- d) Pertussis

Question 12:

The toxin produced by the organisms shown below acts via?

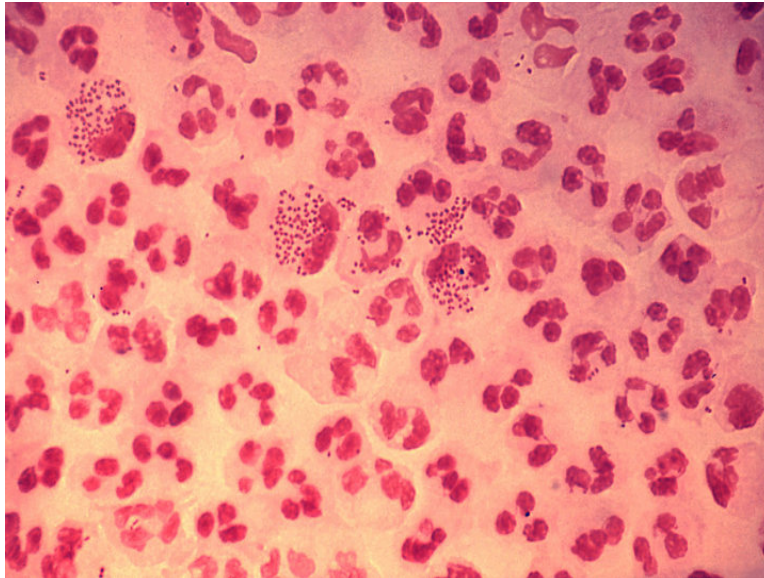


- a) Increase in cAMP
- b) Increase in cGMP

- c) ADP ribosylation of EF2
- d) RNA glycosidase

Question 13:

A 25-year old man presents with a history of urethral discharge for the last three days. A gram stain smear of the discharge is shown below. All of the following are true regarding the likely etiology except?



- a) Catalase and oxidase positive
- b) Pili is a virulence factor
- c) Obligate intracellular organism
- d) Shows twitching motility

Question 14:

The CLED medium is preferred over MacConkey agar for the culture of organisms in case of UTI because:

- a) It inhibits proteus swarming
- b) It promotes the growth of candida, streptococcus and staphylococcus
- c) It differentiates between lactose-fermenting and non-lactose fermenting bacteria
- d) It promotes the growth of pseudomonas

Question 15:

Which of the following organisms cause multiple alveolar-like lesions in the liver?

- a) *Echinococcus multilocularis*
- b) *Echinococcus granulosus*
- c) *Entamoeba histolytica*
- d) *Cysticercus cellulosae*

Question 16:

All of the following can be seen intracellularly in hepatocytes except:

- a) *Toxoplasma*
- b) *Plasmodium*
- c) *Babesia*
- d) *Leishmania*

Question 17:

A Giemsa stain of a thin peripheral smear is prepared. Which of the following cannot be diagnosed?

- a) *Coxiella burnetii*
- b) *Toxoplasma gondii*
- c) *Bartonella henselae*
- d) *Ehrlichia chaffeensis*

Question 18:

Which of the following is false about influenza?

- a) Virus shedding is present before symptoms begins
- b) 5-12 years is a high risk group
- c) Aquatic birds act as reservoir
- d) Secondary attack rate is 5-15%

Question 19:

β 1-3 glucan assay is used to diagnose all except?

- a) Candidiasis
- b) Aspergillosis
- c) Pneumocystis pneumonia
- d) Mucormycosis

Question 20:

Which of the following is a xenodiagnostic method?

- a) Rabbit ileal loop assay for ETEC
- b) Injecting Aedes thorax with blood of a patient with suspected dengue
- c) Injecting a hamster with splenic biopsy for diagnosis of leishmaniasis
- d) Intradermal test on guinea pigs for toxigenicity of Corynebacterium diphtheria

Answer Key

Question No.	Correct Option
1	c
2	b
3	b
4	c
5	c
6	a
7	c
8	c
9	c
10	a
11	a
12	c
13	c
14	b
15	a

16	c
17	a
18	b
19	d
20	b

Detailed Explanations

Solution to Question 1:

The history of a slowly spreading warty, verrucous lesion following a thorn prick injury and the histopathological image showing medlar bodies is suggestive of chromoblastomycosis.

Chromoblastomycosis (chromomycosis) is a subcutaneous mycotic infection that is usually caused by traumatic inoculation. Over months to years, the primary lesion becomes verrucous and wart-like with extension along the draining lymphatics.

Histologically, the lesions are granulomatous and the dark sclerotic bodies may be seen within leukocytes or giant cells called medlar/copper penny bodies.

Subcutaneous mycoses	Causative agent/s
Sporotrichosis	<i>Sporothrix schenckii</i>
Chromoblastomycosis	<i>Phialophora verrucosa</i> <i>Fonsecaea pedrosoi</i> <i>Fonsecaea compacta</i> <i>Rhinocladiella aquaspersa</i> <i>Cladophialophora carrionii</i>
Phaeohyphomycosis	<i>Exophiala jeanselmei</i> <i>Phialophora richardsiae</i> <i>Bipolaris spicifera</i> <i>Wangiella dermatitidis</i>
Mycetoma	<i>Pseudallescheria boydii</i> (Anamorph <i>Scedosporium apiospermum</i>). <i>Madurella mycetomatis</i> <i>Madurella grisea</i> <i>Exophiala jeanselmei</i> <i>Acremonium falciforme</i>

Solution to Question 2:

The absence, not the presence, of the Duffy blood group antigen, is protective against *P. vivax* because the Duffy antigen normally serves as the receptor for *Plasmodium vivax*.

Drug Efflux pumps are the primary mode of resistance for Chloroquine. It is mediated by an ABC transporter encoded by *P. falciparum* multidrug resistance gene 1 (*Pfmdr1*). The resistance of *P. falciparum* to chloroquine increased over time and is present across all regions of the country.

Algid malaria is a rare complication of severe malaria characterized by hemodynamic collapse manifesting as shock with pronounced metabolic changes and hypothermia.

Relapse is seen in cases of *Plasmodium vivax* and *Plasmodium ovale*.

Malaria relapse :

True relapse is caused due to Hypnozoites.

These are resting forms.

In the case of *Plasmodium vivax* and *ovale* two kinds of sporozoites are seen, some of which :

- multiply inside hepatic cells to form - Schizonts
- others persist and remain dormant as - hypnozoites

From time to time some are activated to become schizonts and release merozoites, which go on infecting RBCs producing clinical relapse.

In the case of *Plasmodium falciparum* and *Plasmodium malariae*, the initial tissue phase disappears completely and no hypnozoites are found, hence technically relapse is not seen.

Solution to Question 3:

Diarrhea in an HIV patient is suggestive of a coccidian parasitic infestation. The image shows a blue background and red-stained oocysts, suggestive of modified acid-fast stain positive *Cryptosporidium parvum*.

The modified Ziehl-Neelsen (ZN) method is used for staining *cryptosporidium* and *isospora* species. Unlike the ZN-stain, the modified ZN stain does not require heating of the reagents.

The reagents used in modified ZN stain include:

- Absolute methanol for fixation
- Kinyoun's Carbol fuchsin - Primary stain
- Acid alcohol - Decolourising agent
- Methylene blue or Malachite green - Counterstain

Solution to Question 4:

IgE is elevated in cases of chronic allergy.

Solution to Question 5:

A patient who has had exposure to HBV will have anti-HBc antibody even when he has moved into the chronic or recurrent stages. IgM anti-HBc reappears during acute recurrences while IgG anti-HBc is found in chronic states.

The HBsAg antigen is present in chronic hepatitis B, while the antibody anti-HBs is not.

Solution to Question 6:

The above image shows cutaneous larva migrans. Being a kennel worker is a risk factor for this job as exposure to animal feces is high.

As Larvae are found on sandy beaches, and under dwellings, swimming in natural water bodies would be a risk and not artificial ones like swimming pools.

CLM is caused by various animal hookworm species (e.g., *Ancylostoma brazilienses*), and the parasite is acquired from direct skin contact with soil contaminated by dog or cat feces. The parasitic infection usually begins with an inflamed papule, often located between the toes, where the hookworm species (e.g., *Ancylostoma brazilienses*) larvae enter the skin.

Serpentine or linear single-track lines later mark the course of the larvae as they migrate through the epidermis of the trunk and lower extremities.

Ivermectin is the first-line drug for the treatment of cutaneous larva migrans, taken as a single 200 µg/kg oral dose.

Solution to Question 7:

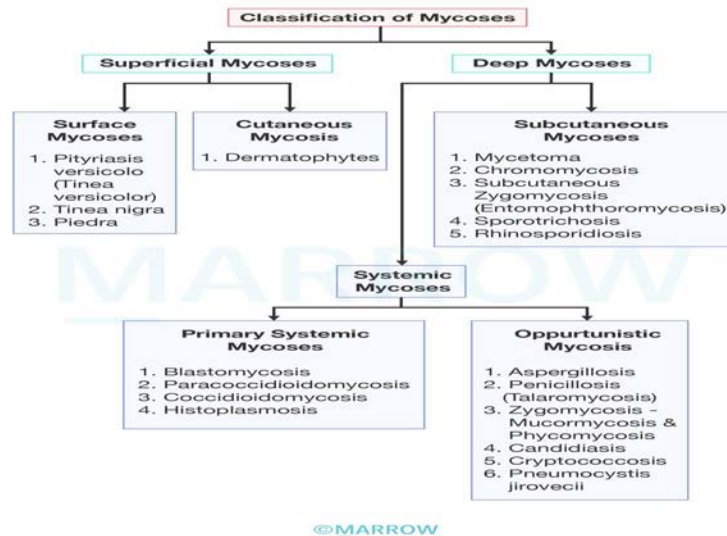
Transmission assessment survey is carried out for filariasis.

Mass Drug Administration is used to reduce infection in the community to levels below a threshold at which mosquitoes are unable to continue spreading the parasites from person to person and thus new infections are prevented.

The transmission assessment survey (TAS) is used to determine when infections have been reduced below target thresholds and Mass Drug Administration can stop. Once MDA has stopped, TAS is used as a surveillance tool to determine that infection levels are sustained below target thresholds.

Solution to Question 8:

Sporothrix schenckii is a subcutaneous fungal agent. The other three mentioned are systemic mycoses.



Solution to Question 9:

Topical mupirocin is the preferred agent for the elimination of nasal carriers of MRSA.

As Mupirocin has no structural similarities with existing systemic antibiotics that might lead to the development of cross-resistance and has minimal toxicity in its topical form, it is approved for the prevention of MRSA outbreaks

Mixed therapy with systemic agents is not preferred as it would cause unnecessary antibiotic exposure potentially paving way for drug resistance.

Solution to Question 10:

Mw vaccine is extracted from *M. indicus pranii*

MIP immunotherapy is used as an adjunct to chemotherapy for category II tuberculosis.

Here, indicus stands for India, where the *M. indicus pranii* was isolated; and pranii stands for the person who discovered the species (Prof. Pran Talwar) and National Institute of Immunology (nii), where trials were conducted (pran + nii).

Administration: 0.1 ml intradermal. It is divided into two equal doses and given on both arms. Booster doses are injected only on one arm.

Solution to Question 11:

The image given above depicts the mechanism of action of botulinum toxin.

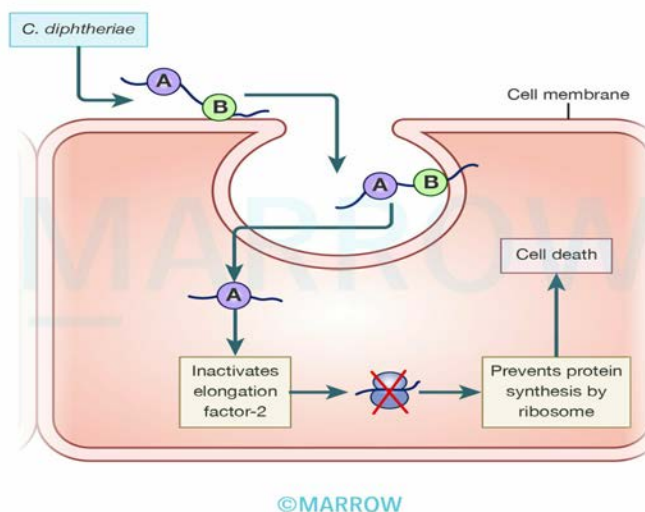
It causes proteolysis of snare proteins. Snare proteins have three components- synaptobrevin, SNAP25 and syntaxin. The snare protein is required for the exocytosis of acetylcholine vesicles. Thus it leads to presynaptic inhibition of neuronal transmission.

There are 7 different strains of clostridium botulinum and each strain acts by a different mechanism. Toxins of C. Botulinum A, C and E cleave SNAP25. Type C also cleaves syntaxin. Type B, D, F and G cleave synaptobrevin.

Solution to Question 12:

The given image shows gram-positive rods with club-shaped ends, beaded appearance, and a Chinese-letter pattern of arrangement indicative of *Corynebacterium diphtheriae*.

The toxin produced by the bacteria - diphtheria toxin - inhibits polypeptide chain elongation and protein synthesis by inactivating the elongation factor-2 (EF-2) by ADP ribosylation. It has two fragments - fragment B helps in binding of the toxin to host cells and fragment A inactivates EF-2.



Solution to Question 13:

The given slide shows gram-negative diplococci within the polymorphs, characteristic of *Neisseria gonorrhoeae* infection. *Neisseria* is not an obligate intracellular organism, it can also occur extracellularly i.e, it is a facultative intracellular organism. The other *neisseria* that are found as normal inhabitants of the human respiratory tract occur extracellularly.

Neisseria gonorrhoeae is a gram-negative, non-sporulating, non-motile diplococcus that is aerobic but may grow anaerobically also. It is oxidase and catalase-positive.

Although *Neisseria* are non-motile organisms, they show twitching motility. This is a flagella-independent form of bacterial translocation over moist surfaces that occurs by the extension and retraction of polar type IV pili. These pili act as virulence factors by promoting attachment to host cells and inhibiting phagocytosis.

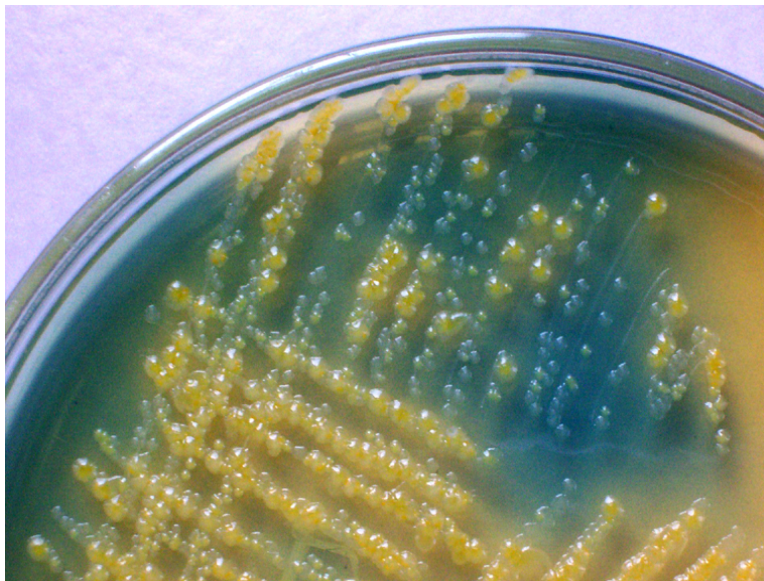
Virulence factors for gonococci:

- Outer Membrane Proteins (OMPs)
- IgA proteases
- Pili
- Lipo-oligosaccharide (LOS)
- Transferrin
- Lactoferrin-binding protein

Solution to Question 14:

CLED (Cysteine lactose electrolyte-deficient) medium is preferred over MacConkey agar for the culture of organisms involved in urinary tract infections because it is less inhibitory and promotes the growth of gram-positive bacteria (except β hemolytic streptococcus) and candida.

A combination of blood agar and MacConkey agar is used for the processing of urine specimens. CLED medium is used as an alternative for this combination. It can also prevent swarming of *Proteus*, and is thus different from blood agar. Like MacConkey agar, it differentiates between lactose fermenters and non-fermenters. Lactose fermenters change the colour of the medium from green to yellow. However, these two facts are not the reason it is better than MacConkey agar for evaluation of UTI.



Solution to Question 15:

Echinococcus multilocularis causes multiple alveolar-like lesions in the liver.

E. multilocularis leads to the formation of a multilocular hydatid cyst, most commonly affecting the liver. It presents as multiple small cysts in the liver.

Echinococcus granulosus causes a hollow fluid-filled unilocular cyst in the liver. The CT below shows water lily sign, which is seen in the hydatid disease of the lung. It is caused by *Echinococcus granulosus*. (Option B)

Water lily sign



The water-lily sign occurs due to the detachment of the endocyst membrane from the cyst wall, giving the appearance of floating membranes within the pericyst. This mimics the appearance of a water lily. It is classically visualised on plain radiographs (mainly chest X-ray) when the collapsed membranes are calcified but may be seen on ultrasound, CT, or MRI.

Solution to Question 16:

Babesia is an intracellular parasite of RBCs, not of the hepatocytes.

Its trophozoites invade the RBCs.

The liver is usually not affected as babesia lack an exoerythrocytic phase.

Solution to Question 17:

The diagnosis of *Coxiella burnetii* cannot be done using a Giemsa stain of a thin peripheral smear.

It is done mainly by serology, which shows a rise in the titer of specific antibodies to *C. burnetii*.

Microorganisms such as *Histoplasma*, *Leishmania*, *Toxoplasma*, and *Pneumocystis* can also be detected with Giemsa, and in gastric tissues, *Helicobacter pylori* (*H. pylori*) appear thin and distinctly blue. Giemsa's stain is frequently used for diagnostic purposes in haematology to differentiate the nuclear and cytoplasmic morphology of platelets, RBCs, WBCs, and parasites.

Solution to Question 18:

Children <5-years, especially those <2-years of age are at the highest risk of serious complications related to influenza infection.

Other high-risk groups for influenza include adults ≥60 years of age, persons with chronic diseases, persons who are immunocompromised.

Option A: The virus is shed in respiratory secretions a day before the onset of illness and for 3-4 days thereafter.

Option C: Birds, especially aquatic birds act as reservoirs of influenza viruses.

Option D: Secondary attack rate is 5- 15%.

Solution to Question 19:

The β-1-3 glucan assay cannot be used to diagnose mucormycosis. This condition is usually diagnosed based on clinical features and culture.

The β-1-3 glucan assay detects the presence of (1-3)-β-d-glucan (BDG), an abundant cell wall polysaccharide found in most fungi. It is a chromogenic, quantitative enzyme immunoassay (EIA) designed to detect beta-D-glucan by using purified, lysed horseshoe crab (*Limulus polyphemus*) amebocytes. 1,3 Beta D Glucan level of ≥80 pg/mL is considered a positive result.

Fungi that do not contain BDG in their cell walls include *Cryptococcus neoformans*, zygomycetes like *rhizopus* and *mucor* and *Blastomyces dermatitidis*. Mucormycosis is an opportunistic mycosis caused by zygomycetes, which do not contain (1-3)-β-d-glucan, and hence, the β 1-3 glucan assay cannot be used for its diagnosis.

Solution to Question 20:

Injecting *Aedes thorax* with blood of a patient with suspected dengue is the xenodiagnostic method among the given options.

Xenodiagnosis (xenos- foreign) is an indirect method of identifying the infectious disease agents (i.e. the pathogen). In xenodiagnosis, the vector of a pathogenic agent is infected and the pathogen is allowed to multiply in the vector. The vector acts as the “foreign” host. Later, the vector is examined for the presence of the pathogen.

This method was first used by Brumpt for the diagnosis of Chagas disease. Laboratory bred *Reduviid* bugs (vector/ foreign host) are allowed to feed on the suspected patients (original host). After 4-5 weeks, the feces of the bugs are microscopically examined for amastigote forms of *Trypanosoma cruzi* (pathogen).

Another method of xenodiagnosis is the intrathoracic inoculation of the mosquitoes (vector) with viruses (pathogen). The virus multiplies in different organs of the mosquito and can be easily demonstrated. *Aedes aegypti*, *Aedes albopictus*, and *Toxorhynchites* mosquitoes are used for the identification of the dengue virus.

In *Trichinella spiralis* infection, muscle biopsy bits from suspected patients are fed to laboratory rats (vector for *T. spiralis*). The larvae of *T. spiralis* can be demonstrated in the muscles of the

infected lab rats.

Advantage of xenodiagnosis: It is useful in cases where a direct microscopic diagnosis is difficult due to the low count of pathogens in the blood of the host/patient.

Other options:

Option A: Rabbit ileal loop assay for ETEC is a method of detection of toxigenicity/ virulence testing.

Option C: Injecting a hamster with splenic biopsy for diagnosis of leishmaniasis is a method of animal inoculation.

Option D: Intradermal test on guinea pigs for toxigenicity of *Corynebacterium diphtheria* is a method of virulence testing.

Microbiology AIIMS 2020 (May and Nov INI-CET)

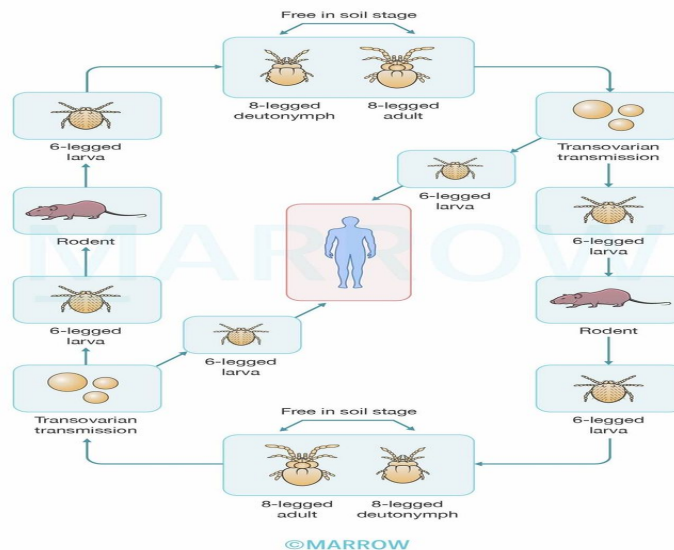
Question 1:

A resident had a needle stick injury while dealing with a Hepatitis B positive patient. Serology was done and HBsAg was negative and the Anti-HBs was >105 mIU/ml. What is the next step of management?

- a) Booster dose of vaccine
- b) Immunoglobulin with full vaccination
- c) Repeat full course of vaccine
- d) No treatment required as he already has Hep B antibody titres from vaccination

Question 2:

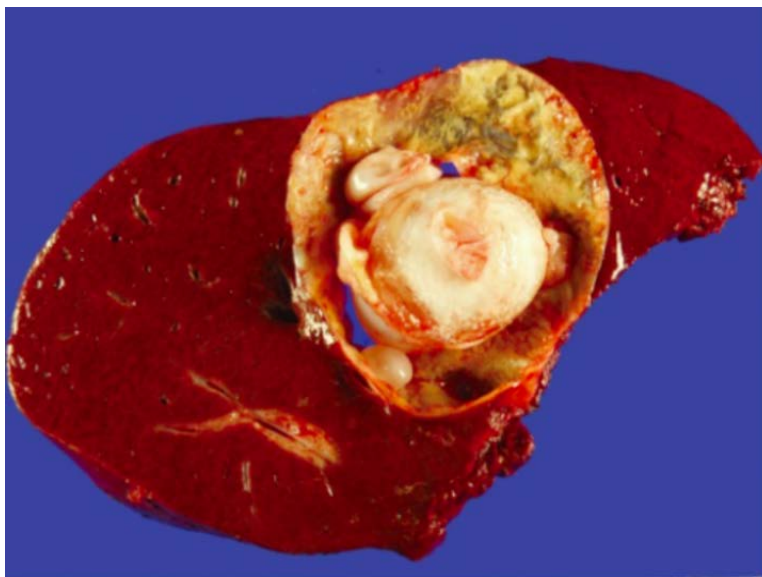
The life cycle of an organism is given below. Identify the vector involved.



- a) Hard tick
- b) Soft tick
- c) Trombiculid mite
- d) Louse

Question 3:

What is the causative organism for the condition shown in the image below?



- a) *Echinococcus granulosus*
- b) *Schistoma japonicum*
- c) *Trypanosoma cruzi*
- d) *Schistosoma mansoni*

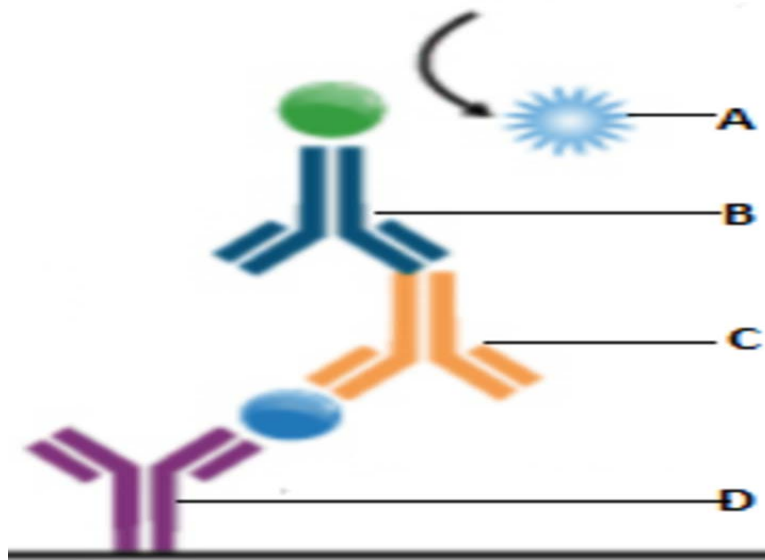
Question 4:

The presence of antibodies against SARS CoV 2 in the blood likely indicates

- a) Previous infection even if patient has been asymptomatic
- b) Presence of acute viral load
- c) A high risk of reinfection
- d) That the patient is in window period

Question 5:

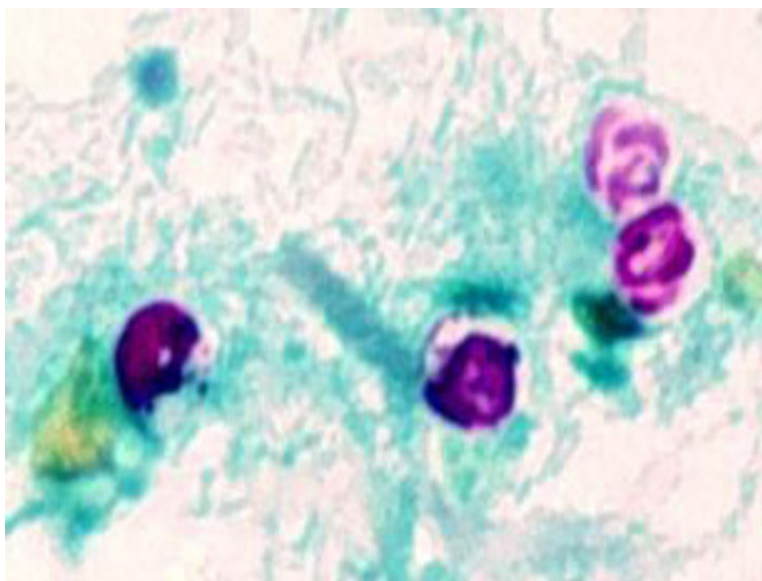
Which of the following is the enzyme conjugate in the given ELISA technique?



- a) A
- b) B
- c) C
- d) D

Question 6:

A patient who underwent a renal transplant 3 months ago comes with complaints of diarrhea for the last 9 days. The stool sample was subject to Kinyoun stain revealing 4-6 μ m structures shown below. Which of the following is true about the organism?



- a) Pathogen does not shows autoinfection

- b) *Cryptosporidium parvum* is the most common organism implicated worldwide
- c) Oocysts are immediately infective after being excreted in stool
- d) It is an obligate intracellular organism

Question 7:

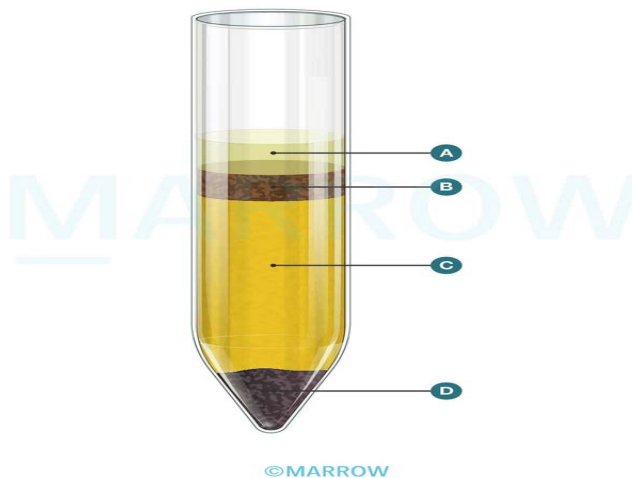
GeneXpert detects mutations in which of the following genes?

- a) inhA
- b) rpoB
- c) KatG
- d) pncA

Question 8:

A stool sample is collected. The specimen is processed for examination and is labeled at different levels as A, B, C, and D. Identify the level at which parasitic ova will be seen when viewed under a microscope.

Formalin-ether sedimentation technique



- a) A
- b) B
- c) C
- d) D

Question 9:

Type IV hypersensitivity is due to?

- a) Cell mediated response
- b) Cell mediated and antibody mediated response
- c) Antibody mediated response
- d) Innate response

Question 10:

A child was brought to the OPD with a swollen, red knee joint that is tender to touch. 2 weeks prior, he had an episode of fever with bloody diarrhea. Which of the following could be the possible organism?

- a) E. coli
- b) Shigella
- c) Giardia
- d) Rotavirus

Question 11:

Specimen handling for molecular testing of SARS CoV 2 would require

- a) BSL1
- b) BSL2
- c) BSL3
- d) BSL4

Question 12:

Which of the following method is employed to measure the viral load?

- a) ELISA
- b) Western blot
- c) Real time PCR
- d) Immunoelectrophoresis

Question 13:

All of the following is true regarding scrub typhus except:

- a) Transmitted by chiggers
- b) Lymphadenopathy is a feature
- c) Produces NLF producing colonies in MacConkey agar
- d) Man is an accidental host

Question 14:

False about the Ebola virus is:

- a) It is a filovirus
- b) Transmitted by mosquito
- c) Bats acts as reservoir
- d) Nosocomial transmission seen

Question 15:

Identify the condenser _____



- a) A
- b) C
- c) B

d) D

Question 16:

A young male presents to the STD clinic with painful lesions as shown in the image. What is the most likely microscopic finding of the lesion?



- a) Tzank cells
- b) Pund cells
- c) Henderson-Patterson bodies
- d) Guarnieri bodies

Question 17:

A patient has complains of fever with chills. A throat examination reveals swollen glands and a thick, whitish membrane over his tonsils. Albert staining is to be done. What are the constituents of Albert's stain?

- a) Toluidine blue 'O' and malachite green
- b) Carbol fuschin and methylene blue
- c) Carbol fuschin and malachite green
- d) Toluidine and methylene blue

Question 18:

Pneumonia due to a gram-negative pleomorphic bacillus growing on chocolate agar which is catalase and oxidase-positive is:

- a) E.coli
- b) Pseudomonas
- c) Haemophilus influenzae
- d) Actinomyces

Question 19:

A 30-year-old male presented to the OPD with painful genital ulcers. Which of the following can cause a painful ulcer on the glans penis?

- a) Syphilis
- b) Chlamydia
- c) Chancroid
- d) Lymphogranuloma venereum

Question 20:

Which protein is present in the HPV vaccine?

- a) E6, E7
- b) L1
- c) E1, E2
- d) L2

Question 21:

A patient presented with meningitis with a history of fever for 4 days and erythematous rashes on the chest. There was no history of travel and tick bite. His CSF findings showed increased protein, normal glucose, mild lymphocytosis and few RBCs. Which of the following is the most likely organism to cause this clinical picture?

- a) West nile virus
- b) Meningococci
- c) Strep pneumoniae
- d) Enterovirus

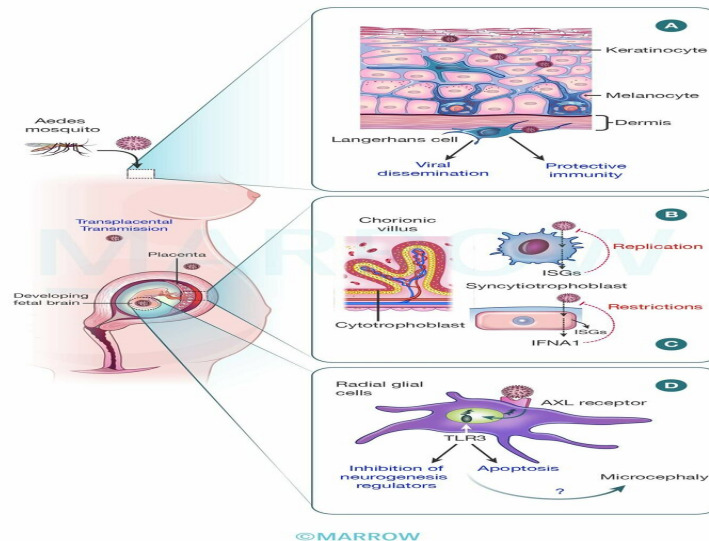
Question 22:

A patient presents with seizures. His CT head reveals multiple parenchymal cysts. Which of the following is correct regarding management in this case?

- a) Steroids followed by antiparasitic drugs
- b) Antiparasitic drugs followed by steroids
- c) Steroids before antiepileptics
- d) Continue antiepileptics

Question 23:

The transplacental transmission of the organism given in the image which causes microcephaly is-



- a) Nipah
- b) Zika virus
- c) CMV
- d) Rubella

Question 24:

The disruption of which of the following is responsible for watery diarrhea in cholera caused by *Vibrio*?

- a) Intercalated discs
- b) Macula densa
- c) Zonula occludens
- d) Hemidesmosomes

Question 25:

Affinity maturation is due to:

- a) Isotype switching
- b) Somatic hypermutation
- c) mRNA splicing
- d) VDJ rearrangement

Question 26:

A farmer presents with a swollen foot and multiple sinuses as shown in the below image. What is the most likely diagnosis?

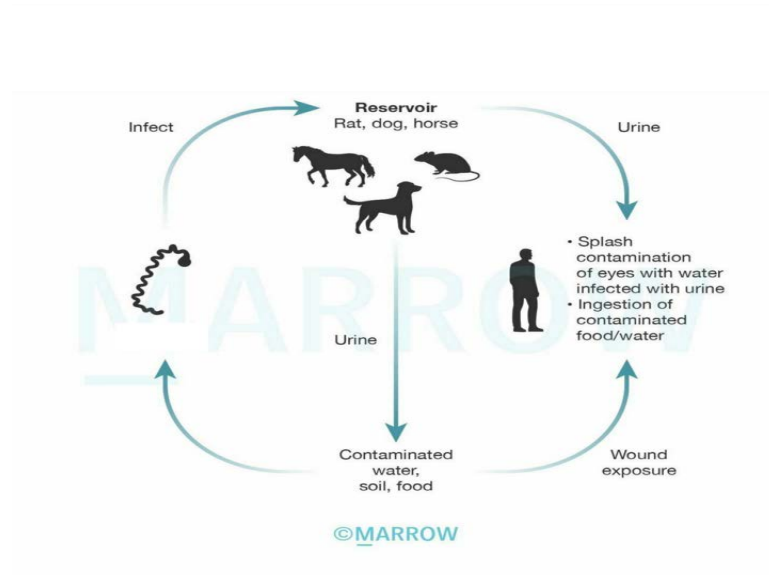


- a) Mycetoma
- b) Squamous cell carcinoma
- c) Nocardia

d) Sporotrichosis

Question 27:

The given life cycle belongs to which of the following organisms?



- a) Leptospira
- b) Hanta virus
- c) Yersinia pestis
- d) Salmonella typhi

Question 28:

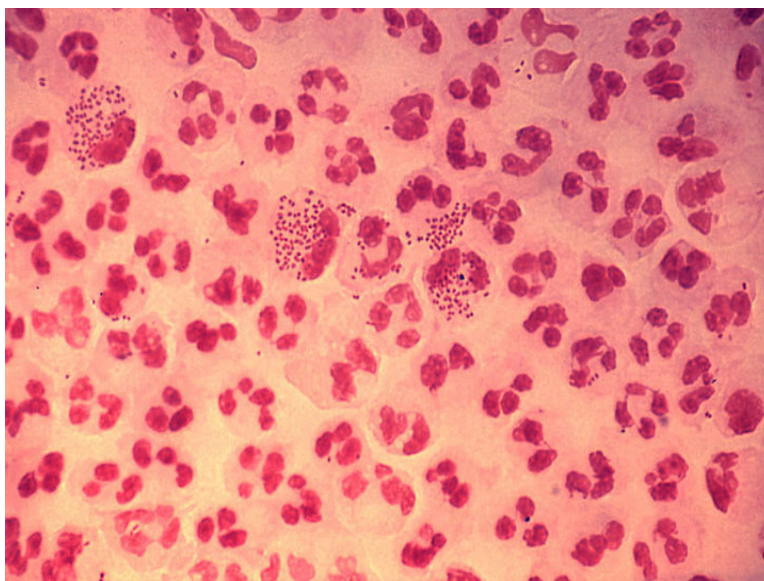
Identify the organism shown in this image.



- a) *Strongyloides stercoralis*
- b) *Enterobius vermicularis*
- c) *Ancylostoma duodenale*
- d) *Ascaris lumbricoides*

Question 29:

A patient presents with urethral discharge. A gram stain smear is made from the discharge sample and the following microscopic image is seen. What is the most likely causative organism?



- a) *Trichomonas*

- b) *Chlamydia trachomatis*
- c) *Neisseria gonorrhoeae*
- d) *Gardnerella*

Question 30:

Which of the following disease has the highest mortality once clinically manifested?

- a) Nipah
- b) Rabies
- c) Ebola
- d) Zika

Question 31:

A patient presents with a mobile, warm to touch nodule in the neck. On the anaerobic culture of the biopsy material from this swelling, it showed colonies as shown in the image which had gram-positive branching rods on staining. Which of the following is the drug of choice in this condition?



- a) Doxycycline
- b) Cotrimoxazole
- c) Penicillin
- d) Surgical drainage

Question 32:

In which of the following situations will you use a sanitizer?

- a) A, B & C only
- b) A & B only
- c) A, B C & D
- d) C & D only

Question 33:

The following are true about chancroid:

- a) C & D only
- b) A, B,C & D
- c) A B & C
- d) A & B only

Question 34:

Match the following E.coli with their respective pathogenesis.

- a) A-4,B-3,C-2,D-1
- b) A-1,B-3,C-2,D-4
- c) A-1,B-2,C-3,D-4
- d) A-4,B-2,C-3.D-1

Question 35:

Match the following:

- a) A-2, B-1, C-3, D-4
- b) A-3, B-2, C-1, D-4
- c) A-4, B-1, C-2, D-3
- d) A-1, B-2, C-4, D-3

Question 36:

Galactomannan in serum is seen in:

- a) 1 & 2 only
- b) 1 & 3 only
- c) 1 only
- d) 1, 2 & 3 only

Question 37:

Match the following parasites with their respective hosts:

- a) A-3, B-2, C-4, D-1
- b) A-2, B-1, C-4, D-3
- c) A-1, B-2, C-3, D-4
- d) A-4, B-3, C-1, D-2

Answer Key

Question No.	Correct Option
1	d
2	c
3	a
4	a
5	b
6	c
7	b
8	d
9	a
10	b
11	b
12	c
13	c
14	b
15	c

16	a
17	a
18	c
19	c
20	b
21	d
22	a
23	b
24	c
25	b
26	a
27	a
28	d
29	c
30	b
31	c
32	a
33	c
34	c
35	b
36	c
37	a

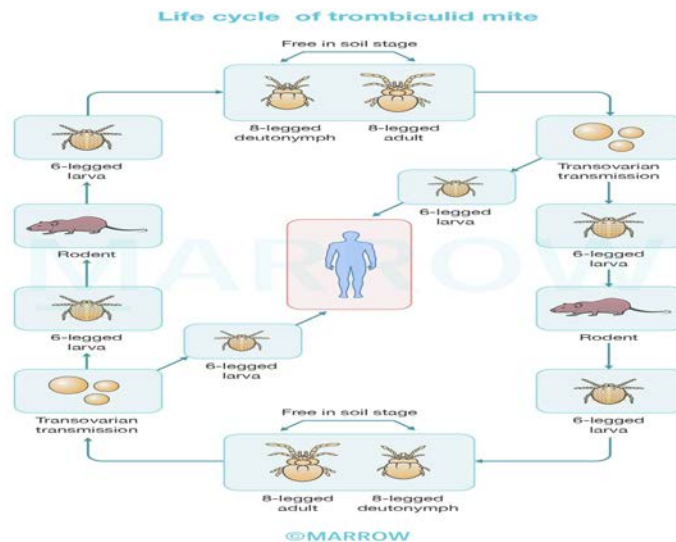
Detailed Explanations

Solution to Question 1:

The given clinical scenario of needle-stick injury shows that HBsAg is negative and the anti-HBs is >105 mIU/ml. This suggests that the resident has had routine vaccination for Hep B infection. Since the antibody titers are more than 10 mIU/ml, no treatment is required.

Solution to Question 2:

The given life cycle is of trombiculid mite which acts as a vector for scrub typhus. Humans are seen to be infected when they are bitten by the mite larvae (chiggers).



Solution to Question 3:

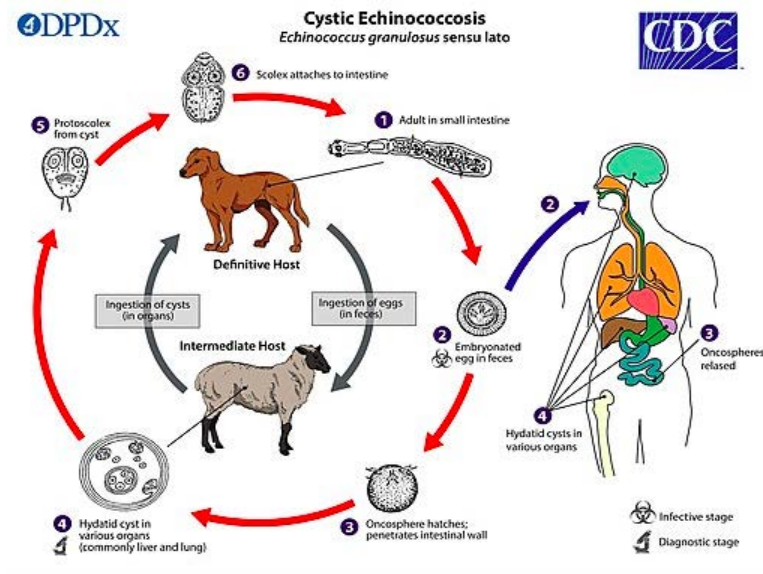
The given image shows a hydatid cyst in the liver. It is caused by *Echinococcus granulosus*.

Echinococcus granulosus is the most common causative agent in hydatid cysts, while *Echinococcus multilocularis* and *Echinococcus ligartus* account for a small number of cases.

Dogs are the definitive hosts of *E. granulosus*. Sheep are the usual intermediate hosts, but humans are accidental intermediate hosts. Humans are end-stage hosts of the parasite since there is no human to human transmission.

Approximately 75% of hydatid cysts are located in the right lobe of the liver and are solitary. The clinical presentation of a hydatid cyst is largely asymptomatic until complications occur. The most common presenting symptoms are abdominal pain, dyspepsia, and vomiting. The most frequent sign is hepatomegaly.

The life cycle of *Echinococcus* is depicted in the image below:



Solution to Question 4:

The presence of antibodies against SARS CoV 2 in the blood likely indicates a previous infection even if the patient has been asymptomatic.

Antibodies most commonly become detectable 1–3 weeks after symptom onset, at which time evidence suggests that infectiousness likely is greatly decreased and that some degree of immunity from future infection has developed. But it is imperative to note that the information on SARS CoV 2 is still emerging.

Solution to Question 5:

The enzyme conjugate in the given ELISA technique is 'B'.

Other options:

A- Substrate - it will be cleaved by the enzyme present on the conjugate antibody.

B - Conjugate - it is an antibody that binds to the detection antibody. It is labeled with an enzyme that cleaves the substrate to form a color product/dye.

C - Detection antibody-binds the antigen. We direct our conjugate against this.

D - Capture Antibody-forms the base, on which antigen (say rotavirus from feces) from the patient sample binds.

Solution to Question 6:

Oocysts are immediately infective after being excreted in the stool.

Cryptosporidium is a coccidian parasite and implicated in causing traveler's diarrhea in immunocompromised patients. The two cryptosporidium species responsible are C.hominis and C.parvum. As the patient has undergone a renal transplant, he/she would most likely be on immunosuppressive medications, making the infection likely. Cryptosporidium hominis is the most common organism implicated worldwide

The infective form is oocysts, excreted in the affected patient's feces, acquired through drinking or recreational water. The oocysts are immediately infectious (unlike Isospora oocysts, which require maturation), and may cause autoinfection or implicated in person to person transmission. Because of its small size (4-5micron), it is difficult to see in routine staining, hence is subjected to acid-fast/kinyoun's stain.

Once ingested, the 4 sporozoites come out of the cyst and attach to the small intestinal mucosa. It causes crampy abdominal pain and non-inflammatory watery diarrhea.

Treatment is mainly supportive, though nitazoxanide has been FDA approved for use in cryptosporidium.

Comparative features of coccidian parasites:

Property	Cryptosporidium	Cyclospora	Cystoisospora
Sporulated oocyst	4-6 µm, round, contains four sporozoites	8-12 µm, round, contains 2 sporocysts each having two sporozoites	23-36 µm, oval, contain two sporocysts, each having four sporozoites.
Acid-fast detection	Uniformly acid-fast	Variable acid-fast	Uniformly acid-fast
Sporulation of the oocyst	Occurs inside the host cell	Occurs in soil	Occurs in soil
Diagnostic form	Sporulated cyst	Unsporulated cyst	Unsporulated oocyst

Solution to Question 7:

Gene Xpert identifies mutations in the rpoB gene to detect rifampicin resistance (RR).

Other options:

inhA and KatG- involved in resistance against isoniazid

pncA- involved in resistance against pyrazinamide

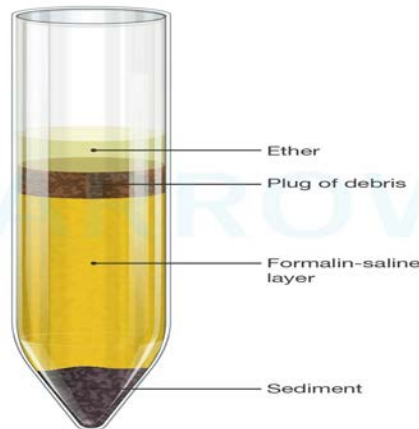
Solution to Question 8:

The parasitic ova will be seen in a specimen taken from level D.

The image shows a formol-ether sedimentation technique to concentrate the ova in the bottom (D) part of the tube. It is used for samples in which the concentration of parasitic ova is scanty. It is useful for all helminth ova/eggs and parasitic cysts. The ether adsorbs the fecal debris and floats. Formalin fixes and preserves the specimen.

This method is useful for all helminth eggs and protozoan cysts.

Formalin-ether sedimentation technique



©Marrow

Solution to Question 9:

Type IV hypersensitivity is due to the cell-mediated response.

Type IV or cell-mediated hypersensitivity reaction: These reactions are induced by sensitized T cells which on contact with the specific antigen releases cytokines that produce biological effects on leucocytes, macrophages, and tissue cells.

Solution to Question 10:

The given clinical scenario points towards a diagnosis of reactive arthritis after shigella infection.

Shigellosis is a spectrum of symptoms caused by shigella consisting of 4 phases:

Other options:

E.coli - it may cause diarrhea associated with fever or blood in stools, but reactive arthritis is not typically seen.

Giardia - it may cause persistent watery diarrhea with waxing and waning course. Fever, blood in stools should prompt alternate diagnosis. Also, reactive arthritis is not seen.

Rotavirus - it is a well known viral cause of fever with diarrhea in children. But not known to have reactive arthritis as a sequela.

Incubation period	1-4 days (up to 8 days)
Watery diarrhea	Fever, watery diarrhea, anorexia
Dysentery	Bloody mucopurulent stools

Incubation period	1-4 days (up to 8 days)
Post-infectious complication	Hemolytic-uremic syndrome Reactive arthritis Toxic megacolon

Solution to Question 11:

Specimen handling for molecular testing of SARS CoV 2 would require BSL-2 or equivalent facilities. Attempts to culture the virus requires minimum BSL-3 facilities.

SARS CoV 2 virus can cause coronavirus disease 2019 (COVID-19)

Microbiology and pathology laboratories performing diagnostic tests on a stool or respiratory specimen should handle potential SARS-CoV specimens using standard biosafety level (BSL)-2.

Note: Biosafety levels include a specific combination of work practices, safety equipment, and facilities that are designed to minimize the exposure of workers and the environment to infectious agents. There are four biosafety levels.

BSL-1: Work can be done on an open bench or table and PPE needed. Eg: Non-pathogenic E.coli.

BSL-2: Work to be done in the biosafety cabinet with PPE. Eg: Staphylococcus aureus.

BSL-3 laboratory has appropriate PPE and respirators might be required. All work performed within BSC(biosafety cabinet). Entrance is through 2 sets of self-closing and locking doors. Eg: when dealing with Mycobacterium tuberculosis.

BSL-4: This is the highest level of biological safety, where containment of microbes is done, which are fatal without treatment and vaccines. Eg: Ebola and Marburg viruses.

Solution to Question 12:

Viral load can be measured by the real time-PCR technique.

Real time-PCR assays are commonly used for viral load quantification of HIV, hepatitis C virus, hepatitis B virus, and CMV.

Other options:

- Western blotting is used for the detection of protein.
- ELISA is an immunosorbent assay used to detect antigens of infectious agents present in clinical specimens.
- Immuno electrophoresis includes electrophoretic separation of antigens followed by an antigen-antibody reaction.

Solution to Question 13:

The organism causing scrub typhus i.e *Orientia tsutsugamushi* is non-cultivable on cell-free media like MacConkey agar.

Scrub Typhus is also known as chigger borne typhus. The vector is trombiculid mite. Humans are infected when they are bitten by the mite larvae known as chiggers.

The incubation period is 1- 3 weeks. The patient typically develops a characteristic eschar at the site of the mite bite, with regional lymphadenopathy and maculopapular rash.

Solution to Question 14:

Ebola virus is not transmitted by mosquito.

Ebola virus is the cause of an emerging disease called Ebola virus disease and is widespread in Africa. It belongs to the Filoviridae family.

Fruit bats act as reservoir hosts of the virus and it is transmitted to humans only accidentally. Human to human transmission is through direct contact with blood or body fluids of an infected symptomatic person.

The incubation period of the disease is 2 to 21 days. Symptoms include fever, intense weakness, vomiting, diarrhea, rashes, and, in extreme cases, internal and external bleeding. The current treatment is mainly supportive. Nosocomial transmission seen.

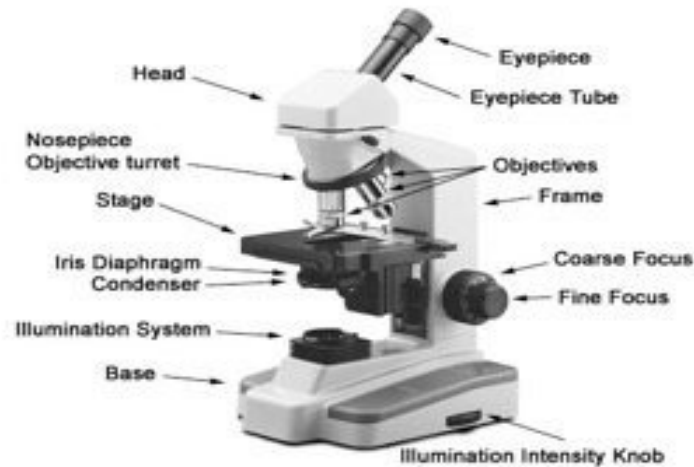
Solution to Question 15:

Condensor lens of microscope is marked as B in the image. The purpose of the condenser lens is to focus the light onto the specimen.

A is coarse focusing knob.

C is objective lens. There are usually 3 or 4 objective lenses on a microscope. They almost always consist of 4X, 10X, 40X and 100X powers.

D is Illuminator. It is a steady light source used in place of a mirror.



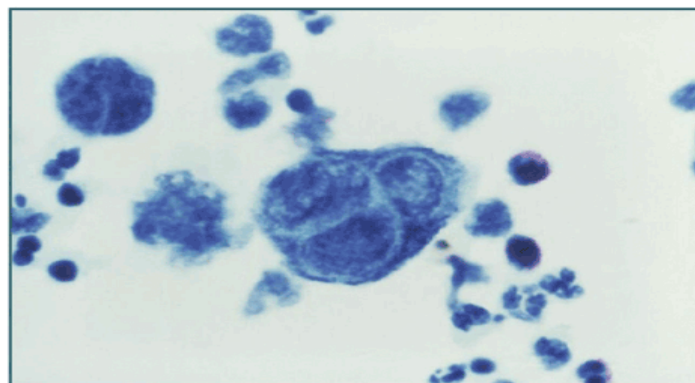
Solution to Question 16:

Clusters of vesicles are seen on an erythematous base in the image, suggesting a diagnosis of genital herpes. Tzanck cells or multinucleated giant cells are present on microscopy.

The Tzanck smear is a rapid, sensitive, and inexpensive diagnostic method for herpes simplex infections. Smears are prepared from the lesions, from the base of vesicles and stained with a 1% aqueous solution of O-toluidine blue for 15 seconds. Multinucleated giant cells with faceted nuclei and homogeneously stained ground glass chromatin (Tzanck cells) are seen.

Mucocutaneous herpetic lesions are thin-walled, umbilicated vesicles, the roof of which breaks down, leaving tiny superficial ulcers which heal without scarring. Painful and pruritic lymphadenopathy may be seen. The most common site of cutaneous lesions is the face - on the cheeks, chin, around the mouth, or on the forehead - and is caused by herpes simplex virus-1. Genital infections are more commonly caused by herpes simplex virus-2.

Tzanck smear



©MARROW

Solution to Question 17:

Albert's stain contains toluidine blue 'O' and malachite green which have an affinity for acidic cytoplasm, both being basic dyes.

Albert's stain, Neisser's and Ponder's stain are used to demonstrate the metachromatic granules (Volutin / Babes Ernst Granules) of *Corynebacterium diphtheria*.

Procedure for Albert's staining:

- A heavy smear of bacterial isolate was prepared on glass slide and allowed to dry without fixing.
- Albert's stain was applied for (3-4) minutes then it was washed with water and Albert's iodine was applied for one minute.
- Wash with water and let dry with blotting paper.
- Examine the slide under oil immersion, it was examined under microscope showed bacilli with numerous metachromatic granules.

Results:

Metachromatic granules: Bluish- black

Bacillary body or protoplasm: Green

Uses:

- Albert staining is useful in the demonstration of metachromatic granules.
- Presence of metachromatic granules helps to distinguish *C.diphtheriae* from diphtheroids which lack them.

Solution to Question 18:

Pneumonia due to a gram-negative pleomorphic bacillus growing on chocolate agar which is catalase and oxidase-positive is most likely *Haemophilus influenza*.

Haemophilus pneumonia typically occurs in infants and is accompanied by empyema and sometimes meningitis as well. In older children and adults, the picture is of lobar pneumonia. Sputum is collected for the detection and isolation of *Hemophilus*. As the bacillus is very sensitive to low temperatures, specimens should never be refrigerated before inoculation. *H.influenzae* is a Gram-negative, non-motile, non-sporing bacillus, exhibiting considerable pleomorphism. In sputum, it usually occurs as clusters of coccobacillary forms.

The bacillus has fastidious growth requirements. The accessory growth factors, named X and V, present in the blood are essential for growth. Catalase and

oxidase reactions are positive. Nitrates are reduced to nitrites. Biotyping has been done on the basis of indole production.

Cultures may be preserved for about a month on chocolate agar slopes. Chocolate agar is made by heating blood agar to 80-90°C for 15-20 minutes. This causes the release of V factor from within the erythrocytes.

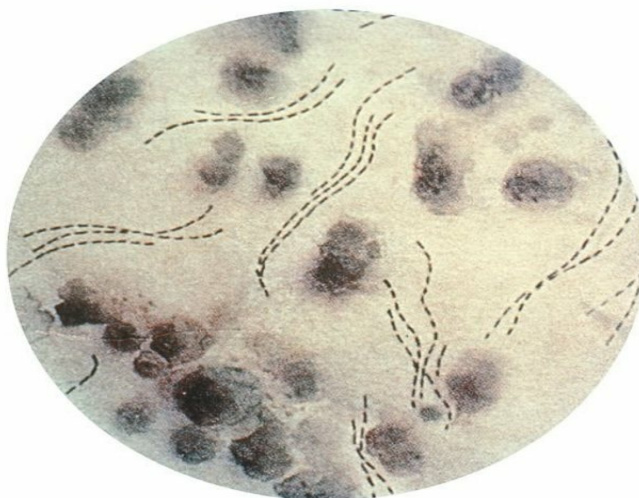
Treatment- Amoxicillin-clavulanate or clarithromycin is more effective in treating respiratory infections by *Haemophilus influenzae*.

Solution to Question 19:

Painful genital ulcers are caused by *Haemophilus ducreyi*, and are known as chancroid.

The regional lymph nodes are enlarged and painful. Microscopic appearance seen in *H. ducreyi* is called the school of fish appearance or railroad appearance. Chocolate agar with isovitaleX is a suitable culture medium for *Haemophilus ducreyi*.

The given image shows the microscopic appearance of *H. ducreyi*.



Solution to Question 20:

The HPV vaccines use virus-like particles (VLPs) that consist of the HPV L1 major capsid protein. The L1 protein self-assembles into VLPs when expressed in eukaryotic cells.

These VLPs contain the same epitopes as the HPV virion. But they do not contain genetic material and cannot transmit infection. The immunogenicity of the HPV vaccines relies on the development of conformational neutralizing antibodies to epitopes displayed on viral capsids.

The nonavalent HPV vaccine works against strains 6,11,16,18,31,33,45,52 and 58. It is usually commercialized in the name of Gardasil 9.

HPV VACCINE	HPV TYPES COVERED
Quadrivalent	6, 11, 16, 18
Bivalent	16, 18
Nonavalent	6, 11, 16, 18, 31, 33, 45, 52, 58

Recent update: Gardasil 9 has received expanded indication which now includes the prevention of oropharyngeal and other head and neck cancers caused by HPV types 16,18,31,33,45,52,58.

Solution to Question 21:

Meningitis with normal glucose, increased protein, and mild lymphocytosis is suggestive of viral meningitis, most probably by an Enterovirus.

Coxsackieviruses are typical enteroviruses. Two groups of Coxsackie viruses A and B, are known. Aseptic meningitis may be caused by a majority of group A and all group B viruses. A maculopapular rash may be present. The disease can also occur as epidemics.

CSF findings in Viral meningitis:

- A mild increase in CSF protein/pressure
- Normal glucose
- Mild increase in lymphocytes in CSF.

Other options:

West Nile virus is endemic in certain regions in North America. But there is no history of travel mentioned in the clinical scenario above. Current hotspots in North America are the California Central Valley, southwestern Arizona, southern Texas, and Louisiana, which have both compatible climates and avian reservoirs for the disease.

	Normal	Bacterial Meningitis	Viral Meningitis	Fungal Meningitis	Parasitic Meningitis	Tuberculous Meningitis
WBC Count (per μ L)	<5	>1000	25-500	40-600	150-2000	25-100
Differential of WBC	60-70% lymphocytes, $\leq$ 30% monocytes/ macrophages	$\uparrow\uparrow$ PMNs ($\geq$ 80%)	Predominantly lymphocytes	Lymphocytes or PMNs, depending on the specific organism	$\uparrow\uparrow$ Eosinophils ($\geq$ 50%)	Predominantly lymphocytes
Gram, stain	Negative	Positive (in > 60% of cases)	Negative	Rarely positive	Negative	Occasionally positive
Glucose (mg/dL)	40-85	<40	Normal	$\downarrow$ to normal	Normal	<50 in 75% of cases

	Normal	Bacterial Meningitis	Viral Meningitis	Fungal Meningitis	Parasitic Meningitis	Tuberculous Meningitis
Protein (mg/dL)	15-45	>100	20-80	150-300	50-200	100-200
Opening pressure (mmH ₂ O)	50-180	>300	100-350	160-340	Normal	150-280

Solution to Question 22:

The given clinical scenario is suggestive of neurocysticercosis caused by larvae of *T. solium* (cysticercosis cellulosae). Glucocorticoid therapy is usually begun before initiating albendazole (antiparasitic drug). This is continued for several days after the institution of therapy to reduce the incidence of side effects from inflammatory reactions to dead and dying cysticerci of *Taenia solium*.

Albendazole is the drug of choice for neurocysticercosis.

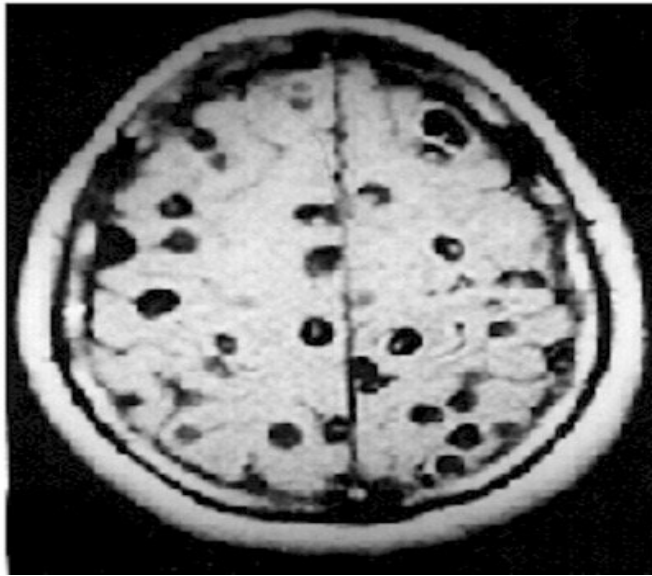
There are various types of cysticercosis based on the site affected such as:

- Subcutaneous nodules (asymptomatic)
- Neurocysticercosis
- Muscular cysticercosis
- Ocular cysticercosis.

Neurocysticercosis is considered the second most common cause of intracranial space-occupying lesion (ICSOL) in India following tuberculosis. Its clinical features include:

- Epilepsy
- Raised intracranial tension
- Hydrocephalus
- Psychiatric disturbances
- Meningoencephalitis
- Transient paresis
- Behavioral disorders
- Aphasia
- Visual disturbances.

The below MRI of the brain shows multiple cysts containing scolices in a patient with neurocysticercosis.



Solution to Question 23:

The image shows transplacental transmission of a virus, leading to microcephaly, with the Aedes mosquito as its vector. Among the given options, Zika virus has the above features.

Zika virus is a mosquito-borne Flavivirus mainly transmitted through *Aedes aegypti*. Zika virus infection presents with mild symptoms of fever, rash, conjunctivitis, malaise, headache and muscle and joint pain. These symptoms usually last for 2-7 days. Maternal infection during pregnancy leads to congenital Zika syndrome or microcephaly.

The complications of Zika virus infection include Guillain-Barré syndrome, neuropathy, and myelitis.

Note: NCDC Delhi and the National Institute of Virology Pune are the only institutes that have the capacity to provide the diagnosis in an acute febrile stage. RT-PCR is the standard test available for the diagnosis of Zika virus infection.

Solution to Question 24:

The disruption of zonula occludens is responsible for watery diarrhea in cholera caused by *Vibrio cholerae*.

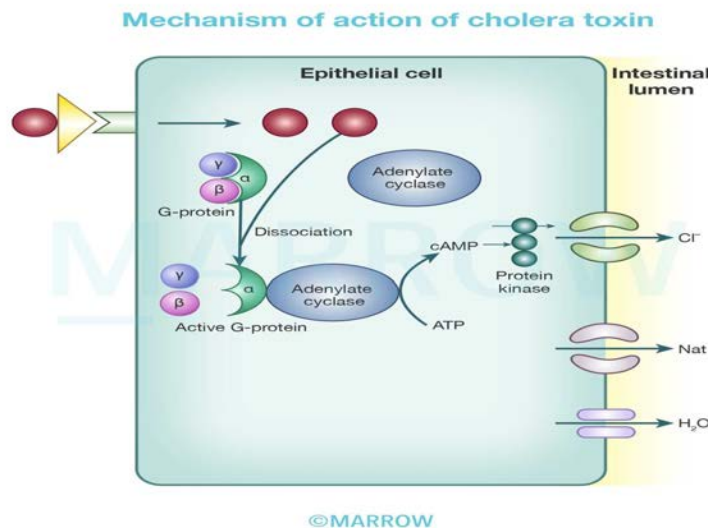
Zonula occludens toxin (Zot) is an enterotoxin elaborated by *Vibrio cholerae* that increases intestinal permeability by interacting with a mammalian cell receptor with subsequent activation of intracellular signaling leading to the disassembly of the intercellular tight junctions.

Mechanism of action of cholera toxin:

- The cholera toxin molecule consists of 1 A and 5 B subunits. The B (binding) units attach to the GM1 ganglioside receptors on the surface of jejunal epithelial cells. The A (active) subunit, on being transported into the enterocyte, dissociates into A1 and A2. A2 fragment only links

biologically active A1 to the B subunit.

- A1 causes prolonged activation of cellular adenylate cyclase and accumulation of cAMP, leading to outpouring into the small intestinal lumen, of large quantities of water and electrolytes. This leads to watery diarrhea which is isotonic with plasma but contains much more potassium and bicarbonate.
- The toxin also inhibits the intestinal absorption of sodium and chloride.
- All clinical manifestations and complications in cholera result from the massive water and electrolyte depletion thus caused.



Solution to Question 25:

Affinity maturation is due to somatic hypermutation.

Affinity maturation is the process by which antibodies gain increased affinity, avidity, and anti-pathogen activity. It is a result of somatic hypermutation (SHM) of immunoglobulin genes in B cells, coupled to selection for antigen binding.

This process occurs in the germinal centers and proceeds for weeks after acute infection or vaccination, or for many cycles during chronic infection. The resulting antibodies can be highly mutated from their germline-encoded counterparts, with increases of several orders of magnitude in affinity for antigen compared to the corresponding naïve B cell receptors (BCRs).

Other options:

Option A: Isotype switching, also known as class switch recombination, is a process where the constant region of the antibody heavy chain is changed, leading to a switch in the antibody isotype (e.g., from IgM to IgG). This process is crucial for the functional properties of antibodies but is not directly involved in affinity maturation.

Option C: mRNA splicing is a cellular process that involves the removal of introns and the joining of exons to generate mature mRNA. While this process is essential for the production of functional antibodies, it is not specifically associated with affinity maturation. Affinity maturation

primarily occurs through somatic hypermutation.

Option D: VDJ rearrangement is the process by which the variable regions of immunoglobulin genes are assembled from gene segments (V, D, and J segments). This process contributes to the diversity of antibodies but is distinct from affinity maturation. Affinity maturation occurs after VDJ rearrangement and is driven by somatic hypermutation in the variable regions.

Solution to Question 26:

The most likely diagnosis here is Mycetoma which is caused by Eumycetoma.

Mycetoma:-

- Mycetoma is a chronic, granulomatous infection of the skin, subcutaneous tissues, fascia and bone, which most often affects the foot or the hand. It is, therefore, commonly known as Maduramycosis or Madura foot. Mycetoma occurs worldwide.
- Etiology: It can be divided into three types, eumycetomas, actinomycetomas and botryomycosis.

Pathogenesis:-

- Triad of symptoms: Infection follows traumatic inoculation of the organism into the subcutaneous tissue from soil or vegetable sources, usually on thorns or splinters and results in tumefactions, deformities and draining sinuses discharging fungal colonies called grains or granules (triad of symptoms).
- Subcutaneous tissues of the feet, lower extremities, hands, and exposed areas are most often involved.
- Grains: Within host tissues the organisms develop to form compacted colonies (grains) 0.5–2 mm in diameter, the color of which depends on the organism responsible.

Laboratory Diagnosis

- A. Direct Examination: The presence of grains in pus collected from draining sinuses or in biopsy material is diagnostic.
- B. Culture Samples should also be cultured, at both 25–30°C and 37°C, on brain-heart infusion agar or blood agar for actinomycetes and on Sabouraud agar (without cycloheximide) for fungi.

Treatment:-

The management of eumycetoma is difficult, involving surgical debridement or excision and chemotherapy. Actinomycetoma responds well to rifampicin in combination with sulfonamides or cotrimoxazole.

Solution to Question 27:

The life cycle shown in the above image is suggestive of leptospirosis.

It is caused by contact with the urine or blood of an infected organism. It is a zoonotic disease, with rats being the most well-known reservoirs.

Illness is classically biphasic after an incubation period of up to 1 month. The stages are:

- Acute leptospiremic phase- organism can be cultured from blood.
- Resolving immune phase- organism can be cultured from urine.

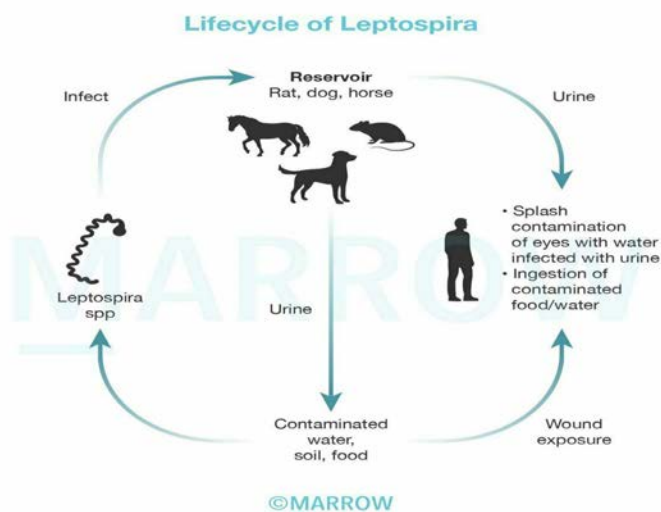
The stages may not be clinically distinguishable.

Most cases are asymptomatic but a few may present with severe manifestations. Clinical features include:

- High-grade fever
- Hepatic dysfunction
- Renal impairment
- Hemorrhages- gastrointestinal, pulmonary (most common), petechiae and purpura

The severe form is known as Weil's disease, with a triad of hemorrhage, jaundice, and acute kidney injury.

Treatment of choice is IV penicillin or IV ceftriaxone.



Solution to Question 28:

The given image shows large cylindrical worms with tapering ends, and thus represents *Ascaris lumbricoides* (roundworm).

Ascaris lumbricoides is the most common of human helminths and is distributed worldwide. It is the largest nematode parasite in the human intestine. The adult worms are large cylindrical worms, with tapering ends. The anterior end is more pointed than the posterior. It appears white outside the body.

Its natural host is humans. There is no intermediate host. Adult worms live in the small intestine - jejunum and ileum. Humans acquire infection by consuming food or water contaminated with nightsoil or human feces containing eggs. The eggs contain the infective rhabditiform larva.

Solution to Question 29:

The above image shows gram⁺negative intracellular diplococci within phagocytes in a stained smear from the urethral discharge sample, which is seen in infection with *Neisseria gonorrhoeae*.

According to CDC recommendations 2021, the first line of treatment in uncomplicated gonococcal infections is a single dose of ceftriaxone 500 mg IM.

Note: BASHH (British Association for Sexual Health and HIV) in the United Kingdom also recommends higher doses of ceftriaxone of about 500 mg along with azithromycin because it would result in fewer failures.

Solution to Question 30:

Rabies has the highest mortality once clinically presented.

Rabies is a vaccine-preventable, zoonotic, viral disease of the central nervous system. Rabies virus is a bullet-shaped rhabdovirus containing ssRNA. It spreads to humans by bites or contact with the saliva of rabid animals (bats, dogs, cats). Once clinical symptoms appear, rabies is virtually 100% fatal. The disease can be divided into 3 phases:

- Prodromal phase: It lasts 2-10 days. Patients present with malaise, headache, photophobia, vomiting, and fever. They may also have parasthesias, pain or pruritus near the site of exposure.
- Acute neurologic phase: It lasts 2-7 days. In the more common encephalitic rabies, Hydrophobia (fear of water) or aerophobia (fear of breeze) can be seen. Swallowing liquids precipitates painful spasms of the throat muscles. Patients may also have nervousness, hallucinations, excessive lacrimation, salivation, perspiration and pupillary dilation. Convulsive seizures may occur
- Coma and death (due to cardiorespiratory arrest)

In about 20% of patients, paralytic rabies is seen. Flaccid muscle weakness is seen, which progresses to quadriparesis and facial palsy.

Note: In rabid animals, the virus depicts two modes of spread:

- Centripetal spread: The virus enters the peripheral neurons. It moves via retrograde fast axonal transport towards the spinal cord or brainstem a.k.a the central nervous system (CNS) at a speed of up to 250 mm/day. Once within the CNS, it spreads quickly to other regions in the CNS through fast axonal transport. It infects neurons primarily, but may rarely infect astrocytes.
- Centrifugal spread: After CNS infection is established, it spreads along the sensory and autonomic nerves to salivary glands, adrenal glands, skin

Rabies virus replicates in acinar cells of the salivary glands and is secreted in the saliva of rabid animals that serve as vectors.

Solution to Question 31:

The given image shows molar tooth colonies that are characteristic of *Actinomyces* species. The given clinical scenario of a mobile, warm to touch nodule in the neck describes cervicofacial actinomycosis also known as 'lumpy jaw syndrome'. Penicillin is the drug of choice for actinomycosis infection.

Actinomyces are Gram-positive anaerobic bacilli that produce molar tooth colonies on incubation for 72 hours on brain-heart infusion agar. They cause a chronic suppurative and granulomatous infection called actinomycosis. It can present as one of the following clinical types:

- Cervicofacial type- Fluctuant mass in the jaw area producing draining fistulas.
- Thoracic type- Pulmonary infection with fever, cough, and purulent sputum.
- Abdominal type- Occurs following ruptured appendix or ulcer. It may spread throughout the peritoneum and involve multiple organs.

Actinomycosis can spread to bones and surrounding tissue and produce multiple discharging sinuses. Treatment requires prolonged administration of penicillin for 6–12 months. In penicillin-allergic patients, clindamycin or erythromycin is effective. Surgical excision and drainage may be required in some cases.

The given image shows cervicofacial actinomycosis.



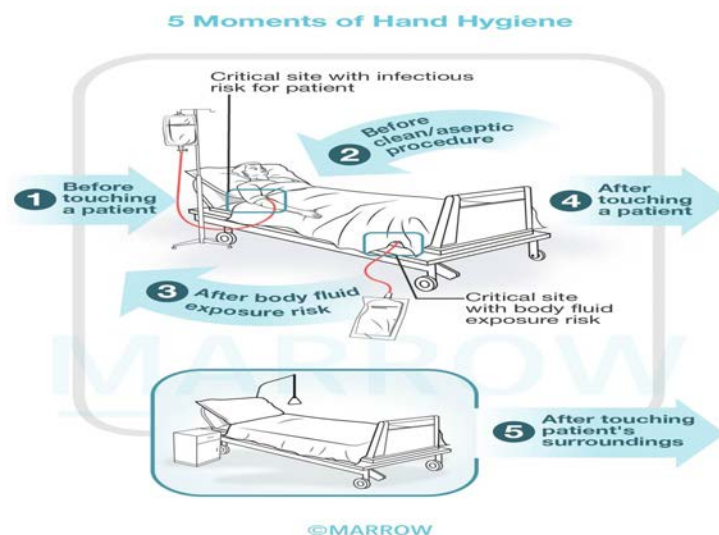
The given image shows a patient with multiple cutaneous lesions caused by *Actinomyces*:



Solution to Question 32:

You should use a sanitiser before and after examining the patient and after touching the patients' surroundings. The hands should be washed with soap and water thoroughly in case they are visibly soiled.

Sanitization is the process through which pathogenic organisms are reduced to safe levels on inanimate objects to reduce the likelihood of cross-infections. It is a component of hand-hygiene, which is demonstrated by the image below.

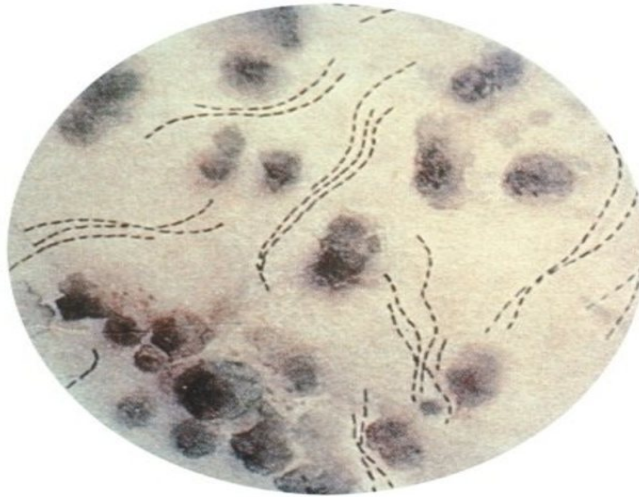


Note: Disinfectants like chlorhexidine and alexidine have a biguanide functional group. They are mostly used as oral disinfectants and hand products. They are bactericidal but not sporicidal. Mycobacteria are also resistant to these compounds.

Solution to Question 33:

Chancroid is a painful genital ulcer caused by *Haemophilus ducreyi*, which is a gram-negative coccobacillus. Microscopic appearance of *H. ducreyi* is called the school of fish appearance or railroad appearance. Chancroid ulcers have a soft base that bleeds easily to touch. They also cause painful lymphadenopathy.

The image below shows the 'school-of-fish' appearance characteristic of *H. ducreyi*.



Note: Groove sign is a characteristic sign seen in lymphogranuloma venereum. It refers to the depression created by the inguinal ligament as it passes between the groups of inflamed inguinal lymph nodes.

Solution to Question 34:

The correctly matched pairs are as follows:

The various strains of *Escherichia coli* are:

- Enterotoxigenic *E. coli* (ETEC) is the most common organism that causes diarrhea in travelers. It produces the following toxins:
 - LT (heat-labile toxin)- acts by $\uparrow$ cAMP
 - ST (heat-stable toxin)- two types of ST are known, ST-A (ST-I) and ST-B (ST-II)
 - ST-A is soluble in methanol and acts by $\uparrow$ cGMP in the intestine.
 - ST-B is insoluble in methanol, and the mode of action is not known.
- Enteroinvasive *E. coli* (EIEC) invade interstitial epithelial cells in the intestine and cause a condition resembling shigellosis. This ability is determined by a plasmid, detection of which can

also be a diagnostic test. Sereny's test was employed to diagnose EIEC but it is not in use anymore.

- Enterohemorrhagic *E. coli* (EHEC) produces verocytotoxin, which targets vascular endothelium. It causes hemorrhagic colitis and hemolytic uremic syndrome.
- Enteropathogenic *E. coli* (EPEC) causes infantile diarrhea in developing countries. It adheres to the mucosal surface of small intestine using EAF and LEE and causes effacement of microvilli. This leads to attachment and effacement lesions of intestinal cells.
- Enteroaggregative *E. coli* (EAEC) produces the stacked brick pattern of adherence on hep-2 cells. EAEC has been associated with persistent diarrhea. Intestinal colonization of EAEC is mediated by AAF (aggregative adherence fimbriae). EAEC produces EAST-1 toxin (enteroaggregative heat-stable toxin) which is considered to be a major virulence factor.

E. coli	Mechanism of action
1. ET EC	A. Produce a heat-stable toxin and cause activation of guanylyl cyclase
2. EI EC	B. Invades intestinal mucosal cells
3. E HEC	C. Produces a toxin against vascular endothelium cells
4. EP EC	D. Adhere to the mucosal surface and cause effacement of microvilli

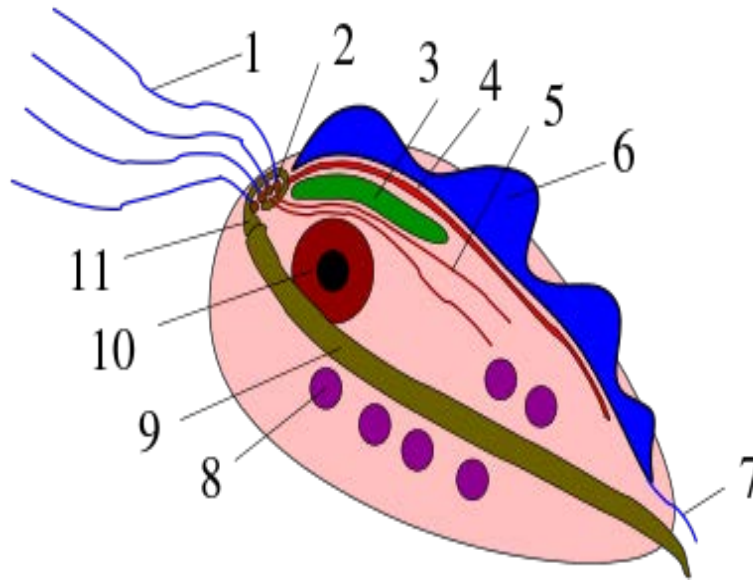
Solution to Question 35:

Identification features of *Trichomonas vaginalis*:

- Pear-shaped or ovoid
- Short undulating membrane reaching upto the middle of the body
- It has four anterior flagella and fifth running along the outer margin of the undulating membrane, which is supported at its base by a flexible rod, costa
- A prominent axostyle runs throughout the length of the body and projects posteriorly like a tail
- The cytoplasm shows prominent siderophilic granules, which are most numerous alongside the axostyle and costa
- Jerky or twitching type of mobility

Parasite	Organ of locomotion
Balamuthia mandrillaris	Pseudopod

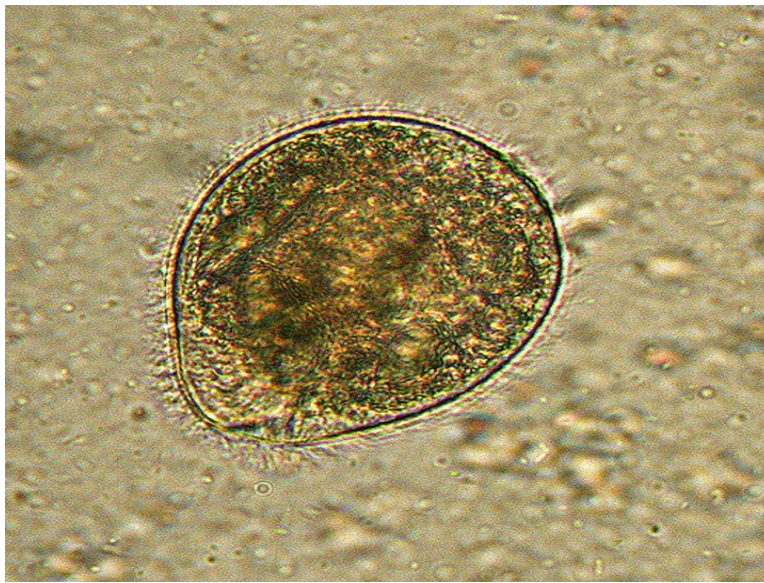
Parasite	Organ of locomotion
<i>Balantidium coli</i>	Cilia
<i>Trichomonas vaginalis</i>	Flagella
<i>Toxoplasma gondi</i>	No Particular organ



The above image shows *Trichomonas vaginalis* : 1-anterior flagella, 2-kinetosomes, 3-parabasal body, 4-costa, 5-parabasal fibers, 6-undulating membrane, 7-posterior flagellum, 8-hydrogenosomes, 9-axostyle, 10-nucleus, 11-pelta.

Obligate parasite - Organism cannot survive outside host

- No cystic stage. Trophozoites are infective forms
- Sexual transmission is the main route of infection. Other modes are fomites, babies during passage through infected birth canal
- Affects squamous epithelium - Intracellular edema and so called chicken-like epithelium, is the most characteristic feature of trichomoniasis
- Habitat: Vagina and cervix in female and urethra in males
- Clinical features: Often asymptomatic in males. In females, it leads to pruritic vaginitis with greenish yellow discharge, strawberry mucosa and dysuria
- Diagnosis: By wet mount microscopy of vaginal or urethral discharge, Culture (gold standard - Cysteine-peptone-liver-maltose (CPLM) medium and plastic envelope medium (PEM)), PCR, and by demonstration of antigen in vaginal smear by ELISA
- Treatment: Metronidazole is the drug of choice and simultaneous treatment of both partners is recommended

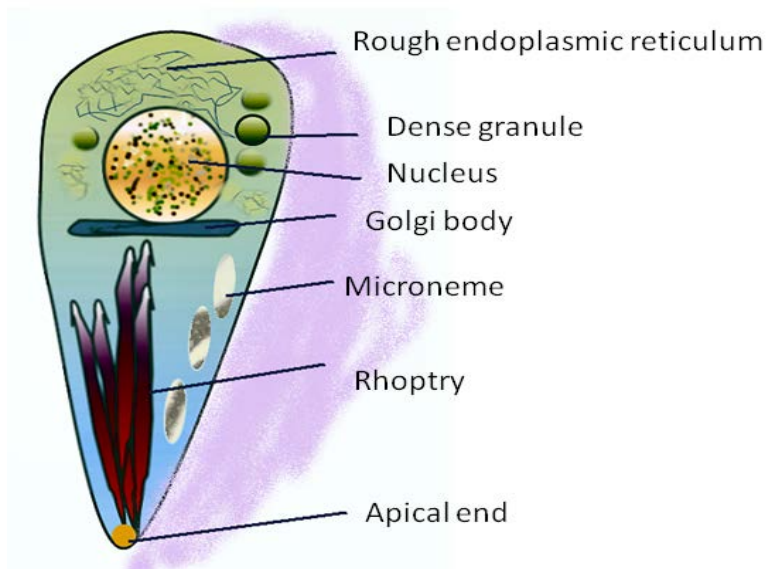


The given image is the trophozoite form of the *Balantidium coli*.

Balantidium coli occurs in two stages

- Trophozoites are oval in shape and contain two nuclei. There is the presence of cilia over the entire surface and are larger around the mouth. Its anterior end is narrow and the posterior end is broad. At the anterior end, there is a groove leading to the mouth (cytostome). Posteriorly, there is a small anal pore (cytopyge).
- Cysts are smaller, round with a micronucleus, a macronucleus and many contractile vacuoles.

Living trophozoites/cysts are yellowish or greenish in color on wet mount.



The image given above shows *Toxoplasma gondii* which has two types of life cycles:

- Enteric cycle
- Exoenteric cycle

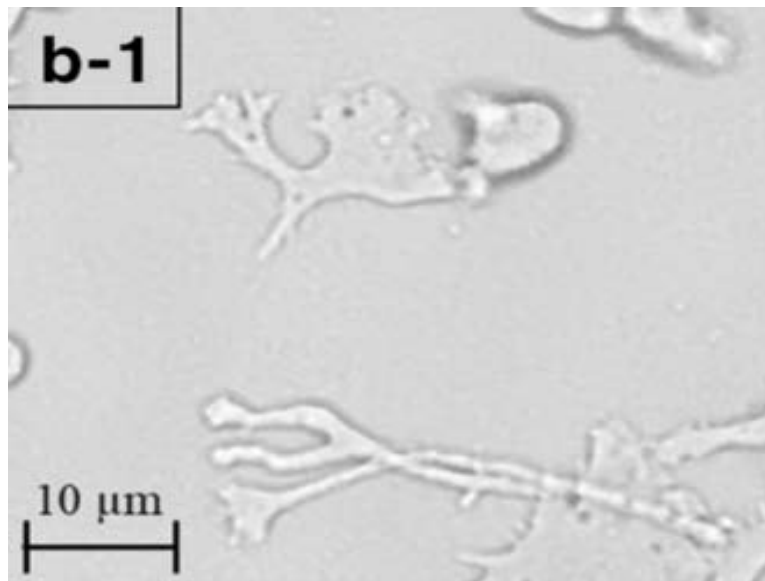
The enteric cycle occurs in definitive hosts such as cats. Exoenteric cycle occurs in intermediate hosts such as humans, mice, rats, etc.

Both sexual reproduction (gametogony) and asexual reproduction (schizogony) occur within the mucosal epithelial cells of the small intestine of the cat.

Sporozoites from the oocysts and bradyzoites from the tissue cysts enter the intestinal mucosa and multiply asexually. Tachyzoites are formed within the intermediate host.

Tachyzoites continue to multiply and spread locally by lymphatic system and blood, while the bradyzoites multiply slowly inside the tissue cyst.

The dormant bradyzoites inside the cyst may be reactivated in immune suppression causing renewed infection in the host.



Images b-1 and b-3 showing the Trophozoite and Cystic form of *Balamuthia mandrillaris* in *Salmonella Shigella* (SS) Agar respectively.

Balamuthia Mandrillaris

B. mandrillaris is a free-living amoeba, which is a newly identified species reported to cause granulomatous amoebic encephalitis (GAE).

Morphology

It exists in amoeboid trophozoite stage. The flagellate stage is absent

- It is relatively large (12-60 μm), irregular in shape and actively motile by broad pseudopodia.
- Cyst of *B. mandrillaris* are usually spherical (6-20 μm), surrounded by a three-layered cyst wall: (1) outer irregular ectocyst, (2) a middle mesocyst and (3) an inner endocyst round wall. Under light microscopy, it appears to have two walls: (1) an outer irregular wall and (2) an inner smooth wall.
- Infection is transmitted through respiratory tract, skin lesions, or eyes.

Solution to Question 36:

The galactomannan assay is used for the diagnosis of invasive (deep-seated) mycoses caused by *Aspergillus* species.

Galactomannan is a component of the cell wall of *Aspergillus*. It can be detected in serum or bronchoalveolar lavage by ELISA for the diagnosis of invasive aspergillosis in adults and children. This test is especially useful in hematopoietic stem cell transplant recipients or patients with hematologic malignancies. Invasive aspergillosis affects the lungs and presents with fever, cough, dyspnea, and hemoptysis. The organism may spread to the eye, gastrointestinal tract, kidney, liver or brain, leading to abscesses and necrotic lesions.

Other diagnostic features of invasive aspergillosis are:

- Positive culture from the affected organ
- Microscopy showing narrow, septate hyphae with branching at 45° and conidiophores
- Halo sign on CT scan

Solution to Question 37:

The correct answer is A-3, B-2, C-4, D-1.

The hosts of the above parasites are given below:

A. Diphyllbothrium	3. Cyclops
B. Paragonimus	2. Crayfish
C. Brugia	4. Mansonia
D. Fasciola	1. Sheep

Parasite	Definitive host	First intermediate host	Second intermediate host
Diphyllobothrium	Humans	Cyclops	Cyclops and fish
Paragonimus	Humans	Snail	Crayfish and crab
Fasciola	Sheep and humans	Snail	Aquatic vegetations
Brugia	Humans	Mansonia mosquito	-

Microbiology NEET 2018

Question 1:

Nude mice are not resistant to xenograft due to the absence of _____

- a) B cell
- b) T cell
- c) Both B and T cell
- d) NK cell

Question 2:

Which of the following is *Burkholderia cepacia* resistant to?

- a) Ceftazidime
- b) Trimethoprim-sulfamethoxazole
- c) Temocillin
- d) Cefotetan

Question 3:

Which of the following organisms causes scombroid fish poisoning?

- a) *Salmonella*
- b) *Staphylococcus*
- c) *Pseudomonas aeruginosa*
- d) *Peptostreptococcus*

Question 4:

Which of the following is the predominant pathogen to infect renal transplant recipients one to four months following transplantation?

- a) Cytomegalovirus
- b) JC virus

- c) BK virus
- d) Epstein-Barr virus

Question 5:

Sabin-Feldman dye test is used for diagnosis of which of the following condition?

- a) Paragonimiasis
- b) Hydatid cyst
- c) Toxoplasmosis
- d) Cryptococcus

Question 6:

Which of the following is called as Australia antigen?

- a) HBsAg
- b) HBeAg
- c) HBcAg
- d) HBV-DNA

Question 7:

Bacterial transduction occurs by

- a) Plasmids
- b) Sex pili
- c) Bacteriophage
- d) Uptake of genetic material by other bacteria.

Question 8:

Which fungus is most commonly associated with orbital cellulitis in patients with diabetic ketoacidosis ?

- a) Candida
- b) Rhizopus

- c) Aspergillus
- d) Trichophyton

Question 9:

The adult form of echinococcus is seen in which of the following?

- a) Dog
- b) Cat
- c) Swine
- d) Sheep

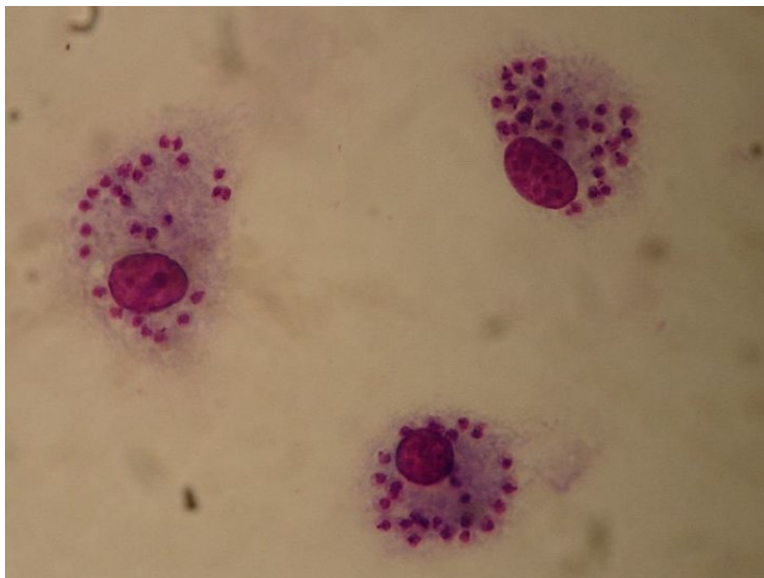
Question 10:

Which of the following can grow in an acidic pH?

- a) Vibrio
- b) Lactobacilli
- c) Salmonella
- d) Shigella

Question 11:

What is the vector for the condition diagnosed by the image shown below?



- a) Anopheles mosquito
- b) Ixodes Tick
- c) Sandfly
- d) Tsetse fly

Question 12:

TB test done only on whole blood sample is _____.

- a) TrueNat TB
- b) Gene xpert
- c) Interferon gamma release assay
- d) All the above

Question 13:

Multiple abscesses with discharging sinuses, demonstrating sulphur granules in pus, are caused by

- a) Actinomyces
- b) Nocardia
- c) Salmonella
- d) Tularemia

Question 14:

Cutaneous larva migrans is most commonly caused by which organism?

- a) Strongyloides stercoralis
- b) Toxocara canis
- c) Ancylostoma braziliense
- d) Necator americanus

Question 15:

Which type of hypersensitivity reaction is involved in the hyperacute rejection of renal transplants?

- a) Type IV
- b) Type III
- c) Type II
- d) Type I

Question 16:

Which of the following organisms produce urease?

- a) *Helicobacter pylori*
- b) *Salmonella*
- c) *Escherichia coli*
- d) *Shigella*

Question 17:

Mycoplasma is inherently resistant to which of the following antibiotics?

- a) Beta lactam antibiotics
- b) Macrolides
- c) Tetracyclines
- d) Flouroquinolones

Question 18:

Pasteurization is done at _____.

- a) 73 degree Celsius for 20 minutes
- b) 63 degree Celsius for 30 minutes
- c) 72 degree Celsius for 30 seconds
- d) 63 degree Celsius for 30 seconds

Answer Key

Question No.	Correct Option
1	b
2	d
3	c
4	a
5	c
6	a
7	c
8	b
9	a
10	b
11	c
12	c
13	a
14	c
15	c
16	a
17	a
18	b

Detailed Explanations

Solution to Question 1:

Nude mice are not resistant to xenograft due to absence of T cell.

A xenograft is a tissue graft or organ transplant from a donor of a different species.

Nude mice lack body hair, hence the name. They lack a thymus and thus cannot generate mature T lymphocytes. The absence of T-lymphocytes makes them unable to mount adaptive immune responses requiring helper T cells and cytotoxic T cells. The absence of functioning T cells also prevents the rejection of allografts & xenografts.

Solution to Question 2:

Burkholderia cepacia is resistant to cefotetan.

B. cepacia complex strains are intrinsically resistant to aminoglycosides, polymixin, first and second-generation cephalosporins (e.g. cefotetan) and carboxypenicillins. This antimicrobial resistance is mediated by efflux pumps, antimicrobial degradation, modifying enzymes or altered

membrane functions.

B. cepacia is a Gram-negative bacillus which can cause severe necrotizing pneumonia in patients with chronic granulomatous disease. It can lead to fatal septicemia in patients with cystic fibrosis.

Antimicrobial agents that are effective against *B. cepacia* complex include meropenem, ceftazidime, trimethoprim-sulfamethoxazole, piperacillin and temocillin.

Solution to Question 3:

Scombroid fish poisoning is associated with *Pseudomonas aeruginosa*.

It follows the consumption of both scombroid (mackerel-like) fish and nonscombroid fish that have not been refrigerated properly. The musculature of these dark- or red-fleshed fish undergoes decomposition by *Morganella morganii*, *Escherichia coli*, *Proteus* species, *Klebsiella* species, and *Pseudomonas aeruginosa*. There is consequent decarboxylation of the amino acid L-histidine to form histamine.

A second compound is thought to play a causative role in this intoxication because large doses of oral histamine do not reproduce the affliction. The toxin or toxins involved are heat stable and are not destroyed by cooking or freezing.

Symptoms develop within 15–90 min of ingestion. Most cases are mild, with tingling of lips and mouth, mild abdominal discomfort, and nausea. The more severe and commonly described presentation includes flushing, conjunctival hyperemia, pruritus, urticaria, and angioedema. Without treatment, the symptoms generally resolve within 8–12 h.

Solution to Question 4:

The predominant pathogen to infect renal transplant recipients in the period around 1 to 4 months following the transplantation is cytomegalovirus (CMV).

CMV is the most common virus to affect recipients after organ transplantation (kidney, heart, lung etc). The maximal risk occurs between 1 and 4 months after transplantation (in the absence of prophylaxis). The most common CMV presentation in kidney transplant patients includes fever, leukopenia, hepatosplenomegaly, myalgia, and arthralgia. Antiviral prophylaxis with ganciclovir or valganciclovir is used in renal transplant recipients whenever indicated to prevent CMV disease.

Given below are the most common opportunistic infections in renal transplant recipients:

Peritransplant (less than 1 month)	Early (1-6 months)	Late (more than 6 months)
Wound infection Herpes virus Oral candidiasis Urinary tract infection	<i>Pneumocystis jirovecii</i> Cytomegalovirus Hepatitis B Listeria	<i>Aspergillus</i> <i>Nocardia</i> BK Virus (polyoma) Herpes Zoster

Solution to Question 5:

Sabin-Feldman dye test is used for the diagnosis of toxoplasmosis.

It is based on the principle that the presence of anti-toxoplasma antibodies in the patient's serum inhibits the staining of trophozoites by alkaline methylene blue dye. Toxoplasma trophozoites and complement are added to the patient's serum which is then incubated for an hour at 37°C. After incubation, methylene blue dye is added.

- If anti-toxoplasma antibodies are present in the serum, these are activated by complement and the parasite membrane is lysed. Thus, the trophozoites are not stained (positive result).
- If there are no antibodies, trophozoites will take up the stain and appear blue under the microscope (negative result).

The dilution of the test serum at which 50% of the tachyzoites are not stained is reported as the antibody titer of the test serum. The test is highly sensitive and specific. However, it cannot differentiate between recent or past infections as it detects only IgG antibodies.

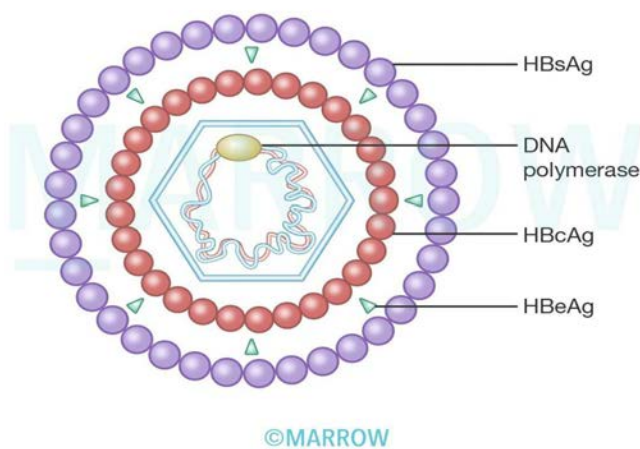
Solution to Question 6:

HBsAg (Hepatitis B surface antigen) is also known as the Australia antigen. It is called so because it was first isolated from the serum of an Australian person.

Sera from hepatitis B patients when observed under an electron microscope show three types of particles:

- A spherical particle: 22nm, most abundant, contains only HBsAg.
- Filamentous or tubular particle: 22nm, contains only HBsAg.
- A double-walled spherical structure: 42nm diameter, is the complete Hepatitis B virus. It was described by Dane and is called Dane particle.

Structure of hepatitis B virus

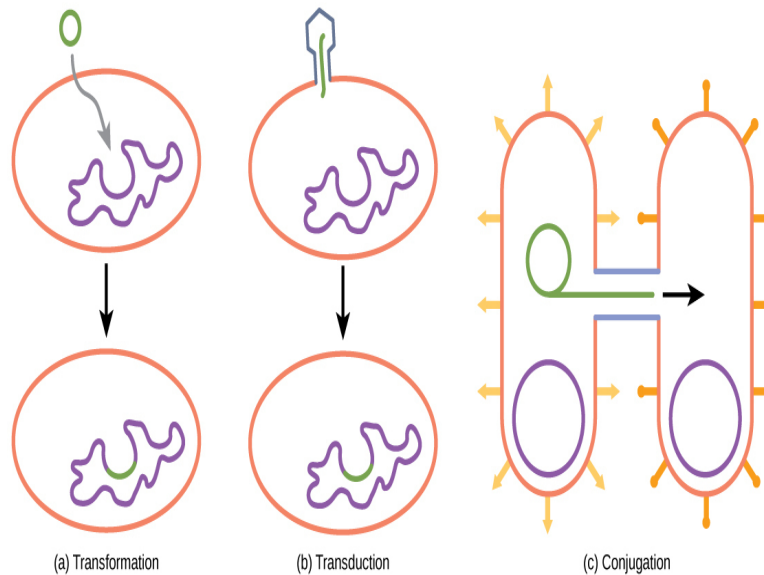


Solution to Question 7:

Bacterial transduction occurs by bacteriophages.

Transduction is a process of acquiring bacterial DNA from a bacteriophage (virus) that has incorporated bacterial DNA from previous host bacteria.

Other methods of exchange of genetic material are transformation and conjugation. Transformation involves the uptake of genetic material from the environment, which is released by other bacteria. Conjugation is the direct transfer of DNA from one bacterial cell to another bacterial cell through a sex pilus or bridge. The transferred DNA is a plasmid, a circle of DNA that is distinct from the main bacterial chromosome.



Solution to Question 8:

Rhizopus is the most common fungus associated with orbital cellulitis in patients with diabetic ketoacidosis.

Diabetic patients are immunocompromised and so prone to develop opportunistic infections like mucormycosis - caused by *Rhizopus* and *Mucor* spp. Rhino-orbital- cerebral disease is the most common form of mucormycosis. It mostly occurs in diabetics. Patients initially present with eye or facial pain, facial numbness, conjunctival suffusion and swelling, and blurry vision. Infection then spreads from the ethmoid sinus to the orbit, leading to proptosis and orbital cellulitis.

Rhizopus is angioinvasive, and thus the infection may spread to the frontal lobe and cavernous sinus, leading to septicemia and thromboses. The sinus tissue in such patients often develops into a black necrotic eschar.

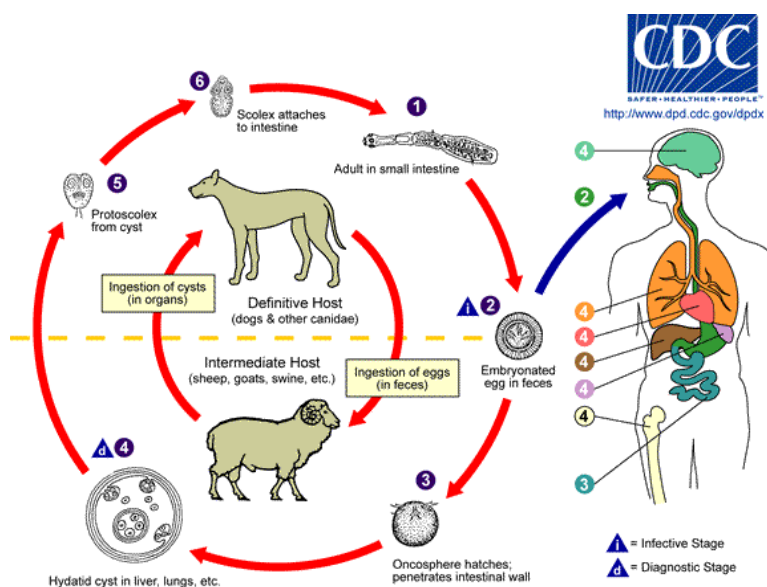
Solution to Question 9:

The adult form of *Echinococcus* is seen in dogs and other canids, which are its definitive hosts.

The larval stage is passed in intermediate hosts such as sheep, cattle, goat, swine, horses, camels, and man. Man acts as an accidental intermediate host (dead end). Human infection follows the ingestion of eggs due to the intimate handling of dogs or by eating raw vegetables or other food contaminated with dog feces.

Note: Humans can become accidental definitive hosts, second intermediate hosts, or paratenic hosts for spirometra. These are cestodes belonging to the Diphyllbothriidae family. They cause a condition called sparagnosis.

The image below shows the life cycle of *Echinococcus granulosus*.



Solution to Question 10:

Among the given options, lactobacilli is the one that can grow in an acidic pH.

It is a gram-positive, facultatively anaerobic, non-motile, non-spore-forming bacterium.

Lactobacillus derives lactic acid from glucose, creating an acidic environment, and can grow in an acidic medium. It is found in the vagina as part of its normal microbiota.

Solution to Question 11:

The image shown is of the Leishman-Donovan bodies whose presence are diagnostic of Kala Azar. The vector of Kala Azar is a sandfly.

Leishman-Donovan bodies are amastigotes with a double dot appearance containing a nucleus and small rod-like kinetoplast. They are present inside the macrophage.

Solution to Question 12:

TB test done on a whole blood sample is the interferon-gamma release assay. It has been used for the diagnosis of latent tuberculosis in patients with HIV(who frequently have a negative Mantoux test).

Interferon-gamma release assays include:

- QuantiFERON-TB Gold
- T-SPOT.TB

QuantiFERON-TB Gold quantitates the amount of IFN- γ produced by CD4+ T cells in response to the antigens from Mycobacterium tuberculosis. QuantiFERON-TB Gold Plus (QFT- Plus) is an improved version of the QuantiFERON-TB Gold test. QuantiFERON-TB Gold detects only TB-specific CD4+ responses. However, QuantiFeron-TB Gold Plus detects both CD8+ and CD4+ T-cell responses to TB-specific antigens.

T-SPOT.TB determines the total number of individual effector T cells expressing IFN- γ .

The TrueNat TB test is a new molecular test that can diagnose TB in one hour as well as test for rifampicin resistance. This test uses a sputum sample taken from the patient. Only about 0.5 ml of the sample is required. Gene Xpert is also a molecular method of testing that cannot use blood as a sample. (Option A and B)

Solution to Question 13:

Multiple abscesses with discharging sinuses, demonstrating sulfur granules in pus, are characteristic of actinomycosis caused by Actinomyces.

Actinomycosis is a chronic granulomatous infection characterized by indurated swellings in the connective tissue with suppuration and discharge of sulfur granules. The most common causative agent is Actinomyces israeli.

Various clinical forms of actinomycosis include cervicofacial, thoracic, abdominal, and pelvic actinomycosis. Among them, oral-cervicofacial actinomycosis is most common, presenting as a soft tissue swelling, abscess, or mass lesion involving the angle of the jaw.

Solution to Question 14:

Cutaneous larva migrans is most commonly caused by Ancylostoma braziliense.

Cutaneous larva migrans or ground itch or creeping eruption is caused by nematode larvae that infect by skin penetration.

The most common cause is non-human species of Hookworm Ancylostoma braziliense and A. caninum.

Necator americanus and Strongyloides sp also cause cutaneous larva migrans, and Toxocara canis usually causes viscreal and ocular larva migrans.

Differences between cutaneous larva migrans and visceral larva migrans:

	Cutaneous larva migrans	Visceral larva migrans
Tissue involved	Skin	Various organs of the body like the liver, lungs, and eyes
Infecting Organism	Mostly by non-human nematodes	Mostly by non-human nematodes (mainly by dog and cat)
Portal of Entry	Penetration of skin	Ingestion of infected eggs
Eosinophilia	Mild	Persistent high
Serodiagnosis	Not developed	Well developed
Treatment	Thiabendazole	Diethylcarbamazine and prednisolone

Solution to Question 15:

Hyperacute graft rejection is an example of type II hypersensitivity reaction.

It is mediated by preformed antibodies. It is a rare event that occurs in kidney transplantation.

Graft rejection	Time taken for rejection	Immune mechanisms involved	Type of hypersensitivity
Hyperacute	Minutes to hours	Preformed antibodies	II
Acute	Weeks to months	Cytotoxic T-cell mediated	IV
Chronic	Months to years	Chronic DTH mediated and Ab mediated	IV

Solution to Question 16:

Helicobacter pylori produces urease.

Urease-producing organisms: (Mnemonic-PUNCH)

- *Proteus*
- *Ureaplasma urealyticum*
- *Nocardia*
- *Cryptococcus*, *Corynebacterium urealyticum*
- *H. pylori*

Solution to Question 17:

Mycoplasma are inherently resistant to beta-lactam antibiotics.

Mycoplasma lack cell wall, and are therefore inherently resistant to antibiotics acting on the bacterial cell wall e.g. beta-lactams (penicillins, cephalosporins).

Solution to Question 18:

Pasteurization: Milk is heated at either

- 63°C for 30 minutes: the Holder method or
- 72°C for 15–20 seconds: the flash process

followed by cooling quickly to 13°C or lower.

All the non-sporing pathogens are killed except *Coxiella burnetti*, which may survive in Holder method.

Microbiology NEET 2019

Question 1:

BCYE medium is used to culture:

- a) *Leptospira*
- b) *Bacillus cereus*
- c) *Campylobacter*
- d) *Legionella*

Question 2:

A 16-year-old boy is admitted with fever, icterus, conjunctival suffusion, and hematuria for 15 days. Which serological test should be done for diagnosis?

- a) Widal test
- b) Microscopic agglutination test
- c) Weil felix reaction
- d) Paul Bunnell test

Question 3:

A patient presents with gastrointestinal problems including abdominal pain and distension, bloody and mucus-filled diarrhea, and tenesmus, with rectal prolapse. A stool examination reveals the presence of typical barrel-shaped eggs. What is the possible causative agent?

- a) *Giardia lamblia*
- b) *Trichuris trichiura*
- c) *Ascaris lumbricoides*
- d) *Ancylostoma duodenale*

Question 4:

Viral infection mimicking erythroblastosis fetalis is:

- a) HSV

- b) Epstein-Barr virus
- c) Picornavirus
- d) Parvovirus B19

Question 5:

An 11-year-old boy was brought to the pediatric OPD with multiple abscesses over his face, chest, and back. The child has a history of recurrent respiratory infections. On examination, he has atopic excoriating skin and multiple cold abscesses on his back. What is the most probable diagnosis?

- a) Hyper IgM syndrome
- b) Wiskott-Aldrich syndrome
- c) Hyper IgE syndrome
- d) Carcinoid syndrome

Question 6:

Cowdry type A inclusions are seen in:

- a) Herpes simplex
- b) Rabies
- c) Molluscum contagiosum
- d) Adenovirus

Question 7:

Which of the following statements is false regarding MHC molecules?

- a) MHC class II presents peptide antigen to CD4 T cells
- b) The action of MHC II occurs via endocytic pathway
- c) MHC Class II presents endogenous antigens
- d) MHC II is present on antigen presenting cells

Question 8:

Which of the following investigations does not use antibody marking?

- a) ELISA
- b) Radioimmunoassay
- c) Immunofluorescence
- d) Hemagglutination

Question 9:

A 3-year-old child presents with a history of repeated sinopulmonary infections caused by encapsulated organisms. Which of the following is most likely to be deficient in this case?

- a) IgG1
- b) IgG2
- c) IgG3
- d) IgG4

Question 10:

A veterinarian developed an ulcer with central black eschar over the forearm as shown in the image. Which of the following is the most probable causative organism?



- a) *Pseudomonas aeruginosa*
- b) *Yersinia pestis*
- c) *Bacillus anthracis*
- d) *Francisella tularensis*

Question 11:

Which of the following is not used as an immediate test in the screening of urinary tract infections (UTI)?

- a) Pour plate culture
- b) Griess Nitrite test
- c) Leukocyte esterase test
- d) Wet mount examination

Question 12:

In which of the following is the disc diffusion method for antibiotic susceptibility testing applied?

- a) Agar dilution method
- b) Kirby-Bauer method
- c) Broth dilution method
- d) PCR based assay

Question 13:

What is the investigation of choice for neurosyphilis?

- a) Venereal disease research laboratory (VDRL) test
- b) FTA-ABS- Fluorescent Treponemal Antibody Absorption test
- c) RPR- Rapid Plasma Reagin test
- d) TPI-Treponema pallidum immobilization test

Question 14:

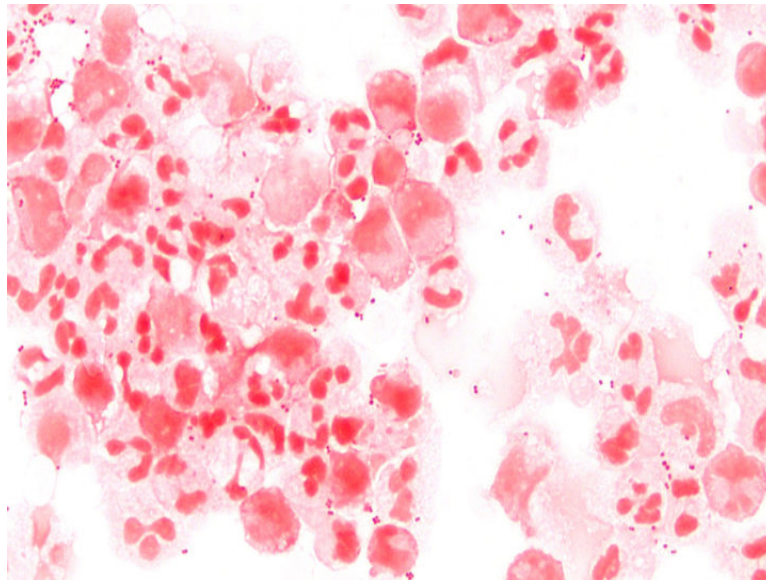
Contact based isolation is employed for which of the following diseases?

- a) Diphtheria
- b) MRSA
- c) Mumps

d) Asthma

Question 15:

An 18 year old female presents with severe headache and high fever of 101°F. On examination, papilloedema and nuchal rigidity was present. Gram staining of the CSF is shown below. Causative organism is



- a) Haemophilus influenza type B
- b) Neisseria meningitidis
- c) Streptococcus pneumoniae
- d) Listeria monocytogenes

Question 16:

What is the stain used for identifying cryptococcus?

- a) Giemsa stain
- b) Ziehl-Nielsen stain
- c) Gram staining
- d) Mucicarmine

Question 17:

Which of the following organisms follow a bird-arthropod-human transmission route?

- a) Malaria
- b) Japanese encephalitis
- c) Plague
- d) Paragonimiasis

Question 18:

A 23-year-old patient has a history of travel to China and the consumption of pickled crab. He now complains of rusty sputum and cough. What is the most likely causative organism?

- a) *Fasciola buski*
- b) *Fasciola hepatica*
- c) *Paragonimus westermani*
- d) *Schistosoma japonicum*

Question 19:

Which type of virus is rubella?

- a) Flavivirus
- b) Togavirus
- c) Paramyxovirus
- d) Orthomyxovirus

Question 20:

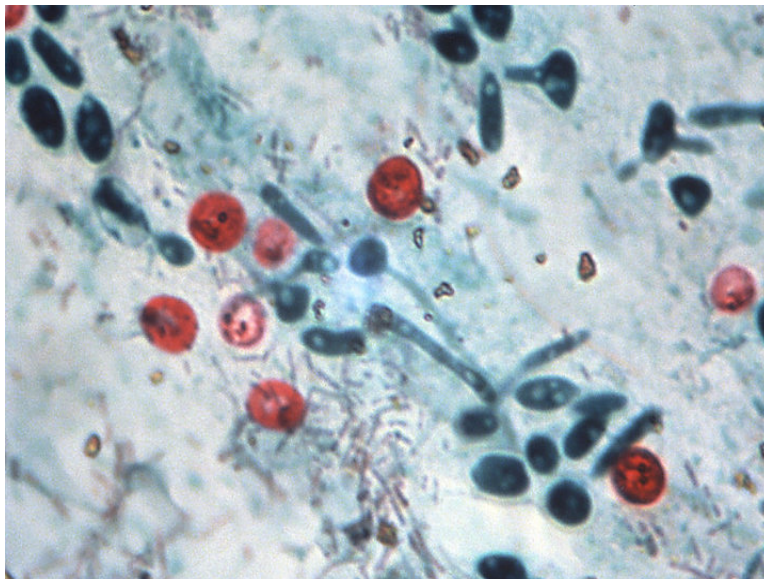
A 30 year old female developed pruritic papules with excoriated plaques in the interdigital web spaces. Later the lesion spread to the groin and under-surface of the breasts. The lady gives a history of intense pruritus which is more severe at night. Given below is the organism responsible for this condition. Identify the wrong statement about the same.



- a) Life history has 4 stages
- b) Larvae have 3 pairs of legs
- c) The entire life cycle takes about 15 days
- d) Male parasite burrows into the epidermis

Question 21:

A HIV patient presents with persistent diarrhea. Microscopic examination of the stool is given below. Identify the causative organism:

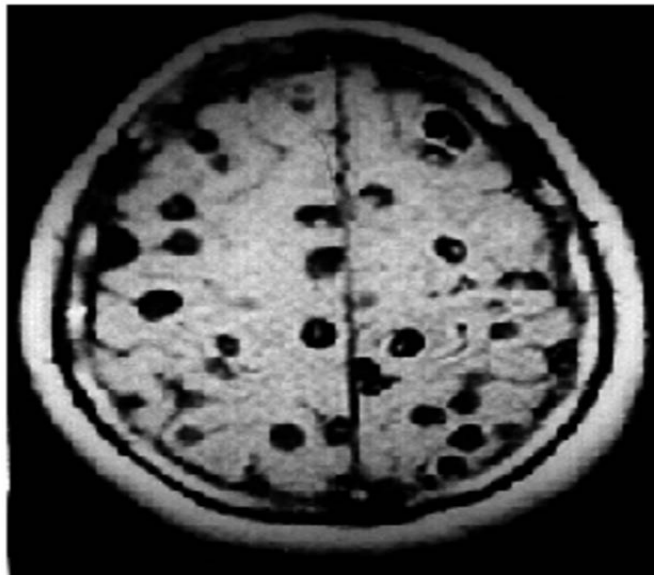


- a) Cryptosporidium

- b) Cyclospora
- c) Microspora
- d) Cystoisospora

Question 22:

A patient is brought to the hospital, with complaints of seizures and the MRI image of the brain is given below. The disease is most probably caused by?



- a) Cysticercosis cellulosae
- b) T. saginata
- c) Cysticercosis bovis
- d) T. asiatica

Question 23:

Vaccine that requires strain change due to evolving organism is?

- a) Influenza
- b) Diphtheria
- c) Polio
- d) Typhoid

Answer Key

Question No.	Correct Option
1	d
2	b
3	b
4	d
5	c
6	a
7	c
8	d
9	b
10	c
11	a
12	b
13	a
14	b
15	b
16	d
17	b
18	c
19	b
20	d
21	a
22	a
23	a

Detailed Explanations

Solution to Question 1:

BCYE medium (buffered charcoal yeast extract agar) is used to culture *Legionella pneumophila*.

Legionella pneumophila has fastidious growth requirements and grows on complex media such as BCYE medium with L-cysteine and antibiotic supplements (e.g., polymyxin B, cefazolin, vancomycin). The charcoal acts as a detoxifying agent. It grows slowly, and the colonies may take 3–6 days to appear.

Other options:

Option A: *Leptospira* is cultured on EMJH (Ellinghausen, McCullough, Johnson and Harris) medium

Option B: *Bacillus cereus* is cultured on PEMBA (Polymyxin pyruvate egg-yolk mannitol–bromothymol blue) agar

Option C: *Campylobacter* is cultured on Skirrow's, Campy BAP or Butzler's agar.

Solution to Question 2:

A history of fever with liver and kidney involvement should arouse suspicion of leptospirosis (Weil's disease). A microscopic agglutination test is used for the diagnosis of leptospirosis.

Antibodies against leptospirosis appear in the blood 5-7 days after infection. Serotype-specific tests for leptospirosis includes:

- Microscopic agglutination test
- Macroscopic agglutination test

Other options:

Option A: Widal test is used to screen for typhoid fever

Option C: Weil-Felix reaction is used to diagnose rickettsial fevers

Option D: Paul-Bunnell test is the investigation of choice for infectious mononucleosis.

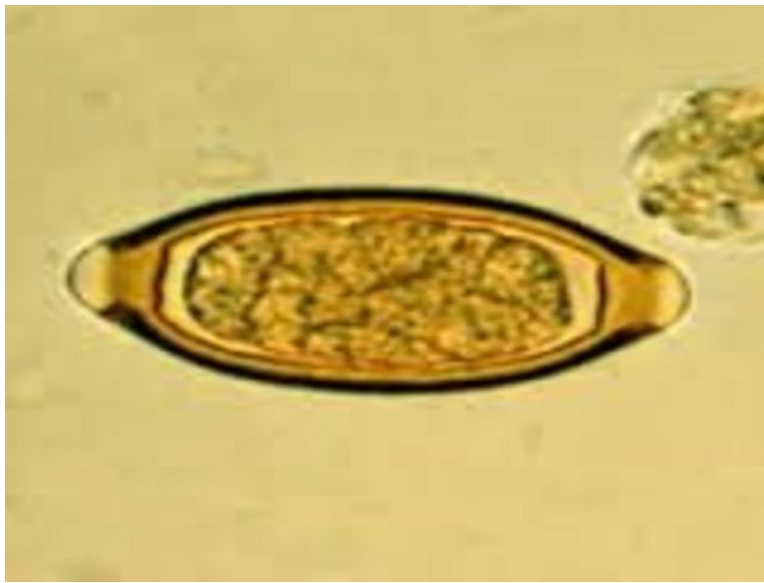
Solution to Question 3:

The given clinical picture and the barrel-shaped eggs are suggestive of infection with *Trichuris trichiura* (whipworm).

The characteristic features of whipworm eggs are:

- Eggs are barrel-shaped in the middle with a projecting mucus plug at each pole containing an unsegmented ovum. The plugs are colorless.
- They float in the saturated salt solution.
- It has a triple shell, the outermost layer of which is stained brown.

The below image shows the egg of *Trichuris trichiura*.



In moderate infections of 20 adult worms or less, the worms are confined to the caecum and ascending colon. The symptoms include epigastric and lower abdominal pain, diarrhea (rarely bloody), vomiting, flatulence and distention, weight loss, and increased blood and iron loss.

In heavy infections of 200 adults worms or more (trichuris dysentery syndrome), the worms are in the distal colon and rectum. The symptoms include bloody diarrhea with profuse mucus, abdominal pain, tenesmus, weight loss leading to cachexia, severe anemia, and rectal prolapse. On examination, clubbing may be present. Blood investigations may show moderate eosinophilia.

Chronic infection leads to growth retardation and nutritional deficiencies. Around 0.005 ml of blood is lost per day per worm. In heavy infections, as much as 4 to 5 ml can be lost per day per worm.

Treatment is mebendazole or albendazole.

Solution to Question 4:

Parvovirus B19 infection mimics erythroblastosis fetalis.

Parvovirus B19, belonging to the parvovirus group consisting of the smallest animal DNA viruses, primarily infect immature cells of the erythroid lineage. It is associated with erythema infectiosum (fifth disease), transient aplastic crisis, pure red cell aplasia and hydrops fetalis.

During pregnancy, infection with parvovirus B19 can lead to severe anemia, non-immune hydrops fetalis and fetal death. This is because parvovirus preferentially infects RBC progenitor cells. This mimics erythroblastosis fetalis, in which maternal antibodies against fetal RBCs cause varying degrees of anemia and immune-mediated hydrops fetalis in severe cases.

Picornavirus, Epstein-Barr virus and HSV infections do not mimic erythroblastosis fetalis.

Solution to Question 5:

From the above clinical history and examination, the most probable diagnosis is hyper IgE syndrome (Job's syndrome).

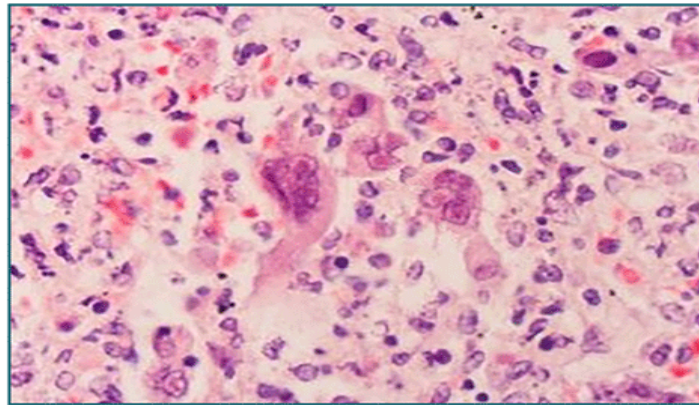
It is an autosomal dominant primary immunodeficiency disease caused by a mutation of STAT3 gene. Neutrophils do not generate an immune response, probably due to a defect in chemotaxis. Patients present with recurrent staphylococcal skin abscesses, recurrent lung infections (pneumatoceles), eczema, eosinophilia and high serum levels of IgE.

Solution to Question 6:

Cowdry type A inclusions are seen in herpes simplex infection.

The classic Cowdry type A (Lipshutz) inclusion appears as a light purple, homogeneous intranuclear structure surrounded by a clear halo. The inclusions stain with antibodies to HSV, permitting a rapid, specific diagnosis of HSV infection.

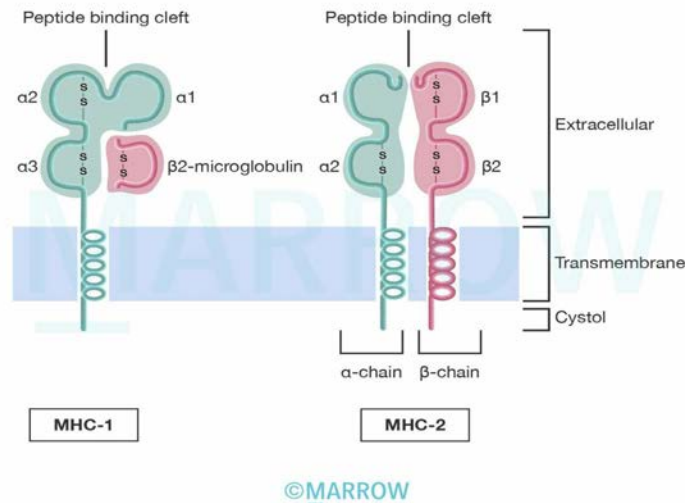
HSV inclusion bodies



©MARROW

Solution to Question 7:

MHC class II presents exogenous antigens while MHC class I presents endogenous antigens.

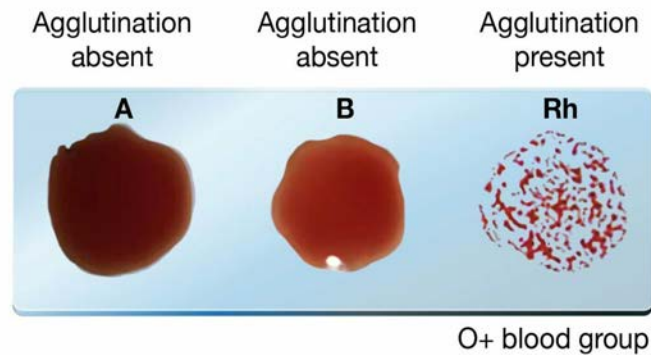


Solution to Question 8:

The hemagglutination test is not based on the principle of antibody marking.

In tests based on the principle of antibody marking, the antigens or antibodies are labelled or marked with enzymes, fluorescent dyes, radioisotopes etc. In radioimmunoassay, fixed amounts of antibody and radiolabelled antigen react in the presence of unlabelled antigen. The labelled and unlabelled antigens compete for binding sites on the antibody. Thus, the concentration of the test antigen can be calculated from the ratio of the bound and total antigen counts using a standard dose-response curve. In immunofluorescence, fluorescent dyes are conjugated to antibodies or antigens. The labelled antibodies or antigens can then be used to identify antigens and antibodies, respectively. In ELISA, an adsorbing material - cellulose, agarose, polystyrene, polyvinyl etc - is used to bind to either antigens or antibodies.

When antibodies bind to antigens on the surface of red blood cells, the resultant clumping reaction is referred to as hemagglutination. It is routinely performed to type red blood cells in ABO blood-typing. The image below shows hemagglutination.



©MARROW

Note: If no hemagglutination occurs it means the patient's serum has anti-haemagglutinin antibodies that bind to hemagglutinin and prevent the clumping reaction. This hemagglutination inhibition reaction is a method of detection of viruses and antiviral-antibodies and is used to make a diagnosis of influenza.

Solution to Question 9:

Deficiency of IgG2 results in recurrent sinopulmonary infections with encapsulated organisms.

IgG is the only maternal immunoglobulin that is normally transported across the placenta and provides natural passive immunity to the newborn. IgG1 is the antibody maximally transported across the placenta, whereas IgG2 has the least transplacental movement.

Deficiencies of other IgG:

- IgG1- bacterial infections (diphtheria, tetanus)(Option A)
- IgG3- viral infections (Option C)
- IgG4- parasitic infections (Option D)

Solution to Question 10:

The given clinical scenario is suggestive of cutaneous anthrax which is caused by *Bacillus anthracis*.

Human anthrax presents in one of three ways:

- Cutaneous Anthrax(hide porter's disease)- Black eschar at the base, with a very prominent inflammatory ring around the eschar, is referred to as a malignant pustule.
- Pulmonary Anthrax(Wool sorter's disease)- Hemorrhagic pneumonia.

- Intestinal Anthrax- Hemorrhagic diarrhoea. It occurs mainly in primitive communities that eat the carcasses of animals dying of anthrax.

Robert Koch validated the germ theory of disease after discovering the anthrax bacillus as the cause of anthrax. He also discovered the causative agents of tuberculosis and cholera. Hence, he is known as the Father of Bacteriology.

Koch's postulates, which link specific microorganisms to specific diseases, state that the microorganism must be:

- Found in abundance in diseased organisms, but not in healthy organisms.
- Isolated from a diseased organism and grown in pure culture
- Able to cause disease when inoculated into a healthy organism
- Re-isolated from the diseased experimental host and identical to the original specific causative agent

Solution to Question 11:

Pour plate culture (quantitative test) is not used as an immediate test for screening of UTI. However it is considered the best diagnostic test for confirmation of UTI due to its high sensitivity and specificity.

The screening tests for UTI are as follows:

- Wet mount examination: It is done to demonstrate pus cells in urine.
- Leukocyte esterase test: It is a rapid and cheaper method, and detects leukocyte esterases secreted by pus cells present in urine.
- Nitrate reduction test (Griess test): Nitrate reducing bacteria like *E. coli* give a positive result.
- Catalase test: *E. coli* and other bacteria that produce the enzyme catalase give a positive result.

Quantitative Culture:

- Standardized loop technique
- Pour plate method

Solution to Question 12:

The disc diffusion method for antibiotic susceptibility testing is applied in the Kirby-Bauer method.

In Kirby-Bauer's disc diffusion method, a suitable standard dilution of a broth culture of the test bacterium is inoculated on the surface of a solid medium (Cation-adjusted Mueller-Hinton agar) as a lawn culture. After drying, various antibiotic discs are placed on its surface. Zones of inhibition around the disc are recorded and interpreted according to the zone diameters available in the tables of the Clinical and Laboratory Standards Institute (CLSI) guidelines. The results obtained are qualitative (resistant, sensitive, or partially sensitive).

The image below shows Kirby-Bauer disc diffusion method.



Laboratory testing for antibiotic sensitivity can be done by:

- Diffusion methods: The drug diffuses through a solid medium so that a gradient is established. The concentration is highest near the site of drug application and decreases with distance. The test bacterium is cultured on the medium and its sensitivity to the antibiotic is determined by the zone of growth inhibition. Examples include Kirby–Bauer method and Stokes method.
- Dilution methods: Serial dilutions of the drug are prepared and inoculated with bacteria. Examples include tube dilution method and agar dilution method
- Diffusion and dilution method: E–test (Epsilometer)

Solution to Question 13:

The investigation of choice for neurosyphilis is VDRL.

The CSF-VDRL test is highly specific and, when reactive, is considered diagnostic of neurosyphilis.

Involvement of the CNS in syphilis is detected by examination of CSF for:

- Mononuclear pleocytosis (>5 white blood cells/ μL).
- Increased protein concentration (>45 mg/dL).
- CSF VDRL reactivity.

Solution to Question 14:

Contact based isolation is employed for patients who are infected with MRSA (methicillin-resistant *Staphylococcus aureus*).

The three categories of transmission-based precautions include:

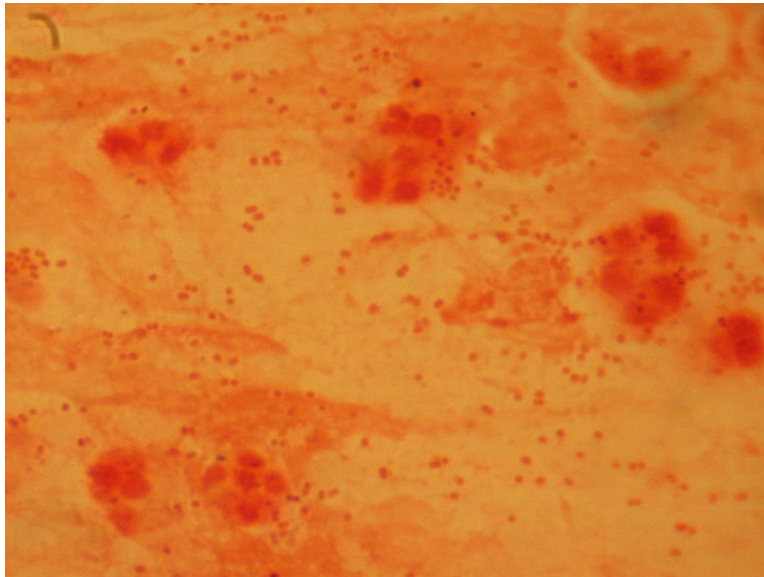
- Contact-based isolation is used for infections, diseases, or germs that are spread by touching the patient or items in the room. Examples include MRSA, vancomycin-resistant enterococci (VRE), diarrheal illnesses, open wounds, and respiratory syncytial virus (RSV).
- Droplet-based isolation is employed for diphtheria and mumps.
- Airborne precautions are required to protect against conditions like measles, severe acute respiratory syndrome (SARS), varicella (chickenpox), and Mycobacterium tuberculosis.

Solution to Question 15:

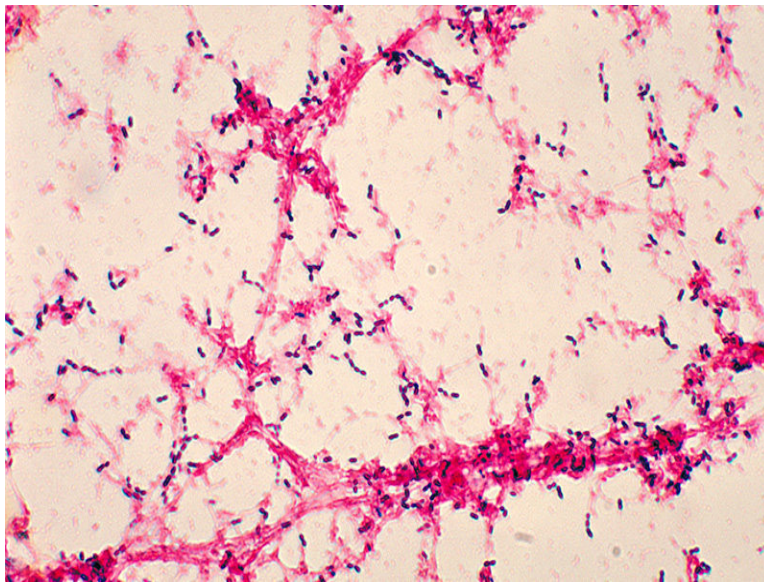
History and physical examination findings are suggestive of meningitis.

The given image shows intracellular gram negative diplococci i.e. *Neisseria meningitidis*.

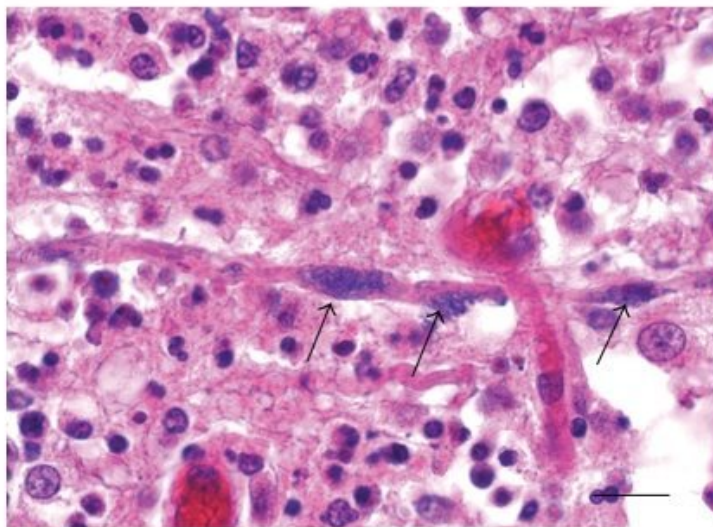
Option a: *Haemophilus influenzae* type B is a gram negative bacillus. In CSF, long bacillary filamentous forms predominate.



Option c: *Streptococcus pneumoniae* is a gram positive, capsulated, flame shaped or lanceolate shaped diplococci.



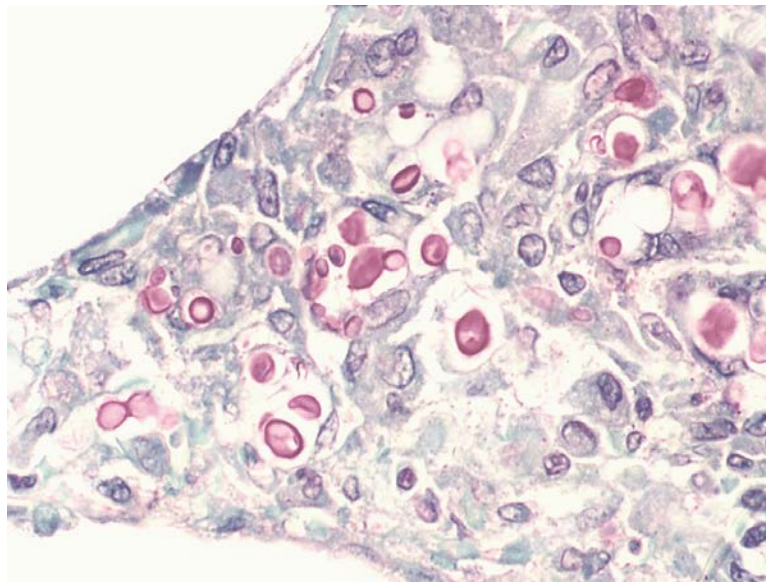
Option d: *Listeria monocytogenes* is a gram positive bacillus.



Solution to Question 16:

Mucicarmine stain is used in the identification of cryptococcus.

Mucicarmine stain helps in the detection of the cryptococcal capsule, as it stains mucins of epithelial origin (PAS, alcian blue). Given below is an image of Mucicarmine stained specimen of lung tissue containing cryptococci:



Solution to Question 17:

Japanese encephalitis follows bird-arthropod-human transmission.

JE virus infects several extra-human hosts, e.g., pigs and birds (like pond herons). The basic cycles of transmission are:

(a) Pig → Mosquito → Pig

(b) Bird → Mosquito → Bird

Humans are incidental "dead-end" hosts and are infected by the bite of infected *Culex* mosquitoes.

Solution to Question 18:

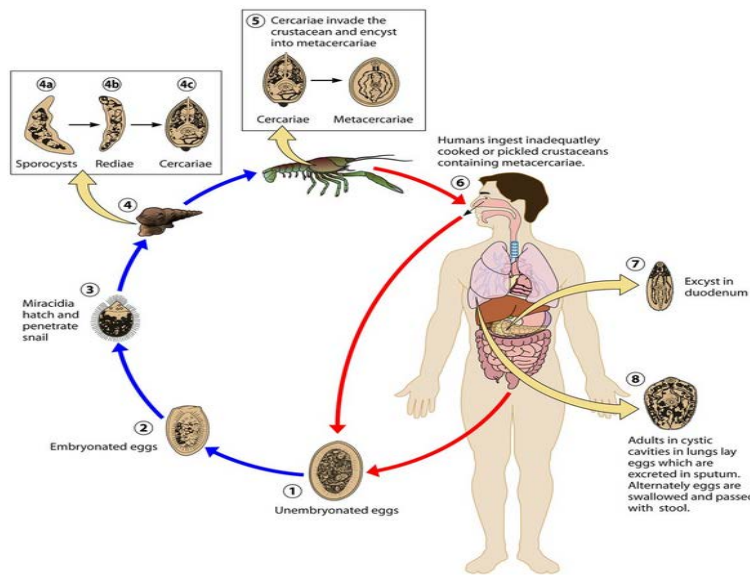
The given clinical scenario is suggestive of infection with *Paragonimus westermani*.

Paragonimus westermani is a trematode. The transmission to humans occurs by eating raw or undercooked crab or crayfish containing metacercariae. The metacercariae excyst in the human gut and young worms migrate to the lungs, where they become encapsulated in lung tissue.

The patient complains of cough, chest pain, and hemoptysis. The viscous sputum is speckled with golden-brown eggs.

Praziquantel (25 mg/ kg TDS for 1-2 days) is the drug of choice.

The image below shows the life cycle of *Paragonimus westermani*.



Solution to Question 19:

Rubella belongs to the Togaviridae family.

Rubella (German measles)

- Causative organism belongs to family Togaviridae
- Period of infectivity is -5 to +6 days after the onset of the rash
- Incubation period: 2-3 weeks
- Rash appears on day 1
- Forchheimer spots are seen, which are tiny, rose-colored lesions in the oropharynx
- Associated with post infectious thrombocytopenia
- Most serious complication: Encephalitis
- Vaccination: 27/3 live attenuated

Congenital rubella syndrome:

- When a woman in the 1st trimester gets infected with rubella, the resulting complications may include miscarriage, fetal death, premature delivery, or live birth with congenital defects.
- The commonest malformations caused by rubella are cardiac defects, cataract, and deafness, which constitute the classical congenital rubella syndrome.
- Thrombocytopenic purpura present in the babies gave rise to the term "blueberry muffin syndrome."

Expanded rubella syndrome:

- Some babies present with several other features like hepatosplenomegaly, thrombocytopenic purpura, myocarditis, and bone lesions constituting the expanded rubella syndrome.
- This term also refers to a baby with metastatic neuroblastoma.

Note: Rubella has been reclassified into the family of Matonaviridae in 2020. This is an MCQ from NEET 2019, hence the best answer out of the given options would be Togaviridae.

Solution to Question 20:

The clinical background is probably suggestive Scabies. The given image depicts Itch mite also called *Sarcoptes scabiei*. The female parasite burrows into the stratum corneum layer of the epidermis and lays eggs in the burrow.

A single female may lay about 30 eggs at a rate of 2 to 3 per day, and ultimately dies at the end of the burrow.

The life cycle has 4 stages: Egg, Larva, Nymph, and Adult. The adult mites live for 1 to 2 months.

Itching is a hypersensitivity reaction to the eggs and dead parasite in the epidermis.

Permethrin cream - 5% is the treatment of choice.

Solution to Question 21:

Diarrhea in an HIV patient is suggestive of a coccidian parasitic infestation. Kinyoun stain - positive oocysts as seen in the given image is suggestive of *Cryptosporidium*.

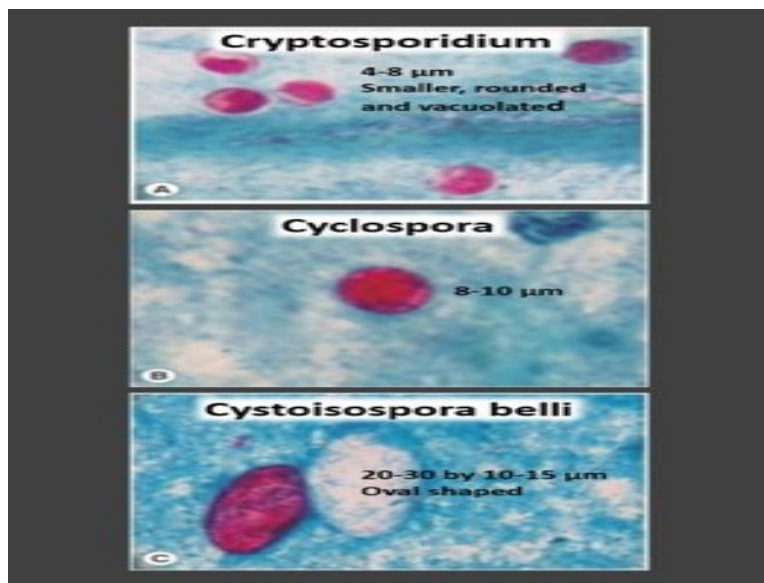
Points in favor:

- HIV patient with diarrhea
- Oocysts are round, contains four sporozoites which is uniformly acid fast on staining.

Comparison of coccidian parasites causing diarrhea in an immunocompromised host:

Treatment is fluid replacement and cotrimoxazole.

Property	<i>Cryptosporidium</i>	<i>Cyclospora</i>	<i>Isospora</i>
Sporulated oocyst	4-6 μm , round contains four sporozoites	8-12 μm , round contains 2 sporocyst each having two sporozoites	23-36 μm , oval contain two sporocyst, each having four sporozoites.
Acid-fast detection	Uniformly acid fast	Variable acid fast	Uniformly acid fast
Sporulation of the oocyst	Occurs inside host cell	Occurs in soil	Occurs in soil
Diagnostic form	Sporulated cyst	Unsporulated cyst	Unsporulated oocyst

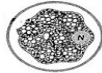


● *Cryptosporidium parvum*



Oocyst with 4 sporozoites
Size: 4 -5 μ m
Staining : Ziehl-Neelsen or Auramin

● *Cyclospora cayetanensis*

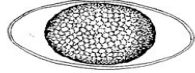


Young oocyst

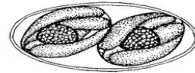


Mature oocyst with 2 sporocysts containing 2 sporozoites each
Size: 8 - 10 μ m
Staining : Ziehl-Neelsen or Auramin

● *Isospora belli*



Young oocyst



Mature oocyst with 2 sporocysts containing 4 sporozoites each
Size: up to 20 x 30 μ m

Solution to Question 22:

The given clinical scenario and MRI brain showing scolex within multiple cystic lesions are suggestive of neurocysticercosis caused by the larval form of *T. solium* called *Cysticercosis cellulosae*.

Neurocysticercosis is the most common central nervous system parasitic infection.

The clinical features of neurocysticercosis include epilepsy, features of raised intracranial tension, and evident hydrocephalus. Patients can also have psychiatric disturbances, transient paresis, and visual disturbances.

The CT and MRI findings seen in neurocysticercosis are:

- Scolex within a cyst

- Cystic lesions with or without ring enhancement
- Nodular calcifications
- Focal enhancing lesions
- Multilobulated cystic lesions in the subarachnoid space.

Note: About 70% of adult-onset epilepsy is due to neurocysticercosis.

Albendazole is the drug of choice for neurocysticercosis. Glucocorticoid therapy is usually begun before initiating albendazole (an antiparasitic drug). This is continued for several days after the institution of therapy to reduce the incidence of side effects from inflammatory reactions to dead and dying cysticerci of *Taenia solium*.

Solution to Question 23:

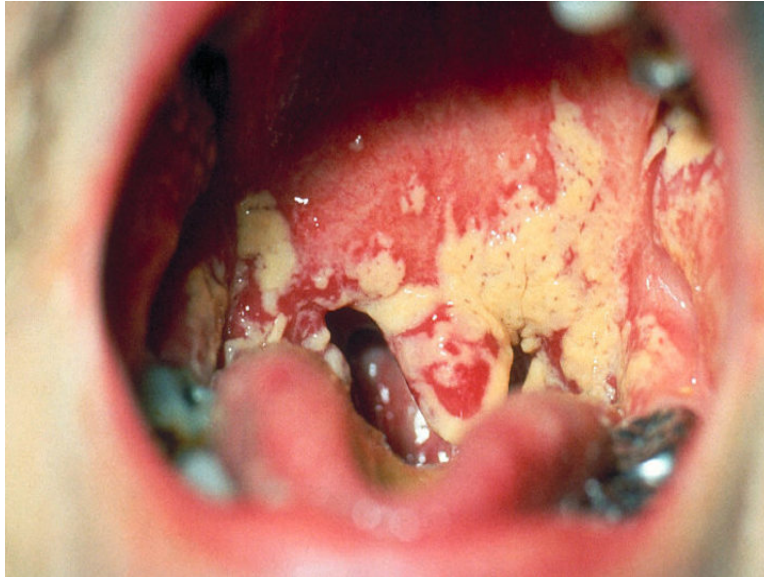
Influenza vaccine requires periodic strain change due to antigenic change in the surface glycoprotein of the virus.

The change may be gradual (antigenic drift) or drastic (antigenic shift).

Microbiology NEET 2020

Question 1:

A 35-year-old HIV positive patient presents with mucosal lesions in the mouth as shown in the image below. On microscopy, budding yeasts and pseudohyphae are seen. What is the likely diagnosis?



- a) Oral candidiasis
- b) Hairy leukoplakia
- c) Lichen planus
- d) Diphtheria

Question 2:

Zika virus is transmitted by:

- a) Aedes aegypti
- b) Culex
- c) Anopheles
- d) Phlebotomus papatasi

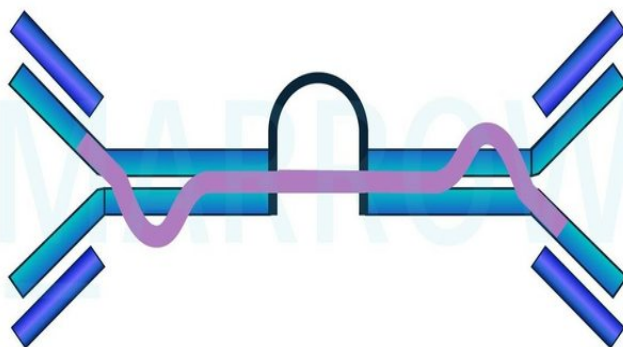
Question 3:

Which of the following organisms causes disease following traumatic inoculation into the skin?

- a) *Sporothrix schenckii*
- b) *Blastomyces dermatitidis*
- c) *Coccidioides immitis*
- d) *Paracoccidioides brasiliensis*

Question 4:

Identify the class of immunoglobulin shown in the image below.



- a) Ig A
- b) Ig G
- c) Ig M
- d) Ig E

Question 5:

A 14-year-old boy presents with a history of vomiting within 3 hours of consumption of food at a party. Which of the following organism is most likely to cause these symptoms?

- a) *Staphylococcus aureus*
- b) *Clostridium botulinum*
- c) *Clostridium perfringens*

d) Salmonella typhi

Question 6:

Which of the following organisms produces superantigen?

- a) Staphylococcus epidermidis**
- b) Streptococcus pyogenes**
- c) Clostridium perfringens**
- d) Vibrio cholera**

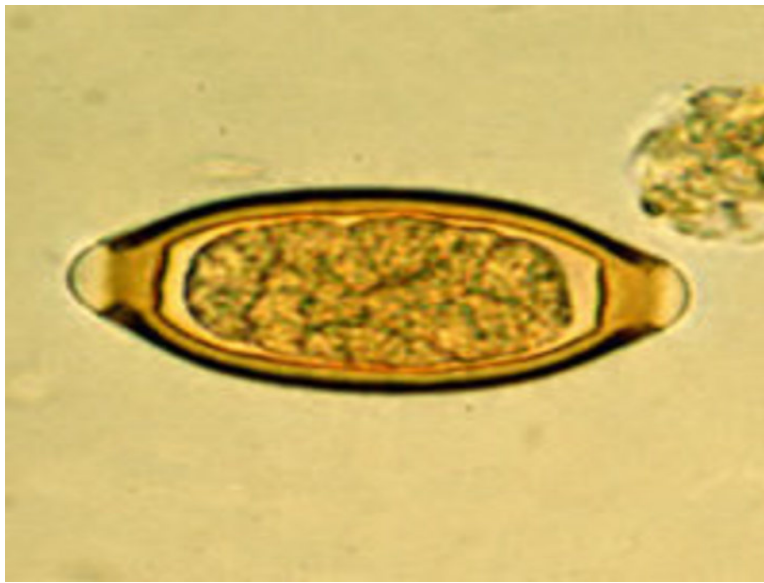
Question 7:

A female patient who is from the north-eastern part of the country came to the hospital with complaints of progressive swelling of the upper limbs. She has a history of coronary artery disease for which she underwent percutaneous coronary intervention. On examination, the upper limbs are more swollen proximally than distally. What is the next best step in the management of this patient?

- a) Echocardiography**
- b) Night blood test for worms**
- c) Duplex venous scan of limbs**
- d) ECG**

Question 8:

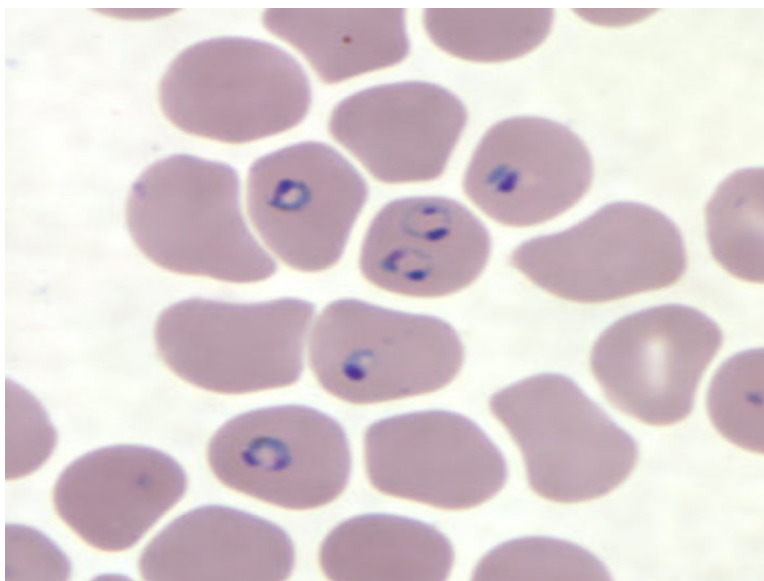
Which of the following helminthic infections, that is commonly transmitted by soil, is characterized by ■ the egg in stool as shown in the image below?



- a) *Trichuris trichiura*
- b) *Ancylostoma duodenale*
- c) *Necator americanus*
- d) *Ascaris lumbricoides*

Question 9:

A 30-year-old female from tribal area presented with fever for the last 3 days. On examination of the peripheral smear, a diagnosis of malaria was made. Which of the following is most likely to be the causative agent?



- a) *P. falciparum*

- b) *P. malariae*
- c) *P. ovale*
- d) *P. vivax*

Question 10:

The test kit shown in the image below works on the principle of:



- a) ELISA
- b) Immunochromatography
- c) Immunofluorescence
- d) Chemiluminiscence

Question 11:

A 22-year-old student presented with high-grade fever, and purpuric rash as shown in the image below. CSF sample showed gram-negative diplococci. Which of the following is the most likely aetiology?



- a) *Neisseria meningitidis*
- b) *Pseudomonas aeruginosa*
- c) *Streptococcus pneumoniae*
- d) *E.coli*

Question 12:

Which of the following is the predominant pathogen to infect renal transplant recipients one to four months following transplantation?

- a) Cytomegalovirus
- b) JC virus
- c) BK virus
- d) Epstein-Barr virus

Question 13:

Which of the following is not true about Japanese encephalitis?

- a) Transmitted by *Culex vishnui*
- b) Pig is the amplifier host
- c) Man acts as the reservoir
- d) Vaccinating pigs is helpful in disease control

Question 14:

Which of the following helps in bacterial adhesion?

- a) Cytoplasmic membrane
- b) Mesosomes
- c) Fimbriae
- d) Lipopolysaccharides

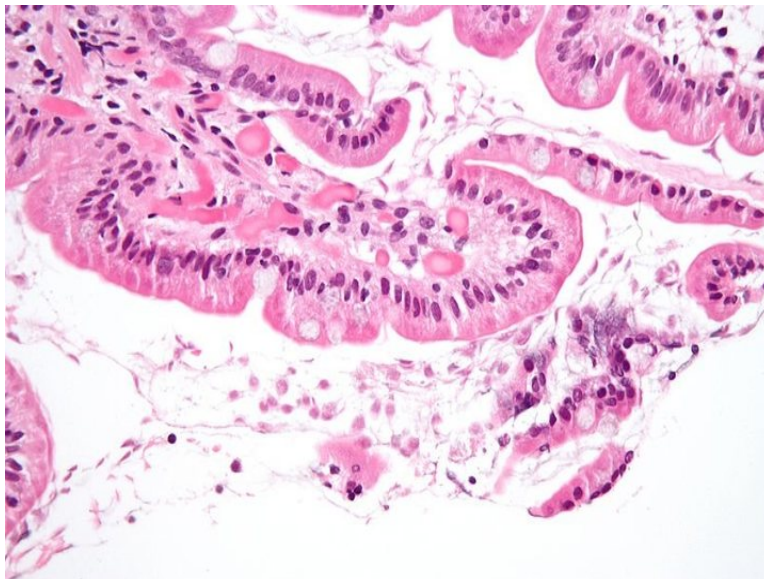
Question 15:

Which of the following is not involved in urethritis?

- a) Trichomonas
- b) Chlamydia
- c) Gonococcus
- d) H.ducreyi

Question 16:

A 30-year-old HIV-positive patient presented with a chronic history of watery diarrhea. Histopathology of the duodenum is shown in the image below. Which of the following is the causative organism?



- a) Giardia lamblia

- b) Entamoeba histolytica
- c) Microsporidia
- d) Cryptosporidia

Question 17:

What is the drug of choice for invasive pulmonary aspergillosis?

- a) Itraconazole
- b) Voriconazole
- c) Lyophilised Amphotericin B
- d) Posaconazole

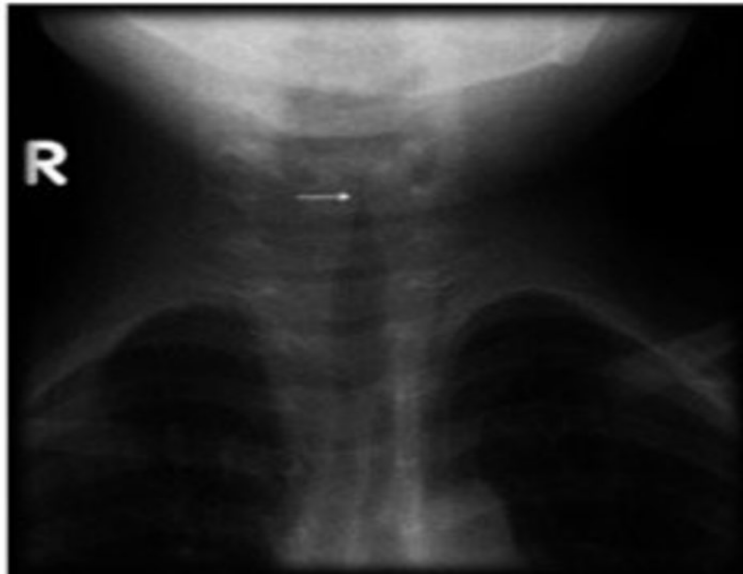
Question 18:

Which of the following culture media is used to differentiate other E.coli species from E.coli O157: H7?

- a) Mannitol salt agar
- b) Xylose lysine deoxycholate agar
- c) Sorbitol MacConkey agar
- d) CLED medium

Question 19:

A 5 year old boy was suffering from low grade fever, inspiratory stridor and barking cough for past 5 days. His chest X-ray is given below. What is the cause of this condition?



- a) Moraxella
- b) Parainfluenza
- c) Hemophilus influenza
- d) Staphylococcus aureus

Question 20:

Which of the following is a bioterrorism category A agent?

- a) Nipah virus
- b) Anthrax
- c) Coxiella
- d) Brucella

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	a
5	a

6	b
7	b
8	a
9	a
10	b
11	a
12	a
13	c
14	c
15	d
16	a
17	b
18	c
19	b
20	b

Detailed Explanations

Solution to Question 1:

An HIV patient with white patches over the tonsils and posterior pharynx showing budding yeasts and pseudohyphae on microscopy suggests a diagnosis of oral candidiasis.

Oral candidiasis or thrush is a patchy, pseudomembranous lesion that develops over the tongue, lips, gum or palate. It is composed of epithelial cells, yeasts, and pseudohyphae and can be scraped off. The causative agent is usually *Candida albicans*. It occurs in almost all individuals with AIDS. Other risk factors include hyperglycemia, cellular immunodeficiency, corticosteroids or antibiotics. It is treated with topical nystatin, oral ketoconazole or fluconazole.

Option B: Hairy leukoplakia is a wart-like growth on the tongue, caused by Epstein-Barr virus. It is commonly seen as a white lesion on the lateral side of the tongue.

Option C: Lichen planus is a pruritic disease characterized by papules commonly seen over the flexor surfaces and trunk. Oral and vulvovaginal lesions may also occur. Oral lesions appear as lacy, white, and raised lesions.

Option D: Diphtheria is characterized by the presence of a grey pseudomembrane covering the tonsils and pharynx. It is caused by *Corynebacterium diphtheriae*.

Solution to Question 2:

Zika virus is transmitted by the bite of infected *Aedes aegypti* and *Aedes albopictus*.

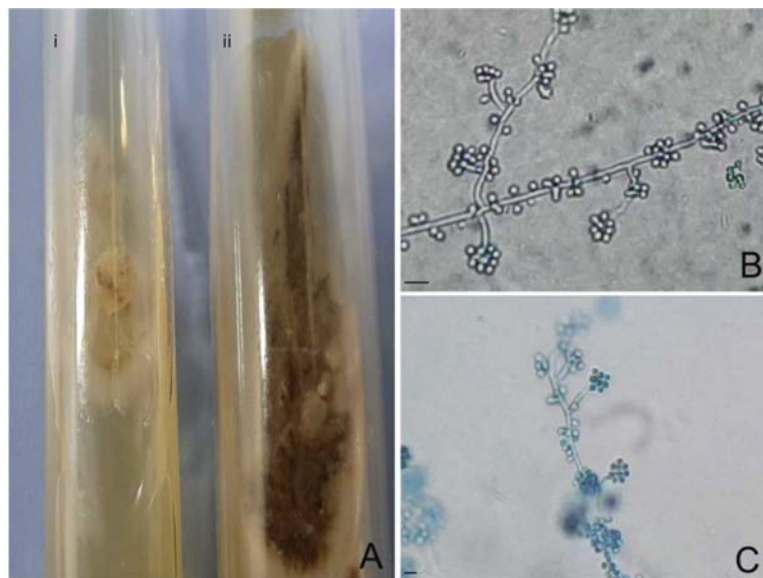
Zika virus is an RNA virus belonging to the Flaviviridae group. The symptoms of Zika virus disease include fever, maculopapular rash, conjunctivitis, muscle and joint pain, malaise and headache. These symptoms are mild and usually last for 2-7 days. It can be misdiagnosed as dengue or influenza. In severe cases, it can lead to complications like Guillain-Barré syndrome, neuropathy, and myelitis.

Zika virus infection during pregnancy may lead to microcephaly or congenital Zika syndrome.

Solution to Question 3:

Sporothrix schenckii causes infection followed by traumatic inoculation into the skin.

It is a dimorphic fungus that lives on vegetation such as grasses, trees, rose bushes, etc. Following traumatic inoculation into the skin as in thorn prick, *S. schenckii* causes sporotrichosis, a chronic granulomatous infection. This is usually further followed by spread along the draining lymphatics and lymph nodes as shown in the image below.



Sporotrichosis



Subcutaneous mycoses are usually caused by traumatic inoculation of the causative fungus. These include:

Subcutaneous mycoses	Causative agent/s
Sporotrichosis	<i>Sporothrix schenckii</i>
Chromoblastomycosis	<i>Phialophora verrucosa</i> <i>Fonsecaea pedrosoi</i> <i>Fonsecaea compacta</i> <i>Rhinocladiella aquaspersa</i> <i>Cladophiala carrionii</i>
Phaeohyphomycosis	<i>Exophiala jeanselmei</i> <i>Phialophora richardsiae</i> <i>Bipolaris spicifera</i> <i>Wangiella dermatitidis</i>
Mycetoma	<i>Pseudallescheria boydii</i> (Anamorph <i>Scedosporium apiospermum</i>) <i>Madurella mycetomatis</i> <i>Madurella grisea</i> <i>Exophiala jeanselmei</i> <i>Acremonium falciforme</i>

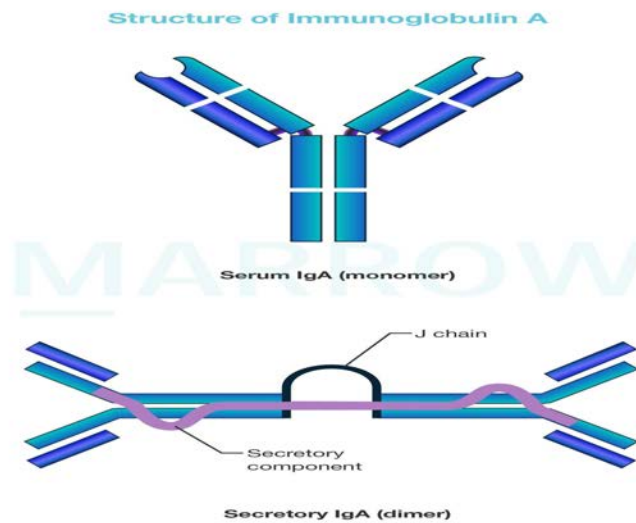
Solution to Question 4:

The above image shows a dimeric structure, characteristic of secretory IgA.

IgA immunoglobulin is the major antibody responsible for mucosal immunity. Its serum levels are usually low. It is predominantly present in the extravascular secretions. It is present in secretions such as milk, saliva, and tears, and in secretions of the respiratory, intestinal, and genital tracts.

When present in the serum, it exists as a monomer similar to IgG. However, when it is secreted, it exists as a dimer. The dimeric structure consists of two monomers, which are stabilized by two additional polypeptides:

- J chain, which stabilizes the molecule.
- Secretory component, which is incorporated into the IgA when transported through the epithelial cell.



Solution to Question 5:

Vomiting 2 hours after the consumption of food is likely food poisoning by *Staphylococcus aureus*.

Staphylococcus aureus can grow in carbohydrate and protein food sources, producing enterotoxins. There are 15 types, all of which are superantigens. They are preformed toxins that are relatively heat stable. They stimulate the vagus nerve, vomiting centre of the brain and intestinal peristaltic activity. Type A enterotoxin is responsible for most cases of food poisoning.

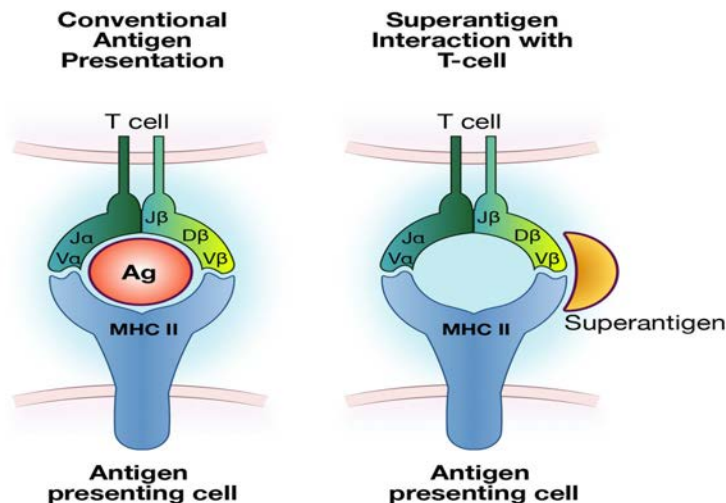
The common sources of infection are the foods that are not cooked after handling. These include sliced meats, puddings, pastries, and sandwiches. Ingestion of such contaminated food causes food poisoning within 30 minutes to 6 hours after consumption. Patients present with nausea, vomiting, stomach cramps, and diarrhoea.

Solution to Question 6:

Streptococcus pyogenes produces streptococcal pyrogenic exotoxin (SPE), which is a superantigen.

Superantigens are protein toxins that stimulate large numbers of T cells, irrespective of antigen specificity. They bind to the class II MHC molecule outside the peptide-binding groove and V β region of the TCR. Some examples of superantigens are staphylococcal enterotoxins, toxic shock syndrome toxin, and group A streptococcal pyrogenic exotoxin.

The below image shows the binding of superantigen to T-cell receptor.



Solution to Question 7:

The given clinical scenario is suggestive of filariasis. The next best step would be to test the night blood for worms since the filarial species is nocturnally periodic (between 10 pm and 4 am) in India.

Solution to Question 8:

The given image shows the egg of *Trichuris trichiura* (whipworm).

The characteristic features of whipworm eggs are:

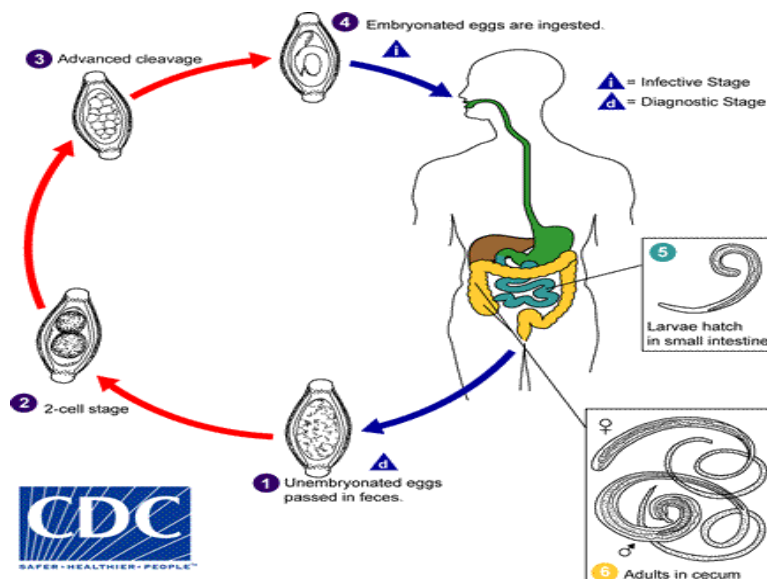
- Barrel-shaped in the middle
- Projecting colourless mucus plug at each pole
- Contains an unsegmented ovum.
- Floast in the saturated salt solution.
- It has a triple shell. The outermost layer is stained brown.

Humans acquire infection by swallowing embryonated eggs in contaminated food or water. The eggs hatch in the small intestine, and the larvae produced pass into the cecum. The larvae become mature adults in 2-3 months. Adult females pass immature un-embryonated eggs in the stool. Thus, diagnosis of whipworm infection is done by microscopic identification of the eggs in a stool specimen.

Most infections are asymptomatic. Moderate whipworm infection (20 adult worms or less), the worms are confined to the caecum and ascending colon. The symptoms include epigastric and lower abdominal pain, diarrhoea (rarely bloody), vomiting, flatulence and distention, weight loss, and increased blood and iron loss. In heavy infections of 200 adult worms or more - trichuriasis

dysentery syndrome -, the worms are in the distal colon and rectum. The symptoms include bloody diarrhoea with profuse mucus, abdominal pain, tenesmus, weight loss leading to cachexia, severe anemia, and rectal prolapse. On examination, clubbing may be present. Blood investigations may show moderate eosinophilia.

Chronic infection leads to growth retardation and nutritional deficiencies. Around 0.005 ml of blood is lost per day per worm. In heavy infections, as much as 4 to 5 ml can be lost per day per worm. Treatment is mebendazole or albendazole.



Solution to Question 9:

The image shows ring forms/ early trophozoites in erythrocyte, which are binucleate (double chromatin) resembling stereo headphones in appearance, characteristic of *P. falciparum*.

There may be several rings present within a single erythrocyte. The rings are often attached to the margins of the red cells and are described as form applique or accolé.

Solution to Question 10:

The above image shows a HbsAg card test kit which works on the principle of immunochromatography.

In immunochromatography, a nitrocellulose membrane is used to absorb the antigen from a specimen. The membrane is impregnated with anti-HBsAg antibody and colloidal gold dye. The specimen (serum) travels through the membrane by capillary action. If the serum contains HBsAg, a coloured reaction (in the form of a band) appears directly on the membrane at the test site. These assays are rapid and include a built-in internal control.

Solution to Question 11:

This patient has a high-grade fever and is a student, so he presumably spends a lot of time in crowded areas. The above image shows a purpuric rash. This patient's features are seen in meningococemia caused by *Neisseria meningitidis*, which are gram-negative diplococci.

The organism enters the body through the nasopharynx by attaching to the epithelial cells with the help of pili. From the nasopharynx, they enter the bloodstream producing bacteremia, and the symptoms may resemble upper respiratory tract infection.

Fulminant meningococemia is the severe form, where the patient presents with high-grade fever and hemorrhagic rash. There is thrombosis of several small blood vessels in various organs with perivascular infiltration and petechial hemorrhages. Some patients with fulminant meningococemia may also develop disseminated intravascular coagulation, and circulatory collapse, a condition known as Waterhouse–Friderichsen syndrome.

Solution to Question 12:

The predominant pathogen to infect renal transplant recipients in the period around 1 to 4 months following the transplantation is cytomegalovirus (CMV).

CMV is the most common virus to affect recipients after organ transplantation (kidney, heart, lung etc). The maximal risk occurs between 1 and 4 months after transplantation (in the absence of prophylaxis). The most common CMV presentation in kidney transplant patients includes fever, leukopenia, hepatosplenomegaly, myalgia, and arthralgia. Antiviral prophylaxis with ganciclovir or valganciclovir is used in renal transplant recipients whenever indicated to prevent CMV disease.

Given below are the most common opportunistic infections in renal transplant recipients:

Peritransplant (less than 1 month)	Early (1-6 months)	Late (more than 6 months)
Wound infection Herpes virus Oral candidiasis Urinary tract infection	Pneumocystis jirovecii Cytomegalovirus Hepatitis B Listeria	Aspergillus Nocardia BK Virus (polyoma) Herpes Zoster

Solution to Question 13:

Humans are the dead-end hosts for the Japanese encephalitis virus.

Japanese encephalitis is an endemic viral infection transmitted by mosquitoes. There is an increased risk in travelers to rural areas. JE has an inactivated vero culture-derived vaccine, given as 2 primary series to adults who are travelling to endemic areas.

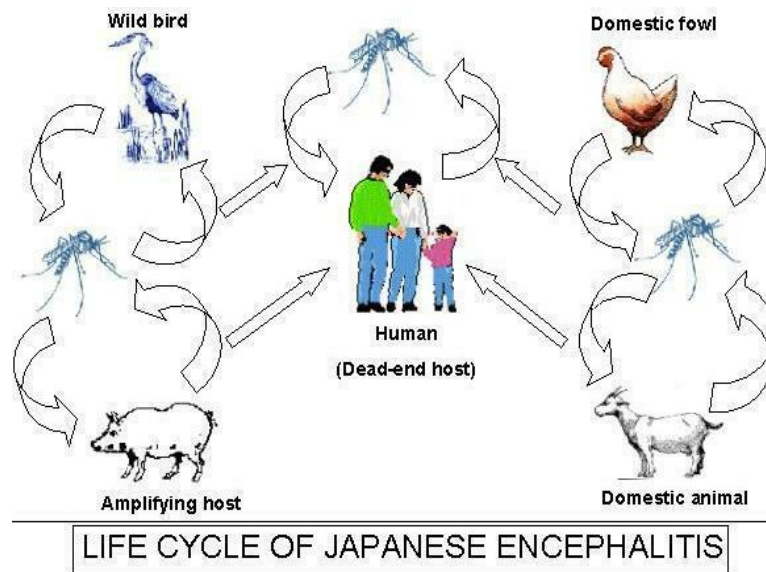
Other options:

Option A: The disease is transmitted by *Culex tritaeniorhynchus* and *Culex vishnui* (common in India) species.

Option B: Herons act as reservoir hosts and pigs as amplifier hosts.

Option D: The control measures include mosquito control, locating piggeries away from the human dwellings and vaccinating pigs.

The image below shows the life cycle of the Japanese B encephalitis virus.



Solution to Question 14:

Fimbriae help in bacterial adhesion.

Fimbriae, also called pili, are hair-like surface appendage that projects from the cell surface as straight filaments. They are primarily seen in Gram-negative bacteria. They are polymers of pilin. Fimbriae are classified based on the function as ordinary pili - for adhesion - and sex pili - involved in bacterial conjugation. The functions of pili are given below:

- Bacterial conjugation through sex pili, which transfer of genetic material occurs.
- Bacterial adhesion of symbiotic and pathogenic bacteria to host cells using adhesins.
- Resistance to phagocytosis.
- Antigenic pilin proteins are responsible for the immune response.
- Twitching motility to bacteria.

Solution to Question 15:

Haemophilus ducreyi does not cause urethritis. It causes chancroid (soft chancre), which presents as a genital ulcer.

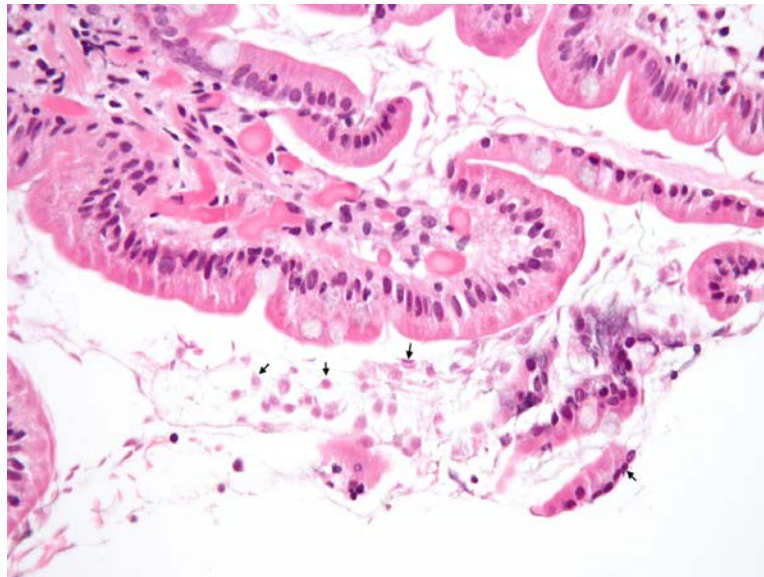
Urethritis is classified into two types:

- Gonococcal urethritis - caused by *Neisseria gonorrhoea*.

- Non-gonococcal urethritis - caused by:
- Chlamydia trachomatis (most common)
- Ureaplasma urealyticum
- Adenovirus
- Haemophilus vaginalis
- Mycoplasma genitalium
- Trichomonas vaginalis (rare)
- Herpes simplex virus (rare)

Solution to Question 16:

The histopathology shows trophozoites of *Giardia lamblia* in the duodenal lumen (marked in arrows).



- *Giardia lamblia*, also called *Giardia duodenalis*, is the causative agent of giardiasis. It is the only common pathogenic protozoan found in the duodenum and jejunum of humans.
- Infection is acquired by the consumption of water or food contaminated with cysts and may also be transmitted by the faeco-oral route.
- It is weakly pathogenic in humans. Secretory IgA and IL-6 are responsible for clearing the infection. Hence, giardiasis usually occurs in immunosuppressed, agammaglobulinemia, or malnourished individuals.
- The symptoms include acute or chronic diarrhea, weight loss, malaise, abdominal cramps, distension, and weakness. The stools may be watery, greasy, bulky, or foul-smelling at various times during the course of the infection.

- Histopathology of the duodenum reveals several heart-shaped or pear-shaped trophozoites with two equally sized nuclei. The small intestinal morphology is usually normal. However, in heavy infections, there may be villous blunting and intraepithelial infiltrates.

Solution to Question 17:

Voriconazole is the drug of choice in the treatment of invasive pulmonary aspergillosis.

Voriconazole can also be used against *Candida* species (including *glabrata* and *krusei*), *Scedosporium* and *Fusarium*. The other options - itraconazole, posaconazole, lyophilised Amphotericin B - can be used as alternatives for the treatment of invasive pulmonary aspergillosis if voriconazole is unavailable.

Solution to Question 18:

E.coli O157: H7 can be differentiated from other *E.coli* species using sorbitol MacConkey agar.

There are more than 150 serotypes of Shiga toxin-producing *E.Coli* (STEC). Of these, O157:H7 is the most common type. Unlike most other *E.Coli* serotypes, the STEC O157:H7 does not use sorbitol and produces clear colonies on the Sorbitol MacConkey agar.

O157: H7 strains also are negative for β -glucuronidase using the substrate 4-methylumbelliferyl- β -glucuronide (MUG) test. More than 90 per cent of other *E.coli* tests positive with this substrate.

Solution to Question 19:

Barking cough is a feature of laryngotracheobronchitis, which is caused by the parainfluenza virus. The X-ray finding is the 'Steeple sign' due to subglottic narrowing.

The age of presentation of laryngotracheobronchitis ranges from 6 months-6 years. The clinical features include barking cough, inspiratory stridor, hoarseness, low-grade fever.

Management:

- Racemic epinephrine nebulization leads to the rapid improvement of upper airway obstruction. Epinephrine constricts pre-capillary arterioles in the upper airway mucosa and decreases capillary hydrostatic pressure, leading to fluid resorption and improvement in airway edema.
- Dexamethasone-by the least invasive route (Oral > IV > IM). It is indicated in all the grades of croup.
- Oxygen should be administered to children who are hypoxemic ($SpO_2 < 92\%$)

Antibiotics are not indicated in laryngotracheobronchitis.

Solution to Question 20:

Bacillus anthracis which causes anthrax is categorized under category A bioterrorism agent.

Microbiology INI-CET 2021

Question 1:

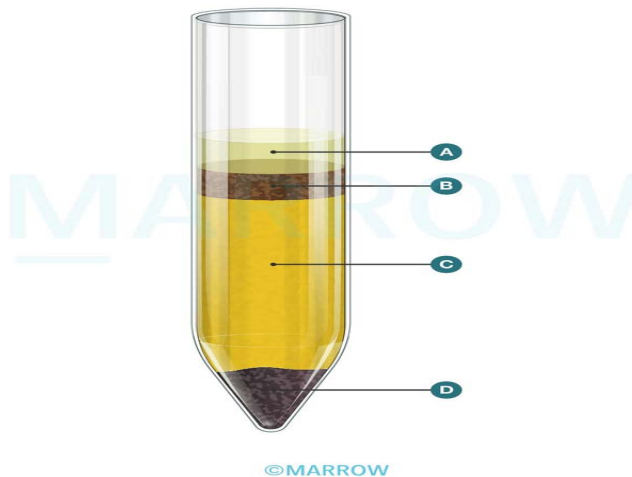
Which of the following causes aplastic crisis in hereditary spherocytosis?

- a) Poxvirus
- b) Epstein barr virus
- c) Adenovirus
- d) Parvovirus

Question 2:

A stool sample is collected. The specimen is processed for examination and is labeled at different levels as A, B, C, and D. Identify the level at which parasitic ova will be seen when viewed under a microscope.

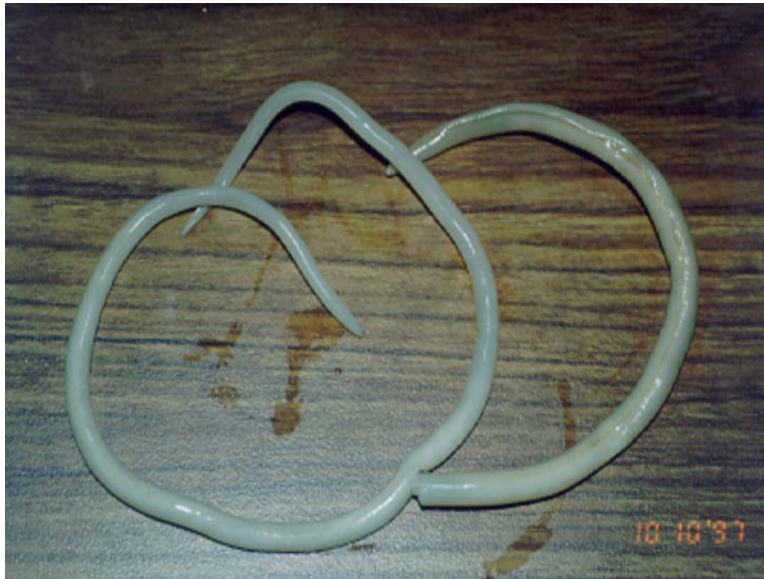
Formalin-ether sedimentation technique



- a) A
- b) B
- c) C
- d) D

Question 3:

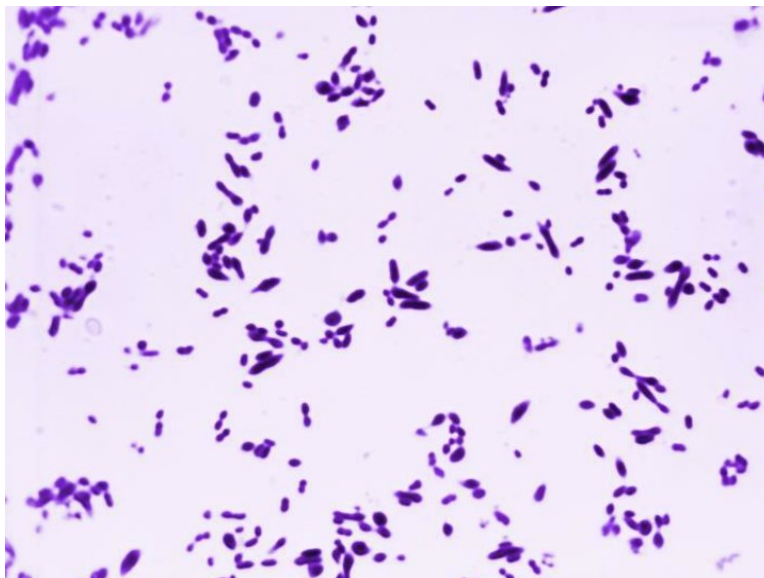
Identify the organism shown in this image.



- a) Strongyloides stercoralis
- b) Enterobius vermicularis
- c) Ancylostoma duodenale
- d) Ascaris lumbricoides

Question 4:

A patient presented with meningitis, and the CSF sample was subjected to gram staining and microscopy and appears as shown in the microscopic image given below. Which of the following features/tests will be characteristic of the organism?



- a) Oxidase and catalase negative, does not ferment inulin
- b) Ferments glucose and maltose but not inulin
- c) Oxidase and catalase positive and does not ferment maltose or glucose
- d) Catalase negative, optochin sensitive

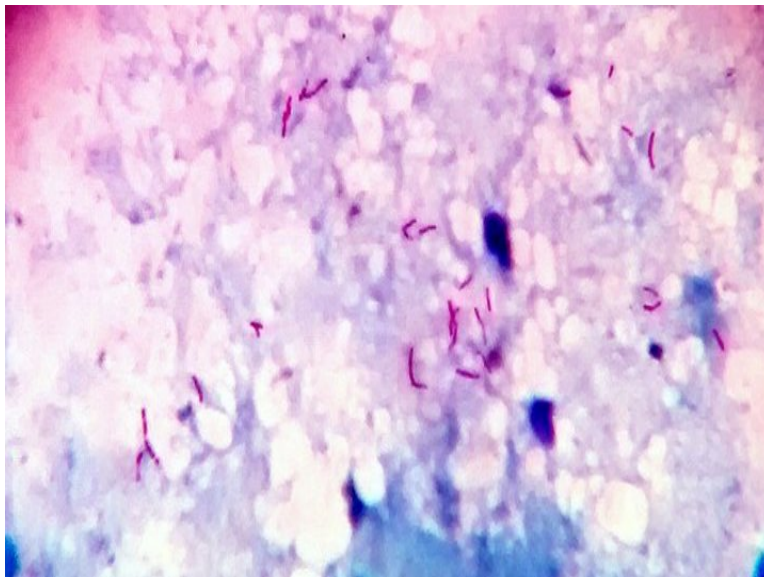
Question 5:

Which cell is involved in type 4/delayed type hypersensitivity?

- a) Cytotoxic T cell
- b) Macrophage
- c) Dendritic cell
- d) Neutrophil

Question 6:

A 32-year-old laborer working at a construction site presented with fever and hemoptysis. The sputum sample collected for examination showed the following. The smear will be stained by which of the following sequences?



- a) Gentian violet - iodine - alcohol - safranin
- b) Methylene blue - malachite green- acetic acid-water
- c) Methanol-methylene blue-acid-water

d) Carbol fuschin - acid-alcohol - methylene blue

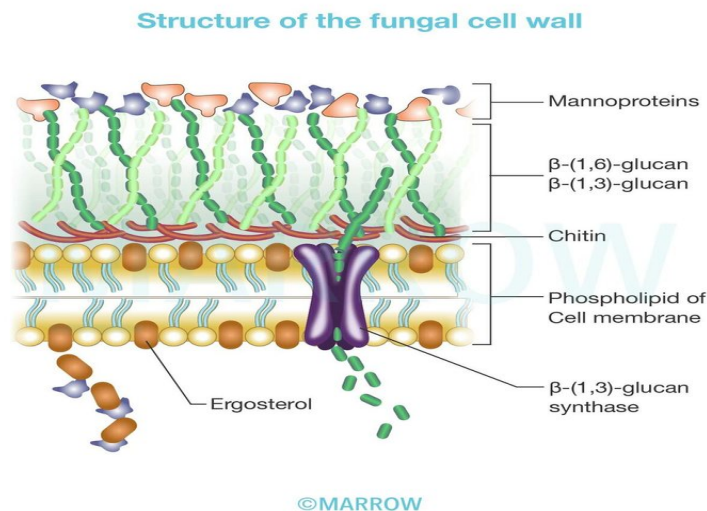
Question 7:

A 55-year-old patient had a tooth extraction three days back. He now presents with a mass in the submandibular region which is otherwise asymptomatic. There is no ulcer on the oral mucosa. On drainage of the discharge, it was foul-smelling with a lot of yellow granules. They showed gram-positive filamentous rods on grams staining. What is the most likely implicated pathogen?

- a) Chromoblastomycosis
- b) Mucormycosis
- c) Actinomyces israeli
- d) Histoplasmosis

Question 8:

Which of the following components of the fungal cell wall is not seen in mucormycosis?



- a) Mannoproteins
- b) Beta 1, 3, glucan
- c) Chitin
- d) Ergosterol

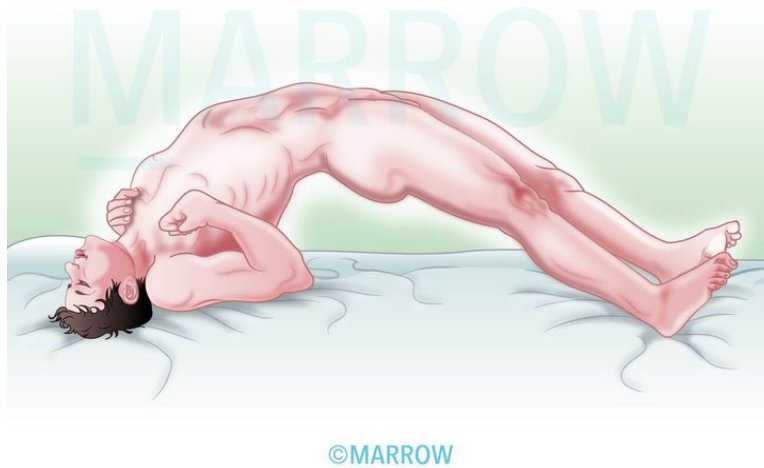
Question 9:

Which of the following statements is True regarding trematodes?

- a) Fasciola hepatica are seen in the intrahepatic biliary radicles
- b) Schistosoma are monoecious
- c) Fasciola hepatica acquired through contaminated water
- d) Triclabendazole is the treatment of choice for all trematode infections

Question 10:

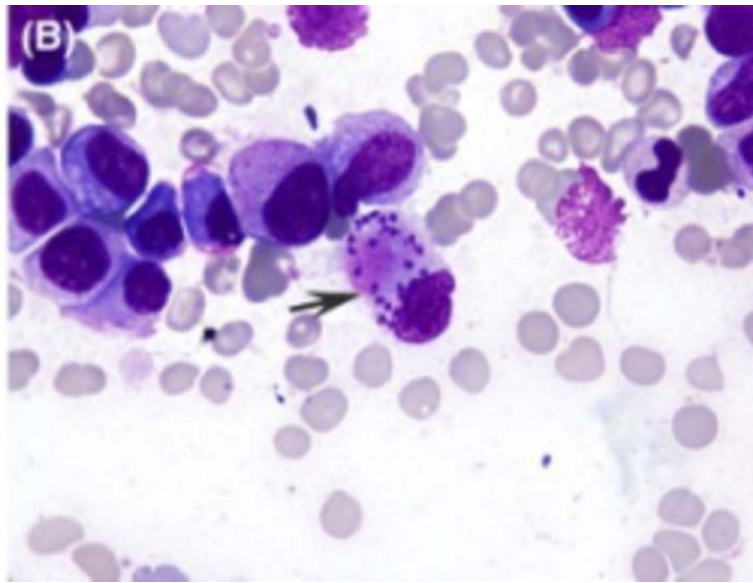
A 45-year-old man developed generalized muscle spasm with arching of his back as shown in the image, a few days after sustaining a puncture wound to his leg. The toxin responsible for this type of abnormal posturing acts by which of the following mechanisms?



- a) Presynaptic inhibition of GABA and glycine release
- b) Presynaptic inhibition of Ach receptors
- c) Postsynaptic inhibition of GABA
- d) Cleavage of SNARE proteins, preventing Ach release

Question 11:

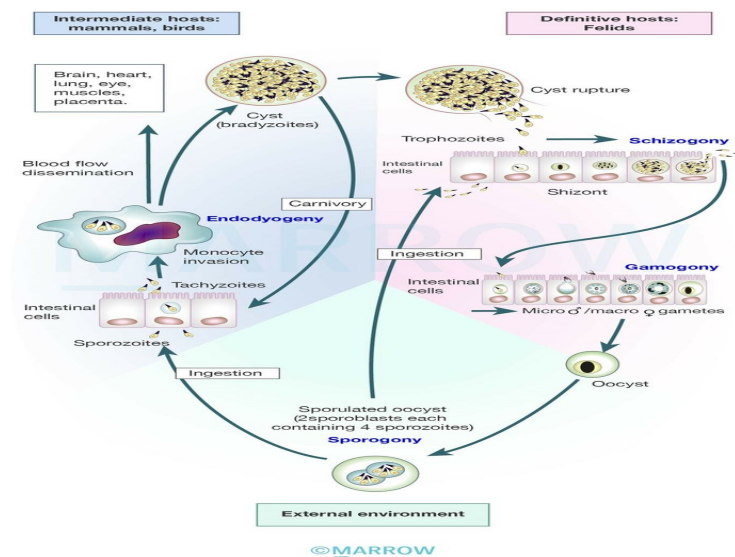
A man with FUO has a job that requires traveling around the world. The bone marrow picture is shown below. Which of the following investigation would be the most appropriate in this case?



- a) rk 39 antigen test, Leishmaniasis
- b) pLDH, Plasmodium malaria
- c) Biopsy, Histoplasmosis
- d) ELISA, HIV

Question 12:

Identify the organism from the lifecycle shown in the image given below.



- a) Cystoisospora
- b) Cryptosporidium

- c) Plasmodium knowlesi
- d) Toxoplasma gondii

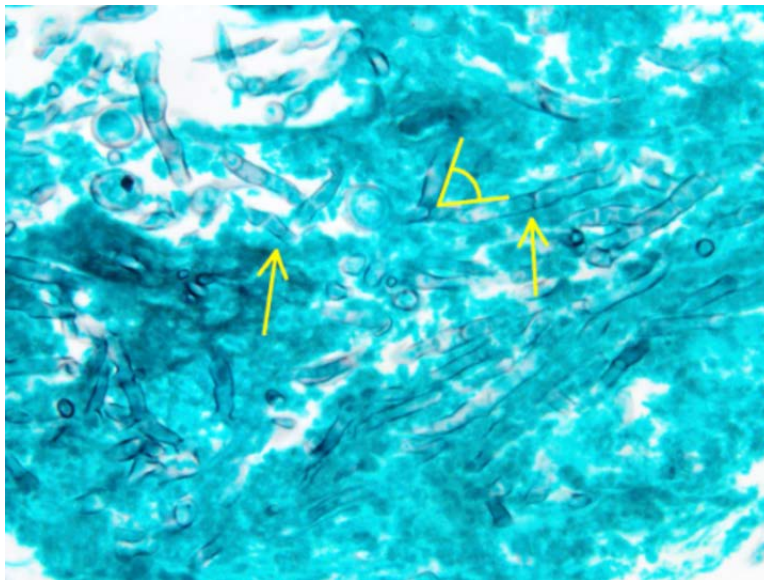
Question 13:

All of the following oncogenic viruses can integrate with the human genome to initiate carcinogenesis except?

- a) HPV
- b) HBV
- c) HTLV
- d) EBV

Question 14:

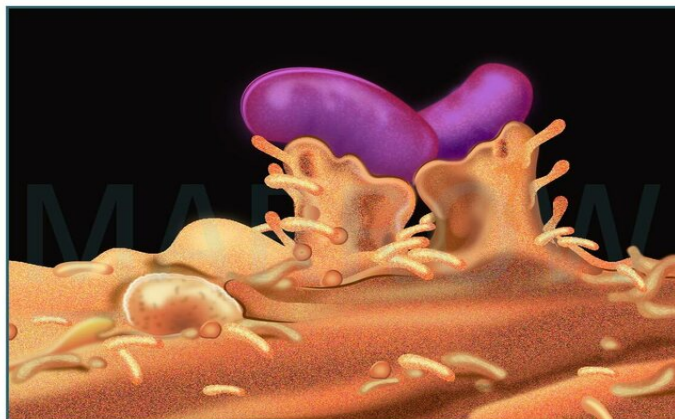
A patient presented with fever, dyspnea, palpitations, and tachycardia. The echocardiography revealed vegetation in the septal leaflets. Tissue sections from them reveal the following findings as shown in the image. What is the most likely implicated agent?



- a) Pigmented septate hyphae at 90 degrees, Mucormycosis
- b) Right angle branching septate hyphae, Fusarium
- c) Acute angle branching hyphae, Rhizopus
- d) Acute angle branching septate hyphae, Aspergillus

Question 15:

Which pathogen causes attachment-effacement lesion in the intestinal mucosa as shown in the image?

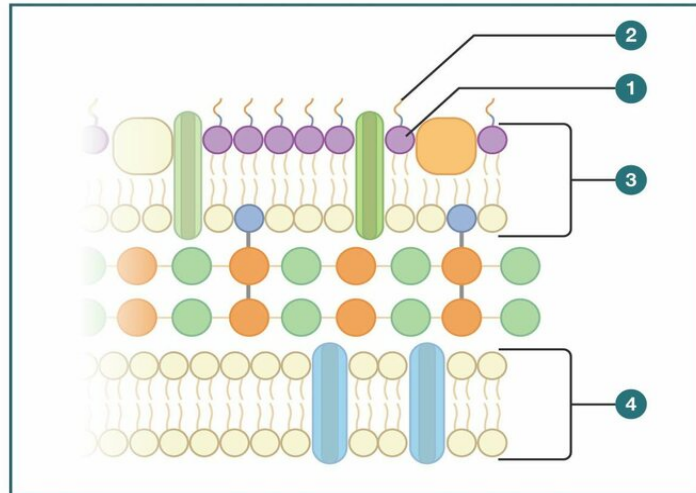


©MARROW

- a) Enterotoxigenic Escherichia coli
- b) Enteropathogenic Escherichia coli
- c) Enteroinvasive Escherichia coli
- d) Diffusely adherent Escherichia coli

Question 16:

Identify the component responsible for endotoxic shock in gram-negative bacteria.



- a) 2
- b) 4
- c) 1
- d) 3

Question 17:

A 6-year old boy complains of cough and throat pain. On examination of the oral cavity, a white pseudomembrane was visualized. A swab was taken from it and stained as given in the image below. What is the procedure for the following staining?

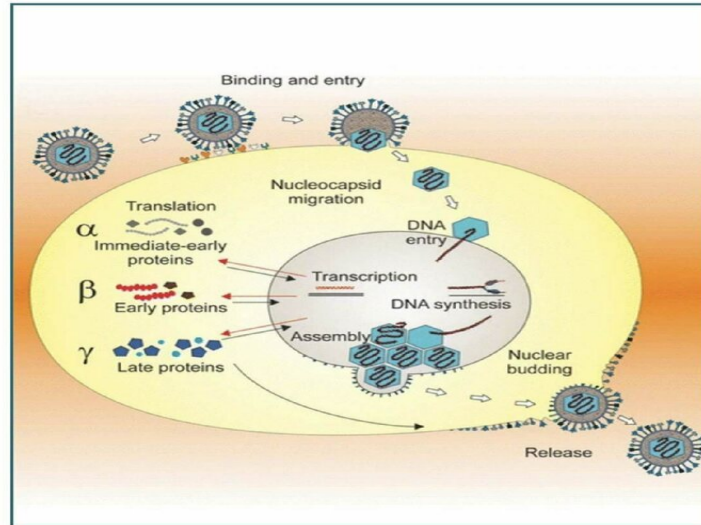


- a) 1, 2, 5, 8

- b) 6, 3, 8, 4
- c) 7, 8, 4, 3
- d) 3, 2, 1, 8

Question 18:

Identify the virus given in the following replication cycle.



- a) Human immunodeficiency virus
- b) Herpes simplex virus
- c) Hepatitis B virus
- d) Coronavirus

Question 19:

Which of the following is a xenodiagnostic method?

- a) Rabbit ileal loop for enterotoxigenic Escherichia coli
- b) Injecting Aedes thorax with blood of a suspected dengue patient
- c) Injecting a hamster with splenic biopsy for diagnosis of leishmaniasis
- d) Intradermal test on guinea pigs for toxigenicity of Corynebacterium diphtheria

Question 20:

A 24-year-old male patient presented to the outpatient department with cervical lymphadenopathy. As a part of the evaluation, the Paul-Bunnell test was done. What is the principle of this test?

- a) Complement fixation
- b) Super-antigens
- c) Heterophile antibodies
- d) Haptens

Question 21:

Match the following in the Japanese encephalitis virus life cycle.

- a) 1-d, 2-c, 3-a, 4-b
- b) 1-c, 2-a, 3-d, 4-b
- c) 1-c, 2-d, 3-a, 4-b
- d) 1-d, 2-c, 3-b, 4-a

Question 22:

Which of the following is a dematiaceous fungus?

- a) Mucor species
- b) Aspergillus niger
- c) Cladosporium
- d) Blastomyces dermatitidis

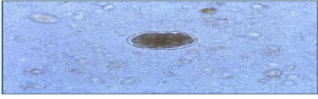
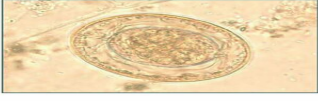
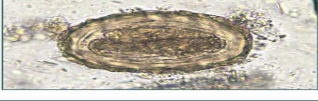
Question 23:

The strict anaerobe among the following is _____.

- a) Burkholderia cepacia
- b) Yersinia enterocolitica
- c) Stenotrophomonas maltophilia
- d) Prevotella

Question 24:

Match the following eggs to the organism.

Organisms	Eggs
1 Hymenolepis nana	A 
2 Ascaris lumbricoides	B 
3 Ancylostoma duodenale	C 
4 Taenia solium	D 

©MARROW

a) 1-D, 2-C, 3-A, 4-B

b) 1-C, 2-D, 3-A, 4-B

c) 1-D, 2-C, 3-B, 4-A

d) 1-C, 2-D, 3-B, 4-A

Question 25:

A man presents with the following lesion after traumatic inoculation. Identify the causative organism.



- a)** *Mycobacterium scrofulaceum*
- b)** *Sporothrix schenckii*
- c)** *Madurella mycetomatis*
- d)** *Phialophora verrucosa*

Answer Key

Question No.	Correct Option
1	d
2	d
3	d
4	d
5	a
6	d
7	c
8	b
9	a
10	a
11	a
12	d
13	c
14	d
15	b
16	c
17	b
18	b
19	b
20	c
21	c
22	c
23	d
24	d
25	b

Detailed Explanations

Solution to Question 1:

Aplastic crisis in a child with hereditary spherocytosis is most likely due to infection of parvovirus. This aplastic crisis is due to the transient arrest of erythropoiesis where the parvovirus infects and kills the red cell precursors. This generally leads to a sudden worsening of anemia. Transfusions may be necessary to support the patient until the immune response clears the infection.

Parvovirus:

Parvovirus-B19 is a non-enveloped single-stranded DNA virus.

Clinically, parvovirus B-19 infection may have several presentations-

- Erythema infectiosum - Also known as the fifth disease. The incubation period after infection is typically around 4-14 days. The hallmark of erythema infectiosum is the characteristic rash, which appears initially as erythematous facial flushing, often described as a “slapped cheek” appearance. Later the rash spreads rapidly or concurrently to the trunk and proximal extremities as diffuse macular erythema. The child may be febrile.
- Transient aplastic crisis - It may be seen in patients with underlying chronic hemolytic anemia like sickle-cell disease, thalassemia, hereditary spherocytosis, and RBC enzyme deficiencies like pyruvate kinase deficiency. Patients with aplastic crisis appear ill with fever, malaise, and pale with tachycardia and tachypnea due to profound anemia. The rash may or may not be present. Children with sickle-cell disease may also develop vaso-occlusive crises.
- Pure red cell aplasia - Persistent B19V infection that results in pure red cell aplasia has been reported in a wide variety of immunosuppressed patients, including patients with congenital immunodeficiency, AIDS, and lymphoproliferative disorders and transplant recipients.
- Primary maternal infection is associated with non-immune fetal hydrops and intrauterine fetal demise

It is a clinical diagnosis not requiring investigations usually. However, in the case of pregnant women or with other manifestations an IgM ELISA or PCR assay may be done. It is self-limited and resolves on its own.

The given image shows the classical slapped cheek appearance of erythema infectiosum caused by parvovirus B19. During the rash period, the patient is generally non-infectious.



Solution to Question 2:

The parasitic ova will be seen in a specimen taken from level D.

The image shows a formol-ether sedimentation technique to concentrate the ova in the bottom (D) part of the tube. It is used for samples in which the concentration of parasitic ova is scanty. It is useful for all helminth ova/eggs and parasitic cysts. The ether adsorbs the fecal debris and floats. Formalin fixes and preserves the specimen.

This method is useful for all helminth eggs and protozoan cysts.

Formalin-ether sedimentation technique



©Marrow

Solution to Question 3:

The given image shows large cylindrical worms with tapering ends, and thus represents *Ascaris lumbricoides* (roundworm).

Ascaris lumbricoides is the most common of human helminths and is distributed worldwide. It is the largest nematode parasite in the human intestine. The adult worms are large cylindrical worms, with tapering ends. The anterior end is more pointed than the posterior. It appears white outside the body.

Its natural host is humans. There is no intermediate host. Adult worms live in the small intestine - jejunum and ileum. Humans acquire infection by consuming food or water contaminated with nightsoil or human feces containing eggs. The eggs contain the infective rhabditiform larva.

Solution to Question 4:

The given clinical vignette of meningitis with the Gram stain slide showing Gram-positive diplococci, some in lancet-shaped and some arranged in chains, narrows down the aetiological/microbial agent to *Pneumococcus* / *Streptococcus pneumoniae*. It is catalase-negative and optochin sensitive.

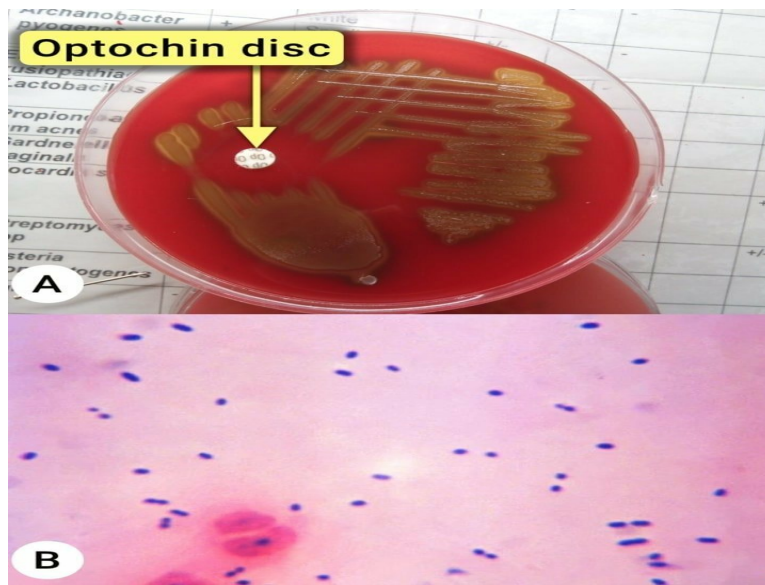
The Gram stain would reveal Gram-positive, lancet-shaped diplococci, single cocci, or chains. *Pneumococci* undergo autolysis with the addition of surface-active agents like ox bile (10%) or sodium deoxycholate (2%). The *Viridans* streptococci do not lyse and are thus easily differentiated from pneumococci. The growth of pneumococci is inhibited by optochin (Ethyl hydrocuprein hydrochloride) while that of *viridans* streptococci is not. It is catalase-negative, ferments inulin, and optochin sensitive.

Pneumococci are commensals of the upper respiratory tract and can cause pneumonia, sinusitis, otitis, bronchitis, bacteremia, meningitis, peritonitis, and other infectious processes. The pneumococcal infection results in inflammation and exudation of fibrinous materials resulting in edema and consolidation. The onset of pneumococcal pneumonia is usually sudden, with fever, chills, and sharp pleural pain. The sputum is similar to the alveolar exudate, being characteristically bloody or rusty colored. Disseminated infection and bacteremia can present as meningitis, endocarditis, and septic arthritis.

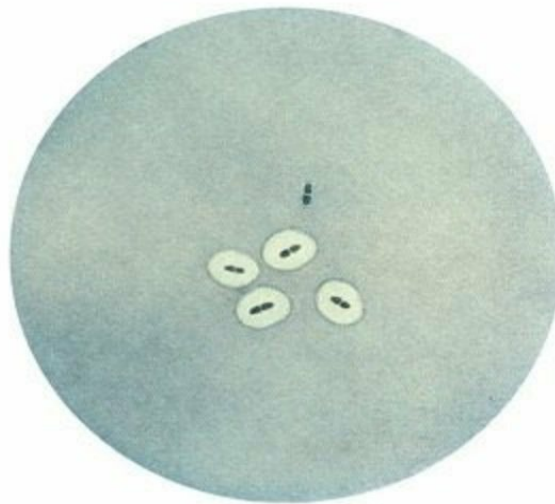
The other diagnostic parameters include the C-polysaccharide found in the cell wall of all *S. pneumoniae* that can be detected in the urine and cerebrospinal fluid (CSF) and a positive Quellung reaction. Quellung reaction/ Neufeld reaction is used to identify capsulated organisms such as pneumococci, *Neisseria meningitidis*, *Haemophilus influenzae*, etc. When a suspension of a capsulated bacterium is mixed with its specific anticapsular serum and examined under the microscope, the capsule becomes very prominent and appears 'swollen' due to an increase in its refractivity.

High-dose penicillin G is effective in treating pneumococcal infection.

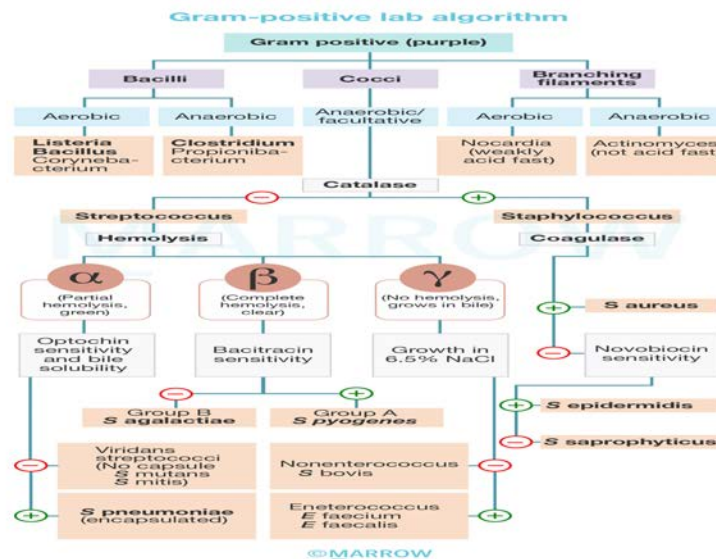
The culture showing alpha hemolysis on blood agar, with optochin (Op) sensitivity, and microscopy showing Gram-positive, lancet-shaped diplococci, suggests *Streptococcus pneumoniae* (pneumococcus).



The image shows a swollen capsule, which is demonstrated by the Quellung reaction



The flowchart demonstrates the algorithm used for identification of gram-positive organisms in the lab.



Solution to Question 5:

Cytotoxic T cells are involved in type 4/delayed type hypersensitivity

Solution to Question 6:

The given clinical vignette of fever and hemoptysis in a labourer with the sputum smear showing acid-fast bacilli is suggestive of tuberculosis. Mycobacterium tuberculosis, detected on sputum smear, is appreciated by Ziehl-Neelsen staining, which would include carbol fuchsin - acid - alcohol - methylene blue.

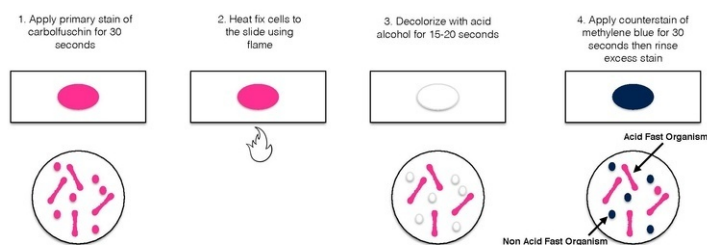
Acid-fast bacteria (like MTB) in a stained Ziehl-Neelsen (ZN) smear appear bright red against a blue background. Steps involved in ZN staining are:

- Place the sample on slide evenly spread it with rotational movement
- Air-dry the slide and fix it by passing it over a flame
- Apply carbol fuchsin stain and heat to fix it.
- Wash with water to remove excess stain
- Decolourise with alcohol
- Apply Methylene blue/Malachite green stain
- Wash with water to remove excess stain
- View under the microscope

Kinyoun modified the classical ZN staining such that heat fixing is not required. It is called the cold method. It involves increasing the concentration of phenol acid and the duration of staining. The slide is then washed with water and decolourized with acid alcohol. Counterstaining with

Loeffler's methylene blue or malachite green is done for a minute and viewed under a microscope. Acid-fast bacteria in a stained smear appear bright red against a blue background.

The image attached below summarizes the steps.



Option A: It depicts the order of Gram staining, which was discovered by Hans Christian Gram in 1884.

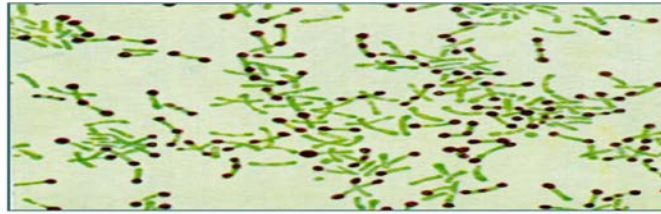
- Primary staining with para-rosaniline dye (crystal violet, methyl violet or gentian violet).
- Application of dilute solution of iodine.
- Decolorisation with an organic solvent (ethanol, acetone, aniline).
- Counterstaining with the dye of contrasting colour (carbol fuchsin, safranin, neutral red).

Option B: It includes the steps for performing Albert's stain which is used for detected *Corynebacterium diphtheriae*.

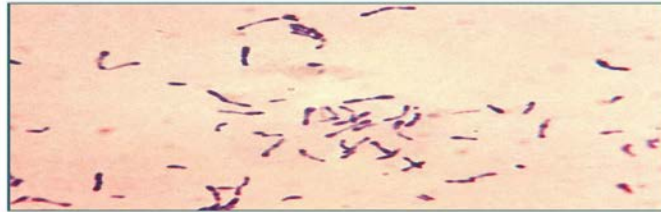
- A heavy smear of the bacterial isolate is prepared on a glass slide and allowed to dry without fixing.
- Albert's stain is applied for (3-4) minutes. Then it is washed with water and Albert's iodine is applied for one minute.
- Wash with water and let dry with blotting paper.
- Examine the slide under oil immersion.

Corynebacterium diphtheriae are slender bacilli with an arrangement resembling the letters V or L. They have metachromatic granules at the poles of bacilli which are also called polar bodies, volutin, or Babes-Ernst granules. Albert staining is useful in the demonstration of metachromatic granules. The presence of metachromatic granules helps to distinguish *C. diphtheriae* from diphtheroids which lack them. The images below show *Corynebacterium diphtheriae*.

Corynebacterium diphtheriae-microscopy



Alberts stain



Gram stain

©Marrow

Solution to Question 7:

This is a case of Cervicofacial actinomycosis caused by *Actinomyces israeli*. It presents as a painless nodule that indurates with purulent discharge that contains sulfur granules giving it a yellow color. Sulfur granules are composed of a matrix of bacteria, calcium phosphate, and host tissue.

Actinomycosis is an indolent, slowly progressive suppurative and granulomatous infection that produces pyogenic lesions with interconnecting sinus tracts that contain sulfur granules. It is an anaerobic or microaerophilic bacteria.

It presents as one of the following clinical types:

- Cervicofacial actinomycosis- (most common site) fluctuant mass in the jaw area producing draining fistulas. (lumpy jaw)
- Thoracic actinomycosis- pulmonary infection with fever, cough, and purulent sputum. It usually occurs following aspiration or trauma leading to inoculation of the infective material.
- Abdominal actinomycosis- following ruptured appendix or ulcer. It may involve multiple organs.
- Pelvic actinomycosis- due to the colonization of intrauterine contraceptive devices.
- Actinomycosis can spread to bones and surrounding tissue and produce multiple discharging sinuses.

Diagnosis is by Gram staining of sulfur granules that shows a dense network of thin gram-positive filaments, surrounded by club-shaped structures. (sunray appearance). Culture shows the "molar tooth" appearance of the colonies. On solid media, *A. israeli* produces spider colonies.

Penicillin G is the drug of choice and the duration of therapy is between 6-12 months. The other drugs that can be used include amoxicillin, erythromycin, tetracycline, doxycycline, clindamycin, and minocycline.

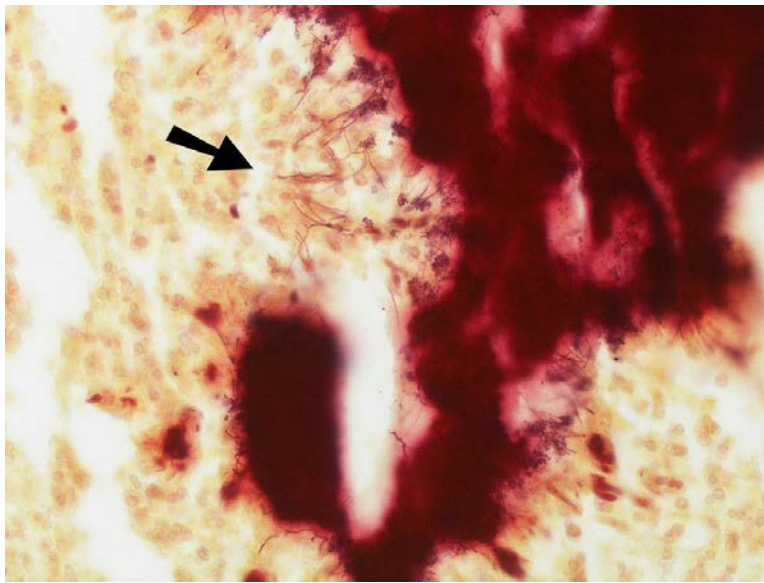
The image below shows a case of cervicofacial actinomycosis with discharging sinuses.



The image below shows the "molar tooth" appearance of the Actinomyces colonies.



The image below shows gram-positive filaments of Actinomyces.



The given image shows a patient with multiple cutaneous lesions caused by actinomycosis:



Solution to Question 8:

Beta-(1,3)-d-glucan (BG), a common component of many pathogenic fungi is absent in Zygomycetes/ Mucormycosis.

Fungal cell wall:

The fungal cell wall is rigid and determines the shape of the fungi. These protect them from osmotic and environmental stresses. They are composed of glycoproteins and lipids. Some sugar polymers such as chitin, glucans, and mannans are also found in the cell wall. All these components remain cross-linked to form a multilayered cell wall matrix. They exhibit pathobiological properties and are responsible for provoking an inflammatory response during

infections. These are also species-specific and release immunodominant antigens that could aid in immunological testing. In addition, some yeasts and molds are described as dematiaceous because their cell walls contain melanin which forms host defenses.

Clinical significance of Beta-1, 3-d-glucan:

G-test:

G-test is done for detection of α -3 glucan. It is helpful in establishing a diagnosis of candidiasis.

Tests for beta-d-glucan:

The presence of an elevated level of β -d-glucan in the serum can be seen in candidiasis, aspergillosis, and pneumonia due to *Pneumocystis carinii* pneumonia. The same test cannot be effectively applied in the evaluation of Mucorales. This assay is also not evaluated in routine in cases of cryptococcosis, not because of the absence of BG but because the capsule captures BG even before it is released into the bloodstream.

Effectiveness of echinocandins:

Echinocandins inhibit the synthesis of glucan synthase, since there is no much of a role for beta-glucans in mucormycosis, this class of drugs is seldom effective in the management of mucormycosis. The drug of choice for the management of mucormycosis is amphotericin-B.

Solution to Question 9:

Of the given statements regarding trematodes, the only true statement is that *Fasciola hepatica* is seen in the intrahepatic biliary radicles.

Schistosomes are dioecious, *Fasciola hepatica* is acquired through metacercariae encysted on raw aquatic vegetation and triclabendazole is not the treatment of choice for all trematode infections as praziquantel is used in the management of Schistosomiasis and lung fluke infections.

Trematodes are leaf-shaped unsegmented, flat, and broad helminths. The males are shorter and stouter than the females. The unique feature of flukes is the presence of two muscular cup-shaped suckers (hence called distomata)- the oral sucker surrounding the mouth at the anterior end and the ventral sucker or acetabulum in the middle, ventrally. Three of the most important pathognomic species include *Schistosoma hematobium*, *Fasciola hepatica*, and *Paragonimus westermani*

Features	<i>Schistosoma hematobium</i>	<i>Fasciola hepatica</i>	<i>Paragonimus westermani</i>
General feature	Male is broader than female	Largest and most common liver fluke.	Endemic foci were seen in Manipur
Sexes	Dioecious (Option-B), sexes are separate	Hermaphroditic parasite	Hermaphroditic parasite
Habitat	Mesenteric venous plexus (<i>S. mansoni</i> and <i>S. japonicum</i>) vesical, and prostatic venous plexus (<i>S. haematobium</i>).	The biliary tract of man.	Cystic spaces in the lung.

Features	Schistosoma hematobium	Fasciola hepatica	Paragonimus westermani
Body	Leaf-like unsegmented body with two cup-like suckers with delicate spines	Large leaf-shaped with a dorsoventrally flattened body.	Egg-shaped, reddish, brown, and covered with a scale-like spine.
Eggs	Nonoperculated eggs containing a ciliated embryo, miracidium	Ovoid, operculated, and bile-stained.	Oval, operculated, and golden brown
Definitive host	Man	Sheep	Man and domestic animals
First intermediate host	Freshwater snails	Freshwater snails (Lymnaea)	Snails of genera Semisulcospira (Melania species)
Second intermediate host	-	Aquatic vegetation	Crab or crayfish
Infective form	Fork-tailed cercariae	Metacercariae encysted on raw aquatic vegetation	Encysted metacercaria in crab or crayfish
Clinical features	Swimmer's itchKatayama feverhematuriaportal hypertension	Acute phase-fever and hepatomegaly.Chronic phase-biliary obstruction, obstructive jaundice, cholelithiasis, and anemia	Peribronchial granuloma and cystic dilation of bronchi.DyspneaHemoptysisPneumonitisBronchiectasisAbscessPneumothoraxExtrapulmonary lesions in the brain and intestine.
Diagnosis	Detection of eggs in urine or stoolbiopsyimagingdetection of antigen and antibody	Detection of eggs in stool and aspirated bileUSGERCP ELISA	Ova in sputumX-ray and CT scan of the chestCFTIH AELISA
Treatment	Praziquantel	Oral triclabendazole	Praziquantel

Solution to Question 10:

The toxin responsible for the abnormal posturing shown in the image is tetanospasmin produced by *Clostridium tetani*. It acts by causing presynaptic inhibition of GABA and glycine release.

Tetanospasmin, after secretion by the bacterium is cleaved into two peptides linked by a disulphide bond. The larger peptide diffuses to terminals of neurons and migrates by the retrograde axonal transport system to the spinal cord and brainstem. It diffuses to the glycinergic interneurons and GABA secreting neurons there. The smaller peptide degrades synaptobrevin and blocks the release of the inhibitory glycine and GABA. This results in hyperreflexia, muscle spasms, and spastic paralysis. Even small amounts of the toxin can be lethal.

The incubation period of *C. tetani* ranges from 4-5 days up to 3 weeks. Muscle spasms usually involve the area of injury first and then the muscles of the jaw causing trismus or lockjaw. Gradually, other voluntary muscles become involved, resulting in tonic spasms. Any external stimulus can precipitate a tetanic generalized muscle spasm. The patient is conscious but may be in pain. Death usually results from respiratory paralysis.

Other options: Proteolysis of target SNARE proteins in the neurons inhibits the release of Ach at the synapse - this is the mechanism of action of Botulinum toxin.

Solution to Question 11:

The given clinical vignette of Pyrexia of Unknown Origin (PUO) / Fever of unknown origin (FUO) with a history of frequent travel is suggestive of an infectious cause of FUO. The given slide shows macrophages infested with intracellular *Leishmania donovani* bodies (arrows) characterized by a kinetoplast and characteristic double dot appearance (Leishman stain). The most likely diagnosis is leishmaniasis and it can be established by the rk 39 antigen test.

Pyrexia/ Fever of unknown origin - criteria:

- Fever ≥ 38.3 degrees C (101 F) on 2 occasions
- Duration $\geq$ 3 weeks
- Diagnosis is not certain after history, physical examination, and routine and obligatory investigations in spite of three OPD visits or after 3 days of hospitalization.

Causes of FUO are enlisted in the table.

Leishmaniasis is a slowly progressing indigenous disease caused by a protozoan parasite of the genus *Leishmania*. It is transmitted by the bite of an infected female *Phlebotomus* sandfly which inoculates the flagellated promastigotes into the skin of the host in developing countries.

In human beings, the disease presents in four different forms with a broad range of clinical manifestations:

- Visceral leishmaniasis, or kala-azar,
- Cutaneous leishmaniasis,
- Mucocutaneous leishmaniasis, and
- Diffuse cutaneous leishmaniasis.

The life cycle of *Leishmania* involves two forms,

- The promastigote which develops and lives extracellularly in the sandfly vector and
- The amastigote multiplies intracellularly in the reticuloendothelial cells of the host.

It lowers immunity, causes persistent fever, anemia, liver and spleen enlargement, and if left untreated, death may occur due to opportunistic infection.

The microscopic demonstration of amastigotes from bone marrow is the gold standard for the diagnosis of Kala-azar.

For microscopic demonstration of the parasite, the materials collected are:

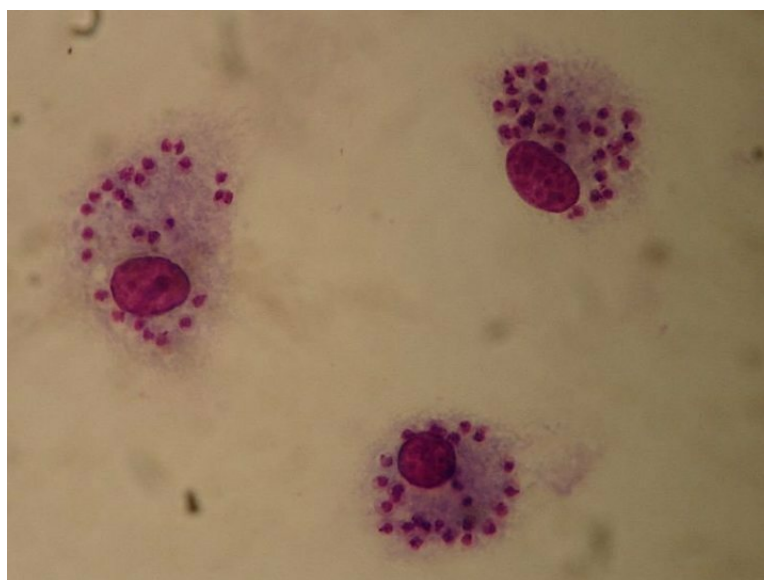
- Peripheral blood
- Bone marrow
- Splenic aspirate
- Enlarged lymph node.

Other diagnostic tests are:

- Rapid diagnostic test: rK39 Rapid diagnostic test and Kala-azar dipstick test strip. This is recommended by National Vector Borne Disease Control Programme (NVBDCP) in India.
- Parasitological diagnosis: Demonstration of LD bodies in buffy coat smear - confirmatory.
- Aldehyde test of Napier: Widely used for surveillance.
- Serological tests for diagnosis: Rapid dipstick test, Direct agglutination test, ELISA, Indirect Fluorescent Antibody Test
- Leishmanin (Montenegro) test: Induration of 5mm or more is considered positive, usually positive 4-6 weeks (delayed-hypersensitivity) after onset in cutaneous leishmaniasis, valuable for distinguishing immune from nonimmune subjects and endemicity.

The given images show amastigotes in the bone marrow.

Infectious cause	Non-infectious cause
Bacterial: <i>M. tuberculosis</i> <i>Salm</i> <i>onella</i> spp <i>Brucella</i> spp <i>Chlam</i> <i>ydias</i> <i>psittaci</i> <i>Leptospira</i> spp <i>Ri</i> <i>ckettsia</i> spp <i>Coxiella burnetti</i> <i>Mycoplasma</i> spp Atypical myc <i>obacteria</i> Viral: Cytomegalovir <i>us</i> Epstein - Barr virus Arbovir <i>uses</i> Enteroviruses HIV Fungal: <i>Candida albicans</i> <i>Cryptococcus</i> <i>neoformans</i> <i>Histoplasma capsulatum</i> <i>Aspergillus</i> spp <i>Coccidioides immitis</i> Parasitic: <i>Plasmodium</i> spp <i>Leishmania</i> spp <i>Trypanosoma</i> spp <i>Toxoplasma gondii</i> <i>Wuchereria bancrofti</i> <i>Brugia malayi</i>	Neoplasm: Lymphoma Leuke <i>mia</i> Myeloma Renal cancer Col <i>on</i> cancer Liver cancer Metabo <i>lic</i> disorder Gout Porphyria Co <i>nnective</i> tissue disorder: Syste <i>mic</i> lupus erythematosus (SL <i>E</i>) Polyarteritis nodosa Granul <i>omatous</i> disease: Crohn's dise <i>ase</i> Sarcoidosis Rheumatoid ar <i>thritis</i> Polymyositis



Drugs used in the treatment of visceral leishmaniasis are sodium stibogluconate, miltefosine, and amphotericin. Liposomal amphotericin single dose 10mg/Kg IV is the first line of treatment in India.

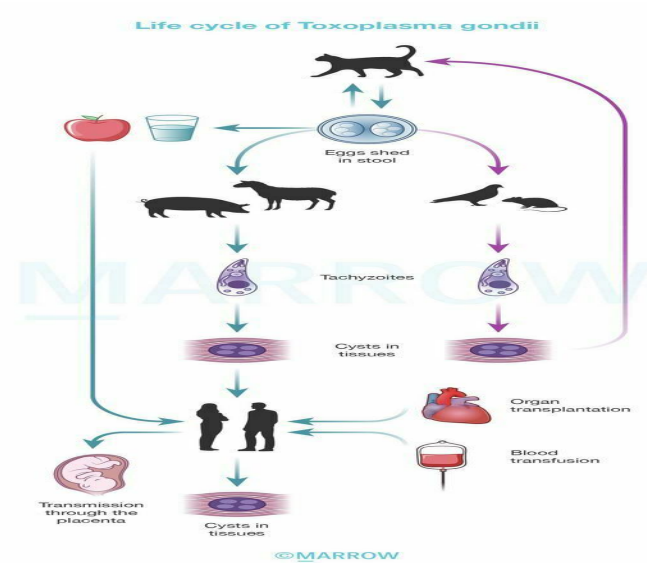
Solution to Question 12:

The given image shows the lifecycle of *Toxoplasma gondii*.

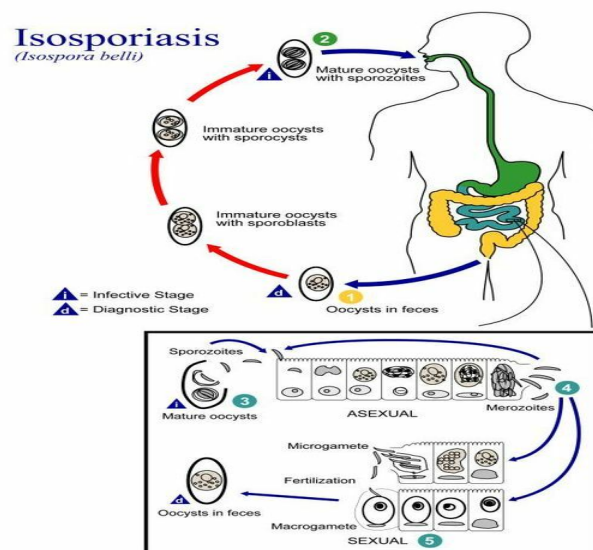
Toxoplasma gondii is an obligate intracellular parasite that causes toxoplasmosis.

- The life cycle of *T. gondii*
- The enteric cycle occurs in definitive hosts such as cats. Exoenteric cycle occurs in intermediate hosts such as humans, mice, rats, etc.
- Both asexual reproduction (schizogony) and sexual reproduction (gametogony) occur within the mucosal epithelial cells of the small intestine of the cat. The tachyzoites and tissue cyst represent stages in asexual multiplication, while the oocyst is formed by sexual reproduction.
- All three forms, trophozoite, tissue cyst, and oocyst occur in the definitive host. Only the asexual forms, tachyzoites, and tissue cysts are present in the intermediate host.
- Asexual stage: Oocysts containing sporozoites or tissue cysts containing bradyzoites are ingested by intermediate hosts like humans, sheep, etc. Bradyzoites and sporozoites are released in the intestinal epithelium and form tachyzoites, which divide rapidly and replicates in all mammalian cells except for red blood cells. Tissue cysts are then formed which can occur in various organs but principally persist in the central nervous system. This chronic stage completes the asexual cycle and thus the parasite completes its life cycle in the intermediate host.
- The definitive host is the cat in which the sexual stage takes place. In this stage, the oocysts are formed, which after sporulation are excreted and if ingested by an intermediate host, cause pathological changes.
- Tachyzoites continue to multiply and spread locally by the lymphatic system and blood, while the bradyzoites multiply slowly inside the tissue cyst. The dormant bradyzoites inside the cyst may be reactivated in immune suppression causing renewed infection in the host.
- Transmission of human *Toxoplasma* infection
- This can be through oral routes from contaminated water or food, etc; it can be transmitted through blood or organs (transfusion or transplantation) and it can be transmitted through the trans-placental route.

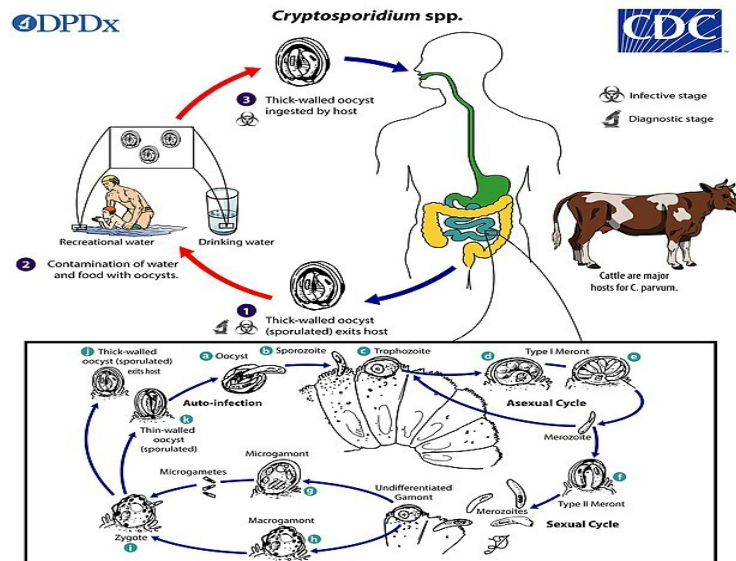
The below image shows the life cycle of *Toxoplasma gondii*.



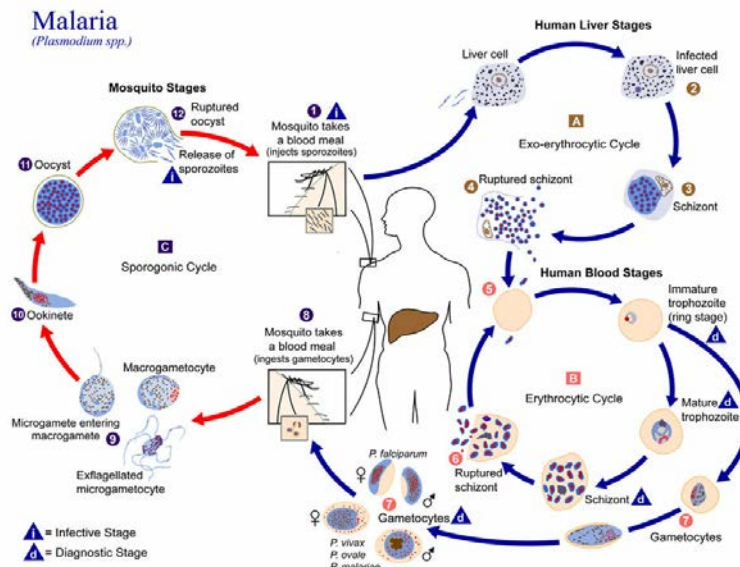
The image below shows the life cycle of *Cystoisospora belli* or *Isospora belli*.



The image below shows the life cycle of *Cryptosporidium parvum*.



The image below shows the life cycle of Plasmodium species.



Solution to Question 13:

HTLV - Human T cell lymphotropic virus causes microbial carcinogenesis by indirect mechanisms. Even, Hepatitis C virus, a flavivirus, does not generate a provirus and appears to induce cancer indirectly.

Oncogenic viruses:

Integration of viral genome into the host cell thereby leading to the formation of a pro-virus is commonly a mechanism of viral microbial carcinogenesis. DNA tumor viruses encode viral oncoproteins that are important for viral replication. Some portion of this may become integrated into the host cell chromosome. For RNA tumor viruses that mostly belong to the retrovirus family

are known to cause microbial carcinogenesis by indirect mechanisms.

Note:

HPV causes carcinogenesis by both integrations and by the E6, E7 protein systems.

Solution to Question 14:

The given clinical vignette is suggestive of infective cardiac valvulitis with mycotic vegetation. The histopathological slide shows fungal hyphae of *Aspergillus* species with non-pigmented, narrow, septate, hyphae with branching at 45° (acute angle).

Invasive aspergillosis is an opportunistic infection seen in the immunosuppressed hosts. *Aspergillus* has a tendency to invade blood vessels. Thus, widespread hematogenous dissemination with involvement of the heart valves and brain is common. *Aspergillus* form fruiting bodies and septate filaments, 5 to 10 µm thick, branching at acute angles (40 degrees). *Aspergillus* hyphae without the distinct fruiting body cannot be distinguished from *Pseudallescheria boydii* and *Fusarium* spp. by morphology alone.

Rhinocerebral *Aspergillus* infection in immunosuppressed individuals resembles that caused by Mucormycoses (e.g., *Mucor* spp., *Rhizopus* spp.). (Option-A) Mucormycetes form broad ribbon-like non-septate or sparsely septate pigmented hyphae of variable width (6 to 50 µm) with frequent right-angle branching, distinct from *Aspergillus* hyphae.

Option B: Disseminated fusariosis differs from disseminated aspergillosis as the former presents with nodular or necrotic painful skin lesions are extremely common with fusariosis. No such history is given in the clinical stem. The organism is difficult to differentiate from *Aspergillus* species in tissues; thus, identification with culture is imperative.

Option C: *Rhizopus* shows aseptate hyphae with sporangium and sporangiophores. It is similar in appearance to that of mucormycosis.

Solution to Question 15:

The attachment-effacement lesion of the intestinal mucosa as shown in the image is caused by EPEC (Enteropathogenic *Escherichia coli*).

E. coli is a gram-negative, flagellated rod. It is a lactose fermenter and hence, grows on MacConkey medium to produce pink colonies.

Clinical infections are of four main types:

- Urinary tract infection
- Diarrhea
- Septicemia, neonatal sepsis and neonatal meningitis
- Pyogenic infections

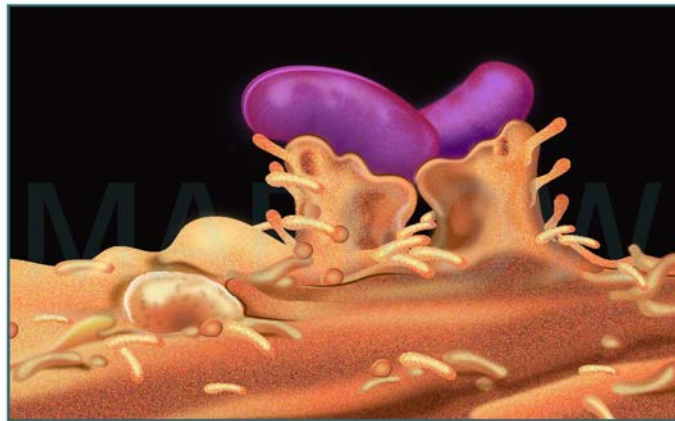
There are mainly six types of diarrheagenic *E. coli*:

- Enteropathogenic *E. coli* (EPEC)

- Enterotoxigenic E.coli (ETEC)
- Enteroinvasive E.coli (EIEC)
- Enterohemorrhagic E.coli (EHEC)
- Enteroaggregative E.coli (EAEC)
- Diffusely adherent E.coli (DAEC)

Enteropathogenic E.coli (EPEC): It causes diarrhea mainly in infants and children. EPEC adherence factor (EAF), a plasmid-encoded protein is responsible for adherence. Another mechanism is adherence to enterocyte membrane by a locus of enterocyte effacement (LEE), which is regulated chromosomally. This promotes tight adherence of the pathogen with loss of microvilli leading to effacement.

The electron micrograph below shows EPEC adherent to effaced microvilli of the small bowel.



©MARROW

Enterotoxigenic E. coli: Causes endemic diarrhoea in developing countries and travellers diarrhoea. ETEC serotypes responsible for diarrhea are mostly O6, O8, O15, O25, O27, O167. It attaches to intestinal mucosa with fimbriae called colonization factor antigens. The pathology is due to the heat-labile toxin (LT) produced by some and heat-stable toxin (ST) produced by others.

Enteroinvasive E.coli (EIEC): The clinical features are similar to shigellosis. It produces disease by invading the intestinal mucosa.

Enteroaggregative E. coli (EAEC): It causes acute and chronic diarrhea (>14 days in duration) in developing countries. These strains appear aggregated in a stacked brick formation. Some strains of EAEC produce ST-like toxins, others produce a plasmid-encoded enterotoxin that produces cellular damage, and still others, a hemolysin.

Enterohemorrhagic E.coli (EHEC): These strains produce two potent toxins, verocytotoxin (VT) or Shiga-like toxin (SLT). Especially in young children and the elderly, they cause diarrhea ranging from mild to fatal hemorrhagic colitis and hemorrhagic uremic syndrome (HUS). EHEC primarily targets vascular endothelial cells leading to HUS. E.coli serotype O157: H7 is associated with EHEC diarrhea and its complications.

Diffusely adherent E.coli (DAEC): These are less established as pathogens.

Solution to Question 16:

The component responsible for endotoxic shock in a gram-negative cell wall is 1, which represents the lipid A part of lipopolysaccharide (LPS).

The numbers labelled in the above image of the cell wall indicate:

- 1- Lipid A part of LPS
- 2- O polysaccharide part of LPS
- 3- Outer membrane
- 4- Plasma membrane

Gram-negative cell wall is divided into 3 regions:

- Outer membrane consisting of LPS, proteins (porins) and phospholipids
- Peptidoglycan
- Plasma membrane consisting of phospholipids

Lipopolysaccharide (LPS) is a part of the outer membrane of the cell walls of gram-negative bacteria. It accounts for their endotoxic activity and O antigen specificity. They were formerly known as the Boivin antigen. LPS consists of:

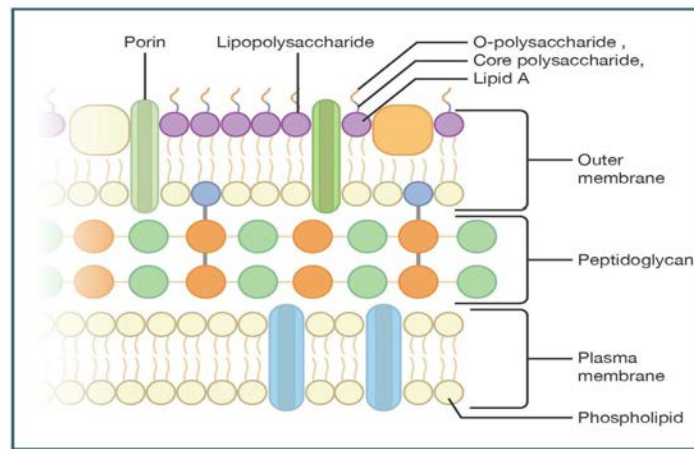
- O polysaccharide portion determining O antigen specificity
- Core polysaccharide
- Lipid A (glycolipid). It is responsible for endotoxic activities such as pyrogenicity, lethal effect, tissue necrosis, anticomplementary activity, B cell mitogenicity, immunoadjuvant property, and antitumor activity.

The differences between exotoxins and endotoxins are summarised below.

Exotoxins	Endotoxins
Actively secreted by cells and diffuse into the surrounding	An integral part of the cell wall of gram-negative bacteria; death and in part during growth; may not need to be released to have biological activity
Produced by both gram-positive and gram-negative bacteria	Found only in gram-negative bacteria
Proteins	Lipopolysaccharide complexes
Heat labile	Heat stable
Highly antigenic, stimulate the formation of high-titer antitoxin. Antitoxin neutralizes the toxin	Weakly antigenic

Exotoxins	Endotoxins
Converted to toxoid, used to immunize (eg, tetanus toxoid)	Not converted to toxoids
Highly toxic	Moderately toxic
Usually bind to specific receptors on cells	Specific receptors not found on cells
Usually do not produce fever in the host	Usually, produce fever in the host

Gram-negative cell wall



Other options:

Option A: 2 is the O polysaccharide portion of the LPS which determines O antigen specificity.

Option B: 4 is the plasma (cytoplasmic) membrane. It is a thin semipermeable layer that lines the inner surface of the cell wall and separates it from the cytoplasm.

Option D: 3 is the outer membrane. It is the outermost layer of the gram-negative bacterial cell wall. It contains various proteins known as outer membrane proteins.

Solution to Question 17:

The image shows Gram's stain of *Corynebacterium diphtheria* and the staining procedure is crystal violet, Gram's iodine, acetone, and safranin.

Corynebacterium diphtheria (also called Klebs-Loeffler bacillus) is a gram-positive bacillus with clubbing at both ends. They possess characteristic polymetaphosphate granules also called metachromatic granules.

Gram's stain procedure:

- Fix the smear using heat or methanol.
- Cover with crystal violet stain for 10–30 seconds.

- Gram's iodine stain used as mordant for 10–30 seconds.
- Decolorize with 30% acetone–alcohol for 10–30 seconds, until the stain cannot flow off the slide.
- Cover with safranin stain for 10–30 seconds.

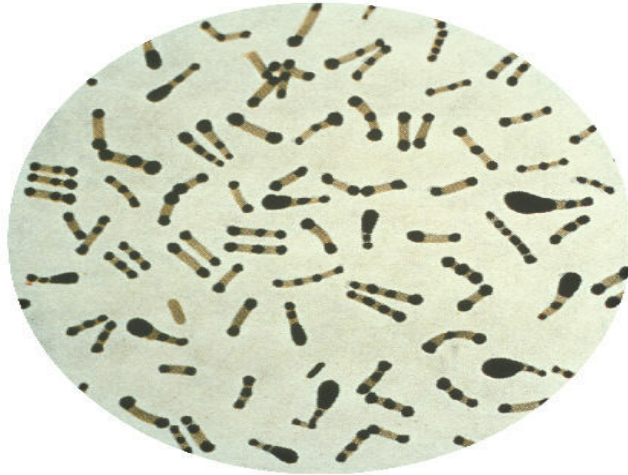
Image shows gram staining of *Corynebacterium diphtheria*



Other stains used for *Corynebacterium diphtheria*:

- Albert's stain shows green bacilli with purple-blue metachromatic granules arranged in typical Chinese letter or cuneiform arrangement. The stain requires two reagents:
- Albert's I solution contains toluidine blue, malachite green.
- Albert's II contains iodine and potassium iodide.
- Loeffler's methylene blue: The granules take up the bluish-purple color.
- Immunofluorescence: This is more specific as the smear is stained with a specific antibody.

Image shows Albert's stain of *C. diphtheria*



The incubation period of diphtheria is usually 2–5 days (range: 1–10 days).

Respiratory diphtheria is characterized by mild fever, sore throat, difficulty swallowing, malaise, loss of appetite, hoarseness (if the larynx is involved). The hallmark is a gray-white pseudomembrane over the mucous lining of the tonsils (most commonly), pharynx, larynx, or nares. It is tightly adherent to the underlying tissue, and any attempt to dislodge the membrane may cause bleeding.

A few patients develop massive swelling of the tonsils and present with bull-neck diphtheria, due to edema of the submandibular and paratracheal region. Cutaneous diphtheria may present as a scaling rash or as ulcers with clearly demarcated edges and membrane.

The images below shows pseudomembrane, bull-neck, and cutaneous diphtheria respectively.





Treatment of diphtheria includes administration of diphtheria anti-toxin. Recommended antibiotics include penicillin G and erythromycin. Prevention is with the diphtheria toxoid vaccine. It is coadministered with tetanus and acellular pertussis (DPT) and should be given according to the National Immunization Schedule.

Solution to Question 18:

The given image shows the life cycle of the herpes simplex virus (HSV).

The herpes virus capsid is icosahedral and encloses a core containing a linear double-stranded DNA genome. The nucleocapsid is surrounded by a lipid envelope derived from the modified host cell nuclear membrane. The envelope carries glycoprotein spikes.

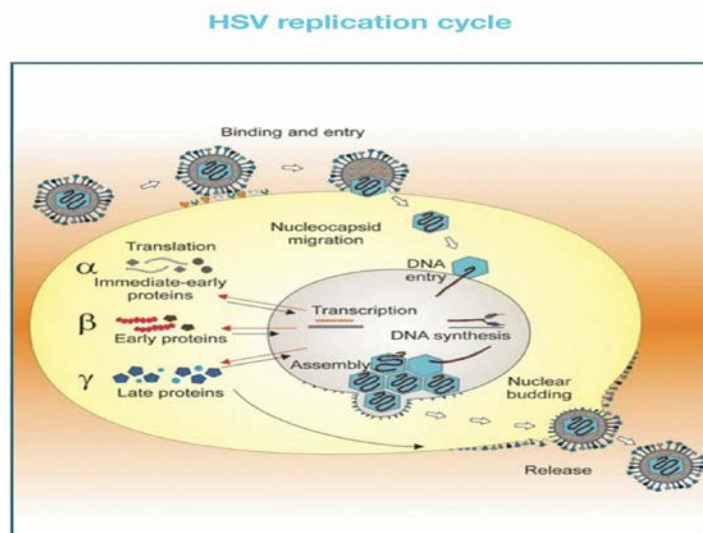
Steps of HSV replication:

The virus enters the cell by fusing with the plasma membrane, and viral DNA is released from the capsid at the nuclear pore. This is followed by circularization of the genome and transcription of immediate-early genes.

α -Proteins are products of immediate-early genes. These proteins stimulate the transcription of early genes. β -proteins are products of early genes. These proteins function in DNA replication, to yield concatemeric DNA. Transcription of late genes gives rise to γ -proteins. These primarily consist of viral structural proteins and participate in virion assembly.

Viral DNA is cleaved from concatemers and then packaged into capsids. Enveloped viral particles collect in the endoplasmic reticulum and are then transported by a vesicular movement to the cell surface.

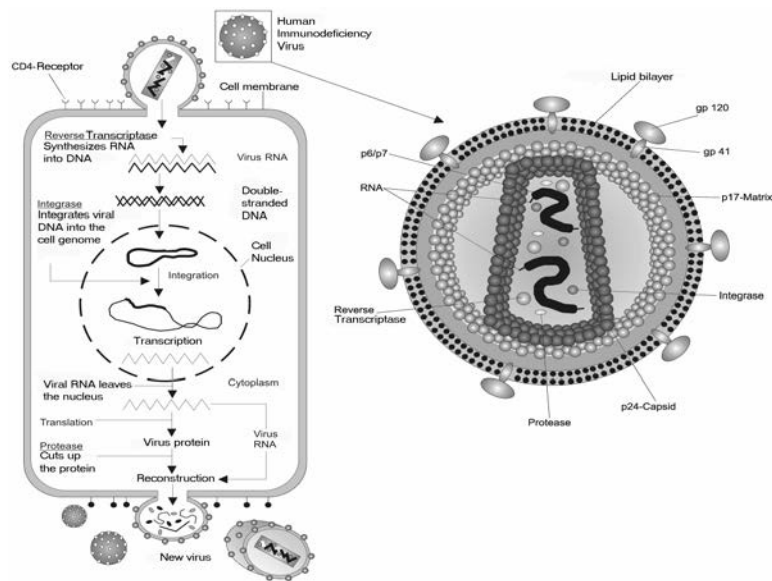
Image shows replication cycle of herpes simplex virus.



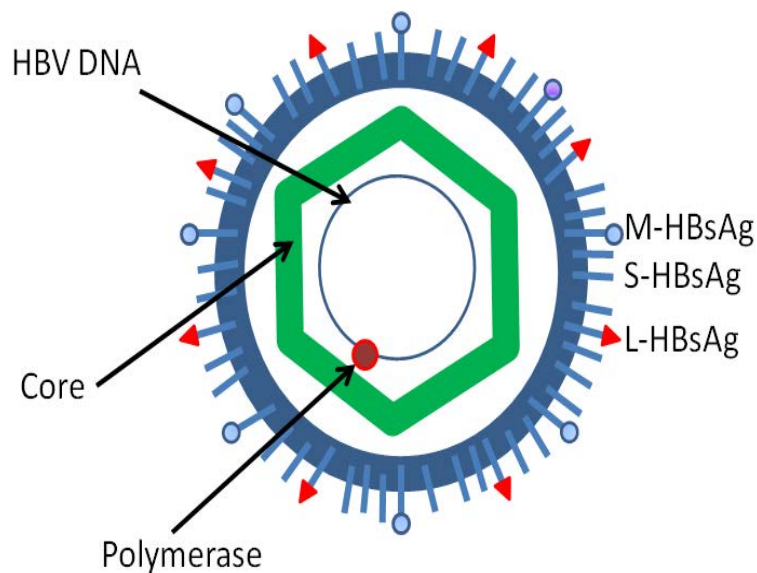
Other options:

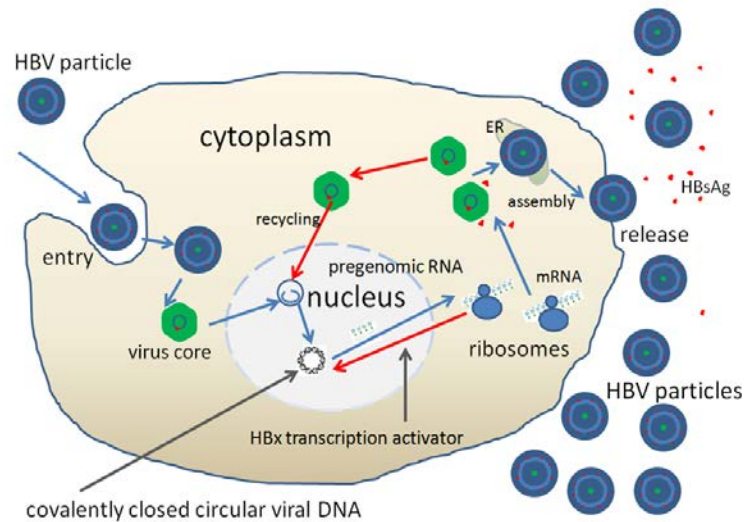
Option A) The human immunodeficiency virus (HIV) virus is spherical, enveloped RNA virus. The nucleocapsid has an icosahedral shell and a cone-shaped core, enclosing the ribonucleoproteins. The genome comprises two identical single-stranded, RNA copies, with the reverse transcriptase enzyme, which is a characteristic feature of retroviruses.

The images below show the structure of the HIV virus and its life cycle.

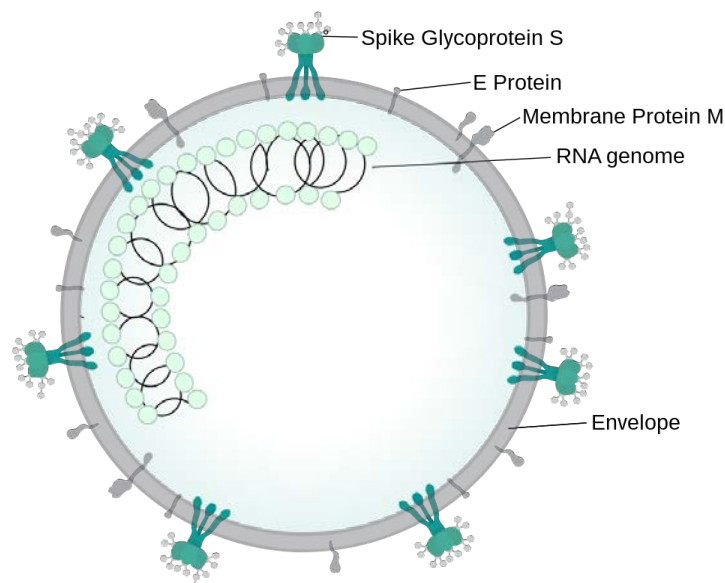


Option C: The hepatitis B virus is a DNA virus with an outer envelope and an inner nucleocapsid. The viral genome consists of two linear strands of DNA in a circular configuration. One of the strands (the plus strand) is incomplete resulting in partially double stranded and partially single-stranded DNA. The images below show the structure and life cycle of the hepatitis B virus.





Option D: Coronavirus are enveloped RNA viruses with a genome consisting of positive sense unsegmented single-stranded RNA. They have petal-shaped surface projections. Coronavirus nucleocapsids develop in the cytoplasm and mature by budding into cytoplasmic vesicles. The images below show the structure and life cycle of coronaviruses.



Option C: Injecting a hamster with splenic biopsy for diagnosis of leishmaniasis is a method of animal inoculation to help in confirmation of the diagnosis.

Option D: Intradermal test on guinea pigs for toxigenicity of *Corynebacterium diphtheriae* is a method of virulence testing.

Solution to Question 20:

The principle of the Paul-Bunnell test is based on heterophile antibodies. Heterophile antibodies have the ability to agglutinate shared antigens.

Paul-Bunnell test is the standard diagnostic procedure for infectious mononucleosis. In infectious mononucleosis, heterophile antibodies agglutinate sheep erythrocytes. However, such antibodies may also occur after injections of sera and sometimes even in normal individuals. Infectious mononucleosis antibodies may be differentiated by absorption tests. An agglutination titer of 100 or above is suggestive of infectious mononucleosis.

The Weil-Felix reaction is also a heterophile agglutination test. It is based on the sharing of a common antigen between typhus rickettsiae and some strains of proteus bacilli. It is used for the serodiagnosis of typhus fevers. Other examples based on the same principle are Streptococcus MG agglutination test for the diagnosis of primary atypical pneumonia and the cold agglutination test for diagnosis of mycoplasmal (primary atypical) pneumonia.

Other options:

Option A: Complement fixation test is based on the ability of antigen-antibody complexes to fix complement. This is a very sensitive test capable of detecting various types of antigens and antibodies.

Option B: Superantigens are certain protein molecules, that activate very large numbers of T cells irrespective of their antigenic specificity such as staphylococcal enterotoxins. Superantigens bind outside the antibody binding groove directly to the lateral aspect of the TCR beta chain. Microbial superantigens are medium-sized proteins characterized by high resistance to proteases and to denaturation by CD4⁺ T cells. They cause the release of cytokines (IL-2) which results in massive proliferation of T lymphocytes. This ultimately leads to further release of a variety of cytokines which can have profound effects on the immune system.

Option D: Haptens are substances that can react specifically with antibodies but are incapable of inducing antibody formation alone. Haptens become immunogenic (capable of inducing antibodies) by combining with a larger molecule carrier. They may be complex or simple. While complex haptens can precipitate with specific antibodies, simple haptens are non-precipitating.

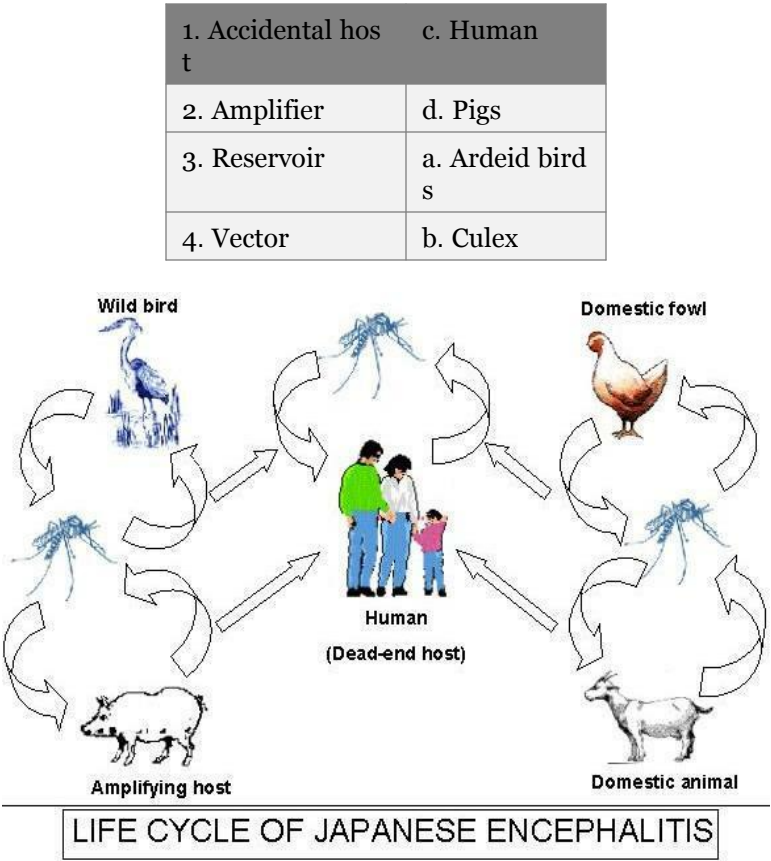
Solution to Question 21:

The correct match is 1-c, 2-d, 3-a, 4-b

Japanese encephalitis virus is a virus from the family Flaviviridae. It is the leading cause of vaccine-preventable encephalitis in Asia and the western Pacific.

Transmission of Japanese encephalitis virus: Natural infection has been demonstrated in Ardeid birds (herons and egrets). Bird-to-bird transmission takes place through the vectors *Culex tritaeniorhynchus* and *Culex vishnui* (common in India). Ardeid birds act as reservoir hosts, while pigs and cattle act as amplifier hosts. Humans are accidental dead-end hosts.

The image below shows the life cycle of Japanese encephalitis virus.



Most people infected with JE do not have symptoms or have only mild symptoms. Less than 1% of people infected with Japanese encephalitis (JE) virus develop clinical illness. The incubation period is typically 5-15 days in persons who develop symptoms. Initial symptoms often include fever, headache, and vomiting. Mental status changes, neurologic symptoms, weakness, and movement disorders might develop over the next few days. Seizures are common, especially among children.

No specific treatments have been found to benefit patients with JE, but hospitalization for supportive care and close observation is generally required. Treatment is symptomatic. Rest, fluids, and the use of pain relievers and medication to reduce fever may relieve some symptoms.

Prevention: In India, inactivated vero cell culture-derived SA-14-14-2 is used. Japanese encephalitis vaccine has been introduced in selected endemic districts. 2 doses of 0.5 ml are given in the left upper arm subcutaneously.

If the first dose is delayed beyond 12 months, a minimum of 3 months gap should be there between the two JE doses.

Vaccine	Due age	Maximum age
Japanese encephalitis - 1	9-12 months	15 years
Japanese encephalitis - 2	16-24 months	15 years

Solution to Question 22:

Cladosporium is the dematiaceous fungus in the given options.

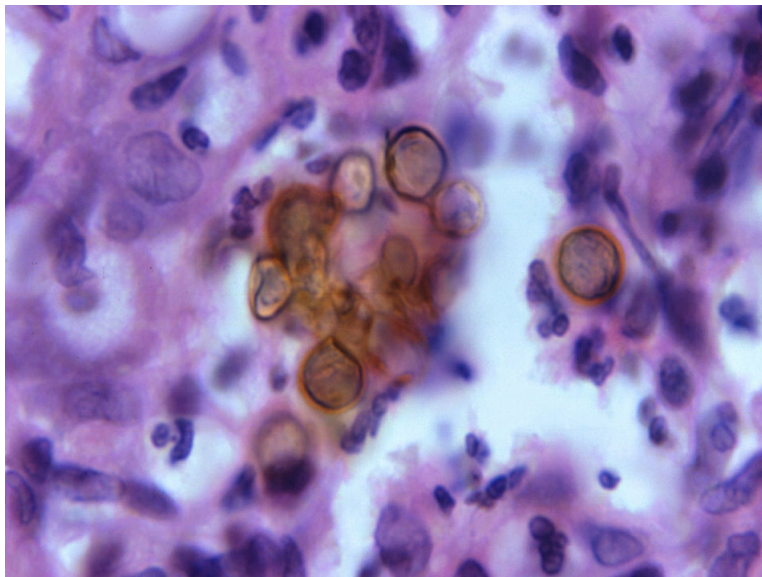
Dematiaceous fungi are fungi whose cell walls contain melanin, which imparts a brown to black color. *Phialophora verrucosa*, *Fonsecaea pedrosoi*, *Fonsecaea compacta*, *Rhinochlamydia aquaspersa*, and *Cladophialophora* (*Cladosporium*) *carrionii* are examples of dematiaceous fungi.

These fungi cause chromoblastomycosis, a chronic subcutaneous infection caused due to traumatic inoculation of fungal agents. The lesion is verrucous and wart-like with extension along the draining lymphatics. Cauliflower-like nodules with crusting abscesses eventually cover the area.

On histological examination, the lesions contain the fungus seen as round or irregular, dark brown, yeast-like bodies with septae, called sclerotic bodies or medlar bodies. Diagnosis can be established by demonstration of these sclerotic bodies in KOH mounts or in tissue sections, and by culture on Sabouraud agar.

Chemotherapy with amphotericin B, thiabendazole, 5-fluorocytosine, itraconazole, and recently voriconazole are useful for the treatment of larger lesions. Small lesions can be excised surgically.

The image below shows sclerotic bodies seen in chromoblastomycosis.



Other options:

Option A: *Mucor* species causes mucormycosis. It is called black fungus because it causes serious angioinvasion leading to tissue necrosis and gangrene. It is not a dematiaceous fungus. They appear as non pigmented (hyaline), broad non-septate hyphae on microscopy.

Option B: *Aspergillus* species usually appear as non pigmented (hyaline), septate hyphae with acute angle branching on microscopy. The conidia of *A.niger* are black in color.

Option D: *Blastomyces dermatitidis* appear as broad-based budding yeasts.

Solution to Question 23:

The strict anaerobe in the given options is *Prevotella*.

Prevotella species are gram-negative rods or coccobacilli. The most commonly isolated *Prevotella* species are *P melaninogenica*, *Prevotella bivia*, and *Prevotella disiens*. *P melaninogenica* is associated with upper respiratory tract infections. *P bivia* and *P disiens* occur in the female genital tract. *Prevotella* species are also found in brain and lung abscesses, empyema, pelvic inflammatory disease and tubo-ovarian abscesses.

Other options

Option A: *Burkholderia cepacia* belongs to the pseudomonad group and is gram-negative, aerobic rods.

Option B: *Yersinia enterocolitica* are gram-negative facultative anaerobes.

Option C: *Stenotrophomonas maltophilia* also belong to the pseudomonad group and are gram-negative, aerobic rods.

Solution to Question 24:

The correct match is 1-C, 2-D, 3-B, 4-A.

The *Hymenolepis nana* egg is roughly spherical or ovoid. It has a thin colorless outer membrane and inner embryophore that encloses the hexacanth oncosphere. They are non bile-stained. The eggs float in saturated salt solution.

The image below shows an egg of *H. nana*.



Ascaris lumbricoides (Roundworm) are round or oval in shape. They are always bile-stained which gives them golden-brown color. They have a thick smooth translucent shell with:

- an outer coarsely mammillated albuminous coat
- a thick transparent middle layer, and
- an inner lipoidal vitelline membrane.

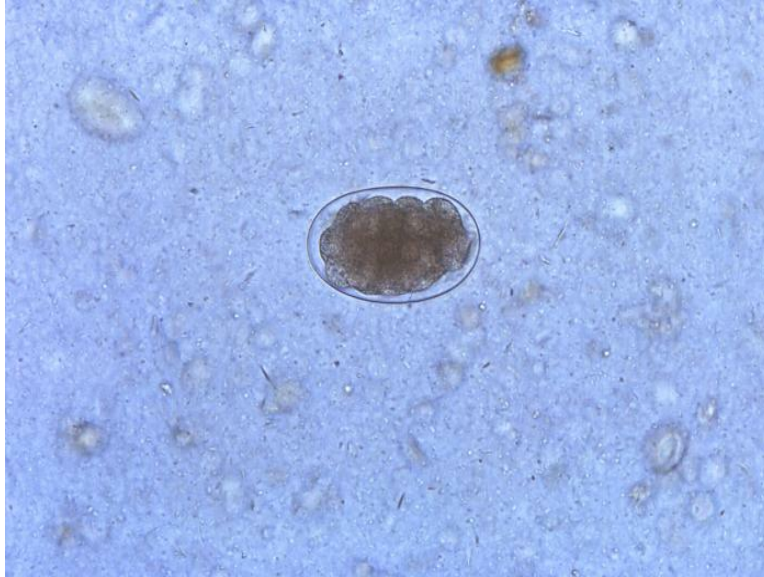
The eggs float in a saturated salt solution. The image below shows the egg of *A. lumbricoides*.



The unfertilized eggs of *Ascaris* are elliptical, narrower, and contain a small atrophied ovum. It has a mass of disorganized refractile granules. The shell is thinner and the egg does not float in a saturated salt solution.

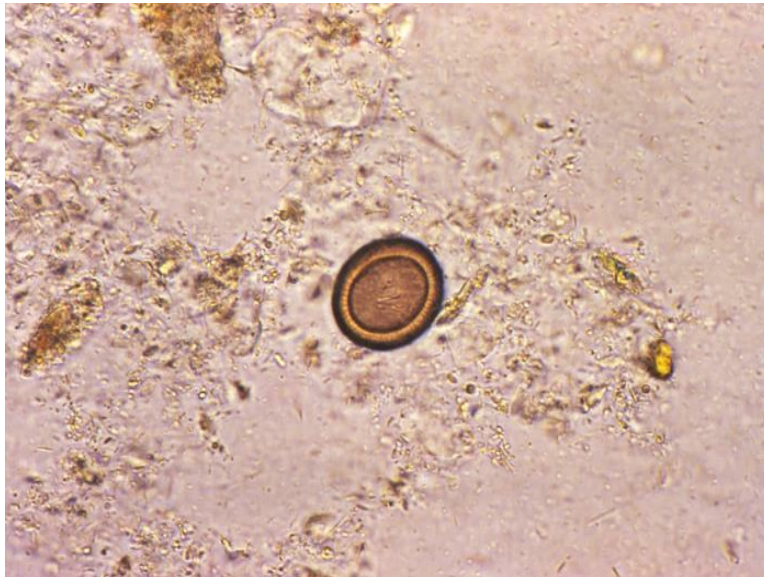
The egg of *Ancylostoma duodenale* or hookworm is oval or elliptical. They are colorless, not bile-stained. The eggs are surrounded by a thin transparent hyaline shell membrane. It contains a segmented ovum, usually with 4-8 blastomeres. There is a clear space between the ovum and the

shell. It floats in saturated salt solution. The image below shows the egg of *A. duodenale*.



The eggs of *Taenia solium* and *Taenia saginata* are indistinguishable. The egg is spherical. The inner embryophore is radially striated and is yellow-brown due to bile staining. A fully-developed embryo (oncosphere) with three pairs of hooklets (hexacanth embryo) is present in the center.

The eggs of *T. saginata* are infective only to cattle and not to humans, whereas the eggs of *T. solium* are infective to pigs and humans. The eggs do not float in a saturated salt solution. The image below shows the egg of *Taenia*.



Solution to Question 25:

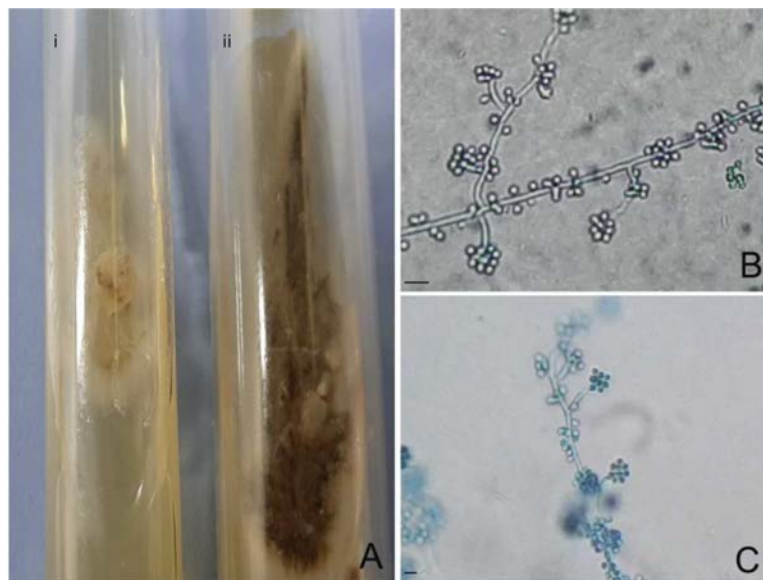
The causative organism of the lesion given in the image is *Sporothrix schenckii* (*S. schenckii*)

S. schenckii causes sporotrichosis, chronic granulomatous disease by traumatic inoculation into the skin. This is usually followed by spread along the draining lymphatics and lymph nodes.

It is a thermally dimorphic fungus. It is in the yeast phase in tissues and in cultures at 37°C, and in the mycelia phase in nature and cultures at 25°C. When it is cultured, the colonies are blackish brown, fuzzy. The organism produces branching, septate hyphae, and microconidia clustered at the ends of conidiophores. It exists as cigar-shaped yeast cells. Stains used for identifying yeast forms:

- Gomori methenamine silver - cell wall appears black
- Periodic acid-Schiff stain- cell wall appears red
- Fluorescent antibody staining

The image given below shows, (A)- (i) culture at 37°C (ii) culture at 25°C. (B) fungi with clustered microconidia with conidiophore. (C) magnified image showing branching and septate hyphae.



Clinical features: The initial infection develops as a nodule, which may break down to form a necrotic or ulcerative lesion. Untreated, the disease progresses and new nodules develop along the lymphatics leading to a chain of nodules. Less commonly it may also cause systemic infection following inhalation, which causes local pulmonary disease or may lead to widely disseminated lesions in joints, meninges, and skin.

Sporotrichosis



Treatment: Itraconazole is the drug of choice for lympho-cutaneous sporotrichosis, also terbinafine can be used. A saturated solution of potassium iodide (SSKI) is a second-line drug.

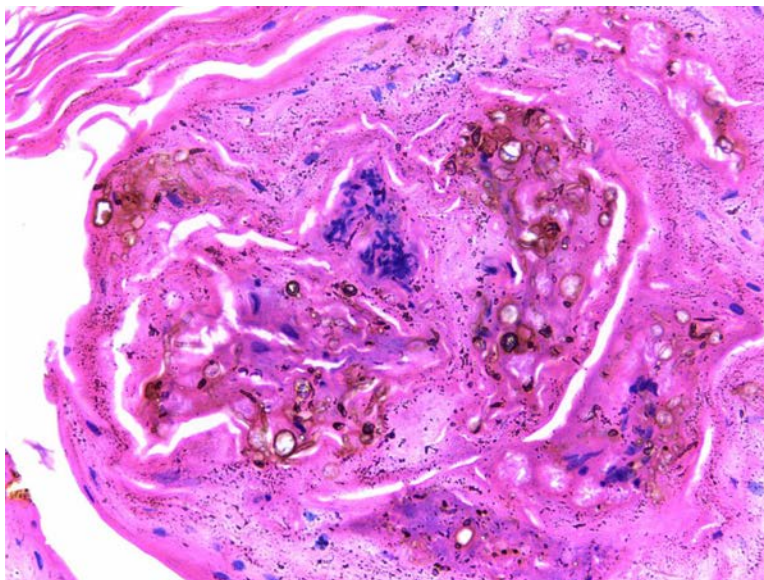
Other options:

Option A: *Mycobacterium scrofulaceum* is a scotochromogen. It causes chronic cervical lymphadenitis in children.

Option C: Eumycetoma is a mycetoma caused by fungi like *Madurella mycetomatis*. A mycetoma is a chronic, post-traumatic infection that begins as a small subcutaneous swelling that enlarges and burrows into the deeper tissues, forming multiple discharging sinuses. It is shown below.



Option D: *Phialophora verrucosa* causes chromoblastomycosis. The lesions appear as warty, cutaneous nodules usually confined to the subcutaneous tissue of the lower limbs. Histologically, the lesions contain the fungus, which is seen as round or irregular, dark brown, yeast-like bodies with septae, known as sclerotic bodies. A skin lesion of chromoblastomycosis is seen below, followed by a histopathological image showing sclerotic bodies.



Microbiology NEET 2021

Question 1:

A patient from Uttar Pradesh presented with fever, pallor, and hepatosplenomegaly. Peripheral smear examination showed pancytopenia. Buffy coat examination showed macrophages laden with organisms with a kinetoplast. What is the vector for the likely disease?

- a) Sandfly
- b) Tse-Tse fly
- c) Triatomine bug
- d) Female anopheles mosquito

Question 2:

A patient who was on steroids for the treatment of chronic urticaria presents to the emergency department with acute worsening of respiratory symptoms. He says that he has been having nocturnal cough for last several weeks. A hemogram revealed elevated eosinophils. Bronchoalveolar lavage (BAL) specimen following bronchoscopy showed the presence of certain larvae as shown in the image below. Identify the causative organism.

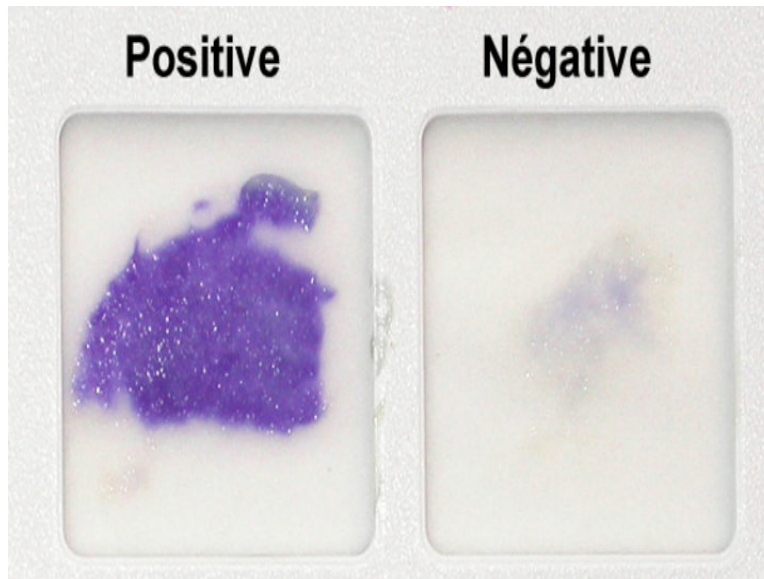


- a) Ancylostoma caninum
- b) Capillaria philippinensis

- c) *Strongyloides stercoralis*
- d) *Enterobius vermicularis*

Question 3:

A patient admitted with 50% burns develops an infection at the burn site. The swab was cultured, the isolate is a strict aerobe and the test that is shown in the image is positive. What is the likely etiology of the burn infection?



- a) *Klebsiella*
- b) *Escherichia coli*
- c) *Pseudomonas aeruginosa*
- d) *Salmonella*

Question 4:

One week after having unprotected sexual intercourse, a 25-year-old male develops urethral discharge as shown in the image below. What is the most likely causative organism for this condition?



- a) *Neisseria gonorrhoeae*
- b) *Ureaplasma urealyticum*
- c) Herpes simplex virus
- d) *Trichomonas vaginalis*

Question 5:

An unimmunized 2-year old child presented with coryza, conjunctivitis, and bluish-white spots in his buccal mucosa near the lower molar teeth. A day later, he developed a maculopapular rash on the face and neck. What is the nature of the causative virus?

- a) Enveloped ss RNA
- b) Naked ss RNA
- c) Naked ds RNA
- d) Enveloped ds RNA

Question 6:

A group of 20 friends kept partying till late at night. Two of them ate pastries from outside. They present with nausea and vomiting in the early morning. What is the most likely causative agent implicated?

- a) Enterotoxin of *Staphylococcus*
- b) Emetic toxin of *Bacillus cereus*
- c) Verocytotoxin from *E. coli*

- d) Shiga toxin from Shigella**

Question 7:

A 3-year old child presents with fever, loose stools, and blood in the stool. What is the enrichment medium used to isolate the organism from the stool sample?

- a) Robertson cooked meat broth**
- b) Alkaline peptone water**
- c) Selenite F broth**
- d) Brain heart infusion broth**

Question 8:

A truck driver came with complaints of a painless indurated genital ulcer after unsafe intercourse. Motility of the most likely causative organism can be detected by which of the following microscopes?

- a) Dark field microscope**
- b) Electron microscope**
- c) Light microscope**
- d) Fluorescent microscope**

Question 9:

Name the parasite whose microfilaria have a sheath with no nuclei at the tail end.

- a) Wuchereria bancrofti**
- b) Brugia malayi**
- c) Brugia timori**
- d) Mansonella perstans**

Question 10:

In a case of tinea capitis, white powdery colonies were isolated from the hair specimen. The hair perforation test was positive on the isolate. Which of the following is the likely etiology?

- a) Epidermophyton floccosum**

- b) *Microsporum audonii*
- c) *Trichophyton mentagrophytes*
- d) *Microsporum gypseum*

Question 11:

A patient was suspected to have acute brucellosis. A serum sample was sent for a standard agglutination test. It came out to be negative initially but after serial dilutions of the serum sample, the test turned out to be positive. Which of the following is responsible for the initial negative test?

- a) Incomplete antibodies
- b) Complement inactivation
- c) Prozone phenomenon
- d) Post-zone phenomenon

Answer Key

Question No.	Correct Option
1	a
2	c
3	c
4	a
5	a
6	a
7	c
8	a
9	a
10	c
11	c

Detailed Explanations

Solution to Question 1:

From the given clinical features, the most likely disease is visceral leishmaniasis or Kala azar caused by *Leishmania donovani*, and the vector is sandfly. A patient from Uttar Pradesh presenting with fever, pallor, and hepatosplenomegaly with a smear examination showing pancytopenia and macrophages containing organisms with kinetoplasts point towards the diagnosis.

Leishmaniasis is transmitted by the bite of an infected female *Phlebotomus* sandfly. In India, it is endemic to Bihar, West Bengal, Jharkhand, and Uttar Pradesh. Patients present with high fever, hepatosplenomegaly, weight loss, and emaciation. The skin becomes dry, rough, and darkly pigmented. There is anemia, leukopenia, and thrombocytopenia.

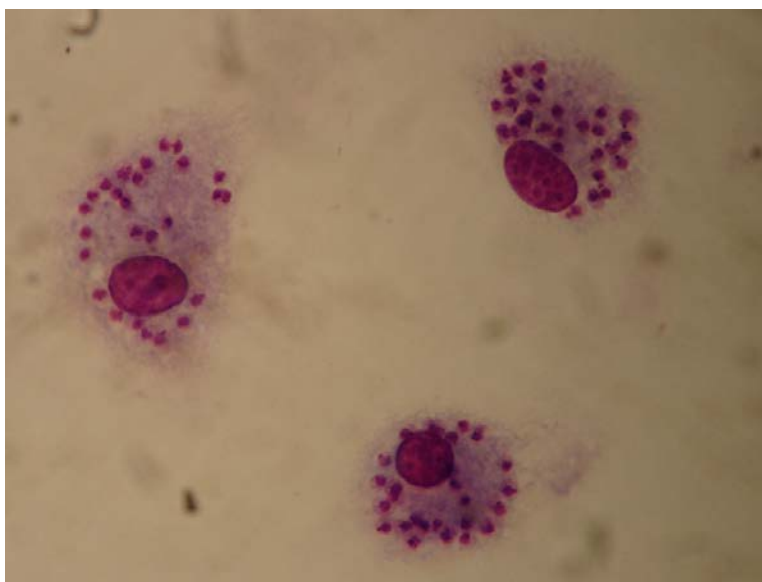
The parasite exists in 2 forms:

- Amastigote form (also called LD bodies)- It is seen within macrophages and contains a kinetoplast.
- Promastigote form- It is the infective form in the sandfly. The kinetoplast is seen at the anterior end.

Kinetoplast are non-nuclear DNA present in addition to nucleus. They consist of blepharoplast and parabasal body. Flagellum originates near the kinetoplast.

Diagnosis of kala-azar is mainly by demonstration of amastigotes within macrophages in tissue aspirates such as blood smears, bone marrow, splenic, or lymph node aspirate. It is the gold standard for diagnosis.

The image below shows amastigotes located intracellularly within macrophages.



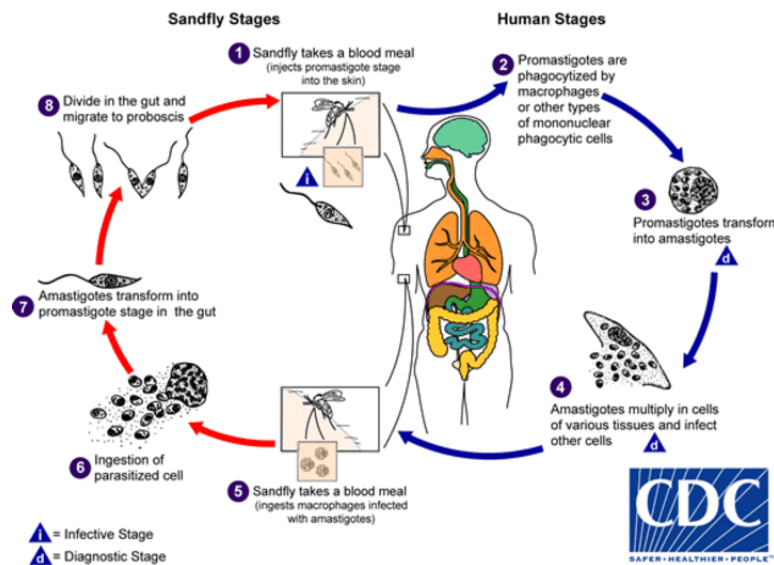
Treatment is with a pentavalent antimonial compound. In Bihar, there is resistance to antimony so amphotericin-B-deoxycholate or miltefosine is preferred.

In human beings, different species of *Leishmania* can cause the following diseases:

- Visceral leishmaniasis, or kala-azar,
- Cutaneous leishmaniasis,
- Mucocutaneous leishmaniasis, and

- Diffuse cutaneous leishmaniasis.

The life cycle of *Leishmania* is shown below.



Other options:

Option B: Tse-Tse fly (*Glossina* species) is involved in the transmission of sleeping sickness or African trypanosomiasis. It is caused by *Trypanosoma brucei*. The patient can have hepatosplenomegaly and lymphadenopathy in the early stages and later on CNS involvement is seen. Blood routine examination would reveal anemia, leucocytosis, and thrombocytopenia.

Option C: Triatomine bug (*Reduviid* bug/ kissing bug) transmits *Trypanosoma cruzi* infection leading to Chaga's disease or American trypanosomiasis. The acute phase of the disease is characterized by a typical subcutaneous lesion at the site of inoculation called chagoma. In the chronic phase, multiorgan involvement occurs leading to cardiac myopathy, megaesophagus, and megacolon.

Option-D: Female anopheles mosquito leads to the transmission of malaria caused by *Plasmodium* species. Malaria can present with fever and hepatosplenomegaly but the morphological forms commonly include the ring forms or the gametocytes.

Solution to Question 2:

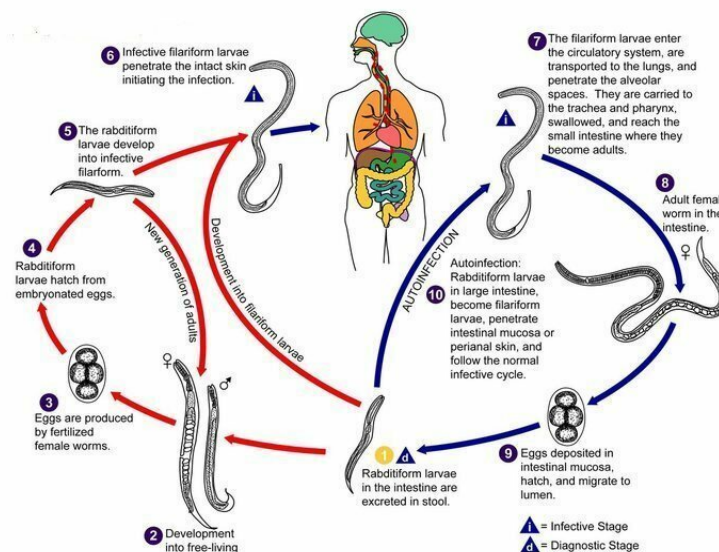
The presence of eosinophilia in an immunosuppressed patient with respiratory symptoms and BAL revealing rhabditiform larvae is suggestive of disseminated strongyloidiasis caused by *Strongyloides stercoralis*.

Strongyloidiasis:

Strongyloides are ovoviviparous. The transmission of infection occurs through penetration of the skin by filariform larvae in the soil. Then, they migrate to the small intestine. The larvae moult twice and become adult female worms in the small intestine. The eggs laid in the mucosa of the intestine hatch immediately releasing the rhabditiform larva (L1 stage). These rhabditiform larvae in the bowel can develop directly into filariform larvae that penetrate the colonic wall or perianal

skin and enter the circulation to repeat the migration. This establishes ongoing internal reinfection.

The life cycle of strongyloides is shown below.



It is generally a benign and asymptomatic condition. In some cases, there may be cutaneous, pulmonary, and intestinal manifestations. Pulmonary manifestations occur when the larva escapes from the pulmonary capillaries into the alveoli leading to bronchopneumonia, chronic bronchitis, and asthmatic symptoms.

The term larva currens (meaning 'racing larvae') has been applied to the rapidly progressing linear or serpiginous urticarial tracks caused by migrating filariform larvae of *Strongyloides stercoralis*. These often follow autoinfection and start perianally. Disseminated Strongyloidiasis is seen in immunodeficient individuals.

Larva currens is shown in the image below.



Visceral larva migrans is caused by the migration of larvae of nonhuman species of nematodes that infect via the oral route.

Etiological agents causing visceral larva migrans:

- Human nematode: *Ascaris lumbricoides*, *Strongyloides stercoralis*.
- Zoophilic nematode: *Toxocara canis*, *Toxocara cati*, *Angiostrongylus cantoniensis*, *Angiostrongylus costaricensis*, *Anisakis*.
- Non human nematode: Filarial species, *Brugia pahangi*, *Brugia pater*

Laboratory diagnosis of strongyloidiasis:

- Eosinophilia and raised IgE levels.
- Microscopic examination of the direct wet mount of the stool demonstrating rhabditiform larvae forms the definitive diagnosis
- Stool culture and serological tests are also done

Treatment: Ivermectin is the preferred agent in the management of strongyloidiasis.

Other options:

Option A: *Ancylostoma caninum* presents with creeping eruption. Cutaneous larva migrans or creeping eruption is caused by nematode larvae that infect by skin penetration.

Option B: *Capillaria philippinensis* can also cause auto-infection. It commonly presents with severe diarrhea.

Option D: *Enterobius vermicularis* or pinworm infection presents as itching (pruritus) and restless sleep (due to nocturnal perianal or perineal pruritus)

Solution to Question 3:

Pseudomonas aeruginosa is the most common cause of infection in the case of a burn. The given image shows a positive oxidase test. *Pseudomonas aeruginosa* is a strict aerobe and shows a positive oxidase reaction.

Pseudomonas aeruginosa is a gram-negative, catalase-positive, oxidase-positive, non-lactose fermenting bacteria that shows swarming motility. They produce a characteristic blue-green pigment called pyocyanin and a greenish-yellow pigment called fluorescein/pyoverdine. Other pigments produced by these organisms include pyorubin (Red), pyomelanin (Brown).

Diseases caused by *Pseudomonas aeruginosa* include-

- Malignant otitis externa in diabetics
- Pneumonia and sepsis in cystic fibrosis
- Wound infection and sepsis in burns
- Hot tub folliculitis/*Pseudomonas aeruginosa* folliculitis/ Jacuzzi syndrome is a type of folliculitis, occurring after hot tub exposure
- Shanghai fever is a febrile illness that manifests as severe enteric disease, sepsis with invasive disease.

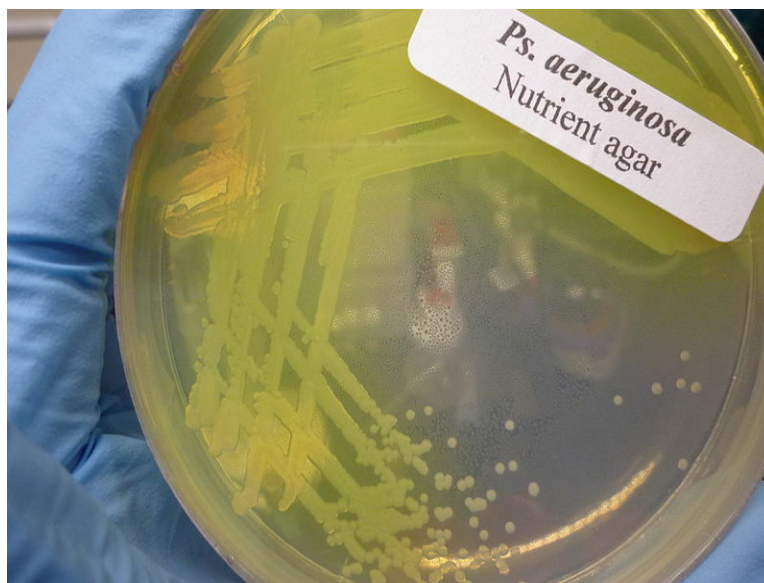
- *Ecthyma gangrenosum*

Pathogenesis - It is an obligate aerobe that exerts its pathogenicity by the production of diphtheria-like exotoxin A and a slime layer. Breach in primary body defenses as in burns predisposes to *Pseudomonas* infection.

Clinical features - It is characterized by blue-green pus. It is notorious for being resistant to antimicrobials and antiseptics and hence is a frequent contaminant in hospital samples. Repeated isolation and associated clinical features should prompt the diagnosis of *Pseudomonas*.

Diagnosis - It grows well on most media and pigment production makes its identification easy. Species identification is made possible by biochemical tests - arginine hydrolysis and oxidase reaction.

Pseudomonas aeruginosa growth in a nutrient agar showing green coloured colonies are shown in the image below.



Oxidase test can be used to distinguish organisms by detecting the presence or absence of a respiratory enzyme, cytochrome oxidase. Cytochrome oxidase catalyzes the oxidation of reduced cytochrome by oxygen. Enterobacteriaceae (*Klebsiella*, *E. coli*, *Salmonella*) are oxidase negative.

Oxidase positive organisms:

- *Pseudomonas*
- *Vibrio*
- *Helicobacter*
- *Campylobacter*
- *Pasteurella*
- *Legionella*
- *Neisseria*
- *Brucella*
- *Moraxella*

- Burkholderia

Solution to Question 4:

The given image shows purulent urethral discharge that developed one week after unprotected sexual intercourse is suggestive of acute urethritis. The most likely causative organism for acute urethritis in males is *Neisseria gonorrhoeae*.

Purulent urethral discharge can be seen in gonorrhea caused by *Neisseria gonorrhoea* or lymphogranuloma venereum caused by *Chlamydia trachomatis*.

Gonorrhea:

Gonorrhea is a disease acquired by sexual contact. The first step in infection is the infection of gonococci to the urethra or other mucosal surfaces. Pili is involved in this adhesion. The cocci penetrate the intracellular spaces and reach subepithelial tissue by the third day of infection. The incubation period is 2-8 days. The major symptoms include urethral discharge and dysuria, usually without urinary frequency or urgency. The specific symptoms seen in males and females are as follows:

In males :

- The disease starts as acute urethritis. This infection may extend along the urethra to involve the prostate, seminal vesicles, and epididymis.
- Chronic urethritis may lead to stricture formation.
- The infection may spread to the periurethral tissues causing abscesses and multiple discharging sinuses (watercan perineum).

In females :

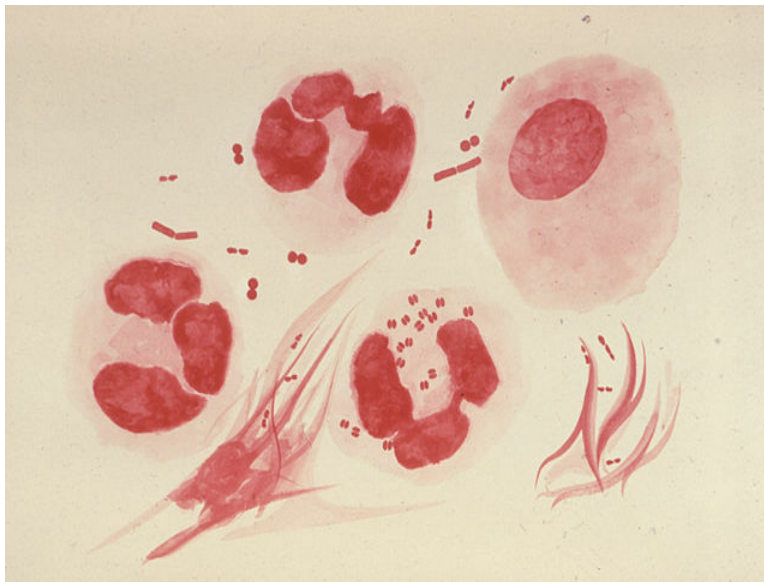
- The initial infection involves the cervix uteri and urethra which may extend to Bartholin glands, endometrium, and fallopian tubes.
- The vaginal mucosa is not usually affected in adults because the stratified squamous epithelium is resistant to infection by the cocci, and also because of the acidic pH of vaginal secretions.
- Pelvic inflammatory disease and salpingitis may lead to sterility.
- Rarely peritonitis may develop with perihepatic inflammation (Fitz Hugh Curtis syndrome)

Lab diagnosis:

The discharge is collected and subjected to Gram's stain which reveals gram-negative intracellular diplococci within phagocytes. Stuart's transport medium is used. Cultures can be obtained from chocolate agar or Mueller-Hinton agar. Specifically, Thayer-Martin agar is used to isolate *Neisseria gonorrhoeae*.

Treatment: Ceftriaxone (250mg, IM) single-dose + azithromycin 1 g PO, single-dose, or doxycycline 100 mg PO BD for 7 days

The image below shows the Gram-stain of a urethral exudate showing typical intracellular gram-negative diplococci and pleomorphic extracellular gram-negative organisms



Non-gonococcal urethritis is caused by:

- Chlamydia trachomatis
- Ureaplasma urealyticum (Option-B)
- Mycoplasma genitalium
- Trichomonas vaginalis (Option-D)
- Herpes simplex virus (rare)
- Adenovirus
- Haemophilus vaginalis
- Mycoplasma genitalium

Other options:

Option B: Ureaplasma urealyticum is the second most common cause of non-gonococcal urethritis next to chlamydia. That is why the initial etiological agent under suspicion always remains gonorrhea.

Option C: Herpes Simplex will be associated with vesicular lesions.

Option D: Trichomonas vaginalis generally presents with a frothy discharge.

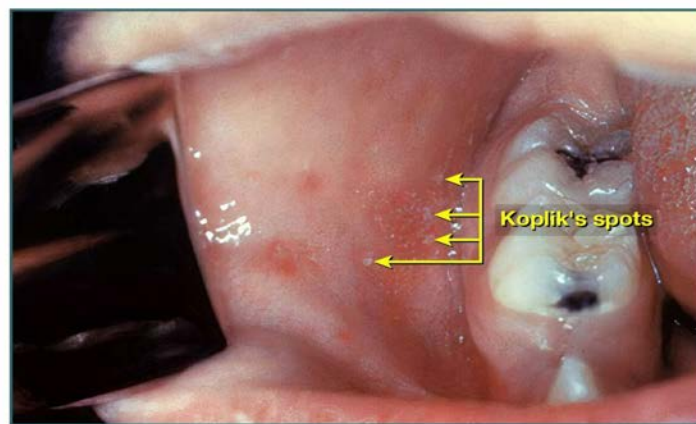
Solution to Question 5:

Fever, coryza, conjunctivitis, maculopapular rash and Koplik spots are suggestive of measles. Measles virus is an enveloped, single-stranded RNA (ssRNA) virus.

Koplik spots are small, bluish-white ulcerations on the inner aspect of the cheek at the level of premolars. These are pathognomic for measles. The measles virus, a paramyxovirus, has an incubation period of 7-18 days. After this, the prodrome phase begins with mild fever, conjunctivitis, upper respiratory tract infection, and pathognomic Koplik spots. This is followed

by the eruptive stage, characterised by the appearance of maculopapular rash on the 4th day of fever. The rash begins on the face and spreads to the torso and extremities. Lymphadenopathy may be present.

These are multinucleated giant cells formed by the fusion of infected cells. The images below show the characteristic rashes and the Koplik spots seen in measles.



©MARROW

The most common complication of measles is acute otitis media. Sinusitis and mastoiditis are also seen. The most common cause of death in measles is pneumonia, which may be giant cell pneumonia. It is caused by the virus directly or can be due to superadded bacterial infection. Black measles or hemorrhagic measles is a fatal complication characterized by hemorrhagic skin eruptions. It is rarely seen nowadays.

Subacute sclerosing panencephalitis is a rare but fatal complication characterized by cognitive decline, spasticity of limbs, and seizures. It is insidious in onset and clinical manifestations usually begin 7-10 years after primary measles infection. It is caused due to persistent infection with a genetically altered variant of the measles virus within the central nervous system. Other

complications include croup, tracheitis, bronchiolitis, diarrhea and vomiting, febrile seizures, and encephalitis.

The diagnosis of measles infection is mainly clinical but may be confirmed by serological demonstration of IgM antibodies. IgM antibodies appear 1-2 days after the rash and are detectable for 1 month. The management is mainly supportive. There is no role for antivirals. Low serum retinol levels in children with measles are associated with higher morbidity and mortality and hence vitamin A therapy is indicated for all patients with measles.

Live attenuated vaccine for measles is safe and effective with no evidence of person-to-person transfer of vaccine strains. It is a freeze-dried product requiring reconstitution. 0.5ml is administered subcutaneously in the right upper arm in 2 doses- at the 9th month and between the 16th - 24th month.

Solution to Question 6:

The most likely causative agent in the given scenario is enterotoxin of *Staphylococcus aureus*. It causes food poisoning within 30 minutes to 6 hours after consumption of the contaminated food.

The common sources of infection are the foods that are not cooked after handling. These include sliced meats, puddings, pastries, and sandwiches. Patients present with nausea, vomiting, stomach cramps, and diarrhea.

Food poisoning is caused by enterotoxin produced by *S. aureus*. It is a preformed toxin that is relatively heat stable while most of the other exotoxins are heat-labile. It stimulates the vagus nerve and vomiting center of the brain and also stimulates intestinal peristaltic activity. Eight antigenic types of enterotoxin are currently known of which Type A toxin is responsible for most cases.

Other options:

Option B: Food poisoning after consumption of fried rice or chinese street food is strongly suggestive of *Bacillus cereus* (Serotypes 1, 3, 5). Two mechanisms of action have been described for the enterotoxin of *B. cereus*, one involving stimulation of the cyclic adenosine monophosphate (cAMP) system and the other independent of it.

Food poisoning after eating food contaminated with *Bacillus cereus* is of two types, namely diarrheal type and emetic type, caused by an enterotoxin and an emetic toxin respectively.

Bacillus cereus	Diarrheal type	Emetic type
Incubation period	8-16 hours	1-5 hours
Toxin	Secreted in intestine	Preformed
Heat	Heat labile	Heat stable
Food items	Meat	Rice
Clinical feature	Diarrhea, abdominal cramps	Vomiting, abdominal cramps

Option C: Verocytotoxin from *E.coli* is also known as Shiga-like toxin and it causes bloody diarrhea. It is produced by enterohemorrhagic *E.Coli*. It can cause illnesses ranging from mild diarrhea to fatal hemorrhagic colitis and hemorrhagic uremic syndrome.

Option D: Shiga toxin acts by inactivating 60s ribosomal subunit thereby inhibiting protein synthesis. *Shigella dysenteriae* causes acute onset of symptoms (fever, abdominal cramps) with dysentery and tenesmus. It also gives a positive Sereny test and is non-lactose fermenting.

Solution to Question 7:

The given clinical vignette of dysentery (bloody stools) and abdominal pain in a child is suggestive of bacillary dysentery most likely caused due to *Shigella*. Selenite F broth is used as an enrichment media for dysentery-causing bacteria like *Shigella*/ *Salmonella*.

Enrichment media:

It is used to suppress commensal bacteria while allowing the pathogen to remain viable and grow. Common examples include:

- Tetrathionate broth – It inhibits coliforms and allows typhoid and paratyphoid bacilli to grow.
- Selenite F broth – used for dysentery causing bacteria.
- Alkaline peptone water (Option-B) - used for vibrio cholera causing watery diarrhea.

Other options:

Robertson's cooked meat broth (Option-A): It is used for the culture of anaerobic organisms.

Brain-heart infusion broth (Option-D): It is an enriched media. It is used for the cultivation of fastidious organisms.

Dysentery/Shigellosis is a type of food poisoning caused by infection with the *Shigella* species. It is spread through faeco-oral transmission. They are catalase-negative, non-lactose-fermenting, gram-negative rods, and are non-motile. It acts by inactivating the 60s ribosomal subunit thereby inhibiting protein synthesis. It causes acute onset of symptoms (fever, abdominal cramps) with dysentery and tenesmus. It also gives a positive Sereny test.

Solution to Question 8:

The clinical vignette of a painless, indurated genital ulcer in a truck driver is most likely suggestive of syphilitic chancre / hard chancre caused by *Treponema pallidum*. The motility of the organism can be demonstrated by dark ground/ field microscopy.

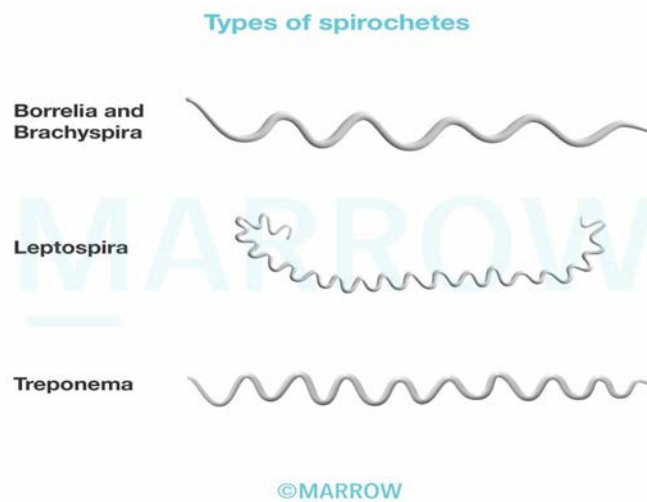
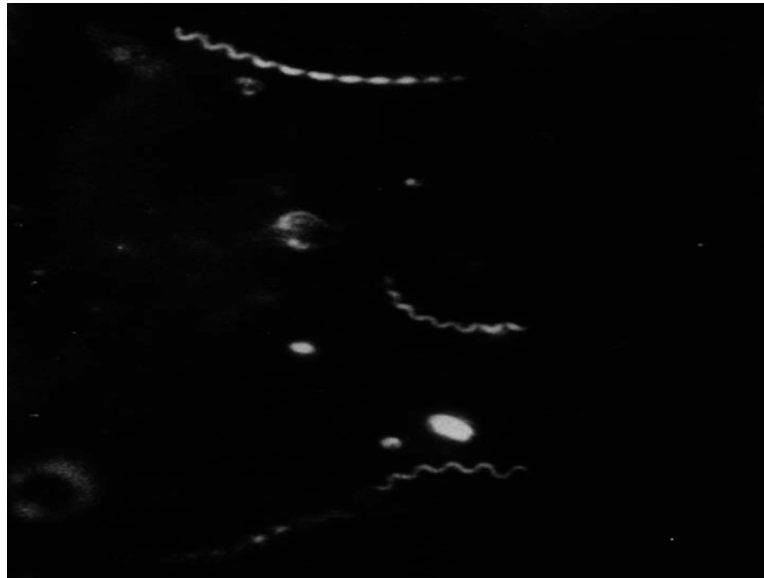
Dark ground microscopy can be used to diagnose syphilis when an active chancre or condyloma lata is present to visualize *Treponema pallidum*. Dark-field microscopy images have a dark background with a refractile organism. The dark ground microscope has an optical system that enhances the contrast of unstained bodies. It is generally less sensitive.

Various types of Spirochetes and their appearances under a dark ground microscope:

- Treponemes generally are short, slender with regular and sharp spirals.

- *Leptospira* have a large number of closely wound spirals and characteristic hooked ends.
- *Borrelia* is large with irregular and wide open coils.

The image given below shows *Treponema pallidum* demonstrated by dark ground microscopy and the various appearances of other spirochetes respectively.



Other options:

Electron microscopy (Option-B) is not commonly used for diagnostics in laboratories. It generally helps in appreciating viral structures.

Light microscopes (Option-C) are commonly used in the labs to visualize the stained smears of various bacteria, fungi in 100x magnification. They are also used during the examination of stool specimens for identifying ova, eggs, or larvae.

Fluorescent microscope (Option-D) uses fluorescent dyes like auramine-rhodamine to appreciate Mycobacterial species and malarial parasites.

Syphilis:

Syphilis is caused by *Treponema pallidum*, a spirochete bacillus.

It shows corkscrew motility which can be demonstrated by darkfield microscopy. Smear specimens are stained using Fontana stain and pathological tissues are stained using Levaditi stain.

Serologic tests for syphilis recommended by the Centers for Disease Control and Prevention (CDC):

- Screening or diagnosis (RPR or VDRL),
- Quantitative measurement of antibody to assess clinical syphilis activity or to monitor response to therapy (RPR or VDRL)
- Confirmation of a syphilis diagnosis in a patient with a reactive RPR or VDRL test (FTA-ABS, TPHA, EIA/CIA)

Treatment of syphilis is with 2.4 million units of benzathine penicillin IM.

Note:

VDRL- Venereal disease Research Laboratory test

RPR- Rapid Plasma Reagin test

TA-ABS- Fluorescent *Treponema* Antibody Absorption test

TPI-*Treponema pallidum* immobilization test

TPPA-*Treponema pallidum* particle agglutination assay

TPHA-*Treponema pallidum* haemagglutination test

EIA-Enzyme-linked ImmunoAssay

Solution to Question 9:

A sheathed microfilaria with a pointed tail free of nuclei is suggestive of *Wuchereria bancrofti*.

Wuchereria bancrofti causes lymphatic filariasis and most cases are seen in India.

Clinical features include massive eosinophilia, hepatosplenomegaly, pulmonary manifestations like dry nocturnal cough, asthmatic wheezing, and dyspnea. It can also cause arthritis, glomerulonephritis, thrombophlebitis, etc. The classical features seen in lymphatic filariasis are usually absent.

Elephantiasis refers to the painful swelling of the legs and genital organs seen in late-stage disease.

Image showing elephantiasis-

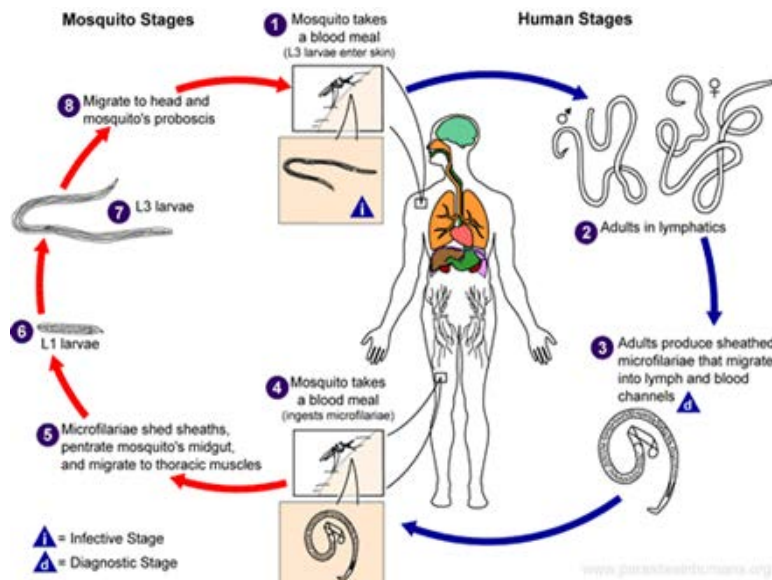


Investigations:

Demonstration of the microfilaria in peripheral blood, chylous urine, hydrocele fluid, etc is done. The specimen of choice is peripheral blood. Night blood samples between 10 pm and 4 am are preferred. This is because the microfilaria shows nocturnal periodicity.

DEC provocation test: In this test, a small dose of DEC is given 20-50 minutes before sample collection so that the microfilaria appears in the blood even during the daytime.

The image below shows the life cycle of *Wuchereria bancrofti*.





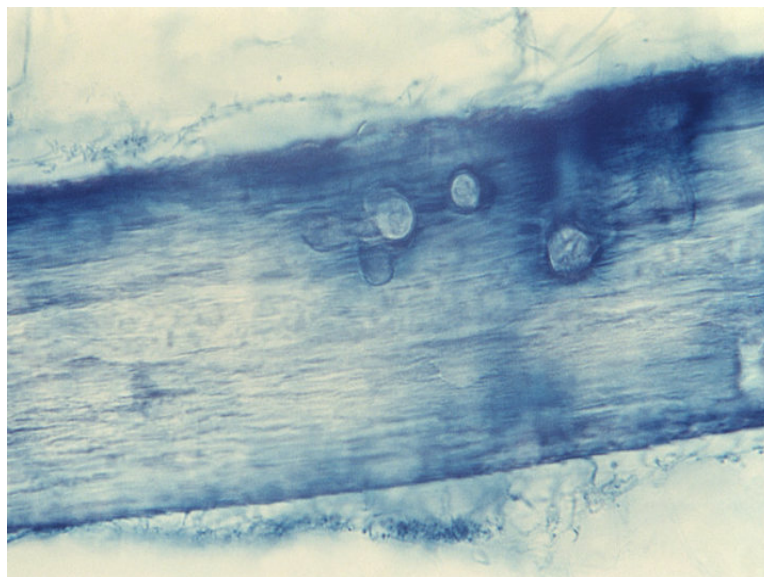
Solution to Question 10:

A positive hair perforation test is caused by *Trichophyton mentagrophytes*.

Hair perforation test:

It is done to differentiate *Trichophyton rubrum* from *T. mentagrophytes*. Few strands of human hair are placed in a petri dish with 20 ml of distilled water and autoclaved. A few drops of sterile 10% yeast extract and a few fragments of the test fungus are added to the hair strands. The entire dish is then incubated at 25°C for 2-3 weeks. The hair is removed and examined microscopically in lactophenol cotton blue-stained wet preparation. *T. mentagrophytes* cause surface erosion of the shaft resulting in the wedge-shaped appearance of the shaft.

The image below shows surface erosion of shaft caused by *Trichophyton mentagrophytes*.



Tinea capitis refers to ringworm of the scalp where there is an invasion of the hair shafts by dermatophyte fungi (Trichophyton, Microsporum, and Epidermophyton). It presents as hair loss with inflammation, with or without itching. Diagnosis is based on clinical features, Wood's lamp, and laboratory investigations such as culture with microscopic examination of the colonies.

- Trichophyton: Colonies may be powdery, velvety, or waxy with characteristic pigment based on the species
- T. rubrum: Red
- T. mentagrophytes: White-tan
- T. tonsurans: Cream-yellow
- Microsporum: Colonies are white, cotton-like, velvety, or powdery
- M. audouinii: Brownish
- M. gypseum: Buff colored
- Epidermophyton: Colonies are greenish-yellow and powdery

Treatment is with antifungals like terbinafine, itraconazole, and griseofulvin.

Solution to Question 11:

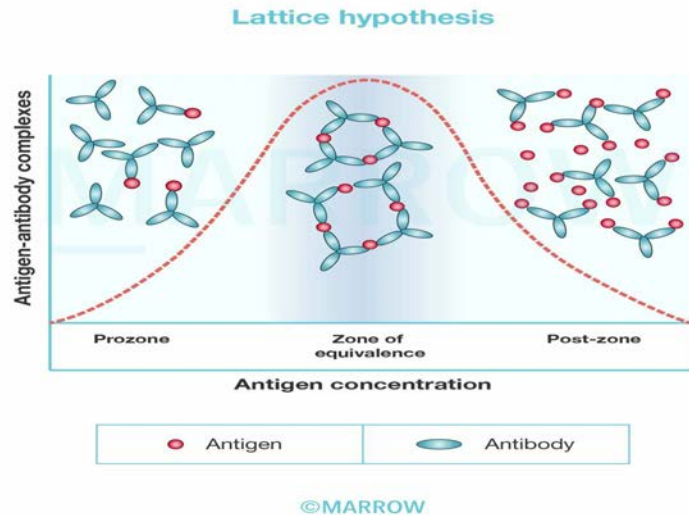
The initial negative test is explained by the prozone phenomenon. The standard agglutination test is a tube agglutination test that mainly identifies IgM antibodies. It is used in establishing a diagnosis of acute brucellosis.

Prozone phenomenon refers to the failure of a visible precipitation reaction due to inhibition of lattice formation by the presence of excessive antibodies. This gives rise to a false-negative test. This excess in antibodies can be overcome by diluting the serum sample, leading to an increase in the antigen-antibody interactions, eventually giving rise to a positive result.

Lattice hypothesis:

The lattice hypothesis explains precipitation reactions. The amount of precipitate formed is greatly influenced by the relative proportions of antigens and antibodies.

- Prozone phenomenon: Caused by excessive antibodies in the test system. (weak/absent precipitation- false-negative test)
- Zone of equivalence: Antigens and antibodies are in optimum proportions and maximum precipitation occurs.
- Post-zone phenomenon: (Option-D) Due to the presence of excessive antigen in the test system. (weaker/absent precipitation)



Standard agglutination test identifies mainly IgM antibodies, hence it is useful in acute brucellosis. Following 7-10 days of infection, both IgM and IgG antibodies appear. The IgM antibodies then gradually fall and only IgG antibodies persist in chronic brucellosis. The complement fixation test (Option-B) is more useful in chronic cases as it detects the IgG antibody also.

Brucellosis/ Undulant fever:

Brucellosis is a zoonotic disease caused by gram-negative, capnophilic coccobacilli. The most important species that are pathologic to humans are *B. melitensis*, *B. suis*, and *B. abortus*. It is transmitted to humans from contact with or consumption of contaminated products.

These bacteria proliferate within the reticuloendothelial system, which accounts for its resistance to simple antimicrobial therapy.

Patients with acute infection may present with undulant fever, arthralgia, muscular pain, sweating, loss of weight, and appetite. CNS symptoms such as fatigue, irritability, and depression might also occur. Chronic infection tends to be episodic with non-specific symptoms.

Laboratory diagnosis of brucellosis includes:

- Polymerase chain reaction
- Blood culture with Castaneda method
- Serological methods include standard agglutination test (SAT), complement fixation test, Coomb's antiglobulin test, and enzyme-linked immunosorbent assay (ELISA).
- Tests for animals and animal products: Milk ring tests and Rose Bengal card tests.
- Milk ring test- A sample of milk is mixed with stained brucella antigen and incubated. If the sample contains antibodies, agglutination occurs seen as a blue ring.

Treatment includes antimicrobial therapy with IM streptomycin and oral doxycycline. Prevention of brucellosis can be achieved by pasteurization of milk and milk products.

The images below show the Castaneda's medium and the milk ring test respectively.

Castaneda's bottle



©MARROW

Milk ring test in Brucellosis



Positive

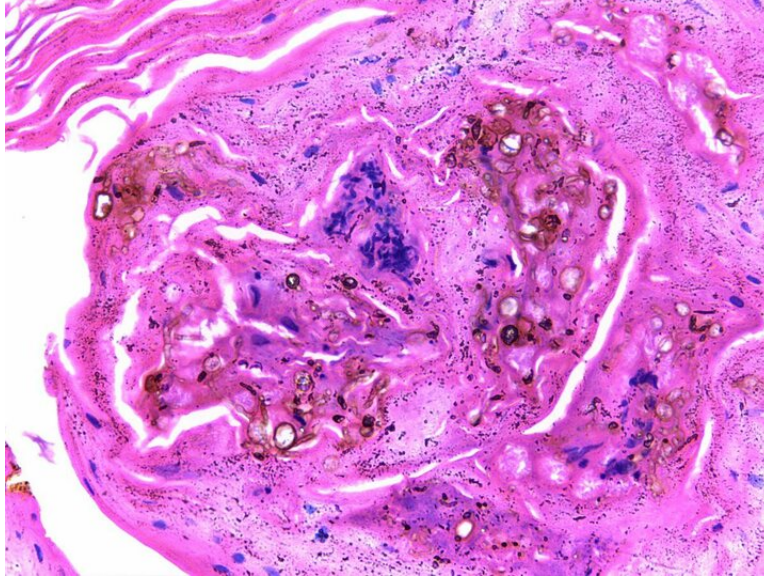
Negative

©MARROW

Microbiology INI-CET 2022

Question 1:

Identify the pathognomonic structure in the slide given below.



- a) Sclerotic body
- b) Budding yeast
- c) Negri body
- d) Asteroid body

Question 2:

Antigen binding to MHC class I activates which of the following cells?

- a) Cytotoxic T cells
- b) B cells
- c) NK cells
- d) Helper cells

Question 3:

Which of the following agents can be used as a disinfectant but not a sanitizer?

- a) Hydrogen peroxide
- b) Penicillin
- c) Methyl paraben
- d) Bleaching powder

Question 4:

Vaccination against which of the following is recommended in the elderly?

- a) 1,2,3,4
- b) 1,3
- c) 1,4
- d) 2,3

Question 5:

Which of the following is wrong regarding primary and secondary immune response ?

- a) 1
- b) 2
- c) 3
- d) 4

Question 6:

Which of the following is true about anti-CMV IgG antibodies?

- a) Denotes latent CMV infection
- b) Denotes chronic CMV infection with immunity to other serotypes
- c) Indicates acute CMV infection
- d) IgG avidity assay helps in differentiating past and primary infection

Question 7:

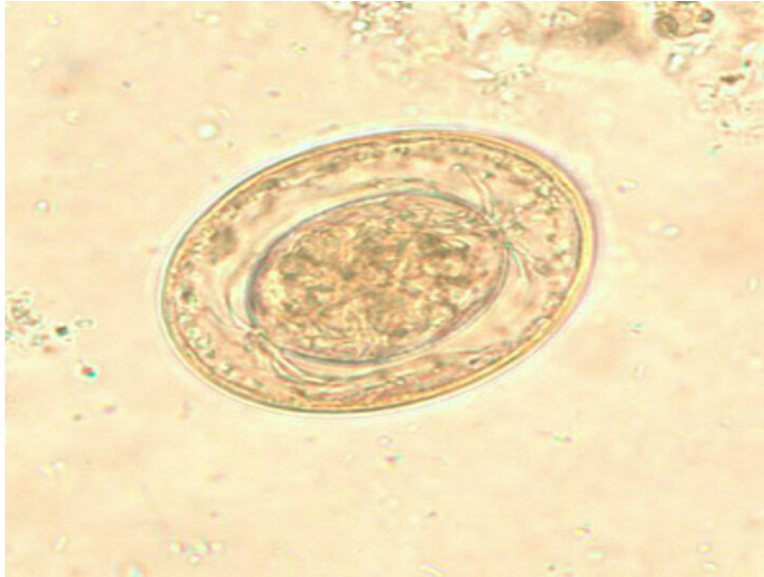
Which of the following is an example of Mollicutes?

- a) Ureaplasma urealyticum

- b) *Orientia tsutsugamushi*
- c) *Chlamydophila pneumoniae*
- d) *Chlamydia trachomatis*

Question 8:

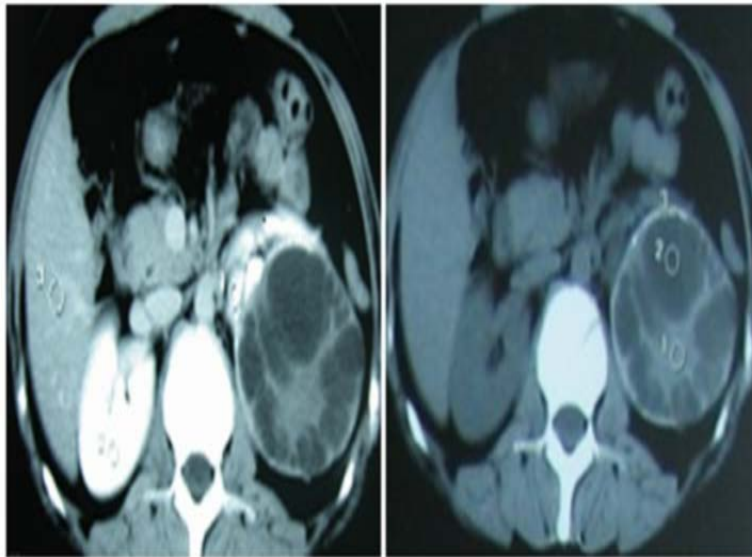
Which of the following statements regarding the given image is correct?



- a) Albendazole is the drug of choice
- b) Majority of infections are asymptomatic in humans
- c) Infection is acquired by ingestion of aquatic plants
- d) Non-segmented leaf like parasite

Question 9:

A patient presents to you with left-sided flank pain. A CT scan was done which is shown below. Which of the following statements is incorrect regarding the condition?



- a) Albendazole can be started preoperatively
- b) Cyst contents can produce anaphylactic reaction
- c) A segmented dog tapeworm is the causative agent
- d) FNAC or biopsy to be done before nephrectomy

Question 10:

Which of the following has been reported to have endemic transmission in India?

- a) Yellow fever
- b) Crimean-Congo hemorrhagic fever
- c) Ebola
- d) Hendra virus

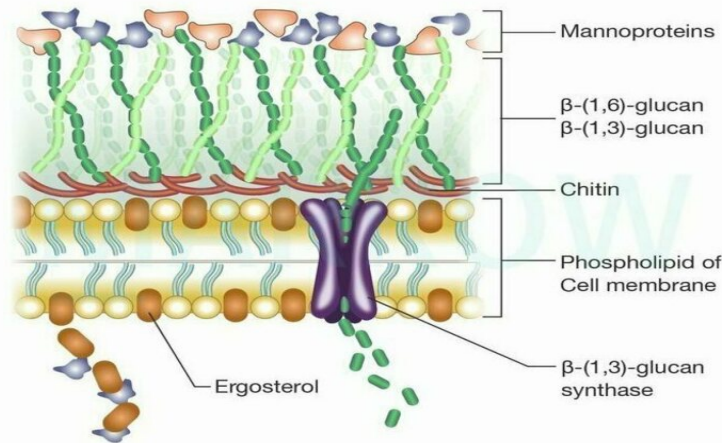
Question 11:

Which of the following are acid-fast staining organisms?

- a) 1, 2, 3, 4, 5
- b) 1, 2, 4, 5
- c) 1, 2, 3
- d) 3, 4, 5

Question 12:

The cell wall depicted below is most likely to belong to which of the following organisms?

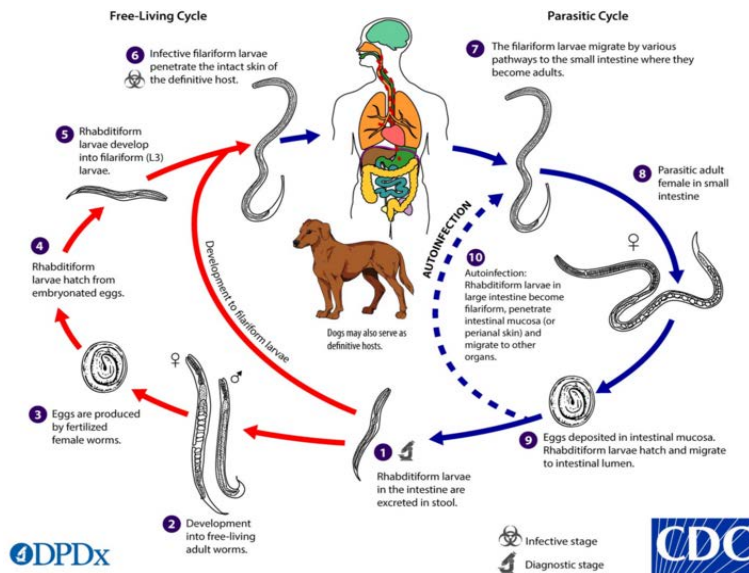


©Marrow

- a) *Aspergillus fumigatus*
- b) *Mycobacterium tuberculosis*
- c) *Mycoplasma pneumoniae*
- d) *Rhinosporidium seeberi*

Question 13:

The organism with following life cycle is:



- a) *Strongyloides stercoralis*
- b) *Ancylostoma duodenale*
- c) *Ascaris lumbricoides*
- d) *Enterobius vermicularis*

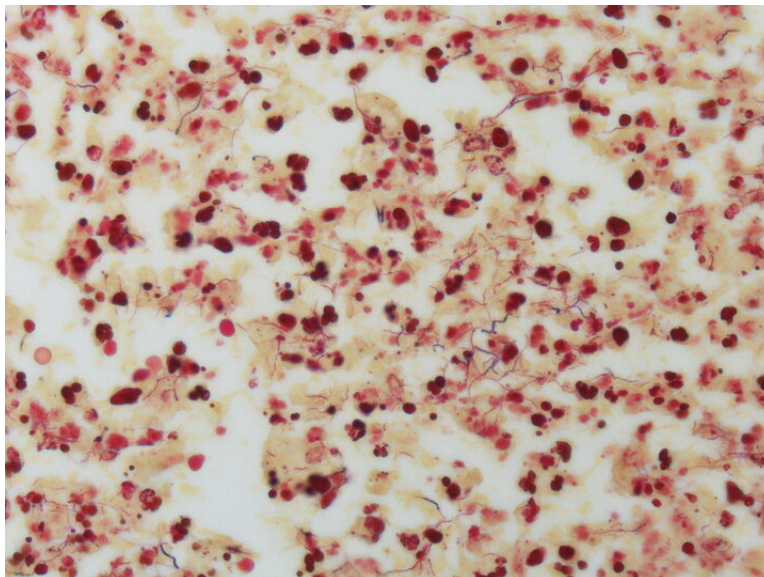
Question 14:

What is the ratio of T cells to B cells in a healthy adult?

- a) 1:1
- b) 3:1
- c) 1:2
- d) 1:3

Question 15:

A microbiologist stained a tissue sample and found thin, filamentous bacteria, which resembled *Nocardia* sp. What other staining technique should be done to confirm the presence of *Nocardia*?



- a) Ziehl-Neelsen staining
- b) Silver impregnation
- c) Albert stain

- d) Modified acid-fast stain

Question 16:

What is the microscopy technique to produce high contrast image of transparent, live bacteria?

- a) Phase contrast microscopy
- b) Dark field microscopy
- c) Electron microscopy
- d) Polarized microscopy

Question 17:

Match the following.

- a) A-4, B-1, C-3, D-2
- b) A-4, B-1, C-2, D-3
- c) A-4, B-2, C-3, D-1
- d) A-2, B-1, C-3, D-4

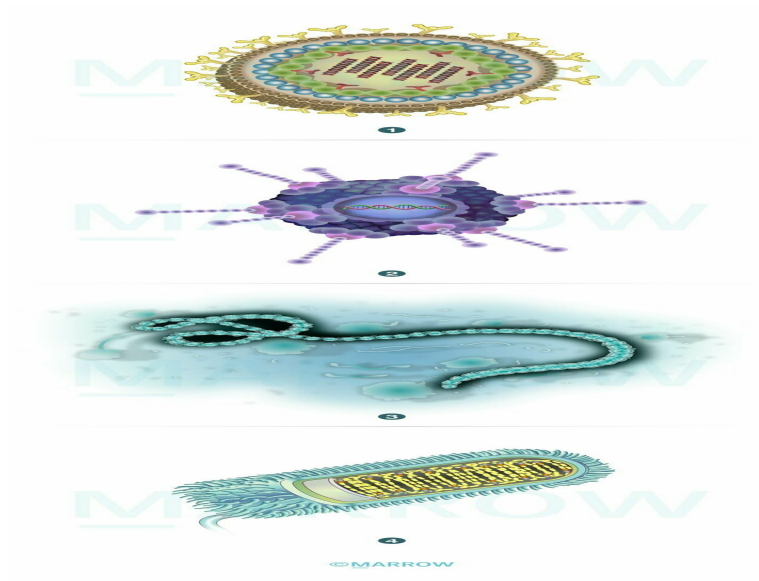
Question 18:

Vesicular rash is seen in _____.

- a) Hand, foot, and mouth disease
- b) Dengue
- c) Zika fever
- d) Roseola infantum

Question 19:

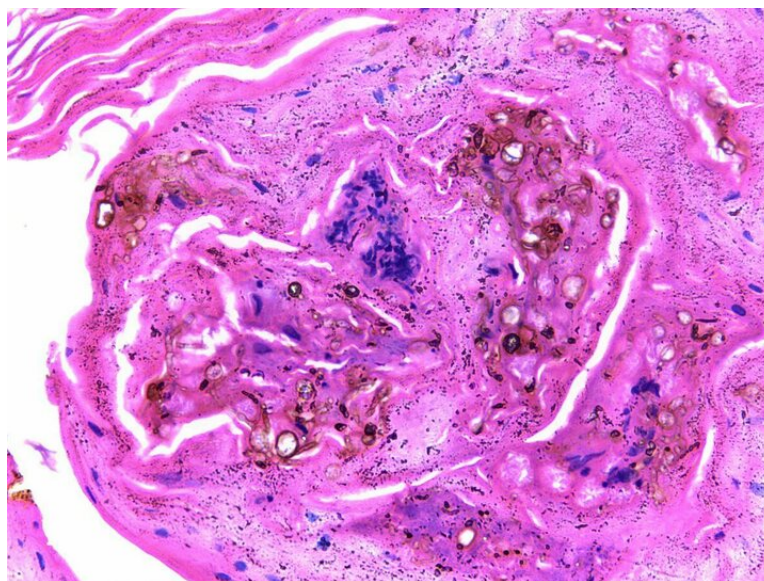
Which of the following viruses show genetic change similar to influenza virus?



- a) 1
- b) 2
- c) 3
- d) 4

Question 20:

A forest worker developed skin lesions over the forearm, which initially started as macules but then became nodules. Histology of the nodules shows the following findings. Which of the following is true regarding this condition?

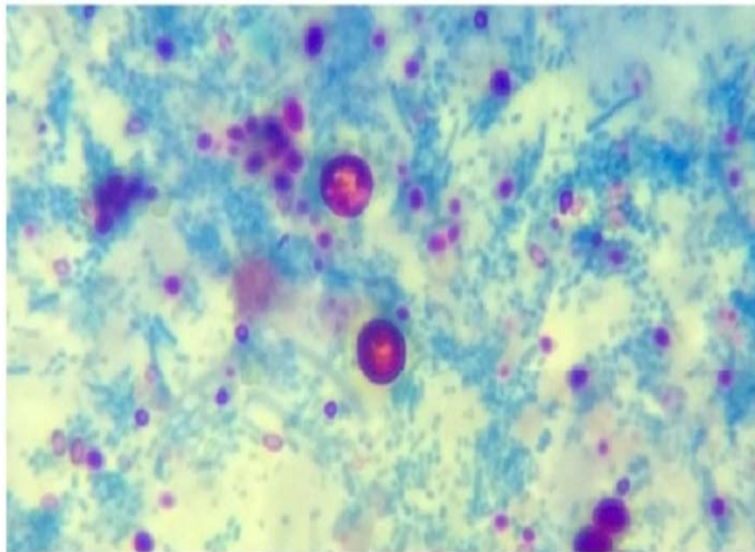


- a) The organism does not invade underlying fascia, tendon, muscle, or joints.

- b) Angioinvasion is common, especially in people with hemolytic anemia.
- c) It is a dimorphic organism, and yeast forms are seen in clinical sample.
- d) These bodies are formed by engulfment of the dead fungi by the macrophages.

Question 21:

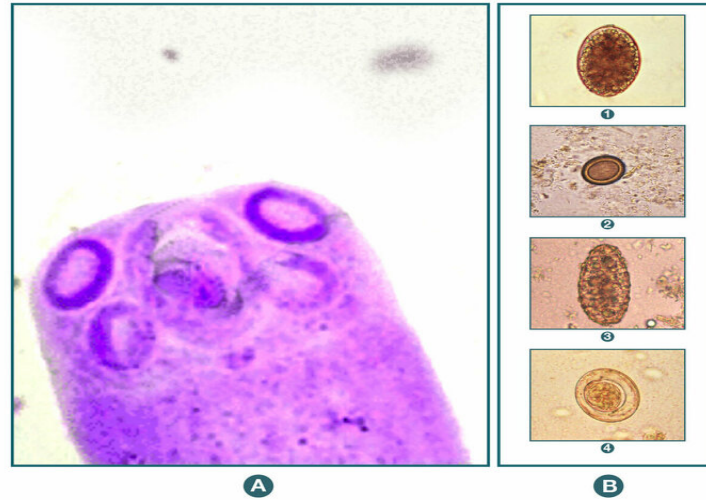
An HIV patient came to the clinic with a history of diarrhea. Stool microscopy showed the oocysts that were 4-8 μm in size. The image is shown below. Identify the organism.



- a) Isospora
- b) Cyclospora
- c) Cryptosporidium
- d) Microspora

Question 22:

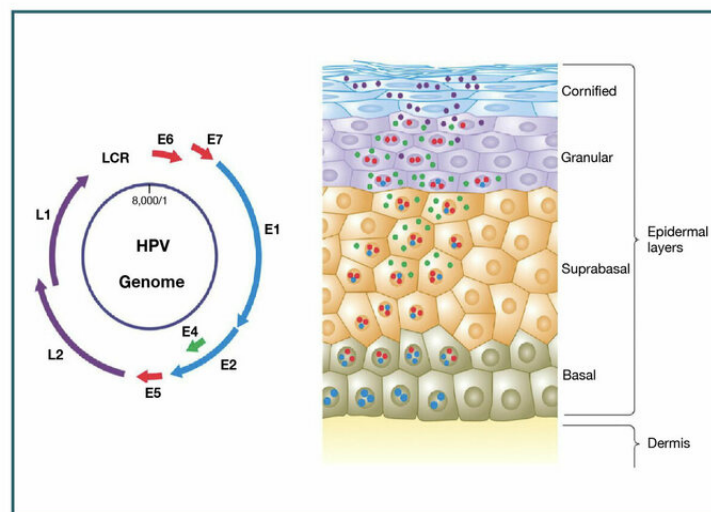
A patient presents to the hospital with complaints of abdominal pain. A stool examination was done, and it showed the presence of the organism shown in image A. The body has segments broader than the length. Which of the eggs shown in column B corresponds to this organism?



- a) 1
- b) 2
- c) 3
- d) 4

Question 23:

The image below shows the life cycle of a virus. Which of the proteins of the virus act as oncogenes?

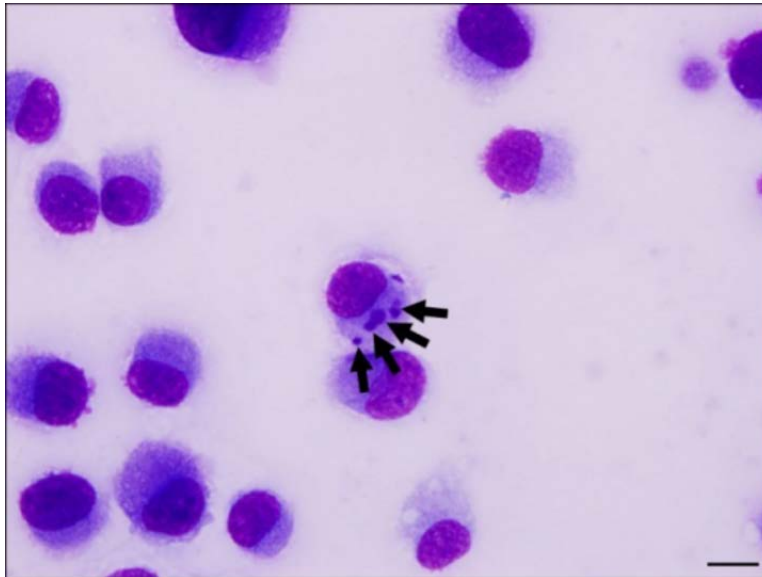


- a) E1, E2, E5

- b) E6, E7
- c) L1, L2
- d) E1, E2

Question 24:

Image of Rickettsial morula in a monocyte is shown below. Identify the organism.



- a) *Rickettsia rickettsii*
- b) *Rickettsia conorii*
- c) *Ehrlichia chaffeensis*
- d) *Anaplasma*

Question 25:

A young man presents with skin lesions as shown in the image below. All of the following organisms can spread through dermal and subcutaneous lymphatics, except _____.



- a) *Sporothrix schenckii*
- b) *Staphylococcus aureus*
- c) *Nocardia asteroides*
- d) *Mycobacterium marinum*

Question 26:

The following are parvovirus-related manifestations. Identify the incorrect pair.

- a) 4
- b) 3
- c) 1
- d) 2

Question 27:

Which virus can be identified by a PCR method and is endemic to India?

- a) Hendra virus
- b) West Nile virus
- c) Ebola virus
- d) Yellow fever

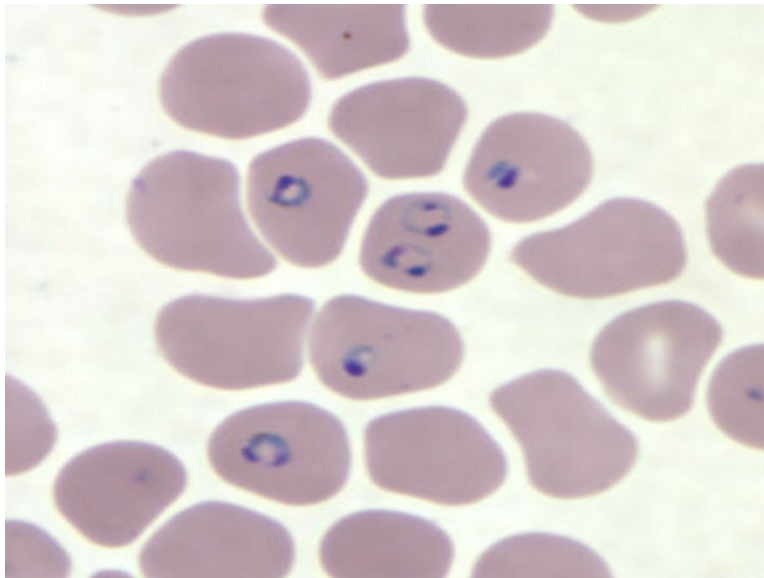
Question 28:

SARS-CoV-2 can be detected by all except _____.

- a) ELISA to detect spike proteins
- b) Real-time PCR to detect viral mRNA
- c) Rapid antigen test to detect spike proteins
- d) Southern blotting

Question 29:

A patient comes with a history of high fever for 5 days and altered sensorium. On further probing, it was found that he had two seizures one day ago. Peripheral smear shows the following findings. How will you manage this patient?



- a) Doxycycline 100 mg intravenously for 7 days
- b) Clindamycin 600 mg intravenously 8 hourly for 7 days
- c) Artesunate 2.4 mg/kg intravenously at 12 and 24 hours, followed by daily once
- d) Chloroquine intramuscularly 8 hourly for 3 days

Question 30:

Delayed hypersensitivity is due to _____.

- a) Interferon gamma

- b) Interleukin-2
- c) Interleukin-12
- d) Interleukin-4

Question 31:

Which disease causes the lesion shown in the image below?



- a) Rocky mountain spotted fever
- b) Malaria
- c) COVID-19
- d) Scrub typhus

Question 32:

Which of the following method of disinfection and sterilization can also kill spores?

- a) 1, 3 and 4
- b) 1, 2 and 4
- c) 2, 3 and 4
- d) 1, 2, 3 and 4

Question 33:

The inclusion body rickettsial morula is found in neutrophils during infection caused by:

- a) *Rickettsia rickettsii*
- b) *Rickettsia conorii*
- c) *Anaplasma phagocytophilum*
- d) *Ehrlichia chaffeensis*

Question 34:

A petechial rash is associated with all the following, except _____.

- a) Zika virus
- b) Roseola infantum
- c) Kyasanur forest disease
- d) West Nile fever

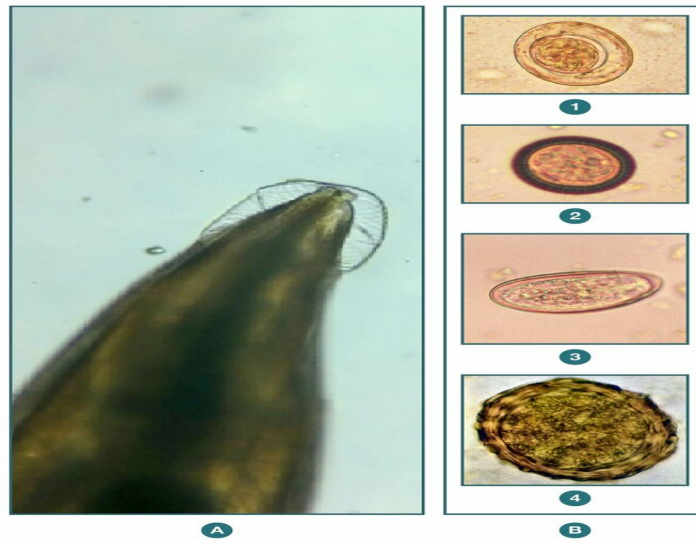
Question 35:

Mode of transmission of abdominal TB includes all except

- a) Direct spread from mesenteric lymph nodes
- b) Hematogenous spread
- c) Feco-oral route
- d) Direct contact with infected animal

Question 36:

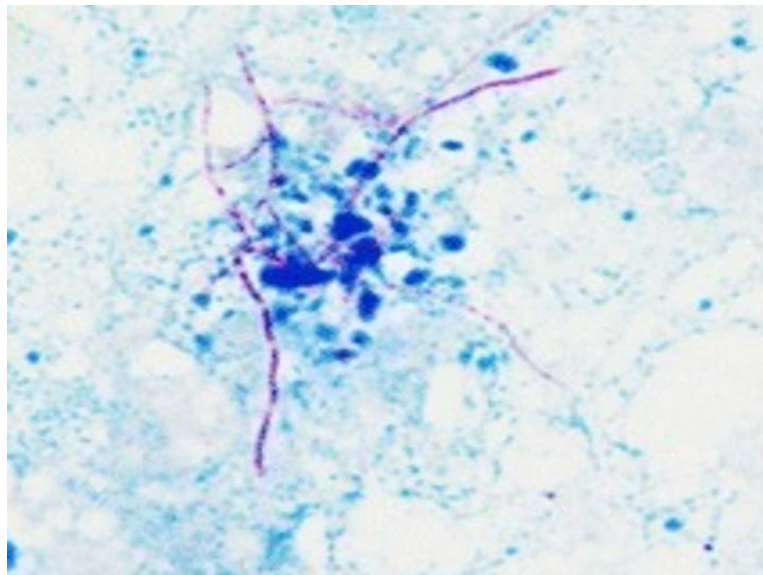
Choose the egg belonging to the following parasite.



- a) 1
- b) 2
- c) 3
- d) 4

Question 37:

Which of the following is a gram-positive organism that shows the following appearance on Ziehl-Neelsen staining?



- a) Nocardia

- b) *Mycobacterium tuberculosis*
- c) *Mycobacterium leprae*
- d) *Actinomyces*

Question 38:

The following are parvovirus-related disease-population pairs. Identify the incorrect pair.

- a) 1
- b) 2
- c) 3
- d) 4

Question 39:

A patient with a history of fever, jaundice, hepatosplenomegaly, and laboratory test result of leptospira IgM reactivity, now presented with breathlessness, fever, and jaundice that worsened after treatment with penicillin. What is the possible diagnosis?

- a) Resistant leptospirosis
- b) Penicillin allergy
- c) Co-infection with dengue
- d) Jarisch-Herxheimer reaction

Question 40:

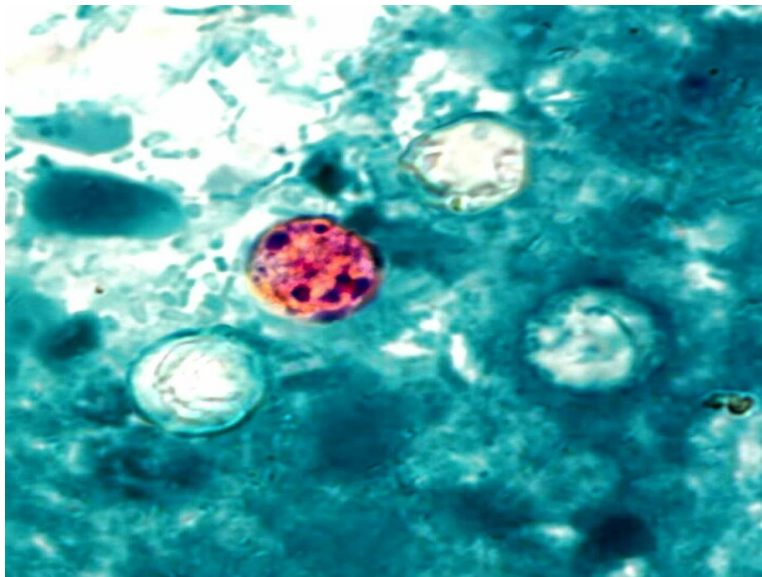
On inoculation in the skin, all of the following organisms can lead to dermal, subcutaneous, and lymphatic involvement except _____



- a) *Sporothrix schenckii*
- b) *Corynebacterium minutissimum*
- c) *Nocardia brasiliensis*
- d) *Mycobacterium marinum*

Question 41:

On Ziehl-Neelsen (ZN) staining oocysts of size 8-10 μm cysts are visible. Identify the organism.

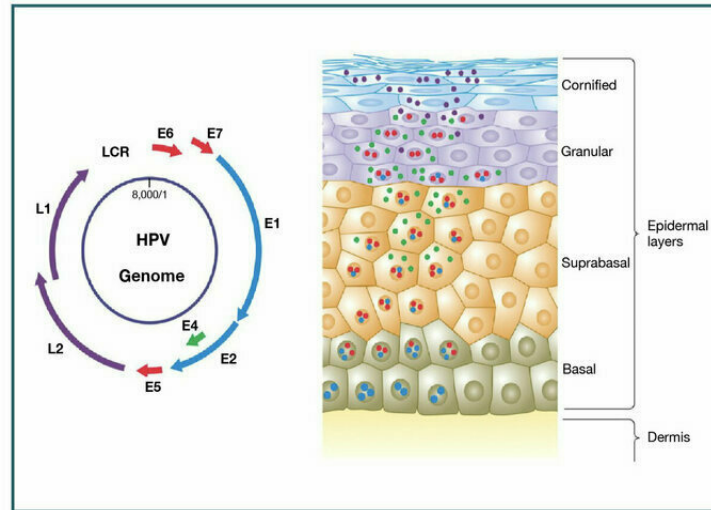


- a) *Isospora*

- b) Cryptosporidium
- c) Cyclospora
- d) Microspora

Question 42:

Vaccine can be prepared against which of the following proteins of the virus?



- a) E6, E7
- b) L1
- c) E1, E2
- d) E3

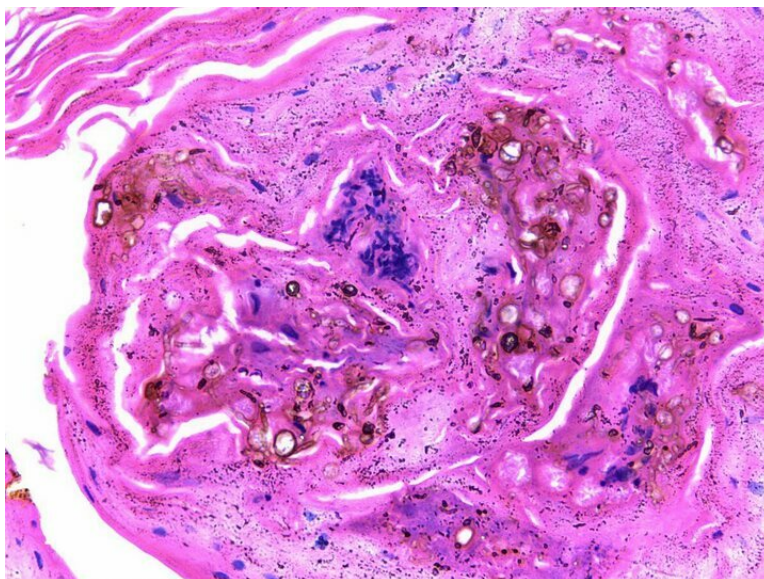
Question 43:

The Epstein-Barr virus interacts with B cells via which of the following?

- a) CD21
- b) CD20
- c) CD19
- d) CD23

Question 44:

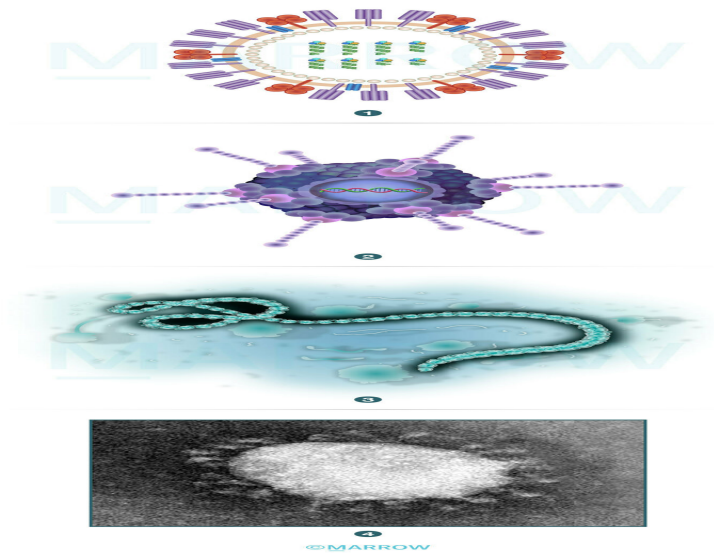
A forest worker developed skin lesions over the forearm, which initially started as macules but then became nodules. Histology of the nodules shows the following findings. Which of the following is true regarding this condition?



- a) It is a dematiaceous fungus
- b) Infection can spread and involve underlying subcutaneous tissue, tendon, muscle or joints
- c) Angioinvasion is common especially in people with hemolytic anemia
- d) These bodies are formed by engulfment of the dead fungi by the macrophages

Question 45:

The viruses of the filoviridae family like Ebola and Marburg resemble which of the following morphologies?



- a) 3
- b) 2
- c) 1
- d) 4

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	c
5	d
6	d
7	a
8	b
9	d
10	b
11	b
12	a
13	a
14	b

15	d
16	a
17	b
18	a
19	a
20	a
21	c
22	d
23	b
24	c
25	b
26	d
27	b
28	d
29	c
30	a
31	d
32	b
33	c
34	b
35	c
36	c
37	a
38	d
39	d
40	b
41	c
42	b
43	a
44	a
45	a

Detailed Explanations

Solution to Question 1:

The given image shows the presence of brown round sclerotic bodies which is characteristically seen in chromoblastomycosis.

Chromoblastomycosis is a chronic fungal infection of the skin and subcutaneous tissue. It is caused by pigmented fungi (most commonly by *Phialophora verrucosa*) and is commonly seen in tropical areas among agricultural workers and woodcutters who walk barefoot.

Patients present with warty cutaneous nodules (cauliflower-like) on the feet and lower legs. Diagnosis is by demonstration of sclerotic bodies in tissue sections. Sclerotic bodies appear as round or irregular, dark brown bodies with septae. It is usually seen in clusters of 4-8 cells due to septation in different planes and delayed separation.

Chromoblastomycosis is managed by:

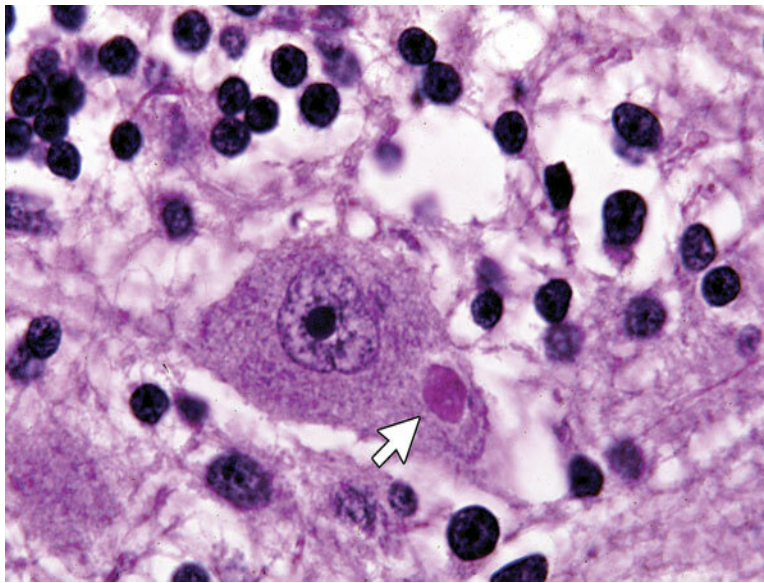
- First-line treatment involves itraconazole (100-200mg daily) or terbinafine (250mg daily)
- Second-line treatment includes the use of cryotherapy or the local application of heat by heat-retaining gels for a period of 2-3 months.
- Surgery is only indicated in very small lesions and should always be combined with chemotherapy
- Flucytosine can also be used in combination with amphotericin B, but should not be used alone due to the development of resistance.

Other options:

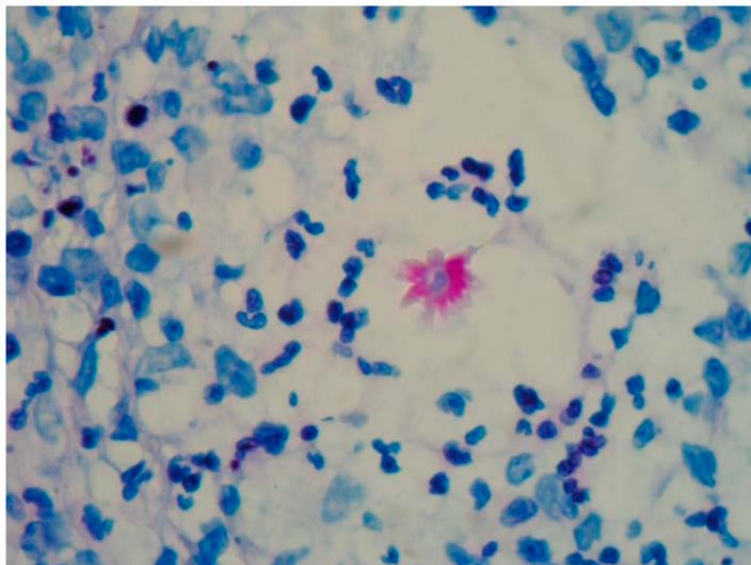
Option B - The below image shows multiple round or oval-shaped yeast cells, pseudohyphae, and buds attached to the parent cell. They are seen in budding yeast.



Option C - Negri bodies are pathognomonic of rabies. It is an acidophilic, intracytoplasmic inclusion body with characteristic basophilic inner granules as shown below.



Option D - Asteroid bodies is seen in sporotrichosis, caused by *Sporothrix schenckii*. It can be seen as an oval, basophilic structure with eosinophilic radiation. These are depositions of antigen-antibody complexes.



Solution to Question 2:

Antigen binding to major histocompatibility complex (MHC) class I activates cytotoxic T cells.

The major histocompatibility complex (MHC), is a genetic locus that encodes a group of antigens. The MHC molecules also bind peptide antigens and present them to the T cells. The human MHC is called the human leukocyte antigen (HLA) complex.

The MHC class I proteins present their antigens to CD8 T cells (cytotoxic T cells). The antigens bound to MHC class II proteins activate the CD4 T cells(T helper cells). There is no cross

recognition between the T cells and MHC classes. This is known as MHC restriction. For example, a CD4 T cell would recognize an antigen presented by a macrophage only if it's bound to MHC class II molecules.

Other options:

Option B - B cells interact with an antigen based on the immunoglobulin surface receptors on the B cell. The receptor along with the antigen is embodied and degraded and the peptides produced are returned to the cell surface. It is then bound to the MHC class II molecules present on the cell surface. This is then recognized by CD4 T cells. Ultimately, the B cells are stimulated to proliferate and differentiate into plasma cells which secrete antibodies.

Option C - Natural killer (NK) cells are granular lymphocytes that confer innate immunity by protecting against viruses and intracellular pathogens.

Option D - T helper cells are responsible for the production of cytokines which activate the humoral and cell-mediated branches of immunity. These cells will recognize only those antigens that are presented along with MHC class II molecules on APC such as macrophages and B lymphocytes.

Solution to Question 3:

Hydrogen peroxide is a disinfectant but cannot be used as a sanitizer.

Sanitization is a process by which levels of pathogenic organisms on inanimate surfaces are brought down to reduce the chances of cross-infection.

Disinfection is a process by which all pathogenic organisms are destroyed.

Sterilization is a process in which all microorganisms in both vegetative and spore state are eliminated from a surface or medium.

Hydrogen peroxide can be used for disinfection or sterilization in the liquid form for cystoscopes, endoscopes, tubings, and metal instruments. The gaseous form can be used for fogging, which has largely replaced fumigation. It has a shorter cycle time and is non-toxic. The mechanism of bactericidal action is the oxidation of the cell wall of the microbe. It is also used in plasma sterilization.

Other options:

Option B - Penicillin, an antibiotic interferes with the synthesis of the bacterial cell wall. It is not used as a disinfectant or sanitizer.

Option C - Methyl parabens are antimicrobials that are commonly used in food and personal care products.

Option D - Household bleaching powder (sodium hypochlorite) is a widely used disinfectant in hospitals. They are oxidizing agents that interfere with cellular protein activity. Diluted solutions are also used for sanitizing surfaces.

Solution to Question 4:

Vaccination against varicella-zoster and pneumococcal meningitis (Options 1 and 4) is recommended in the elderly.

Vaccinations against the following diseases are recommended for the elderly:

- Influenza vaccine
- Pneumococcal vaccine
- Herpes zoster
- Tetanus, diphtheria, and pertussis
- Hepatitis B

Solution to Question 5:

Statement 4 is wrong regarding primary and secondary immune responses. Primary immune response has lower antibody affinity and vice versa.

An immune response is a specific reactivity that occurred in a host as a result of an antigenic stimulus. It is of two types: Humoral or antibody-mediated response and cellular or cell-mediated response.

The response to the antigenic stimulus can be primary or secondary. The features of the primary and secondary immune responses are as follows

Features	Primary immune response	Secondary immune response
1. Lag period	4-10 days	1-3 days
2. Peak response time	7-10 days	3-5 days
3. Antibody response	Depends on the antigen	More potent than the primary immune response
4. Isotype generated	IgM	IgG
5. Antigens	Both thymus-dependent and independent	Depends only on the thymus
6. Antibody affinity	Lower	Higher

Solution to Question 6:

IgG avidity assay helps in differentiating past and primary infection with cytomegalovirus (CMV).

Avidity level measured by avidity index is used to describe the percentage of IgG bound to the antigen after being treated with denaturing agents. These assays are based on the fact that low avidity IgG antibodies are present during the first few months after the infection and avidity increases over time indicating maturation of the immune response. So the presence of high avidity IgG denotes a past infection rather than a primary one. Low avidity IgG signals a primary

infection within the last 3-4 months.

IgM detection can be used as an indicator of acute infection, but it lacks specificity for primary infection. This is because it persists for years in some individuals and is also produced during reactivation.

Cytomegalovirus is a type of herpes virus, which causes severe disease in immunocompromised patients. In most other cases, it is inapparent and marked by prolonged latency with intermittent reactivation.

Solution to Question 7:

Ureaplasma urealyticum belong to the class Mollicutes which are also known as cell-wall-free bacteria. This class contains around 200 species of mycoplasma.

Mycoplasma are the smallest microorganisms that can thrive in a cell-free medium. They are pleomorphic and lack cell walls. There are 4 species that are of importance in humans.

- *Ureaplasma urealyticum*
- *Mycoplasma pneumoniae*
- *Mycoplasma hominis*
- *Mycoplasma genitalium*

It can be grown in fluid or solid media. Colonies typically have a fried egg appearance. They mainly cause pneumonia and genital infections. *Ureaplasma urealyticum* is unique in its ability to hydrolyse urea. They cause non-gonococcal urethritis. It can be resistant to commonly used antibiotics such as tetracycline and doxycycline.

Other options:

Option B - *Orientia tsutsugamushi* belongs to the family rickettsiaceae. It causes scrub typhus, which is spread by mites.

Option C - *Chlamydophila pneumoniae* belongs to the genus Chlamydia. It is a common cause of respiratory diseases such as sinusitis, bronchitis, and pneumonia.

Option D - *Chlamydia trachomatis* also belongs to the genus Chlamydia. It causes trachoma, inclusion conjunctivitis, and genital chlamydiasis.

Solution to Question 8:

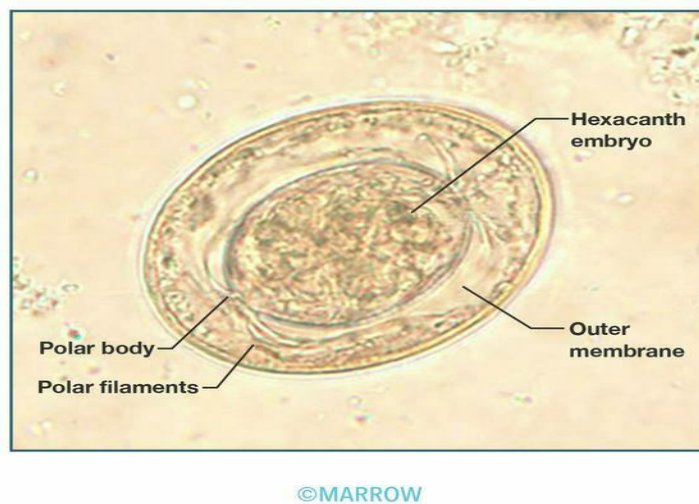
The given image shows a spherical or ovoid, non-bile stained egg which is diagnostic of *Hymenolepis nana* infection. It has a thin colorless outer membrane and a thin embryophore covering the hexacanth embryo. The majority of infections are asymptomatic and are more commonly seen in children.

H. Nana or dwarf tapeworm is the smallest and most common cestode that infects man. There is no intermediate host. The infection is acquired by ingestion of food and water contaminated with eggs.

Patients with heavy infections may present with nausea, abdominal pain, anorexia, diarrhea, and irritability. An allergic response to the parasite can cause pruritus as well.

A single dose of Praziquantel given at the rate of 25mg/kg is the drug of choice. Nitazoxanide (500 mg twice daily for 3 days) may be given as an alternative.

Image below shows egg of *H. Nana*.



Other options:

Option C - Following are the parasites with aquatic plants as the source of infection :

- *Fasciola hepatica*
- *Fasciolopsis buski*
- *Watsonius watsoni*.
- *Gastrodiscoides hominis*

Option D - The adult worm of *Fasciola hepatica* is a flat leaf-shaped liver fluke.

Solution to Question 9:

FNAC or biopsy to be done before nephrectomy is the incorrect statement regarding cystic echinococcosis. The given CT image shows left-sided renal hydatid cyst.

Echinococcosis (hydatidosis) is caused by *Echinococcus granulosus* (larval stage). It is a segmented tapeworm. Dogs are the definitive hosts and pass the eggs in their feces. Intermediate hosts include humans, sheep, and cattle. Patients present with abdominal pain, palpable mass, jaundice, cough, and dyspnea. Rupture of a hydatid cyst can result in fever, pruritus, or anaphylaxis.

Daughter cysts within the larger cyst seen on CT imaging is diagnostic. A biopsy is not recommended because of the risk of fluid leakage which can result in anaphylaxis or dissemination of the infection.

Percutaneous aspiration, infusion of scolicedal agents, and reaspiration (PAIR) is the primary treatment. It is contraindicated in superficial loculated cysts and in cysts that communicate with the biliary tree.

Isolated involvement of the kidneys is rare. Patients can present with nonspecific symptoms, renal colic, and hydatiduria. Due to the lack of an effective systemic scolicedal agent, surgical treatment is necessary when kidneys are involved. Simple excision of the cyst is preferred. If there is extensive involvement, partial or total nephrectomy is warranted. It is recommended to start albendazole one week to one month before the surgery to decrease the risk of spillage.

Solution to Question 10:

Crimean-Congo hemorrhagic fever caused by nairovirus has an endemic transmission in India. An outbreak was recognized in Gujarat in 2011. In India, antibodies against the virus have been detected in human and animal sera. It is transmitted by hyalomma ticks.

Other options:

Option A - Yellow fever is mostly confined to the south and central America, and certain parts of Africa. It is caused by the flavivirus. It is an arthropod-borne hemorrhagic fever. In the urban cycle, *Aedes aegypti* transmits the disease.

Option C - Ebola virus causes a hemorrhagic fever first reported in Sudan and Zaire. A recent outbreak was reported in West Africa.

Option D - Hendra virus was responsible for the death of many horses in Australia.

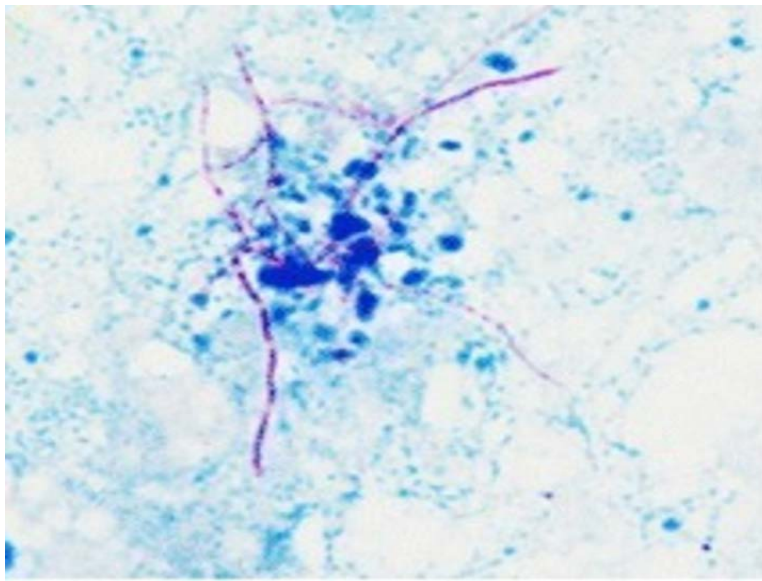
Solution to Question 11:

Nocardia, *Mycobacterium leprae*, *Cryptosporidium parvum*, and *Cytoisospora belli* (Options 1, 2, 4, and 5) are acid-fast staining organisms.

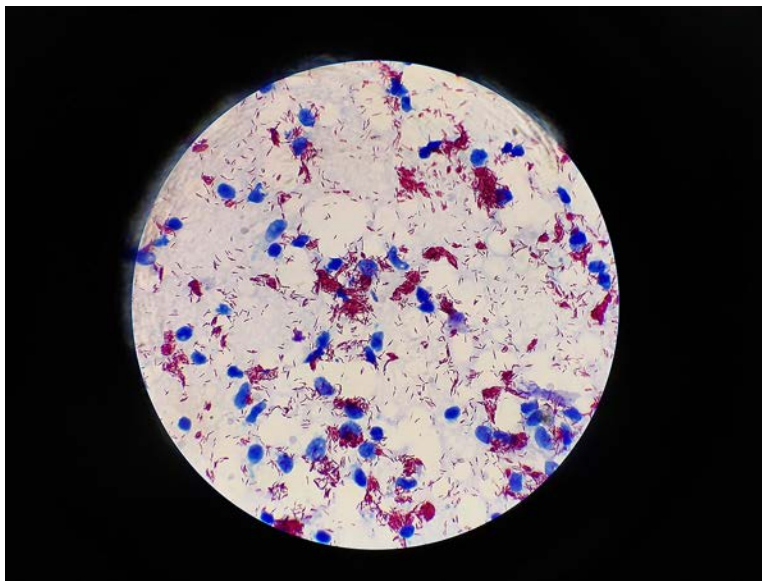
Acid-fast staining is based on the principle that tubercle bacilli resist decolorization with acids after staining with aniline dyes. Acid fastness is attributed to the variety of fatty acids, alcohols, and lipids in the tubercle bacilli. Acid-fast organisms usually contain mycolic acid, which is acid-fast in the free state. The steps to carry out acid-fast staining as modified by Ziehl and Neelson are:

- The slide is stained with carbolfuchsin along with the application of heat.
- It is then decolorized with 20% sulphuric acid. Later methylene blue is used for counterstaining.
- The acid-fast bacteria retain the fuchsin color, while other organisms take up the counterstain.

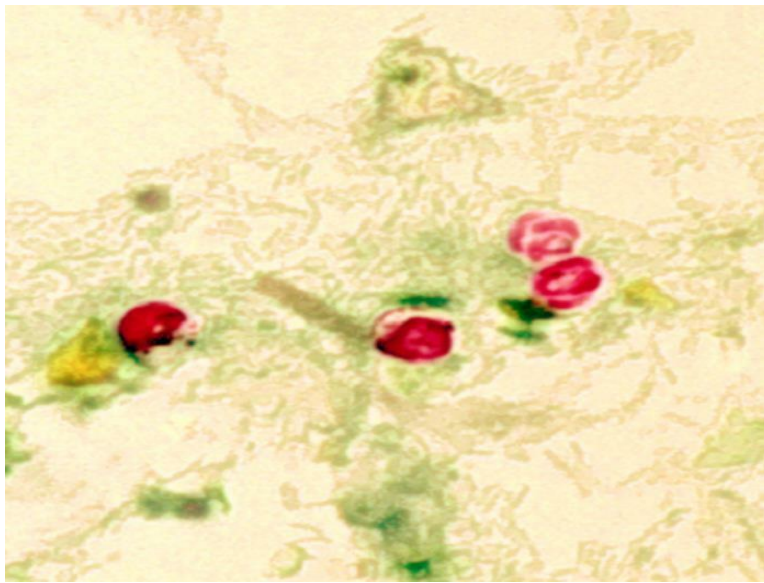
The genus *Nocardia* are aerobic organisms and weakly acid-fast positive rods. They stain with carbolfuchsin and retain it when 1-4% sulfuric acid is used.



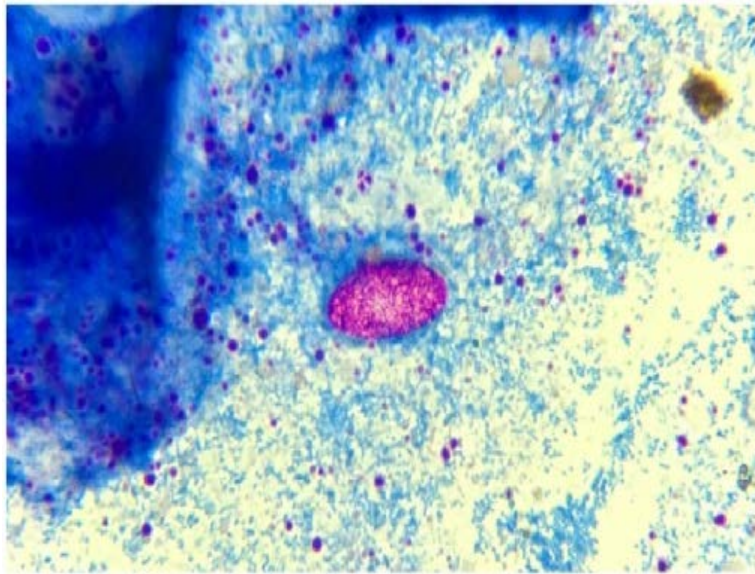
Mycobacterium leprae, the causative agent of leprosy is a gram-positive bacillus. It is acid-fast when 5% sulphuric acid is used instead of 20%.



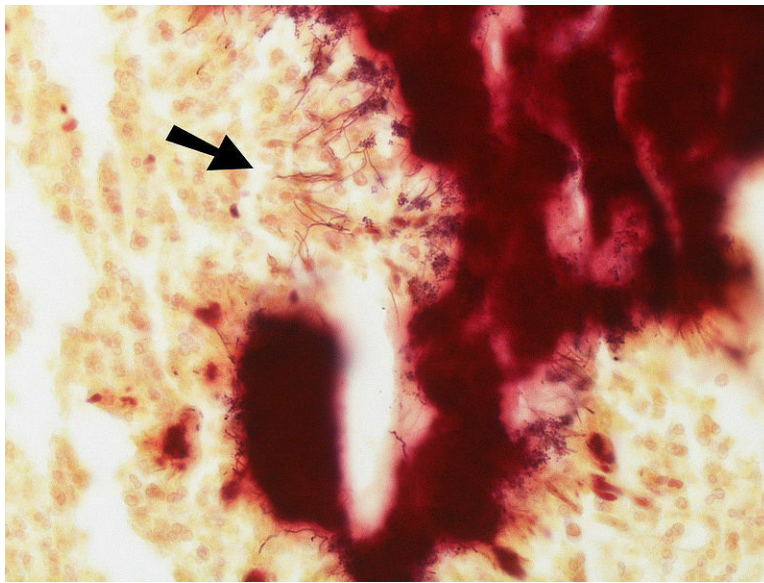
Cryptosporidium parvum, commonly causes diarrhea in immunocompromised individuals. The oocysts in the stool can be visualized by modified Ziehl-Neelsen staining. It appears as red acid-fast spheres. It can also be used for specimens such as sputum and bronchial wash.



Isospora belli is a coccidian parasite that can cause protracted diarrhea in immunocompromised patients. Modified Ziehl-Neelsen staining reveals pink-colored acid-fast oocyst in stool samples.



Actinomyces are gram-positive, anaerobic, and filamentous bacteria that are non-acid-fast in contrast to *Nocardia*.



Solution to Question 12:

The organism most likely to have the cell wall depicted is *Aspergillus fumigatus*.

Fungi usually have a rigid cell wall composed of polysaccharides, glycoproteins, and lipids. Most of them contain sugar polymers such as chitin, glucans, and mannans. The cytoplasmic membrane is usually composed of sterols. Antifungals such as amphotericin and nystatin bind to the ergosterol in the cell membranes which results in membrane damage and leakage.

Aspergillus fumigatus is a filamentous fungi, with hyphae that may have cross-walls or septa. It can cause allergic bronchopulmonary aspergillosis, aspergilloma, and superficial infections of the ear, eye, and sinuses. It causes invasive aspergillosis in immunocompromised individuals.

Other options:

Option B - *Mycobacterium tuberculosis*, are acid-fast staining bacteria, owing to the mycolic acid in the cell wall. They have a three-layered cell wall enclosing a trilaminar plasma membrane.

Option C - *Mycoplasma pneumoniae* lacks cells wall or even cell wall precursors. They are bound by a trilaminar membrane that contains sterols.

Option D - *Rhinosporidium seeberi* causes a granulomatous condition known as rhinosporidiosis. It exists as a trophocyte with clear cytoplasm and a covering of chitin. Mature trophocytes develop into sporangium with an outer chitinous and inner cellulose layer.

Solution to Question 13:

The given life cycle is of *Strongyloides stercoralis*.

The organism has a parasitic cycle and a free-living soil cycle. Humans are the natural host but morphologically indistinguishable strains may be seen in dogs and cats.

Filariform larvae are the infective form and there are two modes of infection:

- Penetration of skin by 3rd stage filariform larva.
- Autoinfection (external and internal).

The female worms present in the intestine lay eggs in the mucosa which hatch immediately, releasing the rhabditiform larva. The larva migrates into the lumen of the intestine and passes down to be released in the feces. The rhabditiform larva may change into filariform larva while passing through the bowel.

The filariform larvae in the bowel do not leave the host and become a source of autoinfection as they penetrate the colonic mucosa or perianal skin.

The rhabditiform larvae passed in the feces may undergo two types of development in the soil:

- Direct development: The rhabditiform larva moults twice on reaching the soil to become the infective filariform larva. The filariform larva can penetrate the skin and enter the circulation.
- Indirect development: The rhabditiform larvae develop into free-living males and females in the soil.

Other options:

Option B: Humans are the only natural host and the life-cycle of *Ancylostoma duodenale* is completed in a single host. The eggs laid by the female worms in the intestine are passed in feces. The embryo develops inside the egg and a rhabditiform larva hatches out in about 2 days.

The rhabditiform larva moults twice to form the filariform larva (infective stage). It penetrates the skin and enters the subcutaneous tissue.

Option C: Humans are the only natural host for *Ascaris lumbricoides* and have no intermediate host. The infection is acquired by ingestion of embryonated eggs (infective stage).

Option D: *Enterobius vermicularis* completes its entire life cycle in the human host and has no intermediate host. Embryonated eggs are the infective form. Infection is acquired by ingestion of the eggs. Autoinfection and reinfection are seen.

Solution to Question 14:

The ratio of T cells to B cells in the blood of a healthy adult is approximately 3:1.

T lymphocytes constitute 60-70% of lymphocytes in the blood while mature B cells account for 10-20%.

T cells are of three types: Helper T cells, cytotoxic T cells, and regulatory T cells. T lymphocytes develop in the thymus.

B lymphocytes develop from precursors present in the bone marrow. They differentiate to form either plasma cells or memory B cells. Plasma cells produce antibodies after stimulation by the antigen.

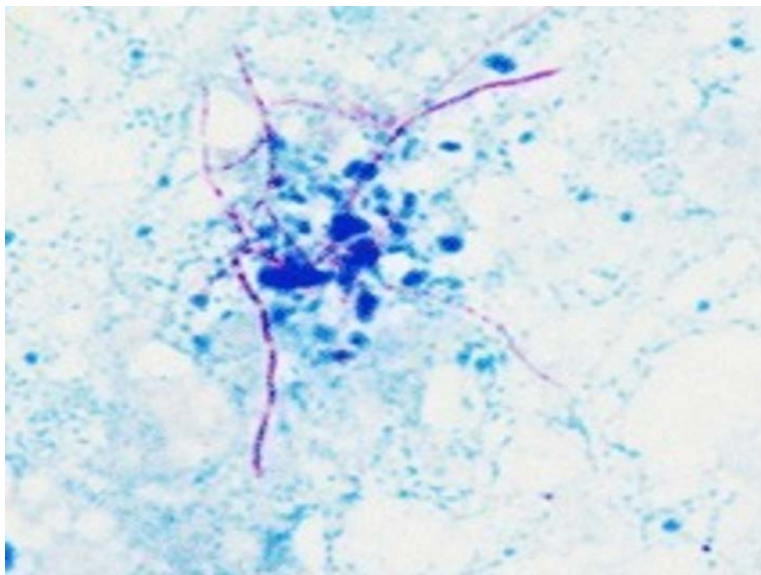
Lymphocyte recirculation is particularly important for T cells, as B cells do not have to move to the site of infection as the antibodies produced are circulated through blood.

Solution to Question 15:

The presence of *Nocardia* species can be confirmed using a modified acid-fast stain.

For microscopy, sputum smears are most commonly used. *Nocardia* species are seen as filamentous organisms, with hyphae-like branching. These bacteria are gram-positive and catalase positive, and they produce urease.

The image below shows acid-fast staining of *Nocardia*.



On direct microscopy of sputum or pus samples, crooked, branching, beaded filaments are seen; these filaments are 1 μm wide and up to 50 μm long.

After incubation at 35–37°C in laboratory media, they develop heaped, irregular, waxy colonies. Strains may vary in their pigmentation from white to orange to red. Blood cultures may be used to detect *Nocardia* species, although it may take up to 2 weeks to detect growth.

Solution to Question 16:

The microscopy technique used to produce high contrast images of transparent, live bacteria is phase contrast microscopy.

It improves the contrast by making use of the difference in thickness or refractive index of structures within and around the cell. The rays passing through different refractive indices are varied by fraction of wavelength, producing a phase difference. This creates a difference in intensity of light, producing contrast and making the structures within the cells and its surroundings clearly visible.

Option B: In Dark field microscope only the reflected light enters the objective lens whereas the unreflected light does not. Hence the specimen gets brightly illuminated against a dark background. It is used for visualizing very slender, live organisms such as spirochetes which are not visible under ordinary illumination.

Option C: Electron microscopy uses a beam of electrons instead of the beam of light. It is used for studying dead and dried objects. Hence, it may lead to distortion in cell morphology.

Option D: Polarized microscopy is used to study the intracellular structures using differences in birefringence.

Solution to Question 17:

The correct pairs are as follows:

Tuberculosis: *Mycobacterium tuberculosis* can be stained with Ziehl-Neelsen (ZN) stain. Steps involved in the ZN stain:

- Place the sample on the slide and evenly spread it with rotational movement.
- Air-dry the slide, and fix it by passing it over a flame.
- Apply carbol fuchsin stain.
- Heat the underside of the slide until steam rises.
- Wash with water to remove excess stain.
- Decolorize with alcohol.
- Apply methylene blue stain.
- Wash with water to remove excess stain.
- View under a microscope.

Malaria: Giemsa stain is used as a gold standard procedure for the detection of malarial parasites. This stain helps visualize Maurer's clefts associated with *Plasmodium falciparum* and Schuffner's dots associated with *Plasmodium vivax* or *Plasmodium ovale*.

Leprosy: Acid fastness in general is related to the carbon chain length of the mycolic acid found in any particular species. Acid and alcohol fastness of the lepra bacilli is less than that of the tubercle bacilli. Fite-Faraco method is the staining procedure of choice in *Mycobacterium leprae*. In this method, alcohol is eliminated from the hydrating and dehydrating steps, and 10% sulphuric acid is used as a decolorizer in place of acid/alcohol solution.

Actinomycosis: *Actinomyces* is a gram-positive bacillus that is filamentous and non-sporing in nature, and the Gram stain is more sensitive in its diagnosis than culture.

Column A	Column B
A. Malaria	1. Giemsa stain
B. Tuberculosis	2. ZN stain
C. Leprosy	3. Fite-Faraco stain
D. Actinomycosis	4. Gram stain

Solution to Question 18:

Vesicular rash is seen in hand, foot, and mouth disease.

It is a mild exanthematous fever usually occurring in children but may also occur in adults. It is characterized by papulovesicular rashes that involve the oral mucosa, palms, and soles.

The virus is shed in the stool and is transmitted by the feco-oral route. It is a self-limiting disease and subsides in 1-2 weeks. Treatment is symptomatic as the rash heals spontaneously.

Coxsackieviruses are enteroviruses and belong to the Picornaviridae family. They are classified into groups A and B. Group A viruses are classified into 24 types and group B viruses are classified into 6 types.

Diseases caused by Coxsackie A viruses:

- Hand, foot, and mouth disease (HFMD)
- Herpangina- It is febrile vesicular pharyngitis. It is characterized by the presence of ulcerating small vesicles on the oral fauces, or palate or uvula, and the posterior pharyngeal wall.
- Aseptic meningitis

Diseases caused by Coxsackie B viruses:

- Pleurodynia or Bornholm disease- It is viral myalgia defined by sudden severe lancinating pain, typically on one side of the chest. It is commonly associated with fever, malaise, and headaches.
- Myocarditis and pericarditis in the newborn.
- Orchitis, juvenile diabetes

The image below shows the rash in the hand, foot, and mouth disease.



Other options:

Option B: Cutaneous and mucosal findings in dengue are erythema, morbilliform eruptions, and hemorrhagic lesions.

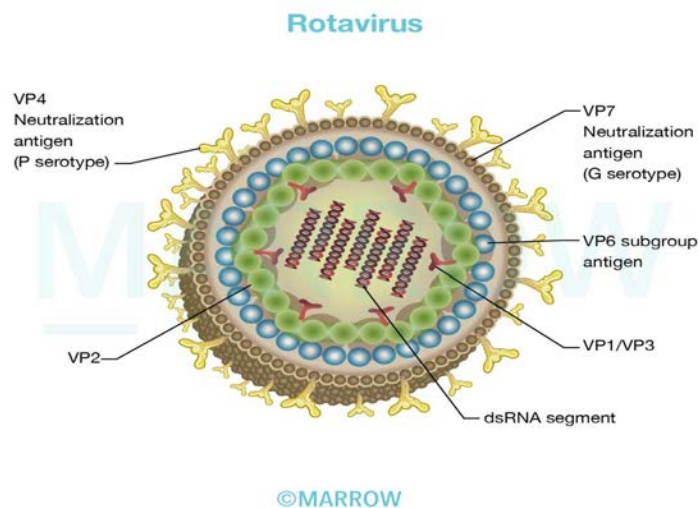
Option C: Zika fever is characterized by fever, maculopapular rash, arthralgia, and conjunctivitis.

Option D: Roseola infantum presents as a nonpruritic, pink papular rash that begins on the trunk.

Solution to Question 19:

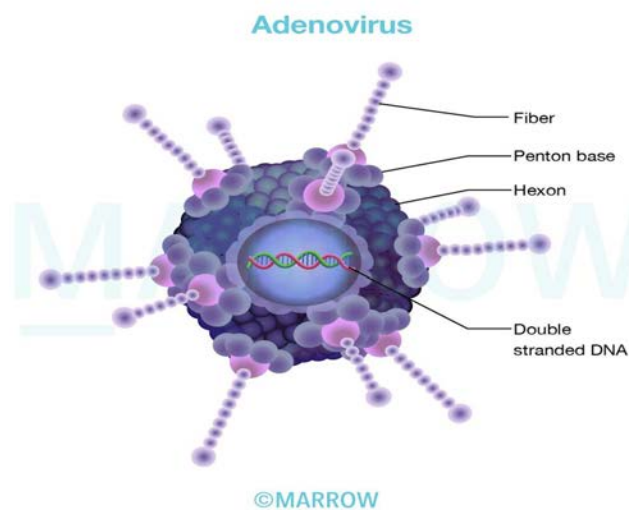
Rotavirus exhibits genetic reassortment/antigenic drift similar to that of the influenza virus.

Rotaviruses appear as cartwheel-shaped structures with short spokes radiating from them. 11 segments of double-stranded RNA constitute the genome. Influenza viruses have a linear, segmented, single-stranded RNA. Reassortment of the genome occurs readily due to the presence of this segmented RNA.

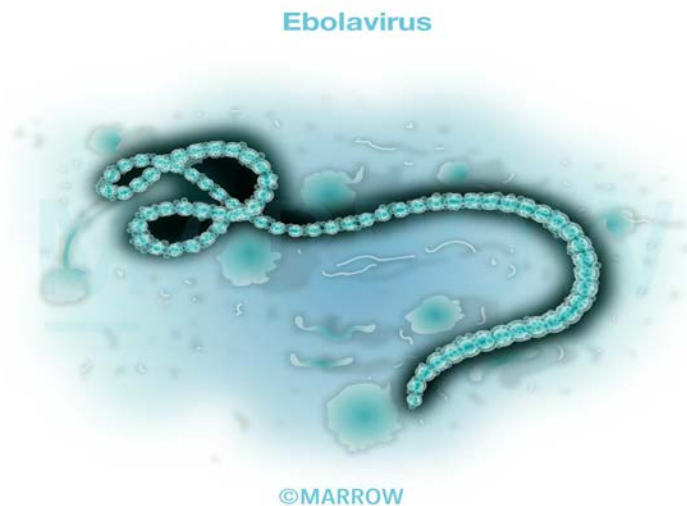


Other options:

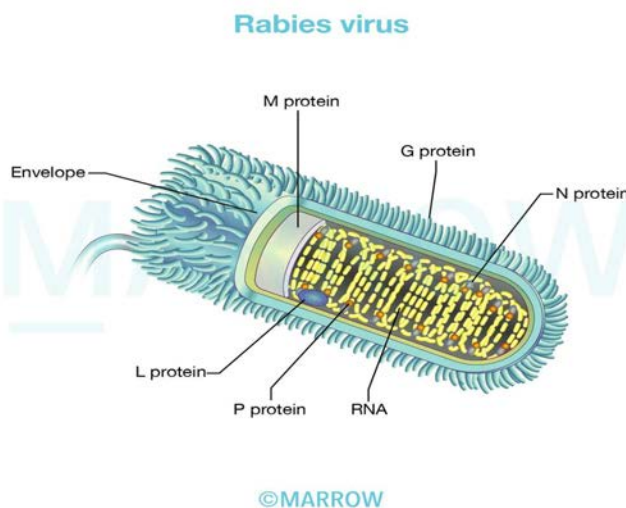
Option B - Adenovirus contains double-stranded DNA. It has icosahedral symmetry, with structures known as fibers extruding from the vertices.



Option C - Ebola viruses are filamentous viruses with nonsegmented single-stranded RNA as genetic material.



Option D - Rabies viruses belonging to the family Rhabdoviridae are bullet-shaped viruses, with nonsegmented single-stranded RNA.



Solution to Question 20:

The most likely diagnosis based on the clinical vignette and the histopathology showing sclerotic bodies is chromoblastomycosis. It is primarily a subcutaneous mycosis and does not commonly invade underlying tissue, tendon, or joints.

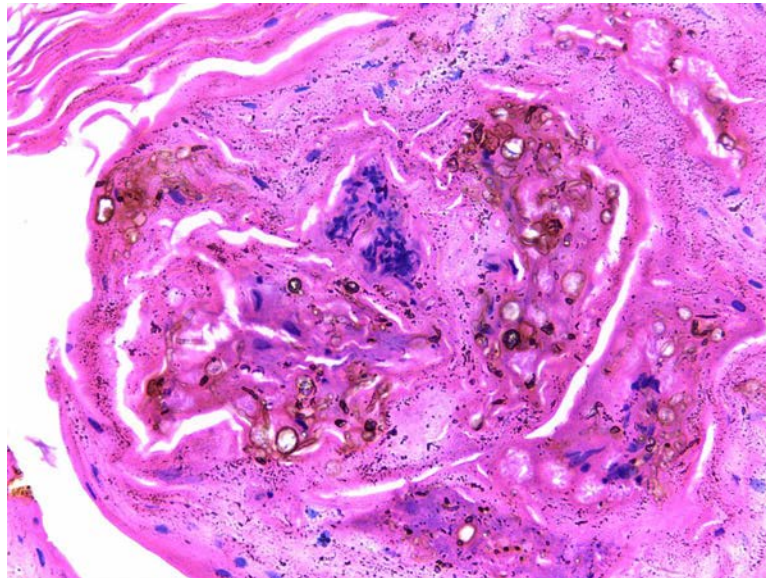
Chromoblastomycosis is caused by dematiaceous fungi, primarily *Fonsecaea* spp and *Cladophialophora* spp. Not all species of dematiaceous fungi are known to be

thermally dimorphic fungi (Option C)

Chromoblastomycosis, the most common chromomycosis, is more commonly seen in tropical areas among agricultural workers and woodcutters who walk barefoot. Traumatic implantation is the mode of infection. Most often, the infection occurs in trauma-prone sites, such as exposed areas of the lower and upper limbs.

Patients present with warty cauliflower-like cutaneous nodules on the feet and lower legs. The disease generally remains localized, with slow peripheral extension. Lymphatic spread can occur.

Diagnosis is made by the demonstration of sclerotic bodies, also called copper-penny bodies, in tissue sections. Sclerotic bodies appear as round or irregular, dark brown bodies with septae.



First-line management includes itraconazole or terbinafine. Flucytosine used on its own or combined with amphotericin B may also be effective. For small lesions, surgical excision is the therapy of choice.

Solution to Question 21:

The given clinical scenario and the stool microscopy show oocysts of *Cryptosporidium*.

Cryptosporidium is a coccidian parasite implicated in causing traveller's diarrhea in immunocompromised patients. The two *cryptosporidium* species responsible are *C. hominis* and *C. parvum*.

The infective form is the oocyst excreted in the patient's feces, which can be acquired through drinking water or recreational water activities. The oocysts are immediately infectious (unlike *isospora* oocysts, which require maturation), and may cause autoinfection.

Because of their small size, they are difficult to see using routine staining and hence, acid-fast staining is done.

Once ingested, the 4 sporozoites are released from each cyst and attach to the small intestinal mucosa. It causes crampy abdominal pain and non-inflammatory watery diarrhea.

Treatment is mainly supportive, though nitazoxanide has been approved for use in *cryptosporidium* infection.

Other options:

Option A: Oocysts of *isospora* are 23-36 μm , oval, and contain two sporocysts, each having four sporozoites. They are uniformly acid-fast.

Option B: Sporulated oocysts of *cyclospora* are 8-12 μm in size, round, and contain 2 sporocysts each having two sporozoites. They are acid-fast.

Option D: *Microspora* oocysts are 2-4 μm in size, oval to cylindrical in shape with a polar filament. Spores are gram-positive and acid-fast.

Solution to Question 22:

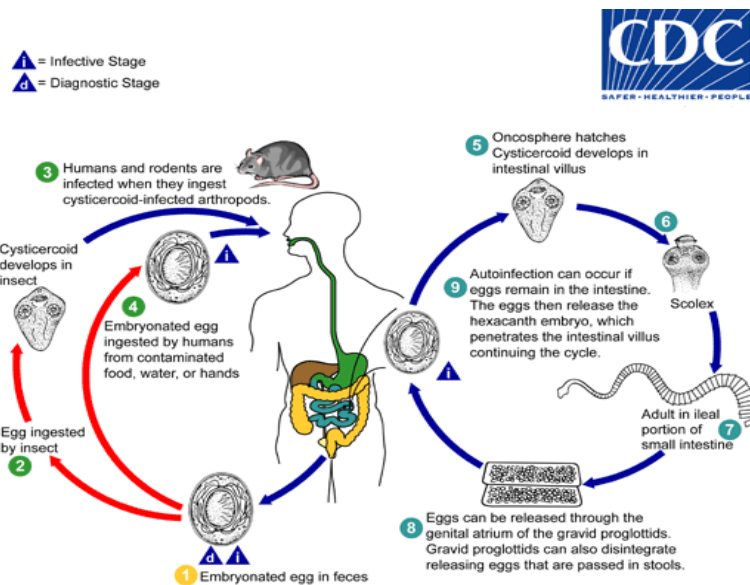
The given image shows a scolex with 4 suckers and a retracted rostellum with a single row of hooklets. The proglottids are broader than long. These features are characteristic of *Hymenolepis nana*.

Option 4 shows the appearance of the egg of *H. nana* on stool microscopy. These eggs have a thin, colorless outer membrane and thread-like polar filaments. They have a hexacanth embryo. The eggs are non-bile stained and float in a saturated salt solution.

H. nana, also called dwarf tapeworm, is the smallest tapeworm. Humans are definitive hosts. Infection is acquired by the ingestion of food and water contaminated with eggs. The egg attaches to the intestine and develops into the larva and adult worm. The adult worm resides in the small intestine. No intermediate host is required. *H. nana* is unusual in that it undergoes multiplication in the body of the definitive host.

Patients are usually asymptomatic. But, in heavy infection, they present with nausea, anorexia, abdominal pain, diarrhea, and irritability. Pruritus may also be present due to an allergic response.

The image given below shows the life cycle of *Hymenolepis nana*.



Solution to Question 23:

The image shows the life cycle of the human papilloma virus, and the E6 and E7 proteins act as oncogenes.

The E6 protein binds to and helps in the degradation of p53 and stimulates the expression of telomerase reverse transcriptase (TERT), which is the catalytic subunit of telomerase, thereby helping in the longevity of the cells.

E7 protein binds to the RB protein and displaces the E2F transcription factors that are normally sequestered by RB, thus promoting progression through the cell cycle.

L1 forms the major portion of the capsid. L2 plays an important role in papillomavirus assembly and the infectious process. E1 and E2 are involved in replication and transcription.

Solution to Question 24:

The organism given in the image is *Ehrlichia chaffeensis*. The peripheral blood smear shows a morula, an intracellular inclusion body in a monocyte.

Ehrlichia chaffeensis is a gram-negative, obligate intracellular bacteria that are transmitted by hard ticks such as the lone star tick.

Clinically, the symptom onset is typically within one-three weeks of exposure. Patients may present with fever, headache, myalgia, nausea, vomiting, loss of appetite, and weight loss. Rashes are generally absent. Severe human monocytic ehrlichiosis is characterized by meningoencephalitis, renal failure, respiratory failure, and shock.

Diagnosis is by demonstration of the characteristic morulae in the mononuclear cells such as monocytes. Serological studies such as indirect fluorescent antibody tests and polymerase chain reaction (PCR) detection can also be used.

The treatment of choice for human monocytic ehrlichiosis is doxycycline.

Other options:

Option A: *Rickettsia rickettsii* causes rocky mountain spotted fever. It is transmitted by ticks. Patients present with fever, headache and rash.

Option B: *Rickettsia conorii* causes Mediterranean spotted fever. The parasite is transmitted by ticks and the patients present with fever, headache, rash, and eschar (tache noire).

Option D: *Anaplasma* causes human granulocyte anaplasmosis. It is transmitted by ticks. Clinical features are fever, headache, and myalgia. Inclusions in granulocytes are seen.

Solution to Question 25:

The given image shows multiple vesicles on hand and multiple eruptions along lymphatics. All of the organisms mentioned in the options can spread through dermal and subcutaneous lymphatics except *Staphylococcus aureus*

Staphylococcus aureus is a gram-positive coccus arranged in grape-like clusters. It causes various localized pyogenic infections like folliculitis, furuncle, abscess, impetigo, paronychia, osteomyelitis, and even septicemia.

Other options:

Option A: *Sporothrix schenckii* is a dimorphic fungus. It causes sporotrichosis, a chronic granulomatous disease, by traumatic inoculation into the skin. This is usually followed by spread along the draining lymphatics and lymph nodes. Itraconazole is the drug of choice for treatment.

Option C: *Nocardia asteroides* is a gram-positive aerobic bacteria. It is frequently found in the soil and infection may be exogenous. Cutaneous infections can present as local abscesses or lymphocutaneous lesions.

Option D: *Mycobacterium marinum* is a non-tuberculous mycobacterium. It causes swimming pool granuloma. The lesion begins as a papule and breaks down to form an indolent ulcer. It usually occurs after abrasions and is therefore seen on the prominences like elbows, knees, ankles, or toes.

The image below shows sporotrichosis.

Sporotrichosis



Solution to Question 26:

Polymyalgia rheumatica is not associated with parvovirus B19, and more commonly occurs in women aged 50 and above. It is closely associated with giant cell arteritis.

The main manifestation of symptomatic parvovirus B19V infection is erythema infectiosum, also known as fifth disease or slapped-cheek disease, which most commonly occurs in children of early school age or infants. Fever and mild constitutional symptoms may accompany the rash, which has a typical “slapped cheek” appearance.

Pure red cell aplasia and chronic hemolytic anemia have been reported in patients with immunosuppression due to a wide range of clinical settings, including AIDS, lymphoproliferative disorders, and transplantation.

Parvovirus B19V infection during pregnancy can lead to hydrops fetalis and/or fetal loss. The risk of transplacental fetal infection is 30%, and the risk of fetal loss, predominantly early in the second trimester, is 9%.

Solution to Question 27:

West Nile virus can be identified by polymerase chain reaction (PCR) and is endemic to India.

West Nile virus is a flavivirus that is transmitted by Culex mosquitoes. It can be subclinical or may present as a mild influenza-like illness. Encephalitis occurs in severe forms. Reverse-transcriptase PCR has been used to identify the virus.

Other options:

Option A- Hendra virus is endemic to Australia, and mostly affects horses. Some human cases are also reported.

Option C - Ebola virus belongs to the genus Filovirus. It causes hemorrhagic fever. It is endemic to Africa.

Option D - Yellow fever is mostly confined to South and Central America, and certain parts of Africa. It is caused by flavivirus. It is an arthropod-borne hemorrhagic fever. Aedes aegypti transmits the disease.

Solution to Question 28:

SARS-CoV-2 can be detected by all except Southern blotting.

Southern blotting cannot be used to detect SARS-CoV-2 because there is usually not enough DNA in the collected sample to obtain a pattern on gel electrophoresis.

Real-time PCR uses nasopharyngeal secretions to detect viral genomic RNA or DNA in respiratory secretions. It is the most sensitive and specific test available.

Rapid-antigen test detects spike proteins in nasopharyngeal secretions.

Enzyme-linked immunosorbent assay (ELISA) against spike protein is also found to be sensitive and specific for SARS-CoV-2.

Solution to Question 29:

The given scenario and image from the peripheral smear showing rings (headphones appearance) point towards the diagnosis of severe Plasmodium falciparum malaria, and the treatment of choice for this condition is artesunate 2.4 mg/kg IV at 12 and 24 hours, followed by daily once for 4 days.

Major manifestations of severe falciparum malaria include more than two convulsions in 24 hours, unarousable coma/ cerebral malaria (failure to localize or respond appropriately to stimuli or coma persisting for >30 min after generalized convulsion), acidosis, severe normocytic normochromic anemia, renal failure, pulmonary edema, hypoglycemia, shock, and disseminated intravascular coagulation.

Other manifestations include hemoglobinuria, hyperparasitemia, jaundice, and extreme weakness.

The treatment of choice is artesunate given at a dose of 2.4 mg/kg stat IV, followed by 2.4 mg/kg at 12 and 24 hours, and then daily if necessary.

If artesunate is unavailable, artemether is given at a dose of 3.2 mg/kg stat IM, followed by 1.6 mg/kg QD, or quinine dihydrochloride given at a dose of 20 mg of salt/kg infused over 4 hours, followed by 10 mg of salt/kg infused over 2–8 hours.

Solution to Question 30:

Delayed hypersensitivity is due to interferon-gamma (IFN-gamma).

Delayed hypersensitivity is also called cell-mediated/type 4 hypersensitivity reaction. It is mediated by T cells and occurs by 2 mechanisms:

- CD4+ T helper cells mediated
- CD8+ T helper cells mediated

CD4+ T helper cells mediated pathway:

Antigen peptides recognized by CD4+ T cells secrete interleukin -2. Interleukin -2 helps in further proliferation of T cells that respond to the antigen peptide. These T cells differentiate into Th1 or Th17 cells as a result of the cytokines produced from the antigen presenting cells (APC).

APCs also secrete interleukin - 12 which enhances differentiation of the T cells to the Th1 subset. Inflammatory cytokines like interleukins -1, 6, 23 induce differentiation of T cells to the Th17 subset. These responses lead to the release of interferon gamma that alters the macrophages in several ways to ensure eliminating the antigen.

Interferon-gamma-induced changes in the macrophages:

- Augmented ability of the macrophages to phagocytose and kill micro-organisms.
- Increased expression of class 2 MHC molecules to enhance antigen presentation.
- Increased secretion in presence of tumor necrosis factor, interleukin-1, and other inflammatory chemokines.
- Amplification of the Th1 response by producing more interleukin-12.

CD8+ T helper cells mediated pathway:

CD8+ T cells destroy antigen-presenting target cells using perforins and granzymes. These mediators are preformed in the lysosomes of a cell. The perforin-granzyme complex enters the target cell by endocytosis. Granzymes cause the activation of caspases inducing apoptosis of the target cell. CD8+ T cells also produce interferon gamma following viral infections and contact allergens. Interferon-gamma then proceeds to stimulate the production of inflammatory mediators and bring about the hypersensitivity reaction.

Examples of delayed hypersensitivity reactions include the tuberculin reaction and contact dermatitis.

Interferon-gamma is an important effector involved in both immune mechanisms of delayed hypersensitivity reactions and hence is the best answer for this question.

Solution to Question 31:

The lesion shown in the image is eschar. It is seen in scrub typhus.

Scrub typhus is caused by *Orientia tsutsugamushi* and transmitted by trombiculid mites. Humans are infected when they are bitten by the mite larvae known as chiggers. Transovarian transmission of microbes occurs in mites.

The incubation period is 1-3 weeks. The patient typically develops a characteristic eschar which is a punched-out ulcer covered with a blackened scab at the site of the mite bite, with regional lymphadenopathy and maculopapular rash.

Other options:

Option A: Rocky Mountain spotted fever is a rickettsial infection transmitted by ticks (*Dermacentor andersoni*). It is the most serious type of fever. Symptoms include fever, headache, Rash, muscle pain and lack of appetite. Early treatment with doxycycline is recommended to prevent death and severe illness.

Option B: Malaria is a protozoal disease caused by *Plasmodium* species. It is transmitted by the female anopheles mosquito. Here, the definitive host is the anopheles mosquito and humans are the intermediate hosts. Clinical features are periodic bouts of fever with chills and rigors, followed by anemia, splenomegaly, and hepatomegaly.

Option D: COVID-19 is a highly contagious viral disease caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). The primary mode of transmission is through respiratory droplets. The median incubation period is around 5 days. Patients commonly present with fever, cough, and shortness of breath and less commonly with a sore throat, anosmia, anorexia, nausea, malaise, myalgias, and diarrhea.

Solution to Question 32:

Ethylene oxide, glutaraldehyde, and ortho-phthalaldehyde are disinfectants with sporicidal activity.

Ethylene oxide is a gas used to sterilize heat-sensitive articles such as syringes, needles, catheters, and gloves. *Bacillus atrophaeus* and *Bacillus subtilis* are used as sterilization control.

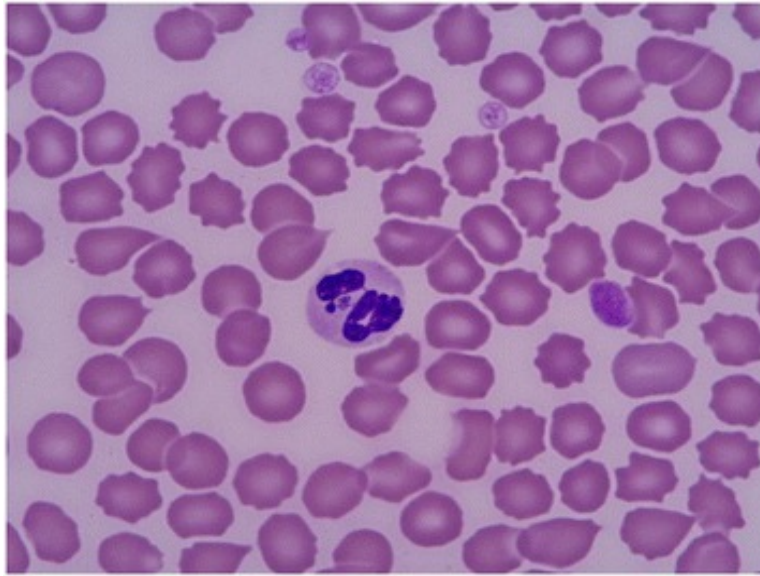
Glutaraldehyde and ortho-phthalaldehyde are liquid-chemical sterilizing agents. They are used for the disinfection and sterilization of cystoscopes, bronchoscopes, rubber anesthetic tubes, endotracheal tubes, polythene tubing, and other metal instruments.

Pasteurization is when milk is heated at either 63°C for 30 minutes (the holder method) or at 72°C for 15–20 seconds (the flash process), followed by cooling quickly to 13°C or lower. It has no sporicidal activity and kills non-spore-forming bacteria like *Mycobacteria*, *Brucella*, and *Salmonella*. *Coxiella burnetii* is relatively heat resistant and may survive the holder method.

Solution to Question 33:

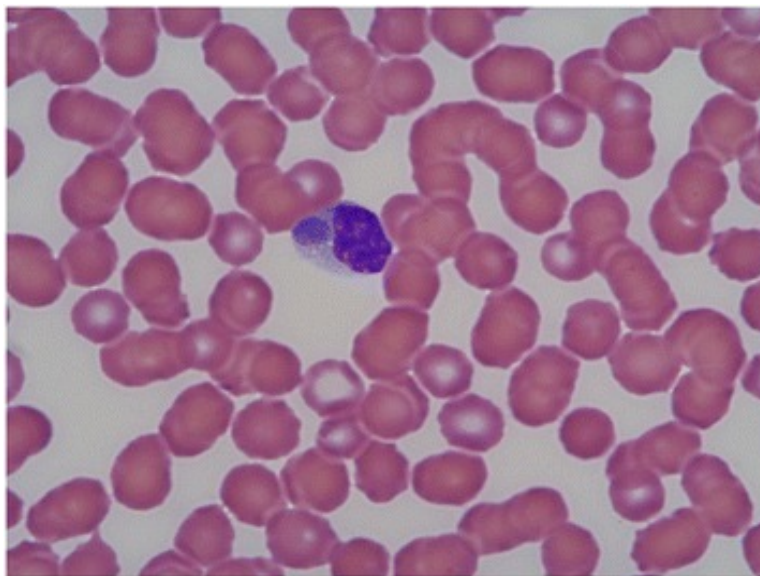
The inclusion body rickettsial morula is found in granulocytes (neutrophils) during infection caused by *Anaplasma phagocytophilum*.

Anaplasma causes human granulocyte anaplasmosis. It is transmitted by ticks. Clinical features are fever, headache, confusion, vomiting and myalgia. Inclusions in granulocytes are seen. The image below shows a morula of *Anaplasma* in a neutrophil:



Ehrlichia chaffeensis is a gram-negative, obligate intracellular bacterium that is transmitted by hard ticks such as the lone star tick. Clinical presentation of *Ehrlichia chaffeensis* infection is similar to that of *Anaplasma* infection. Severe human monocytic ehrlichiosis is characterized by meningoencephalitis, renal failure, respiratory failure, and shock. The treatment of choice for human monocytic ehrlichiosis is doxycycline.

The image below shows a morula of *Ehrlichia* in a monocyte:



Note: Infection by both *Anaplasma* and *Ehrlichia* can lead to the formation of rickettsial morulas in leukocytes. However, *Anaplasma* leads to the formation of morulas predominantly in the granulocytes, whereas *Ehrlichia* predominantly affects the monocytes.

Other options:

Option A: *Rickettsia rickettsii* causes rocky mountain spotted fever. It is transmitted by ticks. Patients present with fever, headache, and rash.

Option B: *Rickettsia conorii* causes Mediterranean spotted fever. The parasite is transmitted by ticks, and the patients present with fever, headache, rash, and eschar (tache noire).

Solution to Question 34:

Roseola infantum is not associated with petechial rashes.

Roseola infantum, also known as exanthem subitum or sixth disease, is caused by human herpes viruses 6 and 7 and is an acute febrile illness that presents with a rose-pink maculopapular eruption after the fever subsides. It initially develops on the neck and trunk and may spread to the arms, face, and leg. They are surrounded by pallor and rarely become vesicular.

The causative agents of Zika, Kyasanur Forest disease, and West Nile fever belong to Flaviviridae group of viruses, which are known to cause hemorrhagic fever and present with petechial rashes.

Note: Although the Zika virus classically presents with a maculopapular rash. A minority of cases present with petechial rashes as well. Therefore, the better answer to this question is roseola infantum.

Solution to Question 35:

Abdominal tuberculosis (TB) is not transmitted by feco-oral route.

The terminal ileum and the cecum are the sites most commonly involved in abdominal TB. It causes abdominal pain (at times similar to that associated with appendicitis) and obstruction, hematochezia, palpable mass in the abdomen along with fever, weight loss, anorexia, and night sweats.

Modes of spread:

- By ingestion of infected food or milk (primary intestinal TB) or infected sputum (secondary intestinal TB)
- Contagious spread from infected adjacent foci
- Hematogenous spread from distant TB focus
- Through lymphatic channels
- Direct contact with an infected animal (*Mycobacterium bovis*)

There is no evidence of spread of TB through the feco-oral route.

Solution to Question 36:

The correctly matched egg for the given parasite - *Enterobius vermicularis* is 3

The picture on the left shows a worm with a pointed end. The mouth is surrounded by wing-like cuticular expansions- cervical alae which are transversely striated suggesting it is *Enterobius vermicularis*. The eggs are colourless, non-bile stained, and elongated with a

characteristic planoconvex shape-flattened on one side and convex on the other. It has an outer albuminous layer making the eggs stick to each other, to clothing and to other objects.

It presents as intense irritation and pruritus of the perianal and perineal area which is more at night when the worm crawls out of the anus to lay eggs. In females, the worm may crawl into the genital tract causing irritation, mucoid discharge and symptoms of chronic salpingitis.

The eggs are infrequently found in stools as they are deposited on the perianal and perineal skin. They are diagnosed using NIH swabs (National Institutes of Health). The treatment of choice is albendazole, mebendazole, or pyrantel pamoate.

Solution to Question 37:

The image showing filamentous, branching acid-fast rods is suggestive of *Nocardia*.

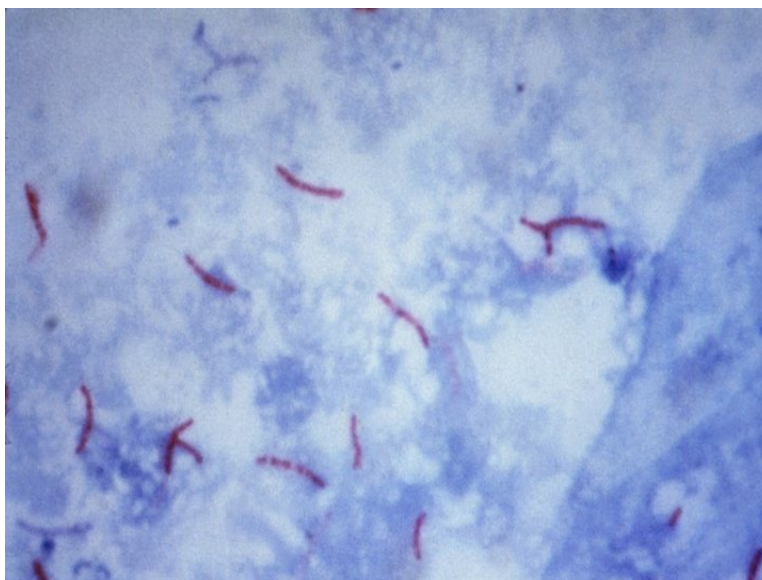
Nocardia appears as filamentous organisms with extensive hyphae-like branching. They are gram-positive, catalase-positive, and produce urease. They are weakly acid-fast because they stain with carbolfuchsin (routine acid-fast stain) and retain when decolorized with 1-4% sulfuric acid instead of the stronger acid-alcohol decolorant.

Acid-fast staining is based on the principle that tubercle bacilli resist decolorization with acids after staining with aniline dyes. Acid fastness is attributed to the variety of fatty acids, alcohols, and lipids in the tubercle bacilli. Acid-fast organisms usually contain mycolic acid, which is acid-fast in the free state. The steps to carry out acid-fast staining as modified by Ziehl and Neelsen are:

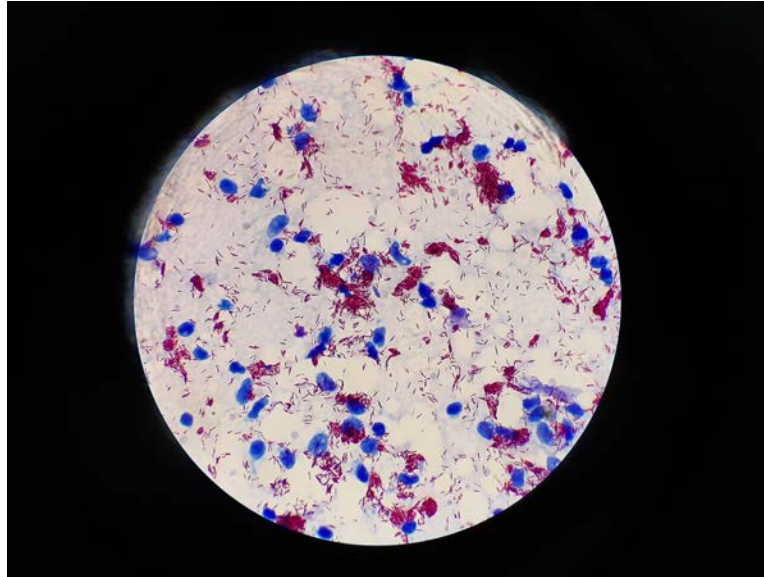
- The slide is stained with carbolfuchsin along with the application of heat.
- It is then decolorized with 20% sulphuric acid. Later methylene blue is used for counterstaining.
- The acid-fast bacteria retain the fuchsin color, while other organisms take up the counterstain.

Other options:

Option B: The image given below is a smear with Ziehl-Neelsen staining showing non-branching and acid-fast *Mycobacterium tuberculosis* bacilli.

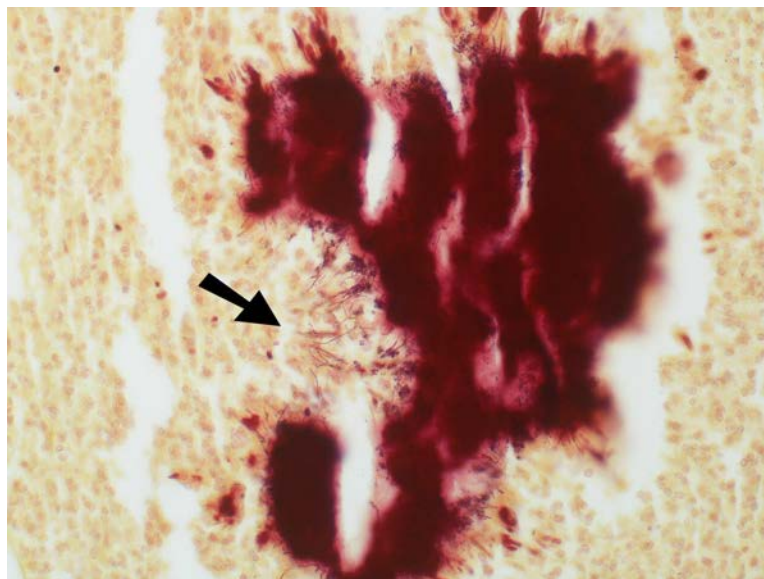


Option C- *Mycobacterium leprae* bacilli are less acid and alcohol fast than the tubercle bacillus; therefore, alcohol is removed from the hydrating and dehydrating steps and 5% sulphuric acid is used as a decolorizer in place of acid/alcohol solution. The image below shows *Mycobacterium leprae* on ZN staining.



Option D: Actinomyces are gram-positive, anaerobic, and filamentous bacteria that are non-acid-fast in contrast to *Nocardia*.

The following image shows the filamentous gram-positive stained actinomyces:



Solution to Question 38:

The incorrectly matched pair is Non-erosive arthritis - Patients of SLE.

Polyarthropathy occurs in ~50% of adults infected with parvovirus B19 who are healthy without underlying SLE. Although parvovirus B19 sometimes mimics rheumatoid arthritis with detectable rheumatoid factor, it has not shown any significant association with SLE.

Parvovirus B19 infection occurs usually in children 5–10 years causing erythema infectiosum. It is also known as fifth disease. It begins as a prominent erythema of the cheeks- slapped cheek disease, spreading to the trunk and limbs, followed by lymphadenopathy and arthralgia. Fever and mild constitutional symptoms may accompany the rash.

Parvovirus B19 replicates in immature erythroid cells and causes a transient aplastic crisis in cases of chronic hemolytic anemia, such as sickle cell disease, thalassemias, and acquired hemolytic anemias in adults.

Similarly, infection during pregnancy targets fetal red blood cells causing hemolysis leading to hydrops fetalis and/or fetal loss. Since the hydrops is not caused by antigen-antibody incompatibility, it is termed as non-immune hydrops.

It can also cause pure red cell aplasia in immunocompromised patients manifesting as chronic anemia and bone marrow suppression.

The following is an image of erythema infectiosum affecting a child:



Solution to Question 39:

The possible diagnosis based on the given clinical scenario is Jarisch-Herxheimer reaction.

JHR is a condition that occurs in patients infected by spirochetes who undergo antibiotic treatment. The reaction occurs within 24 hours of antibiotic treatment. Clinical features are fever, chills, nausea and vomiting, tachycardia, hypotension, flushing, myalgia, and exacerbation of skin lesions. JHR is due to IL6, TNF-alpha, and IL 8.

Syphilis is the leading spirochetal infection associated with JHR. Other infections are Lyme disease, louse-borne relapsing fever, leptospirosis, and tick-borne relapsing fever. Antibiotics that have been most associated are penicillins, tetracyclines, and erythromycin.

Mild reactions are self-limiting and often resolve spontaneously within 24 hours. Prevention of the JHR is by giving TNF-alpha antibodies and steroids.

Other options:

Option A: All of the strains of leptospira are found to be sensitive to ampicillin, cefotaxime, ciprofloxacin, doxycycline, erythromycin, and streptomycin. Resistance is shown to amphotericin B, 5-fluorouracil, trimethoprim, sulfamethoxazole, and vancomycin.

Option B: Penicillin belongs to the beta-lactam group of antibiotics. Hypersensitivity is the most important adverse reaction, resulting in nausea, vomiting, pruritus, wheezing, laryngeal edema, and, ultimately, cardiovascular collapse.

Option C: Dengue fever presents with headache, musculoskeletal pain, and occasional biphasic fever.

Solution to Question 40:

The given image shows multiple vesicles on hand and multiple eruptions along lymphatics characteristic of sporotrichoid lymphocutaneous infection. All of the organisms mentioned in the options can cause this except *Corynebacterium minutissimum*.

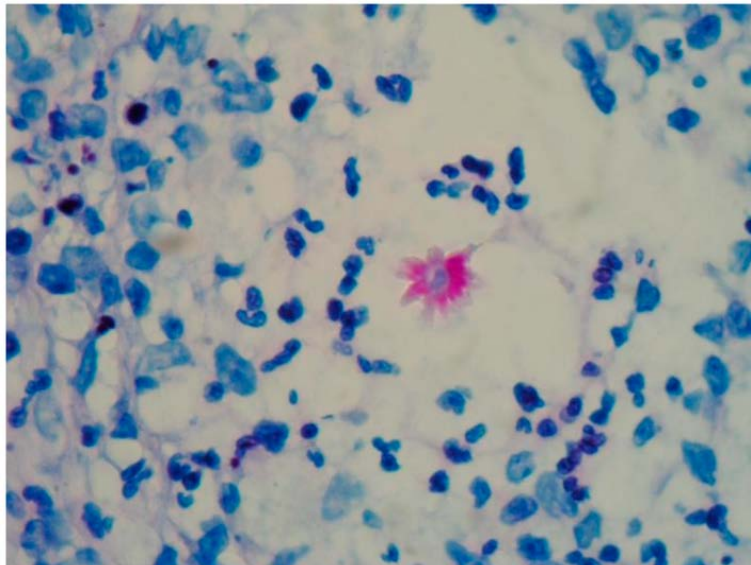
Corynebacterium minutissimum causes a localized infection of the stratum corneum called erythrasma. It presents as superficial skin lesions in the axilla and groin. This is a lipophilic corynebacterium and can be grown readily in media containing 20% fetal calf serum.

Sporotrichoid lymphocutaneous infection classically presents as superficial cutaneous lesions following cutaneous inoculation or trauma and progresses along dermal and subcutaneous lymphatics. Also known as nodular lymphangitis, the most common causes include:

- *Sporothrix schenckii*
- *Nocardia brasiliensis*
- *Mycobacterium marinum*
- *Leishmania brasiliensis*

Option A: *Sporothrix schenckii* is a dimorphic fungus. It causes sporotrichosis, a chronic granulomatous disease, by traumatic inoculation into the skin. This is usually followed by spread along the draining lymphatics and lymph nodes. Microscopy shows asteroid bodies. It can be seen as an oval, basophilic structure with eosinophilic radiation. Itraconazole is the drug of choice for treatment.

The image below shows asteroid bodies.



Option C: *Nocardia brasiliensis* is a gram-positive aerobic bacteria frequently found in the soil and infection may be exogenous. Cutaneous infections can present as local abscesses, cellulitis, or lymphocutaneous lesions. It causes mycetoma which is a localized chronic, granulomatous condition involving subcutaneous and deeper tissues, commonly affecting the foot. It presents as a tumor with multiple discharging sinuses.

Option D: *Mycobacterium marinum* is a non-tuberculous mycobacterium that causes swimming pool granuloma. The lesion begins as a papule which breaks down to form an indolent ulcer. It usually occurs after abrasions and is seen on the prominences like elbows, knees, ankles, or toes.

Solution to Question 41:

The organism with oocysts of size 8-10 μm is *Cyclospora*.

Cyclospora is a coccidian parasite. The sporulated oocysts are infectious to humans. They reside in the enterocytes in the small intestine. The infection causes prolonged diarrhea, abdominal pain, and vomiting, which is more severe in immunocompromised individuals.

On a wet mount of the feces, oocysts are seen as red spheres when stained with acid-fast stain. Sporulated oocysts of *Cyclospora* are 8-10 μm in size, round, and contain 2 sporocysts, each having two sporozoites. Treatment is by giving cotrimoxazole twice daily for seven days.

Other options:

Option A: Oocysts of *Isospora* are 23-36 μm , oval, and contain two sporocysts, each having four sporozoites. They are uniformly acid-fast.

Option B: Sporulated oocysts of *Cryptosporidium* are spherical, 4-5 μm in size, and acid-fast. It is implicated in causing traveler's diarrhea and chronic persistent diarrhea in immunocompromised patients. The two *cryptosporidium* species responsible are *Cryptosporidium hominis* and *Cryptosporidium parvum*.

Option D: *Microspora* oocysts are 2-4 μm in size, oval to cylindrical in shape with a polar filament. Spores are gram-positive and acid-fast. They cause a wide range of opportunistic

infections in immunocompromised patients.

Solution to Question 42:

L1 protein is used in the formulation of HPV vaccines.

Viral proteins can be divided into early and late proteins.

- Early proteins are called non-structural proteins as they do not form any structural component of the virus. They play a role within the infected cell during virus replication or act in the regulation of virus replication or assembly.
- Late proteins are also called structural proteins as they form viral components like capsids and envelopes.

Human papillomavirus (HPV) is a DNA virus that is spread by close contact. Serotypes 6 and 11 cause anogenital warts and laryngeal papilloma. The risk of these lesions can be reduced by vaccination against HPV. Three vaccines approved by FDA include bivalent (Cervarix-types 16 and 18), quadrivalent (Gardasil- 6, 11,16, 18), and 9-valent (Gardasil 9- additional 31, 33, 45, 52, and 58). All three provide protection against infection and the development of precancerous lesions.

The vaccine is prepared by using the L1 capsid protein. It is given to girls aged 11 to 12 years (can start at 9 years of age) and can also be given to boys.

E6 and E7 are oncogenic proteins that interact with p53 and Rb genes respectively. E1, E2, and E3 are other early proteins which help in viral replication and are not used to formulate vaccines.

Solution to Question 43:

The Epstein-Barr virus (EBV) binds to CD21 and acts as a B-cell mitogen.

It affects adolescents and younger adults more frequently but may occur from childhood to old age. It is transmitted primarily through close contact with oropharyngeal secretions. The incubation period of EBV infectious mononucleosis is 4-6 weeks.

Pathogenesis of Epstein Barr virus infection:

- The virus infects nasopharyngeal epithelial cells, salivary tissues, and lymphoid tissues.
- Lymphocytosis is caused by a proliferation of these EBV-infected B lymphocytes.
- There is a production of atypical reactive T cells which are CD8+ cytotoxic T cells, that are also called Downey cells.
- Heterophile antibodies are also produced due to B cell mitogenesis.

Clinically, patients may present with

- Fever, malaise, nausea, and anorexia.
- Lymphadenopathy- bilateral and symmetrical. Bilateral posterior cervical lymphadenopathy is characteristic.
- Pharyngitis.

- Tonsillar enlargement.
- Hepatomegaly and splenomegaly. Splenic rupture may occur even with minor trauma and is highly fatal.
- A drug-induced rash may develop on using beta-lactam antibiotics.

Solution to Question 44:

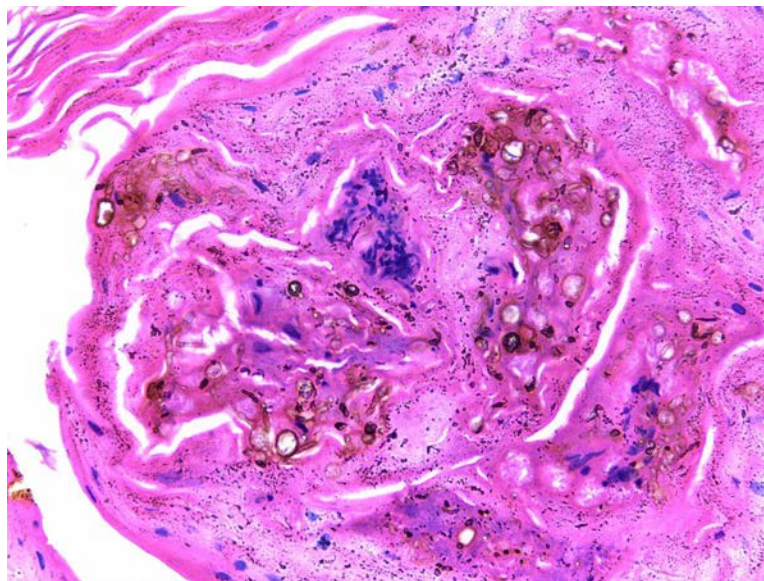
The most likely diagnosis based on the clinical vignette and the histopathology showing sclerotic bodies is chromoblastomycosis. The true statement regarding the condition is that it is caused by a dematiaceous fungus, which means that this fungus produces a brown pigment.

The various dimorphic pigmented fungi are *Phialophora verrucosa*, *Fonsecaea pedrosoi*, *Fonsecaea compacta*, and *Cladophialophora carrionii*.

Chromoblastomycosis, the most common chromomycosis, is more commonly seen in tropical areas among agricultural workers and woodcutters who walk barefoot. Traumatic implantation is the mode of infection.

Patients present with warty cauliflower-like cutaneous nodules on the feet and lower legs. The disease generally remains localized with slow peripheral extension. Lymphatic spread can occur.

Diagnosis is made by the demonstration of sclerotic bodies, also called copper-penny bodies in tissue sections. Sclerotic bodies appear as round or irregular, dark brown bodies with septae.





First-line management includes itraconazole or terbinafine. Flucytosine used on its own or combined with amphotericin B may also be effective. For small lesions, surgical excision is the therapy of choice.

Solution to Question 45:

Viruses belonging to the filoviridae family, such as Ebola and Marburg, resemble the morphology shown in image 3.

Image 1- Influenza virus - Influenza viruses have a linear, segmented, single-stranded RNA.



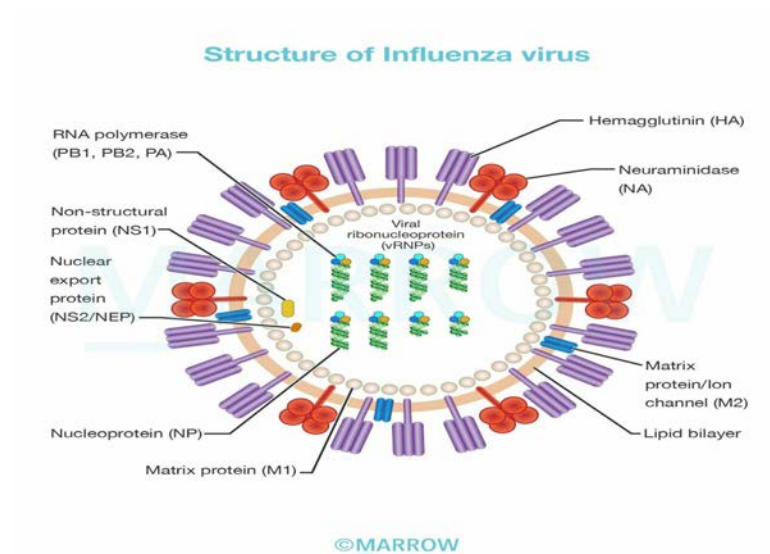
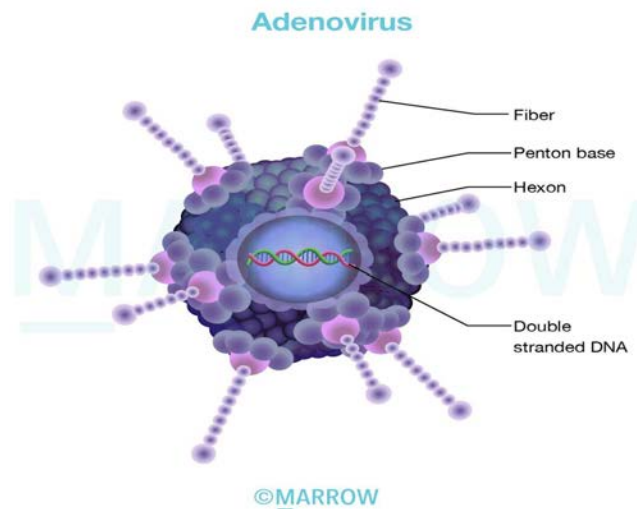


Image 2- Adenovirus contains double-stranded DNA. They have an icosahedral symmetry with structures known as fibers extruding from the vertices.



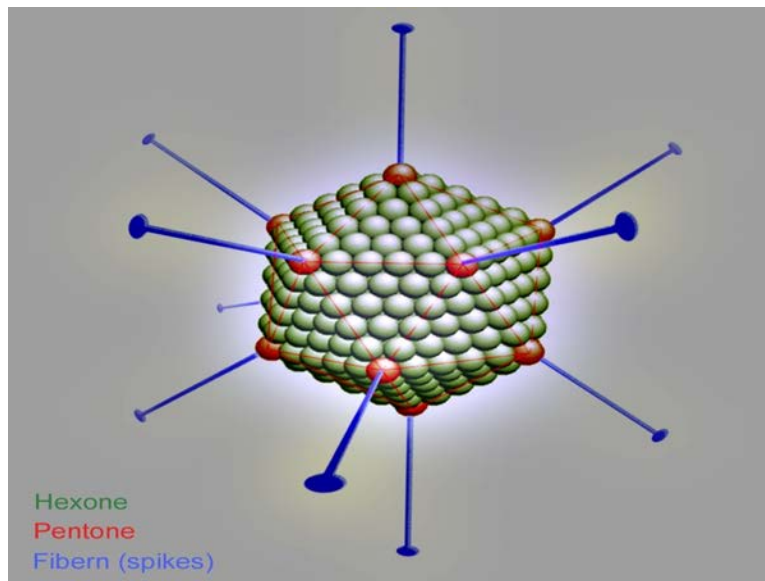


Image 3 - Ebola viruses are filamentous viruses with nonsegmented single-stranded RNA as genetic material.

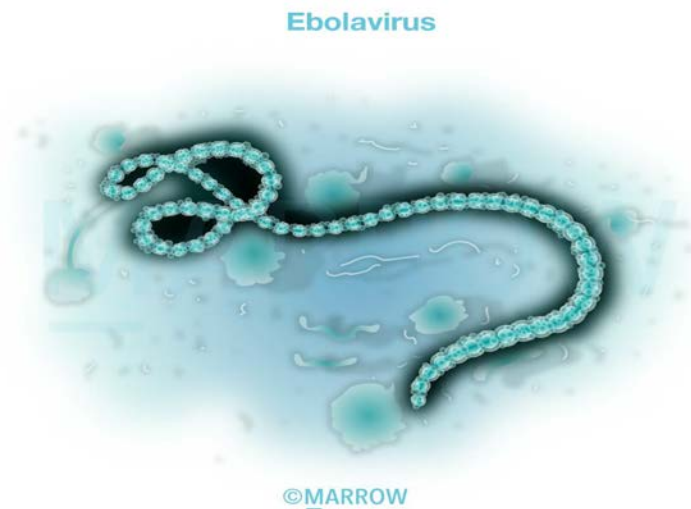
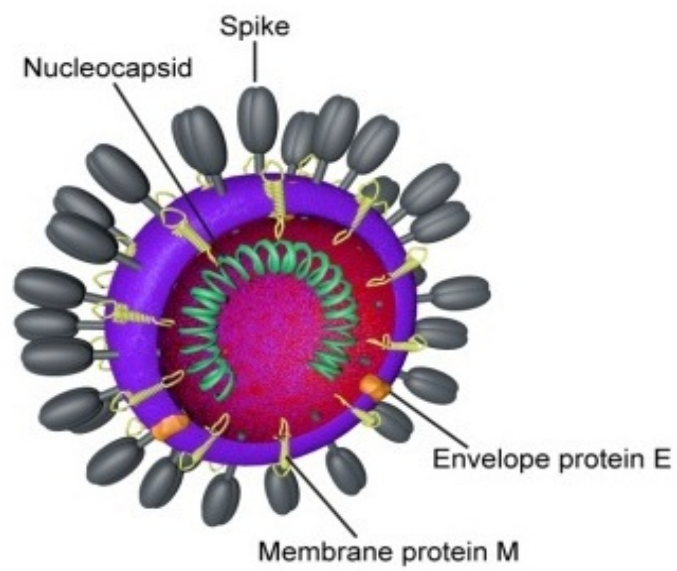
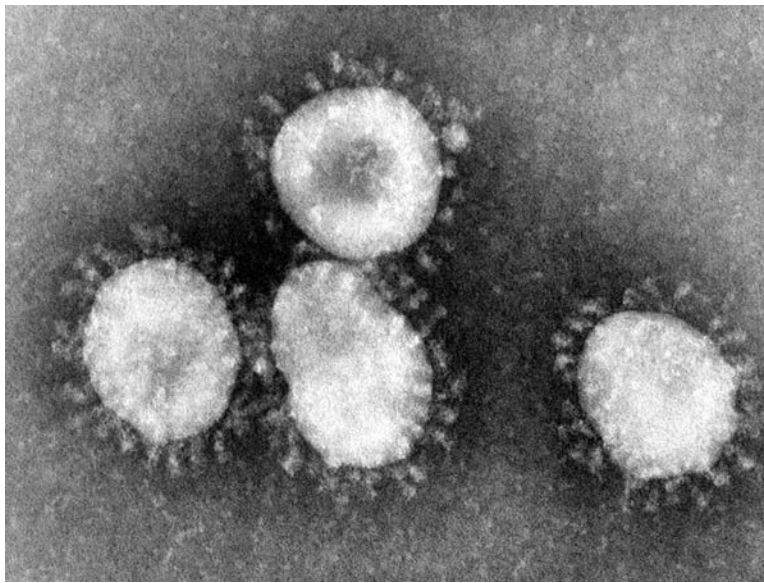


Image 4 - Corona virus belonging to the family are shaped viruses with nonsegmented single-stranded RNA. Corona virus has petal/club-shaped peplomers radiating like sunrays.



Microbiology NEET 2022

Question 1:

A photographer who recently returned from Africa presents to the emergency department with abdominal pain, hepatomegaly, and hemorrhagic manifestations. He died despite treatment, and an autopsy revealed intranuclear Torres bodies in the liver. Which of the following vaccines could have prevented it?

- a) 17D
- b) Nakayama vaccine
- c) Weigl's vaccine
- d) Jeryl Lynn strain

Question 2:

A militant presents with rashes all over his body sparing the palms and soles. On examination, he was febrile and lice were noted. Which of the following is responsible for his condition?

- a) *Rickettsia typhi*
- b) *Rickettsia prowazekii*
- c) *Rickettsia akari*
- d) *Rickettsia conorii*

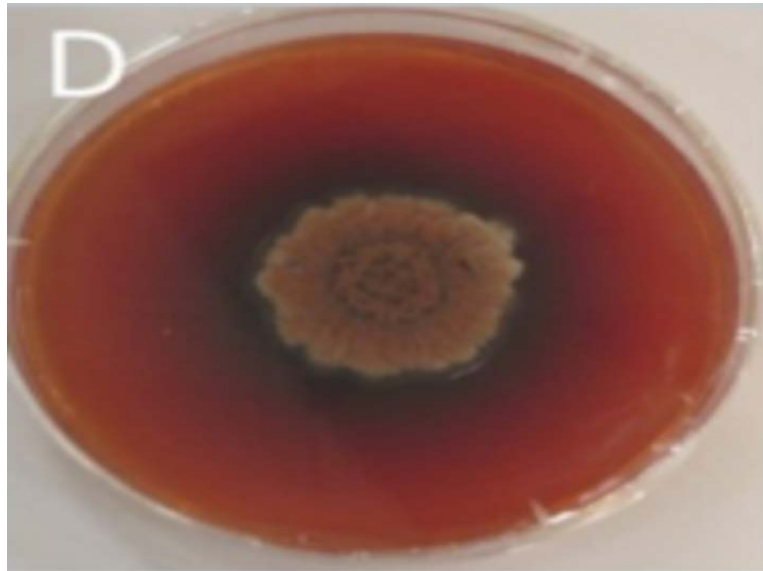
Question 3:

Microfilariae with sheathed tail and two nuclei at the tail tip is suggestive of?

- a) *Wuchereria bancrofti*
- b) *Brugia malayi*
- c) *Loa loa*
- d) *Onchocerca volvulus*

Question 4:

A patient presents with cervical lymphadenopathy and is found to be retropositive. A fungal culture depicts a velvety growth with red diffusible pigment on the underside, as shown below. Which is the most likely causative organism?



- a) *Talaromyces marneffei*
- b) *Blastomyces*
- c) *Aspergillus*
- d) *Pneumocystis jirovecii*

Question 5:

A patient presents with itching in the axilla. On microscopic examination of the skin scrapings, a red pigment-producing fungus with pencil-shaped macroconidia is seen. Which of the following is the most likely organism?



- a) *Trichophyton violaceum*
- b) *Trichophyton rubrum*
- c) *Trichophyton tonsurans*
- d) *Trichophyton schoenleinii*

Question 6:

A female patient presents with complaints of thick white vaginal secretions. Which of the following can be used to identify the likely species of the causative agent?

- a) Bird seed agar
- b) Brain heart infusion agar
- c) CHROMagar
- d) Sabouraud dextrose agar

Question 7:

A 12-year-old boy presents with right upper quadrant pain, calf pain, conjunctival suffusion, icterus, and fever. Examination shows tender hepatomegaly. What is the most likely diagnosis?

- a) Chikungunya
- b) Leptospirosis
- c) Dengue hemorrhagic fever

- d) Encephalopathy caused by hepatitis A

Question 8:

A newly joined teacher develops rice water stools. The causative agent acts on which of the following receptors?

- a) GM1 ganglioside receptor
- b) GM2 ganglioside receptor
- c) Sphingomyelin
- d) Cerebroganglioside

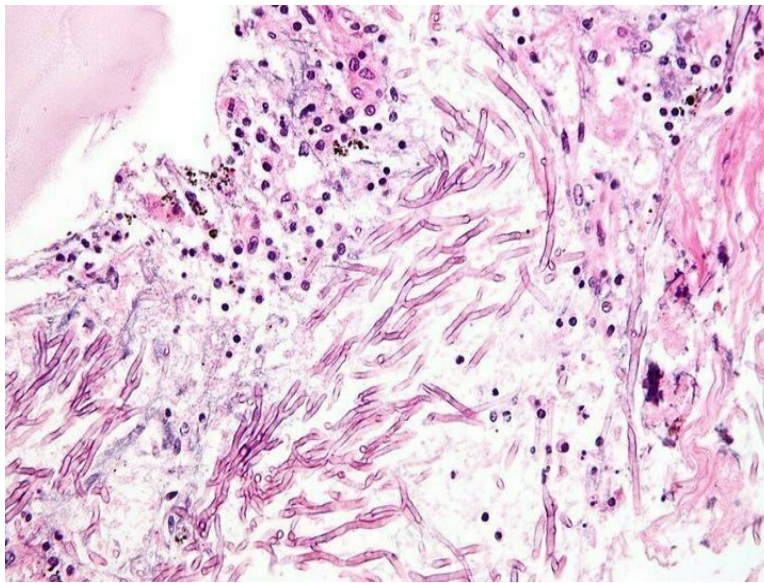
Question 9:

Which of the following is considered to be the best for the detection of *Clostridium difficile*?

- a) Aerobic culture of stool
- b) Glutamate dehydrogenase and toxin assay
- c) Glutamate dehydrogenase assay
- d) None of the above

Question 10:

A diabetic truck driver presents to you with cough and breathlessness. A diagnosis of pneumonia was confirmed, and histopathological examination revealed dichotomous branching, as shown below. What is the likely causative agent?



- a) Rhizopus
- b) Mucor
- c) Candida
- d) Aspergillus

Question 11:

A woman with recurrent diarrhea is prescribed a broad-spectrum antibiotic. Which of the following is not true regarding *Clostridium difficile* infection?

- a) Oral fidaxomicin is used for treatment
- b) It is toxin mediated
- c) IgM assay is used to confirm the diagnosis
- d) Pseudomembrane consists of leukocytes, fibrin, and cellular debris

Answer Key

Question No.	Correct Option
1	a
2	b
3	b
4	a

5	b
6	c
7	b
8	a
9	b
10	d
11	c

Detailed Explanations

Solution to Question 1:

The above clinical vignette is suggestive of yellow fever and the use of 17D vaccine could have prevented this.

17D vaccine is a live-attenuated vaccine administered through the subcutaneous route. It is contraindicated in infants less than 9 months of age, in pregnancy, and in individuals with egg allergy.

Yellow fever is caused by flavivirus and is endemic to certain parts of Africa and America. It occurs in two patterns: the urban cycle and the sylvatic cycle. *Aedes aegypti* is the vector in the urban cycle, while *Aedes africanus* is responsible for transmission in the sylvatic cycle.

Patients present with acute onset fever, headache, and vomiting. Jaundice and hemorrhagic manifestations can also be present. Ultimately, hepatic or renal failure can result in death.

Torres bodies are intranuclear eosinophilic inclusion bodies in hepatocytes. These are rarely seen in yellow fever. To prevent the establishment of the disease in India, strict vaccination and quarantine protocols for travelers from endemic areas are enforced.

Other options:

Option B: Nakayama vaccine is used for immunization against Japanese encephalitis. *Culex tritaeniorhynchus* is the principal vector. There is a sudden onset of fever and vomiting, which later progresses to encephalitis. In India, it was first reported in Vellore and continues to be active in the country.

Option C: Weigl's vaccine is the earliest used vaccine against epidemic typhus caused by *Rickettsia prowazekii*. The main foci of the disease are Europe, Africa, South America, and Asia. Kashmir is the endemic spot in India. *Pediculus humanus corporis* (human body louse) is the vector. Symptoms include fever and chills. A characteristic rash on the trunk and limbs, sparing the face, palms, and soles, may also be seen.

Option D: Jeryl-Lynn strain of the mumps virus is used against the disease as a live-attenuated vaccine. It is contraindicated in pregnancy and individuals allergic to eggs. Mumps present as parotid swelling associated with fever and tenderness.

Solution to Question 2:

The given clinical scenario is suggestive of epidemic or louse-borne typhus caused by *Rickettsia prowazekii*.

Pediculus corporis (human body louse) is the vector for the parasite, although the head louse may also transmit the disease. After the incubation period of 5 - 15 days, the disease typically starts with fever and chills. By around 4 - 5th day of fever, a characteristic rash appears, which starts on the trunk and spreads over the limbs but spares the face, palms, and soles.

The infection can be transmitted by

- Contaminated louse feces rubbed through abrasions caused by scratching
- Inhalation of aerosols of dried louse feces or directly through the conjunctiva

The disease is often severe with the fever lasting for over 2 weeks and is more fatal in patients older than 40 years of age.

Other options:

Option A: *R. typhi* causes murine (flea-borne) or endemic typhus. Rats are the reservoirs of infection and the parasite is transmitted by *Xenopsylla cheopis* (flea). The infection is transmitted when the flea feces is rubbed into the bite wound. The clinical features are similar to that of epidemic typhus but are milder and rarely fatal. Patients can present with fever, headache, myalgia, and rash.

Option C: *R. akari* causes rickettsial pox, which is a mild disease resembling varicella. The disease is transmitted by mites and the patients may present with fever, headache, and a vesicular rash.

Option D: *R. conorii* causes the Mediterranean spotted fever. The parasite is transmitted by ticks and the patients present with fever, headache, rash, and eschar (tache noire).

Solution to Question 3:

The most likely organism based on the description of the microfilariae is *Brugia malayi*.

The microfilariae of *B. malayi* are sheathed and have kinks and secondary curves. The tail tip has two nuclei - a terminal and a subterminal one. It has nocturnal periodicity.

The female filarial worms, which belong to the phylum Nematoda, give birth to larvae known as microfilariae. Some of the microfilariae retain their egg membranes and are known as sheathed microfilariae. They can be further differentiated on the basis of the arrangement of nuclei in the tail.



Classical filariasis is due to obstruction of lymphatics by the adult worms resulting in lymph stasis. This leads to leakage of lymph into the tissues causing hard pitting (brawny) edema. It can present as acute adenolymphangitis, hydrocele, or elephantiasis which is shown below.



Occult filariasis (Meyers-Kouwenaar syndrome) results from a hypersensitivity reaction to the microfilarial antigens. Here, the microfilariae are not found in the blood. Symptoms include hepatosplenomegaly, nocturnal cough, or arthritis. Tropical pulmonary eosinophilia is a manifestation of occult filariasis which presents as low-grade fever, weight loss, and dry nocturnal cough. There is marked eosinophilia and raised serum IgE levels. It is managed with diethylcarbamazine.

Other options:

Option A - Microfilariae of *Wuchereria bancrofti* are sheathed and the tail tip lacks nuclei. It has nocturnal periodicity.

Option C - *Loa loa* causes subcutaneous filariasis which presents as swellings on the extremities (Calabar/fugitive swellings). It can also cause conjunctival granulomas and proptosis. The nuclei are arranged in a row up to the tail tip of the sheathed microfilariae. It exhibits diurnal periodicity.

Option D - *Onchocerca volvulus* (blinding filaria) causes onchocerciasis. The microfilariae are unsheathed and the nuclei do not extend to the tail tip. Patients present with subcutaneous nodules (onchocercoma), dermatitis, photophobia, or even blindness. It is non-periodic.

Solution to Question 4:

The most likely causative organism based on the history and image given is *Talaromyces marneffe* (*Penicillium marneffe*).

T. marneffe is a dimorphic fungus that causes talaromycosis (penicilliosis). It is commonly seen in individuals with advanced AIDS. It is diagnosed by blood culture, which shows a mold with a powdery surface and characteristic red pigment that diffuses into the agar. Histopathological examination reveals yeast-like organisms with central septation.

Clinical features include fever, weight loss, lymphadenopathy, and hepatosplenomegaly. Umbilicated papules that resemble molluscum contagiosum may also be seen.

Mild and moderate infections can be treated with itraconazole. Amphotericin B is recommended for severe infections.

Other options:

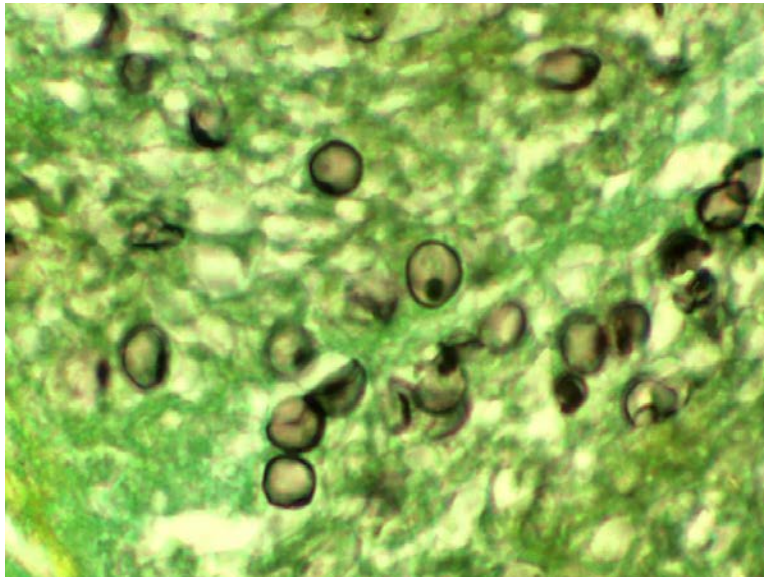
Option B - *Blastomyces* are thermally dimorphic fungi and cause blastomycosis. Primary infection usually affects the lungs, which can later disseminate through the blood to form multiple abscesses at sites such as skin and bone. Culture at 25–30 degrees Celsius promotes hyphal growth. Microscopic examination of stained specimens shows broad-based budding yeast cells.

Option C - *Aspergillus* colonies have a velvety powdery surface, of which the color varies according to the species. *Aspergillus fumigatus* is seen as dark green colonies, while *Aspergillus niger* appears black colored. Yellow-green growth is seen in the case of *Aspergillus flavus*. On microscopy, dichotomous branching at acute angles seen after staining is characteristic. It can cause invasive aspergillosis in immunocompromised patients.

The image below shows *Aspergillus niger* colonies.



Option D - *Pneumocystis jirovecii* cannot be cultured. On being stained by methenamine silver, the cyst wall appears black under the microscope. Trophozoites can also be seen after staining. Clinical features include dyspnea, fever, and cough. It can cause lethal pneumonia in immunocompromised individuals. Diffuse bilateral perihilar interstitial infiltrates can be seen on chest radiograph.



Solution to Question 5:

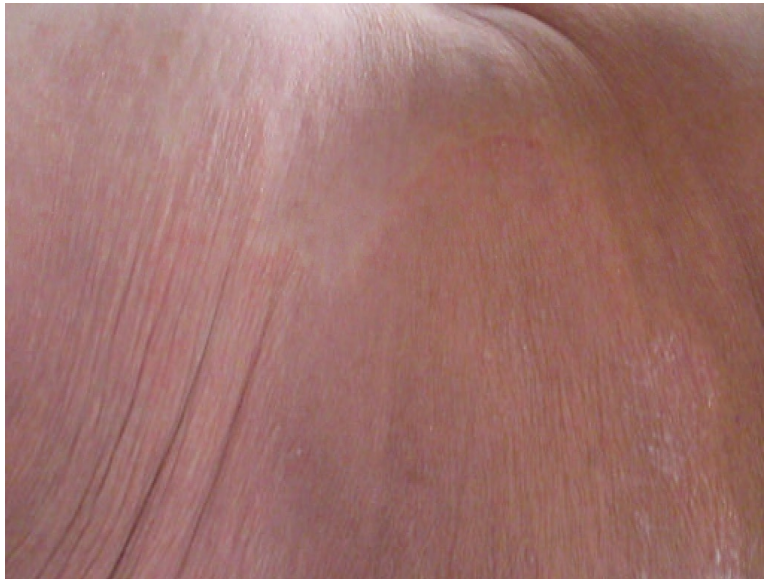
The most likely organism based on the given history and image is *Trichophyton rubrum*. The colonies appear velvety and a deep red, non-diffusible pigment is seen when viewed from the reverse side of the colony. A few long and pencil-shaped macroconidia can be seen.

Trichophyton is a dermatophytic fungi that can infect hair, skin, or nails. Dermatophytes infect only the keratinized tissue and have keratinases that can digest keratin. There are 3 genera of

dermatophytes:

- Trichophyton - The colonies may appear powdery, waxy, or velvety with pigmentation characteristics of different species.
- Epidermophyton - It infects the skin and nails, but not the hair. The colonies appear powdery and greenish-yellow.
- Microsporum - It infects the hair and skin, but not the nails. The colonies may appear cotton-like, velvety, or powdery, with white to brown pigmentation.

T. rubrum causes tinea corporis. It results from the invasion and proliferation of the causal fungi in the stratum corneum, with lesions on the trunk and limbs. The lesions are usually well-defined and circular with a raised edge. It can also cause tinea pedis, tinea cruris, and onychomycosis.

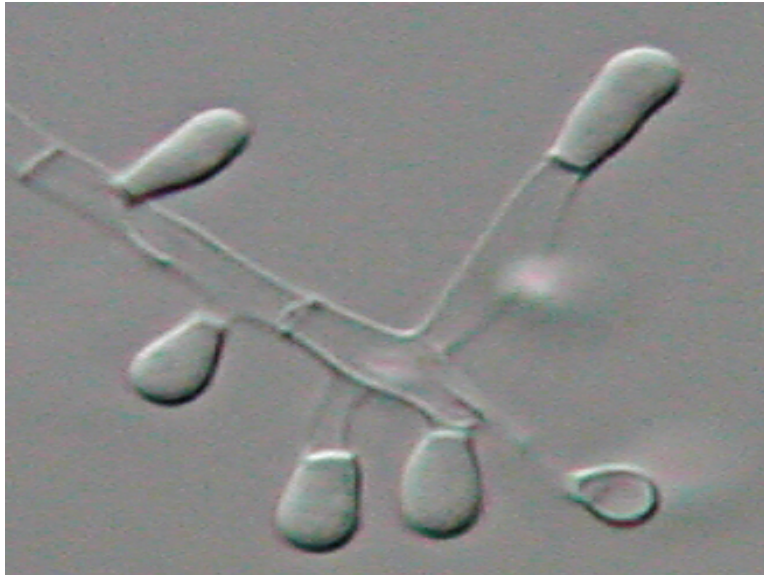


Other options:

Option A: The colonies of *T. violaceus* are very slow-growing, waxy with a violet/purple pigment. They have distorted hyphae, with conidia seen rarely.



Option C: *T. tonsurans* has a cream or yellowish colony with central furrows. Microconidia are abundantly seen. Thick-walled, irregular macroconidia are also seen.



Option D: Colonies of *T. schoenleinii* are smooth, waxy and brownish. Hyphal swellings are seen along with chlamydospores and favic chandelier.



Solution to Question 6:

The clinical scenario points to a diagnosis of candidiasis. CHROMagar can be used for the rapid identification of *Candida* species. It is based on the fungal enzymatic action on various chromogenic substrates in the medium.

After incubation on CHROMagar for 1–4 days, colonies are seen: *Candida albicans* colonies appear green, as shown below; *Candida glabrata* colonies appear dark purple; and *Candida parapsilosis* colonies have a pinkish hue.



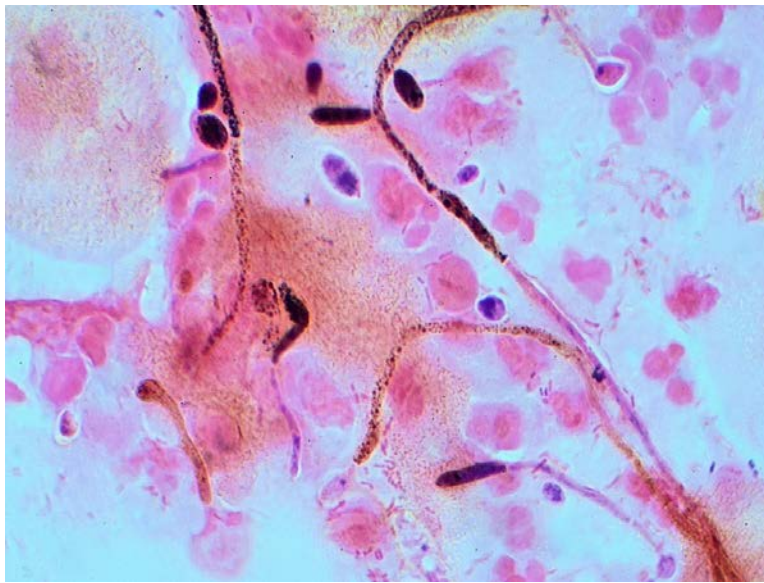
The most common species implicated in candidal infections are *C. albicans*, *C. parapsilosis*, and *C. glabrata*. Vaginal mucosal invasion by the organism causes vulvovaginitis, marked by pruritus and vaginal discharge. A white pseudomembranous patchy lesion called thrush can occur on the tongue, lips, or gums.



Immunocompromised individuals are at an increased risk of systemic candidiasis, which includes candidal endocarditis and urinary tract infections.

It appears as budding gram-positive cells on wet films and smears. Mycelial forms suggest colonization. *Candida albicans*, the most commonly implicated pathogenic species form chlamydospores on corn meal agar.

The image below shows candidal hyphae and chlamydospores.



Mucocutaneous manifestations are managed with topical nystatin or oral ketoconazole.

Other options:

Options A and B: Bird seed agar and brain heart infusion agar are non-specific media used in fungal culture.

Option D: Sabouraud dextrose agar is the traditional mycological medium, which contains glucose and modified peptones. It supports the growth of fungi, while restricting bacterial growth due to the addition of antibacterials.

Solution to Question 7:

The most likely diagnosis based on the clinical vignette is leptospirosis.

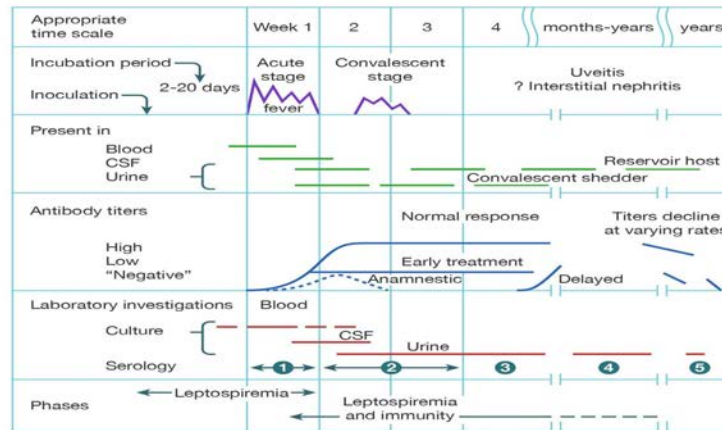
Leptospirosis is a zoonosis caused by spirochetes of the genus *Leptospira*. Patients with mild disease may present with flu-like symptoms of sudden onset, fever with chills, headache, abdominal pain, conjunctival suffusion, and myalgia. The muscle pain usually affects the calves, abdomen, and back. Mild cases can be treated with oral doxycycline, ampicillin, or amoxicillin.

Severe leptospirosis is rapidly progressive and is associated with high fatality. The classic presentation of the severe disease called Weil's syndrome includes the triad of hemorrhage, jaundice, and acute kidney injury. Jaundice when present can be profound and physical examination may reveal an enlarged, tender liver. Temperature-pulse dissociation /relative bradycardia can be seen. Moderate to severe cases are treated with intravenous penicillin, ampicillin, or ceftriaxone.

The image below shows icterus and conjunctival suffusion in a patient with leptospirosis.



Progression of leptospirosis

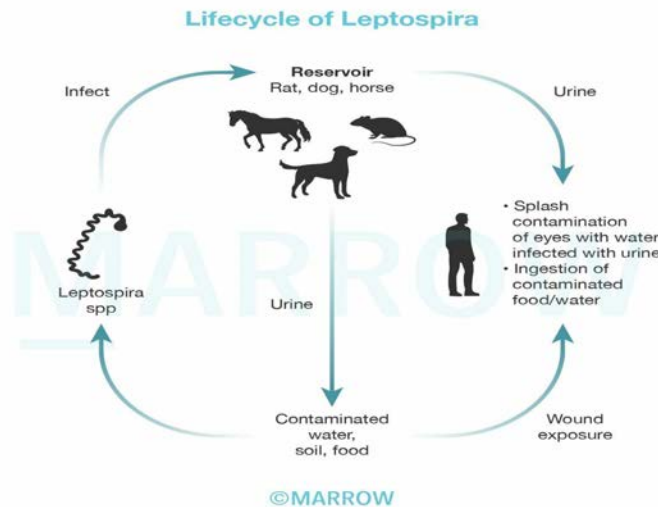


©Marrow

Leptospira are actively motile organisms and are best seen using a dark-field microscope. They are thin, tightly coiled, and flexible spirochetes with one end forming a hook. Diagnosis is by microscopic agglutination test.

Transmission occurs through cuts or breaks in the skin and mucous membranes (conjunctival and oral mucosa). Several outbreaks have been reported that resulted from exposure to floodwaters contaminated by urine from infected animals. The organism enters the bloodstream and lodges in the parenchymatous organs (liver and kidneys) leading to hemorrhage and necrosis of the tissue causing dysfunction of the organs.

The lifecycle of Leptospira is shown below.



Illness is classically biphasic after an incubation period of up to 1 month. The stages are:

- Acute leptospiremic phase- the organism can be cultured from blood.
- Resolving immune phase- the organism can be cultured from urine.

The stages may not be clinically distinguishable.

Other options:

Option A - Chikungunya is a viral disease that presents with saddle-back fever, conjunctival suffusion, and photophobia. Migratory arthralgia, especially involving small joints may also be seen. A maculopapular or petechial rash is frequently seen on the trunk.

Option C - Dengue hemorrhagic fever is a form of severe dengue with bleeding manifestations. Evidence of plasma leakage and thrombocytopenia is also generally seen.

Option D - Encephalitis due to Hepatitis A is seen in fulminant hepatitis, a rare complication. Patients present with confusion, disorientation, ascites, and edema.

Solution to Question 8:

The given clinical scenario of rice-water stools is suggestive of cholera, caused by *Vibrio cholerae*. The toxin produced by the organism acts on GM1 ganglioside receptors.

Vibrio cholerae is a comma-shaped, gram-negative bacillus with darting motility. It has a single polar flagellum. It causes cholera, an acute diarrheal disease. It is marked by painless, profuse, watery diarrhea with flecks of mucus (rice water stools) associated with effortless vomiting. Pulmonary edema and arrhythmias are some of the complications.

Infection is acquired by ingestion of contaminated food and water. The organisms produce an enterotoxin known as cholera toxin (CT). It has 1 A and 5 B subunits. The B subunit (binding units) fixes to the GM1 ganglioside receptors on jejunal epithelial cells. The A subunit dissociates into the A2 fragment, which attaches the A1 fragment to the B subunit. This results in prolonged accumulation of cAMP, which causes an outpouring of large amounts of water and electrolytes

into the intestinal lumen. Ultimately, it results in profuse watery diarrhea.

In addition to the cholera toxin, additional toxins such as zona occludens toxin and accessory cholera toxin are also produced. The former alters intestinal permeability by its action on intestinal epithelial cells.

The characteristic darting mobility and its hindrance by antiserum can be used for rapid microscopic diagnosis.

Fluid resuscitation by rice-based ORS is recommended. Medical management includes erythromycin or azithromycin.

Solution to Question 9:

A combination of glutamate dehydrogenase and toxin assay can be used for the detection of *Clostridium difficile*.

Clostridium difficile is a gram-positive anaerobic bacillus that causes *Clostridioides* (formerly *Clostridium*) *difficile* infection after ingestion of spores of the organism. The more severe form of the condition is known as pseudomembranous colitis (PMC). Enzyme immunoassay to detect glutamate dehydrogenase is less specific compared to toxin detection for diagnosis. So a combination of the two can be used for diagnostic confirmation.

The condition is most commonly associated with prolonged antimicrobial use, which interferes with the commensal colonic microflora. The most commonly implicated antibiotics are cephalosporins, ampicillin, and clindamycin. The spores are spread by contact with the feces of affected individuals or asymptomatic carriers. It can persist on surfaces for months. Risk factors include poor hand hygiene, old age, and gastrointestinal surgery.

The spores germinate in the small intestine and produce two toxins—toxin A (enterotoxin) and toxin B (cytotoxin). These interfere with the epithelial-cell barrier function and result in diarrhea and pseudomembrane formation. The pseudomembranes appear as small whitish-yellow plaques initially which later coalesce to involve the entire colon wall. It contains necrotic leukocytes, fibrin, and cellular debris. The anamnestic IgG response determines which patients develop the disease or remain asymptomatic.

Symptoms include diarrhea, fever, and abdominal pain. The diagnosis is based on a combination of clinical criteria and detection of toxin or the organism in stool samples. This can be done by nucleic acid amplification testing (NAAT) methods such as polymerase chain reaction (PCR) or by culture.

Discontinuation of any ongoing antimicrobials and adequate hydration are general measures that are recommended. Oral fidaxomicin or vancomycin is the first-line treatment. Oral metronidazole can be used when these are not available.

Other options:

Option A: Aerobic culture of stool is the most sensitive test and is highly specific. It is not used practically as the turnaround time is slow.

Solution to Question 10:

The most likely causative agent based on the clinical features and image showing dichotomous branching at acute angles given is *Aspergillus*.

Aspergillus fumigatus is the most common pathogenic species. It can cause invasive aspergillosis in immunocompromised patients. This can be due to prolonged treatment with steroids or retro-positive status. It starts as pneumonia, which can later spread to involve other organs. Voriconazole is the drug of choice.

Other manifestations include allergic bronchopulmonary aspergillosis where central bronchiectasis is usually seen. Aspergilloma and superficial infections like otomycosis or mycotic keratitis are the other presentations.

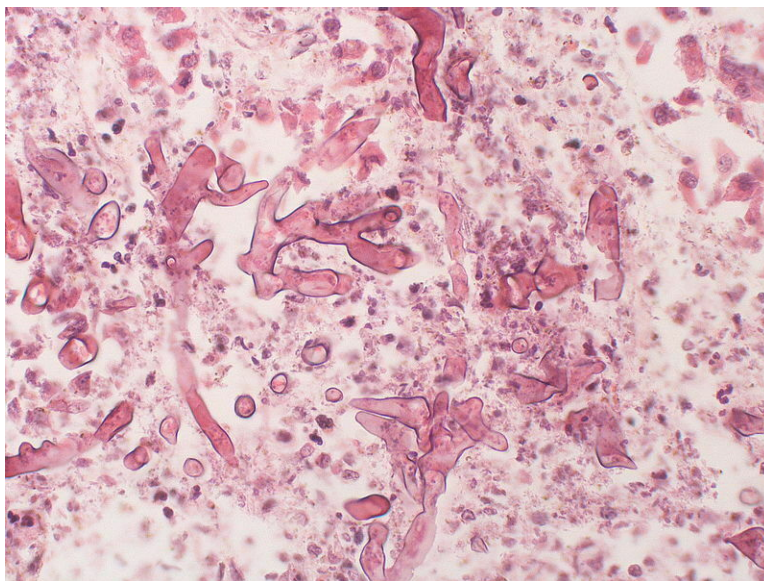
Histopathological examination of the tissue section shows characteristic dichotomous branching at an acute angle. The hyphae stain poorly with hematoxylin and eosin, so periodic acid–Schiff (PAS) stain is used.

Other methods for diagnosis include detection of galactomannan (*Aspergillus* cell-wall component) and culture isolation. *Aspergillus fumigatus* grows dark green colonies, while *Aspergillus niger* appears black. Yellow-green colonies are seen in case of *Aspergillus flavus*. Microscopic examination of colonies stained with lactophenol cotton blue would show septate hyphae and conidiophores.

Other options:

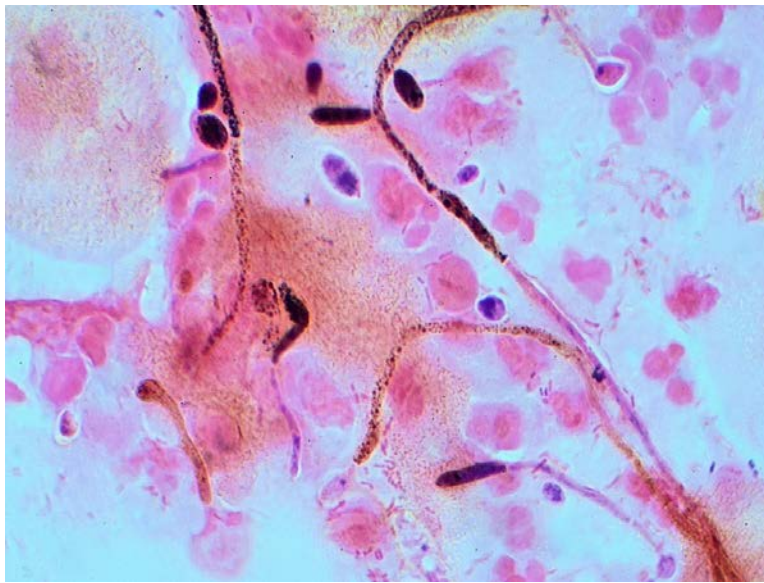
Options A and B - *Rhizopus* and *Mucor* are zygomycetes that are responsible for mucormycosis. Non-septate, broad hyphae are seen on tissue samples stained with hematoxylin and eosin.

The image below shows broad aseptate and irregularly-branched non-septate hyphae of *Mucor*.



Option C - *Candida*, the causative agent for candidiasis, appear as budding gram-positive cells on wet films and smears. Mycelial forms suggest colonization. Culture colonies are creamy-white and have a yeasty odor. *Candida albicans*, the most commonly implicated pathogenic species, form chlamydospores on corn meal agar.

The image below shows candidal hyphae and chlamydospores.



Solution to Question 11:

IgM assay cannot be used to confirm a diagnosis of *Clostridium difficile* infection. The diagnosis is based on a combination of clinical criteria and detection of toxin or the organism in stool samples.

C. difficile, a gram-positive anaerobic bacillus, causes *Clostridioides* (formerly *Clostridium*) *difficile* infection (CDI) after ingestion of spores of the organism. The more severe form of the condition is known as pseudomembranous colitis (PMC). It is most commonly associated with prolonged antimicrobial use which interferes with the commensal colonic microflora. The most commonly implicated antibiotics are cephalosporins, ampicillin, and clindamycin.

The spores are spread by contact with the feces of affected individuals or asymptomatic carriers. It can persist on surfaces for months. Risk factors include poor hand hygiene, old age, and gastrointestinal surgery.

The spores germinate in the small intestine and produce two toxins - toxin A (enterotoxin) and toxin B (cytotoxin). These interfere with the epithelial-cell barrier function and result in diarrhea and pseudomembrane formation. The pseudomembranes appear as small whitish-yellow plaques initially which later coalesce to involve the entire colon wall. It contains necrotic leukocytes, fibrin, and cellular debris. The anamnestic IgG response determines which patients develop the disease or remain asymptomatic.

Symptoms include diarrhea, fever, and abdominal pain. The diagnosis is based on a combination of clinical criteria and detection of toxin or the organism in stool samples. This can be done by nucleic acid amplification testing (NAAT) methods such as polymerase chain reaction (PCR) or by culture. Some cases of PMC can be visualized through endoscopy.

Discontinuation of any ongoing antimicrobials and hydration are general measures that are recommended. Oral fidaxomicin or vancomycin is the first-line treatment. Oral metronidazole can be used when these are not available.

Microbiology NEET 2023

Question 1:

A patient hailing from Delhi presents with fever, arthralgia, and extensive petechial rash for 3 days. Lab investigations revealed a hemoglobin of 9 g/dL, a white blood cell count of 9000 cells/mm³, a platelet count of 20000 cells/mm³, and a prolonged bleeding time. The clotting time was normal. What is the most likely diagnosis?

- a) Dengue
- b) Malaria
- c) Scrub typhus
- d) Typhoid

Question 2:

Which of the following statements is true about *Trichomonas vaginalis*?

- a) It cannot be cultured
- b) Twitching motility is seen on wet saline mount
- c) Cysts are seen on wet saline mount of vaginal secretions
- d) It is not a sexually transmitted infection

Question 3:

A lady from West Rajasthan presented with an ulcer surrounded by erythema on the right leg. Microscopy of the biopsy from the edge of the ulcer showed organisms with dark staining nuclei and kinetoplast. What is the most likely causative agent?

- a) *Leishmania tropica*
- b) *Babesia*
- c) *Trypanosoma*
- d) *Histoplasma*

Question 4:

A child presents with a fever and a rash. Urine examination showed cells with owl's eye appearance. What is the most likely diagnosis?

- a) Cytomegalovirus
- b) Epstein-Barr virus
- c) Herpes simplex virus
- d) Toxoplasma gondii

Question 5:

Infection with *Clonorchis sinensis* is associated with an increased risk of _____.

- a) Cervical cancer
- b) Cholangiocarcinoma
- c) Gastric carcinoma
- d) Bladder carcinoma

Question 6:

A patient presented with complaints of persistent cough and weight loss. He had a history of contact with multiple sexual partners. He was diagnosed as HIV positive 6 months back and is poorly adherent to his medications. The fungal culture of sputum was negative. On Gomori methamine silver staining, the cells showed darkly stained crescent-shaped cysts. What is the most likely causative organism?

- a) *Paracoccidioides brasiliensis*
- b) *Histoplasma capsulatum*
- c) *Coccidioides immitis*
- d) *Pneumocystis jirovecii*

Question 7:

A sewage worker presented with abdominal pain, jaundice, conjunctival suffusion, and blood in the urine for the past 5 days. Which of the following is the investigation of choice?

- a) Widal test
- b) Microscopic agglutination test
- c) Weil-Felix reaction

d) Paul-Bunnell test

Question 8:

Which of the following laboratory test is used to diagnose rotavirus diarrhea?

- a) Antigen detection in the stool sample**
- b) Antigen detection in blood**
- c) Antibody detection in serum**
- d) Light microscopy of stool specimen**

Question 9:

A boy presented with a fever and chills. Rapid test was positive for specific antigen HRP-2. Which of the following species of *Plasmodium* is the most likely causative agent?

- a) *Plasmodium falciparum***
- b) *Plasmodium malariae***
- c) *Plasmodium vivax***
- d) *Plasmodium ovale***

Question 10:

A child was brought with complaints of high fever, multiple seizures, headache, and neck rigidity. CSF analysis showed low glucose, high protein, and many polymorphs. The CSF culture showed pleomorphic gram-negative bacilli. What is true about the causative organism?

- a) It is bacitracin sensitive.**
- b) Produces non-lactose fermenting colonies on MacConkey agar**
- c) Exhibits satellitism around *Staphylococcus aureus* colonies**
- d) It exhibits alpha hemolysis on sheep blood agar**

Question 11:

A patient presented with a high fever, altered sensorium, headache, neck rigidity, and seizures. He was diagnosed with HSV encephalitis. Which of the following tests can be used

to confirm the diagnosis?

- a) IgM in CSF
- b) Tzanck smear of CSF
- c) CSF PCR for viral DNA
- d) CSF culture on chick embryo lines

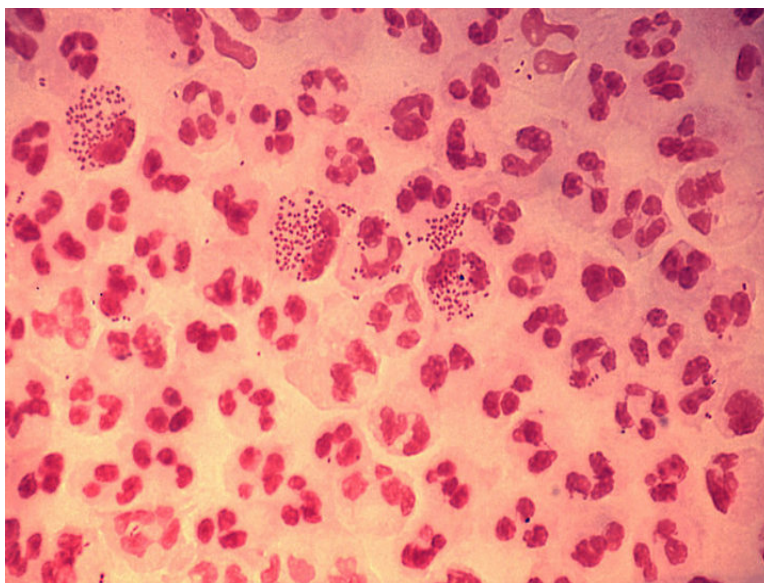
Question 12:

In a patient presented with a fever and a positive filarial antigen test, what is the next appropriate method of management?

- a) Bone marrow biopsy
- b) DEC provocation test
- c) Detection of microfilariae in the blood smear
- d) Ultrasound of the scrotum

Question 13:

A man presents with a history of dysuria and urethral discharge after having unprotected sexual intercourse. Gram-staining of the purulent discharge is seen in the following image. What is the treatment of choice for this infection?



- a) Erythromycin
- b) Azithromycin

- c) Ceftriaxone
- d) Penicillin G

Answer Key

Question No.	Correct Option
1	a
2	b
3	a
4	a
5	b
6	d
7	b
8	a
9	a
10	c
11	c
12	c
13	c

Detailed Explanations

Solution to Question 1:

The given clinical vignette is suggestive of dengue fever.

Dengue is an acute febrile disease caused by the mosquito-borne dengue viruses (DENVs), consisting of four serotypes (DENV 1 to 4), that are members of the Flaviviridae family, genus Flavivirus.

Dengue virus is transmitted from person to person by *Aedes aegypti* mosquitoes.

After an incubation period of 3–15 days, classical dengue fever begins with an abrupt onset of high fever. The condition is self-limiting though debilitating with headache, retro-orbital pain, myalgia, arthralgia, petechiae rash, and leucopenia. A macular-papular recovery rash appears 3–4 days after the onset of fever, and it usually starts on the trunk before spreading peripherally. The rash in dengue appears as 'white islands in a red sea' due to erythema with petechiae.

Shown below is an image of the rash in dengue:



Solution to Question 2:

Trichomonas vaginalis has a jerky or twitching type of mobility on the wet saline mount. Option B is the correct answer.

T. vaginalis is the only protozoan with one stage in its life cycle, i.e., trophozoite form (no cyst stage is seen). It is an obligate parasite and cannot live without close association with vaginal, urethral, or prostatic tissues. Sexual transmission is the main route of infection. Other modes are fomites, and babies during passage through the infected birth canal.

The incubation period ranges from 4 days to 4 weeks.

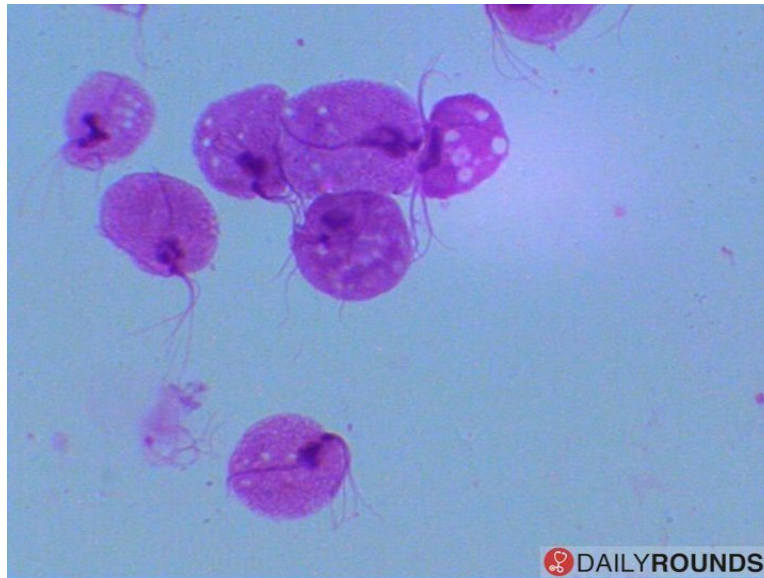
The common presentation of trichomoniasis is foul-smelling frothy vaginal discharge, itching, dysuria, and dyspareunia. Petechial hemorrhage and mucosal capillary dilation give the appearance of a strawberry cervix. Chicken-like epithelium occurs due to intracellular edema

which is also a characteristic feature.

The gold standard for diagnosis is culture on cysteine-peptone-liver-maltose (CPLM) medium or plastic envelope medium (PEM). The treatment of choice is metronidazole 2 g orally as a single dose.

T. vaginalis infection in women may lead to infertility, preterm birth, and low-birth-weight infants. In men, trichomoniasis has been also associated with infertility through inflammatory damage to the genitourinary tract or interference with sperm function.

The below image shows the trophozoite form of *Trichomonas vaginalis*.



Solution to Question 3:

The above clinical vignette is suggestive of cutaneous leishmaniasis.

Leishmaniasis is a slowly progressing indigenous disease caused by a protozoan parasite of the genus *Leishmania*. It is transmitted by the bite of an infected female *Phlebotomus* sandfly which inoculates the flagellated promastigotes into the skin of the host in developing countries. In India, it is endemic to Bihar, West Bengal, Jharkhand, and Uttar Pradesh.

Causative organisms include:

- Old world leishmaniasis: *Leishmania donovani*, *Leishmania major*, and *Leishmania tropica*
- New world leishmaniasis: *Leishmania mexicana* complex and *Leishmania braziliensis* complex

The image given below shows a *phlebotomus* sandfly.



The image given below shows Chiclero ulcer/New World cutaneous leishmaniasis.

Cutaneous New World leishmaniasis
(Chiclero ulcer)



The image below shows Lutzomyia sandfly responsible for the New World cutaneous leishmaniasis.



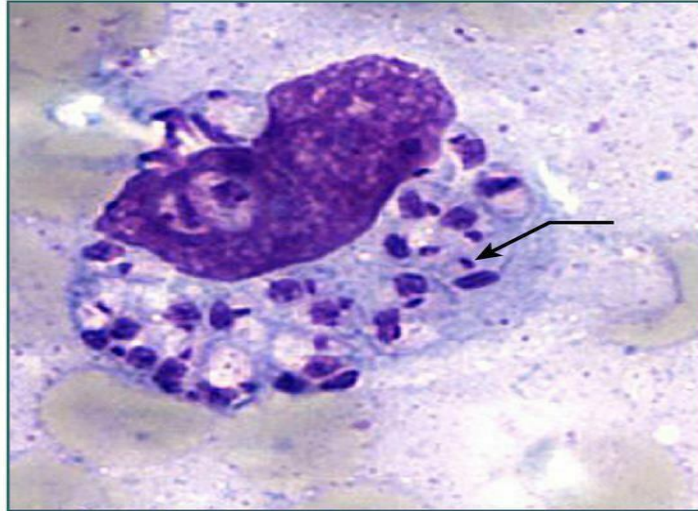
In human beings, the disease presents in four different forms with a broad range of clinical manifestations:

- Visceral leishmaniasis, or kala-azar,
- Cutaneous leishmaniasis,
- Mucocutaneous leishmaniasis, and
- Diffuse cutaneous leishmaniasis.

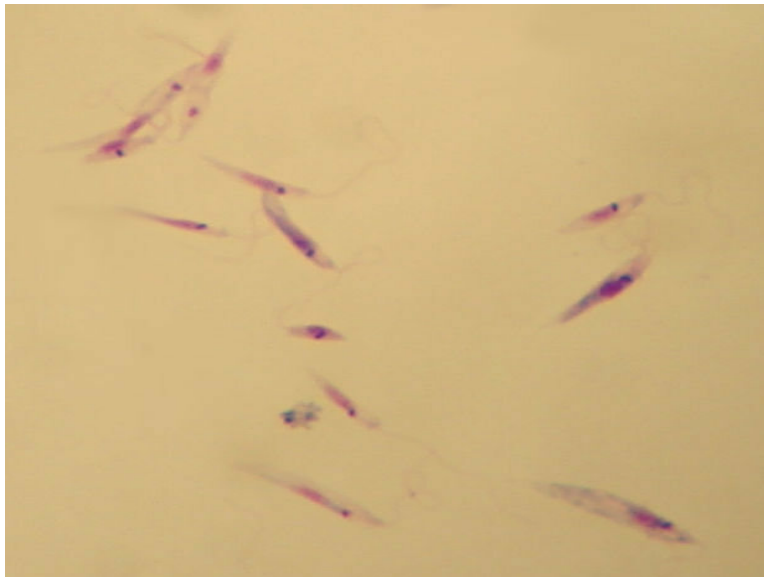
The life cycle of *Leishmania* involves two forms:

- The promastigote which develops and lives extracellularly in the sandfly vector and
- The amastigote multiplies intracellularly in the reticuloendothelial cells of the host.

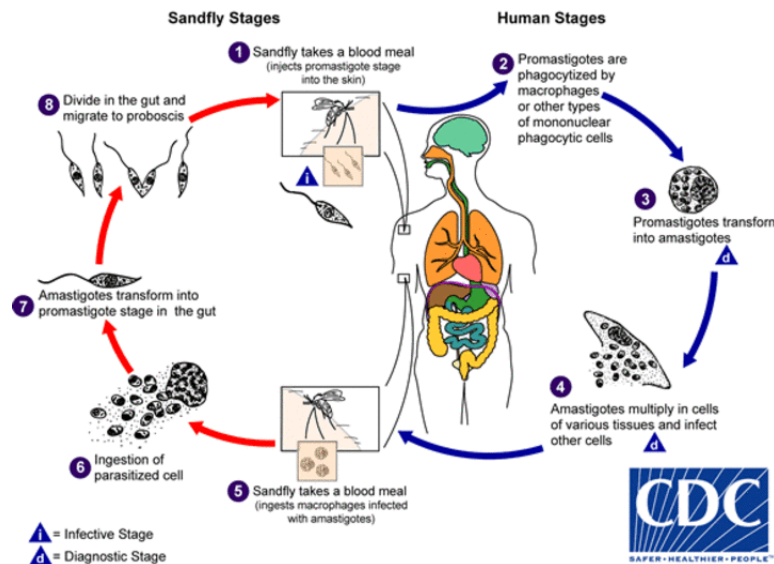
When the infected sandfly bites a person, infectious promastigotes are released into the host dermis along with the sandfly saliva, which potentiates parasite infectivity. Amastigote form (as shown in the image below) is also known as Leishman-Donovan (LD) body. It is an intracellular form found inside macrophages, monocytes, and reticuloendothelial cells. It contains a kinetoplast, which consists of a parabasal body and a blepharoplast. Flagellum is absent.



Promastigote form (as shown in the below image) is present in the midgut of the female sandfly. The kinetoplast lies transversely near the anterior end. A single flagellum is seen. It is the infective form of *L. donovani* for humans.



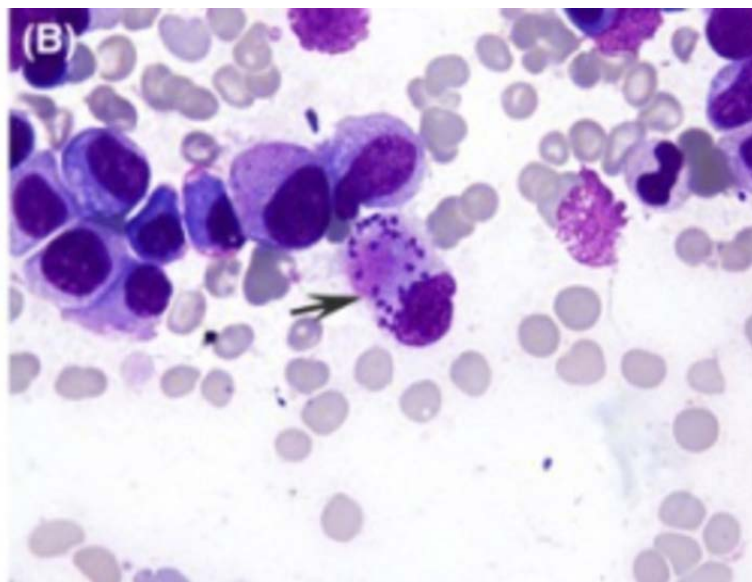
Given below is the lifecycle of *L. donovani*.



This condition causes fever with rigor and chills associated with splenomegaly and hepatomegaly. It lowers immunity, causes persistent fever, anemia, and liver and spleen enlargement, and if left untreated, death may occur due to opportunistic infection.

Diagnosis of kala-azar is mainly by demonstration of amastigotes within macrophages in tissue aspirates such as blood smears, bone marrow, splenic, or lymph node aspirate. It is the gold standard for diagnosis.

The image below shows amastigotes located intracellularly within macrophages.



Tests to detect leishmaniasis include:

- Rapid diagnostic test: rK39 Rapid diagnostic test and Kala-azar dipstick test strip. This is recommended by National Vector Borne Disease Control Programme (NVBDCP) in India.
- Parasitological diagnosis: Demonstration of LD bodies in buffy coat smear - confirmatory.
- Aldehyde test of Napier: Widely used for surveillance.

- Serological tests for diagnosis: Rapid dipstick test, Direct agglutination test, ELISA, indirect fluorescent antibody test
- Leishmanin (Montenegro) test: Induration of 5mm or more is considered positive, usually positive 4-6 weeks (delayed-hypersensitivity) after onset in cutaneous leishmaniasis, valuable for distinguishing immune from nonimmune subjects and endemicity.

Drugs used in the treatment of visceral leishmaniasis are

- Sodium stibogluconate
- Miltefosine
- Amphotericin B

In Bihar, there is resistance to antimony so amphotericin-B-deoxycholate or miltefosine is preferred.

Other options:

Option B: Babesia is a protozoan parasite that causes babesiosis, a tick-borne disease. Babesiosis manifests as a malaria-like illness with fever, chills, and hemolytic anemia.

Option C: Trypanosoma is a genus of protozoan parasites responsible for diseases like African trypanosomiasis (sleeping sickness) and American trypanosomiasis (Chagas disease). These diseases present with different clinical features, such as fever, swollen lymph nodes, and neurological symptoms in African trypanosomiasis, and fever, swelling at the infection site, and cardiomyopathy in Chagas disease.

Option D: Histoplasma is a dimorphic fungus that causes histoplasmosis, a systemic mycosis. It primarily affects the lungs and can disseminate to other organs in immunocompromised individuals. The clinical presentation of histoplasmosis is variable, but it does not typically present as a skin ulcer with erythema as described in the clinical scenario.

Solution to Question 4:

The most likely causative agent is cytomegalovirus (CMV) based on the clinical symptoms and the presence of the owl's eye appearance on urinary examination.

CMV is the largest virus in the herpes family of viruses. It causes the following conditions:

- Immunocompetent individuals - asymptomatic or mononucleosis-like infection such as fever and rash
- Neonates - devastating systemic infections
- Immunosuppressed individuals - serious, life-threatening, disseminated CMV infections of the lungs (pneumonitis) and gastrointestinal tract (colitis)

CMV spreads through contact with salivary secretions and sexual contact, blood transfusion, and organ transplantation.

Laboratory diagnosis is by recovery of the virus from urine, saliva, or other bodily fluids into human fibroblast cells. Antibody demonstration using ELISA is used for diagnosis of primary infection.

A cytomegalic pneumocyte containing characteristic intranuclear basophilic inclusion surrounded by a clear halo known as an owl's eye appearance.

Other options:

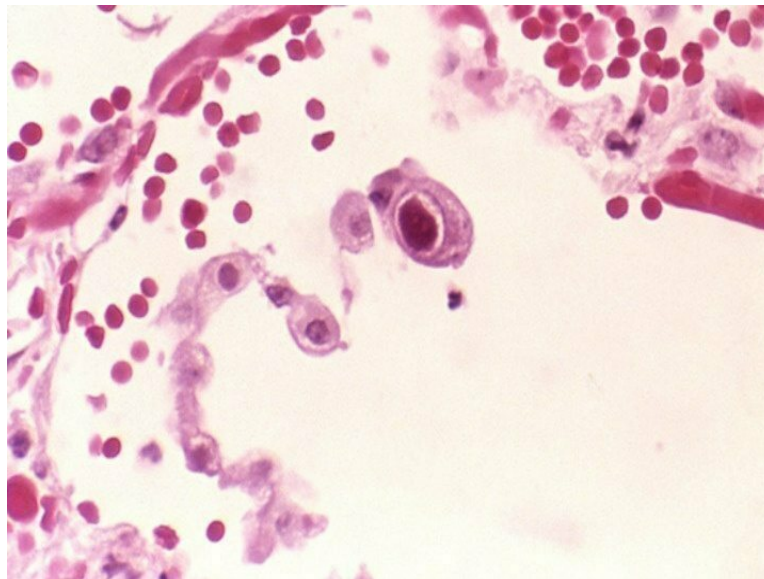
Option B: Epstein Barr virus: Conditions associated with EBV infection are

- Glandular fever (infectious mononucleosis): fever, pharyngitis, and posterior cervical lymphadenopathy
- Hodgkin lymphoma
- Non-Hodgkin lymphoma
- Burkitt's lymphoma
- Nasopharyngeal carcinoma
- EBV-associated lymphoproliferative disease

Option C: Herpes simplex virus causes cutaneous infections like herpes labialis caused by viral reactivation in febrile patients. It presents as blisters around the mouth, cheeks, and forehead. Gingivostomatitis and pharyngitis are the most frequent manifestations of primary herpes infections. HSV esophagitis presents as dysphagia, sub-sternal pain, and weight loss.

Option D: Acute congenital toxoplasmosis may manifest as fever, jaundice, petechial rashes, microphthalmia, cataract, glaucoma, lymphadenopathy, hepatosplenomegaly, myocarditis, cerebral calcifications, and chorioretinitis.

Most infections in adults who are immunocompetent are usually asymptomatic. Active progression of infection is more likely in immunocompromised individuals.



Solution to Question 5:

Clonorchis sinensis or Chinese liver fluke causes clonorchiasis. It is also associated with an increased risk of cholangiocarcinoma.

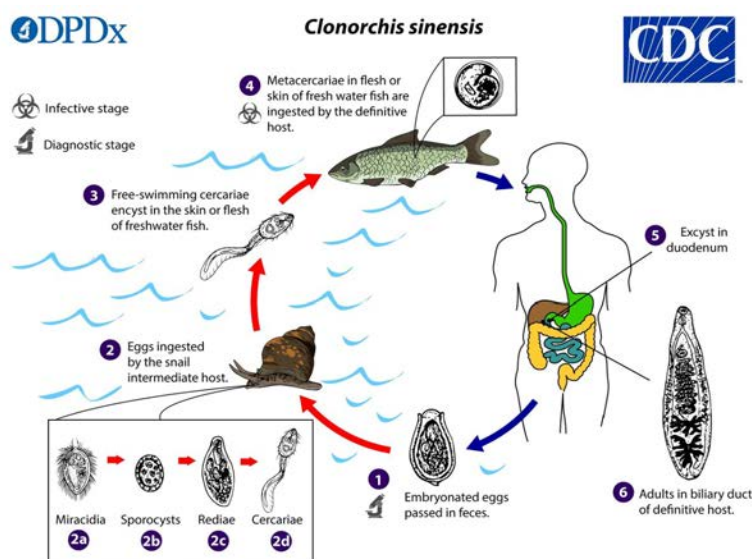
Clonorchis sinensis infection is endemic in south China, Japan, Korea, and Taiwan due to a long tradition of consuming raw freshwater fish or shellfish. Humans are infected by eating raw or undercooked fish containing encysted larvae (metacercariae).

The immature flukes after excystation in the duodenum enter the biliary ducts and differentiate into adults.

The hermaphroditic adults produce eggs that are excreted in the feces. Upon reaching freshwater the eggs are ingested by snails, which are the first intermediate hosts. The eggs hatch within the gut and differentiate first into larvae (rediae) and then into many free-swimming cercariae. Cercariae encyst to form metacercariae under the scales of certain freshwater fish (second intermediate host) which are then ingested by humans.

In humans, it infects the common bile duct and gallbladder. Liver imaging shows a dilated biliary tract. It is a risk factor for cholangiocarcinoma.

The life cycle of *Clonorchis sinensis* is given below:



Other options:

Option A: Cervical cancer - Human papilloma virus (HPV) infection, particularly with high-risk strains like HPV-16 and 18, is the main risk factor for cervical cancer. *Clonorchis sinensis* is not associated with cervical cancer.

Option C: Gastric carcinoma - *Helicobacter pylori* infection is a major risk factor for the development of gastric carcinoma. *Clonorchis sinensis* is not linked to gastric carcinoma.

Option D: Bladder carcinoma - The vesical blood fluke *Schistosoma haematobium* primarily infects the urinary bladder and increases the risk of squamous cell carcinoma of the bladder. *Clonorchis sinensis* is not associated with bladder carcinoma.

Solution to Question 6:

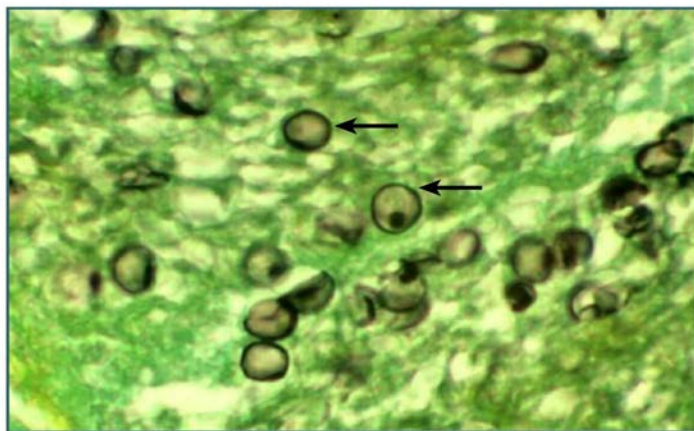
The given clinical vignette is suggestive of pneumonia caused by *Pneumocystis jirovecii*.

P. jirovecii pneumonia (PCP) is an opportunistic infection in immunocompromised patients. It usually occurs when the CD4 count falls below 200 cells/ μ L. Patients usually present with fever and nonproductive cough with progressive shortness of breath. It can also affect other organs like lymph nodes, spleen, and liver.

Laboratory and radiographic findings are also non-specific and show an increase in lactate dehydrogenase levels and symmetric, perihilar, and diffuse bilateral interstitial infiltrates, respectively. The classic radiographic appearance of symptomatic *Pneumocystis pneumonia* consists of diffuse bilateral symmetric interstitial infiltrates that are perihilar. CT will show diffuse ground-glass opacities. Giemsa stain, toluidine blue, methenamine silver, and calcofluor white stains demonstrate trophozoites. The cyst wall appears black with Gomori's methenamine silver stain.

Shown below is an image of the staining showing the black-stained crescent-shaped cysts resembling the crushed ping-pong balls of *P. jirovecii*. The stain used for staining the causative organism is Gomori's methenamine silver stain.

Pneumocystis jirovecii



Trimethoprim-sulfamethoxazole combination (commonly known as co-trimoxazole) is the treatment of choice for PCP, and it is administered for 21 days, followed by a course for secondary prophylaxis. Concomitantly, glucocorticoids should also be administered to reduce pulmonary inflammation.

Patients who are susceptible to PCP are also given co-trimoxazole as prophylaxis. For patients who are unable to tolerate co-trimoxazole, IV pentamidine is the drug of choice.

Prophylaxis is indicated for other opportunistic infections like *Mycobacterium avium* complex (MAC) infection if CD4 counts fall below 50 cells/ μ L; *Coccidioides*, if CD count is $<250/\mu$ L; and *Cryptococcus*, when the CD count is $<100/\mu$ L.

Other options:

Option A: *Paracoccidioides brasiliensis* causes paracoccidioidomycosis, which usually presents with ulcerations in the oral and nasal mucosa, lymphadenopathy, and pulmonary lesions, but it does not form crescent-shaped cysts.

Option B: *Histoplasma capsulatum* causes histoplasmosis and may be associated with similar symptoms as *Pneumocystis jirovecii*, but it does not show crescent-shaped cysts on Gomori methenamine silver staining.

Option C: *Coccidioides immitis* causes coccidioidomycosis, which is characterized by respiratory symptoms, skin lesions, and joint pain. It also does not show crescent-shaped cysts on Gomori methenamine silver staining.

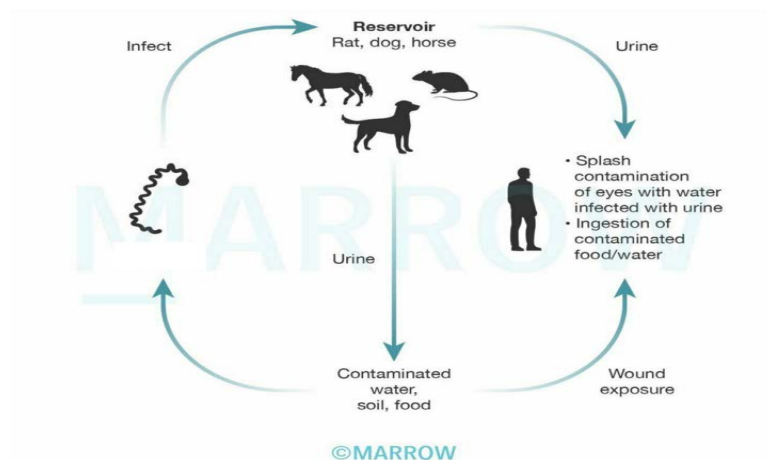
Solution to Question 7:

The given clinical scenario is suggestive of leptospirosis (Weil's disease). A microscopic agglutination test is the investigation of choice for the diagnosis of leptospirosis.

Leptospirosis is a zoonotic disease caused by *Leptospira interrogans*. Humans are infected when the leptospirae in water contaminated with the urine of carrier animals enter the body through cuts or abrasions on the skin or through the intact mucosa of the mouth, nose, or conjunctiva. Hence, a sewage worker is at an increased risk of it.

Clinical features include fever, jaundice, vomiting, headache, and intense injection of the eyes. Some may present as aseptic meningitis or with hepatorenal damage (Weil's disease). There are two clinical types of leptospirosis: icteric and non-icteric. It can also manifest as aseptic meningitis.

The life cycle of *Leptospira* is depicted below:



Solution to Question 8:

Antigen detection in the stool sample is the laboratory test used to detect rotavirus diarrhea.

Rotavirus is the most common identifiable viral cause of diarrhea in children below the age of 5 years and is most frequent between 6-24 months of age. The incubation period is 1-3 days. Most

cases are asymptomatic or with mild symptoms like watery diarrhea, vomiting, low-grade fever, and temporary lactose intolerance.

Diagnosis can be made by rapid antigen detection of rotavirus in stool specimens. Laboratory diagnosis depends on the demonstration of the virus in stool samples which is collected at an earlier course of illness. Antigen enzyme immunoassays (EIAs) and polymerase chain reaction (PCR) are used to detect the virus in the stool. The polymerase chain reaction is the most sensitive detection method for genotyping of rotavirus nucleic acids from stool specimens.

Treatment is supportive including the correction of fluid and electrolyte imbalance. The main goals of treatment are avoiding and treating dehydration. Maintaining nutrition is a secondary goal.

Solution to Question 9:

The given clinical scenario and the HRP-2 positivity are suggestive of *P. falciparum* malaria.

Malaria is a parasitic infection caused by the *Plasmodium* species transmitted by the *Anopheles* mosquito. It is characterized by many nonspecific symptoms, including but not limited to tachycardia, tachypnea, chills, malaise, fatigue, diaphoresis (sweating), and headache. The fever of malaria is characterized by periodicity depending on the infective *Plasmodium* species. The fever returns every other day for *P. vivax*, *P. ovale*, and *P. falciparum*, and every three days for *P. malariae*.

The newer tests include rapid antigen detection tests that employ immunochromatographic methods. The parasite-F test is one of them. This test is based on the detection of histidine-rich protein-2 (HRP-2) which is produced only by the asexual forms of *Plasmodium falciparum* that are present on the surface of RBCs. The test strip has monoclonal antibodies against the HRP-2 antigen. This test has very high sensitivity and specificity. The disadvantages of the test include that it can detect only *Plasmodium falciparum* HRP-2 antigen but not the other three species of *Plasmodium*. Also, the test remains positive until 2 weeks after the cure.

Another diagnostic test of significance is the dual antigen test. This test detects both the parasite lactate dehydrogenase of trophozoites and gametocytes of all the *Plasmodium* species and HRP-2 produced by *P. falciparum* simultaneously.

Solution to Question 10:

This clinical vignette is suggestive of meningitis caused by *Haemophilus influenzae*. *H. influenzae* is a gram-negative pleomorphic bacillus that is non-motile and non-spore-forming. The correct statement about this causative organism is that it exhibits satellitism around *Staphylococcus* colonies, i.e option C.

When present in the sputum, *H. influenzae* appears as a coccobacillus, but in cerebrospinal fluid (CSF), it appears as a filamentous bacillus.

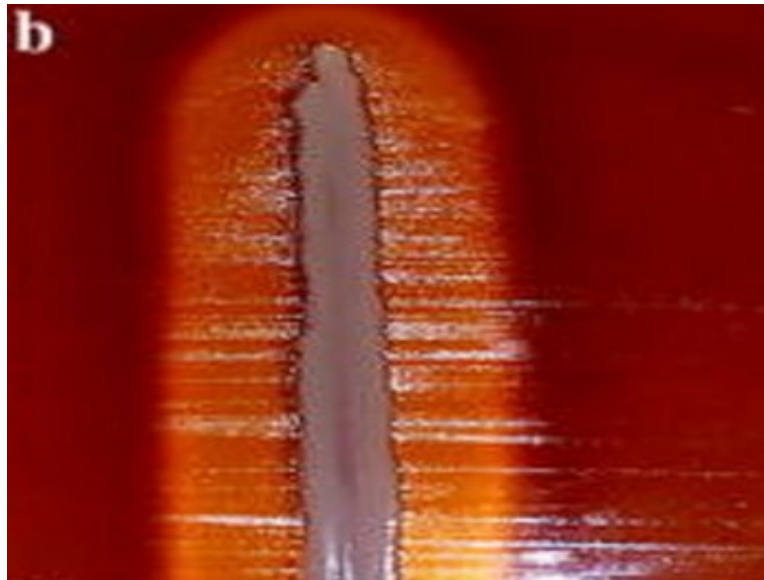
H. influenzae requires growth factors V and X that are present in the blood.

- Factor V is a nicotinamide adenine dinucleotide (NAD), a coenzyme, and is heat labile.

- Factor X is hemin. It is required for cytochrome synthesis and the function of enzymes like catalase and peroxidase. It is heat stable.

When *H. influenzae* is cultured on the same plate as *Staphylococcus aureus*. The colonies of *H. influenzae* that are closer to *S. aureus*, have larger growth. This is termed satellitism (Option C). This occurs because factor V is released by the hemolytic action of *S. aureus*.

The image below shows satellitism.



©MARROW

H. influenzae has a capsular polysaccharide, which is a major antigenic determinant. Capsular typing is done to identify strains. Importantly, the type b capsular polysaccharide has a polyribosyl ribitol phosphate (PRP) antigen. PRP is targeted during immunization.

H. influenzae can also cause meningitis, laryngo-epiglottitis, pneumonia, bronchitis, and suppurative lesions like arthritis, endocarditis, and pericarditis.

Cefotaxime or ceftazidime is the drug of choice for the treatment of *Haemophilus meningitis*. *H. influenzae* produces beta-lactamase, hence resistant to beta-lactam antibiotics like ampicillin.

Solution to Question 11:

HSV-DNA demonstration in CSF by PCR is the investigation of choice for HSV encephalitis.

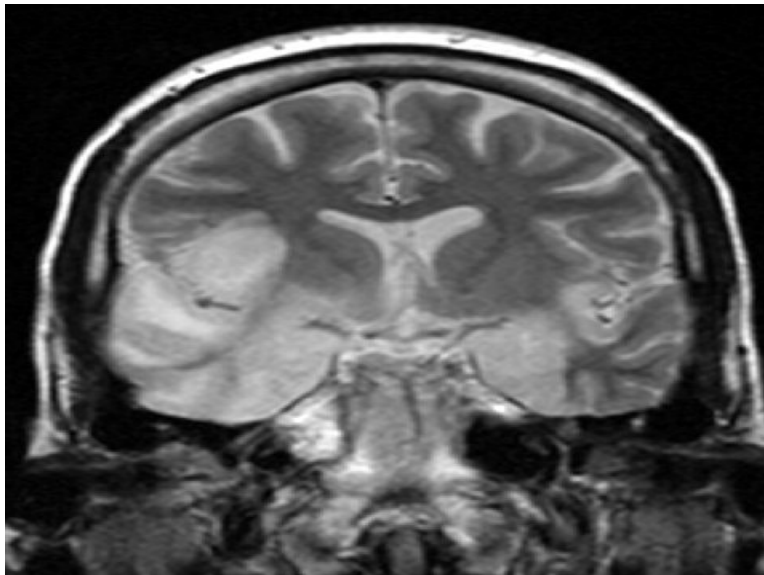
Herpes simplex virus (HSV) encephalitis is defined as inflammation of brain parenchyma caused by the herpes virus, and it typically presents as an acute febrile illness.

Patients typically present with acute onset of fever, altered levels of consciousness, personality changes, and evidence of either focal or diffuse neurologic signs and symptoms, especially in the temporal lobe. Focal or generalized seizures occur in many patients with HSV encephalitis. Other most commonly encountered focal findings include aphasia, ataxia, upper motor neuron or lower motor neuron type lesions, and cranial nerve deficits.

HSV-DNA demonstration in CSF by PCR is the investigation of choice for HSV encephalitis. Other investigations that can help include CSF examination, CSF culture, serological studies, MRI, CT, EEG, and brain biopsy.

The T2-weighted MRI of the brain will show hyperintensity in the temporal lobe in most of the patients. Bilateral involvement of the temporal lobe is more common in HSV encephalitis.

The image below is a coronal T2-weighted MR image showing a high signal intensity in the temporal lobes in a patient with HSV encephalitis.



Management includes supportive therapy (monitoring of intracranial pressure, fluid restriction, avoidance of hypotonic IV fluids), anticonvulsants for seizures, and intravenous acyclovir is beneficial in the treatment of HSV encephalitis.

Other options:

Option A: IgM antibodies in CSF - Serological tests to detect antibodies are used for primary herpes infections.

Option B: Tzanck smear of CSF - A positive smear shows multinucleated giant cells with faceted nuclei and ground glass appearance. It is a rapid, inexpensive, and fairly sensitive diagnostic

test. The Tzanck smear is typically used to diagnose cutaneous herpes infections, such as herpes simplex or varicella-zoster virus. It is not an appropriate diagnostic test for HSV encephalitis, as it does not provide definitive confirmation of the presence of HSV in the CSF.

Option D: CSF culture in chick embryos - It is not a sensitive test used for the diagnosis of HSV encephalitis. Negative CSF viral cultures have no value in ruling out HSV encephalitis. Although this method can be used to isolate HSV, it is not as sensitive or specific as PCR for viral DNA and it can take a longer time to obtain results.

Solution to Question 12:

A patient with fever and filarial antigen test positive is suggestive of filariasis. The next most appropriate step for this patient is the detection of microfilariae in the blood smear.

Classical filariasis occurs due to the obstruction of lymphatics by the adult worms, which results in lymph stasis. The disease initially causes acute attacks of lymphangitis and lymphadenitis. As time goes by the lymphedema worsens and is associated with the thickening of the skin and fibrosis of lymph vessels or blockage by adult worms. The lymphatic vessels of the male genitalia are most commonly affected in bancroftian filariasis, producing episodic funiculitis, epididymitis, and orchitis. Hydrocele or elephantiasis is a very common sign of chronic bancroftian filariasis.

The standard method for diagnosing active infection is the identification of microfilariae in a blood smear by microscopic examination. The microfilariae that cause lymphatic filariasis circulate in the blood at night showing a nocturnal periodicity. Microfilariae can be detected in blood smears of samples obtained at night between 10 PM to 2 AM. A thick smear should be made and stained with Giemsa or hematoxylin and eosin. For increased sensitivity, concentration techniques like membrane filter concentration can be used.

DEC provocation test (Option B) refers to the administration of Diethylcarbamazine (DEC) at a dose of 100 mg or 2 mg/kg body weight orally and subsequently collecting blood samples. This causes the mobilization of microfilariae into the peripheral blood and thus microfilariae can be detected 20-50 minutes after the DEC administration in the sample of peripheral blood.

Although the DEC provocation test has the advantage of blood sample collection even during day time, the guidelines from CDC, NVBDCP, and IAPSM suggest blood sample collection at night, followed by identification of microfilariae in a blood smear by microscopic examination as the standard method for diagnosis (Option C).

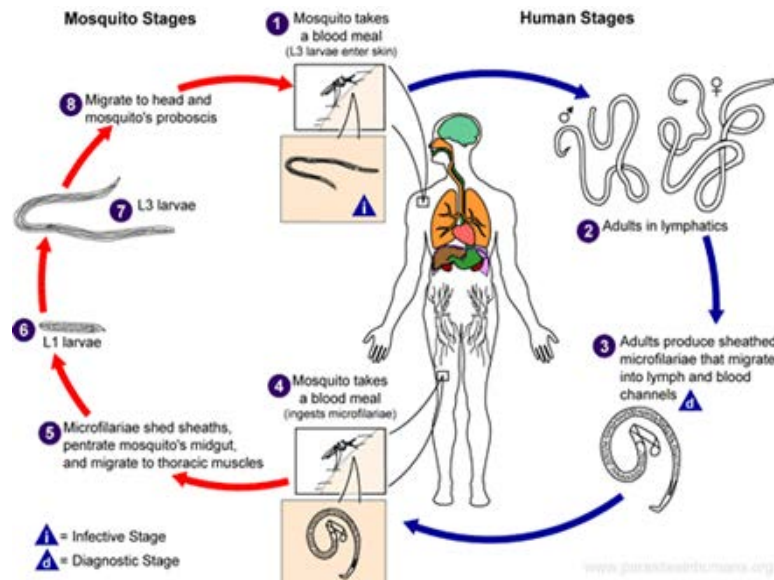
Methods to detect adult worms include:

- Lymph node biopsy to detect adult worms. Bone marrow biopsy is not a standard diagnostic procedure for filariasis, as adult worms typically reside in the lymphatic system rather than the bone marrow (Option A).
- X-ray (if worms are calcified)
- High-frequency ultrasound and doppler of the scrotum (Option D).

Serologic techniques provide an alternative to the microscopic detection of microfilariae for the diagnosis of lymphatic filariasis. Patients with active filarial infection typically have elevated levels of anti-filarial IgG4 in the blood and these can be detected using routine assays.

- Antigen detection by enzyme-linked immunosorbent assay (ELISA) and immunochromatographic test (ICT)
- Antibody detection tests have low sensitivity and specificity
- Molecular diagnosis is done by PCR (polymerase chain reaction). This test is positive only when microfilaria are present in peripheral blood. It is negative in chronic filariasis.

Diethylcarbamazine is the drug of choice. WHO has recently recommended a triple-drug combination of ivermectin, DEC, and albendazole as a faster and more effective alternative to expedite the elimination of filariasis.



Solution to Question 13:

This clinical scenario along with the microscopic findings of gram-negative intracellular diplococci indicates infection with *Neisseria gonorrhoeae*. The treatment of choice for the infection is ceftriaxone.

Gonorrhea is a disease acquired by sexual contact. The first step in infection is the adhesion of gonococci to the urethra or other mucosal surfaces. The incubation period is 2-8 days. The major symptoms include purulent urethral discharge and dysuria.

In males, the disease starts as acute urethritis presenting with purulent discharge. The patients may also have penile lymphangitis when the soft tissue inflammation is extensive. This infection may extend along the urethra to involve the prostate, seminal vesicles, and epididymis.

In females, the initial infection involves the cervix uteri and urethra which may extend to Bartholin glands, endometrium, and fallopian tubes. Pelvic inflammatory disease and salpingitis may lead to sterility. Rarely peritonitis may develop with perihepatic inflammation (Fitz-Hugh-Curtis syndrome).

The discharge is collected and subjected to Gram stain, revealing gram-negative intracellular diplococci within phagocytes. Stuart's transport medium is used. Cultures can be obtained from chocolate agar or Mueller-Hinton agar. Specifically, Thayer-Martin agar is used to

isolate *N. gonorrhoeae*.

Single-agent therapy with ceftriaxone is preferred for the treatment of gonococcal infections.

According to CDC guidelines (2021), treatment of uncomplicated gonococcal infections is as follows:

- Weight <150 kg – Ceftriaxone 500 mg intramuscular (IM) in a single dose
- Weight ≥150 kg – Ceftriaxone 1 g IM in a single dose

The doses of ceftriaxone are higher than the previous guidelines as the gonococcal minimum inhibitory concentrations (MICs) are rising worldwide. The British BASHH guidelines also recommend a high dose of 1g.

Concurrent chlamydial infection must also be treated unless it has been excluded by molecular testing.

First-line drug administered is

- Oral doxycycline 100 mg BD for 7 days

Partner treatment is also required.

Other options:

Option A: Erythromycin: This is not the preferred treatment for gonorrhea, as it is less effective and has a higher rate of resistance compared to ceftriaxone.

Option B: Azithromycin: Although it can be used in combination with ceftriaxone for the treatment of gonorrhea, it is not the treatment of choice as a single agent.

Option D: Penicillin G: This was once a treatment option for gonorrhea, but it is no longer recommended due to widespread resistance.

Microbiology INI-CET 2023

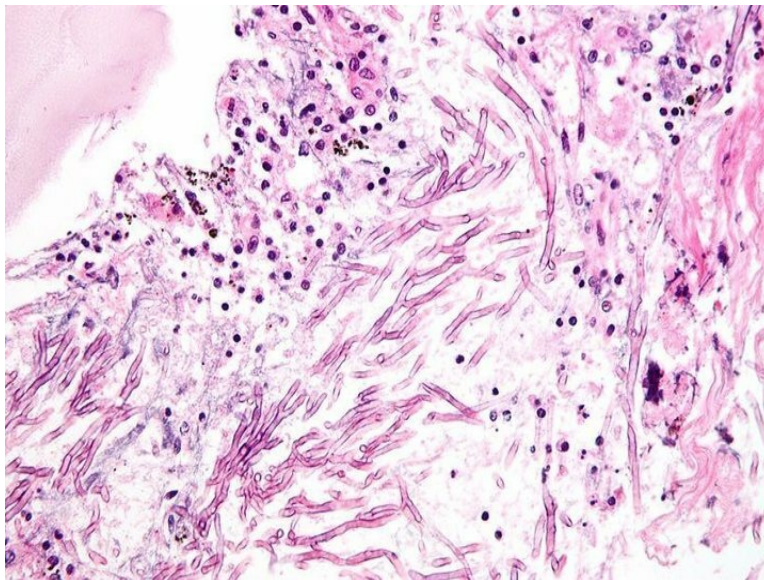
Question 1:

A microorganism is resistant to meropenem and aminoglycoside and sensitive to ticarcillin-clavulanate, and cotrimoxazole. What is the likely microorganism?

- a) *Stenotrophomonas maltophilia*
- b) *Burkholderia cepacia*
- c) *Pseudomonas aeruginosa*
- d) *Acinetobacter baumannii*

Question 2:

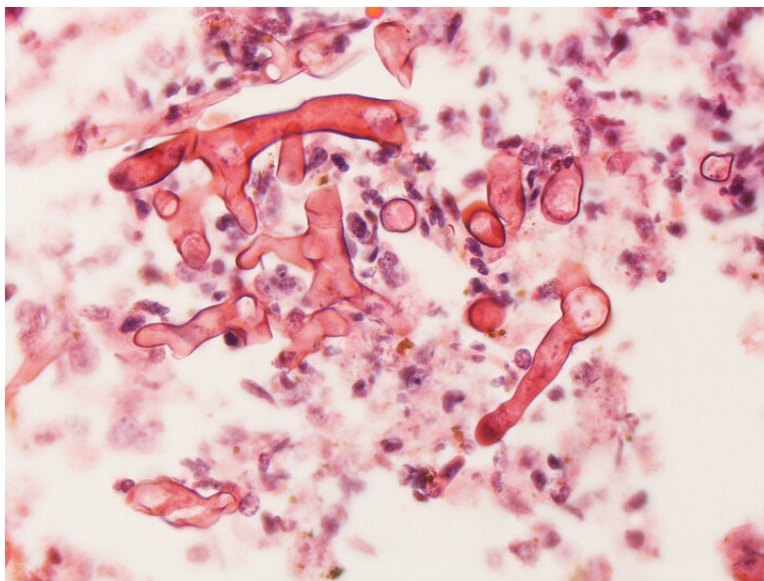
A patient with a history of bone marrow transplant presented with a lesion in the lung. On microscopic examination, the lesion is shown below. What is the likely causative agent?



- a) Histoplasmosis
- b) Cryptococcosis
- c) Mucormycosis
- d) Aspergillosis

Question 3:

Identify the organism in the given image.



- a) Aspergillus
- b) Mucor
- c) Talaromyces
- d) Blastomyces

Question 4:

A pregnant woman comes for an antenatal check-up. She has a history of stillbirth at 28 weeks in her previous pregnancy. She also has a history of multiple sexual partners and vaguely remembers having a painless ulcer on her vulva prior to her last pregnancy. What would be the first step of investigation for this lady?

- a) HIV ELISA
- b) VDRL
- c) Dark ground microscopy
- d) Culture

Question 5:

Match the diseases with their maximum incubation period.

- a) 1-a, 2-b, 3-c, 4-d
- b) 1-d, 2-c, 3-b, 4-a

- c) 1-d, 2-c, 3-a, 4-b
- d) 1-b, 2-a, 3-d, 4-c

Question 6:

A CSF sample of an elderly male presenting with meningitis is subjected to gram staining and revealed gram-positive bacilli. Which of the following is true regarding the motility of the causative agent?

- a) Motile at 22°C and 37°C
- b) Motile at 37°C, non motile at 22°C
- c) Motile at 25°C, non motile at 37°C
- d) Nonmotile at 22°C and 37°C

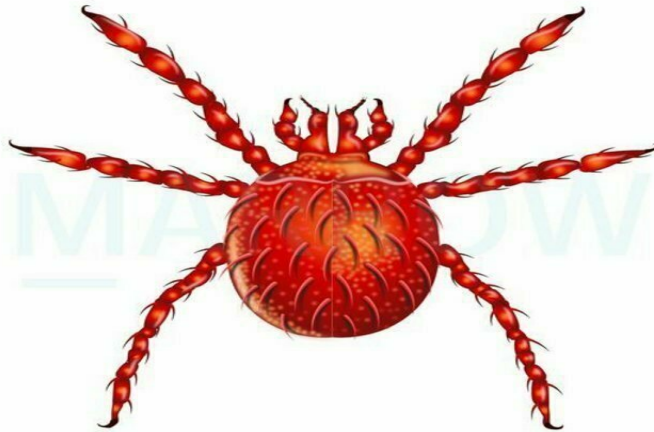
Question 7:

Which of the following is a protozoan parasite, that can be seen inside Kupffer cells?

- a) Toxoplasma
- b) Leishmania donovani
- c) Babesia
- d) Plasmodium vivax

Question 8:

Which of the following organism is transmitted by the vector shown in the image below?



©MARROW

- a) *Orientia tsutsugamushi*
- b) Kyasanur forest disease
- c) Chandipura virus
- d) *Anaplasma*

Question 9:

A lady cleans her cat litter regularly to avoid infection by toxoplasma during her pregnancy. What is the infective stage of this organism in the cat feces?

- a) Bradyzoites
- b) Tissue cysts
- c) Tachyzoites
- d) Oocysts

Question 10:

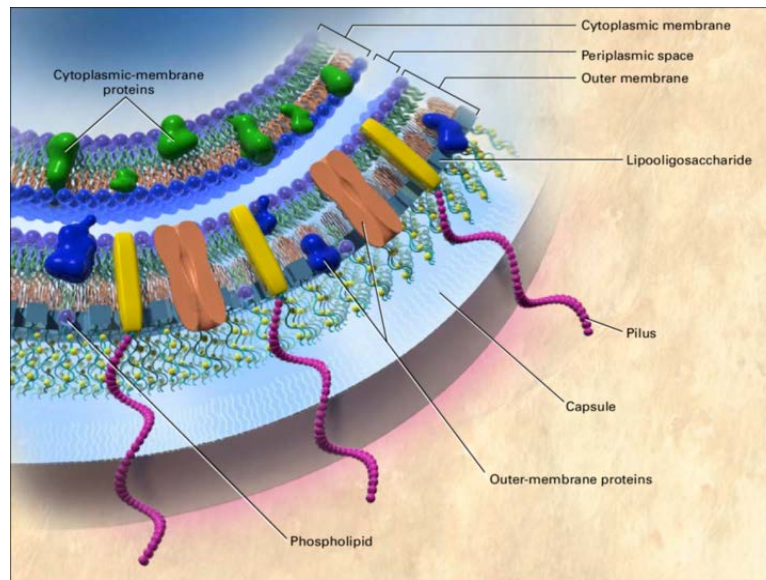
Which of the following is a false statement according to the CDC classification of bioterrorism agents?

- a) Category A - Easily transmitted and are associated with high mortality rates
- b) Category D - Re-emerging diseases that are high priority agents
- c) Category C - Emerging diseases that are easy to disseminate and are associated with high mortality rates

- d) Category B - Moderately easy to disseminate and are associated with low mortality rates

Question 11:

The given image of the cell wall belongs to which of the following organism?



- a) *Staphylococcus aureus*
- b) *Escherichia coli*
- c) *Neisseria meningitidis*
- d) *Pseudomonas aeruginosa*

Question 12:

A patient presents with PUO. He is suspected to be suffering from brucellosis. Multiple dilutions of his serum sample are put up for a standard agglutination test. The results came as follows: Negative agglutination for the serum dilutions 1:20 and 1:40; But 1:80 and 1:160 were positive for agglutination. Which of the following is responsible for the negative test in the first 2 dilutions?

- a) Incorrect antigen
- b) Incorrect antibody
- c) Antibody excess
- d) Antigen excess

Question 13:

A patient presented with a lesion on the left buttock. He has a history of heart transplant 15 years back. Aspirate from the lesion shows gram-positive filamentous bacteria. What is the likely causative agent?

- a) Nocardia
- b) Bacillus anthracis
- c) Mycobacterium leprae
- d) Mycobacterium tuberculosis

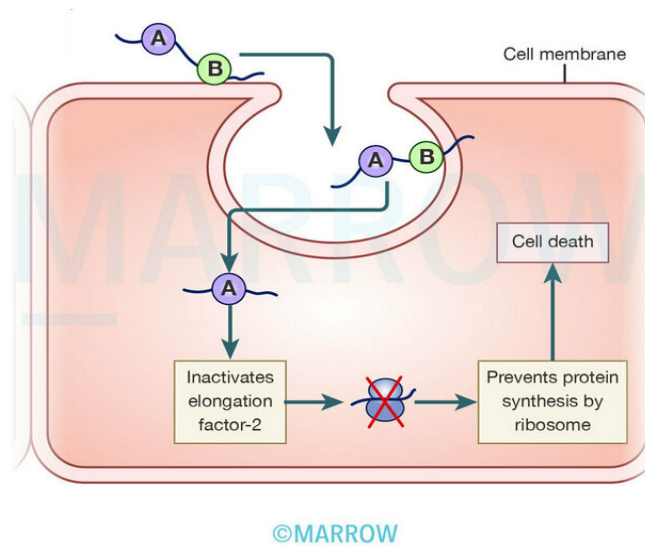
Question 14:

An elderly male patient who underwent a kidney transplant 10 years back is on immunosuppressive therapy. He gives a history of cough and dyspnea for the past 1 week, and the sputum microscopy shows gram-positive budding yeasts cells and pseudohyphae. What is the likely diagnosis?

- a) Invasive mucormycosis
- b) Invasive aspergillosis
- c) Invasive candidiasis
- d) Information is not enough for the diagnosis

Question 15:

The toxin produced by which of the following organisms acts by the mechanism shown in the image below?



- a) *Vibrio cholerae*
- b) *Corynebacterium diphtheriae*
- c) *Bacillus anthracis*
- d) *Shigella*

Question 16:

A 13-year-old girl presented with fever and retro-orbital pain for 2 days. The tourniquet test and NS1 antigen test were positive. Which of the following features is considered a criterion for the severe form of this disease?

- a) AST and ALT >1000
- b) Leucopenia plus thrombocytopenia
- c) Persistent vomiting
- d) Hepatomegaly $>1\text{cm}$

Question 17:

A 25-year-old patient presented with a 3-day history of fever, headache, myalgia and retro-orbital pain with no signs of bleeding. On examination, maculopapular rash is seen. Which is the next appropriate step in management?

- a) NS1 antigen detection
- b) Widal test
- c) Rapid influenza antigen test

d) IgM ELISA

Question 18:

Which of the following diseases is not transmitted by the vector shown in the image below?



- a) Ehrlichia
- b) Babesiosis
- c) Orientia
- d) Crimean-Congo haemorrhagic fever (CCHF)

Question 19:

Which of the following is a biosafety level-3 infection?

- a) Extensively drug-resistant (XDR)- TB
- b) Ebola Virus
- c) Salmonella typhi
- d) Crimean-Congo haemorrhagic fever (CCHF) virus

Question 20:

Which of the following statements are advantages of saline wet mount over iodine wet mount?

- a) 1 and 2
- b) 1 and 3
- c) 2 and 3
- d) 1, 2 and 3

Question 21:

Which of the following helps the Human immunodeficiency virus (HIV) to penetrate into the host cell?

- a) gp 120
- b) gp 41
- c) CCR5
- d) CXCR4

Question 22:

An HIV-positive patient presented with headache, and neck stiffness. Fundoscopy revealed papilledema. India ink staining of the CSF showed an encapsulated organism. Which of the following statements is true regarding this condition?

- a) Start ART after 2-3 weeks
- b) Opening pressure will be normal
- c) Start amphotericin monotherapy
- d) 80% patients will have positive radiological findings

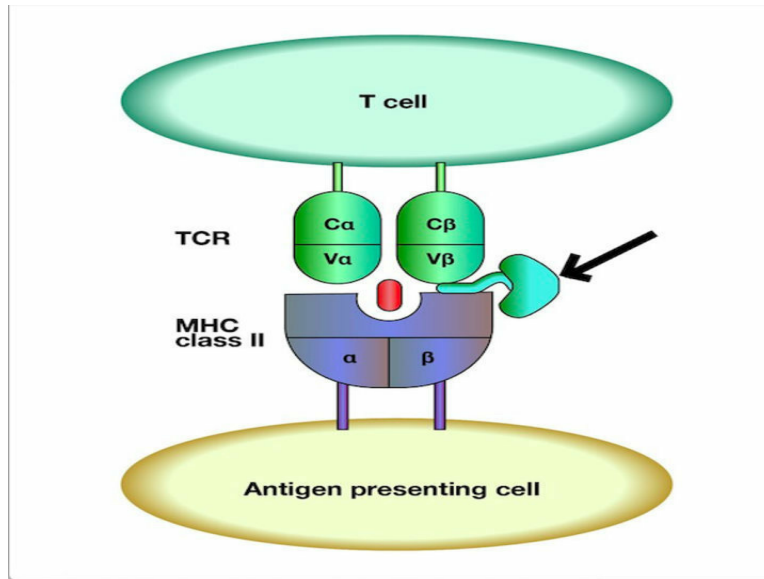
Question 23:

Which of the following techniques is used to detect autoantibodies?

- a) ELISA
- b) Indirect immunofluorescence
- c) Flow cytometry
- d) Fluorescent in-situ hybridization

Question 24:

Which of the following toxins acts by the mechanism shown below?



- a) Clostridium difficile exotoxin
- b) Diphtheria toxin
- c) Botulinum toxin
- d) Streptococcal pyrogenic exotoxin

Question 25:

In which of the following organisms' genetic material is adenine not equal to thymine?

- a) HIV
- b) E coli
- c) Human mitochondrial cells
- d) Herpes virus

Question 26:

An 18-month-old child presents with fever, irritability, neck stiffness and Kernig's sign positive. The mother gives a history of painful genital ulcers a year ago which had resolved spontaneously. What is the most likely causative agent?

- a) Herpes simplex virus-2
- b) Cytomegalovirus (CMV)
- c) Human papilloma virus (HPV)

d) Varicella zoster virus

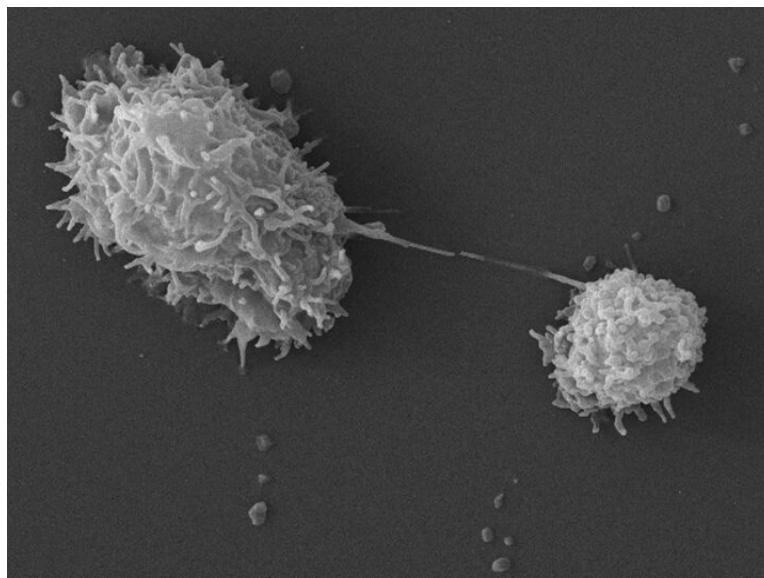
Question 27:

Which of the following organisms exhibits pathogenesis by cell-to-cell transfer, facilitated by actin filaments, eliminating the need for extracellular transport?

- a) Enterohemorrhagic *E. coli*
- b) *Salmonella typhi*
- c) *Listeria monocytogenes*
- d) *H. pylori*

Question 28:

A 19-year-old boy from a rural area presents with a 12-day history of altered mental status, decreased vision, ataxia, multiple cranial nerve palsies and hemiplegia. The electron microscopy image of the organism is given below. What is the likely organism?



- a) *Acanthamoeba*
- b) *Naegleria fowleri*
- c) *Sappinia*
- d) *Balamuthia*

Question 29:

An immunocompromised patient presents with a white corrugated plaque on the lateral border of the tongue that cannot be rubbed off. Which of the following is the causative agent?

- a) Candida
- b) Human herpes virus-8
- c) Herpes simplex virus (HSV)-1
- d) Epstein Barr virus

Question 30:

Which of the following organisms requires the use of reverse transcriptase in PCR?

- a) Parvovirus B19
- b) Adenovirus
- c) Cytomegalovirus (CMV)
- d) Respiratory Syncytial virus (RSV)

Question 31:

A farmer presented with fever and a skin lesion as shown in the image below. Blood investigations revealed creatinine levels of 1.4 mg/dl and platelet levels of 86,000 cells/mm³. What is the likely diagnosis?

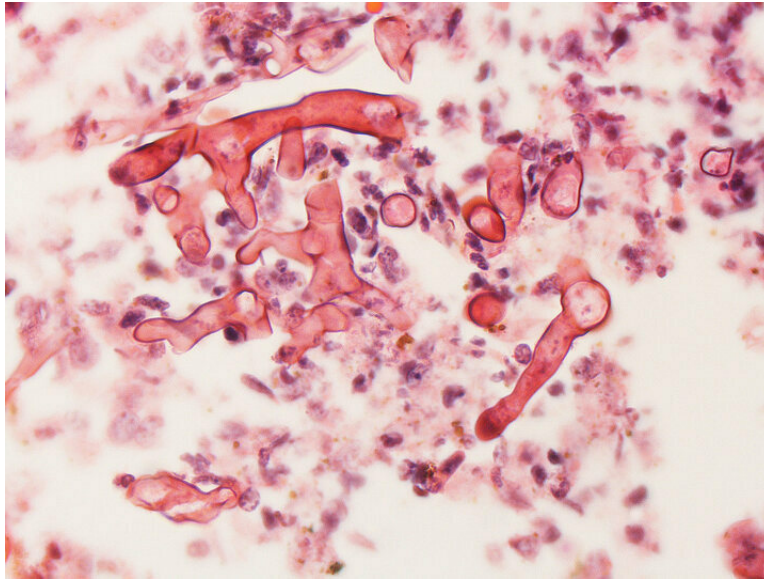


- a) Anthrax
- b) Leptospirosis

- c) Brucellosis
- d) Scrub typhus

Question 32:

A COVID-19 patient who was treated with steroids, presented with blood-tinged nasal discharge, 3 weeks after recovery. The histopathological examination of the discharge showed the following findings. Which of the following organisms is likely to be involved?



- a) Aspergillus
- b) Mucor
- c) Candida
- d) Coccidioides

Question 33:

You suspect dengue in a patient who presented with a 3-day history of fever, headache, and retro-orbital pain. Blood investigations revealed Hb=14 g/dl, WBC= 4000 cells/ mm³ and platelet count = 8900 cells/mm³. Which of the following tests will not be useful in confirming the diagnosis of this condition?

- a) NS1 antigen detection
- b) Viral culture and isolation in aedes albopictus c6/36 cell line
- c) RT-PCR
- d) IgM ELISA

Question 34:

Which of the following viruses has the least teratogenic effect?

- a) Chickenpox
- b) Rubella
- c) Cytomegalovirus (CMV)
- d) Herpes simplex virus (HSV)-2

Answer Key

Question No.	Correct Option
1	a
2	d
3	b
4	b
5	c
6	c
7	b
8	a
9	d
10	b
11	c
12	c
13	a
14	c
15	b
16	a
17	a
18	c
19	a
20	b
21	a
22	a

23	b
24	d
25	a
26	a
27	c
28	a
29	d
30	d
31	d
32	b
33	d
34	d

Detailed Explanations

Solution to Question 1:

Stenotrophomonas maltophilia, formerly called *Pseudomonas maltophilia*, is an opportunistic pathogen that causes wound infection, urinary tract infection, and septicemia. It is sensitive to cotrimoxazole.

Intrinsic resistance is the innate ability of a bacterial species to resist the activity of a particular antimicrobial agent through its inherent structural or functional characteristics, which allow tolerance of a particular drug or antimicrobial class.

Burkholderia cepacia, formerly called *Pseudomonas cepacia*, is an opportunistic pathogen that causes fatal necrotizing pneumonia. *B. cepacia* complex strains are intrinsically resistant to aminoglycosides, polymyxin, and first and second-generation cephalosporins but sensitive to meropenem, co-trimoxazole, doxycycline, minocycline, chloramphenicol and ceftazidime.

Solution to Question 2:

The given clinical scenario of lung lesion and the image showing septate hyphae with branching at acute angles is suggestive of aspergillosis.

Invasive aspergillosis occurs due to infection with *Aspergillus* species. It is often seen in immunocompromised patients. Most cases of invasive aspergillosis involve the lungs. The clinical features include fever, cough which may be productive, chest discomfort, hemoptysis, and shortness of breath. Voriconazole is the drug of choice.

Other manifestations include allergic bronchopulmonary aspergillosis where central bronchiectasis is usually seen. Aspergilloma and superficial infections like otomycosis or mycotic keratitis are the other presentations.

Histopathological examination of the tissue section shows characteristic dichotomous branching at an acute angle. The hyphae stain poorly with hematoxylin and eosin, so periodic acid–Schiff (PAS) stain is used.

Other methods for diagnosis include detection of galactomannan (*Aspergillus* cell-wall component) and culture isolation. *Aspergillus fumigatus* grows dark green colonies, while *Aspergillus niger* appears black. Yellow-green colonies are seen in case of *Aspergillus flavus*. Microscopic examination of colonies stained with lactophenol cotton blue would show septate hyphae and conidiophores.

Solution to Question 3:

Microscopy showing broad, aseptate, ribbon-like hyphae with irregular branching is suggestive of mucor.

Mucormycosis/ Zygomycosis is an opportunistic infection caused principally by species of rhizopus (most prevalent), mucor, and absidia. The common clinical form is rhino-orbital-cerebral mucormycosis in which there is an invasion of the blood vessels in the nasal cavity, eyes, and brain causing thrombosis, infarction, and necrosis. The blackish discoloration is due to arterial and venous thrombosis leading to gangrene.

Diagnosis can be made by staining them with 10% potassium hydroxide and culture. On Sabouraud agar at 30-37°C, they appear as grey-white colonies with a thick cottony, fluffy surface. Under the microscope, they appear as irregularly branched broad hyphae with aerial sporangioophores which end in a sac containing spores called sporangium.

Other options:

Options A and C: *Aspergillus* and *Talaromyces* (previously known as *Penicillium*) both demonstrate septate hyphae with conidiophores.

Option D: *Blastomyces* is a dimorphic fungus that exists in the yeast form in body tissues and cultures maintained at 37°C and in the filamentous mold form in cultures at room temperature (25°C). In the yeast form, it shows thick-walled round yeast cells with single broad-based budding, giving a figure of 8 appearance.

Solution to Question 4:

Venereal disease research laboratory test (VDRL test) is the first investigation in this patient with a history of stillbirth and painless genital ulcer, which is likely suggestive of syphilis.

Syphilis is caused by *Treponema pallidum*, a spirochete that is actively motile. It exhibits characteristic corkscrew-like motility.

Primary syphilis occurs subsequent to inoculation of *treponema pallidum* following sexual contact. It manifests as a painless, non-tender, single papule over the genitalia, that rapidly becomes eroded and usually becomes indurated, and is called a hard chancre. It can be accompanied by lymphadenopathy. The chancre generally heals within 10-40 days even without treatment, leaving a scar.

The obstetric history in an untreated syphilitic woman typically gets better with subsequent pregnancies. Initial few pregnancies can result in abortions and stillbirths, subsequent pregnancies can result in live births of infants with stigmata of syphilis. Finally, after a few pregnancies, healthy infants are born.

It can be diagnosed using VDRL, and RPR tests. It can be confirmed by FTA – ABS testing or TPHA, TPI testing.

Dark ground microscopy is the most specific and sensitive technique to diagnose syphilis when an active chancre or condyloma lata is present. Since the ulcer is not active in this patient, it is not useful in this scenario.

Smear specimens can be stained using Fontana stain and pathological tissues are stained using Levaditi stain. Silver impregnation technique can also be used.

Solution to Question 5:

The correct match is:

The incubation period is the time interval between the invasion of the infectious agent and the appearance of the first symptom or signs in the body.

The incubation period of syphilis is 10-90 days and varies inversely with the size of the Spirochaete inoculum.

The incubation period of SARS infection ranges from 5 to 10 days.

The incubation period of Hepatitis A infection ranges from 10 to 50 days (usually 14-28 days).

Chickenpox is caused by varicella zoster virus (HSV-3). The incubation period is between 10 to 21 days.

Diseases	Incubation period
1. Hepatitis A	d. 50 days
2. Chickenpox	c. 21 days
3. SARS-CoV-2	a. 10 days
4. Syphilis	b. 90 days

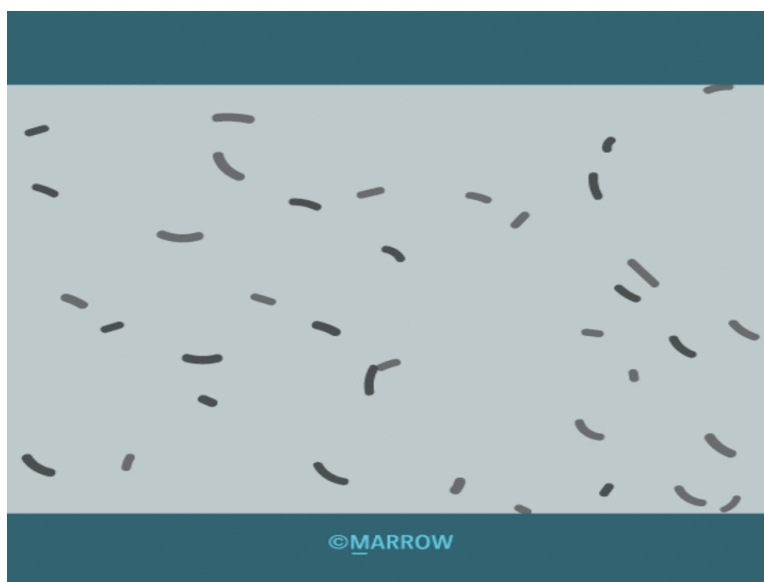
Solution to Question 6:

The given scenario of an elderly male with meningitis and smear showing gram-positive bacillus is suggestive of Listeria infection. Listeria is motile at 25°C and non-motile at 37°C.

Listeria exhibits a characteristic slow tumbling end-over-end motility at 22–28°C but is non-motile at 37°C. This is because the peritrichous flagella are produced by the bacillus optimally between 20°C and 30°C. This is known as differential motility as the motility pattern changes with a change in temperature.

Listeria monocytogenes is a gram-positive bacillus. It can survive at refrigerator temperatures (4°C), under conditions of low pH and high salt. So it is able to overcome food preservation and safety barriers, making it an important foodborne pathogen. The virulence factor produced by *Listeria* is called listeriolysin O. This helps the bacteria to escape and causes septicemia. They are catalase-positive, will not produce H₂S, and shows beta hemolysis. This helps to differentiate *Listeria* from other gram-positive bacilli.

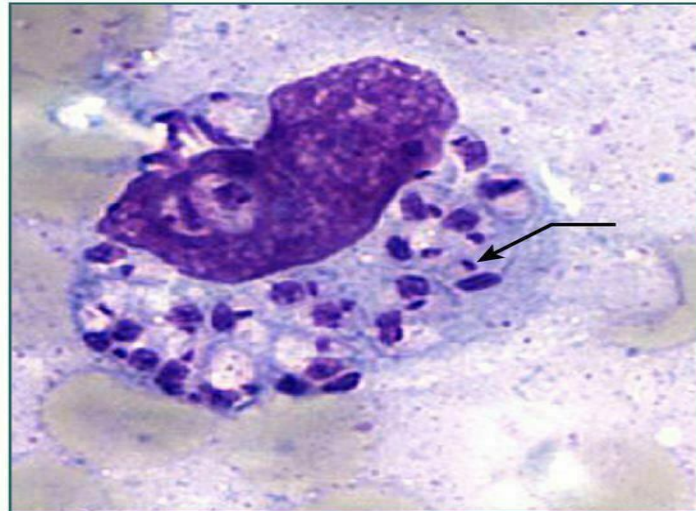
L. monocytogenes enters the body after ingestion of contaminated foods such as refrigerated milk, cheese, fruit, or vegetables. Healthy persons exposed to *L. monocytogenes* in food may develop mild, self-limiting febrile gastroenteritis lasting 1–3 days. This develops after an incubation period of 6–48 hours. Symptoms include fever, chills, headache, myalgias, abdominal pain, and diarrhea.



Solution to Question 7:

The protozoan parasite, that can be seen inside Kupffer cells is *Leishmania donovani*.

Kupffer cells are a component of the mononuclear phagocytic system found in the liver. The amastigote form of *leishmania* also known as the Leishman-Donovan (LD) body (shown in the image below) is an intracellular form found inside macrophages, monocytes, and other reticuloendothelial cells. It contains a kinetoplast, which consists of a parabasal body and a blepharoplast. Flagellum is absent.



Leishmaniasis is a slowly progressing indigenous disease caused by a protozoan parasite of the genus *Leishmania*.

The life cycle of *Leishmania* involves two forms:

- The promastigote develops and lives extracellularly in the sandfly vector, and
- The amastigote multiplies intracellularly in the reticuloendothelial cells of the host

It is transmitted by the bite of an infected female *Phlebotomus* sandfly which inoculates the flagellated promastigotes into the skin of the host. This condition causes fever with rigor and chills associated with splenomegaly and hepatomegaly. It lowers immunity, causes persistent fever, anemia, and liver and spleen enlargement, and if left untreated, death may occur due to opportunistic infection.

Diagnosis of leishmaniasis is mainly by demonstration of amastigotes within macrophages in tissue aspirates such as blood smears, bone marrow, splenic, or lymph node aspirate. It is the gold standard for diagnosis.

Solution to Question 8:

The image shows the trombiculid mite which is responsible for the transmission of *Orientia tsutsugamushi*.

Scrub typhus, also known as chigger-borne typhus, is caused by *Orientia tsutsugamushi*. The vector is trombiculid mite. Humans are infected when they are bitten by the mite larvae known as chiggers. The incubation period is 1- 3 weeks. Patients develop a characteristic eschar at the site of the mite bite, along with regional lymphadenopathy and maculopapular rash. It can be associated with fever, headache, and conjunctival injection. Meningoencephalitis, myocarditis, and pneumonia can be seen in severe cases.

Serologic assays are used for diagnosis. Weil-Felix reaction is a heterophile agglutination reaction used for diagnosis, wherein the sera are tested for agglutinins to the O antigens of certain *Proteus* strains OX19, OX2, and OXK. OXK is agglutinated only in scrub typhus. Doxycycline is the

treatment of choice for both adults and children.

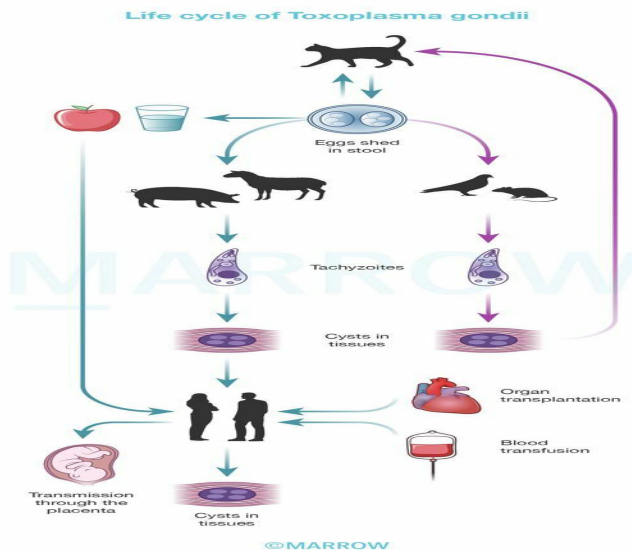
Solution to Question 9:

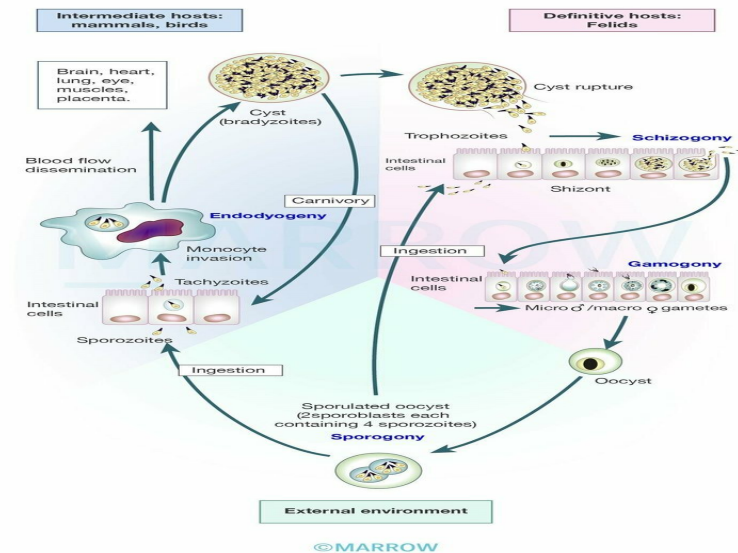
The infective stage of *Toxoplasma* found in the cat feces is oocysts.

The oocysts develop from the sexual reproduction of *Toxoplasma* in the gut lumen of the cat and pass out via the feces. It takes 1 to 5 days for the oocysts to become infective after being excreted in the feces. Hence, daily removal of cat litter and safe disposal can prevent infection.

Both asexual reproduction (schizogony) and sexual reproduction (gametogony) occur within definitive hosts such as cats. The tachyzoites and tissue cysts represent stages in asexual multiplication, while the oocyst is formed by sexual reproduction, which occurs only in the definitive hosts such as cats.

Only the asexual forms i.e., tachyzoites, and tissue cysts are present in intermediate hosts such as humans, mice, rats, etc.





Humans acquire the infection upon consumption of meat containing tissue cysts, food or water containing mature oocysts, or fingers contaminated by cat feces.

Manifestations are mostly asymptomatic and if symptomatic can be congenital or acquired. Features such as fever, headache, myalgia, splenomegaly, and lymphadenopathy commonly affect the cervical lymph nodes.

Central nervous system involvement is observed in toxoplasma infection affecting immunocompromised individuals. Toxoplasma infection in pregnancy can lead to congenital toxoplasmosis that most frequently presents with chorioretinitis.

Solution to Question 10:

According to CDC, the agents of bioterrorism are classified only into 3 categories (Category A, B, and C), there is no category D.

Bioterrorism is the intentional release or threat of release of biological agents (i.e. viruses, bacteria, fungi, or their toxins) in order to cause disease or death among the human population or food crops and livestock to terrorize a civilian population or manipulate the government.

The agents of bioterrorism are classified into 3 categories:

Category A bioterrorism agents are those which are easily transmitted from one person to the other and are associated with high mortality rates and thus have the potential to cause major public health impacts.

Category B bioterrorism agents are those which are moderately easy to disseminate and are associated with moderate morbidity and low mortality rates.

Category C bioterrorism agents are those which are emerging pathogens such as Nipah and Hantavirus, that could be engineered for mass dissemination in the future due to ease of availability, high mortality and morbidity rates, and ease of production and dissemination.

Solution to Question 11:

The image showing a cell wall with a capsule and lipooligosaccharides (LOS) is suggestive of *Neisseria meningitidis*.

Unlike lipopolysaccharide (LPS), LOS does not have a long O-antigen side chain. It is an important virulence factor and is found in a few organisms such as *N. gonorrhoeae*, *N. meningitidis*, *H. ducreyi*, etc.

The genus *Neisseria* consists of Gram-negative, aerobic, non-sporulating, non-motile, oxidase-positive cocci arranged typically in pairs (diplococci). It includes two pathogens, *Neisseria meningitidis*, and *Neisseria gonorrhoeae*. The important difference is the presence of a capsule in *N. meningitidis*.

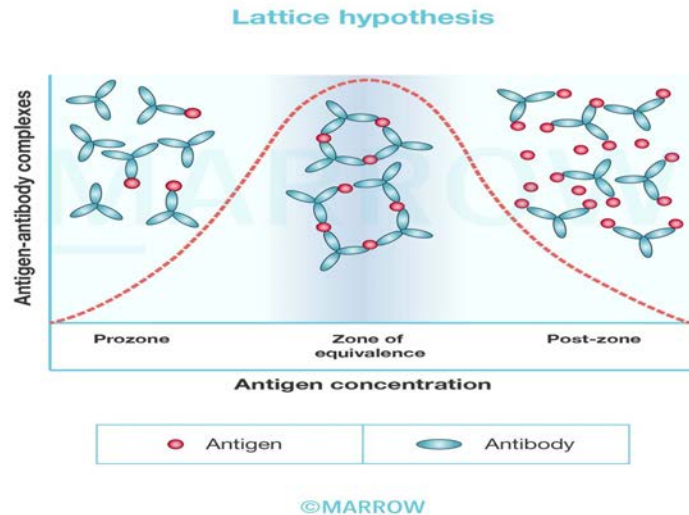
Solution to Question 12:

The negative test in the first 2 tubes is due to the prozone phenomenon. Here, antibodies are in excess compared to antigens leading to weak or absent precipitation.

The lattice hypothesis explains precipitation reactions and agglutination reactions. The amount of precipitate formed is greatly influenced by the relative proportions of antigens and antibodies. Precipitation occurs when a lattice consisting of alternating antigen and antibody molecules is formed.

- The prozone phenomenon is caused by excessive antibodies in the test system. There is weak/absent precipitation, giving a false-negative test.
- In the zone of equivalence, antigen and antibody are in optimum proportions and maximum precipitation occurs. Lattice formation is enhanced.
- Post-zone phenomenon is due to the presence of excessive antigens in the test system. There is weaker/absent precipitation.

Prozone phenomenon and post-zone phenomenon can give rise to false-negative results in precipitation reactions. This can be overcome by making serial dilutions of serum, thereby optimizing the concentrations of antigens and antibodies in the test system.



Solution to Question 13:

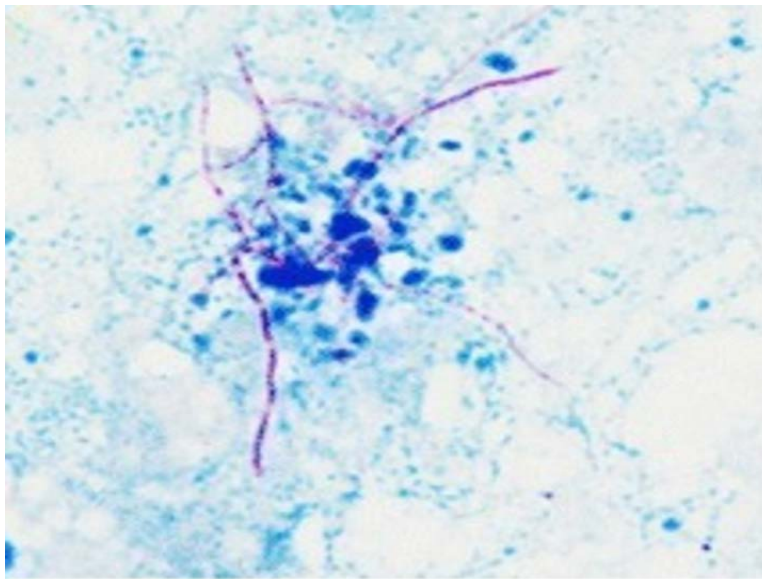
The given clinical scenario of local abscess and aspirate showing filamentous gram-positive bacteria are suggestive of *Nocardia*.

Nocardia species appear as filamentous organisms with extensive hyphae-like branching. They are aerobic, gram-positive, catalase-positive, and urease positive. They are weakly acid-fast because they stain with carbolfuchsin and retain when decolorized with 1–4% sulfuric acid instead of the stronger acid-alcohol decolorant.

Infections caused by *Nocardia* include cutaneous, subcutaneous, or systemic lesions. It may lead to local abscesses, cellulitis, or lymphocutaneous lesions. Systemic nocardiosis occurs more often in immunodeficient persons.

Systemic nocardiosis usually caused by *N. asteroides* manifests primarily as pulmonary disease, pneumonia, lung abscess, or other lesions resembling tuberculosis. Extrapulmonary manifestation may involve the brain, kidneys, and other organs.

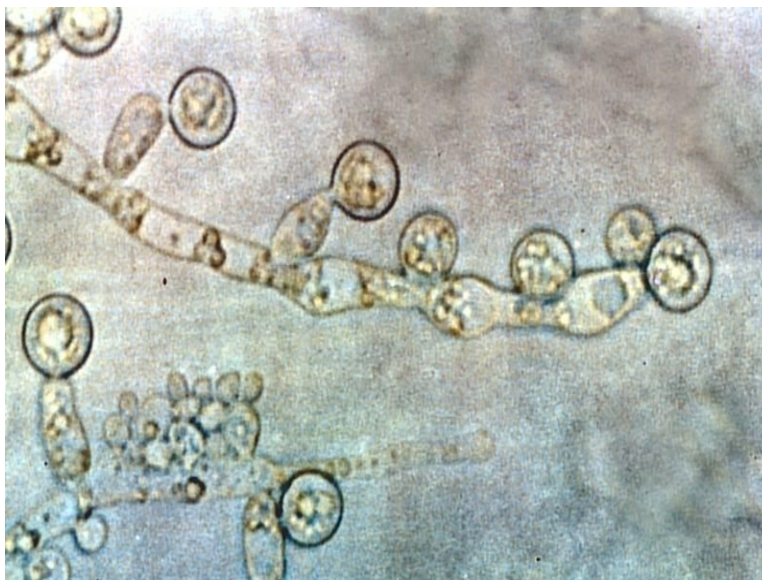
The image given below is a smear of modified acid-fast Kinyoun staining showing filamentous red-stained partially acid-fast rods suggestive of *Nocardia*.



Solution to Question 14:

Cough and dyspnea in an immunosuppressed patient with microscopy showing gram-positive budding yeasts cells and pseudohyphae suggest a diagnosis of invasive candidiasis.

Candida is the most common cause of opportunistic mycoses in the world. Immunocompromised individuals are at an increased risk of systemic candidiasis. The most common species implicated in candidal infections are *C. albicans*, *C. parapsilosis*, and *C. glabrata*. Morphologically, they are yeast-like fungi. They exist as yeast and pseudohyphae (image below). It appears as budding gram-positive cells on wet films and smears. Mycelial forms suggest colonization. *Candida albicans* form chlamydospores on corn meal agar.



When colonies of *Candida albicans* are inoculated in serum at 37°C, the yeast cells form elongations called germ tubes within 2 hours. The image below shows germ tube formation, also

known as the Reynolds Braude phenomenon.

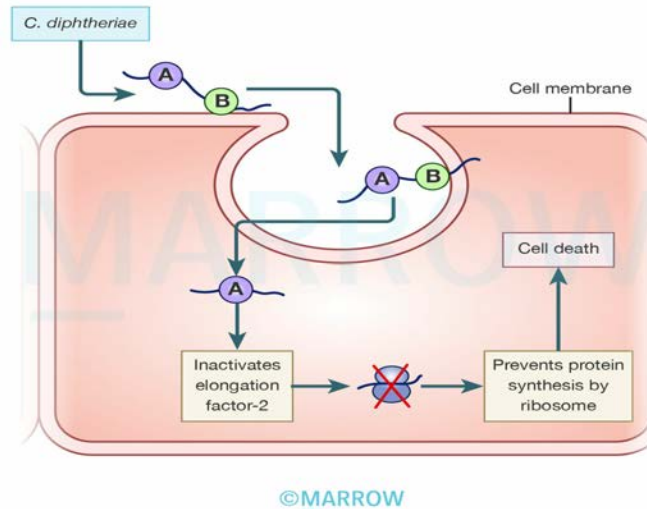


Clinically, infection with *Candida* spp. can present as cutaneous and mucosal candidiasis, systemic candidiasis, or rarely, as chronic mucocutaneous candidiasis. Mucocutaneous infections manifest as thrush affecting the tongue, lips, gums, and palate. It is seen as a patchy to confluent whitish pseudomembranous lesion and is commonly associated with AIDS. *Candida* may also invade the vaginal mucosa, leading to vulvovaginitis, and the nails and nail plate, leading to onychomycosis. Systemic candidiasis occurs when *Candida* enters the bloodstream. It can manifest as kidney infections, urinary tract infections, infective endocarditis, arthritis, meningitis, endophthalmitis etc.

Solution to Question 15:

The above image depicts the inactivation of elongation factor-2 which is the mechanism of action of diphtheria toxin.

Corynebacterium diphtheriae produces the diphtheria toxin, which is an AB subunit toxin. While the B subunit helps in attachment to cell surface receptors, the A subunit causes ADP-ribosylation of elongation factor-2 (EF-2) resulting in the inhibition of protein synthesis.

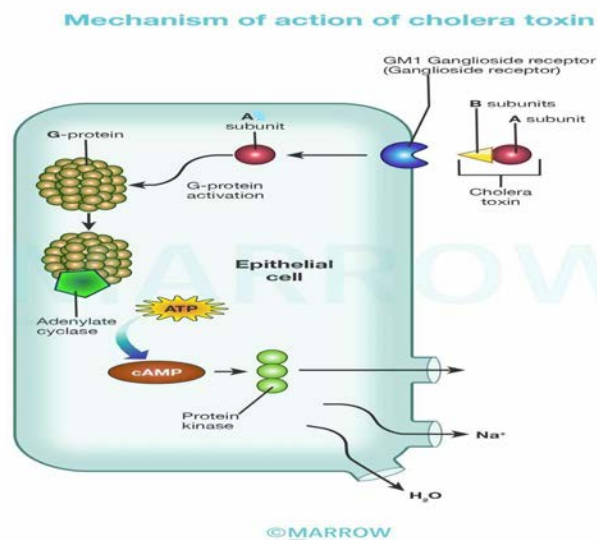


The exotoxin A produced by *Pseudomonas aeruginosa*, also an AB subunit toxin, acts by the same mechanism to cause inhibition of protein synthesis.

This infection presents as fever and sore throat, which progresses to dyspnea due to obstruction by the pseudomembrane. Bull's neck appearance is seen due to cervical lymphadenitis. Complications like damage to the heart (myocarditis), kidney (tubular necrosis), liver, and adrenal glands can develop due to the toxin.

Other options:

Option A: Cholera toxin is an enterotoxin produced by *Vibrio cholerae*. It is an AB subunit toxin where the B subunit binds to GM1 ganglioside receptors on the surface of jejunal epithelial cells. The A subunit is internalized and causes prolonged activation of adenylate cyclase resulting in increased cAMP activity which results in watery diarrhea.



Option C: *Bacillus anthracis* produces toxins that are made of 3 proteins-

- Protective antigen (PA) binds to specific cell receptors and mediates entry of edema factor and lethal factor into the cell.
- Edema factor (EF) is an adenylate cyclase, which is activated only inside the target cells, leading to intracellular accumulation of cyclic AMP and causing tissue edema.
- Lethal factor (LF) causes tissue death.

Option D: Shiga toxin or verotoxin is an exotoxin produced by *Shigella* species. It consists of two subunits A and B. The B subunit binds to globotriaosylceramide receptors on the target cell, and then, the A subunit gets internalized. The A subunit leads to inhibition of 60S ribosome, leading to shut down of protein synthesis.

Solution to Question 16:

The given clinical scenario of a 13-year-old girl presenting with fever and retro-orbital pain with a positive tourniquet and NS1 antigen tests is suggestive of dengue fever. AST or ALT ≥ 1000 is a criterion for severe dengue.

Dengue is an acute febrile disease caused by the mosquito-borne dengue viruses (DENVs), consisting of four serotypes (DENV 1 to 4), that are members of the Flaviviridae family, genus *Flavivirus*. It is transmitted from person to person by *Aedes aegypti* mosquitoes.

The World Health Organisation released guidelines in 2009 regarding the diagnosis of probable dengue, dengue with warning signs, and a category known as severe dengue. In addition to the classic DHF/DSS syndrome, rare cases of encephalitis or encephalopathy, liver damage, or myocardial damage are also included in the category of severe dengue.

Criteria for severe dengue include:

- Severe plasma leakage leading to shock (DSS) and fluid accumulation with respiratory distress
- Severe bleeding
- Severe organ involvement- Liver: AST or ALT ≥ 1000 , CNS: Impaired consciousness, heart and other organs.

Probable dengue includes those people who live in/ travel to dengue-endemic areas. Fever and 2 of the following criteria: nausea, vomiting, rash, aches and pain, Tourniquet test positive, leukopenia and any warning sign.

Warning signs include:

- Abdominal pain or tenderness
- Persistent vomiting
- Clinical fluid accumulation
- Mucosal bleed
- Lethargy, restlessness
- Hepatomegaly ≥ 2 cm
- Laboratory: increase in HCT concurrent with thrombocytopenia.

After an incubation period of 3–15 days, classical dengue fever begins with an abrupt onset of high fever. The condition is self-limiting though debilitating with headache, retro-orbital pain, myalgia, arthralgia, petechiae rash, and leucopenia. A macular-papular recovery rash appears 3–4 days after the onset of fever, and it usually starts on the trunk before spreading peripherally. The rash in dengue appears as 'white islands in a red sea' due to erythema with petechiae.

Shown below is an image of the rash in dengue:



Solution to Question 17:

The given clinical scenario of a 25-year-old patient presenting with a 3-day history of fever, headache, myalgia, retro-orbital pain with no signs of bleeding and a maculopapular rash on examination is suggestive of dengue fever. The next appropriate step in management is getting an NS1 antigen detection test done.

Dengue is an acute febrile illness caused by the dengue virus that is transmitted by the *Aedes aegypti* mosquito.

It presents as sudden-onset fever, retroorbital pain, myalgia, arthralgia, and a rash. The initial rash in dengue may appear as flushing erythema of the face in the first 2 days of symptoms. A second rash occurs 3-6 days after the onset of fever and appears maculopapular or morbilliform. Sometimes, lesions may coalesce and appear as 'white islands in a red sea' due to generalized erythema with petechiae and islands of sparing.

Before day 5 of illness, during the febrile period, dengue infections may be diagnosed by detection of viral antigens like non-structural viral protein antigen(NS 1) by ELISA or rapid tests. This is considered the mainstay of diagnosis.

Virus isolation in cell culture and viral RNA detection by nucleic acid amplification tests (NAAT) can also be done.

Serological tests, such as IgM ELISA (option D), are performed after day 5 when dengue viruses and antigens are no longer present in the blood due to the appearance of specific antibodies.

Other options:

Option B - Widal test is used to diagnose typhoid fever caused by *Salmonella Typhi* and *Paratyphi*. It is characterized by gradual onset headache, malaise, step ladder pyrexia and gastrointestinal symptoms like abdominal pain with either constipation or diarrhoea. They also present with rose spots on the skin during the 2nd to 3rd week, that fade on applying pressure.

The following image depicts rose spots in enteric fever.



Option C: The rapid influenza antigen test uses immunofluorescence to demonstrate the influenza viral antigen on the surface of nasopharyngeal cells. The typical clinical disease has a sudden onset of fever, headache, and generalised myalgia. The presence of respiratory symptoms is prominent. It is not associated with retro-orbital pain and maculopapular rash, as given in the above scenario.

Solution to Question 18:

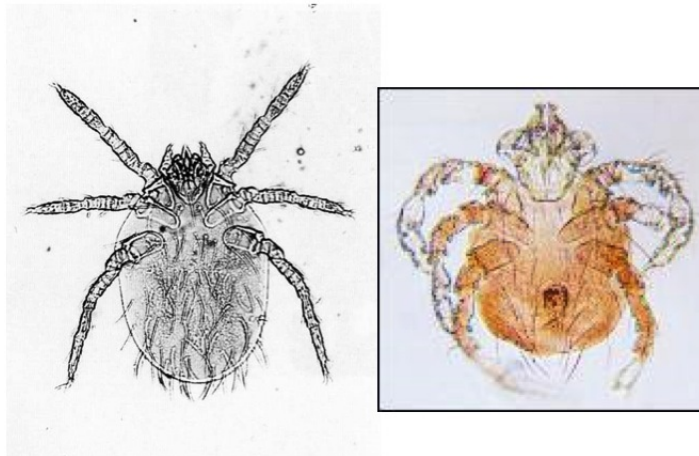
The vector shown in the image is hard tick. Scrub typhus caused by *Orientia tsutsugamushi* is transmitted by trombiculid mites.

Hard tick has an oval-shaped body, no antennae with 4 pairs of legs and is covered by a chitinous shield on its dorsal surface (scutum). It transmits diseases such as *Ehrlichia*, babesiosis, viral hemorrhagic fevers like Crimean-Congo haemorrhagic fever (CCHF) and Kyasanur forest disease (KFD), tick typhus, viral encephalitis (e.g., Russian spring-summer encephalitis), viral fevers (e.g., Colorado tick fever), tularemia, tick paralysis.

The following image shows a soft tick.



Orientia tsutsugamushi is transmitted by trombiculid mite which is as shown below.



Solution to Question 19:

Extensively drug-resistant(XDR)-TB is a biosafety level-3 infection.

Biohazard levels, also known as biosafety levels, are classifications of safety precautions required in the clinical microbiology laboratory based on specific pathogens handled when performing laboratory procedures.

There are 4 classifications of biosafety levels (BSLs). Each level contains specific recommendations for a clinical microbiology laboratory, with a focus on laboratory practises, safety equipment, and facility construction. As each level progresses, it includes additional biosafety considerations from the previous level.

Microorganisms classified as biosafety level-3 can be lethal to humans, but there are vaccines and other treatments available for them. They include Mycobacterium tuberculosis, anthrax, many types of viral encephalitis, hantavirus, Rift Valley fever, malaria, Rocky Mountain spotted fever and yellow fever.

GRO UP	DEFINITION	BACTERIA	VIRUS	FUNGI
Group 1	Biological agents unlikely to cause human disease	No pathogenic organisms	-	-
Group 2	Biological agents that can cause human disease and may be hazardous to workers but are unlikely to spread to the community; effective treatment or prophylaxis is usually available.	Bacillus (except anthrax) Streptococci Staphylococci E. coli Clostridium species Corynebacterium diphtheriae Salmonella typhi	Adenovirus Calicivirus Coronavirus Herpesvirus Influenza virus	Cryptococcus Candida Dermatophytes Aspergillus
Group 3	Biological agents that can cause severe human disease and are a serious hazard to workers; they may spread to the community; but effective treatment or prophylaxis is usually available.	B. Anthracis Brucella Coxiella burnetii Tularemia M. tuberculosis	Lymphocytic choriomeningitis Haanta virus St. Louis encephalitis virus Japanese encephalitis virus West Nile encephalitis virus	
Group 4	Same as group 3 except that effective treatment or prophylaxis is usually not available		Lassa Ebola Marburg Herpes simiae Crimean-Congo hemorrhagic fever (CCHF) virus	

Solution to Question 20:

The advantages of saline wet mount over iodine wet mount include its ability to demonstrate the motility of trophozoites and to differentiate between bile and non-bile-stained eggs.

Saline wet mount is an unstained preparation made by using physiological saline, while iodine wet mount is prepared either using diluted Lugol's iodine or Dobell and O'Connor iodine solution.

Saline wet mount aids in illustrating the motility of trophozoites such as *E. histolytica*, *Balantidium coli*, and *Giardia lamblia* because it does not dry up as quickly as iodine wet mount.

It also helps differentiate between bile-stained and non-bile-stained eggs. Bile stained eggs appear brown, while non-bile stained eggs appear colourless. However in iodine wet mounts, it is challenging to distinguish between bile-stained and non-bile-stained eggs because they are both stained brown.

Since iodine clearly stains the glycogen vacuoles and nuclei, the iodine wet mount helps in confirming the identification of cysts. Iodine-stained protozoan cysts display pale refractile nuclei, brown glycogen material, and yellow-gold cytoplasm.

Solution to Question 21:

The HIV virus can penetrate the host's CD4⁺ cells with the help of the viral glycoprotein gp120, also known as the spike antigen which is present in its envelope.

HIV-1 virion is spherical and contains an electron-dense, cone-shaped core surrounded by a lipid envelope derived from the host cell membrane. Target cells of HIV include CD4⁺ helper T cells (worst affected), monocyte-macrophages, and dendritic cells.

The viral envelope is studded by two viral glycoproteins, gp120 and gp41, which are coded by the env gene. Gp120 also known as the principal envelope antigen forms surface spikes while gp41 functions as a transmembrane anchoring protein. Both of these are critical for HIV infection of cells.

Viral glycoprotein gp120 binds to CD4⁺ cells in the host. Binding to the CD4⁺ receptor requires the participation of a co-receptor molecule, which has been identified as CXCR4 for the T cell-tropic HIV strains (X4 strains) and CCR5 for the macrophage-tropic strains (R5 strains).

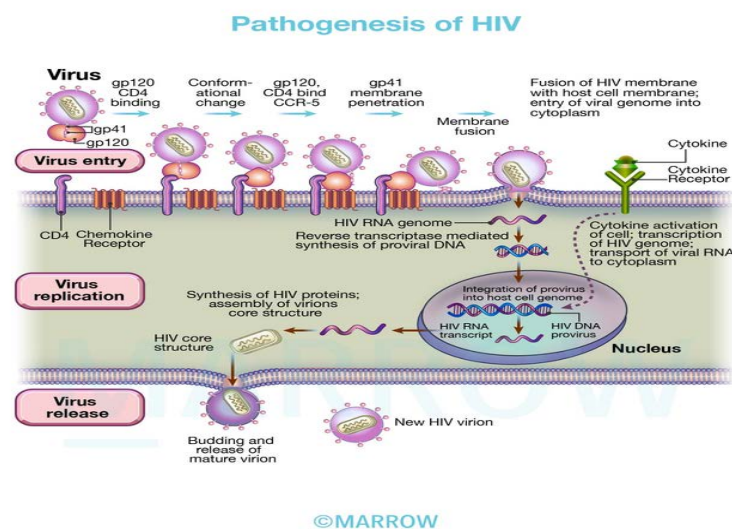
This binding leads to a conformational change in gp41 causing exposure of a hydrophobic region called the fusion peptide at the tip of gp41. This peptide inserts into the cell membrane of the target cells (e.g., T cells or macrophages), leading to the fusion of the virus with the host cell.

The virus core contains:

- Major capsid protein p24
- Nucleocapsid protein p7/p9
- Two copies of viral genomic RNA
- The three viral enzymes (protease, reverse transcriptase, and integrase)

The viral core is surrounded by a matrix protein called p17, which lies underneath the virion envelope.

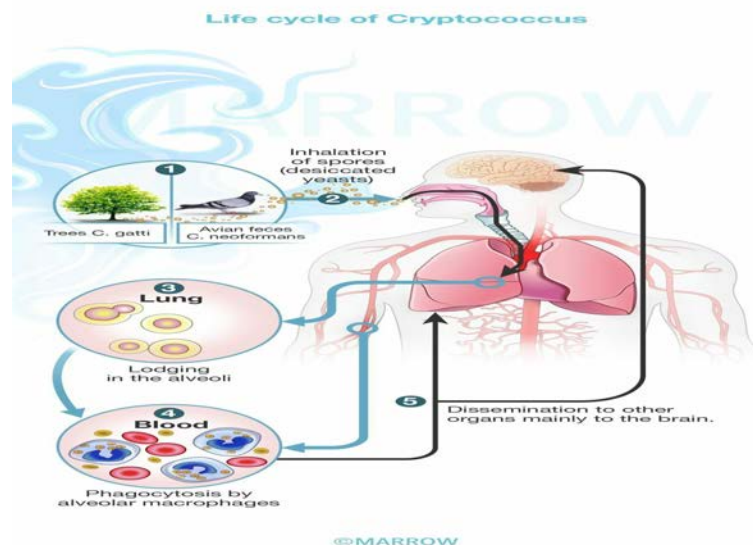
The image below shows the pathogenesis of HIV infection.



Solution to Question 22:

The above clinical scenario and the stain findings are suggestive of cryptococcosis and the causative organism is *Cryptococcus neoformans*. In a patient with *Cryptococcus* infection, antiretroviral therapy should be started 2-3 weeks after initiation of treatment. This is due to a risk of immune reconstitution which can lead to severe inflammation and further worsen symptoms.

C. neoformans spreads from inhalation of the yeast cells in pigeon faeces causing infection in HIV/AIDS and other immunosuppressive conditions. It causes influenza-like respiratory infection in immunocompetent persons. In immunocompromised it disseminates to CNS causing cryptococcal meningoencephalitis and patients may complain of headaches, neck stiffness, and disorientation as in this case. Other common sites of dissemination include the skin, adrenals, bone, eye, and prostate gland.



Diagnosis is made by demonstration of yeast cells or cryptococcal antigen in CSF. *Cryptococcus* presents with raised intracranial pressure so precautions must be taken before taking a CSF sample. (Option B) India ink stain is used to demonstrate capsulated budding yeast cells in CSF as they have a distinctive appearance because their capsules exclude ink particles. Rapid detection of cryptococcal antigen in CSF and blood can also be done. Colonies develop within a few days on most media at room temperature or 37°C. Media with cycloheximide should be avoided as it inhibits *Cryptococcus*. MRIs and CT scans can be used but they're only reliable in about 21-27% of cases (Option D).

Combination therapy of amphotericin B and flucytosine has been considered the standard treatment for cryptococcal meningitis (Option C).

Solution to Question 23:

Circulating autoantibodies in serum are best detected by indirect immunofluorescence.

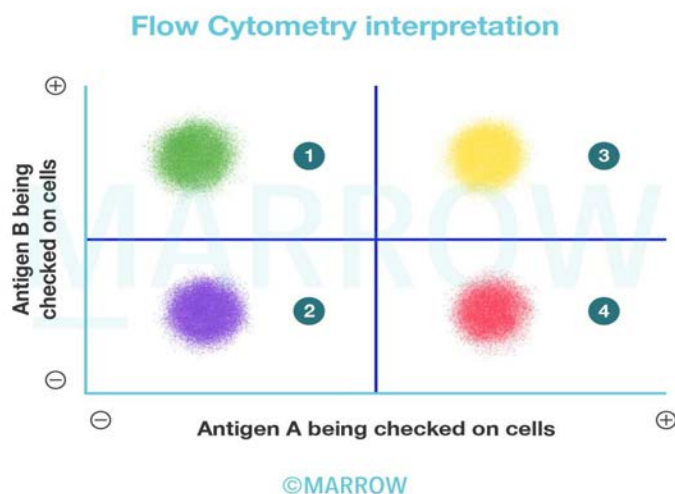
The two-step indirect immunofluorescence assay is used to find out if a serum specimen contains antibodies specific to a particular organism. A known antigen is first attached to a slide. After incubating the slide with a serum sample, the sample is removed, the slide is washed, and a second fluorescein-labelled anti-immunoglobulin is added. If the patient's serum contains the organism-specific antibodies, the organism will appear green under the fluorescence microscope.

This assay is the accepted method for identifying autoantibodies in autoimmune disorders (such as antinuclear antibodies) and detecting antibodies to certain microorganisms (eg, *T. pallidum*)

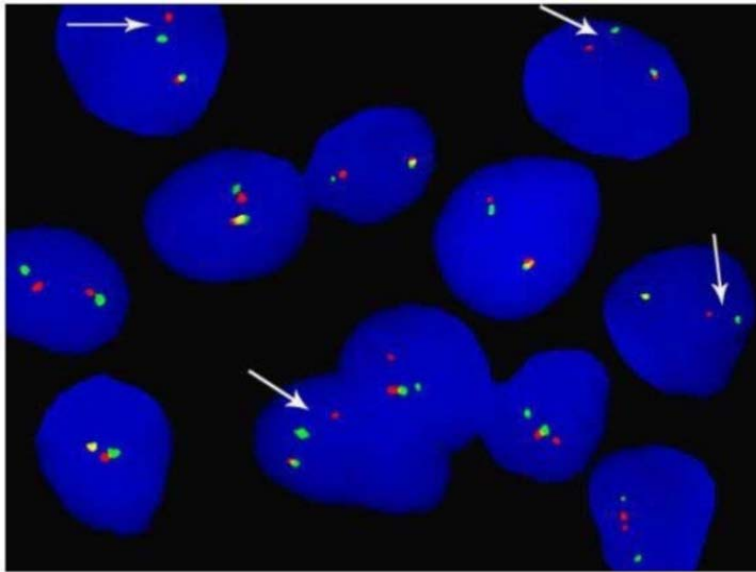
Other options:

Option A: Enzyme-linked Immunosorbent Assay (ELISA) is a type of heterogenous enzyme immunoassay that involves the use of an immunosorbent, an absorbing material specific for one of the components of the reaction: the antigen or antibody. It mainly detects antigens of infectious agents present in clinical specimens like rotavirus detection in fecal specimens and HIV detection.

Option C: Flow cytometry is a laser-based method that analyzes the cells and cell components based on the presence of certain cell surface molecules which are named as per cluster of differentiation (CD) nomenclature. It is very effective for the study of the immune system and its response to infectious diseases and cancer.



Option D: Fluorescent In Situ Hybridisation (FISH) is a simple technique for the detection of specific DNA sequences in cells/tissues. It uses fluorescent labelled probes to bind to complementary DNA. FISH can be performed on prenatal samples, peripheral blood cells, touch preparations from cancer biopsies, and even fixed archival tissue sections.

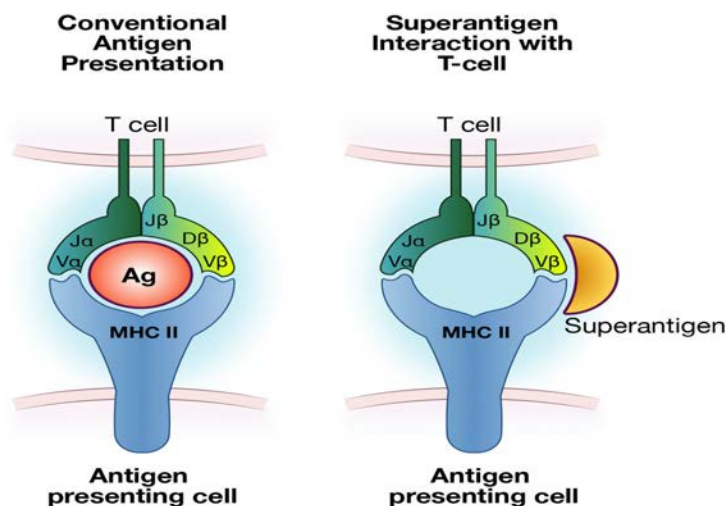


Solution to Question 24:

The given image depicts a superantigen which binds to the variable β region of the T-cell receptor. Streptococcal pyrogenic exotoxin (SPE) is a superantigen.

Superantigens are a third variety of the biological class of antigens that activate a very large number of T cells, irrespective of antigen specificity. This means they can activate T-cells directly without being processed by antigen-presenting cells (APCs).

- The variable β region of the T-cell receptor ($v\beta$ of TCR) appears to be the receptor for superantigens.
- They directly bridge non-specifically between MHC-II of APCs and T-cells.



The overproduction of cytokines, particularly IFN- γ , is a result of the massive activation of T cells. IFN- γ in turn activates macrophages to produce IL-1, IL-6, and TNF- α , all of which may contribute to a “cytokine storm” causing severe symptoms of shock and multiple organ failure.

Examples of superantigens:

- Bacterial
- Staphylococcal toxic shock syndrome toxin-1
- Streptococcal toxin
- Mycoplasma arthritidis mitogen 1
- Yersinia enterocolitica
- Yersinia pseudotuberculosis
- Viral
- Epstein–Barr virus (EBV)
- Cytomegalovirus (CMV))
- Rabies nucleocapsid
- HIV-encoded negative regulatory factor.
- Fungal
- Malassezia furfur.

Solution to Question 25:

The genetic material of HIV consists of a single-stranded RNA and therefore adenine will not be equal to thymine.

The pyrimidine components of RNA can differ from those of DNA. Although RNA contains the ribonucleotides adenine (A), guanine (G), and cytosine (C), it contains uracil (U) instead of thymine (T).

Because RNA is a single strand that is complementary to only one of two strands of a gene, its adenine content will not be equal to its uracil content and its guanine will not be equal to its cytosine content.

Chargaff's rule states that double-stranded DNA, such as that found in *E. coli*, human mitochondrial cells and herpes virus, will always have equal concentrations of (A = T) and (G = C).

The pairings of purine and pyrimidine nucleotides on opposite strands are very specific and rely on hydrogen bonds between A and T and G and C. The base pairs A-T and G-C are frequently referred to as Watson-Crick base pairs.

Solution to Question 26:

The given clinical scenario of an 18-month-old child presenting with fever, irritability, neck stiffness and positive Kernig's sign (CNS manifestations indicating meningitis) with the mother giving a past history of painful genital ulcers, is suggestive of neonatal herpes due to herpes simplex virus (HSV)-2. CMV, HPV and varicella zoster virus do not present with painful genital ulcers and can be easily ruled out.

Painful genital ulcers are commonly seen in genital herpes and chancroid due to *Hemophilus ducreyi* ((ducreyi makes you cry).

Herpes infection can be caused by HSV-1 and HSV-2 viruses. HSV-1 causes orofacial lesions, i.e. herpes labialis, while HSV-2 mainly causes genital herpes which is characterised by vesiculoulcerative lesions of the penis of the male or of the cervix, vulva, vagina, and perineum of the female.

The typical herpes lesions are thin-walled, umbilicated vesicles, the roof of which breaks down, leaving tiny superficial ulcers which heal without scarring. The most common site of cutaneous lesions is in the face, cheeks, chin, around the mouth and forehead. A cesarean section is performed in a case of active genital herpes at the time of labour in order to prevent neonatal herpes infection.

One of the following three disease patterns can be seen in infants infected during delivery or in the postpartum period: (1) disease limited to the skin, eyes, or mouth; (2) encephalitis with or without skin, eye, and mouth disease; or (3) disseminated infection affecting several organs, such as the brain, lungs, liver, heart, and adrenal glands.

Infants suffering from encephalitis usually exhibit clinical symptoms that point to bacterial meningitis at 8- 17 days of life. These symptoms include seizures, poor feeding, lethargy, neck stiffness, irritability, and positive Kernig's sign. When left untreated, HSV encephalitis results in 50% of infant deaths and severe neurologic sequelae in the majority of survivors.

Acyclovir at a high dose of 60 mg/kg/day, administered intravenously every eight hours, should be the first line of treatment for any infant with a confirmed or suspected neonatal HSV infection.

Other options

Option B - The transplacental spread of cytomegalovirus (CMV) primarily affects the foetus, resulting in microcephaly, intracranial calcification, hepatosplenomegaly, chorioretinitis, mental retardation, and sensorineural deafness. It is not associated with meningitis.

Option C - Adolescents who are sexually active can contract genital HPV infection, which is typically caused by HPV types 6 and 11. It can develop as a result of inoculation during childbirth through an infected birth canal or as a result of unintentional dissemination from cutaneous warts. It results in moist, fleshy, verrucous growths on the skin of the labia majora, perineum, perianal region, and vulva which are known as genital warts/venereal warts/condyloma acuminata.

Option D - Congenital varicella syndrome (CVS) is characterized by hypoplasia of limbs, psychomotor retardation, IUGR, chorioretinal scarring, cataracts, microcephaly and cutaneous scarring. It is not associated with meningitis.

Solution to Question 27:

Listeria monocytogenes exhibit pathogenesis by cell-to-cell transfer with the help of the host cell's actin filaments.

L. monocytogenes is a short, gram-positive, non-spore-forming bacteria. It enters the body through the GIT after ingestion of contaminated food such as cheese or vegetables. It has two cell-surface proteins, internalins A and B, which interact with the epithelial E-cadherin and promote its own phagocytosis into the gut epithelial cells, leading to the formation of the phagolysosome.

Low pH activates the bacterium to produce listeriolysin O, which lyses the phagolysosome and allows *Listeria* to escape into the cytoplasm of the epithelial cell. Here, the organisms proliferate and produce ActA, a listerial protein that induces host cell actin polymerization, which propels them to the host cell membrane.

While pushing against the host cell membrane, they produce elongated protrusions called filopods, which are ingested by adjacent epithelial cells, macrophages, and hepatocytes, resulting in the release of listeriae and the cycle starts again.

In this way, *Listeria* moves from cell-to-cell using the host's actin framework without being exposed to antibodies, complement systems, or neutrophils. Gram-negative bacteria like *Shigella flexneri* and *rickettsiae* also make use of the host cells' actin and contractile system to spread their infections, just like *Listeria*.

Other options:

Option A: Enterohemorrhagic *E. coli* (EHEC) is one of the six different types of diarrheagenic *E. coli*. They produce verocytotoxin (VT) or Shiga-like toxin (SLT), which can lead to a spectrum of illnesses, from mild diarrhea to more severe, life-threatening conditions such as hemorrhagic colitis (bleeding in the colon) and hemolytic uremic syndrome (HUS), which can cause kidney failure.

Option B: *Salmonella typhi* is a non-spore-forming, facultative anaerobic, gram-negative bacilli. Following ingestion of contaminated foods or beverages, salmonellae enter the small intestine, where they pass through the epithelium via specialised M-cells overlying the Peyer's patches, and then enter the intestinal lymphatics, with subsequent invasion into the bloodstream. Unlike *Listeria*, it does not utilise the actin filament of the host cell.

Option D: *H. pylori* is a gram-negative, flagellated, helical bacterium. Virulence factors of *H. pylori* include:

- Flagella- which allows motility in viscous gastric mucous
- Urease production- urease generates ammonia and bicarbonate from endogenous urea, creating an alkaline microenvironment that protects it from gastric acidic pH.
- Adhesins- for bacterial attachment to surface foveolar cells
- Toxins- such as cytotoxin-associated gene A (CagA) and Vacuolization cytotoxin (VacA). CagA is translocated into the gastric epithelial cells where it interferes with the epithelial cells' cytoskeletal structure and induces IL-8 production, leading to neutrophil attraction. VacA causes gastric mucosal injury by inhibiting T cells.

Solution to Question 28:

In the given clinical scenario of a 19-year-old boy presenting with a 12-day history of altered mental status, decreased vision, ataxia, multiple cranial nerve palsies and hemiplegia with electron microscopy showing an organism with characteristic spine-like pseudopodia (acanthopodia) is suggestive of infection caused by *Acanthamoeba* species.

Acanthamoeba exists as an active trophozoite form and a resistant cystic form, both of which are infective. The large, 20–50 µm trophozoite form is distinguished by spine-like pseudopodia called acanthopodia.

Human beings acquire infection by inhalation of cysts or trophozoites, ingestion of cysts or through traumatized skin or eyes. Following inhalation, the trophozoites reach the lungs and invade the central nervous system through the bloodstream, producing granulomatous amebic encephalitis (GAE).

It presents chiefly as two chronic conditions: (1) keratitis and (2) encephalitis.

With a slow relapsing course, keratitis resembles severe herpetic keratitis; however, in the case of an amebic infection, the eye is extremely painful. The initial symptoms are unilateral photophobia, excessive tearing, redness and foreign body sensation. Keratitis and uveitis can cause permanent vision loss or blindness. Diagnosis includes demonstrating the cyst in corneal scrapings using wet mount, histology, and culture.

Granulomatous amebic encephalitis is a serious infection of the brain and spinal cord that usually affects people who have a weakened immune system. The clinical picture is that of intracranial space-occupying lesions with seizures, pareses, and mental deterioration. It is diagnosed by demonstration of trophozoites and cysts in brain biopsy, culture and immunofluorescence microscopy using monoclonal antibodies.

No effective treatment is available for GAE. With limited success, multidrug combinations such as pentamidine, sulfadiazine, rifampicin, and fluconazole are used.

Other options:

Option B: The electron microscopy image of the infective form of *Naegleria fowleri* shows rounded pseudopodia called lobopodia. The infection usually follows swimming in freshwater rivers or ponds. The amoebae invade the nasal mucosa and pass through the olfactory nerve branches in the cribriform plate into the meninges.

It is mainly associated with primary amoebic meningoencephalitis. The disease advances rapidly, causing fever, headache, vomiting, stiff neck, ataxia, seizure and coma.

Option C: *Sappinia*, a newly discovered human pathogen of the central nervous system (CNS), can cause amoebic encephalitis in humans. In the single case of *Sappinia* amoebic encephalitis that has been reported, a sinus infection occurred prior to the onset of symptoms. *Sappinia* can be distinguished from other free-living amoeba by the presence of a distinctive double nucleus in which the 2 nuclei are closely apposed with a central flattening.

Option D: *Balamuthia* is a leptomixid free-living amoeba, reported to cause GAE. It exists in the ameboid trophozoite stage and is not associated with acanthopodia as seen in *Acanthamoeba* species.

Solution to Question 29:

The given clinical scenario of an immunocompromised patient presenting with a white corrugated plaque on the lateral border of the tongue that cannot be rubbed off is suggestive of oral hairy leukoplakia caused by Epstein Barr Virus (EBV).

Hairy leukoplakia is a distinct oral lesion on the lateral border of the tongue caused by EBV that typically occurs in immunocompromised patients. It appears as white, confluent patches of fluffy or hairy, hyperkeratotic thickenings, almost always on the lateral border of the tongue. Unlike candida-caused oral thrush, the lesion cannot be rubbed away.

Predisposing factors include Immunodeficiency, haematological malignancies, and steroid abuse.

The Epstein-Barr virus replicates in maturing epithelial cells but not in basal or parabasal keratinocytes. On microscopy, the upper spinous layer shows the characteristic hyperparakeratosis and acanthosis with balloon cells.

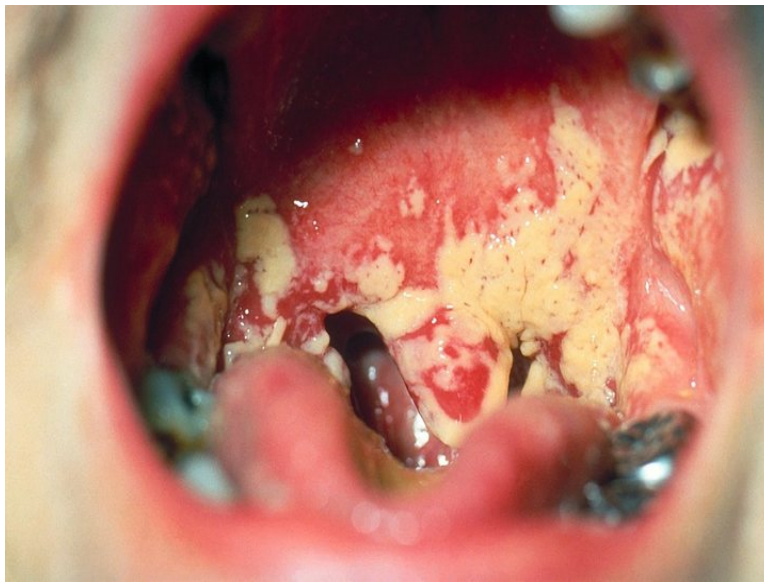
Investigations show high levels of EBV DNA in the blood with circulating antibodies against viral capsid antigen (VCA) and early antigen (EA)■restricted proteins.

Oral hairy leukoplakia



Other options:

Option A: Oral Thrush/ oral candidiasis associated with *Candida albicans* presents as white plaque-like lesions with cheesy exudate on an erythematous mucosa. They are typically painful and can be easily brushed off.



Option B: AIDS-associated Kaposi sarcoma (KS) due to human herpes virus-8 (HHV-8) is an AIDS-defining illness and is the most common malignancy associated with HIV infection. Kaposi sarcoma mainly affects the hard palate mucosa in particular. The gingiva, tongue and buccal mucosa are also involved. The initial lesion is a reddish-purple macule, which develops into a purple nodule swelling that could be widespread and ulcerated.



Option C: Herpes simplex virus (HSV) infection is common in and around the mouth. HSV-1 is the primary cause of gingivostomatitis, which is most common in children. The gums are swollen, inflamed and bleed easily. The mucous membranes of the tongue, pharynx, palate, and buccal region are covered in vesicles that appear as white plaques. The plaques turn into ulcers with a pseudomembrane that is yellow in colour. The regional lymph nodes are enlarged and tender.

Solution to Question 30:

Respiratory Syncytial virus (RSV) which is the only RNA virus in the given list of viruses, requires the use of reverse transcriptase in PCR.

Reverse Transcriptase PCR (RT-PCR) is a molecular analysis method used for the detection of RNA and thus helps in detecting RNA-containing viruses. It can be used to quantify mRNA levels by generating complementary DNA (cDNA) copies of the mRNA using the reverse transcriptase enzyme.

Respiratory Syncytial virus belongs to the Pneumovirus genera of the Paramyxoviridae family. They contain a linear, single-stranded, nonsegmented, negative-sense RNA.

The northern blot technique can also be used to detect RNAs. Here, the mixture of mRNA molecules is separated by electrophoresis and then transferred to a nitrocellulose membrane and hybridized with radiolabeled probes.

Other options:

Option A: Parvovirus B19 is a member of the Parvoviridae family, which consists of very small viruses with a particle size ranging from 18 to 26 nm. They have single-stranded, linear DNA in them.

Option B: Adenoviruses are members of the Adenoviridae family and have double-stranded, linear DNA in their genomes.

Option C: The Herpesviridae family includes the cytomegalovirus (CMV), which has a double-stranded, linear DNA genome.

All DNA viruses can be detected by polymerase chain reaction (PCR). It is a technique used to amplify a selected DNA sequence. It can be used to synthesize millions of copies of the selected DNA sequence in a few hours, enabling the detection of even trace amounts of DNA in a sample.

Solution to Question 31:

The given clinical scenario of a farmer presenting with fever and a black eschar lesion with elevated creatinine levels of 1.4 mg/dl, indicating kidney injury, is suggestive of scrub typhus.

Scrub typhus is caused by *Orientia tsutsugamushi* and transmitted by trombiculid mites. Humans are infected when they are bitten by the mite larvae known as chiggers. Transovarian transmission of microbes occurs in mites.

The incubation period is 1-3 weeks. The patient typically develops a characteristic eschar which is a punched-out ulcer covered with a blackened scab at the site of the mite bite, with regional lymphadenopathy and maculopapular rash.

Renal involvement is a common systemic manifestation of scrub typhus. The severity of renal involvement can range from minor urinary abnormalities such as albuminuria and active urinary sediments such as cellular casts, pyuria, and hematuria to acute kidney injury (AKI) necessitating the immediate initiation of renal replacement therapies.

Other options:

Option A: Anthrax- Cutaneous anthrax (Hide porter's disease) present as a black eschar at the site of entry. At the site of entry of organism or spores, a papule develops in about 1–7 days. This develops into a vesicle that later breaks down into an ulcer with a black, central, necrotic eschar

surrounded by edematous skin. The disease used to be common in dock workers, carrying loads of hides and skins on their bare backs. Renal involvement is unlikely to be seen in this condition.



Leptospirosis (option B) and Brucellosis (option C) can be easily ruled out as they do not present with the black eschar lesions as shown in the image.

Solution to Question 32:

The given clinical vignette of a COVID-19 patient who was treated with steroids, presenting with blood-tinged nasal discharge, 3 weeks after recovery and histopathological examination showing broad, aseptate, ribbon-like hyphae with right angle branching is suggestive of mucor.

Mucormycosis is a fungal infection of the nose and sinuses caused by *Mucor*. It is an opportunistic infection seen even in mild to moderate COVID-19 patients with strong predisposing factors like uncontrolled diabetes and those immunosuppressed or on steroids.

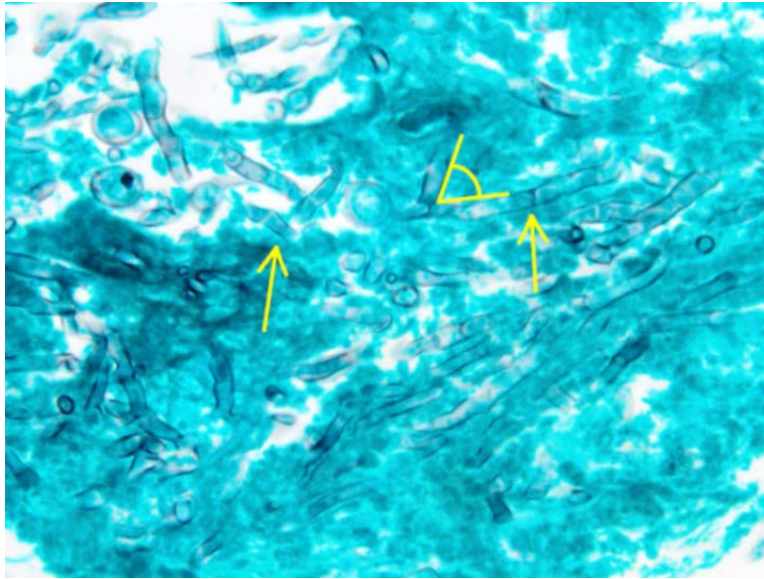
The common clinical form is rhino-orbital-cerebral mucormycosis which occurs due to invasion of the blood vessels in the nasal cavity, eyes, and brain causing thrombosis, infarction, and necrosis. The typical finding is black necrotic material filling the nasal cavity and eroding the nasal septum, hard palate and turbinates due to arterial and venous thrombosis leading to gangrene.

Diagnosis can be made by staining them with 10% potassium hydroxide and culture. On Sabouraud agar at 30-37°C, they appear as grey-white colonies with a thick cottony, fluffy surface. Under the microscope, they appear as broad, ribbon-like, non-septate hyphae with hyphal elements branching at right angles. They also contain aerial sporangiophores which end in a sac containing spores called sporangium.

The drug of choice is the antifungal Amphotericin, administered either as Amphotericin B 0.6-1 mg/kg/day or Liposomal Amphotericin B 5 mg/kg/day which has higher CNS penetration. Surgical debridement of all dead necrotic tissues should be done endoscopically.

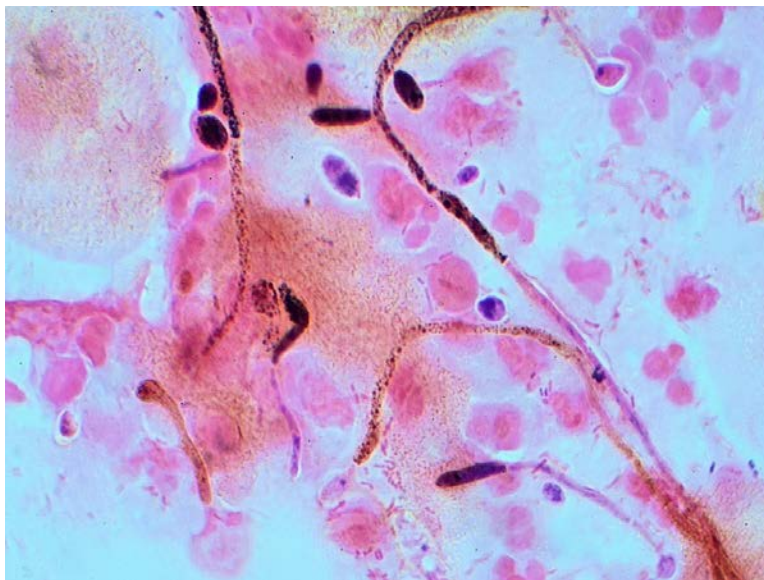
Other options:

Options A: *Aspergillus* shows narrow, septate hyphae with typical dichotomous branching at acute angles (45°) on microscopy.



Option C: *Candida*, the causative agent for candidiasis, appears as budding gram-positive cells on wet films and smears. Culture colonies are creamy-white and have a yeasty odour. *Candida albicans*, the most commonly implicated pathogenic species, form chlamydospores on corn meal agar. They exist as yeast and pseudohyphae (buds do not separate from parent cells) and true hyphae form.

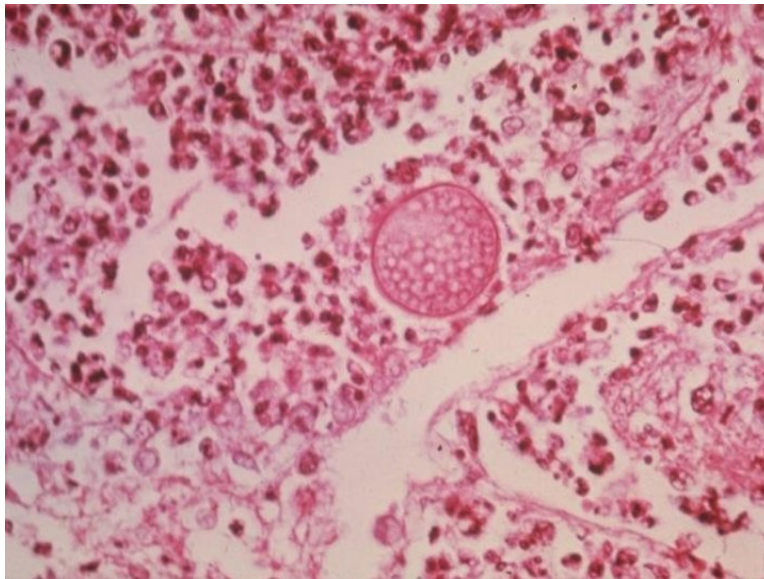
The image below shows candidal hyphae and chlamydospores.



Option D: *Coccidioides immitis* is a dimorphic fungus, known to cause Valley Fever. It exists in two distinct forms, saprophytic and parasitic. In the soil, the organism exists as a mold with septate hyphae resembling a shape of a barrel.



When entering a host, the arthroconidia (spores) break off from the hyphae and evolve into round structures called spherules. Spherules are large sac-like structures filled with endospores as shown below.



Solution to Question 33:

In a suspected case of dengue, where a patient presented with a 3-day history of symptoms like fever, headache, and retro-orbital pain, all the above investigations can be done except for IgM-ELISA, which is used to identify antibodies against the dengue virus.

Dengue viruses belong to flaviviruses and present with sudden onset of fever (saddleback type), frontal headache, retro-orbital pain, back pain, and severe myalgias. It is transmitted from person to person by the bite of the *Aedes aegypti* mosquitoes.

The virus can be found in serum, plasma, circulating blood cells, and other tissues for 4-5 days following the onset of the illness. Virus isolation, nucleic acid detection, or antigen detection can be used to diagnose the infection in its early stages. Serology is the preferred method of diagnosis at the end of the acute phase of infection.

Before day 5 of illness, during the febrile period, diagnosis can be made by the following tests:

- Virus isolation can be done during the period of viremia i.e. usually before day 5. The mosquito cell line C6/36 (cloned from *Aedes albopictus*) or AP61 are the host cells of choice for routine isolation of the dengue virus.
- RT-PCR - They have higher sensitivity than virus isolation and a much faster turnaround time.
- NS1 antigen can be detected up to 7-10 days after the onset of illness.

After day 5, dengue viruses and antigens disappear from the blood with the appearance of specific antibodies. The presence of IgM antibody or rising titres of IgG antibody on ELISA is used for detecting and distinguishing primary and secondary dengue infections.

Solution to Question 34:

In the given list of viruses, herpes simplex virus (HSV)-2 has the least teratogenic effect.

HSV-2 presents as a genital tract infection transmitted via sexual contact. Premature labour or IUGR may occur if the primary infection is acquired during the last trimester. As the transplacental route of spread is uncommon, it is considered to have a less teratogenic effect on the baby.

Caesarean delivery is recommended in cases of active primary genital HSV infection because the virus shed from the cervix or lower genital tract during vaginal delivery is a major source of infection in the fetus.

Babies with neonatal herpes exhibit three categories of disease: (1) localized lesions of the skin, eye, and mouth, (2) encephalitis with or without skin involvement and (3) disseminated disease. It is treated with IV acyclovir.

Other options:

Option A: The varicella-zoster virus, which causes chickenpox, can pass through the placenta and result in congenital varicella syndrome (CVS). Hypoplasia of the limbs, psychomotor retardation, IUGR, chorioretinal scarring, cataracts and microcephaly are the characteristics of CVS. If maternal infection develops after 20 weeks, the risk of congenital malformation is almost absent. Varicella zoster immunoglobulin (VZIG) should also be given to newborns exposed within 5 days of delivery.

Option B: The transplacental spread of the rubella virus is known to cause congenital rubella syndrome (CRS) in the baby. When this infection develops in the first, second or third months, there is a 60%, 25%, or 10% chance of significant abnormalities, respectively. Congenital rubella syndrome predominantly includes sensorineural deafness, patent ductus arteriosus (PDA), cataracts and retinopathy. The virus predominantly affects the fetus and is extremely teratogenic if contracted within the first trimester.

Option C: Cytomegalovirus (CMV) replicates in respiratory epithelial cells, salivary glands, and kidneys and persists in lymphocytes. The transplacental route primarily affects the foetus, resulting in microcephaly, intracranial calcification, hepatosplenomegaly, chorioretinitis, mental retardation, and sensorineural deafness. Unlike rubella, CMV may damage fetal organs throughout gestation.

Microbiology INI-CET 2024

Question 1:

A 24-year-old female comes to the OPD with dysuria. She has a non-irritant profuse white discharge from her vagina. On examination, her labia were inflamed and swollen. The openings of her urethra and Bartholin's glands were congested. On further collection of her endocervical discharge, intracellular gram-negative diplococci were found. Which of the following is the most likely diagnosis?

- a) Neisseria Gonorrhea
- b) Chlamydia Trachomatis
- c) Treponema Pallidum
- d) Hemophilus Ducreyi

Question 2:

Drug resistance is seen with which of the following STD?

- a) Syphilis
- b) Gonorrhoea
- c) Chancroid
- d) Donovanosis

Question 3:

Which of the following parasites reproduce directly from egg to larva without fertilisation through parthenogenesis?

- a) Strongyloides stercoralis
- b) Ancylostoma doudenale
- c) Enterobius
- d) Trichuris trichura

Question 4:

Which is the best test used for Malarial Parasite speciation?

- a) Acridine orange stained thin smear
- b) Giemsa-stained thin smear
- c) Giemsa-stained thick smear
- d) Quantitative Buffy Coat (QBC)

Question 5:

Assertion(A): Late trophozoite and schizont stage in *Plasmodium falciparum* are not seen in blood smears. Reason(R): These two stages are trapped in the capillary endothelium of internal organs due to cytoadherence.

- a) Both Assertion (A) and Reason (R) are true and reason is Reason (R) is the correct explanation of Assertion (A)
- b) Both Assertion (A) and Reason (R) are true and reason is Reason (R) is not the correct explanation of Assertion (A)
- c) Assertion (A) is true and Reason (R) is false
- d) Assertion (A) is false and Reason (R) is true

Question 6:

Which of the following pathological features is incorrectly matched with the disease?

- a) No reaction: Immunodeficiency disease
- b) Cytopathic and cytoproliferative effect : Chickenpox
- c) Mononuclear and granulomatous lesions: Berylliosis
- d) Suppurative and purulent lesions: Syphilis

Question 7:

Out of the given lab sterilisation techniques, which of the following can be used for a disposable pipette tip?

- a) A only
- b) D only
- c) A,B
- d) C,D

Question 8:

Which of the following is transmitted by all the components of blood?

- a) Syphilis
- b) Malaria
- c) Filaria
- d) Toxoplasma

Question 9:

The following statements are connected with 'because'

- a) Statement 1 and Statement 2 are true and Statement 2 is correct explanation of Statement 1
- b) Statement 1 and Statement 2 are true but Statement 2 is not the correct explanation of Statement 1
- c) Statement 1 is true and statement 2 is false
- d) Both the statements 1 and 2 are false

Question 10:

A construction worker presents with a lock jaw and muscle rigidity. He gives a history of exposure to a rusted iron nail through an existing wound. Which of the following is the mechanism of action of the toxin responsible for this presentation ?

- a) Inhibition of Acetyl choline release
- b) Inhibition of glycine and GABA release
- c) Inhibition of epinephrine
- d) Inhibition of Nor-epinephrine

Question 11:

Which of the following immunodeficiency diseases predisposes to infection by catalase-positive organisms?

- a) Severe combined immunodeficiency (SCID)
- b) Hyper IgE syndrome

- c) X-linked agammaglobulinemia
- d) Chronic granulomatous disease

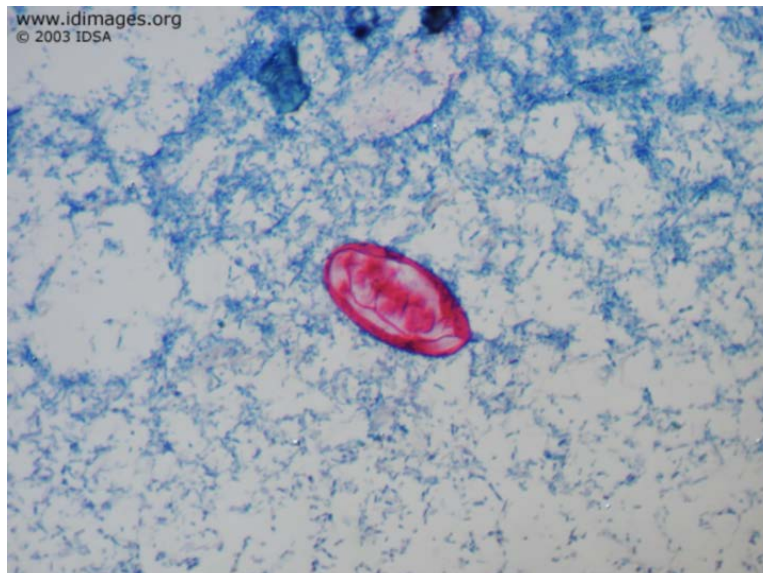
Question 12:

Which of the following is not a clinical manifestation of Parvovirus B19 infection?

- a) Hydrops fetalis
- b) Persistent anemia or pure red cell aplasia in immunodeficient individuals
- c) Transient aplastic crisis in children with hemolytic anemia
- d) Febrile illness with petechial rash

Question 13:

A 50-year-old patient who underwent a renal transplant recently came with complaints of watery diarrhoea. The stool sample revealed oocysts of size 30x15 microns. Identify the organism



- a) Cyclospora cayatenensis
- b) Cystoisospora
- c) Cryptosporidium
- d) Microsporidium

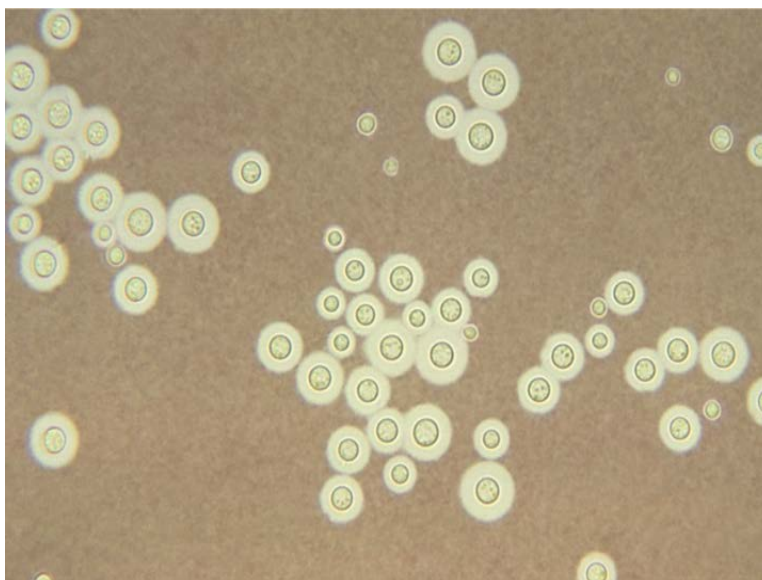
Question 14:

An 18-day-old baby presents with altered sensorium, and neck rigidity with vesicular lesions on the anterior abdomen and the forearm. The mother revealed a history of treatment for painful genital ulcers a year ago and the baby was born through vaginal delivery. CSF examination reveals multinucleate giant cells. Which of the following is the most likely involved organism?

- a) HSV - 2
- b) Coxsackie virus
- c) Enterovirus
- d) VZV

Question 15:

Which of the following is true with respect to the organism shown below?



- a) Most commonly involves the lungs
- b) Associated with budding yeast cells and pseudohyphae
- c) Only reservoir is man
- d) Opportunistic infections are seen in patients with CD4 count < 900

Question 16:

Which of the following is most commonly associated with *S.agalactiae* ?

- a) Neonatal sepsis

- b)** Pharyngitis with or without tonsillitis
- c)** Dental caries
- d)** Endocarditis

Question 17:

Match the following:

- a)** 1-b, 2-a, 3-c, 4-d
- b)** 1-a, 2-c, 3-b, 4-d
- c)** 1-d, 2-c, 3-a, 4-b
- d)** 1-c, 2-a, 3-d, 4-b

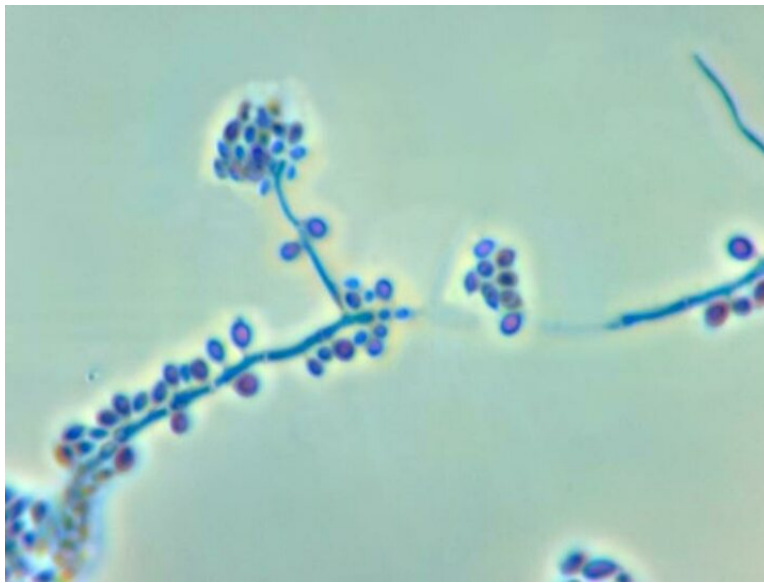
Question 18:

Assertion (A) - Mycobacterium is detected in sputum samples with N-acetyl cysteine (NAC) and NaOH. Reason(R) - The sputum samples are treated by NAC to kill commensals and NaOH to liquefy the sputum.

- a)** Assertion(A) is true but Reason(R) is false
- b)** Both assertion(A) and Reason(R) are true and Reason(R) is the correct explanation of assertion(A)
- c)** Assertion(A) is false but Reason(R) is true
- d)** Both assertion(A) and Reason(R) are true but Reason(R) is not the correct explanation of assertion(A)

Question 19:

A gardener in Himachal notices a black ulcer in the right upper arm. Axillary lymph nodes are enlarged and matted. Aspirate from the lymph nodes shows elongated yeasts 3-5 micrometres. Mycelium was isolated from the aspirate and grown at 22 degrees Celsius in SDA, and the image is given below. What is this condition?



- a) Coccidiomycosis
- b) Histoplasmosis
- c) Sporothrix schenkeii
- d) Blastomycosis

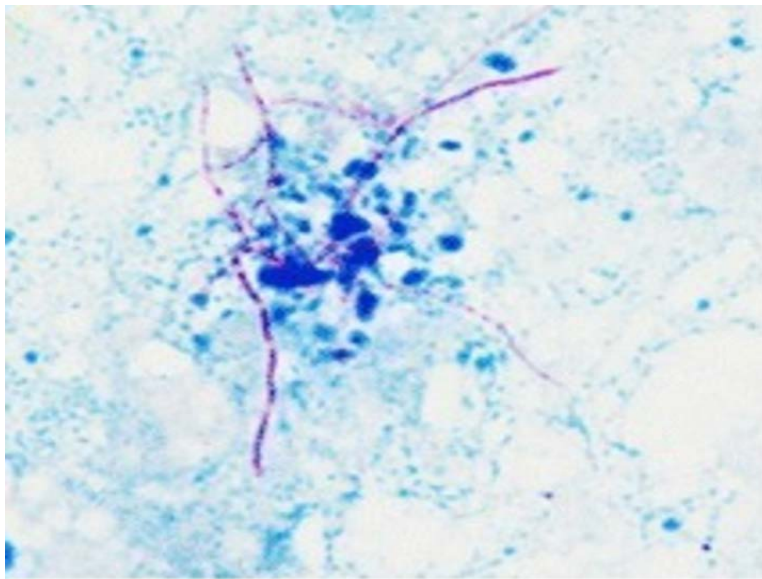
Question 20:

A 2-year-old male child presents with fever, hematuria, thrombocytopenia, and diarrhea for the last 3 days. What is isolated from the stool sample?

- a) *Shigella flexneri*
- b) *Salmonella enteritis*
- c) *Aeromonas*
- d) *E. Coli* O157:H7

Question 21:

A long-beaded, filamentous, branching, weakly acid-fast organism is isolated from a lesion caused by trauma to the chest by a blunt farming tool. What is the isolated organism?



- a) Bacillus anthrax
- b) Listeria
- c) Nocardia
- d) Mycobacterium tuberculosis

Question 22:

What is not true regarding naive T-cell interaction with CD4 effector T-cells?

- a) T cell interaction with APC
- b) Interaction between CD 80 and CD 28
- c) PD-1 on T cell interacts with PD ligand 1 on APC
- d) Cytokine mediated proliferation and differentiation

Question 23:

Which of the following is a method of antibiotic sensitivity testing?

- a) E test
- b) Elek test
- c) Latex agglutination test
- d) CAMP test

Question 24:

Which of the following is transmitted by walking barefoot in soil?

- a) *Madurella mycetomatis*
- b) *Curvularia*
- c) *Coccidioides immitis*
- d) *Aspergillus fumigatus*

Question 25:

Which of the following cancers can be prevented by vaccination?

- a) A,B,C,D
- b) A and C only
- c) A and D only
- d) C and D only

Question 26:

Which of the following statements is true about *Cryptococcus*?

- a) 1,2,3
- b) 2,3,4
- c) 2,4
- d) 1,2

Question 27:

What is the culture medium for *M. tuberculosis*?

- a) Lowenstein-Jenson medium
- b) Blood agar
- c) Novy-MacNeal-Nicolle medium
- d) Sabouraud dextrose agar

Question 28:

A 2-year-old presents with excessive watery diarrhoea and severe dehydration. What is the causative organism?

- a) Trichuris Trichura
- b) Shigella
- c) Rotavirus
- d) Yersenia

Question 29:

Match the causative organism with the disease.

- a) i-c,ii-b,iii-d,iv-a
- b) i-b,ii-c,iii-d,iv-a
- c) i-b,ii-c,iii-a,iv-d
- d) i-d,ii-a,iii-b,iv-c

Question 30:

Which of the following does not confer passive immunity?

- a) Toxoid
- b) Antitoxin
- c) IgA
- d) Maternal monoclonal antibody

Question 31:

Which of the following is not a cold sterilization technique?

- a) Plasma sterilization
- b) Autoclave
- c) UV radiation
- d) Gamma radiation

Question 32:

Which of the following is not true regarding *Ascaris lumbricoides*?

- a) Most infected humans are asymptomatic
- b) Larvae in the lungs produce asthma-like symptoms
- c) High larval load can cause rectal prolapse
- d) Unembryonated eggs in stool are not infective and embryonated eggs in soil are infective

Question 33:

What is the drug of choice for *fasciola hepatica*?

- a) Triclabendazole
- b) Praziquantel
- c) Mebendazole
- d) Albendazole

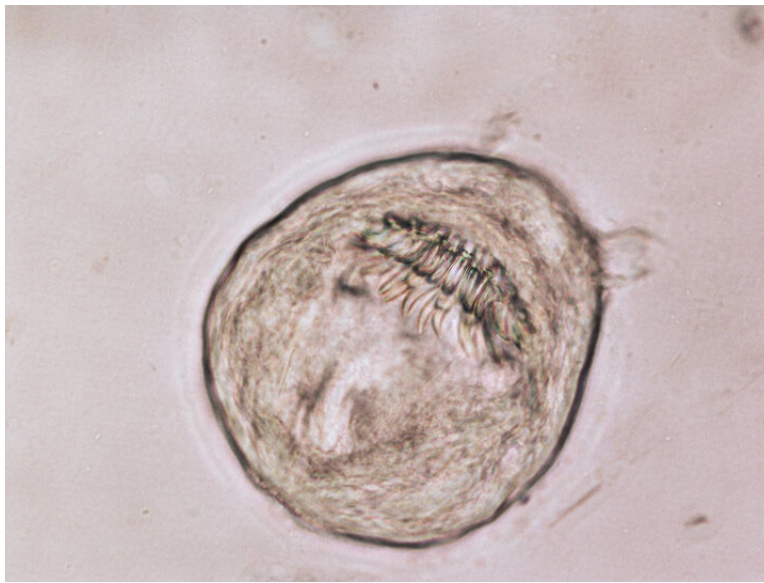
Question 34:

A 20-year-old male presented with a pustule that developed into an ulcer which was diagnosed to be a Necrotising fascitis. A sample of the ruptured pustule showed gram-positive cocci in chains. Identify the organism.

- a) *Streptococcus pyogenes*
- b) *Staphylococcus aureus*
- c) *Enterococcus*
- d) *Micrococcus*

Question 35:

A 25-year-old male presented with abdominal pain. On examination, a painful growing mass is felt in the right hypochondrium. Aspiration of the cyst is shown below. Identify the organism.



- a) Echinococcus granulosus
- b) Hymenolopsis nana
- c) T.solium
- d) T.saginata

Question 36:

Which of the following drugs is not given in uncomplicated falciparum malaria?

- a) Chloroquine
- b) Primaquine
- c) Arthemeter
- d) Pyrimethamine Sulfadoxine

Question 37:

How do you sterilise bacteriological loops used for inoculation?

- a) Heat till metal turns red hot
- b) Pass over flame 3 times
- c) Autoclave
- d) Microwave

Answer Key

Question No.	Correct Option
1	a
2	b
3	a
4	b
5	a
6	d
7	a
8	b
9	c
10	b
11	d
12	d
13	b
14	a
15	a
16	a
17	c
18	a
19	c
20	d
21	c
22	c
23	a
24	a
25	c
26	c
27	a
28	c
29	b
30	a
31	b
32	c
33	a
34	a

35	a
36	a
37	a

Detailed Explanations

Solution to Question 1:

The above clinical scenario and endocervical examination of gram-negative diplococci denote a diagnosis of *Neisseria gonorrhoea*.

Gonorrhoea is a sexually transmitted disease where infection occurs due to the adhesion of gonococci to the urethra or other mucosal surfaces with the help of pili. By the third day of infection, the cocci have penetrated intracellular spaces and reached subepithelial tissue. The incubation period ranges from 2 to 8 days. The main symptoms are urethral discharge and dysuria.

In females :

The infection begins in the cervix uteri and urethra and may spread to the Bartholin glands, endometrium, and fallopian tubes. The vaginal mucosa is usually unaffected in adults because the stratified squamous epithelium is resistant to cocci infection and also because vaginal secretions have an acidic pH. Rarely peritonitis may develop with perihepatic inflammation (Fitz Hugh Curtis syndrome).

In males :

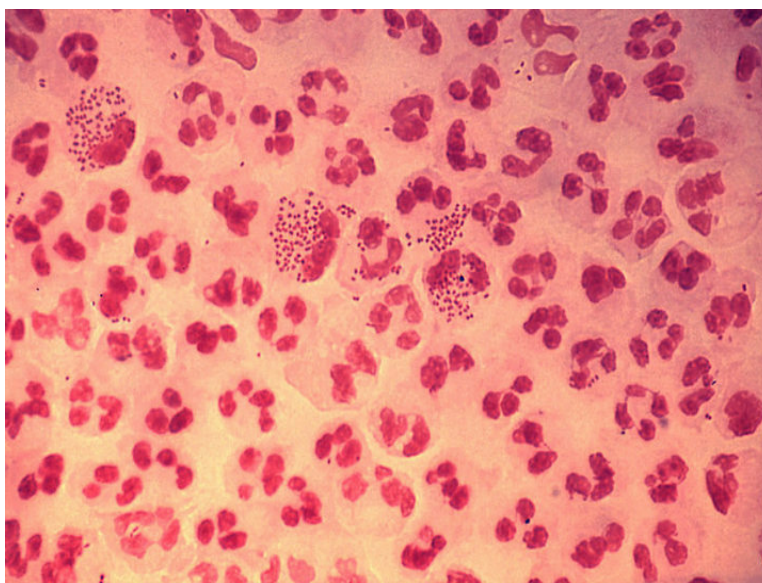
The disease starts as acute urethritis. This infection has the potential to spread down the urethra and into the prostate, seminal vesicles, and epididymis. Chronic urethritis may lead to stricture formation. Infections in the periurethral tissues can lead to abscesses and multiple discharging sinuses (watercan perineum).

Lab diagnosis:

The discharge is collected and subjected to Gram's stain which reveals gram-negative intracellular diplococci within phagocytes. Stuart's transport medium is used. Specifically, Thayer-Martin agar is used to isolate *Neisseria gonorrhoeae*.

Treatment: Ceftriaxone (250 mg, IM) single-dose + azithromycin 1 g PO, single-dose, or doxycycline 100 mg PO BD for 7 days.

The image below shows the Gram-stain of a urethral exudate showing typical intracellular gram-negative diplococci and pleomorphic extracellular gram-negative organisms



Solution to Question 2:

Neisseria gonorrhoea endures maximum drug resistance among the given organisms causing sexually transmitted diseases.

The ability of *N. gonorrhoea* to change its antigenic properties and adapt to different environments has rendered it resistant to the majority of antibiotics. Gonococci develop complete resistance to antibiotics, either through chromosomal mutations or by acquiring R factors, which are plasmids.

- Often β -Lactamase producing Gonococci and tetracycline-resistant strains occur together. Hence Penicillin, ampicillin, and tetracycline are no longer reliable for the treatment of gonorrhea and should not be used.
- Changes in DNA gyrase and topoisomerase IV have been identified as mechanisms contributing to fluoroquinolone resistance.
- Mechanisms leading to enhanced drug efflux pump and reduced drug influx through PorB, collectively contribute to resistance against third-generation cephalosporins.

Currently, the effective first-line treatment of choice for uncomplicated Gonococcal infection remains Ceftriaxone (500 mg IM, single dose) plus Azithromycin (1 g PO, single dose) .

Other options :

A) Syphilis : The drug of choice for treating syphilis is benzathine penicillin G. Penicillin has been the mainstay of treatment for syphilis for many years due to its effectiveness in curing the infection and its relatively low cost.

c) Chancroid: The drug of choice for treating chancroid, a sexually transmitted infection caused by the bacterium *Haemophilus ducreyi*, is azithromycin or ceftriaxone. These antibiotics are effective in treating chancroid and preventing its complications. The choice between azithromycin and ceftriaxone may depend on factors such as patient allergies, local resistance patterns, and individual patient factors.

d) Donovanosis : The drug of choice for treating donovanosis, also known as granuloma inguinale, is typically azithromycin or doxycycline. Donovanosis is a sexually transmitted infection caused by the bacterium *Klebsiella granulomatis*, and prompt treatment is essential to prevent complications and transmission to others.

Solution to Question 3:

The females of *Strongyloides stercoralis* are known to be parthenogenetic i.e. capable of generating offspring without the need for fertilisation by a male.

Strongyloides stercoralis has a parasitic cycle and a free-living cycle in soil. Humans are the natural host but morphologically indistinguishable strains may be seen in dogs and cats. Filariform larvae are the infective form and there are two modes of infection: penetration of the skin by 3rd stage filariform larva, and autoinfection (external and internal).

The female worms present in the intestine lay eggs in the mucosa which hatches immediately, releasing the rhabditiform larva. The larva migrates into the lumen of the intestine and passes down to be released into the faeces. The rhabditiform larva may change into a filariform larva while passing through the bowel.

The filariform larvae in the bowel do not leave the host and become a source of autoinfection as they penetrate the colonic mucosa or perianal skin. The rhabditiform larvae passed in the faeces may undergo two types of development in the soil:

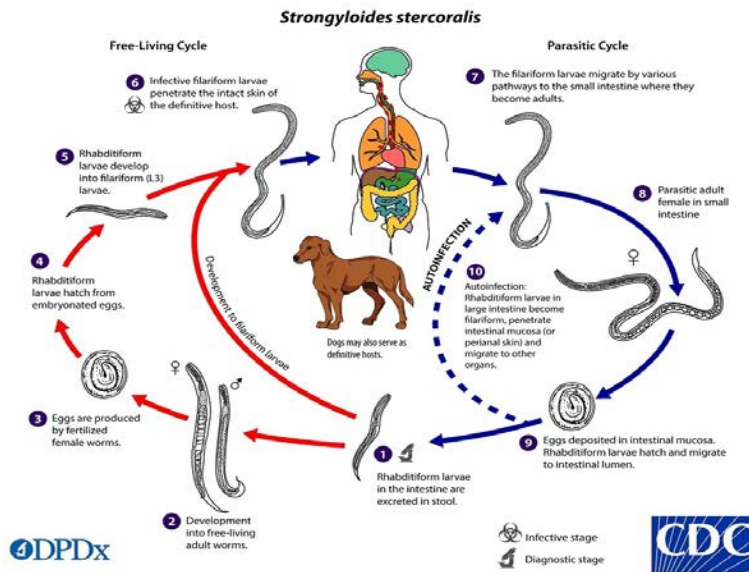
- Direct development: The rhabditiform larva moults twice on reaching the soil to become the infective filariform larva. The filariform larva can penetrate the skin and enter the circulation.
- Indirect development: The rhabditiform larvae develop into free-living males and females in the soil.

Other options :

Option B: *Ancylostoma duodenale* are dioecious parasites, there is no multiplication within the host, and a single infective larva matures into either a male or a female adult to undergo mating through copulation. It typically takes around 6 weeks from the point of infection for the adult worms to reach sexual maturity and initiate egg-laying.

Option C: *Enterobius vermicularis* is dioecious. The male is usually unseen as it does not migrate from caecum or appendix. It usually dies after mating and is passed in the faeces. The gravid female migrates down the colon to the rectum and a single worm lays from 5000 to 17,000 eggs

Option D: *Trichuris trichuria* or the whipworm has a distinct male and female parasite and is typically dioecious. The fertilised female lays about 5000 eggs per day.



Solution to Question 4:

Malarial parasite speciation is best done by a Giemsa-stained thin smear.

Microscopy is the best method for the diagnosis of malaria due to its high sensitivity and specificity. A diluted Giemsa stain is applied to both thick and thin films and left to dry for thirty minutes to two hours. Afterwards, the slide is rinsed and dried.

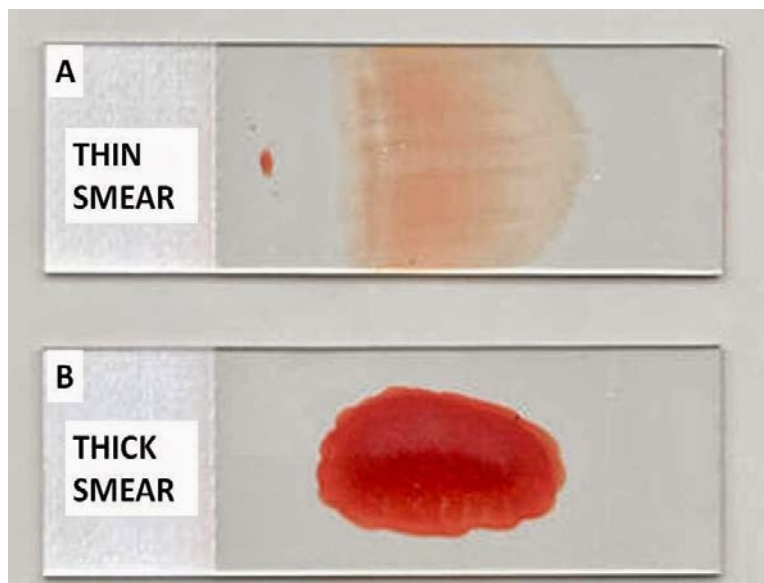
Thin smear is used to identify the particular infecting species of Plasmodium. It is prepared by air drying for 10 minutes and fixed in methanol. Thus, the plasmodia morphology is not distorted.

Thick smear is used to detect infection and estimate parasite concentration. The correct thickness is the thickness through which a newsprint is barely visible. It is prepared by slow-drying for 30 minutes, which leads to hemolysis and distortion of the plasmodia morphology. The thick film is not fixed using methanol as it can become dehaemoglobinised. Hence, species cannot be detected.

For rapid staining, especially beneficial for fieldwork, the Field's stain is utilized. In India, the frequently employed rapid staining method is the JSB stain, named after Jaswant Singh and Bhattacharji.

The following image shows the difference between thick and thin smear.

- When properly prepared, the thick film will just permit printed letters to be visible when viewed through it.
- A well-prepared thin film will consist of a continuous smear of a single layer of red cells, terminating in a tapered tongue end that falls slightly before reaching the edge of the slide.



Other options:

Option D: The QBC method is a rapid test for malaria which is highly sensitive and specific. In this method, a small amount of blood (50 to 110 μ l) is centrifuged in the QBC centrifuge, concentrating parasites near the tip of the red blood cell column.

Option A: Acridine orange is a fluorescent stain used in the QBC test of malaria. RBC tubes are pre-coated with acridine orange, and parasites fluoresce, making them easily visible under an oil immersion microscope. It is hence not the ideal stain used for speciation.

Solution to Question 5:

Late trophozoite, early and mature schizonts are stages in the asexual life cycle of *P. falciparum*. These stages are usually not seen in the peripheral smear unless the malaria is exceptionally severe or pernicious. In regular conditions, These two stages are trapped in the capillary endothelium of internal organs due to cytoadherence.

Plasmodium falciparum is the most highly pathogenic strain of malaria, causing fatal complications. Hence, it is also called malignant tertian or pernicious malaria. The life cycle of *P. falciparum* can be divided into three primary stages based on whether parasites are in the liver, the bloodstream or the mosquito's stomach.

Exo-erythrocytic cycle-Human liver :

- The female anopheles mosquito injects parasites (sporozoites) into the human after a blood meal. The sporozoites are sickle-shaped and invade into hepatocytes.
- Sporozoites mature into schizonts which release merozoites

Erythrocytic cycle:

- Merozoites leave the liver and invade red blood cells, in which they reproduce asexually and develop into trophozoites.

- Early ring trophozoites are often seen attached along the margin of the red cell, the so-called form applique or accolé. Binucleate rings are common resembling stereo headphones in appearance. Several rings may be seen within a single erythrocyte.
- Typically, the trophozoites vanish from the bloodstream within approximately 24 hours.
- By this time, a strain-specific high molecular weight antigen emerges on the surface of infected red cells resembling a knob-like projection on the erythrocyte membrane.
- Infected red cells with presenting antigen, exit the bloodstream and stick to the walls of small blood vessels (cytoadherence) of internal organs, including the brain, heart, kidneys, lungs, spleen, intestines, bone marrow, and placenta.
- This adherence leads to the sequestration of infected cells in these areas, contributing to severe complications like cerebral malaria.

Sporogonic cycle- Mosquito :

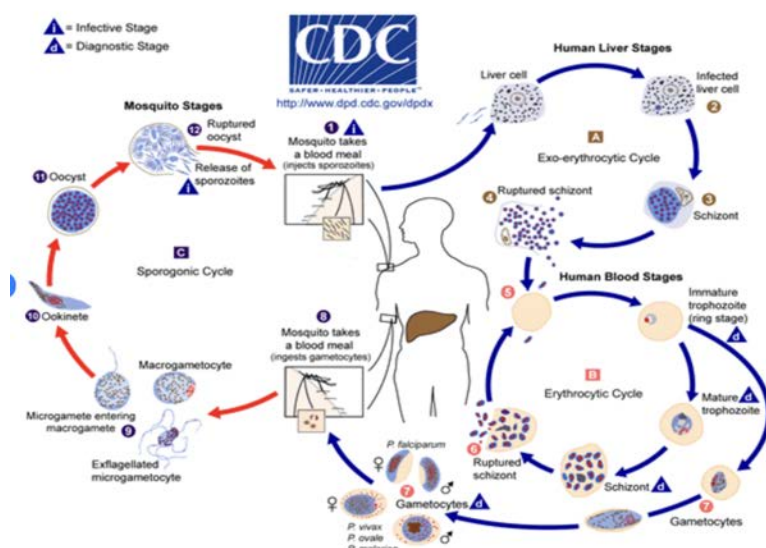
- Gametocytes are taken up by a mosquito when it feeds. (9-12) Gametocytes undergo sexual reproduction in the midgut of the mosquito and develop into sporozoites, which migrate to the salivary glands.
- The mosquito injects sporozoites through saliva into another human source.

Major manifestations of severe falciparum malaria include more than two convulsions in 24 hours, unarousable coma/ cerebral malaria (failure to localise or respond appropriately to stimuli or coma persisting for >30 min after generalised convulsion), acidosis, severe normocytic normochromic anaemia, renal failure, pulmonary oedema, hypoglycemia, shock, and disseminated intravascular coagulation.

Other manifestations include hemoglobinuria, hyperparasitemia, jaundice, and extreme weakness.

The treatment of choice is artesunate given at a dose of 2.4 mg/kg stat IV, followed by 2.4 mg/kg at 12 and 24 hours, and then daily if necessary.

The life cycle of plasmodium falciparum is given below :



Solution to Question 6:

Syphilis is not known to cause suppurative and purulent lesions. It causes macular, papular or mixed maculopapular lesions.

Syphilis presents with the following lesions as it progresses through different stages :

- Primary syphilis - Hard chancre, which is painless papule that becomes eroded and indurated.
- Secondary Syphilis - Skin rashes on palms and soles which may be macular, papular, papulosquamous and nodular, but never vesicular or bullous.
- Condylomata lata- mucocutaneous papules that coalesce to form large pink to grey lesions in warm moist intertriginous areas)
- Generalized lymphadenopathy is seen
- Tertiary Syphilis - Gummas, which are locally destructive granulomatous lesions. They are most commonly seen in bone and skin.

Other options :

Option A - Immunodeficient individuals lack the normal response to antigens. This is the reason for the false negative Mantoux test seen in such individuals.

Option B - Chickenpox belongs to the alpha subfamily of herpes group of viruses which has a cytopathic effect of formation of multinucleated giant cells, with rounding and ballooning of infected cells

Option C - Berryliosis is an occupational lung disease, causing a chronic granulomatous inflammatory disease similar to sarcoidosis. Biopsy reveals non-caseating granulomas with monocyctic infiltration.

Solution to Question 7:

The most common and effective method to sterilise disposable pipette tips is by using an autoclave.

Autoclaving involves subjecting the pipette tips to high-pressure steam at temperatures above 121°C (250°F) for around 20 minutes. This process effectively kills microorganisms and sterilises the pipette tips for reuse. However, it's important to check the manufacturer's instructions to ensure that the pipette tips are autoclavable and to follow proper sterilisation protocols to maintain their integrity and functionality.

Solution to Question 8:

Among the options provided, Malaria is the only disease transmitted by all components of blood.

Malaria is primarily spread through the bite of infected Anopheles mosquitoes, which inject sporozoites into the bloodstream but can also be transmitted by transfusion of blood and blood products.

other options:

Option A : Syphilis is primarily transmitted through direct contact with syphilitic sores (chancres), which can occur on the genitals, lips, mouth, or anus during sexual activity. In most cases, transmission via blood transfusion is rare.

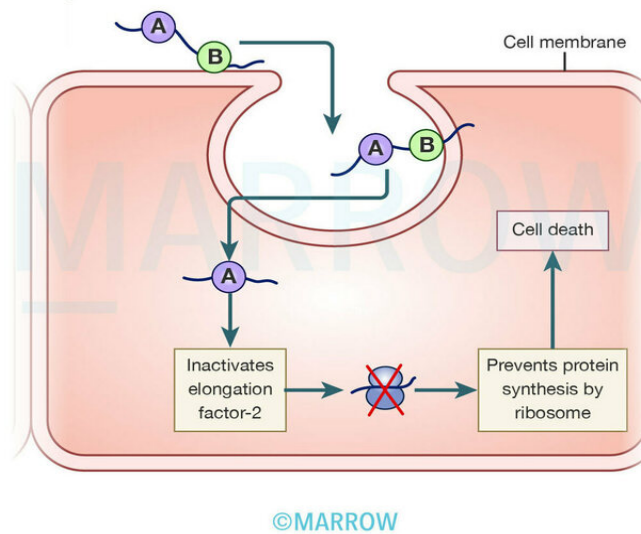
Option C: Filarial parasites are primarily transmitted through the bite of infected mosquitoes. However, they can also be found in the blood of infected individuals, which can potentially transmit the infection through blood transfusion. However, plasma and platelets may not typically harbor filarial parasites unless there's concurrent infection or contamination.

Option D: Toxoplasma is transmitted through various routes, including ingestion of contaminated food or water, contact with infected cat feces, and transplacental transmission from mother to fetus. However, it is not primarily transmitted through blood

Solution to Question 9:

The statement Faucial diphtheria can cause death due to myocarditis is true. However, *Corynebacterium diphtheriae* causes myocarditis due to toxemia and not bacteremia.

Corynebacterium diphtheriae produces the diphtheria toxin, which is an AB subunit toxin. While the B subunit helps in attachment to cell surface receptors, the A subunit causes ADP-ribosylation of elongation factor-2 (EF-2) resulting in the inhibition of protein synthesis.



The exotoxin A produced by *Pseudomonas aeruginosa*, also an AB subunit toxin, acts by the same mechanism to cause inhibition of protein synthesis.

This infection presents as fever and sore throat, which progresses to dyspnea due to obstruction by the pseudomembrane. Bull's neck appearance is seen due to cervical lymphadenitis.

Complications of *C. diphtheriae*:

- damage to the heart (myocarditis)
- kidney (tubular necrosis)

- liver and adrenal glands can be affected due to the toxin

In respiratory diphtheria patients, cardiac complications, particularly myocarditis, arise in around 10-25% of cases. Cardiac involvement is the leading cause of death in diphtheria, with a mortality rate of 50-75%. Manifestations of cardiac involvement vary and may encompass myocardial dysfunction, rhythm disturbances such as bradyarrhythmias or tachyarrhythmias, and heart blocks necessitating pacing. Elevated cardiac enzymes and ECG abnormalities may signal asymptomatic myocardial involvement.

Solution to Question 10:

The above clinical scenario, coupled with a history of contact with a rusty iron nail, suggests a diagnosis of tetanus, which is caused by the anaerobic bacterium *Clostridium tetani*. The toxin responsible for muscle rigidity and lockjaw is tetanospasmin produced by *Clostridium tetani*. It acts by causing presynaptic inhibition of GABA and glycine release.

Tetanospasmin, after secretion by the bacterium is cleaved into two peptides linked by a disulphide bond. The larger peptide diffuses to terminals of neurons and migrates by the retrograde axonal transport system to the spinal cord and brainstem. It diffuses to the glycinergic interneurons and GABA secreting neurons there. The smaller peptide degrades synaptobrevin and blocks the release of the inhibitory glycine and GABA. This results in hyperreflexia, muscle spasms, and spastic paralysis. Even small amounts of the toxin can be lethal.

The incubation period of *C. tetani* ranges from 4-5 days up to 3 weeks. Muscle spasms usually involve the area of injury first and then the muscles of the jaw causing trismus or lockjaw. Gradually, other voluntary muscles become involved, resulting in tonic spasms. Any external stimulus can precipitate a tetanic generalized muscle spasm. The patient is conscious but may be in pain. Death usually results from respiratory paralysis.

Other options:

Option A : Botulinum toxin targets the presynaptic terminals of motor neurons and inhibit the release of acetylcholine. As a result, muscle paralysis occurs, leading to the temporary weakening or cessation of muscle activity in the affected area.

Option D: The venom of the black widow spider, specifically the component α -latrotoxin, has been found to induce the massive release of neurotransmitters, including acetylcholine, norepinephrine, dopamine, and others, from presynaptic nerve terminals rather than inhibition of nor-epinephrine.

Solution to Question 11:

Chronic granulomatous disease (CGD) is a rare inherited immune system disorder predisposing individuals to infection by catalase-positive organisms. It is mostly acquired as X-linked recessive seen in boys or very rarely autosomal recessive seen in girls.

The most common organism is *S. aureus*, but any catalase-positive microorganism may be involved. Other organisms frequently causing infections include *Serratia marcescens*, *B. cepacia*, *Aspergillus*, *Candida albicans*, *Nocardia*, and *Salmonella*. There may also be increased

susceptibility to mycobacteria, including the bacille Calmette-Guérin vaccine. Catalase-negative bacteria like streptococci and pneumococci are efficiently handled by the immune system.

This disease manifests itself as chronic suppurative granulomatous lesions of the skin and lymph nodes, along with hepatosplenomegaly, progressive lung infiltrates and granulomatous septic osteomyelitis.

CGD is usually diagnosed in early childhood:

- Phagocytes cannot efficiently produce reactive oxygen species (ROS), which are crucial for destroying pathogens engulfed by these cells.
- Reduced hydrogen peroxide (H₂O₂) production seems to be the primary cause of the impaired bactericidal function.
- Failure of leucocyte degranulation after phagocytosis, leading to delayed granule rupture and faulty release of myeloperoxidase further contributes to ineffective bactericidal activity.
- Bacteria proliferate within the cells, shielded from antibodies and antibiotics due to their intracellular location, leading to the establishment of chronic suppurative infection.
- Humoral and cellular immune response is normal

Nitroblue-tetrazolium test: This is used as a screening method for the diagnosis of chronic granulomatous disease. Leucocytes of individuals with CGD fail to reduce nitroblue tetrazolium in CGD, due to a defect in the NADPH oxidase complex, neutrophils are unable to perform this reduction, leading to a negative or deficient result in the NBT test.

Treatment typically involves long-term antibiotic and antifungal therapy to prevent and manage infections. In some cases, bone marrow transplantation may be considered a potential cure for CGD. Corticosteroids may be useful for the treatment of children with antral and urethral obstruction or severe granulomatous colitis.

Solution to Question 12:

The characteristic clinical manifestation of Parvovirus is febrile illness with an erythematous rash. Petechial rash is seen only in some patients.

Parvovirus B19 is the smallest single-stranded DNA virus belonging to the family Parvoviridae. It is a non-enveloped virus with an icosahedral capsid. The virus is primarily transmitted through respiratory secretions, such as saliva and respiratory droplets. It can also be transmitted vertically from mother to fetus during pregnancy, leading to congenital infection.

Clinical manifestations:

Erythema infectiosum (Fifth Disease): Characterized by a "slapped cheek" rash on the face, the most common manifestation of Parvovirus B19 infection. This is due to erythematous flushing of the cheeks. This is followed by a lacy, macular erythematous rash on the trunk and limbs. The rash typically resolves within 1-3 weeks but may recur with exposure to sunlight or heat.

Arthropathy: Transient arthritis, predominantly affecting the joints of the hands, wrists, knees, and ankles. Arthropathy typically occurs in adults, especially women, and may persist for several weeks to months.

Aplastic Crisis: In individuals with underlying hemolytic disorders, such as sickle cell disease or hereditary spherocytosis, Parvovirus B19 infection can precipitate a transient aplastic crisis. This results in a severe reduction in red blood cell production, leading to anaemia.

Hydrops fetalis: Vertical transmission of Parvovirus B19 during pregnancy can lead to severe complications in the fetus, including hydrops fetalis.

Hydrops fetalis is characterised by fetal oedema, ascites, pleural effusion, and pericardial effusion, often resulting in intrauterine death or neonatal complications.

Parvovirus B19 mainly attacks the erythroid progenitor cells and leads to transient arrest of erythropoiesis. However, immunocompromised individuals are at risk of persistent Parvovirus B19 infection which leads to chronic suppression of bone marrow, resulting in persistent anemia or pure red cell aplasia.

Diagnosing Parvovirus B19 infection is primarily based on clinical presentation and confirmed through serological testing, including detecting specific IgM antibodies. Polymerase chain reaction (PCR) assays can also detect viral DNA in blood or other tissues.

The following image shows a slapped cheek appearance seen in Erythema infectiosum (Fifth Disease):



Note: This is a best-out-of-four type of question where all 4 options are factually correct. Petechial rash is seen in several viral infections including Parvovirus B19 infection. However, the characteristic rash of this infection is erythematous. This makes option D the best possible answer.

Solution to Question 13:

The given clinical scenario of a patient with a history of renal transplant and complaints of watery diarrhoea along with a Kinyoun stain positive image of pink coloured, acid-fast oocyst with the size of 30x15 microns is suggestive of infection due to *Cystoisospora*.

Cystoisospora usually causes a self-limiting but protracted diarrhoea, lasting for several years, majorly in immunocompromised persons and particularly among the HIV infected. Human infection is acquired by ingestion of mature oocysts in food or drink. The parasites generally dwell in the epithelial cells of the small intestine .

Diagnosis involves identifying oocysts in fecal smears, which may sometimes be sparse and overlooked due to their transparent appearance in unstained films. They stain red using the cold acid-fast technique. Concentration methods like zinc sulfate or formol-ether techniques can be used.

The first line of management for isosporiasis is cotrimoxazole for 7-10 days. Patients who do not tolerate cotrimoxazole are treated with nitazoxanide or ciprofloxacin.

A comparison of these coccidian parasites is given in the table below:

Property	Cryptosporidium	Cyclospora	Isospora
Sporulated Oocyst	4-6 µm, round contains four sporozoites.	8-12 µm, round contains 2 sporocysts each having two sporozoites.	23-36 µm, oval contains two sporocysts, each having four sporozoites
Acid-fast detection	Uniformly acid-fast	Variable acid-fast	Uniformly acid-fast
Sporulation of the oocyst	Occurs inside the host cell	Occurs in soil	Occurs in soil
Diagnostic form	Sporulated cyst	Unsporulated cyst	Unsporulated cyst

Solution to Question 14:

The given clinical scenario of an 18-day-old child presenting with altered sensorium, neck rigidity, and vesicular lesions on the anterior abdomen and the forearm with the mother giving a history of painful genital ulcers, is suggestive of neonatal herpes due to herpes simplex virus (HSV)-2. CSF examination with multinucleate giant cells with CNS manifestations indicates neonatal meningitis to the DNA virus.

Painful genital ulcers are commonly seen in genital herpes and chancroid due to *Haemophilus ducreyi* ((ducreyi makes you cry).

Herpes infection can be caused by HSV-1 and HSV-2 viruses. HSV-1 causes orofacial lesions, i.e. herpes labialis, while HSV-2 mainly causes genital herpes which is characterised by vesiculo ulcerative lesions of the penis of the male or of the cervix, vulva, vagina, and perineum of the female.

The typical herpes lesions are thin-walled, umbilicated vesicles, the roof of which breaks down, leaving tiny superficial ulcers which heal without scarring. The most common site of cutaneous lesions is in the face, cheeks, chin, around the mouth and forehead. A cesarean section is performed in a case of active genital herpes at the time of labour in order to prevent neonatal herpes infection.

One of the following three disease patterns can be seen in infants infected during delivery or in the postpartum period: (1) disease limited to the skin, eyes, or mouth; (2) encephalitis with or without skin, eye, and mouth disease; or (3) disseminated infection affecting several organs, such

as the brain, lungs, liver, heart, and adrenal glands.

Infants suffering from encephalitis usually exhibit clinical symptoms that point to bacterial meningitis at 8- 17 days of life. These symptoms include seizures, poor feeding, lethargy, neck stiffness, irritability, and positive Kernig's sign. When left untreated, HSV encephalitis results in 50% of infant deaths and severe neurologic sequelae in the majority of survivors.

Acyclovir at a high dose of 60 mg/kg/day, administered intravenously every eight hours, should be the first line of treatment for any infant with a confirmed or suspected neonatal HSV infection.

Other options:

Coxsackie virus, Enterovirus and VZV are not attributed to painful genital ulcers and hence ruled out.

Solution to Question 15:

The above image shows *Cryptococcus neoformans* stained with India ink and its infection most commonly involves the lungs. Indian ink is a type of negative stain that is not picked up by the capsule, making it evident. Although Cryptococcosis usually presents clinically as chronic meningo-encephalitis, the primary infection always involves the lungs.

Cryptococcus is an opportunistic fungus that is found worldwide as a saprophyte in soil enriched with pigeon droppings.

Pathogenesis:

- Infection is initiated by inhalation of the yeast cells.
- The primary pulmonary infection may be asymptomatic or may mimic an influenza-like respiratory infection, often resolving spontaneously.
- In immunocompromised, the yeasts may multiply and disseminate to other parts of the body, but preferentially to the CNS, causing cryptococcal meningoencephalitis.
- Other common sites of dissemination include the skin, adrenals, bone, eye, and prostate gland.

Clinical findings:

- Chronic meningitis (Headache, nuchal rigidity, disorientation)
- CSF pressure, protein concentration and cell count are elevated
- Cutaneous ulcers, granulomata.

Diagnosis:

- Specimens - CSF, tissue, exudates, sputum, blood, urine, skin scrapings
- Microscopy - Gram staining shows gram-positive round budding yeast cells
- India ink preparation reveals capsulated budding yeast cells; capsules are prominent in this staining.
- Culture - Colonies on SDA at 37°C appear as smooth, mucoid and creamy white.
- Serology - Demonstration of cryptococcal antigen by latex agglutination test is a sensitive, specific and rapid method for identification of cryptococcal meningoencephalitis.

Treatment:

- Cryptococcosis without CNS involvement: Fluconazole is the drug of choice
- HIV patient with CNS involvement: Induction: Liposomal Amphotericin B with Flucytosine for at least 2 weeks

Maintenance: Oral Fluconazole

Other Options:

Option B: *Cryptococcus neoformans* is a round or ovoid budding yeast cell, 4-20 µm in diameter, with a prominent polysaccharide capsule. Although the fungus grows primarily as a yeast, hyphae are produced during the sexual phase and during a process called monokaryotic fruiting, which is also believed to involve sexual reproduction, but no pseudohyphae are produced during normal circumstances.

Option C: *C. neoformans* is frequently found in soils contaminated with avian excreta and can easily be recovered from shaded and humid soils contaminated with pigeon droppings. Hence, can be found in more than one reservoir other than man.

Option D: *C. neoformans* usually affects patients with advanced HIV infection and CD4+ T lymphocyte counts of as low as <200/µL.

Solution to Question 16:

S. agalactiae belonging to Group B streptococci are the most common cause of meningitis and neonatal sepsis followed by *E. coli*.

S. agalactiae is a capsulated Gram-positive coccus occurring in chains. It causes meningitis in the neonatal period. Infection is commonly acquired from the maternal vagina during birth. There are two clinical types seen in neonates:

- Early-onset type (more common) presents within a week of birth as meningitis or septicemia and is often fatal. Early-onset infection presents similarly to other types of neonatal sepsis, with common symptoms such as respiratory distress, lethargy, and hypotension. Almost all infants with this condition have bacteremia, while one-third to one-half also experience pneumonia, respiratory distress syndrome, and one-third develop meningitis.
- Late-onset type presents between the second to twelfth week of life commonly as meningitis, and in most cases is associated with a strain of capsular type III.

Group B streptococci can also lead to infections in non-pregnant immunocompromised adults. Common infections include sepsis, skin and soft tissue infections, respiratory tract infections, and urinary tract infections in women.

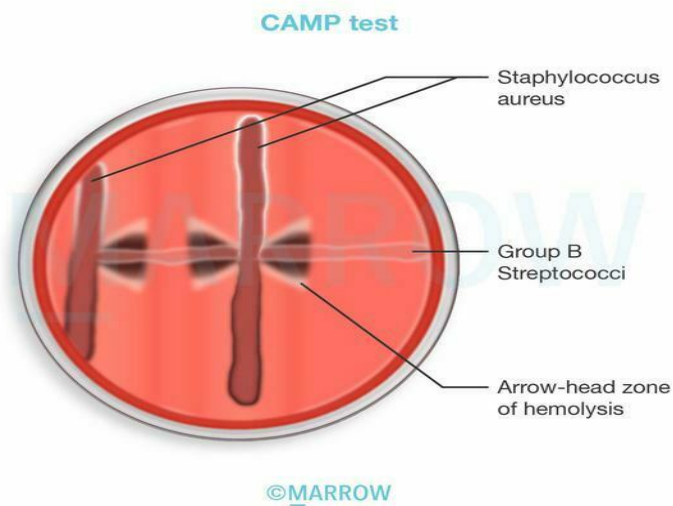
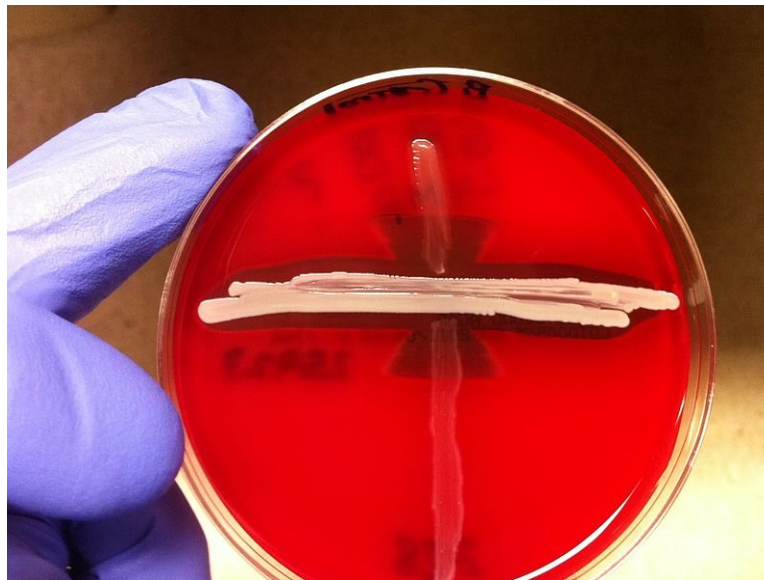
Identification :

A streptococcal isolate can be classified presumptively as GBS based on biochemical tests, including hydrolysis of sodium hippurate, hydrolysis of bile esculin, and bacitracin susceptibility.

The CAMP test (Christie, Atkins and Munch-Peterson) is used to identify group B β-hemolytic-streptococci based on their ability to form CAMP factor. A characteristic arrowhead zone of hemolysis is seen when *Streptococcus agalactiae* is inoculated perpendicular to

Staphylococcus aureus grown on blood agar. The group B streptococci hydrolyse sodium hippurate and give a positive CAMP test.

The image given below shows a positive CAMP test



Solution to Question 17:

The correct answer is option C: 1-d, 2-c, 3-a, 4-b

1. SARS COV -2	d. Single-stranded R NA
2. Rotavirus	c. Double-stranded R NA

1. SARS COV -2	d. Single-stranded R NA
3. Human Papilloma Virus	a. Double-stranded D NA
4. Parvovirus	b. Single-stranded D NA

Solution to Question 18:

After sputum sample collection, Mycobacterium is detected in sputum samples with N-acetyl cysteine (NAC) and NaOH where N-acetyl cysteine is used for liquefaction of sputum and NaOH kills the contaminating bacteria. Therefore in the above question assertion(A) is true but reason(R) is false.

Sputum microscopy is the most reliable single method in the diagnosis and control of tuberculosis. NALC (N-acetyl cysteine) with 2% NaOH is superior for detecting bacilli in microscopy.

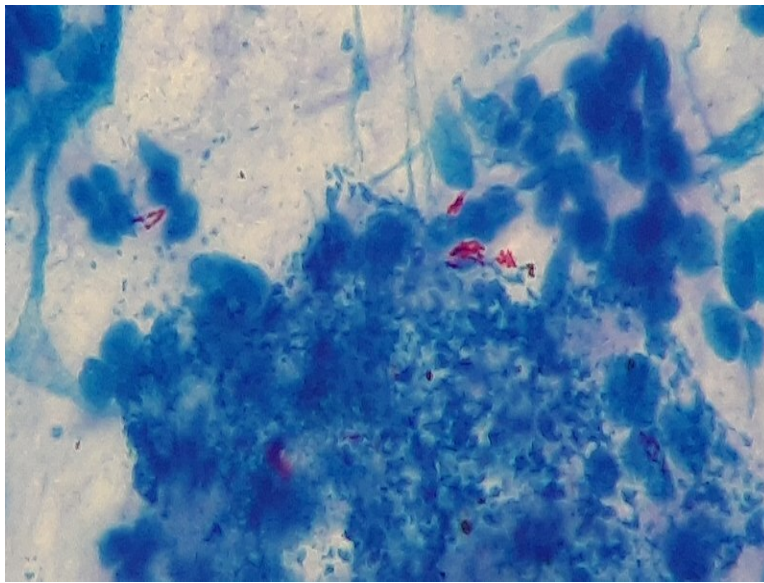
Sample collection:

- Sputum Collection: Sputum yields abundant bacilli in caseation, less in lesions organised lesion not connected to airways. Best collected in the morning pre-meal; scanty samples may be pooled over 24 hours. Sampling over three days enhances detection.
- Laryngeal aspiration and bronchial washing is done in situations where sputum is not available
- In children who are unable to expectorate sputum, gastric lavage is an ideal method for sample collection.

Collected Sputum and non-sterile samples need initial pre-treatment to prevent microbial overgrowth during incubation. Processes such as liquefaction, decontamination, and concentration are essential to enhance the yield from the sputum samples. Hence, techniques preserving bacilli viability are necessary. Always follow Class II biosafety methods while performing any procedures related to sputum.

- Petroff's method: In this method, sputum is incubated with 4% sodium hydroxide at 37°C, shaken till clear (~20 mins), then centrifuged (3000 rpm, 20 mins). The sediment is neutralised with N/10 HCl for further use. Avoid excessive alkali exposure.
- NALC (N-acetyl cysteine) with 2% NaOH: This method is superior to Petroff's, NALC liquefies sputum while NaOH eliminates contaminating bacteria. The sample is then, neutralised with buffer and centrifugation follows. This method is compatible with automated culture systems."

The image given below shows a sputum smear stained with Ziehl-Neelsen stain showing Mycobacterium tuberculosis.



Solution to Question 19:

The presence of ulcerative eruptions along the lymphatics in a gardener from the sub-Himalayan region and the microscopic finding of flower-like clusters of conidia are suggestive of sporotrichosis caused by *Sporothrix schenckii*. It is a thermally dimorphic fungus that lives on vegetation.

The lesions occur following traumatic inoculation of the fungus. A granulomatous nodule is first formed at the site of inoculation and may progress to form a necrotic or ulcerative lesion. The infection can spread via lymphatic channels from the skin and subcutaneous tissue to the lymph nodes, resulting in the formation of secondary ulcers on the lymph nodes. This condition is known as lymphocutaneous sporotrichosis. This disease is commonly encountered among gardeners, forest workers, and manual labourers.

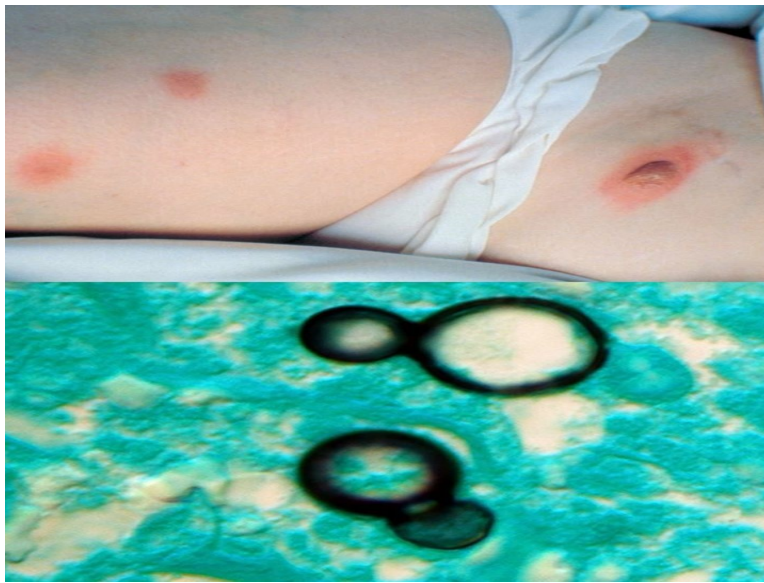
Other options:

Option A: Coccidiomycosis is caused by *Coccidioides immitis* which is a thermally dimorphic fungus. It occurs as a spherule in tissues and culture at 37°C and as the mycelial form in soil and culture at room temperature. They are endemic to the arid desert regions of the southwestern USA and California. Infection is acquired by inhaling dust containing arthrospores of the fungus.

Option B: Histoplasmosis is caused by *Histoplasma capsulatum*. It has tiny oval yeast cells with narrow-based budding within the macrophages with an underlying granulomatous response. This is a dimorphic fungus that exists in the yeast form in body tissues and cultures maintained at 37°C, and in the filamentous mold form in cultures at room temperature (25°C).

Option D: *Blastomyces* cause North American Blastomycosis/ Chicago disease. Pulmonary involvement is the most common form followed by skin. Histopathological staining shows thick-walled yeast, with single broad-based budding (figure 8 appearance).

The image given below is that of *Blastomyces* which is a thick-walled yeast, with single broad-based budding (figure 8 appearance).



Solution to Question 20:

The given scenario is indicative of hemolytic uremic syndrome, and *E. Coli* O157:H7 is the likely causative organism.

O157: H7 is a strain of *E. coli* called enterohemorrhagic *E. coli* (EHEC), which produces verocytotoxin or Shiga-like toxin. They produce two antigenic forms of the cytotoxic toxin known as Shiga-like toxin 1 and Shiga-like toxin 2. Shiga-like toxin 1 is identical to the Shiga toxin of *Shigella dysenteriae* type 1. *E. coli* O157:H7 has a small infective dose (around 100 bacilli) and spreads easily from child to child by the fecal-oral route.

It can cause mild non-bloody diarrhoea to hemorrhagic colitis, or a severe form of diarrhoea, with hemolytic uremic syndrome, a disease resulting in acute renal failure, microangiopathic hemolytic anemia, and thrombocytopenia. The most common cause of hemolytic uremic syndrome in the world is *E. coli* O157:H7. The most common cause of hemolytic uremic syndrome in India is *Shigella dysenteriae* type I.

EHEC (O157: H7) dysentery is commonly self-limiting, and management is conservative, with the mainstay being the correction of dehydration and electrolyte imbalances and other symptomatic supportive treatment such as blood transfusion in case of severe blood loss anaemia, but antibiotics are not to be used. Therefore, the treatment of HUS is primarily supportive care, which may include:

- Intravenous fluids to prevent dehydration
- Blood transfusions to correct anemia and thrombocytopenia
- Dialysis to treat kidney failure

In some cases, plasma exchange or eculizumab may be used to treat HUS. Plasma exchange is a procedure that removes harmful substances from the blood. Eculizumab is a medication that blocks the complement system, which is part of the body's immune system.

Other options:

Option A: *Shigella flexneri* produces Shigella enterotoxin (ShET). It enhances local vascular damage of the intestine, kidney, and brain. It is commonly implicated in reactive arthritis, also called Reiter syndrome (post-infectious arthritis).

Option B: *Salmonella enteritidis* is commonly implicated in osteomyelitis in patients with sickle cell anemia.

Option C: *Aeromonas* is implicated in inflammatory diarrhoea and does not present with fever, hematuria, and thrombocytopenia.

Solution to Question 21:

Nocardia is the isolated organism in this patient based on the given histopathological features.

Nocardia species appear as filamentous organisms with extensive hyphae-like branching. They are aerobic, gram-positive, catalase-positive, and urease-positive. They are weakly acid-fast because they stain with carbolfuchsin and retain when decolourized with 1–4% sulfuric acid instead of the stronger acid-alcohol decolorant.

Infections caused by *Nocardia* include cutaneous, subcutaneous, or systemic lesions. It may lead to local abscesses, cellulitis, or lymphocutaneous lesions. Systemic nocardiosis occurs more often in immunodeficient persons.

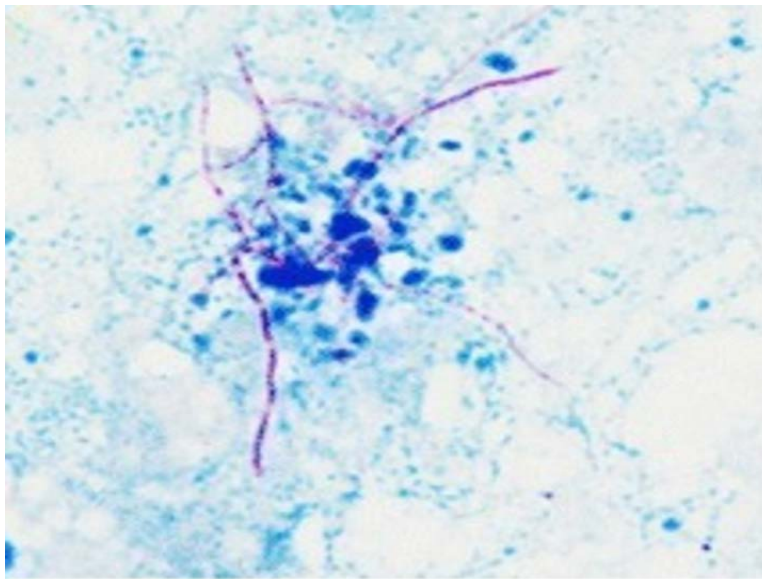
Infection follows traumatic inoculation of the organism into the subcutaneous tissue from soil or vegetable sources. They can present with a characteristic powdery substance that appears as gritty or necrotic material on the skin or within soft tissues. Accompanying this are multiple discharging sinuses, which are abnormal openings that form as the body attempts to expel infected material. These sinuses can drain purulent, often foul-smelling discharge typically associated with localised abscesses, and the discharge may be viscous, sometimes containing necrotic tissue.

Systemic nocardiosis usually caused by *N. asteroides* manifests primarily as pulmonary disease, pneumonia, lung abscess, or other lesions resembling tuberculosis. Extrapulmonary manifestation may involve the brain, kidneys, and other organs.

Diagnosis is by microscopy of sputum, bronchoalveolar lavage specimens, and tissue biopsy. Gram stain shows beaded gram-positive filamentous rods. In Modified ZN staining they are weakly acid-fast.

The drug of choice for nocardiosis is cotrimoxazole. In severe cases, 3rd generation cephalosporins/ carbapenems and amikacin are recommended.

The image given below is a smear of modified acid-fast Kinyoun staining showing filamentous red-stained partially acid-fast rods suggestive of *Nocardia*.

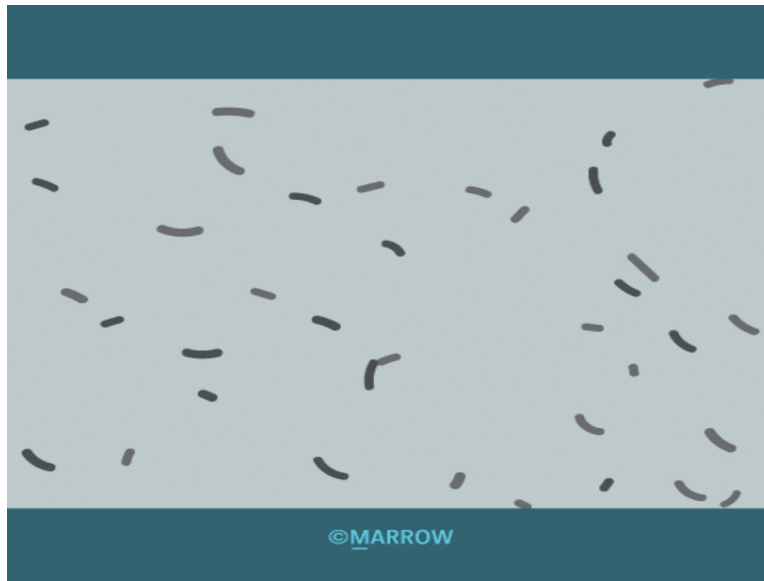


Other options:

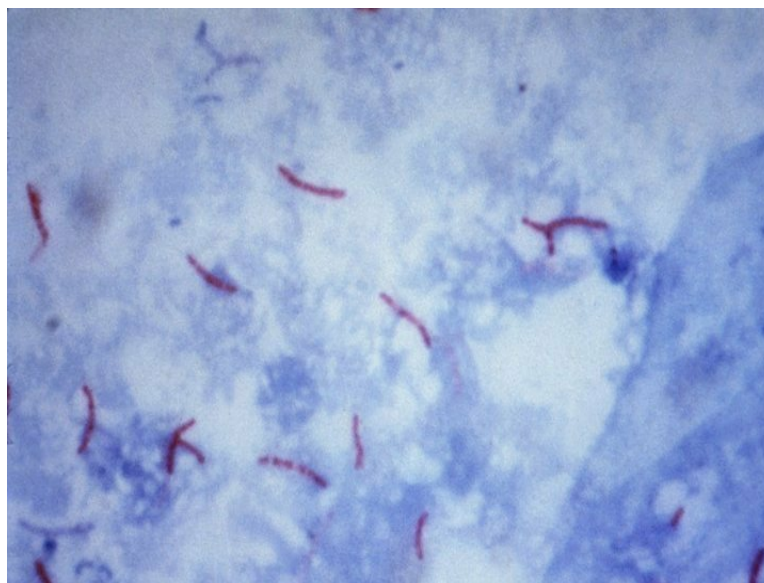
Option A: Gram's staining of *Bacillus anthrax* shows bacilli arranged end to end in the form of long chains. This is called bamboo stick appearance.



Option B: *Listeria monocytogenes* is a short, non sporing, gram positive bacillus. It shows characteristic tumbling motility at 25°C, but is non motile at 37°C.



Option D: The image given below is a smear with Ziehl-Neelsen staining showing non-branching and acid-fast Mycobacterium tuberculosis bacilli.



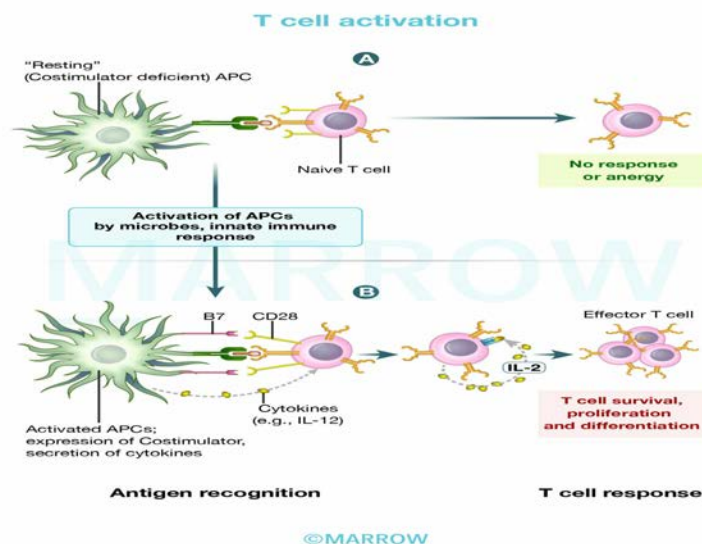
Solution to Question 22:

PD-1 on T cell interacts with PD ligand 1 on APC is not true regarding naive T cell interaction with CD4 effector T-cells.

T-cell activation requires two signals that are delivered by APCs (Option A). They include:

- Antigens displayed by APCs in the form of peptides bound to MHC molecules, that are recognised by T-cell receptors
- Costimulatory signal (as it stimulates T cells in conjunction with antigen) provided by molecules on APCs that engage particular costimulatory receptors on T cells

CD80 (B7-1) and CD86 (B7-2) are the costimulatory molecules present on antigen-presenting cells (APCs). They bind to the CD28 protein present on T cells (Option B). This promotes the expansion of antigen-stimulated T cells and their differentiation into effector and memory cells. (Option D).



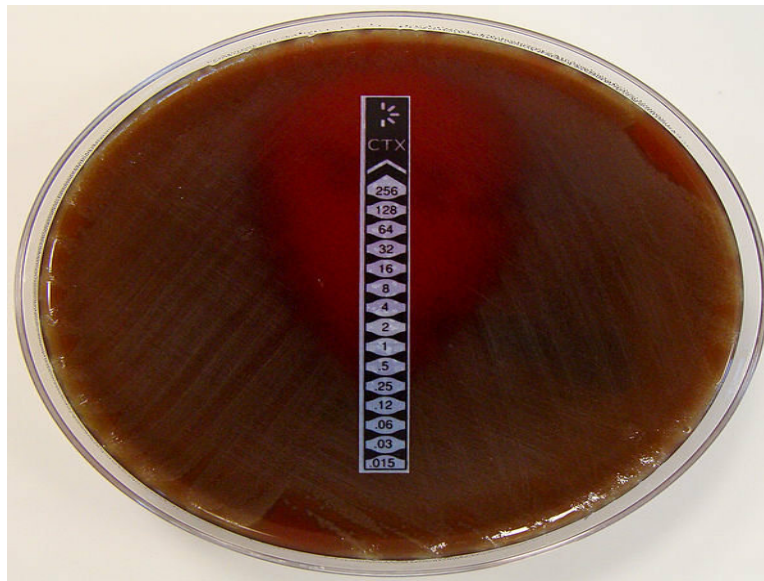
Solution to Question 23:

E-test is a method of antibiotic sensitivity testing.

E-test (epsilometer) is a diffusion and dilution method of testing antibiotic sensitivity. It is a quantitative technique for determining antimicrobial susceptibility and minimum inhibitory concentration (MIC).

An absorbent strip containing a predefined gradient (serial dilution) of antibiotic concentration immobilized along its length is placed on a lawn inoculum of a bacterium. An elliptical zone of inhibition is produced surrounding the strip. The antibiotic concentration at which the ellipse edge intersects the strip is taken as the MIC value.

The given image shows the E-test.



The antibiotic sensitivity tests are classified as follows:

1. Diffusion methods:

- Kirby–Bauer disk diffusion method
- Stokes disk diffusion method

2. Dilution methods:

- Broth dilution
- Agar dilution

3. Diffusion and dilution method:

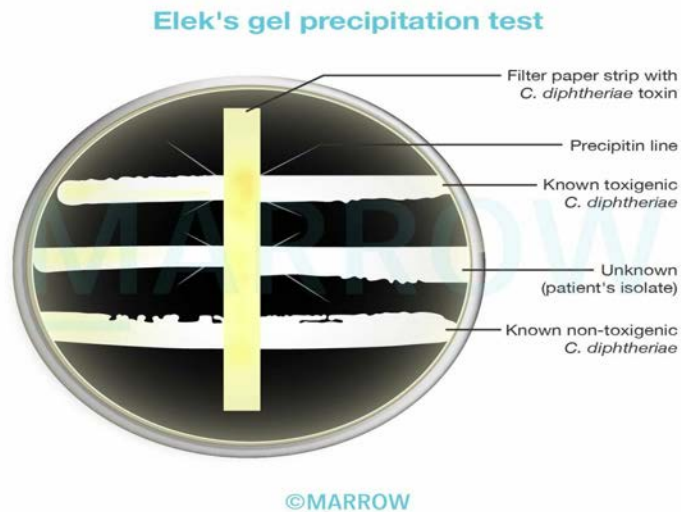
- E–test (Epsilonometer)

4. Automated methods:

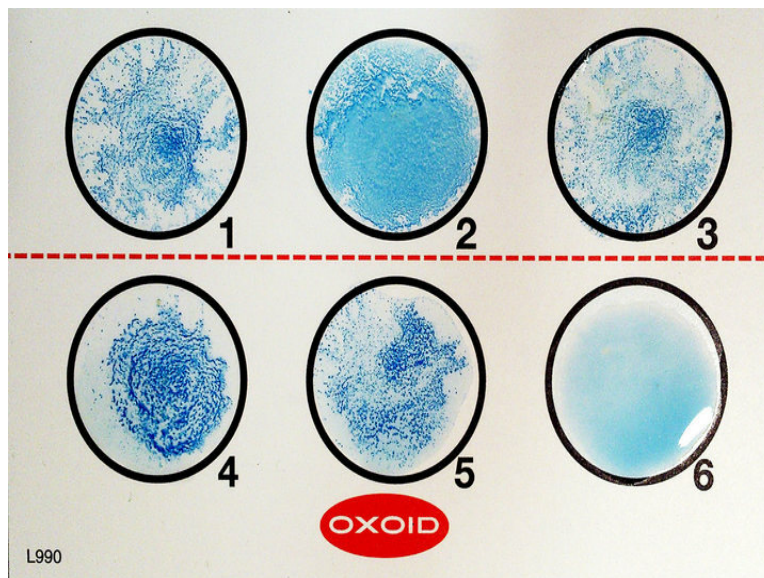
- VITEK 2
- Phoenix system

Other options:

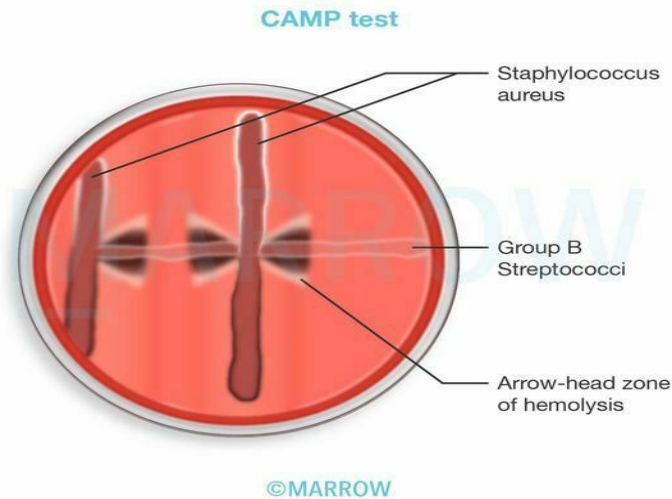
Option B: Elek test is an in vitro immunoprecipitation (immunodiffusion) test used to determine whether or not a strain of *Corynebacterium diphtheriae* is toxigenic.



Option C: In latex agglutination tests, an antigen-specific antibody (either polyclonal or monoclonal) is fixed to latex beads. When the clinical specimen from the patient is added to a suspension of the latex beads, the antibodies bind to the antigens on the microorganism forming a lattice structure, and agglutination of the beads occurs. Latex agglutination tests are mainly used for the detection of carbohydrate antigens of encapsulated microorganisms.



Option D: The Christie-Atkins-Munch-Peterson (CAMP) test is used to identify group B, β -streptococci based on their ability to form CAMP factor, which causes synergistic hemolysis with β -hemolysin formed by *Staphylococcus aureus*. When *Streptococcus agalactiae* is inoculated perpendicular to *Staphylococcus aureus* grown on blood agar, a characteristic arrowhead zone of hemolysis is seen. The group B streptococci hydrolyze sodium hippurate and give a positive CAMP test.



Solution to Question 24:

Madurella mycetomatis is transmitted by walking barefoot in soil. It causes Eumycetoma.

Mycetoma is a chronic, granulomatous infection of the skin, subcutaneous tissues, fascia, and bone, which most often affects the foot or the hand. It can be divided into three types- eumycetomas, actinomycetomas, and botryomycosis.

Infection follows traumatic inoculation of the organism into the subcutaneous tissue from soil or vegetable sources. Within host tissues, the organisms develop to form compacted colonies (grains) 0.5–2 mm in diameter, the colour of which depends on the organism responsible. It results in tumefactions, deformities, and draining sinuses, discharging these grains or granules.

Laboratory Diagnosis:

A. Direct Examination: The presence of grains in pus collected from draining sinuses or in biopsy material is diagnostic.

B. Culture Samples should also be cultured, at both 25–30°C and 37°C, on brain-heart infusion agar or blood agar for actinomycetes and on Sabouraud agar (without cycloheximide) for fungi.

Treatment:

The management of eumycetoma is difficult, involving surgical debridement or excision and chemotherapy. Actinomycetoma, on the other hand, responds well to rifampicin in combination with sulfonamides or cotrimoxazole.

Curvularia, *Coccidioides immitis*, and *Aspergillus fumigatus* are not transmitted by walking barefoot in soil.

The image below shows a mycetoma of the foot:



Solution to Question 25:

A and D can be prevented by vaccination.

Cervical cancer is caused by Human papillomavirus (HPV). Currently, there are four vaccines against HPV approved for clinical use which are:

- Cervarix is a bivalent vaccine active against HPV types 16 and 18. Anogenital warts are commonly caused by subtypes 6 and 11. Thus, Cervarix does not confer protection against anogenital warts.
- Gardasil is quadrivalent and additionally protects against HPV 6 and 11.
- Gardasil-9 is nonavalent and protects against 9 strains.
- Cervavac is India's first indigenously prepared quadrivalent HPV (qHPV) vaccine against HPV strains 6, 11, 16, and 18.

Hepatitis B and C can cause Hepatocellular carcinoma. Hepatitis B vaccine is an example of a subunit vaccine made using HBsAg. The first dose (0.5 mL, intramuscular) is given at birth or as early as possible within 24 hours on the anterolateral side of the mid-thigh. Subsequent doses are given as a part of the pentavalent vaccine at 6 weeks, 10 weeks, and 14 weeks (can be given until one year of age).

Burkitt lymphoma and Merkel cell carcinoma cannot be prevented by vaccination.

Solution to Question 26:

Statements 2,4 are true about Cryptococcus.

Cryptococcus is an opportunistic fungus that is found worldwide as a saprophyte in soil enriched with pigeon droppings. C.gatti sources from Eucalyptus trees, not bird Guano. (Statement 3)

Pathogenesis:

- Infection is initiated by inhalation of the yeast cells.
- The primary pulmonary infection may be asymptomatic or may mimic an influenza-like respiratory infection, often resolving spontaneously.
- In immunocompromised, the yeasts may multiply and disseminate to other parts of the body, but preferentially to the CNS, causing cryptococcal meningoencephalitis.
- Other common dissemination sites include the skin, adrenals, bone, eye, and prostate gland.
- Extrapulmonary cryptococcosis is an AIDS-defining illness (Statement 2)
- It is seen when CD4 <200 cells/microliter (Statement 1)

Clinical findings:

- Chronic meningitis (Headache, nuchal rigidity, disorientation)
- CSF pressure, protein concentration, and cell count are elevated
- Cutaneous ulcers, granulomata.

Diagnosis:

- Specimens - CSF, tissue, exudates, sputum, blood, urine, skin scrapings
- Microscopy - Gram staining shows gram-positive, urease-positive (Statement 4) round budding yeast cells
- India ink preparation reveals capsulated budding yeast cells; capsules are prominent in this staining.
- Culture - Colonies on SDA at 37°C appear as smooth, mucoid and creamy white.
- Serology - Demonstration of cryptococcal antigen by latex agglutination test is a sensitive, specific and rapid method for identification of cryptococcal meningoencephalitis.

Treatment:

- Cryptococcosis without CNS involvement: Fluconazole is the drug of choice
- HIV patient with CNS involvement: Induction: Liposomal Amphotericin B with Flucytosine for at least 2 weeks
- Maintenance: Oral Fluconazole

Solution to Question 27:

Lowenstein-Jenson medium is used for *M. tuberculosis*.

Lowenstein-Jenson medium is a selective medium used for the culture of *Mycobacterium tuberculosis* which causes tuberculosis. The presence of malachite green imparts a green colour and makes Lowenstein Jenson medium a selective media by inhibiting other bacteria.

Components of LJ medium:

- Egg: Provide fatty acids and protein required for metabolism.

- Malachite green: Prevents growth of contaminants.
- Mineral salt solution: Buffer.
- Glycerol: Source of carbon, Favours growth of *M. tuberculosis*, but not *M. Bovis*.
- L-asparagine: Source of Nitrogen.

Colony characteristics on LJ Medium:

Mycobacterium tuberculosis colonies on LJ medium are rough, raised, irregular colonies that are creamy white, and buff-colored. (rough, tough, buff). They do not emulsify in a saline solution.

Culture media used to cultivate *Mycobacterium tuberculosis*:

- Solid culture media: Lowenstein–Jensen (A commonly used, egg-based, selective media).
- Liquid culture media: Dubos' media, Middle Brook's media. Virulent strains tend to form long serpentine cords in liquid media, while avirulent strains grow in a more dispersed manner.
- Automated Culture Media:

BACTEC 460: Uses radioisotopes

BacT/ALERT: A colorimetric method

BACTEC MGIT: A fluorometric detection method.

The culture medium shown in the image is the Lowenstein-Jensen.



Other options:

Option B: Blood agar is a differential medium as it helps in differentiating various organisms based on the degree of hemolysis observed.

- Alpha hemolysis- Partial lysis of red blood cells and haemoglobin
- Beta hemolysis- Complete lysis of the red blood cells and haemoglobin
- Gamma hemolysis- No hemolysis

Option C: Novy-MacNeal-Nicolle medium is used for isolating parasites like *Leishmania* and *Trypanosoma cruzi*.

Option D: Sabouraud dextrose agar is the traditional mycological medium, which contains glucose and modified peptones. It supports the growth of fungi while restricting bacterial growth due to the addition of antibacterials.

Solution to Question 28:

The given clinical scenario with complaints of non-bloody diarrhoea and dehydration in a child suggests rotavirus.

Rotavirus is the most common identifiable viral cause of diarrhoea in children below the age of 5 years and is most frequent between 6-24 months of age. The incubation period is 1-3 days. Most cases are asymptomatic or with mild symptoms like watery diarrhoea, vomiting, low-grade fever, and temporary lactose intolerance. Diagnosis can be made by rapid antigen detection of rotavirus in stool specimens.

Treatment is supportive, including hydration and correction of fluid and electrolyte imbalance. The main goals of treatment are avoiding and treating dehydration. Maintaining nutrition is a secondary goal.

Other options:

Option A: *Trichuris trichiura* causes rectal prolapse in children (characteristic feature), chronic dysentery, anemia, poor growth, and developmental defects. The disease may result either due to the mechanical effect of the worms or an allergic reaction to the parasite.

Option B: *Shigella* causes Shigellosis, which is a spectrum of symptoms consisting of 4 phases:

1. Incubation
2. Watery diarrhoea: Fever, watery diarrhoea, anorexia
3. Dysentery: Bloody mucopurulent stools
4. Post-infectious complication: Hemolytic-uremic syndrome, reactive arthritis, and toxic megacolon.

Option D: *Yersinia enterocolitica* causes diarrhoea with symptoms resembling appendicitis (pseudoappendicitis). The patients present with abdominal pain, which can be localised to the right iliac fossa, thus mimicking appendicitis.

Solution to Question 29:

i-b,ii-c,iii-d,iv-a is the correct match of the causative organism with the disease.

i. <i>Klebsiella</i> Granulomatosis	b. <i>Donovanosis</i>
ii. <i>Bartonella bacilliformis</i>	c. Oroya fever
iii. <i>Treponema carateum</i>	d. Pinta disease

i. Klebsiella Granulomatosis	b. Donovanosis
iv. Streptobacillus moniliformis	a. Rat-bite fever

Solution to Question 30:

Toxoid does not confer passive immunity but plays a role in active immunity.

Toxoid vaccines are a type of subunit vaccine prepared by detoxifying the toxins produced by organisms like tetanus bacilli. In general, they are highly efficacious and safe immunising agents.

Passive immunity is the resistance that is transmitted passively to a recipient in readymade/preformed antibodies. Here, the recipient's immune system plays no active role, and there is no antigenic stimulus.

Types of passive immunity:

- Natural passive immunity: It is the immunity passively transferred from mother to baby. Maternal IgG antibodies (Option D) are transmitted through the placenta. The human colostrum is rich in IgA (Option C) antibodies and gives protection to the neonate.
- Artificial passive immunity: It is the resistance passively transferred to a recipient by the administration of antibodies. For example, antitoxins (Option B) like anti-tetanus serum and anti-diphtheria serum.

Solution to Question 31:

Autoclave is not a cold sterilization technique

Autoclaving is a form of moist heat sterilization that creates an increased temperature of steam under pressure. This method of sterilization is effective against spores also. Autoclaving is done for instruments that cannot withstand high temperatures of hot air oven or dry heat. Surgical instruments are sterilized at 121° C for 10-15 minutes. The given image shows an autoclave.



Cold sterilisation is a process in which sterilisation is carried out at low temperatures with the help of chemicals, radiation, membranes (filters) and all other means excluding high temperatures.

Cold sterilisation techniques:

1. Ionising Radiation:

- X-ray
- Gamma rays (Option D)
- Cosmic rays

2. Non-ionising Radiation:

- Infrared
- Ultraviolet (Option C)

Other options:

Option A: Plasma sterilisation is a new method of sterilisation for heat-sensitive articles. A radio frequency energy is applied to create an electromagnetic field. Into this, hydrogen peroxide vapours are introduced, which generate a state of plasma containing free radicals of hydrogen and oxygen. This state renders the articles sterile by denaturing all microorganisms. Arthroscopes, urethroscopes, etc., are sterilised by plasma sterilisation.

Solution to Question 32:

A high larval load can cause rectal prolapse is not true regarding *Ascaris lumbricoides*.

Ascaris lumbricoides (roundworm) reside in the small intestine and not the large intestine to cause rectal prolapse. *Trichuris Trichura* (whipworm) cause rectal prolapse.

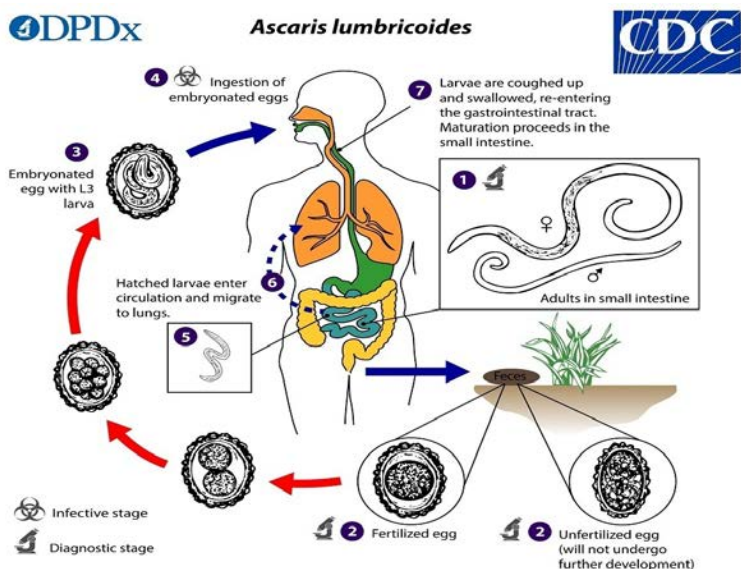
Ascaris lumbricoides is an intestinal nematode that causes ascariasis. Infection occurs through food or water contaminated with nightsoil or human faeces having eggs of *Ascaris* that contain the infective rhabditiform larva. Man is the only natural host. There is no intermediate host. Most infected humans are asymptomatic. (Option A)

The life cycle is as follows:

- Adult worms live in the lumen of the small intestine. A female produces eggs, which are passed with the faeces.
- Unfertilized eggs in the stool may be ingested but are not infective. The infective form of *Ascaris* is the embryonated egg in the soil (Option D). Infection typically arises from consuming food or water contaminated with embryonated eggs. These eggs harbour the rhabditiform larva.
- After infective eggs are swallowed, the larvae hatch.
- The larvae invade the intestinal mucosa and are carried through the portal system, then systemic circulation to the lungs.
- The larvae mature further in the lungs (10 to 14 days), penetrate the alveolar walls, ascend the bronchial tree to the throat, and are swallowed. This may present with asthma-like symptoms (Option B).
- Upon reaching the small intestine, they develop into adult worms. Between 2 and 3 months are required from ingestion of the infective eggs to oviposition by the adult female.
- Adult worms can live for 1 to 2 years.
- Adult worms can be occasionally detected in the stool or sputum of the patient. A definitive diagnosis is made by demonstration of eggs in faeces. Fertilised eggs are always bile-stained.

Effective and safe drugs for the treatment of ascariasis are Pyrantel pamoate 11 mg/kg once, Albendazole 400 mg once, Mebendazole 100 mg twice daily for 3 days, Ivermectin 150-200 mcg/kg once.

The image below shows the life cycle of *Ascaris lumbricoides*:



Solution to Question 33:

Triclabendazole is the drug of choice for fasciola hepatica.

Triclabendazole belongs to the benzimidazole group. The primary mechanism of action of benzimidazoles is by inhibition of microtubule polymerization. Benzimidazoles bind to beta-tubulin and cause inhibition of microtubule polymerization.

Drugs belonging to the benzimidazole group are:

- Albendazole
- Mebendazole
- Triclabendazole
- Thiabendazole

Mechanism of action:

- Inhibition of mitochondrial fumarate reductase
- Reduced glucose transport
- Uncoupling of oxidative phosphorylation

Other options:

Option B: Praziquantel is the drug of choice for all trematodes except Fasciola hepatica and Fasciola gigantica.

Option C: Mebendazole is used in ascariasis, trichuriasis, hookworm, pinworm infections, and other helminthic infections.

Option D: Albendazole is the drug of choice for the treatment of hydatid disease and cysticercosis. It is also used in the treatment of pinworm, hookworm, ascariasis, trichuriasis, and strongyloidiasis.

Solution to Question 34:

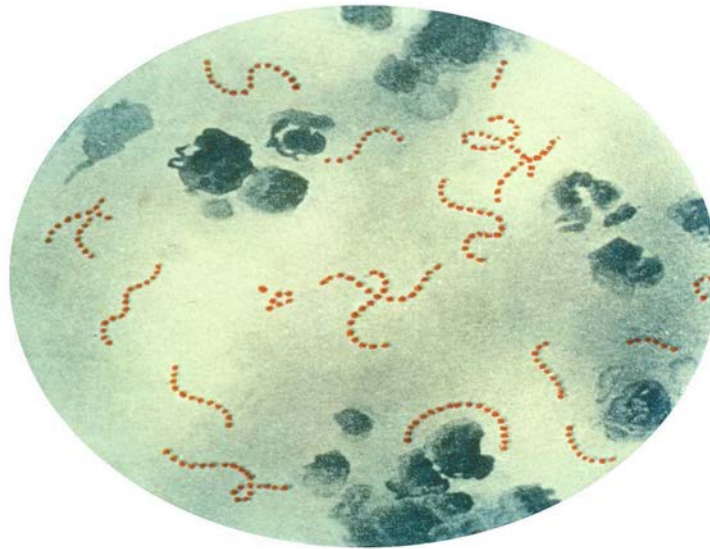
The gram-positive cocci in chains is Streptococcus pyogenes (Group A β -hemolytic Streptococcus), which causes Necrotizing fasciitis.

Necrotising fasciitis is a rapidly progressive necrosis of the subcutaneous tissues and fascia. The source of the infection is either the skin following trauma or a surgical incision. Diabetes mellitus is the most common comorbidity associated with the condition.

Local symptoms and signs are pain, erythema, edema, warmth, crepitus, blisters, bullae, greyish drainage ('dishwater pus'), necrosis, and gangrene. Systemic features are fever, tachycardia, tachypnea, shock, coagulopathy, and multiorgan failure.

Treatment involves appropriate intravenous antibiotics with urgent radical surgical debridement.

The image given below is of Streptococcus pyogenes, which is a gram-positive cocci in chains.



Other options:

Option B: Gram-positive cocci arranged in clusters is suggestive of *Staphylococcus aureus*. It causes toxic shock syndrome and staphylococcal scalded skin syndrome.

Option C: *Enterococcus* are catalase-negative, bile-esculin-positive, gram-positive cocci that grow in chains and are characterised by their ability to grow in the presence of 6.5% NaCl (salt tolerance). They are implicated in Infective endocarditis.

Option D: *Micrococcus* is a non-pathogenic skin commensal organism that appears as gram-positive tetrads on microscopy.

Solution to Question 35:

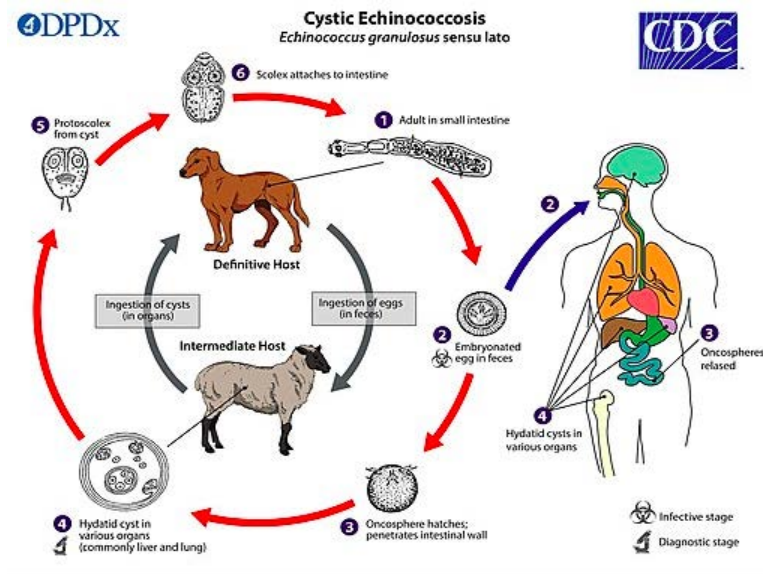
The patient presenting with abdominal pain and a growing mass in the right hypochondrium has a hydatid cyst of the liver. Aspirate of the cyst shows *Echinococcus granulosus* (with hooks).

Echinococcus granulosus is the most common causative agent in hydatid cysts, while *Echinococcus multilocularis* and *Echinococcus ligartus* account for a small number of cases.

Dogs are the definitive hosts of *E. granulosus*. Sheep are the usual intermediate hosts, but humans are accidental intermediate hosts. Humans are end-stage hosts of the parasite since there is no human-to-human transmission.

Approximately 75% of hydatid cysts are located in the right lobe of the liver and are solitary. The clinical presentation of a hydatid cyst is largely asymptomatic until complications occur. The most common presenting symptoms are abdominal pain, dyspepsia, and vomiting. The most frequent sign is hepatomegaly.

The life cycle of *Echinococcus* is depicted in the image below:



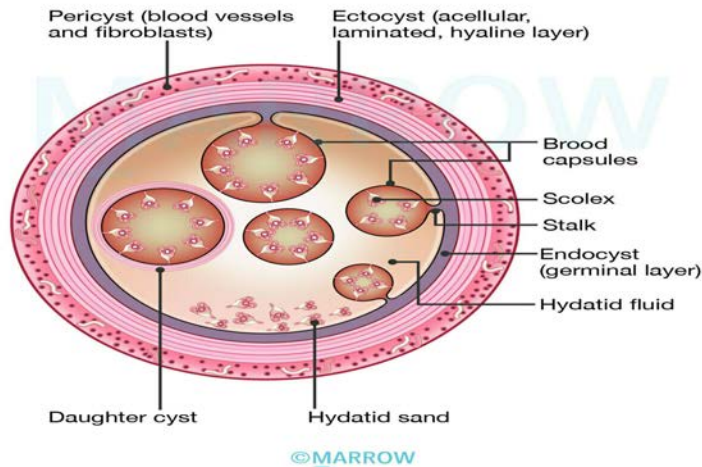
The hydatid cyst wall consists of 3 layers:

- Pericyst- the outer dense fibrous capsule that is formed as a result of host inflammatory response.
- Ectocyst - the intermediate layer consisting of acellular, chitinous, laminated hyaline material.
- Endocyst - the inner germinal layer that is cellular and consists of several nuclei embedded in protoplasmic mass and is extremely thin. It is the vital layer of the cyst, and it is the site of asexual reproduction, giving rise to brood capsules with scolices.

The brood capsules are cellular masses in the hydatid fluid, which consists of scoleces. In a definitive host, these scoleces develop into adult tapeworms, while in the intermediate hosts, these develop into new hydatid cysts. The brood capsules are attached to the endocyst by a pedicle or stalk.

The brood capsules may mature and detach from the mother cyst and become the daughter cysts, which are replicas of the mother cysts. The freed brood capsules and scoleces present in the hydatid fluid are called hydatid sand.

Structure of the hydatid cyst (Cross section)



Other options:

Option B: *Hymenolepis nana* (dwarf tapeworm) is the smallest tapeworm. Humans are the definitive host. It has no intermediate host. It occurs more commonly in children, and the patient acquires this infection by ingesting food or water contaminated with eggs. The egg attaches to the intestine and develops into the larva and adult worm. The adult worm resides in the small intestine. Patients are usually asymptomatic. However, in heavy infections, they present with nausea, anorexia, abdominal pain, diarrhoea, and irritability. Pruritus may also be present due to an allergic response.

Option C: *Taenia solium* is a parasitic tapeworm transmitted through undercooked pork, causing intestinal infection (taeniasis) or more severe cyst formation in tissues, including the brain (cysticercosis). It's prevalent in regions with poor sanitation and where pork is consumed.

Option D: *Taenia saginata* is a parasitic tapeworm transmitted through undercooked beef containing cysticerci ("measly" beef). Humans are the definitive host, and cows are the intermediate hosts. *Taenia solium* and *Taenia saginata* eggs are morphologically indistinguishable.

Solution to Question 36:

Chloroquine is not given in uncomplicated falciparum malaria.

NVBDCP guidelines for the treatment of uncomplicated falciparum cases:

- North-eastern states: Artemether-lumefantrine-based ACT (Option C)
- Rest of the country: Artesunate-sulfadoxine+pyrimethamine (Option D)

The artemisinin-based combination therapy (ACT) recommended by NVBDCP in the northeastern states is a combination of artemether and lumefantrine. In the northeastern states (Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, and Tripura), the presently recommended ACT for *P. falciparum* malaria is a fixed-dose combination of artemether-lumefantrine due to the recent reports of late treatment failures with the current

combination of AS+SP.

Primaquine 0.75 mg/kg on day 2 is given as a single dose in uncomplicated falciparum malaria.(Option B)

Solution to Question 37:

Bacteriological loops used for inoculation are sterilised by heating till metal turns red hot.

Sterilisation of bacteriological inoculating loops, the tip of spatulas and forceps is done by placing them over a Bunsen flame until they get red hot. Inoculation loops carrying infective material should be dipped in a disinfectant before flaming to prevent spattering.

Other options:

Option B: Disinfection of slides and cover slips are done by passing over the flame 3 times.

Option C: Autoclaving is a method of heat sterilisation. The autoclave is a device that uses steam under pressure to sterilise linen, instruments, laboratory ware, media, pharmaceutical products, aqueous solutions, surgical instruments, cardiac catheters, and implants.

Option D: In Microwave irradiation, microorganisms are killed using high-frequency microwaves. This cannot be used for animal or human body parts, metal items or toxic or radioactive material.

Microbiology NEET 2024

Question 1:

Which of the following is used for syndromic diagnosis of infected cases like meningitis?

- a) Real time PCR
- b) Nested PCR
- c) Reverse transcriptase PCR
- d) Multiplex PCR

Question 2:

A patient receiving chemotherapy develops pneumonia with skin lesions. Microscopy of specimen from skin lesions showed broad based budding with a thick capsule. What is the likely diagnosis?

- a) Blastomycosis
- b) Histoplasmosis
- c) Coccidioidomycosis
- d) Candidiasis

Question 3:

A 30-year-old man came to the OPD with a febrile illness of 2 days, associated with abdominal tenderness, malaise and diarrhoea. He had a fever with bradycardia and BP was noted as 110/70 mmHg. He lives in a crowded neighborhood with poor sanitation where many others have reported similar complaints recently. Which of the following investigations would be the most appropriate choice for this patient?

- a) Stool culture
- b) Widal test
- c) Urine culture
- d) Blood culture

Question 4:

A child is brought with complaints of abdominal pain, diarrhoea and perianal itching for the last 3 days. The mother said she noticed a white thread-like structure in the child's night clothes. She collected the worm and brought it to the clinician. A perianal swab was collected. Microscopy of the swab and the worm are shown in the images. What is the likely causative organism?



- a) *Enterobius vermicularis*
- b) *Ascaris lumbricoides*
- c) *Ancylostoma duodenale*
- d) *Strongyloides stercoralis*

Question 5:

A male patient comes to you with a history of UTI, the cause of which is suspected to be bacteria. Which of the following is used by the bacteria for adherence?

- a) Cytoplasmic membrane
- b) Cilia
- c) Fimbriae
- d) Flagella

Question 6:

A thalassemia patient has received multiple blood transfusions. Aplastic crisis can occur in this patient due to infection from?

- a) Parvovirus
- b) Epstein barr virus
- c) Cytomegalovirus
- d) Herpes simplex virus

Question 7:

A 45-year-old farmer comes with complaints of high-grade fever with chills, retro-orbital pain and conjunctival suffusion. On examination, he has severe calf tenderness. Lab examination shows hypokalemia and neutropenia. What is the likely diagnosis?

- a) Chikungunya
- b) Leptospirosis
- c) Dengue
- d) Malaria

Question 8:

A patient with a history of consumption of oysters comes with complaints of nausea, vomiting and crampy abdominal pain. The culture of the organism showed the Kanagawa phenomenon. What is the causative organism?

- a) Vibrio cholera
- b) Salmonella typhi
- c) Vibrio parahaemolyticus
- d) E coli

Question 9:

A man is having petechial rashes with a previous history of meningitis. He was vaccinated at the age of 5 years. Now, Brudzinski and Kernig's signs were positive. Microscopy showed gram negative diplococci. What is the likely causative organism?

- a) Hemophilus influenzae
- b) Neisseria Meningitidis
- c) Cytomegalovirus
- d) Klebsiella

Question 10:

A patient who has a central line on long-term IV antibiotics develops a fever. Blood cultures show *Staphylococcus epidermidis* along with biofilm formation. Which of the following describes its features of antibiotic resistance?

- a) Biofilm increased susceptibility to antibiotics
- b) Biofilm offers resistance so catheter is to be removed
- c) Biofilm does not affect susceptibility to antibiotics
- d) Biofilm can be washed off from the catheter

Question 11:

Laboratory investigations in a patient with desert rheumatism showed dimorphic and thermal fungi with arthrospores. What is the causative organism of desert rheumatism?

- a) *Histoplasma capsulatum*
- b) *Paracoccidioides brasiliensis*
- c) *Blastomyces dermatitidis*
- d) *Coccidioides immitis*

Question 12:

A farmer from South East India presents with fever, respiratory distress and neurological deficits. The vector of the causative organism is shown below. what is the associated organism?



- a) Ebola virus
- b) Zika virus
- c) Malaria
- d) Nipah virus

Question 13:

A 21-year-old female complains of fever, headache and vomiting from the past 3 days. She has been living in the army recruitment camp for the last three weeks. On examination, she has neck stiffness and a maculopapular rash was noted over the trunk and extremities. Which of the following is false about the disease she's likely suffering from?

- a) Carriers of this disease can be treated with Rifampicin
- b) It can lead to adrenal hemorrhage
- c) Terminal complement deficiency increases the risk of this disease
- d) Metronidazole is the drug of choice

Question 14:

A young sexually active female presented with a symptomatic UTI. Urinalysis and cultures showed novobiocin-resistant coagulase negative staphylococcus with a count of 10^4 cells/HPF. What is true about this condition?

- a) Asymptomatic bacteriuria
- b) Test must be repeated with a fresh specimen

- c) Diagnose and treat as urinary tract infection
- d) Normal skin commensal grown on culture that needed to be proved

Question 15:

A group of hostel boys consumed hamburgers from a local street food shop following which they developed abdominal pain, diarrhoea, vomiting, and fever. They were diagnosed with hemolytic uremic syndrome (HUS). *Escherichia coli* O157:H7 was detected in the stool samples and the ham. What does H represent in H7?

- a) Fimbriae
- b) Capsule
- c) Flagella
- d) Lipopolysaccharide

Question 16:

An elderly female with a central line undergoes surgery. On postoperative day 3, she develops fever and tachycardia. A culture from the tip of the central line revealed brown buff colonies. Microscopy showed gram-positive organisms that reproduce by budding. What is the likely causative organism?

- a) *Staphylococcus epidermidis*
- b) *Streptococcus*
- c) *Candida*
- d) *Aspergillus*

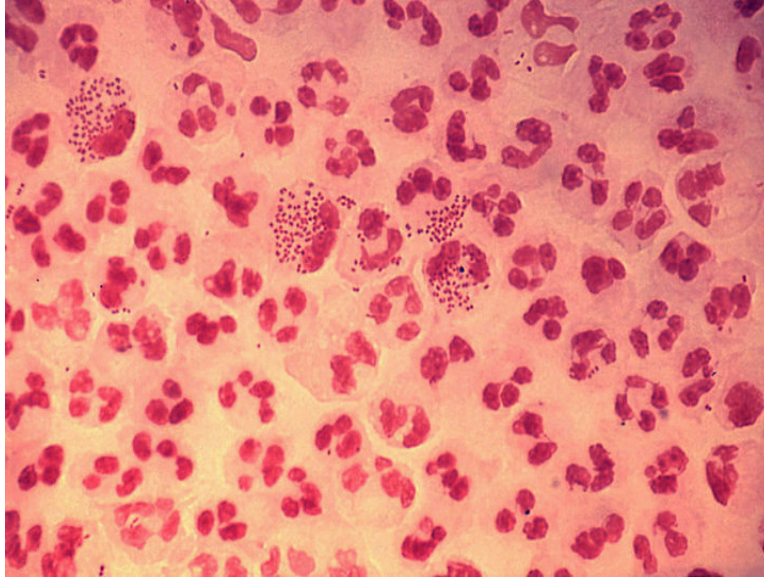
Question 17:

A 22-year-old farmer develops an ulcer on his foot with a white powdery substance and multiple discharging sinuses. Microscopy reveals a gram-positive organism with radiating filaments. What is the most likely organism isolated from the discharging sinus?

- a) *Nocardia*
- b) *Staphylococcus aureus*
- c) Mucormycosis
- d) *Sporothrix schenckii*

Question 18:

A middle-aged man with a history of unprotected sexual intercourse presents with complaints of painful urination and itching. On examination, Gram-negative intracellular cocci are observed. What is the most likely causative organism?



- a) *Staphylococcus aureus*
- b) *Neisseria gonorrhoeae*
- c) *Proteus vulgaris*
- d) *Chlamydia trachomatis*

Question 19:

A patient diagnosed with HIV presents with complaints of recurrent diarrhoea. The stool sample reveals only larvae and no ova. Which of the following is the most probable causative organism?

- a) *Ancylostoma*
- b) *Ascaris*
- c) *Strongyloides*
- d) *Enterobius*

Question 20:

A young man who has just returned from the USA develops fever, cough, and breathlessness. A chest radiograph shows bilateral hilar lymphadenopathy and mediastinal lymphadenopathy. Culture on Sabouraud Dextrose Agar (SDA) reveals white to light brown puffy colonies. Lactophenol blue staining shows the findings as depicted in the image. What is the most likely causative organism?



- a) Cryptococcosis
- b) Histoplasmosis
- c) Blastomycosis
- d) Mucormycosis

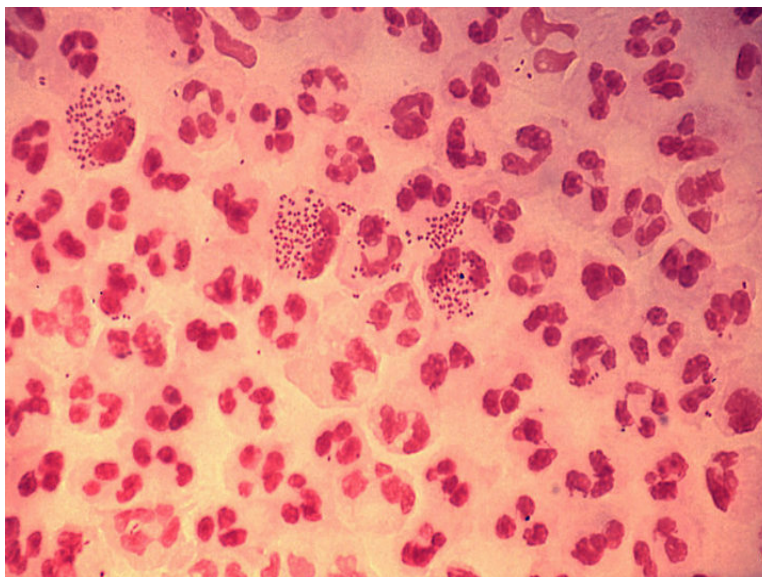
Question 21:

A middle-aged man presents with complaints of itching around the perianal region and unilateral lower limb lymphadenopathy. Larvae of approximately 200 microns in size were identified, and these larvae lacked nuclei at their tail tips. What is the most likely causative organism?

- a) *Wuchereria bancrofti*
- b) *Brugia malayi*
- c) *Brugia timori*
- d) *Mansonella perstans*

Question 22:

A middle-aged man presented to the OPD with dysuria and purulent urethral discharge for the last 3 days. Gram stain of the discharge is shown below. He admits to having unprotected sexual intercourse one week ago. What is the most appropriate treatment?



- a) Tab. Acyclovir 500 mg TDS for 1 week
- b) Tab. Doxycycline 100 mg BD for 14 days
- c) Tab. Cefixime 400 mg Stat
- d) Inj. Benzathaine Penicillin 2.4 MU IM Stat

Question 23:

A young man presents with burning and painful micturition. Microscopy showed gram-negative intracellular kidney bean-shaped organisms. Which of the following culture media is used to isolate the organism?

- a) Blood agar
- b) Thayer-Martin media
- c) MacConkey agar
- d) TCBS agar

Question 24:

A 35-year-old man presented with high fever, bleeding gums, myalgia, and hepatosplenomegaly. His TLC is 3100, Hb- 11 g/dl, hematocrit- 55%, and platelets - 25,000 cells. What is the most important prognostic factor in this condition?

- a) Hematocrit**
- b) Total Leukocyte Count**
- c) Hemoglobin**
- d) Platelets**

Answer Key

Question No.	Correct Option
1	d
2	a
3	d
4	a
5	c
6	a
7	b
8	c
9	b
10	b
11	d
12	d
13	d
14	c
15	c
16	c
17	a
18	b
19	c
20	b
21	a
22	c
23	b
24	a

Detailed Explanations

Solution to Question 1:

Multiplex PCR can be used for syndromic diagnosis of meningitis.

A syndromic diagnosis is a method of diagnosing infectious diseases that uses a single test to screen for multiple pathogens that could be causing a specific set of symptoms.

Multiplex PCR (Polymerase Chain Reaction) is a variation of the standard PCR technique that allows simultaneous amplification of multiple target sequences in a single reaction. It uses more than one set of primers (specific to different target sequences) to amplify different DNA fragments at the same time.

Other options:

Option A: Real-time PCR helps quantify the PCR and gives an estimate of the pathogen DNA load in the specimen. It uses fluorescent dyes or specific DNA probes. It is also known as qPCR (Quantitative PCR).

Option B: Nested PCR involves the sequential use of two primer sets. The first primer set is used for amplification and the amplicon obtained is then used as a target sequence for second amplification. This ensures specificity. Used in GeneXpert to detect Mycobacterium TB and resistance to rifampicin.

Option C: Reverse transcriptase PCR: The enzyme reverse transcriptase (RT) is used to synthesize a complimentary copy of DNA (cDNA) from RNA and this cDNA is then used as a template for PCR. Unlike conventional PCR which can only amplify DNA, this one can even amplify RNA. It is used to detect H1N1 from Nasopharyngeal swab.

Solution to Question 2:

In this case of an immunocompromised patient with pneumonia and skin lesions, the microscopy of specimens from skin lesions showing broad-based budding with a thick capsule suggests the diagnosis of Blastomycosis caused by *Blastomyces dermatidis*.

Blastomycosis is a chronic infection, characterized by the formation of suppurative and granulomatous lesions, especially, in the lungs and skin.

Blastomycosis predominantly affects individuals in regions where the fungus *Blastomyces* is endemic, such as the Midwest and southeastern United States. It is most common in North America and so is also called North American Blastomycosis.

The disease has a broad demographic impact, occurring in both immunocompetent and immunocompromised individuals. However, those with impaired immune systems—such as patients with AIDS, those on glucocorticoid therapy, or those receiving TNF- α inhibitors—are at higher risk for severe or disseminated disease involving the skin, bone, or central nervous system. Healthy individuals may experience a range of symptoms from mild pulmonary illness to more severe forms of pneumonia. It is more commonly observed in males aged 30-50 years, possibly due to occupational or recreational exposures.

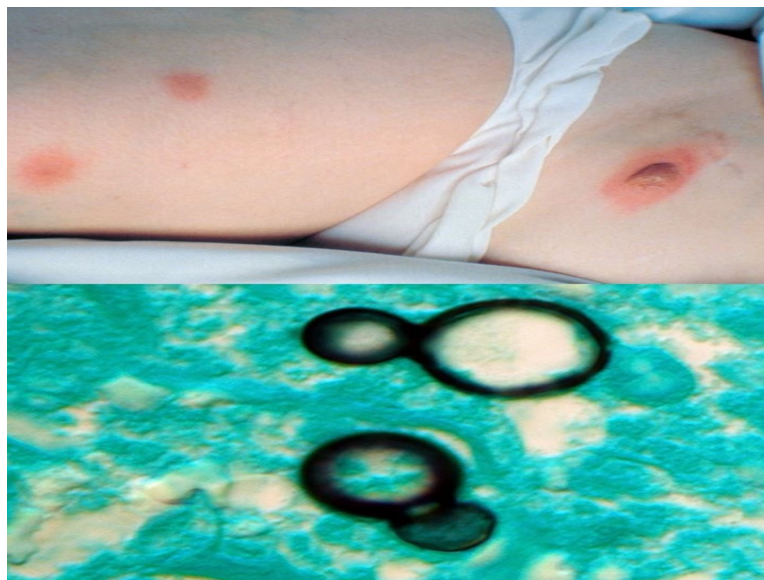
Blastomycosis can present in three forms:

- Pulmonary disease:
 - Presents with cough, sputum production, chest pain, and fever
 - Hilar lymphadenopathy may be present
 - Focal or diffuse consolidation, miliary lesions, or abscess formation may occur.
 - Resembles TB and Histoplasmosis
- Cutaneous disease:
 - Involvement of face and exposed parts of the skin
 - Skin lesion: ulcerated verrucous granulomas with an advancing border and central scarring. The border is filled with microabscesses and has a sharp, sloping edge.
- Disseminated disease:
 - Multiple skin lesions are the most common presentation
 - Highly fatal
 - The genitourinary tract (especially the prostate) is frequently involved
 - Bone involvement resembles chronic osteomyelitis.

It can be diagnosed based on culture findings. Characteristic findings include the presence of thick-walled round yeast cells with a single broad-based bud when cultured at 37°C. (figure of 8 appearance)

Treatment of localised mild/moderate disease includes Itraconazole (6 months) and for disseminated/ severe disease- amphotericin B.

The image given below shows the skin lesions and microscopy of broad-based budding by *Blastomyces dermatitidis*.



Other options

Option B: Histoplasmosis is caused by *Histoplasma capsulatum*. It has tiny oval yeast cells with narrow-based budding within the macrophages with an underlying granulomatous response. This

is a dimorphic fungus that exists in the yeast form in body tissues and cultures maintained at 37°C, and in the filamentous mould form in cultures at room temperature (25°C).

Option C: Coccidiomycosis is caused by *Coccidioides immitis* which is a thermally dimorphic fungus. It occurs as a spherule in tissues and culture at 37°C and as the mycelial form in soil and culture at room temperature. They are endemic to the arid desert regions of the southwestern USA and California. Infection is acquired by inhaling dust containing arthrospores of the fungus.

Option D: Candidiasis is caused by candida species which are yeast-like fungi. They exist as yeast and pseudohyphae. It appears as budding gram-positive cells on wet films and smears. Mycelial forms suggest colonization. *Candida albicans* form chlamydospores on corn meal agar.

Solution to Question 3:

This patient's history of fever with relative bradycardia, abdominal tenderness, and similar cases in the neighbourhood suggests enteric fever. It is caused by *Salmonella typhi* or paratyphi. Out of the given options, blood culture is the most sensitive test in this patient.

Diagnosis of enteric fever:

Blood culture: It is more sensitive during 1st week than in the subsequent weeks. It rapidly becomes negative on treatment with antibiotics.

Antibody testing: WIDAL test to detect rising titres of H and O agglutinins against *salmonella typhi* and paratyphi. Agglutinins appear by the end of the first week and rise steadily up to the 3rd or 4th week, after which they disappear.

Stool culture: It yields positive results from the 3rd week onwards.

Urine culture: Results may be positive after the 2nd week.

Bone marrow culture remains positive for 5 days even after the start of antibiotics and is the most sensitive test in the first week. The culture of duodenal aspirate may yield positive results when bone marrow culture is inconclusive.

Relative bradycardia is seen when patients have a lower heart rate than expected for a given rise in body temperature. (Faget's sign)

Infectious causes of relative bradycardia include:

- *Salmonella typhi*
- *Francisella tularensis*
- *Coxiella burnetii* (Q fever)
- *Leptospira interrogans*
- *Legionella pneumophila*
- *Rickettsia* spp.

Noninfectious causes include:

- Drug fever
- Beta-blocker use

- Central nervous system lesions
- Malignant lymphoma
- Factitious fever

Solution to Question 4:

The likely causative organism for the child's abdominal pain, diarrhoea, perianal itching, and the white thread-like worm found in the night clothes is *Enterobius vermicularis*, as evidenced by the worm's pointed end and wing-like cuticular expansions (cervical alae) shown in the left image, and the colourless, non-bile stained, planoconvex eggs with an outer albuminous layer that stick to each other and other objects.

Enterobius vermicularis completes its entire life cycle in the human host. They are oviparous. Man acquires infection by ingesting embryonated eggs containing larva through contaminated fingers and by autoinfection.

At night, the gravid female worm comes out through the anus and lays its sticky eggs on the perianal skin. This leads to pruritus and the patient scratches the affected perianal area. The worm can crawl into the vulva and vagina causing irritation and a mucoïd discharge. It can also cause chronic salpingitis, cervicitis, peritonitis, recurrent urinary tract infections, and appendicitis.

Since pinworms are rarely released in faeces, the faecal examination is not useful in the diagnosis.

The NIH swab (National Institutes of Health, USA) consists of a glass rod with a piece of transparent cellophane attached at one end with a rubber band. The cellophane part is used for swabbing by rolling over the perianal area. The cellophane piece is spread over a glass slide and examined microscopically.

The image given below shows the NIH swab.



Another method of demonstrating eggs is the scotch tape method. Here, the tape is firmly pressed against the anal margin, covering all sides, and transferred to a glass slide. A drop of toluene is added and examined under the microscope.

The below image shows the scotch tape method.

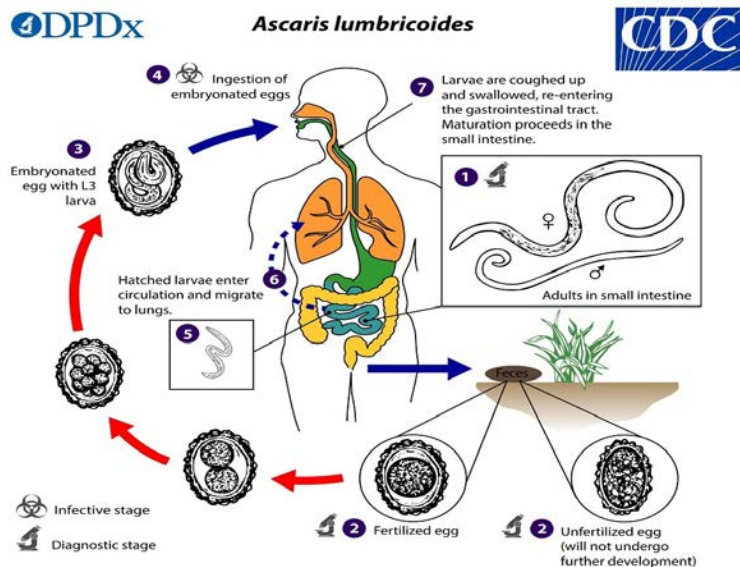


The adult worms may sometimes be noticed on the surface of stools, can be found crawling out of the anus, or may be detected in stools collected after an enema.

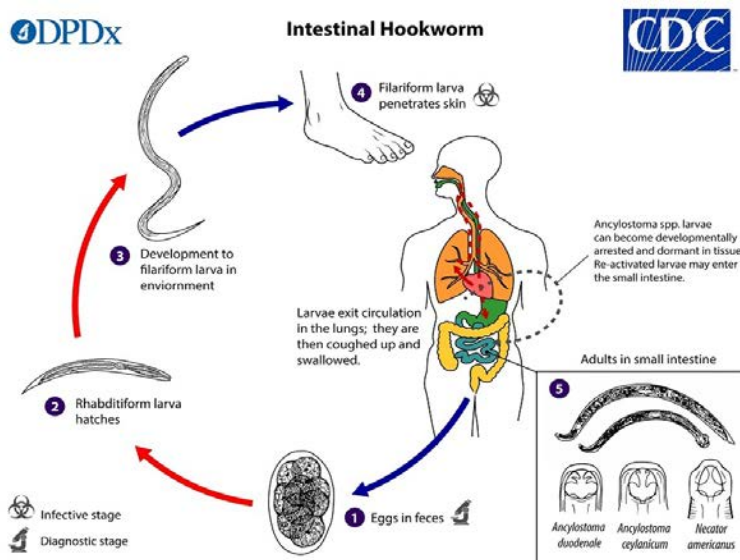
Frequent hand washing, fingernail cleaning, regular bathing, and frequent washing of night clothes, and bed linen should be done. Infection can be treated with pyrantel pamoate, albendazole, or mebendazole as single-dose therapy.

Other options:

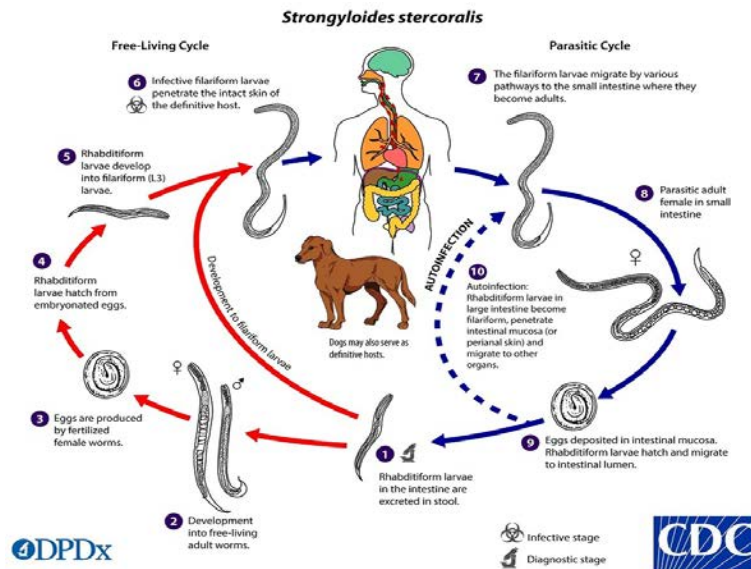
Option B: *Ascaris lumbricoides* causes infection when the egg containing the infective rhabditiform larva is swallowed. The eggs hatch in the duodenum to release the rhabditiform larvae, which then migrate to the liver via the portal vein, heart, and then to the lungs, pass up the respiratory tract, and are swallowed. They finally develop into adults in the small intestine. Here, the gravid females lay eggs, which are shed via the feces, and the cycle repeats. Definitive diagnosis is by demonstration of eggs in feces. The life cycle is shown below.



Option C: *Ancylostoma duodenale* or hookworm infects a person by direct penetration of the skin by filariform larvae. They develop into adult worms in the small intestine. Segmented eggs are passed in feces and these develop into filariform larvae in the soil. They can cause severe local itching, cutaneous larva migrans, and respiratory symptoms. Adult hookworm disease is characterized by anaemia, epigastric pain, dyspepsia, vomiting, and reddish black stools. Diagnosis is by demonstration of eggs or larvae in feces. The life cycle is shown below.



Option D: *Strongyloides stercoralis* infection is by direct penetration of the skin by the filariform larva or by autoinfection. The larvae enter lay eggs in the small intestine mucosa. The eggs hatch and release rhabditiform larvae which pass through feces into the soil and start another cycle. External autoinfection occurs when the larvae penetrate the perianal skin. Internal autoinfection occurs in immunocompromised patients. Here, the larvae mature to the infective form in the bowel itself. Diagnosis is by demonstration of rhabditiform larvae in feces.



Solution to Question 5:

Bacteria utilise fimbriae for adherence.

Fimbriae, also called pili are hair-like surface appendage that projects from the cell surface as straight filaments. Fimbriae are classified based on the function of ordinary pili and sex pili.

Functions of pili:

- Bacterial adhesion of symbiotic and pathogenic bacteria to host cells using adhesins.
- Resistance to phagocytosis.
- Antigenic pilin proteins are responsible for the immune response.
- Twitching motility to bacteria.
- Bacterial conjugation through sex pili, in which transfer of genetic material occurs.

Other options:

Option A: The cytoplasmic membrane, or plasma membrane, is a fundamental structure of all cells, including bacteria. It acts as a selective barrier, regulating the entry and exit of substances.

Option B: In organisms like protozoa, cilia are used for locomotion and feeding. These organisms use their cilia to move through water and to sweep food particles towards their oral groove.

Option D: Flagella are thread-like appendages composed of protein. They are the organs of locomotion but motility via pili is completely different from flagellar motion because of the helical arrangement of pilin molecules.

Solution to Question 6:

Aplastic crisis in this patient is triggered by parvovirus infection.

Parvovirus infects and kills the red cell precursors. This creates a transient arrest of erythropoiesis i.e. aplastic crisis as the life span of RBCs is shortened. The immune response clears the virus in 1-2 weeks. It may be seen in patients with underlying chronic hemolytic anaemia like sickle-cell disease, thalassemia, hereditary spherocytosis, and RBC enzyme deficiencies like pyruvate kinase deficiency. Patients with aplastic crisis appear ill with fever, malaise, and paleness with tachycardia and tachypnea due to profound anaemia. The rash may or may not be present.

Parvovirus B19 infection occurs usually in children 5–10 years causing erythema infectiosum. It is also known as the fifth disease. It begins as a prominent erythema of the cheeks- slapped cheek disease, spreading to the trunk and limbs, followed by lymphadenopathy and arthralgia. Fever and mild constitutional symptoms may accompany the rash.



Other options:

Option B: Epstein Barr virus affects adolescents and younger adults more frequently but may occur from childhood to old age. It is transmitted primarily through close contact with oropharyngeal secretions. The incubation period of EBV infectious mononucleosis is 4-6 weeks. Clinically, patients may present with, fever, malaise, nausea, anorexia, lymphadenopathy- bilateral and symmetrical, pharyngitis, tonsillar enlargement, hepatomegaly and splenomegaly. A drug-induced rash may develop on using beta-lactam. antibiotics.

Option C: Cytomegalovirus causes mononucleosis in healthy individuals and disseminated diseases like leukopenia, hepatitis, colitis, pneumonitis, esophagitis, gastritis, and retinitis in immunocompromised patients.

Option D: Herpes simplex virus type 1 which belongs to the subfamily alpha viruses causes Herpes labialis (also known as cold sores). HSV type 1 (human herpes virus type 1 or HHV type 1) is isolated from lesions in and around the mouth and is transmitted by direct contact or droplet spread from cases or carriers. The most common site is the face- around the mouth, on the cheeks, chin, or the forehead.

Solution to Question 7:

The most likely diagnosis based on the clinical vignette of high-grade fever with chills, retro-orbital pain, conjunctival suffusion, severe calf tenderness, and laboratory findings of hypokalemia and neutropenia is suggestive of leptospirosis.

Leptospirosis is a zoonosis caused by spirochetes of the genus *Leptospira*. Patients with mild disease may present with flu-like symptoms of sudden onset, fever with chills, headache, abdominal pain, conjunctival suffusion, and myalgia. The muscle pain usually affects the calves, abdomen, and back. Severe cases of leptospirosis, which can progress rapidly and have a high fatality rate, are associated with Weil's syndrome, characterized by triad of:

- Hemorrhage
- Jaundice
- Acute kidney injury

Jaundice, when present, can be profound, and physical examination may reveal an enlarged, tender liver. Temperature-pulse dissociation or relative bradycardia may also be observed.

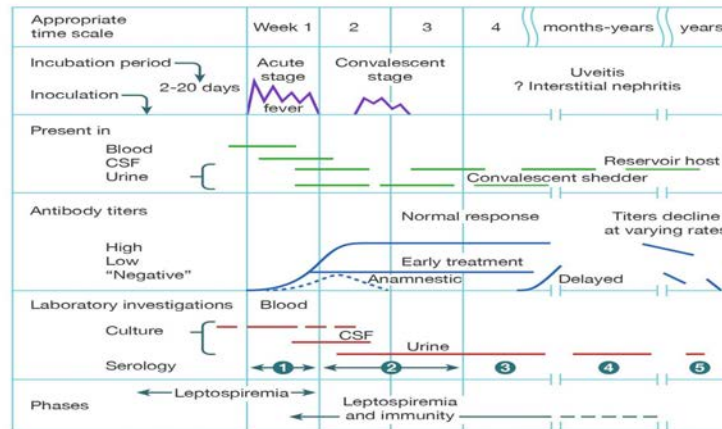
The image below shows icterus and conjunctival suffusion in a patient with leptospirosis:



Leptospira are actively motile organisms and are best seen using a dark-field microscope. They are thin, tightly coiled, and flexible spirochetes with one end forming a hook. Diagnosis is by microscopic agglutination test.

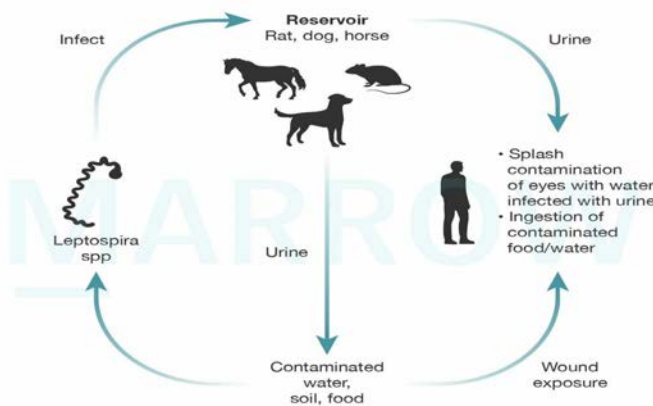
Transmission occurs through cuts or breaks in the skin and mucous membranes (conjunctival and oral mucosa). Several outbreaks have been reported that resulted from exposure to floodwaters contaminated by urine from infected animals. The organism enters the bloodstream and lodges in the parenchymatous organs (liver and kidneys) leading to haemorrhage and necrosis of the tissue causing dysfunction of the organs. The progression and lifecycle of *Leptospira* are shown below.

Progression of leptospirosis



©Marrow

Lifecycle of Leptospira



©Marrow

Treatment for leptospirosis is based on the severity of the disease. In mild cases, oral doxycycline, ampicillin, or amoxicillin is preferred and in severe cases, intravenous penicillin, ampicillin, or ceftriaxone is commonly used.

Other options:

Option A: Chikungunya is a viral disease that presents with saddle-back fever, conjunctival suffusion, and photophobia. Migratory arthralgia, especially involving small joints may also be seen. A maculopapular or petechial rash is frequently seen on the trunk.

Option C: Dengue hemorrhagic fever is a form of severe dengue with bleeding manifestations. Evidence of plasma leakage and thrombocytopenia is also generally seen.

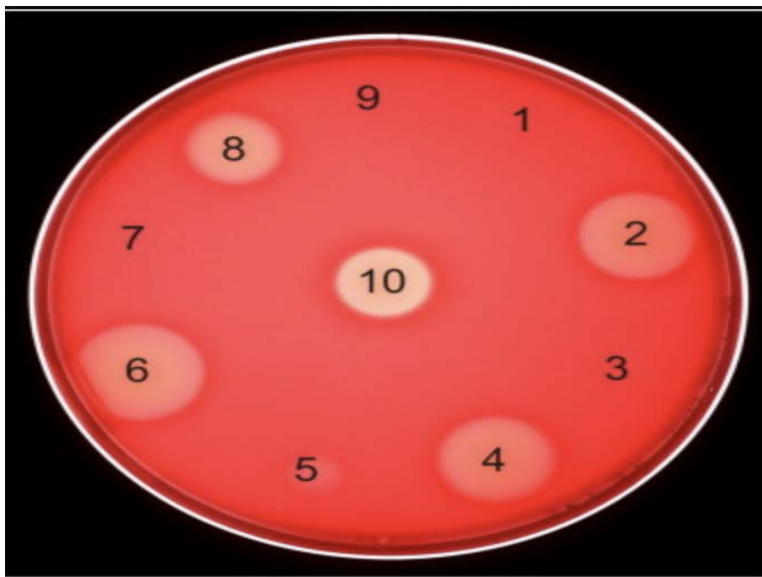
Option D: Malaria is a parasitic infection caused by the Plasmodium species transmitted by the Anopheles mosquito. It is characterized by many nonspecific symptoms, including but not limited to tachycardia, tachypnea, chills, malaise, fatigue, diaphoresis (sweating), and headache. The fever of malaria is characterized by periodicity depending on the infective Plasmodium species. The

fever returns every other day for *Plasmodium vivax*, *Plasmodium ovale*, and *Plasmodium falciparum*, and every three days for *P. malariae*.

Solution to Question 8:

Kanagawa phenomenon is shown by *Vibrio parahemolyticus*.

On Wagatsuma agar (blood agar with 2% salt), pathogenic strains of *V. parahemolyticus* are hemolytic, while non-pathogenic strains are non-hemolytic. This is due to a heat-stable hemolysin. They produce green-coloured colonies on the TCBS agar as they do not ferment sucrose.



On staining with Giemsa stain or methylene blue (Wayson's stain), the bacteria has bipolar staining with two ends densely stained and the central area clear. This is the characteristic 'safety-pin appearance' seen in *Vibrio parahemolyticus*.

Other options:

Option A: *Vibrio cholerae* is a facultative anaerobe, which produces yellow-colored colonies on thiosulfate citrate bile salts sucrose agar (TCBS). It is a selective and differential medium used for the culture of *Vibrio* species.

Option B: Wilson and Blair's medium is selective for *Salmonella* species. It is a type of selective and indicator media. *Salmonella typhi* reduces sulphite to sulphide in the presence of glucose and produces jet-black colonies with a metallic sheen due to the production of H_2S on Wilson and Blair bismuth sulphite medium.

Option D: *E. coli* is a gram-negative, flagellated rod. It is the most frequent cause of acute complicated urinary tract infections. It is a lactose fermenter and hence, grows on MacConkey medium to produce pink colonies.

Solution to Question 9:

The clinical vignette is suggestive of fulminant meningococcemia. The organism causing this condition is *Neisseria meningitidis*.

Neisseria meningitidis is an aerobic, gram-negative, nonmotile diplococcus that is catalase and oxidase-positive. It oxidizes glucose and maltose but not lactose or sucrose. The bacteria typically spread from the nasopharynx to the meninges via the perineural sheath of the olfactory nerve following an initial flu-like infection.

Meningococcemia often starts with acute fever, chills, malaise, and a petechial rash. As the disease advances, symptoms of meningitis, such as severe headache, vomiting, photophobia, confusion, and a stiff neck, may develop, potentially progressing to coma within a few hours. In its fulminant form, meningococcemia presents with high fever and a hemorrhagic rash and can lead to severe complications like disseminated intravascular coagulation, circulatory collapse, and Waterhouse-Friderichsen syndrome, characterized by bilateral adrenal gland necrosis and subsequent adrenal failure.

Meningitis can rapidly progress to fulminant meningococcemia, evolving from acute illness to severe disease in hours. Classic meningitis symptoms include fever, headache, neck stiffness, and decreased consciousness. Nuchal rigidity, along with Kernig's and Brudzinski's signs, may indicate meningeal irritation, though their sensitivity can vary. Seizures, occurring in 15-40% of patients, may result from complications like raised intracranial pressure, leading to severe outcomes such as cerebral herniation. The rash associated with meningococcemia evolves quickly from a diffuse maculopapular appearance to petechiae on the trunk, extremities, mucous membranes, and sometimes palms and soles.

Other options:

Option A: *Haemophilus influenzae* is a gram-negative, non-motile, and non-spore-forming coccobacillus. *H. influenzae* causes meningitis, a potentially fatal disease that is most common in children of two months to 3 years old. It also causes laryngo-epiglottitis, pneumonia, suppurative lesions, and bronchitis.

Option C: Cytomegalovirus infection is the most common congenital infection. The clinical manifestations are hepatosplenomegaly, petechial rashes, jaundice, chorioretinitis, and microcephaly. One of the long-term sequelae is sensorineural hearing loss.

Option D: *Klebsiella pneumoniae* causes fatal pneumonia, which occurs in middle-aged or older persons with risk factors such as alcoholism, chronic bronchopulmonary disease or diabetes mellitus. Gram stain and culture are used for diagnosis.

Solution to Question 10:

In a patient with a central line on long-term IV antibiotics along with culture showing biofilm formation, the correct measure is to remove the catheter as biofilm offers resistance.

Biofilms are structured microcolonies of organisms that are enclosed in a self-produced glycocalyx matrix. Bacteria within biofilms are usually metabolically less active.

Biofilms are aggregates of bacteria that are enclosed in an exopolysaccharide matrix that is produced by the bacteria itself. The biofilms are attached to solid surfaces and form a slimy layer over them. Organisms that produce biofilm include:

- *Staphylococcus epidermidis* on medical devices and implants
- *Streptococcus viridians* on heart valves
- *Pseudomonas aeruginosa* lung infections
- *Staphylococcus aureus* on catheters
- *Legionella pneumophila* colonization of hospital water systems

The exopolysaccharide helps in protection against the host's immunity. It acts as a diffusion barrier for antimicrobials and makes the organism relatively insusceptible to drug action (Option A and C). Bacteria embedded in biofilms frequently become metabolically less active or dormant. It is thus harder to eradicate these organisms with antibiotics.

These bacteria within biofilms also produce small molecules, such as homoserine lactones. These molecules are taken up by adjacent bacteria and function as a communication system called Quorum Sensing and help turn on certain genes at a particular time.

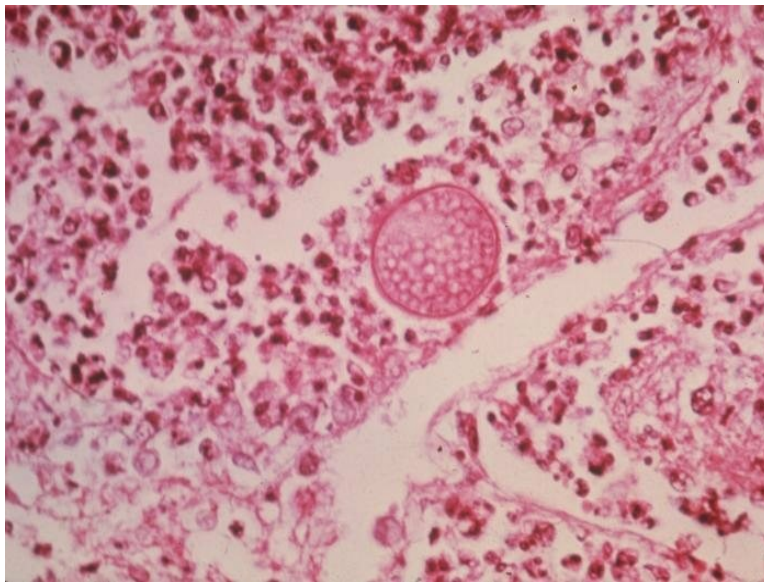
Domboro's solution, containing 0.25% acetic acid, helps prevent biofilm formation but is not effective at removing established biofilm. Biofilm, once established, adheres strongly to surfaces and is resistant to simple washing or flushing (option D). More aggressive methods, such as mechanical debridement or specialized treatments, are required to remove it.

Solution to Question 11:

Coccidioidomycosis, commonly known as Valley fever or desert rheumatism, is a mammalian fungal disease caused by *Coccidioides immitis* or *Coccidioides posadasii*.

The primary infection is often respiratory, with symptoms resembling bronchitis or pneumonia that resolve over a few weeks. The classic triad of coccidioidomycosis, known as 'desert rheumatism', includes the combination of fever, joint pains, and erythema nodosum.

Fluconazole and intravenous amphotericin B are used in progressive or disseminated disease, or immunocompromised individuals. Itraconazole or ketoconazole may be used for milder disease. Fluconazole is the preferred medication for coccidioidal meningitis, due to its penetration into CSF. Intrathecal or intraventricular amphotericin B therapy is used if infection persists after fluconazole treatment. The image showing mature spherules of *Coccidioides* is given below.



Other options

Option A: Histoplasma causes histoplasmosis/ Darling's disease/ reticuloendotheliosis. It has tiny oval yeast cells with narrow-based budding within the macrophages with an underlying granulomatous response. This is a dimorphic fungus that exists in the yeast form in body tissues and cultures maintained at 37°C, and in the filamentous mould form in cultures at room temperature (25°C).

Option B: Paracoccidioidomycosis, also known as Brazilian blastomycosis, South American blastomycosis, Lutz-Splendore-de Almeida disease and paracoccidioidal granuloma, is a fungal infection caused by the fungus *Paracoccidioides brasiliensis*.

Option C: Blastomycosis is a chronic infection caused by a dimorphic fungus called *Blastomyces dermatitidis*. It is characterized by the formation of suppurative and granulomatous lesions in the lungs and skin. It is most common in North America and is also called North American Blastomycosis.

Solution to Question 12:

The given clinical scenario along with the vector bat is suggestive of infection from the nipah virus.

Nipah virus is an RNA virus that belongs to the family Paramyxoviridae, genus Henipavirus. It is a zoonotic virus that shows both animal-to-human (bats, pigs) and human-to-human transmission. The natural hosts are fruit bats of the Pteropodidae family.

The clinical manifestations include asymptomatic subclinical infection, acute respiratory infection, and fatal encephalitis. There is no treatment or vaccine available for either people or animals. The primary treatment for humans is supportive care. Nipah virus is on the WHO list of Blueprint priority diseases.

The life cycle of nipah virus is shown below:



Other options

Option A: Ebola virus belongs to the family Filoviridae and is transmitted by African fruit bats. Human-to-human transmission through blood or body fluids can occur. It causes fever, abdominal pain, diarrhoea, and haemorrhage

Option B: Zika virus is transmitted to human beings by the bite of an infected *Aedes* mosquito. It can also be transmitted trans-placentally, causing microcephaly and other severe fetal brain defects.

Option C: Malaria is a parasitic infection caused by the plasmodium species and is transmitted by the female anopheles mosquito. Symptoms include fever, chills, and myalgia with periodicity differing for each species. The fever returns every two days for *P. ovale*, *P. vivax*, and *P. falciparum* and every three days for *P. malariae*.

Solution to Question 13:

The clinical features are suggestive of meningococcal meningitis and metronidazole is not the drug of choice. The drug of choice is a third-generation cephalosporin such as ceftriaxone. Chloramphenicol is used in patients with penicillin allergy.

It is caused by *Neisseria meningitidis*, a gram-negative, oxidase-positive, catalase-positive diplococci. Outbreaks usually occur in mass gatherings such as in army camps or hostel dorms.

The risk of infection is increased in patients with terminal complement deficiency, hypogammaglobulinemia and hyposplenism. This is because opsonization by complement is a major mechanism by which the bacteria is cleared from the blood.

Characteristic clinical findings include:

- Maculopapular or purpuric rash
- Shock due to capillary leak
- Bilateral adrenal hemorrhage (Waterhouse–Friderichsen syndrome)

Rifampicin, ciprofloxacin and cephalosporins can be used to treat carriers.

Solution to Question 14:

A young sexually active female presenting with a symptomatic UTI and urinalysis and cultures showing novobiocin-resistant coagulase-negative staphylococcus with a count of 10^4 cells/HPF should be diagnosed as a urinary tract infection and should be treated.

Staphylococcus saprophyticus is a gram-positive, coagulase-negative staphylococcus. It is non-pigmented, novobiocin resistant, and non-hemolytic on blood agar. It is a leading cause of cystitis in women. It is a part of the normal flora of human skin and periurethral area and thus causes cystitis in sexually active women.

Other options

Option A: A colony count of $>10^5$ /mL of a clean catch sample is considered significant bacteriuria with no clinical symptoms of UTI. In men, a count of 100,000 CFU/ml is sufficient to diagnose a UTI, while for women a UTI is diagnosed when the count is 100,000 CFU/ml in two consecutive midstream urine samples.

Option B: As the patient already has symptoms of UTI and the culture is showing a significant count redoing the test is not required.

Option D: Though *staphylococcus saprophyticus* is a normal commensal of skin since the patient already has symptomatic UTI with a proven culture, it need not be proved again.

Solution to Question 15:

H in *E. coli* O157:H7 represents flagellar antigen.

O represents somatic antigen.

E. coli O157:H7 is an enterohemorrhagic bacterial strain that causes hemorrhagic diarrhoea and cramping in the abdomen. It is transmitted through the fecal-oral route after consumption of contaminated, undercooked liquids and foods.

E. coli O157:H7 uses the fimbriae to adhere to the microvilli of intestinal epithelial cells. When bacteria attach to an epithelial layer, they rearrange the cytoskeletal actin, which flattens the microvilli and interferes with normal intestinal cell function. This is known as an effacing lesion.

Three days after exposure to the contaminated food specimen, diarrhoea—often the hemorrhagic kind—occurs due to exposure to the Shiga toxin. The patients will experience bloody diarrhoea after three days of symptoms. After seven days, the symptoms of hemorrhagic diarrhoea usually go away on their own for 85% of patients; the remaining 15%, who are usually children, will experience systemic manifestations, most commonly HUS.

EHEC are gram-negative bacilli which are lactose fermenting. Unlike most other *E. coli* serotypes, O157:H7 does not use sorbitol and produces clear colonies on the Sorbitol MacConkey agar, whereas other serotypes produce pink colonies. O157:H7 is a strain of *E. coli* called enterohemorrhagic *E. coli* (EHEC) which produces verocytotoxin or Shiga-like-toxin. Shiga toxin

acts by inactivating the 60S ribosomal subunit thereby inhibiting protein synthesis. It is absorbed from the inflamed gastrointestinal mucosa into the circulation and causes endothelial dysfunction, and causes platelet activation and aggregation. Children and older adults are at the highest risk. It causes mild non-bloody diarrhea to hemorrhagic colitis, or a severe form of dysentery, with hemolytic uremic syndrome, a disease resulting in acute renal failure, microangiopathic hemolytic anemia, and thrombocytopenia.

Solution to Question 16:

The given clinical scenario along with microscopy showing gram-positive budding yeasts cells and brown buff colonies is suggestive of invasive candidiasis.

Candida species are yeast-like fungi that can cause bloodstream infections, particularly in patients with central lines or other invasive devices. It is a common cause of central line-associated bloodstream infections (CLABSI) and frequently presents with fever and signs of systemic infection..

The most common species implicated in candidal infections are *C. albicans*, *C. parapsilosis*, and *C. glabrata*. Clinically, infection with candida can present as cutaneous & mucosal candidiasis, systemic candidiasis, or rarely, as chronic mucocutaneous candidiasis.

Cutaneous and mucosal infections manifest as thrush affecting the tongue, lips, gums, and palate. It is seen as a patchy to confluent whitish pseudomembranous lesion and is commonly associated with AIDS. Candida may also invade the vaginal mucosa leading to vulvovaginitis, and the nails, and around the nail plate leading to onychomycosis.

Morphologically, they are yeast-like fungi. They exist as yeast and pseudohyphae (Shown in the image below). It appears as budding gram-positive cells on wet films and smears. Mycelial forms suggest colonization. Candida albicans form chlamydospores on corn meal agar.



When colonies of *Candida albicans* are inoculated in serum, the yeast cells start to form elongations called germ tubes within 2 hours, when kept at 37 °C. The image below

shows germ tube formation, also known as the Reynolds Braude phenomenon, seen in *Candida albicans*.



Other options:

Option A: *Staphylococcus epidermidis* is a gram-positive cocci that commonly colonizes the skin and can cause infections related to indwelling devices, such as central lines. However, it does not reproduce by budding, and the description of brown buff colonies does not fit.

Option B: *Streptococcus* species are gram-positive cocci that are typically found in chains. They do not reproduce by budding, and the brown buff colonies seen in the culture do not align with the typical appearance of *Streptococcus* in culture.

Option D: *Aspergillus* is a mold that forms septate hyphae, not yeast-like cells, and does not reproduce by budding. Additionally, *Aspergillus* forms colonies with a different characteristic appearance, usually described as greenish or fuzzy, rather than brown buff.

Solution to Question 17:

The given clinical scenario of multiple discharging sinuses and aspirate showing filamentous gram-positive bacteria are suggestive of *Nocardia*.

Nocardia species appear as filamentous organisms with extensive hyphae-like branching. They are aerobic, gram-positive, catalase-positive, and urease-positive. They are weakly acid-fast because they stain with carbolfuchsin and retain when decolorized with 1–4% sulfuric acid instead of the stronger acid-alcohol decolorant.

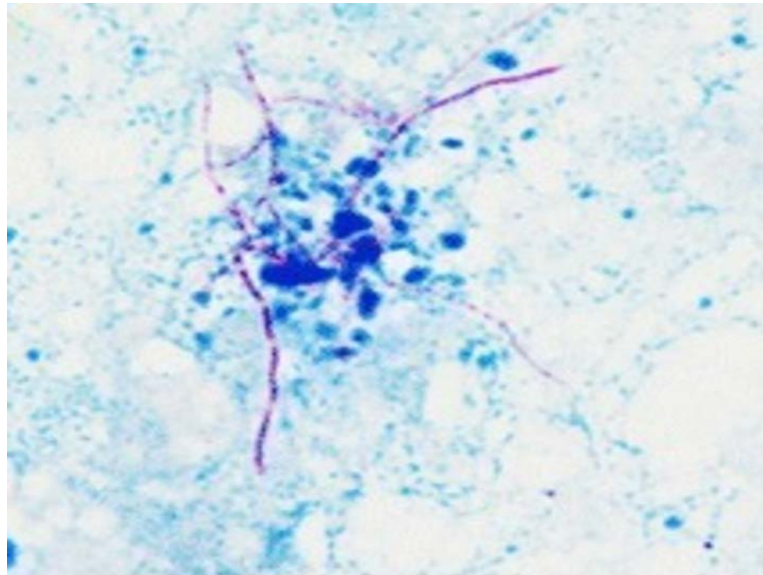
Infections caused by *Nocardia* include cutaneous, subcutaneous, or systemic lesions. It may lead to local abscesses, cellulitis, or lymphocutaneous lesions. Systemic nocardiosis occurs more often in immunodeficient persons.

They can present with a characteristic powdery substance that appears as gritty or necrotic material on the skin or within soft tissues. Accompanying this are multiple discharging sinuses, which are abnormal openings that form as the body attempts to expel infected material; these

sinuses can drain purulent, often foul-smelling discharge typically associated with localized abscesses, and the discharge may be viscous, sometimes containing necrotic tissue.

Systemic nocardiosis usually caused by *N. asteroides* manifests primarily as pulmonary disease, pneumonia, lung abscess, or other lesions resembling tuberculosis. Extrapulmonary manifestation may involve the brain, kidneys, and other organs.

The image given below is a smear of modified acid-fast Kinyoun staining showing filamentous red-stained partially acid-fast rods suggestive of *Nocardia*.



Other options:

Option B: *Staphylococcus aureus* is a common cause of skin and soft tissue infections, but it appears as gram-positive cocci in clusters, not as branching filaments. It would not typically be associated with discharging sinuses and white powdery discharge.

Option C: Mucormycosis is a fungal infection that typically affects immunocompromised individuals, particularly those with diabetes or undergoing chemotherapy. It presents with necrotizing infections but does not exhibit branching filaments in microscopy and is not associated with white powdery discharge from sinuses.

Option D: *Sporothrix schenckii* is a dimorphic fungus that causes sporotrichosis, typically following exposure to soil or plants. It can cause skin ulcers, but it appears as cigar-shaped yeast under microscopy, not as gram-positive filamentous bacteria.

Solution to Question 18:

The above clinical scenario along with the Gram stain showing gram-negative intracellular diplococci within phagocytes is suggestive of infection by *Neisseria gonorrhea*.

Gonorrhea is a sexually transmitted disease caused by *Neisseria gonorrhea*. It uses pili in adhesion to the urethra or other mucosal surfaces. The cocci penetrate the intracellular spaces to reach the subepithelial tissue by the third day of infection. The incubation period is 2-8 days.

The major symptoms include vaginal discharge, dyspareunia, and dysuria, usually without urinary frequency or urgency. The specific symptoms seen in males and females are as follows:

In female patients:

About half the female patients remain asymptomatic. But, patients typically present with urethritis, cervical discharge, pruritus, dysuria, and intermenstrual bleeding.

The ectocervix and vagina are not usually affected in adults as the stratified squamous epithelium is resistant to infection by the cocci. The acidic pH of vaginal secretions is also protective. The principal sites of invasion for gonococcal infection are the columnar and transitional epithelium.

Pelvic inflammatory disease and salpingitis may lead to sterility. Other complications are tubo-ovarian abscesses and ectopic pregnancies. Rarely peritonitis may develop with perihepatic inflammation (Fitz Hugh Curtis syndrome). The patient presents with right upper quadrant pain and tenderness with abnormal liver function tests.

In male patients :

- Acute urethritis
- Genital discharge: It is mucopurulent that can be described as that resembling the flow of seed (semen).
- Strictures
- Water-bug perineum: The infection may spread to the periurethral tissues causing abscesses and multiple discharging sinuses.

Lab diagnosis:

The discharge is collected and subjected to Gram's stain which reveals gram-negative intracellular diplococci within phagocytes. It is non-sporulating, non-motile and aerobic but may grow anaerobically also. It acidifies glucose but not maltose in serum sugars. Stuart's transport medium is used. Cultures can be obtained from chocolate agar or Mueller-Hinton agar. Specifically, Thayer-Martin agar is used to isolate *Neisseria gonorrhoeae*.

Treatment:

According to CDC guidelines (2021), treatment of uncomplicated gonococcal infections is as follows:

- Weight <150 kg – Ceftriaxone 500 mg intramuscular (IM) in a single dose
- Weight ≥150 kg – Ceftriaxone 1 g intramuscular (IM) in a single dose

Concurrent chlamydial infection must also be treated unless it has been excluded by molecular testing. First-line drugs used are as follows:

- Oral azithromycin 1g single dose
- Oral doxycycline 100 mg BD for 7 days (non-pregnant patients only)

The sexual partners of the patient must be treated to reduce the risk of recurrence. Expedited partner therapy refers to the index patient delivering therapy to one's own partners. However, this type of empirical therapy ignores the importance of direct counseling.

Other options:

Option A: *Staphylococcus aureus* is a gram-positive bacterium that can cause various infections, but it is not typically associated with sexually transmitted infections (STIs) or intracellular gram-negative cocci.

Option C: *Proteus vulgaris* is a gram-negative bacillus and is not typically associated with sexually transmitted infections. It is more often linked to urinary tract infections but would not appear as gram-negative intracellular cocci.

Option D: *Chlamydia trachomatis* is another common cause of sexually transmitted infections. It is an obligate intracellular bacterium that is difficult to visualize using standard Gram staining.

Solution to Question 19:

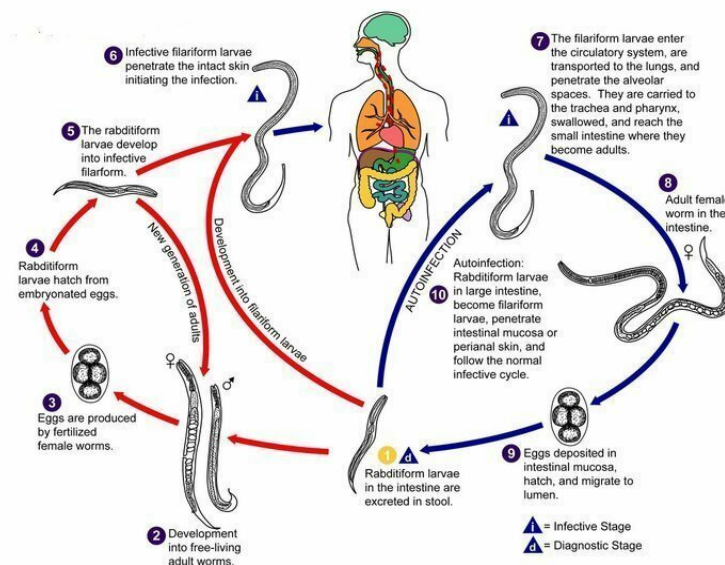
The presence of recurrent diarrhea in HIV patient along with the presence of only larvae in stool is suggestive of infection by *Strongyloides stercoralis*.

Strongyloides stercoralis is a parasitic nematode known for causing chronic diarrhea, particularly in immunocompromised individuals such as those with HIV. In this case, only larvae (and not ova) are seen in the stool, which is characteristic of *Strongyloides* infection due to its autoinfective cycle, where the larvae hatch inside the intestine and are passed in the stool.

Strongyloidiasis:

Strongyloides are ovoviviparous. The transmission of infection occurs through penetration of the skin by filariform larvae in the soil. Then, they migrate to the small intestine. The larvae moult twice and become adult female worms in the small intestine. The eggs laid in the mucosa of the intestine hatch immediately releasing the rhabditiform larva (L1 stage). These rhabditiform larvae in the bowel can develop directly into filariform larvae that penetrate the colonic wall or perianal skin and enter the circulation to repeat the migration. This establishes ongoing internal reinfection.

The life cycle of strongyloides is shown below.



It is generally an asymptomatic condition. There may be cutaneous, pulmonary, and intestinal manifestations in some cases. Pulmonary manifestations occur when the larva escapes from the pulmonary capillaries into the alveoli leading to bronchopneumonia, chronic bronchitis, and asthmatic symptoms.

The term larva currens (meaning 'racing larvae') has been applied to the rapidly progressing linear or serpiginous urticarial tracks caused by migrating filariform larvae of *Strongyloides stercoralis*. These often follow autoinfection and start perianally. Disseminated Strongyloidiasis is seen in immunodeficient individuals.

Larva currens is shown in the image below.



Visceral larva migrans is caused by the migration of larvae of nonhuman species of nematodes that infect via the oral route.

Etiological agents causing visceral larva migrans:

- Human nematode: *Ascaris lumbricoides*, *Strongyloides stercoralis*.
- Zoophilic nematode: *Toxocara canis*, *Toxocara cati*, *Angiostrongylus cantoniensis*, *Angiostrongylus costaricensis*, *Anisakis*.
- Non human nematode: Filarial species, *Brugia pahangi*, *Brugia pater*

Laboratory diagnosis of strongyloidiasis:

- Eosinophilia and raised IgE levels.
- Microscopic examination of the direct wet mount of the stool demonstrating rhabditiform larvae forms the definitive diagnosis
- Stool culture and serological tests are also done

Treatment: Ivermectin is the preferred agent in the management of strongyloidiasis.

Other options:

Option A: *Ancylostoma* presents with creeping eruption. Cutaneous larva migrans or creeping eruption is caused by nematode larvae that are infected by skin penetration.

Option B: *Ascaris* is a large intestinal worm, and its ova are typically seen in the stool, not larvae. It is not commonly associated with chronic diarrhoea, especially in immunocompromised patients.

Option D: *Enterobius* or pinworm infection presents as itching (pruritus) and restless sleep (due to nocturnal perianal or perineal pruritus)

Solution to Question 20:

The image shows the tuberculate macroconidia, with finger-like projections which, along with the history, is a characteristic feature of *Histoplasma*.

Histoplasmosis (Darling's disease) is a granulomatous disease caused by a dimorphic fungus, *Histoplasma capsulatum*. It is endemic in the USA, in the states bordering the Ohio River valley and the lower Mississippi River. In India, it is reported from the region of West Bengal, along the Ganga River. The fungus inhabits humid and acidic soil that contains a large amount of bird or bat droppings.

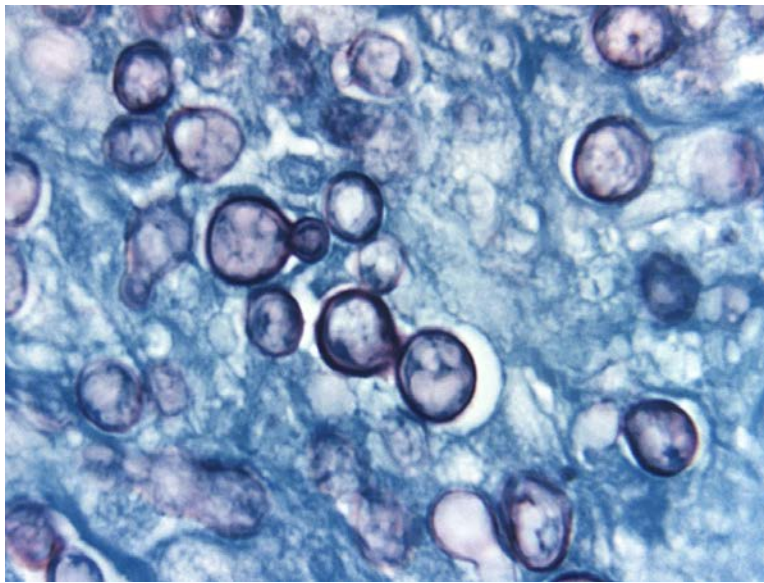
H. capsulatum is transmitted by inhalation of spores (i.e. microconidia) which usually circulate in the air after the contaminated soil is disturbed. After the spores enter into the lungs, they are engulfed inside the alveolar macrophages and then transform into yeast forms.

Clinically, the classical histoplasmosis ranges from asymptomatic infection (in immunocompetent people) to life-threatening illness seen in people with low CMI. The various clinical types include:

- Pulmonary histoplasmosis - Acute form presents with mild flu-like illness, pulmonary infiltrates in the chest X-ray with hilar or mediastinal lymphadenopathy. Chronic cavitary histoplasmosis may be seen in smokers with underlying structural lung disease.
- Mucocutaneous histoplasmosis - Skin and oral mucosal lesions may develop secondary to pulmonary infection. Oral lesions are particularly seen in Indian patients.
- Disseminated histoplasmosis - It develops if CMI is very low (e.g. untreated HIV-infected people or following organ transplantation). The common sites affected are bone marrow, spleen, liver, eyes, and adrenal glands.

Diagnosis :

Direct microscopy - Histopathological staining of the specimens reveals tiny oval yeast cells (2–4 μm size) with narrow-based budding within the macrophages with an underlying granulomatous response.



Culture - It is the gold standard method of diagnosis. Specimens should be inoculated onto media such as SDA, blood agar, and BHI agar in duplicate and incubated simultaneously at 25°C and 37°C.

At 25°C it forms a mycelial phase and produces white to buff brown colonies that consist of two types of conidia or spores - tuberculate macroconidia (with typical thick walls and finger-like projections which is a characteristic feature) and microconidia(smaller, thin, and smooth-walled).

At 37°C it gets converted into yeast form (creamy white colonies), which is best developed in special media like Kelley's media.

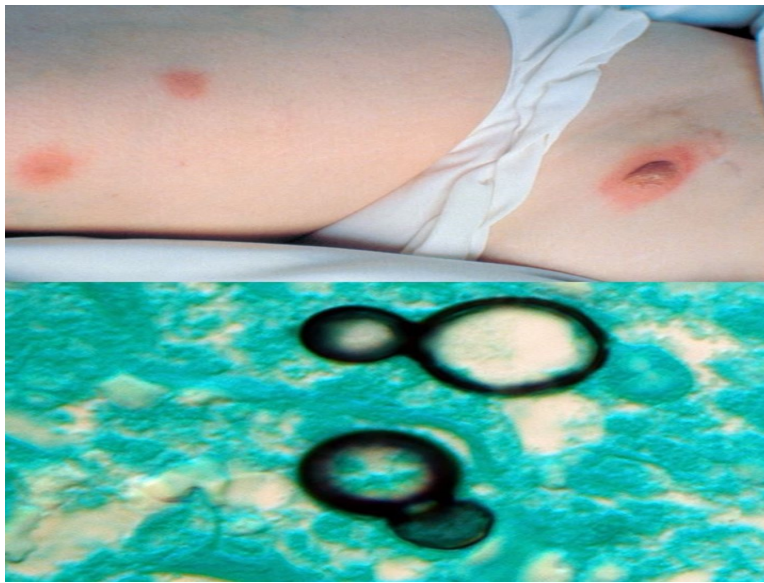
Treatment :

Liposomal amphotericin B is the antifungal agent of choice in acute pulmonary and disseminated histoplasmosis. Itraconazole is recommended for chronic cavitary pulmonary histoplasmosis.

Other options :

Option A - Cryptococcosis: Cryptococcus neoformans typically presents as a lung infection or meningitis in immunocompromised individuals. On SDA, Cryptococcus forms mucoid colonies, and Lactophenol blue would show encapsulated yeast, which differs from the presentation in this case.

Option C - Blastomyces causes North American Blastomycosis/ Chicago disease. Pulmonary involvement is the most common form followed by skin. Histopathological staining shows thick-walled yeast, with single broad-based budding (figure 8 appearance).



Option D: Mucormycosis typically affects immunocompromised patients and presents with invasive disease, often causing necrotizing infections in the sinuses or lungs. The fungal hyphae seen on Lactophenol blue are broad, non-septate, and ribbon-like, which does not match the culture description here.

Solution to Question 21:

The given clinical presentation of perianal itching and unilateral lower limb lymphadenopathy is suggestive of *Wuchereria bancrofti*.

It is a major cause of lymphatic filariasis, often presenting with lymphadenopathy and lower limb swelling. Its microfilariae are approximately 200-300 microns long and are characterized by the absence of nuclei at the tail tip. Clinical features include massive eosinophilia, hepatosplenomegaly, pulmonary manifestations like dry nocturnal cough, asthmatic wheezing, and dyspnea. It can also cause arthritis, glomerulonephritis, thrombophlebitis, etc. The classical features seen in lymphatic filariasis are usually absent.

Elephantiasis refers to the painful swelling of the legs and genital organs seen in late-stage disease.

Image showing elephantiasis-

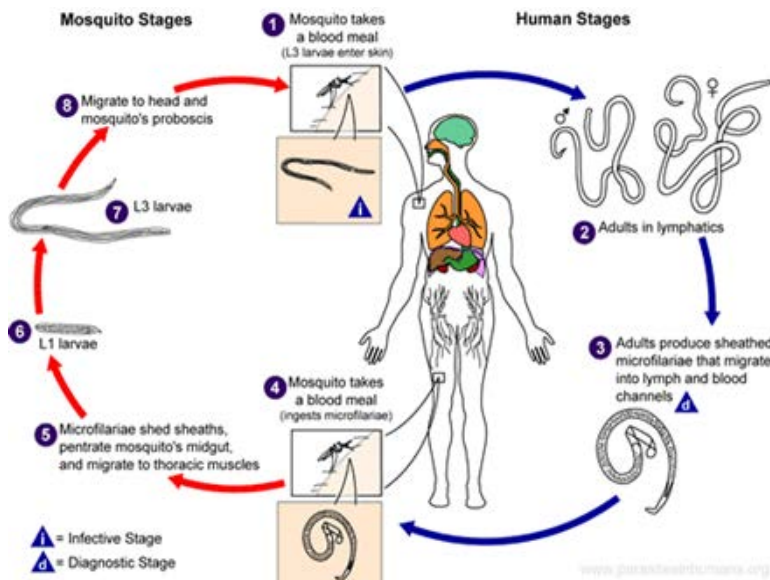


Investigations:

Demonstration of the microfilaria in peripheral blood, chylous urine, hydrocele fluid, etc is done. The specimen of choice is peripheral blood. Night blood samples between 10 pm and 4 am are preferred. This is because the microfilaria shows nocturnal periodicity.

DEC provocation test: In this test, a small dose of DEC is given 20-50 minutes before sample collection so that the microfilaria appears in the blood even during the daytime.

The image below shows the life cycle of *Wuchereria bancrofti*.





Solution to Question 22:

This clinical scenario along with the microscopic findings of gram[−]negative intracellular diplococci indicates infection with *Neisseria gonorrhoeae*. The treatment of choice for the infection among the given options would be Tab. Cefixime 400 mg Stat.

Gonorrhea is a disease acquired by sexual contact. The first step in infection is the adhesion of gonococci to the urethra or other mucosal surfaces. The incubation period is 2-8 days. The major symptoms include purulent urethral discharge and dysuria.

In males, the disease starts as acute urethritis presenting with purulent discharge. The patients may also have penile lymphangitis when the soft tissue inflammation is extensive. This infection may extend along the urethra to involve the prostate, seminal vesicles, and epididymis.

In females, the initial infection involves the cervix uteri and urethra which may extend to Bartholin glands, endometrium, and fallopian tubes. Pelvic inflammatory disease and salpingitis may lead to sterility. Rarely peritonitis may develop with perihepatic inflammation (Fitz-Hugh-Curtis syndrome).

The discharge is collected and subjected to Gram stain, revealing gram[−]negative intracellular diplococci within phagocytes. Stuart's transport medium is used. Cultures can be obtained from chocolate agar or Mueller-Hinton agar. Specifically, Thayer-Martin agar is used to isolate *N. gonorrhoeae*.

Single-agent therapy with ceftriaxone is preferred for the treatment of gonococcal infections.

According to CDC guidelines (2021), treatment of uncomplicated gonococcal infections is as follows:

- Weight <150 kg – Ceftriaxone 500 mg intramuscular (IM) in a single dose
- Weight ≥150 kg – Ceftriaxone 1 g IM in a single dose

The doses of ceftriaxone are higher than the previous guidelines as the gonococcal minimum inhibitory concentrations (MICs) are rising worldwide. The British BASHH guidelines also

recommend a high dose of 1g. Other cephalosporins like Cefixime can be given orally.

Concurrent chlamydial infection must also be treated unless it has been excluded by molecular testing.

First-line drug administered is

- Oral doxycycline 100 mg BD for 7 days

As per Suraksha clinic, syndromic management of Urethral discharge is done by Gray kit which contains Tab. Azithromycin 1gm stat and Tab. Cefixime 400mg stat. Partner treatment is also required.

Other options:

Option A: Acyclovir is given for viral infections. It has no role in gonococcal urethritis

Option B: Doxycycline is given alongside Ceftriaxone or cefixime to concurrently treat chlamydial infection along with gonorrhoea.

Option D: Benzathine Penicillin was once a treatment option for gonorrhoea, but it is no longer recommended due to widespread resistance.

Solution to Question 23:

The clinical scenario is suggestive of gonococcal urethritis caused by *Neisseria gonorrhoeae*. Thayer-Martin and modified Thayer-Martin media are used for the culture of this organism.

Gonorrhoea may manifest as a mucopurulent genital discharge which can be described as resembling the 'flow of seed' (semen). The discharge is collected and subjected to Gram's stain which reveals gram-negative intracellular diplococci within phagocytes. Stuart's media is used as transport media and Thayer Martin media is used for culture. Thayer-Martin medium contains vancomycin, colistin and nystatin.

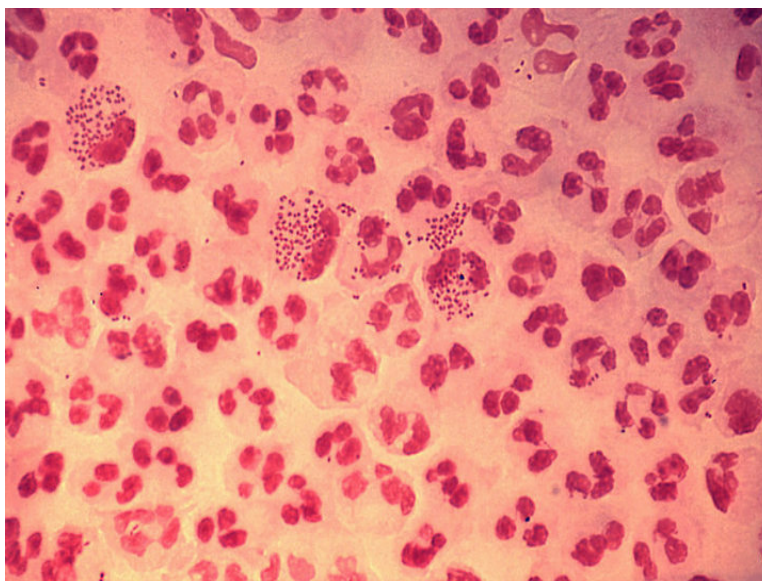
Gonorrhoea is a sexually transmitted disease where infection occurs due to the adhesion of gonococci to the urethra or other mucosal surfaces with the help of pili. By the third day of infection, the cocci have penetrated intracellular spaces and reached subepithelial tissue. The incubation period ranges from 2 to 8 days. The main symptoms are urethral discharge and dysuria.

In females: The infection begins in the cervix uteri and urethra and may spread to the Bartholin glands, endometrium, and fallopian tubes. The vaginal mucosa is usually unaffected in adults because the stratified squamous epithelium is resistant to cocci infection and also because vaginal secretions have an acidic pH. Rarely peritonitis may develop with perihepatic inflammation (Fitz Hugh Curtis syndrome).

In males: The disease starts as acute urethritis. This infection has the potential to spread down the urethra and into the prostate, seminal vesicles, and epididymis. Chronic urethritis may lead to stricture formation. Infections in the periurethral tissues can lead to abscesses and multiple discharging sinuses (watercan perineum)

Treatment: Ceftriaxone (250 mg, IM) single-dose + azithromycin 1 g PO, single-dose, or doxycycline 100 mg PO BD for 7 days.

The image below shows the Gram-stain of a urethral exudate showing typical intracellular gram-negative diplococci and pleomorphic extracellular gram-negative organism



Solution to Question 24:

The given clinical history of a patient with a fever, myalgia, gum bleeding, hepatosplenomegaly, and laboratory investigations showing anaemia, leukopenia, raised hematocrit and low platelets is suggestive of dengue with warning signs and hematocrit is the most important prognostic factor in this patient.

Hematocrit is the percentage of red blood cells in a given blood volume, and a rising hematocrit is a sign of hemoconcentration due to plasma leakage. As plasma leakage signifies the severity of the disease, hematocrit is monitored to assess prognosis.

Dengue is an arbovirus disease transmitted by *Aedes aegypti* and causes a clinical spectrum:

- Probable dengue: fever with myalgia, nausea, vomiting, leukopenia or a positive tourniquet test.
- Dengue with warning signs: abdominal pain, tenderness, persistent vomiting, fluid accumulation (pleural effusion, ascites), mucosal bleed, hepatomegaly(>2cm), rapid rise in hematocrit and fall in platelet count
- Severe dengue: severe plasma leakage causing dengue shock syndrome, respiratory distress, severe hemorrhagic manifestations, severe organ involvement (AST or ALT >1000U/L, CNS manifestations, AKI, pancreatitis, DIC)

Other options:

Options B and C- Although dengue causes anaemia and leukopenia, they do not determine the prognosis of the disease.

Option D- Reduced platelet count is a warning sign but does not correlate with plasma leakage. Therefore hematocrit monitoring would be a better option.